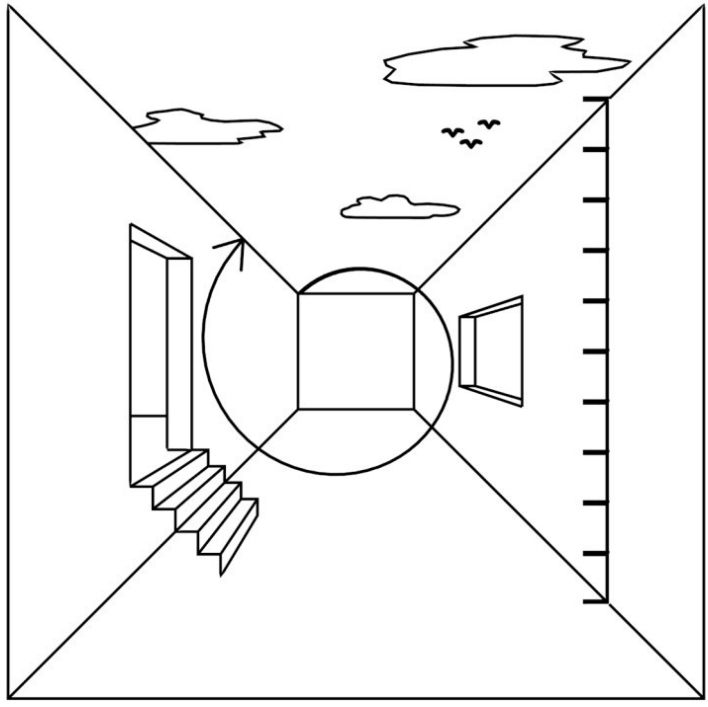
ועודות דעירובין

Eruvin Math



An Algebraic Analysis of Masechta Eruvin And Additional Notes

חיים אברהם גולדסאן
Bumy Goldson

Eruvin Math

by Bumy (Chaim Avrohom) Goldson
Beitar Illit, Israel

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ועודות דעירובין – חיים אברהם גולדסאן
ביתר עילית, ארץ ישראל

ה' תשפ"ו

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בס"ד

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Dear Friends,

I have seen portions of the book "Eruvin Math" by Rabbi Chaim Avrohom (Bumy) Goldson. The Tractate Eruvin is one of the most difficult ones to grasp. Its complicated cases are difficult to picture and many topics depend on intricate mathematical calculations.

Rabbi Goldson with his mastery of mathematics and ability to present diagrams and pictures makes the topics much more easily comprehensible. He also uses his mathematical expertise to explain various topics shedding new light in solving the difficulties that are inherent in their true understanding.

I stand in awe of Rabbi Goldson's knowledge of both Torah and mathematics and commend him for a very interesting and quality presentation.

I pray that Hashem Yisborach bless him and his family with life, health and the wherewithal to continue to merit the community.

Sincerely,
With Torah blessings

Rabbi Zev Leff

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About This Sefer

I am not an Eruvin expert.

And I am not a math expert.

I'm just trying to learn masechta Eruvin and I'm finding many applications of high school math. I started keeping elaborate notes, and my chavrusa suggested I put it together in a pamphlet. So I did. So the pamphlet grew from the originally intended 3 page fold-out to a full 500 page book. Very nice.

What happened was that I got stuck on something, some piece of Gemara, some Tosfos, and I had to dig around, various Rishonim, Achronim, Artscroll, Mesivta. Sometimes I found answers and sometimes I didn't. I clarified the sugya as much as I could and I wrote up notes to make it easier for myself for the next time I went through.

This was not written as a peirush, per se. In other words, I didn't sit down and think "OK, how can I explain Eruvin to the uninitiated?"

This is a collection of my notes as I went through the masechta for the first time. There are probably many gaps and obvious omissions throughout. I'm writing this up into a sefer in order to clarify to myself the various sugyos that I delved into and I hope that some people out there will find it useful and enjoyable. If you look at it as personal notes, that would explain the somewhat tedious and technical nature of this sefer, in case you were wondering why we need to calculate everything and drag everything out unto the Nth degree...

Like they say the difference between Tosfos and Tosfos Harosh is that Tosfos was written as shorthand notes for the students to remember what they learned, as opposed to Tosfos Harosh was written more as a peirush for the masses, so it's much clearer.

So too I guess is how this was written. I wrote it as my own personal notes so there's a lot of unspoken assumptions and understandings. As putting it together came along, and I saw it becoming an actual book, I tried to guide the style more to be clear to others, but often not.

I may have missed an obvious Rashi or Tosfos, or even misread the plain reading of the Gemara; it happens. This is not a professional sefer in that sense. Just some fun geshmake ideas and personal notes which I think many people can gain from.

I'm not presenting necessarily THE pshat; it's just a working pshat that we (my chavrusa and I) used to get through the sugya.

You may disagree with many of my interpretations and that's fine. This is not intended to be any kind of definitive peirush of the Masechta. It is a lot of exploration and suggested possibilities.

Please don't take my word for anything, not the math, not the pshat, and certainly not the halacha. A good chunk of this sefer came about by me coming across some idea, and checking it out myself and finding that something wasn't adding up. The

point here is to CHECK IT ONE'S SELF. Use this sefer as a springboard for your own learning.

Neither is this sefer comprehensive. Pirush Chai does a good job, as do Artscroll and Mesivta, etc. Don't expect all topics to be represented.

I focused on specific sugyos with mathematical depth and/or other interesting points that I felt my kochos could bring out something unique or a chiddush, or ideas that I really related to, was excited about, or enjoyed.

This sefer is not meant for a daf yomi style pace since there are many daf I didn't touch at all and there are some that I spent many pages on (see, for example, daf 14, 56, and 83).

This should be seen more of a companion as you go through and learn the masechta, with interesting or helpful comments on the side.

This is intended for someone who is going through the masechta but I tried for some pieces to give enough background information so that anyone can follow, even if they never learned Eruvin before. For each piece that could be understood by itself without going through the Gemara first, I put a "*" symbol on the right side of the "opening gate" symbol that each piece starts with.

Limeisah, as I was doing chazara on Eruvin, I got stuck a few times and I found the smoothing out of the sugya in this sefer very helpful.

I tried to make the ideas available and accessible to those not learning the sugya, and perhaps even those not familiar with talmud study, and those without a strong background in math, although these factors will greatly enhance appreciation of the pieces, methinks.

Perhaps a good use of this sefer would be a Yeshiva High School, where they anyway are learning this math stuff without any real world application, and at the same time are trying to wrap their heads around how Gemaras work.

Overly technical equations that might be beyond the interests of the average reader are placed against grey background or in an appendix in the back.

I like to quantify everything and put things into formulas

Sometimes, to make a sugya more clear to myself, I wrote up a chart/table thingy. The point of the chart is to organize the flow of the Gemara into a visually simple format, to get it clear in the mind and to make it easy to remember.

The style of the sefer is somewhat unique. The way I organize on paper is with interwoven text and pictures; I had to find a way to do that in printed form.

The "voice" of the sefer is written in a very candid style, like the way I would speak to a chavrusa, or would want a chavrusa to speak to me. In particular by the

appendixes, I threw out all pretenses of formality out the window and took great liberty with the candidness there, assuming that if you had made it that far in the sefer with me you would be forgiving enough to go with me to the end... (Agav, there were a lot of ideas squashed into the appendixes, so sometimes it might seem a little... squashed.)

There's a bit of an inconsistency of style throughout the sefer since the style itself developed as I wrote, for example, writing words in Hebrew vs. transliteration.

Regarding transliteration, basically, words that transliterate easily and unambiguously are done so. I didn't really develop a consistent style of transliterations until I was well on in the sefer. Towards the end of the sefer (besides the appendixes) I left many words untransliterated.

I generally chose clarity of expression over traditional convention, sometimes having to make up my own words (circlify) or combine multiple languages in a single word like דוחקness.

For spelling, I chose for each example what makes more sense to me, as long as it is an acceptable way (exept on page 37, wir I kept the footnote in my own spelling systm. I had a very strong yetzer hara to ryt the entyr sefer in cut speling, but I held myself bak in the intrest of clarity. This is one of those batls Im just gonna hav to leav to other pipl). For example, color (US) and travelling (British). Also, I put the period after the closing quotation marks "like this". Just makes more sense to me, and seeing how there's no official body that controls the English language, I figured I could do what I want.

I refer to Tosfos in the singular, even though there were hundreds of baalei tosfos and the Hebrew miforshim refer to Tosfos as "them". This (to refer to Tosfos in the singular) is a style that is somewhat common in Yeshivos. One of the justifications for this is that there was generally only one Baal Tosfos who wrote any particular piece of commentary.

Initially, I wanted to call the sefer "Vi'Odos Di'Eruvin", which is a play on words on the Gemara's play on words when Rebbe said a certain shiur is 2 eggs "Vi'od", "and a little bit more" (see page 314). So I wanted to add "a little bit more" to the discussion on masechta Eruvin, plus (!) the implication of math. But despite the poetry, I thought that the name is too difficult to remember, or spell correctly in a search engine. Nor does it give the casual passerby any inkling of what the sefer is about. So I went with straight simplicity and got "Eruvin Math".

Programs used: Microsoft Word and Excel, Sketchup, Blender, Photoshop, MS Paint, Inkscape, Graph, Python, PDF24.

I want to thank all my friends and family who supported me and encouraged me to continue with this project to the end, especially my wife. Thank you Rabbi Zev Leff for your beautiful letter of recommendation.

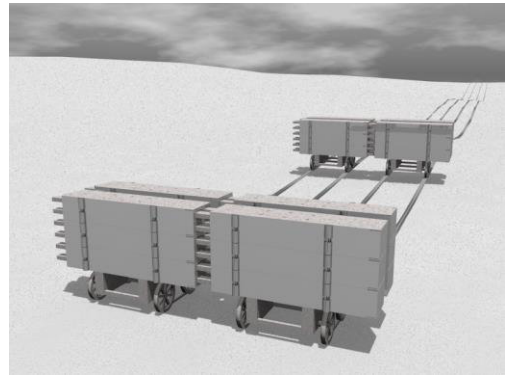
Please send any feedback, mistakes, corrections, suggestions, comments, questions, complaints, criticism, and compliments by email to bumygoldson@gmail.com

Eruvin Math

*

ב. - Source for 16 Amah Width For Reshus Harabim

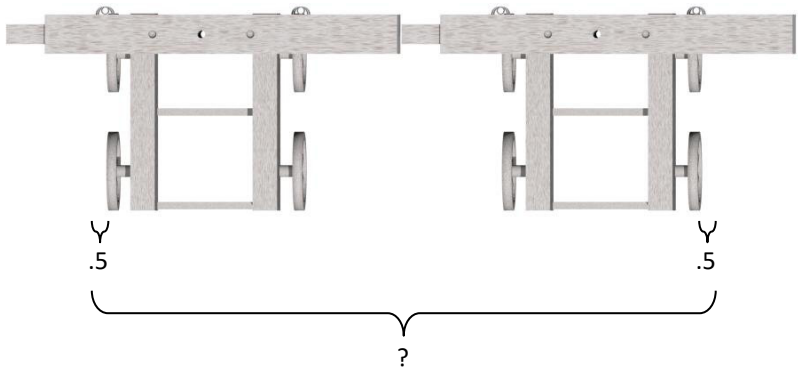
Rashi[1] brings the Gemara in Masechta Shabbos that all the laws of the melachos of Shabbos are learned from the Mishkan.



The laws of "carrying", specifically, are learned from the wagons that were used to transport the beams of the Mishkan. The wagons travelled in pairs. As the wagons travelled, they would effectively be delineating a road, a reshus harabim, with the space they took up.

Therefore, the width of a reshus harabim is the total width on the desert floor that the wagons took up, measured from the outer wheels of one wagon to the outer wheels of the other wagon, plus one more amah for a Levi to pass by to make sure the beams stayed in place, half an amah on one side and half on the other.

Thus, we have to take into account the width of the wagons, the length of the beams, and 2 spaces of half an amah each at the side of the wagon wheels.



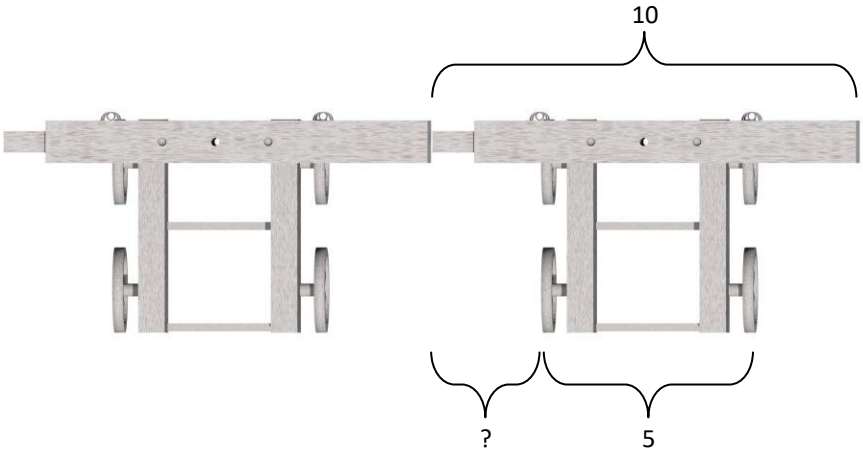
This is a top view. Most of the beams have been removed for simplicity.

[1] ד"ה מבוי

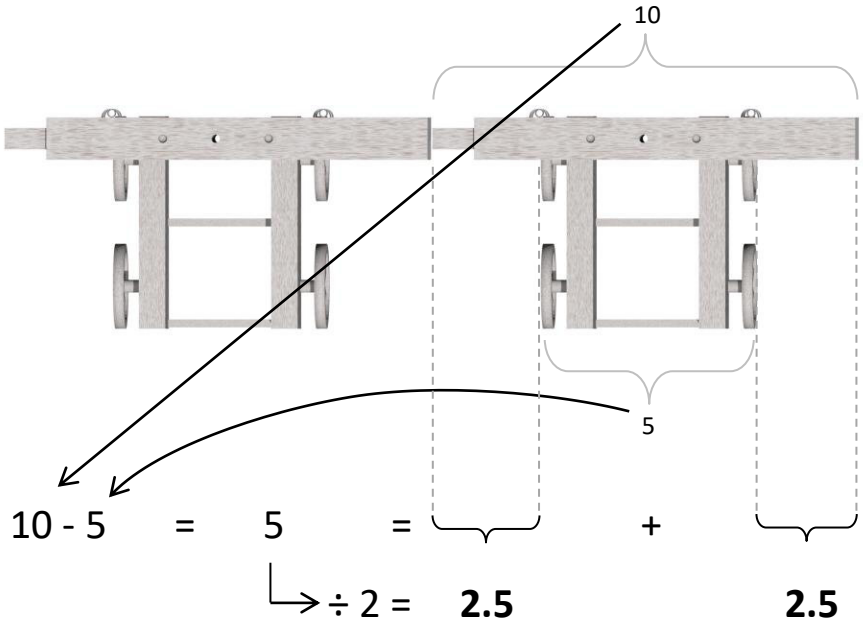
First, let's figure out how far the beams stuck out past the wheels:

The wagons were 5 amos wide (measured from the outside of the wheels).

The beams were 10 amos long.



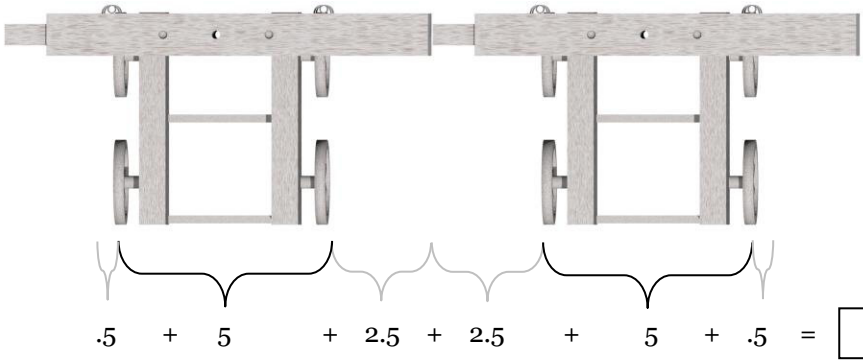
If we subtract the wagon width from the beam length we'll get the total amount that each beam juts out beyond the wheels. If we divide that by 2 we will get the amount each beam juts out on each side.



Now that we know how far each beam jutted out beyond the wheel, we can add all the relevant spaces together to get the full width.

- Half an amah on each side for a Levi to stabilize the beams .5 + .5 = 1
- 2 wagon-widths of 5 amos each 5 + 5 = 10
- Beams sticking out towards each other, form each side 2.5 + 2.5 = 5

16



That is the source for the 16-amah width for a reshus harabim.

*

ב. - Every Doorway Associated with Beis Hamikdash

It is fascinating that the Gemara's definition of "a doorway" is derived either from the Beis Hamikdash or from the palaces of kings. Every time a Shabbos-conscious Jew walks through a doorway, especially on Shabbos, he compares this doorway with those of the Beis Hamikdash and of kings.

Yirmiyahu[2] says, "It shall be that if you truly listen to me, the word of Hashem, not to bring a burden into the gates of this city on the Shabbos day... then kings and princes... will enter the gates of this city..."

I take this to mean that *anyone* who keeps this mitzva properly will be considered a king in the eyes of Hashem, and not a burden-bearing slave.

See Rav Hirsch's piece on "Carrying" in Appendix W.

[2] 17:24-25

*

ב. - Eruvin Is So Simple?!

The Gemara differentiates between the laws of Sukkah and the laws of Eruvin, saying, "Sukkah has a lot of laws[3]... Eruvin has a few laws[4]..."

I guess what the Gemara means is that the headings of laws are more by Sukkah than by Eruvin. Rashi[5] has a similar pshat.

That might very well be what the Gemara means because, after all, the Gemara is talking about the Mishna, which is more of a table of contents than a law book.

Or, perhaps more simply, our Mishna has only one דין in it whereas the Mishna in Sukkah has several. This would be more of a stylistic answer, as Rashi there in Sukkah[6] says that one should teach his students in a concise manner.

ב. - Tosfos Sukkah ≃ Rabeinu Peretz

From the words of Tosfos, I had trouble understanding what was the question on Rashi. After seeing Rabeinu Peretz, I thought that perhaps that's what Tosfos meant. So, let me explain Rabeinu Peretz and afterwards I'm going to try to put that pshat into the words of Tosfos.

This is the flow of the Gemara according to Rashi as explained by Rabeinu Peretz:

The Mishna by us, when talking about a problematic mavoy, teaches us how to fix it, the תקנתא.

---

[3] literally: a lot of things
[4] literally: a few things
[5] ד"ה פסיק ותני. Rashi in סוכה ב'. ד"ה מבוי לא מפיש מילי seems to be addressing my problem when he writes "באותה משנה".
[6] ב'. ד"ה סוכה דנפישי מילתה

The Mishna in Sukkah, when talking about a problematic sukkah, teaches us the law, that it is pasul.

According to Rashi, the Gemara at this point really has two questions, each one answered by the Gemara's two answers.

Q&A #1

> Q: Why didn't the Mishna teach mavoy with a lashon of "pesulah"?
>
> Subject under attack: Mavoy
>
> Assumption of term priority: Pesulah
>
> A: We can't use the word "Pesulah" in reference to a topic we never heard of before. Since the Mishnayos are the first place we're hearing about mavoy and its laws, we teach its "takanta".

Q&A #2

> Q: Why didn't the Mishna teach Sukkah with a lashon of "takanta"?
>
> Subject under attack: Sukkah
>
> Assumption of term priority: Takanta
>
> A: We can't use the word "takanta" in reference to Sukkah since it has many laws and if we spoke about takanta we would have to enumerate every takana for every possible problem. So, we just lumped everything together and labeled them all "pasul".

Rabeinu Peretz finds it difficult that the Gemara would be providing two different types of answers, each one's subject under discussion and assumptions of term priorities opposite from each other.

Now, to put the pshat of Rabeinu Peretz into the words of Tosfos:

> Tosfos: "Rashi explains that it's possible to teach "pasul" [by sukkah]... even though... it could've said "takanta" [but it chose "pasul" because that is the preferable term].
>
> In the איבעית אימא, [the Gemara says that] even [sukkah]... could've been taught with a lashon of "takanta"... [These words of the Gemara] don't read well according to Rashi's peirush because even the first [answer could've]... said "takanta". All the Gemara meant [in the first answer] was that it's possible to say "pasul" [even though it could've said "takanta"].

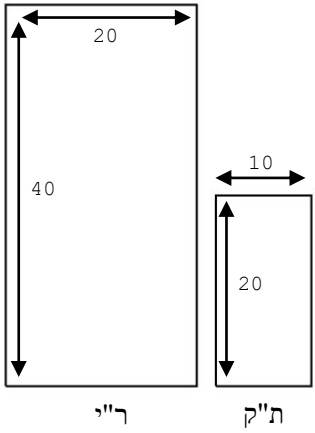
In other words, according to Rashi, the first answer of the Gemara already acknowledged that sukkah could've been taught with the term "takanta". The Mishna just chose not to, for the reason given in the Gemara. How then can the Gemara follow this with an "alternative" answer that seems to introduce the possibility of sukkah being taught with the term "takanta"?

Isn't this the same question as Rabeinu Peretz?

*

ב: - **Door Width:Height Ratio**

The Gemara is discussing the allowable (or necessary) width and height of a mavoy doorway.



The Tanna Kama holds that the doorway must be no greater than 20 amos tall and no wider than 10 amos wide[7], like the entrance to the Haychal in the Beis Hamikdash.

Rebbe Yehuda holds that it can be 40 tall and 20 wide[8], like the Ulam of the Beis Hamikdash.

(This whole discussion is according to Rav, who says that the tannayim are learning from the entrances to the Haychal and the Ulam.)

[7] These are the maximum dimensions of a mavoy that could be rendered permitted to carry within it when a korah (beam) is placed along the top of its entrance.

[8] At this stage of the Gemara. Later, an amorah will hold $13\frac{1}{3}$ wide, ע' תוס' ד"ה אמר.

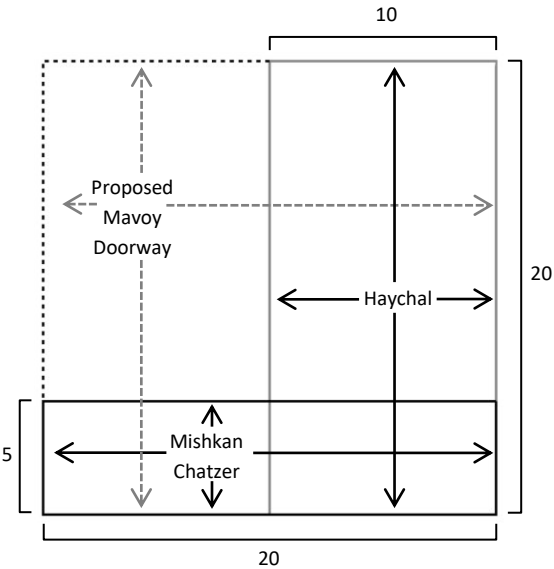
The Gemara then asks, "[Why learn the dimensions from the Haychal or Ulam?] Learn them from the gateway of the Mishkan's courtyard, which was 5 amos high and 20 amos wide!"

The Gemara asks this question both according to the Tanna Kama and according to Rebbe Yehuda.

Let us discuss the Tanna Kama first.

We had said that the Tanna Kama learns the doorway dimensions from the Haychal, which was 20 amos high and 10 amos wide.

The Gemara now asks, "Why doesn't the Tanna Kama learn from the chatzer of the Mishkan, which was 5 amos high and 20 amos wide?"



According to Rashi[9], the Gemara's question is why are we limiting the width to 10 amos? Learn from the Mishkan's chatzer that it can be as wide as 20! In other words, from the two sources, we should learn that the doorway can be 20 amos high *and* 20 amos wide!

According to this understanding, it becomes a little difficult rendering the words of the Gemara when it says, "Just like by the Mishkan's chatzer it was 5 high by 20 wide, so too by mavoy, it should be 5 high by 20 wide!" According to Rashi, the Gemara should not mention the words "5 high" at all. Not by chatzer since we are not learning from the chatzer's height, and not by mavoy since we are not changing the allowable height of the mavoy.

Because of this difficulty, some[10] take out these words in Rashi's reading.

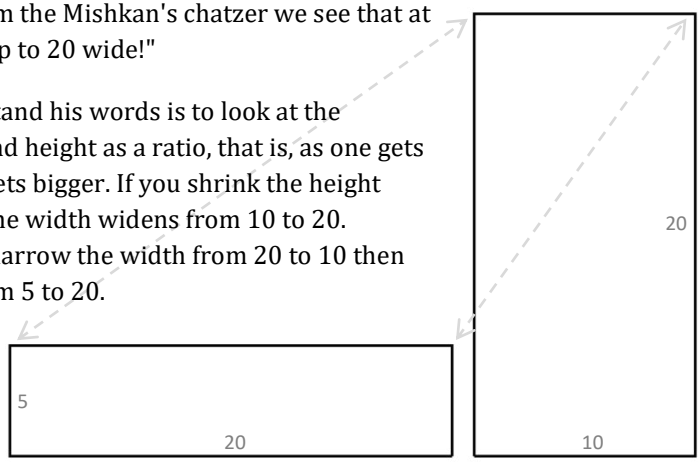
[9] ד"ה לילפו

[10] עיין בהערה ב"עוז והדר" על מילים אלו

However, the Riva, brought by the Rashba, somewhat justifies the Gemara's lashon with Rashi's pshat, like so: "If by the Mishkan's chatzer, where it's much wider than its height, and still it can be as wide as 20, then by the mavoy, where it's not overly wide compared to its height, certainly it can be 20 wide!"

This doesn't answer the second use of the words "5 high"[11] but we can answer that it is a borrowed term, i.e., since we use that term in the beginning of the line we use it at the end as well. Rashi gives a similar pshat on ד"ה כאן שעירבו :ז, where the Gemara's terms are inexact, but parallel to each other. See there.

Tosfos[12] brings the Ritzba who understands the Gemara's question differently than Rashi. According to him, each source for the mavoy's dimensions is a unique case that is independent of the other one. To paraphrase: "Good, at 20 high you can only go 10 wide, but from the Mishkan's chatzer we see that at 5 high you can go up to 20 wide!"



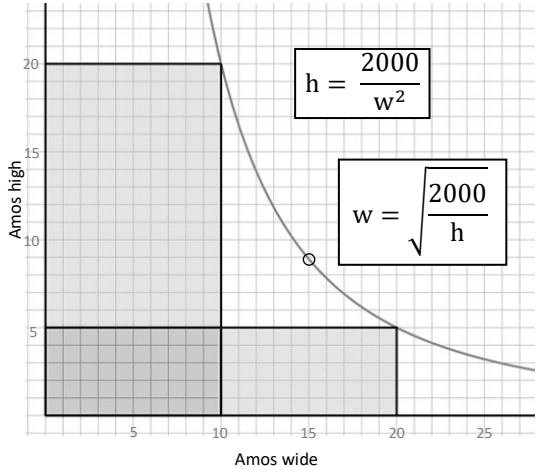
One way to understand his words is to look at the doorway's width and height as a ratio, that is, as one gets smaller the other gets bigger. If you shrink the height from 20 to 5 then the width widens from 10 to 20. Conversely, if you narrow the width from 20 to 10 then the height rises from 5 to 20.

It seems to me that the סברה is that it has to do with how obvious the doorway is. When the walls are close to each other, you can easily see them as part of the doorway and then the height need not be so low in order to be conspicuous. When the walls are far from each other, they are not so obvious and the height needs to be lower down so that you will notice it as part of the doorway, thus enabling you to "see" the whole doorway better[13].

[11] Where we say, "So too by mavoy, it needs to be **5 high** by 20 wide"
[12] ד"ה ואיבעית אימא
[13] Especially when there is a beam across the top

What would be the דין with doorways in between these dimensions, or beyond these dimensions? Could we interpolate and extrapolate from these two cases an array of kosher doorways? For example, what if the doorway was 15 amos wide, how high can we put the korah? (Or, put another way: What's the maximum height of a mavoy that would allow a width of 15 amos?)

One way I thought of looking at this problem is to notice that when we doubled the width of the doorway (when we went from the Haychal to the Mishkan's chatzer) we halved the area (the width times the height) of the doorway. Or you could say when we doubled the width we quartered the height. The technical term for this is "inverse squared". This would create a curve on a graph that would show us for any given width how high the korah can be.



h = height
w = width

According to this approach, a doorway that was 15 amos wide could have the korah up to $8\frac{8}{9}$ amos high.

Keep in mind that we cannot have the mavoy doorway be lower than 10 tefachim nor narrower than 4 tefachim[14].

If so then the widest the mavoy can be is about 34.6 amos $(\sqrt{1200})$.

If the mavoy was 4 tefachim wide then we are saying that the korah can be 4,500 amos high! That doesn't seem so מסתבר.

[14] גמ' ה.

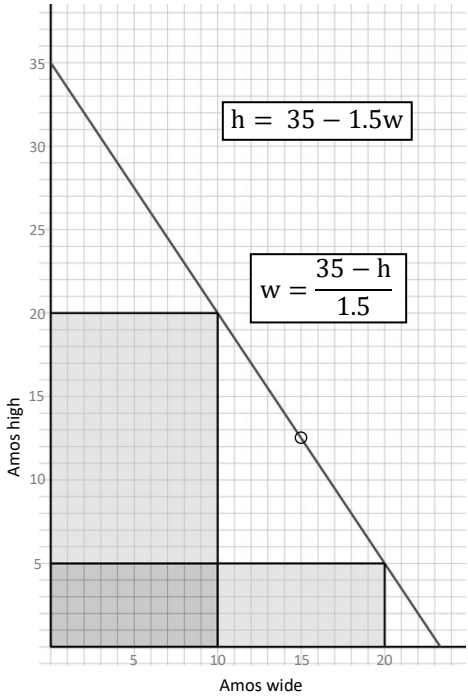
My friend, Yishai Rasowsky had a different approach. Draw a straight line between and beyond the two kosher doorways as they appear on a graph, like so:

h = height
w = width

According to this approach, a doorway that was 15 amos wide could have the korah up to $12\frac{1}{2}$ amos high.

Using these formulas, we can find the highest and widest doorways by plugging in the values of the narrowest and lowest doorways, respectively.

We said the narrowest doorway is 4T wide, which is $\frac{2}{3}$ of an amah. That gives us a beam height of 34A.

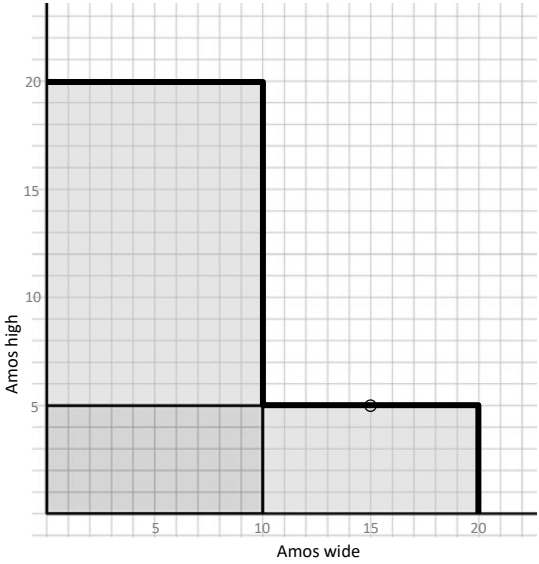


We said the lowest doorway is 10T high, which is $1\frac{2}{3}$A (10 ÷ 6). That gives us a width of $22\frac{2}{9}$A

These last two approaches assume that there's a gradual ratio between the doorway height and width.

What seems to me to be more mistaber is to say that it's not a smooth line or curve at all but rather we can only go as far as we know. That is, when the doorway is wider than 10A we can only put the beam 5A high, because that is the only permitted height we know of for wider than 10A.

For a doorway that is less than 10A wide we can only go up to 20A high for the same reason, since we only have a source for that height and no more.

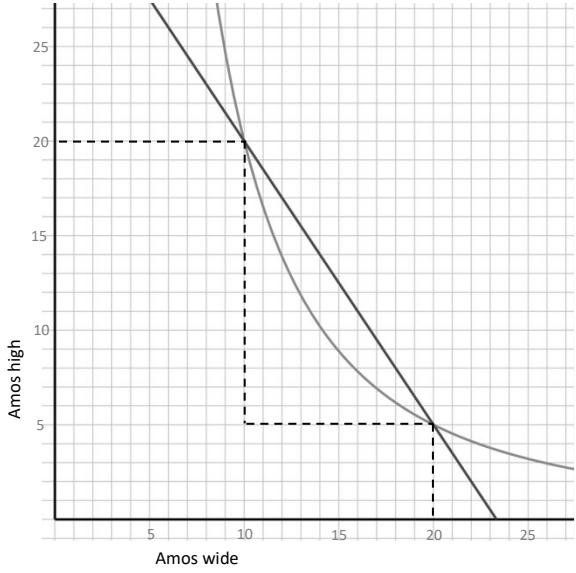


According to this, if we have a doorway that is 15A wide then the korah beam cannot be more than 5A high.

At 4T wide the beam can be 20A high.

A doorway 10T high can be up to 20A wide.

In summary, I have 3 interpretations for what the Ritzba might mean.

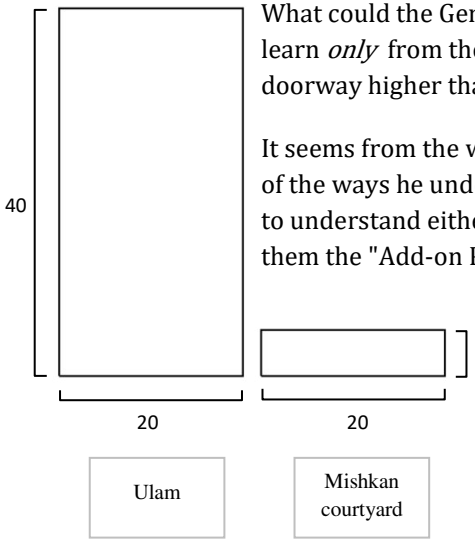


So far, we have discussed the opinions of Rashi and the Ritzba.

There is another way to understand the Gemara's question. When the Gemara suggested learning from the chatzer of the Mishkan, it meant learn *only* from there and not from Haychal! In other words, when you have two sources from which to learn a דין, and those sources contradict each other, how do we know from which one to learn?

It would seem to me that this way of understanding the Gemara's question is indicated in the fact that the Gemara asked its question on Rebbe Yehuda as well.

Rebbe Yehuda learns from the Ulam that the doorway of the mavoy can be 40A tall and up to 20A wide. The Mishkan's courtyard entrance is 5 amos tall and 20 amos wide.



What could the Gemara be asking on Rebbe Yehuda if not to learn *only* from the Mishkan and to thereby passul a doorway higher than 5?

It seems from the words of the Gilyon Hashas that this is one of the ways he understood the Gemara's question. He seems to understand either like Rashi or like this way. We can call them the "Add-on Pshat" and the "Exclusive Pshat".

What the Gilyon writes is in the form of a question on the Gemara.

To paraphrase the Gilyon's question: "What can the Gemara mean by asking on Rebbe Yehuda? Nothing is gained by learning from the chatzer of the Mishkan because we already know the mavoy can be 20 wide from the Ulam. And besides, we need to learn from the Ulam to know we can go above 20 amos."

From the first part of his words, where he says, "Nothing is gained by learning from the Mishkan since we already know from Ulam that the mavoy can be 20 wide" it sounds like he's learning like Rashi. That is, that each source is **adding** to the other (what we called the "Add-on Pshat").

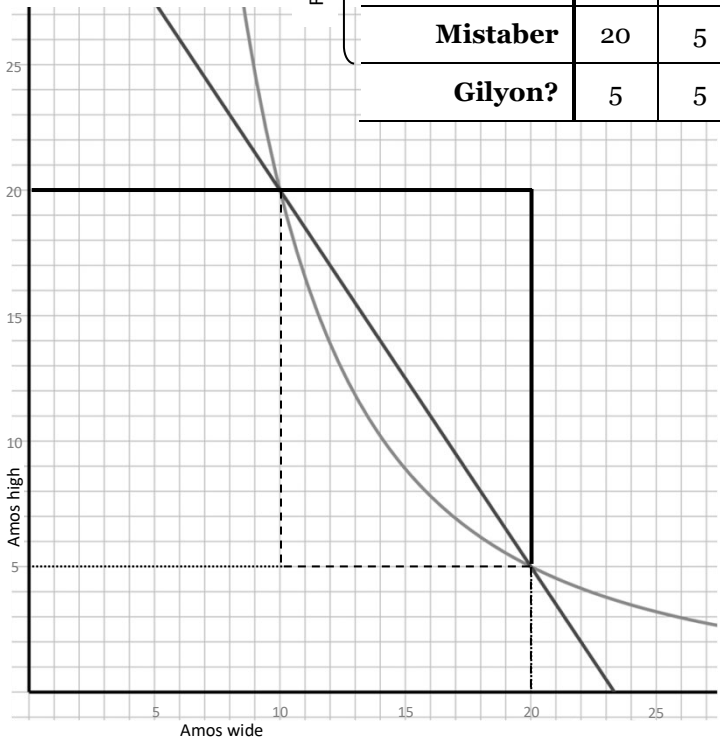
However, from the second part of his words, where he says, "Besides, we need to learn from the Ulam that the mavoy can be higher than 20 amos" it sounds like he's learning the "Exclusive Pshat". That is, learn the dimensions from the Mishkan and **not** from the Ulam!

Perhaps.

In summary of all the pshatim:

	Allowable height for mavoy			Graph Shape
Given width	**10**	**15**	**20**	
Rashi (+Tos')	20	20	20	
Bumy	20	$8\frac{8}{9}$	5	
Yishai	20	12.5	5	
Mistaber	20	5	5	
Gilyon?	5	5	5	

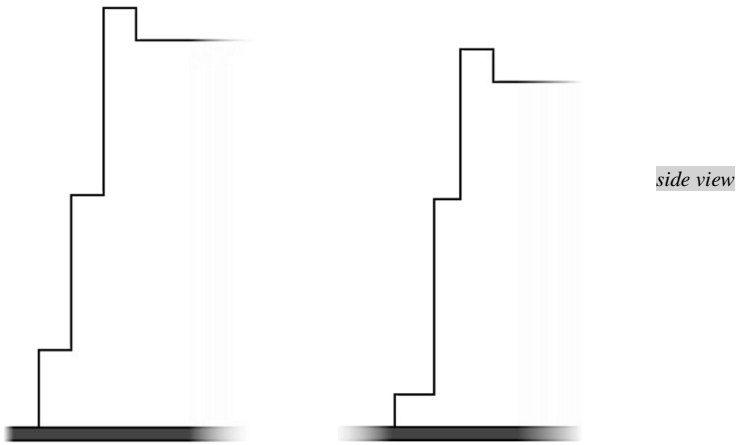
Ritzba



ד. - **Mizbeach Amos**

I was a little confused when I saw this Gemara and I finally figured out that the Gemara is quoting Yechezkel, which is talking about the 3rd Beis Hamikdash, whereas Rashi is referring to the Mishna in Middos, which is talking about the 2nd Beis Hamikdash.

The two different mizbeachs have slightly different dimensions.



side view

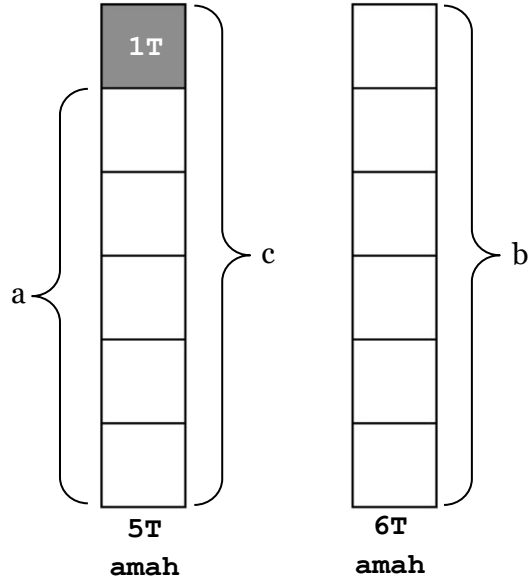
Although the actual number of amos differs by both, the *types* of amos used is going to be the same by both mizbeachs. Each part of the mizbeach uses a different kind of amah, sometimes a 5-tefach amah and sometimes a 6-tefach amah. Whatever part uses whatever amah by one mizbeach, the other mizbeach will use the same type of amah by the same corresponding part.

Let's see the passuk inside, find the key components of the mizbeach, and then map out the measurements for Yechezkel's mizbeach (from the 3rd Beis Hamikdash) and for the mizbeach from Middos (2nd Beis Hamikdash), while paying attention to whether we are using 5-tefach amos or 6-tefach amos.

יחזקאל מ"ג:י"ג "וְאֵלֶּה מִדּוֹת הַמִּזְבֵּחַ בָּאַמּוֹת אַמָּה אַמָּה וָטֹפַח וְחֵיק הָאַמָּה
וְאַמָּה רֹחַב וּגְבוּלָהּ אֶל שְׂפָתָהּ סָבִיב זֶרֶת הָאֶחָד וְזֶה גַּב הַמִּזְבֵּחַ:"

(Artscroll translation) "Now these are the dimensions of the Alter in cubits, each cubit being a cubit and a handbreadth - except for the base of one cubit [wide]; the cubit of the [ledge's] width; and the border [pieces] upon its edge all around of one span each; and this [too] for the top of the [Inner] altar."

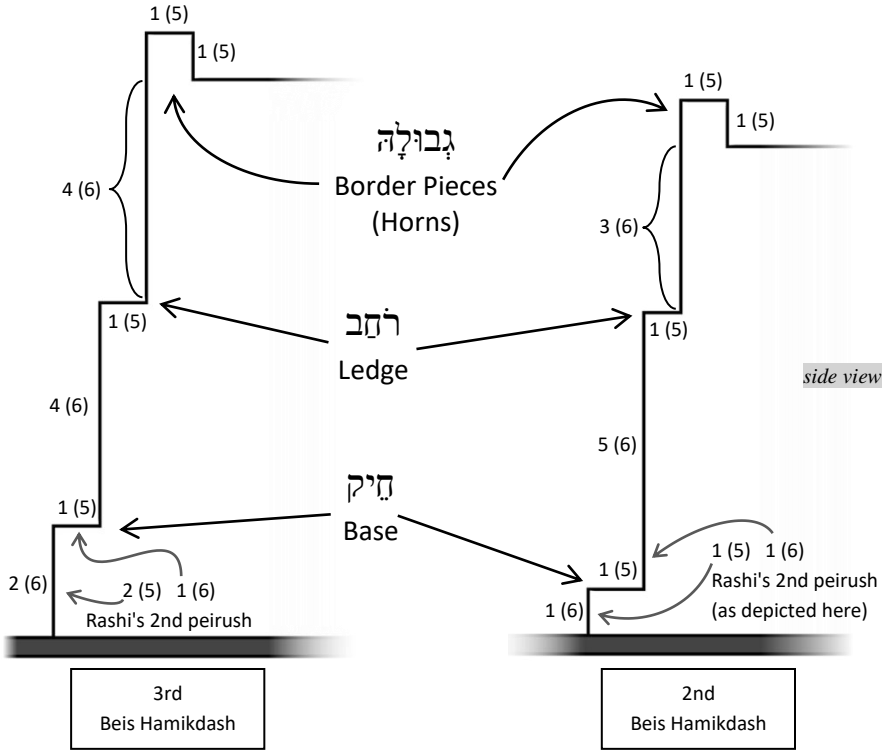
I boxed in the main components that we're going to focus on. But first let's look at Rashi[15] alongside the picture comparing the two different kinds of amos. The amah on the left is made up of 5 tefachim (not including the grey square); on the right is a 6-tefach amah.

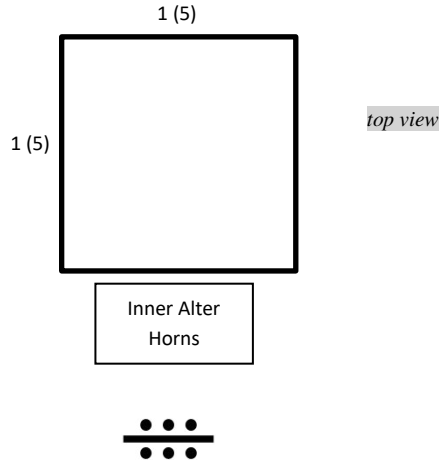


Rashi: "The (5T) amah[a] [when compared to a regular] (6T) amah[b] [would need] a (5T) amah and a tefach[c] [to complete the regular amah[b]]."

a = 5T amah
b = 6T amah
c = 5T + 1T

[15] ד"ה באמה אמה אמה וטופח


1 (5)
1 (5)
גְּבוּלָה
Border Pieces
(Horns)
4 (6)
1 (5)
1 (5)
3 (6)
1 (5)
רַחַב
Ledge
1 (5)
side view
4 (6)
5 (6)
חֵיק
Base
1 (5)
2 (6)
2 (5) 1 (6)
Rashi's 2nd peirush
1 (5) 1 (6)
Rashi's 2nd peirush
(as depicted here)
1 (5)
1 (6)
3rd
Beis Hamikdash
2nd
Beis Hamikdash


1 (5)
1 (5)
top view
Inner Alter
Horns

*

ד: - Shechina Height

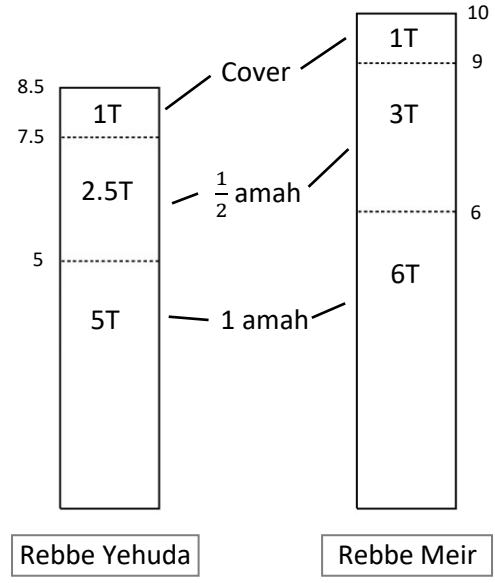
The Gemara begins on עמוד א' with a statement from Rav Chiya bar Ashi in the name of Rav that "Mechitzos", among other things, are known solely through tradition, הלכה למשה מסיני. The term "Mechitzos" in this context refers to the fact that they need to be at least 10 tefachim high.

The Gemara then brings a machlokes between Rebbe Meir and Rebbe Yehuda regarding the height of the Aron.

The passuk[16] describes the Aron as "1.5 amos tall". How many tefachim is that?

According to Rebbe Meir, there's 6 tefachim per amah for the keilim in the Beis Hamikdash. The Aron itself is therefore 9 tefachim tall (6T * 1.5 amos) plus 1 tefach for the cover for a total of 10 tefachim.

According to Rebbe Yehuda, there's only 5 tefachim per amah for the keilim in the Beis Hamikdash. The Aron itself is therefore 7.5T tall (5T * 1.5A) plus 1 tefach for the cover for a total of 8.5T.



[16] שמות כ"ה:י

Rashi[17] brings a Gemara in Sukkah that proves that the Shechina never comes below 10 tefachim, from the fact that one passuk says, השמים שמים לה' והארץ נתן לבני אדם, which means that Hashem is in His domain and man is in his own. At what point do these domains meet? Well, if we find an instance where Hashem comes down to a known height then we know that that is still considered שמים, i.e., a separate domain from ארץ.

The passuk by the Aron says, "ודברתי אתך מעל הכפורת". So we see the Shechina comes down to at least the top of the Aron. So if we know the height of the Aron then we know what height is considered "שמים".

According to Rebbe Meir, the height is 10 and so that is the דאורייתא source for the height that is considered a separate domain.

It is on this opinion of Rebbe Meir that the Gemara asks why do we need to rely on a tradition of הלכה למשה מסיני when we could infer a mechitza's height of 10 tefachim from the Aron.

The Gemara doesn't have a question on Rebbe Yehuda because according to him the top of the Aron was only 8.5T above the ground. Since 10T cannot be learned from the Aron therefore we need to rely on the הלכה למשה מסיני.

The Rashash asks, Rebbe Yehuda agrees that the minimum height for a mechitza is 10T. If so, that means that the Aron is not a valid source from which to learn the height of mechitza (since according to him the top of the Aron was only 8.5T above the ground). If so, then Rebbe Meir cannot either learn from the Aron, since we see that the Shechina could theoretically come down below 10, as Rebbe Yehuda says!

In other words, you can't say that something is a source for a din if that din is still true even when that something isn't saying it.

For example, you can't say, "This light switch controls the light" if the light remains on even when the light switch is off.

[17] ד"ה ארון תשעה וכפורת טפח

One could answer the Rashash's question by saying that Rebbe Meir and Rebbe Yehuda argue about this point as well, about whether you can learn from Aron or not.

Perhaps the Rashash doesn't suggest this because that would be unnecessarily expanding the machlokes between the two tannayim.

It is interesting that according to Rebbe Yehuda (and Rebbe Meir, according to the Rashash) the Shechina can and does come down below 10T, although this seems to contradict the drasha in Sukkah.

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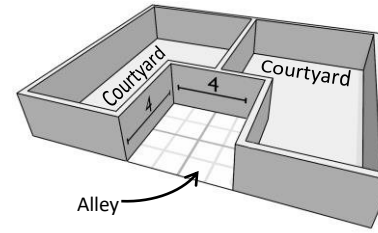
ה. - Tiny Alley Doorways - Diagonal of 4 Tefachim

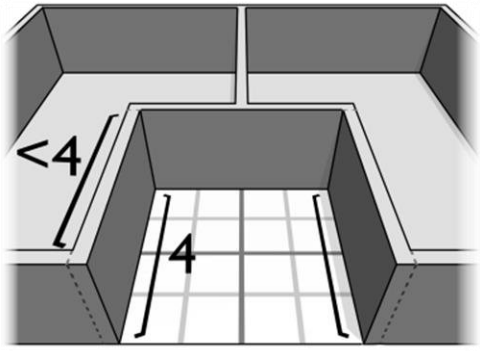
The Gemara discusses an alley whose dimensions are 4 tefachim long by about 4 tefachim wide[18]. There is a back wall and two side walls. The front is open.

There are two courtyards behind the alley, one on each side.



To make openings in the wall of the alley to the courtyards we would need a space of at least 4 tefachim wide, the smallest width for a halachic doorway.

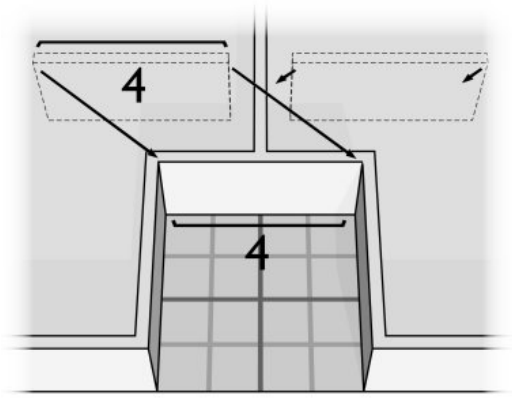
[18] The Gemara said that a mavoy must be longer than its width. Rashi ד"ה ארכו יתר על רחבו says that this mavoy is "4T minus a bit". For our purposes in this piece, this difference is negligible.



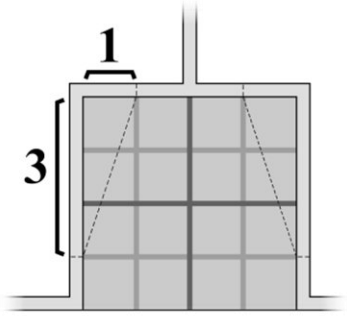
There's not enough space in the side walls for the doorways since the 4 tefachim wide doorways would take up the entire side walls. There wouldn't be any space left to have a post or a wall or something to actually delineate the doorway.

The front walls of the alley (the sides at the bottom of the picture) would have to have some thickness to them, so the doorways would, perforce, be less than 4 tefachim wide. So that doesn't work.

Putting the openings into the back wall wouldn't work either because that wall is only 4 tefachim wide and we need to put in two openings, each 4 tefachim wide for a total of at least 8 tefachim.

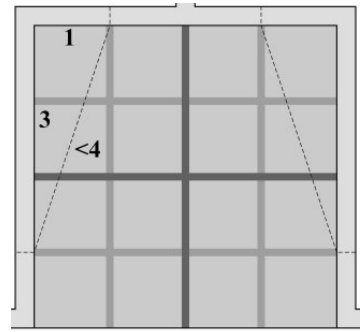


The Gemara therefore suggests making the openings at an angle into the back corners of the alley.



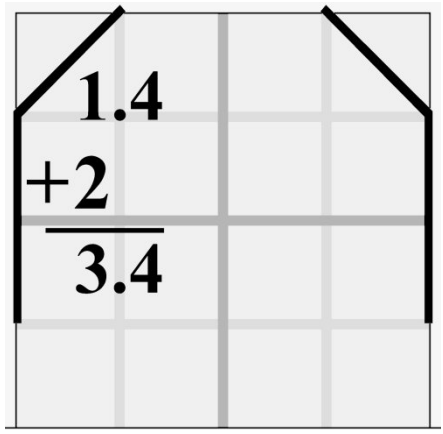
Rashi says, "3 tefachim of the side wall and 1 tefach of the back wall."

Tosfos, as understood by the Maharshal, asks on Rashi that the diagonal of 3 x 1 tefachim is less than 4 tefachim (the minimum width necessary for an opening).



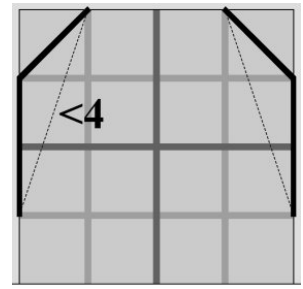
This can be proven with the following thought experiment:

The Gemara elsewhere[19] approximates the diagonal of a 1 x 1 square as being 1.4.



If we were to measure Rashi's 3 x 1 diagonal by drawing a straight line along the side of the first 2 tefachim and then across diagonally the last tefach we would arrive at a total length of 3.4 tefachim (2 for the straight line and 1.4 for the diagonal).

And now for the proof: If measured in this roundabout way we get a value that is less than 4 then certainly if we measured directly the whole 3 x 1 diagonal we would get a value that is less than 4 (since a straight line is the shortest path between two points)!



[19] 57a & 76b; See Appendix G regarding the Gemara's value of a diagonal.

The Rashba is seemingly bothered by the same question.

He says that with Rashi's values for the diagonal (3 * 1) we "won't even get 3.4 on the diagonal."

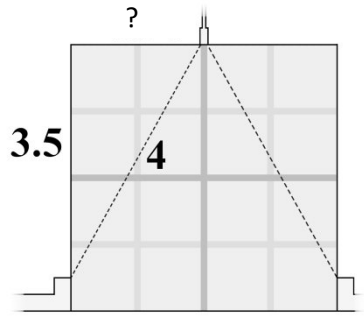
Presumably, the number "3.4" is a result of the same proof of Tosfos, as we explained according to the Maharshal.

The Rashba offers slightly different numbers than Rashi: 3.5 * "almost 2".

Let's calculate exactly the value of the Rashba's "almost 2".

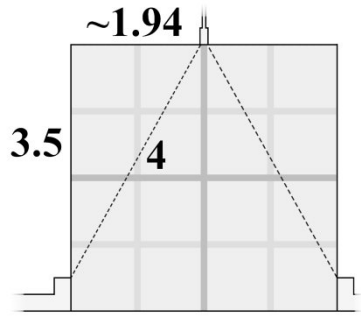
The formula for right triangles[20a] (since this diagonal is essentially such) is

$$a^2 + b^2 = c^2.$$



We know b and c are 3.5 and 4, respectively. So we can work it out as follows:

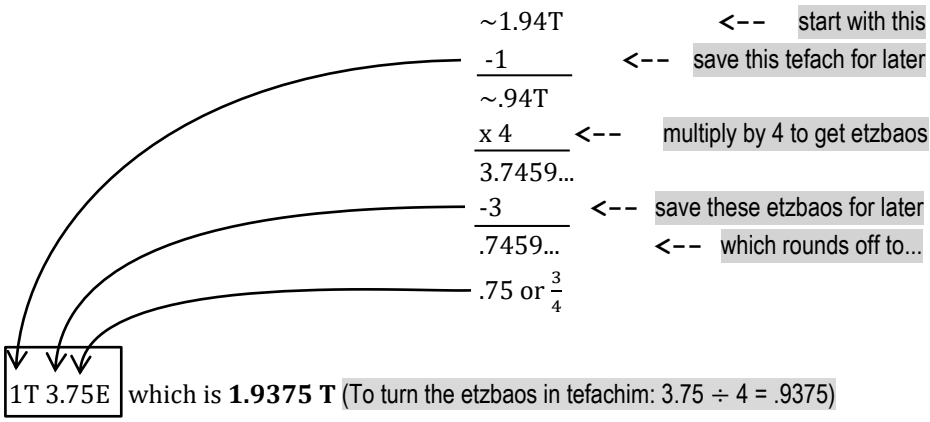
- $a^2 = c^2 - b^2$
- $a^2 = 4^2 - 3.5^2$
- $a^2 = 16 - 12.25$
- $a^2 = 3.75$
- $a = \sqrt{3.75}$
- $a = 1.93649...$



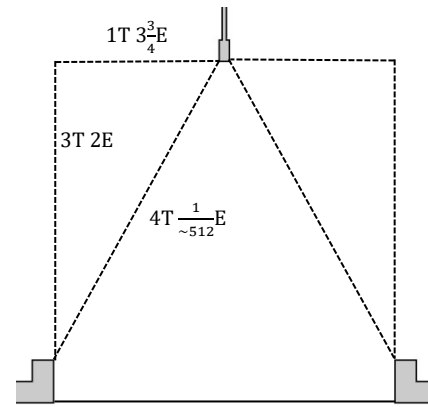
[20a] See Appendix D on Pythagorean Theorem

The style of the Gemara and the miforshim is to give measurements in units of amos, tefachim, etzbaos, and/or simple fractions of them.

If I were to round off the Rashba's "~1.94" tefachim to the nearest simple fraction I would end up with 1T 3.75E, or 7.75E. Like this:

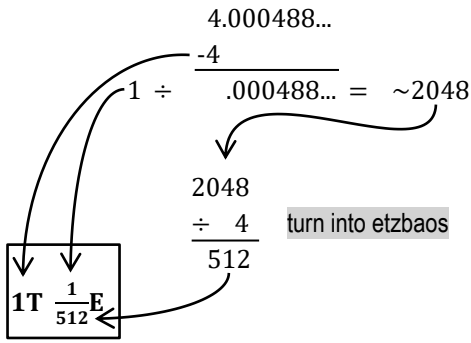


Plugging in this number for the back wall, and the Rashba's "3.5" for the side wall, gives us a diagonal of 4.000488...



(because $\sqrt{3.5^2 + 1.9375^2} = 4.000488$)

which in fractions would be:



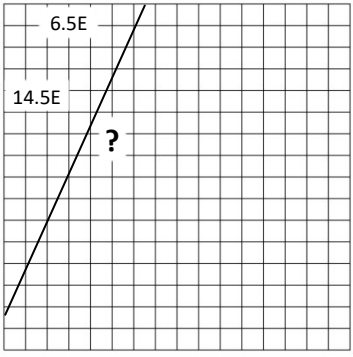
Let's look at the accuracy of this rounded Rashba's number another way.

Let's do it as a percent. To do that, we take the diagonal that we got, which was 4.000488... and divide that by our target diagonal of 4:

$$\begin{array}{r} 4.000488... \\ \div \quad 4 \\ \hline 1.000122062... \end{array}$$

or %100.0122062... which is $\boxed{.\mathbf{0122}...}$ of a percent above our target.

The sefer Dvash Tamar gives different numbers for the mavoy's diagonal doorways. First, he quotes Tosfos' question of Rashi (without specifying how he understood Tosfos) and says that to make the openings in the wall we need 3.5T .5E (14.5E) from the side wall and 1.5T .5E (6.5E) from the back wall, and he writes that this will give you a diagonal of exactly 4T (16E).



$\sqrt{14.5^2 + 6.5^2} = 15.89...$

$\div 4$ <-- turn into tefachim

$\boxed{\mathbf{3.97256}...}$

Accuracy in fractions:

3.97256...

-3 <-- save for later

.97256...

x4 <-- turn into etzbaos

3.89...

-3 <-- save for later

1 - .89 = .11...

1 ÷ .11.. = 9 <-- rounded off

$1 - \frac{1}{9} = \frac{8}{9}$

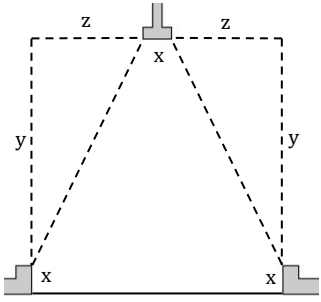
$\mathbf{3T\ 3\frac{8}{9}E}$

Dvash Tamar accuracy in percentage:

Diagonal = 3.972...T
÷ 4 target
.99314... or %99.314.

Short of target $\boxed{.6859\ldots}$ of a percent.

Taking a cue from the Rashba, I wondered if I could make a cheshbon that would yield a diagonal of exactly 4, with the amount of back wall remaining and of the side wall remaining being the same amount, as that would seem to me to be the most logical way to build an alley.



Let's lay down some terms:

x is the size of the back wall width (the solid part) as well as the side wall length

y is the amount of side wall taken off

z is the amount of back wall taken off

The entire width of the back wall could be defined as $\mathbf{2z + x = 4}$

The entire length of the side wall could be defined as $\mathbf{y + x = 4}$

Since both walls = 4 we can put those two formulas together like so:

$\mathbf{2z + x = y + x}$ and then we can factor out the "x"s

$\mathbf{2z = y}$

And we also know that $\sqrt{\mathbf{y^2 + z^2}} = \mathbf{4}$, since that's the formula for the diagonal.

OK, now let's do some cheshbonos.

Let's start by calculating z.

We said that $\sqrt{y^2 + z^2} = 4$, and since we also said $2z = y$ so we could replace the "y" in the above square root with "2z" and we get:

▶ $\sqrt{(2z)^2 + z^2} = 4$

simplify the paratheses

⬇ $\sqrt{4z^2 + z^2} = 4$

add the $4z^2$ and the one z^2 for a total of $5z^2$

⬇ $\sqrt{5z^2} = 4$

square both sides

⬇ $5z^2 = 16$

divide both sides by 5

⬇ $z^2 = 3.2$

square root both sides

■ $z = \sqrt{3.2} = \boxed{1.78885\ldots}$ tefachim

Now let's round that decimel into etzbaos:

1.78885...

-1 <-- tefach

.78885

x4 <-- turn into etzbaos

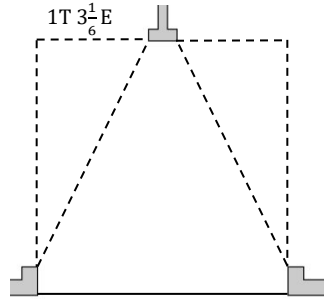
3.1554...

-3 <-- etzbaos

$1 \div .1554 = 6.43$

$\frac{1}{6}$ <-- rounded off

z = 1T $3\frac{1}{6}$E



To get the value of y we could just double z (since we said 2z = y) but since we rounded it off it's safer to double-check it by making a separate cheshbon for each number.

So, let's do for y what we did for z.

$$\sqrt{y^2 + z^2} = 4$$

since 2z = y we could replace z with .5y

$$\sqrt{y^2 + (.5y)^2} = 4$$

simplify the parantheses

$$\sqrt{y^2 + .25y^2} = 4$$

add the one y^2 with the $.25y^2$ to get $1.25y^2$

$$\sqrt{1.25y^2} = 4$$

square both sides

$$1.25y^2 = 16$$

divide both sides by 1.25

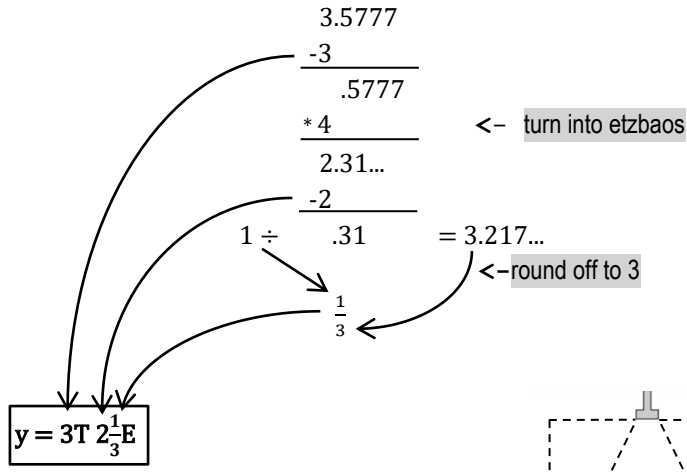
$$y^2 = 12.8$$

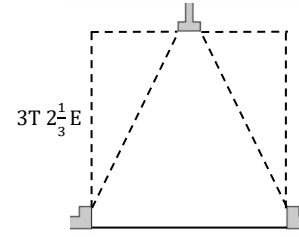
square root both sides

$$y = \sqrt{12.8}$$

$$y = 3.5777...$$

Now, let's round that into etzbaos:





Now, let's just check y and z together to see if it matches.

We came up with:

$z = 1T\ 3\frac{1}{6}E$

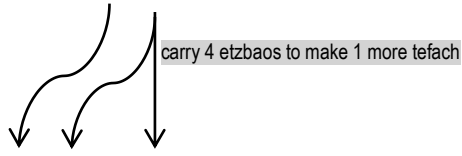
$y = 3T\ 2\frac{1}{3}E$

And we said that 2z = y. Does that work out?

$z = 1T\ 3\frac{1}{6}E$

$*2$

$2T\ 6\frac{2}{6}E$



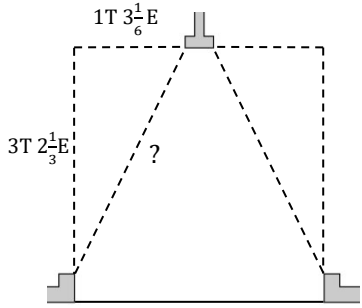
$2T + 1T + 2\frac{2}{6}E = 3T\ 2\frac{1}{3}E$

Great!

Now that we know y or z we can get x pretty easily. We said that $y + x = 4(T)$ so:

- $y + x = 4T$
- $3T\ 2\frac{1}{3}E + x = 4T$ — move 3T... to the right side
- $x = 4T - 3T\ 2\frac{1}{3}E$ — take off 3T from 4T
- $x = 1T - 2\frac{1}{3}E$ — 1T = 4E
- $x = 4E - 2\frac{1}{3}E$
- $\boxed{x = 1\frac{2}{3}E}$

Now let's check how accurate this approach is.



$$\text{diagonal} = \sqrt{\left(3\text{T } 2\tfrac{1}{3}\text{E}\right)^2 + \left(1\text{T } 3\tfrac{1}{6}\text{E}\right)^2}$$

which is

$$\sqrt{\left(14\tfrac{1}{3}\text{E}\right)^2 + \left(7\tfrac{1}{6}\text{E}\right)^2}$$

= 16.02515...

- 16

.02515 = 39.755...

1 ÷ $\frac{1}{40}$ <--rounded off

16E ÷ 4 = 4 tefachim

4T $\frac{1}{40}$E

Now let's check accuracy in percentage:

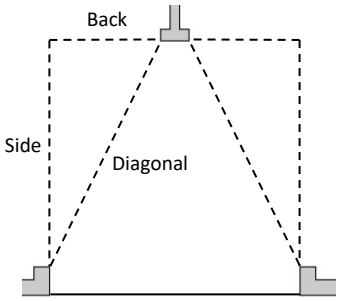
Diagonal = 16.02515...E

÷ 16 4T in etzbaos

1.00157... or %100.157

Above target by **.157...** of a percent.

	Side	Back	Diagonal	Remainder	
				fraction	percentage
Rashba	3.5T	$1T\ 3\frac{3}{4}E$	4.000488...	$\frac{1}{\sim 2048}T$	0.0122...%
	(14E)	$(7\frac{3}{4}E)$		$\left(\frac{1}{\sim 512}E\right)$	
Dvash Tamar	$3T\ 2\frac{1}{2}E$	$1T\ 2\frac{1}{2}E$	3.97256...	$-\frac{1}{\sim 36}T$	-0.685...%
	$\left(14\frac{1}{2}E\right)$	$\left(6\frac{1}{2}E\right)$		$\left(-\frac{1}{\sim 9}E\right)$	
B.G.	$3T\ 2\frac{1}{3}E$	$1T\ 3\frac{1}{6}E$	4.006288...	$\frac{1}{\sim 159}T$	0.157...%
	$\left(14\frac{1}{3}E\right)$	$\left(7\frac{1}{6}E\right)$		$\left(\frac{1}{\sim 40}E\right)$	



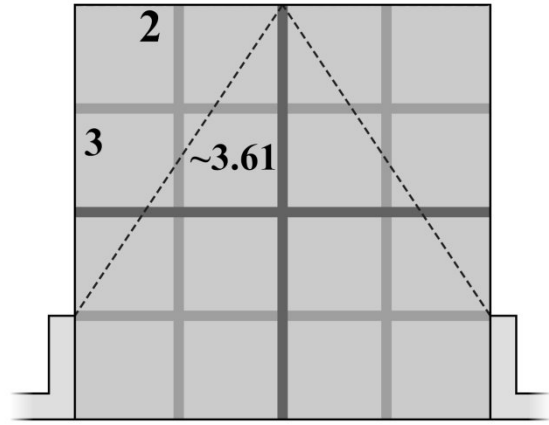
Until now, we have been discussing adjusting Rashi's values of 3x1 by changing both the side wall and the back wall.

In a slightly different approach, the Maharsha says that we need to change either the side wall or the back wall of Rashi's 3x1.

The side wall is easy enough to figure out. If the space of the back wall is 1 tefach and the diagonal is 4 then the side wall is $\sqrt{4^2 - 1^2}$, which is $\sqrt{16-1}$, which is $\sqrt{15}$, which is about 3.87298... tefachim, or about 3T 3.5E.

The back wall is harder to understand. The most we can take from the back wall is just under 2T, since we need to take from both the right and the left side, in addition to the width of the back wall, which must be some tiny amount, at least.

Even if we took 2T from the back wall and left nothing for the wall itself, we would still be left with a triangle of 2 by 3 which produces a diagonal of about 3.61.



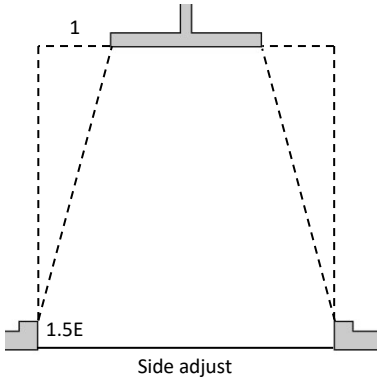
$$\sqrt{2^2 + 3^2} = 3.605551 \ldots$$

We need a diagonal of 4.

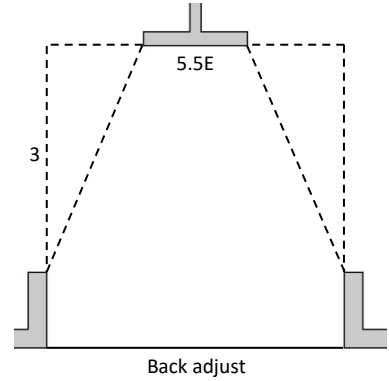
What can the Maharsha mean?

I saw in the Oz Vihadar Gemara that they quoted the Dvash Tamar (that we brought above) as an explanation of the Maharsha. In the version of Oz Vihadar that I have (from 5766) it says like this:

Maharsha: "We have to explain that the side wall [i.e., the solid part] is not a tefach [as Rashi implied] (but rather 1.5 etzbaos -Dvash Tamar) or that the [solid part of the] back wall is not 2 tefachim [as Rashi implies](rather, it should only be 5.5 etzbaos -Dvash Tamar)"



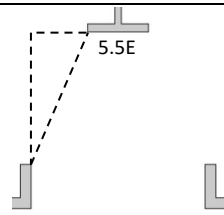
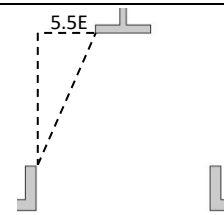
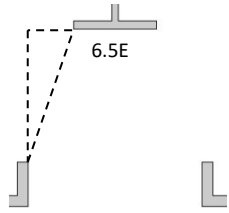
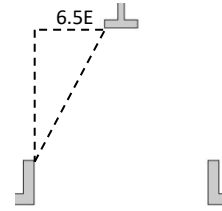
Side adjust



Back adjust

There's two things I don't understand about this pshat. First, the Dvash Tamar never mentioned 5.5E. Perhaps there was a typing error in the Oz Vihadar and they meant to write 6.5E. Secondly, the Dvash Tamar gave values for the space to be taken from the back wall for one of the doorways, whereas the Maharsha is giving us the amount of solid wall left over, after the openings are made.

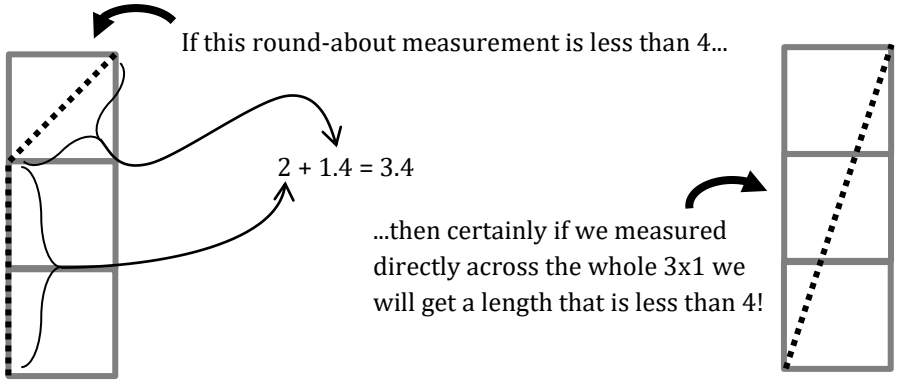
However, we will see that any combination of these parameters are not sufficient to explain the Maharsha since none of them are close to the required 16E diagonal.

	Solid part of wall	Space taken from wall
5.5E	 diagonal = 13.098...	 diagonal = 13.2...
6.5E	 diagonal = 12.9...	 diagonal = 13.647...

When we see my explanation of the Maharsha we will see if that's what the Oz Vihadar meant.

I want to explain the Maharsha as follows. The Maharsha is not trying to explain the Gemara or Rashi. All he's trying to do is take away the <u>proof</u> that Tosfos brought against Rashi.

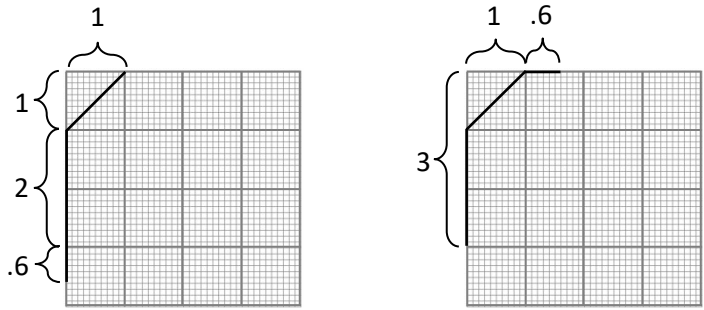
The proof was that if we took a 3x1 tefach rectangle and measured along the side of the first 2 tefachim and then diagonally across the last tefach we would arrive at a length of 3.4T (assuming the diagonal of the last square tefach is 1.4).



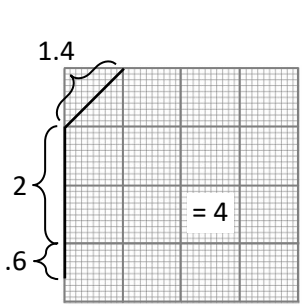
It is to this proof that the Maharsha suggests taking more from the side wall or from the back wall. Tosfos proved that the diagonal was, at most, 3.4 tefachim, which is .6 of a tefach short of the target length of 4. What the Maharsha is suggesting is to add .6 of a tefach to Rashi's values of 3x1, either to the 3 (the side) or to the 1 (the back).

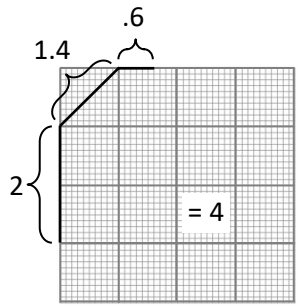
Adding this ".6" to the side measurement would result in taking 3.6 tefachim from the side wall.

Alternatively, adding this ".6" to the back measurement would result in taking 1.6 tefachim from the back wall.



Whichever way we measure it, we will get a total length of 4.





Taking an additional .6 from the side wall leaves the solid part of the side wall **.4 tefachim** $\left(\mathbf{1\frac{3}{5}E}\right)$.

Taking an additional .6 from the back wall leaves the solid part of the back wall **.8 tefachim** $\left(\mathbf{3\frac{1}{5}E}\right)$. (Since we took 1.6 of the back wall from either side of the middle for a total of 3.2 subtracted from 4 = .8)

We can now put this pshat into the words of the Maharsha: "We have to explain that the side wall [i.e., the solid part] is not a tefach [as Rashi implied] (but rather $1\frac{3}{5}$ etzbaos - B.G.) or that the [solid part of the] back wall is not 2 tefachim [as Rashi implies](rather, it should only be $3\frac{1}{5}$ etzbaos - B.G.)"

One might counter that the diagonal straight across is still less than 4 since it is a straight line and therefore is shorter than this round-about line.

While that is true, this method of proof that Tosfos brought doesn't quanitify how much less the straight line would be and therefore Tosfos' proof falls away. Besides, we can still add some amount in the Maharsha's method.

It is for the same reason that it doesn't matter that the diagonal of the Maharsha's numbers are actually less than 4 when measured since we are only attempting to undermine the proof that Tosfos brought.

I thought perhaps that this was the intention of the Oz Vihadar, the only difference being that they've rounded the numbers to the nearest half-etzba.

I explained the Maharsha's "side wall not being a tefach" as $1\frac{3}{5}$E. Rounded to the nearest half-etzba would yield $\mathbf{1\frac{1}{2}E}$, which is the number of the Dvash Tamar's side wall.

I explained the Maharsha's "back wall not being a tefach" as $3\frac{1}{5}$E. The space left on each side is $6\frac{2}{5}$E.

(16 etzbaos of the total alley width minus $3\frac{1}{5}$ etzbaos for the back wall equals $12\frac{4}{5}$ etzbaos. Divided in half is $6\frac{2}{5}$ etzbaos.)

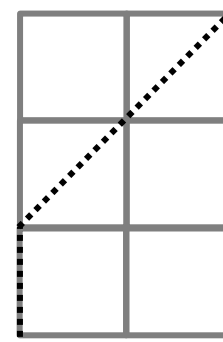
Rounded to the nearest half-etzba would yield $\mathbf{6\frac{1}{2}E}$, which is the number of the Dvash Tamar's back wall.

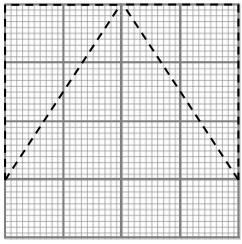
Either way, I owe a hakoras hatov to the Oz Vihadar since the only reason I came up with my pshat was because I was trying to figure out what the Oz Vihadar meant!

Even if we say that this is the pshat in the Maharsha, Tosfos can come back with another proof.

The most you could take from the back wall is 2T for each side of the mavoy. That gives you a triangle of 3x2.

Since the diagonal of 1x1 is 1.4 then the diagonal of 2x2 is 2.8. If we measure along the side of the first tefach and then diagonally across the next 2 tefachim we will arrive at a value of 3.8. And like before: if measured in this round-about way we get a value of less than 4 then certainly if we measure directly across the diagonal of 3x2 we will get a value of less than 4!



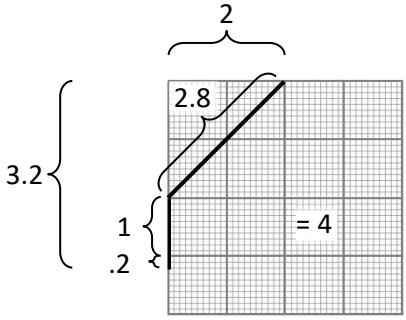


And in this example, we can't add any space from the back wall since we took the whole wall already!

So, אין הכא נמי, to this proof we'd have to adjust the side wall as well.

Since we proved that this diagonal of 3 x 2 will be less than 3.8 tefachim, we need to add at least an additional .2 of a tefach from the side wall.

In total, we'll have to make the doorway from 3.2 x 2 of the alley.



Stam, I was wondering, maybe Rashi holds that when you need a halachic distance in a diagonal, like here, you measure along the sides and add them up. So 3 x 1 = 4 because 3 + 1 = 4.

This would be true except by a square since Chazal give us the value of the diagonal of a square as 1.4.

This would explain Rashi's pirush in many places in Shas, where he gives the value of a rectangle's diagonal as the sum of the short and long side of the rectangle. For example, on ע"ח. where he explains that a ladder 14 tefachim long whose feet are placed 4 tefachim away from the wall will reach the top of a 10 tefach high wall.

(Perhaps this is based on shitas Rebbe Yochanan on 76a; See page 285.)

ז'. - ח'. - Alley Opening Into Rachba

I did not get absolute clarity in this piece so please consider it more of a springboard than of a pirush.

I had a lot of trouble getting the whole שקלא וטריא of this Gemara straight and so I wrote up this chart (or table) to help me[20b]. I will first explain how the chart works, and then we'll see the chart, and then I'll try to explain the Gemara's שקלא וטריא as we keep our eye on the chart.

The top row of the chart is the apparent discrepancy between the two amorayim in what they held about two seemingly comparable cases.

The middle row of the chart is the Gemara's הבא אמינא in understanding what exactly the machlokes is.

The bottom row is the Gemara's מסקנא (not the final מסקנא, just how the Gemara first resolves the difference between the two amorayim.)

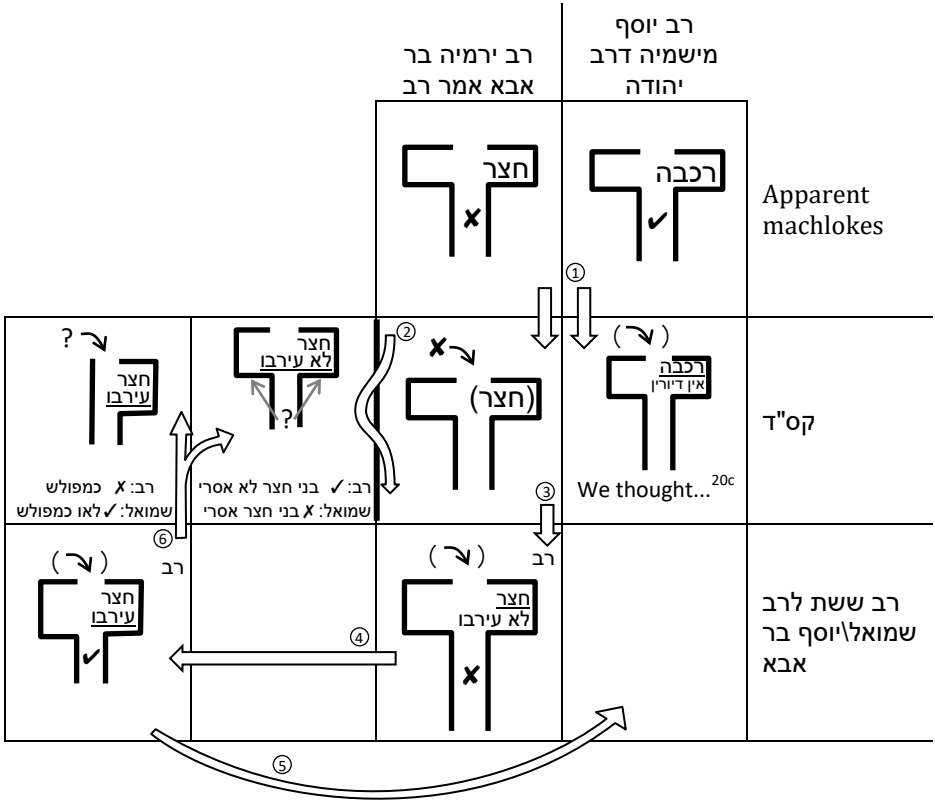
The thick arrows ⟹ show the flow of the Gemara, as it appears in the text (as opposed to what the Gemara was thinking at the time).

[20b] My handwritten notes before the chart say (spelling in shorthand):

> 31.10.18 Graet-Grandmother חנה בת חיים יארצייט
>
> I'm trying to lern this סוגיא on (עירובין) ז': - ח'. and it's frustrating becaus I dont noe wats going on. Rashi says a lot of things but I dont noe wat hes tauking about. I'm trying to see other ראשונים but they ar somtyms arguing with רש"י, somtyms not, it's hard to noe wen, and anyway it just makes it that much mor complicated.

After writing out the chart I wrote:

> OK, it's much better now.



Rav Yosef said in the name of Rav Yehuda that a mavoy that opens into a rachba (a kind of backyard) doesn't need any kind of a tikkun[20d] on that end[21].

Abaya says to Rav Yosef that this statement must be that of Shmuel and not Rav[22], since Rav said a (seemingly) contradictory statement elsewhere.

[20c] We thought Rav Yehuda was davka when he said רכבה but later Rav Yosef will say that was just the story.

[20d] like a lechi or korah

[21] Picture on the top right of the chart

[22] This statement of Rav Yosef had come right after a list of statements in the name of Rav

Rav Yirmiya bar Abba had said in the name of Rav that a mavoy that opens into a chatzer is אסור[23] (without some kind of tikkun).

Now, follow the thick arrows labeled "1" to the next 2 pictures below, on the row labeled on the right ' קס"ד '. There's 2 contradictions between these two opinions.

1. We assume the reason that Rav said the mavoy is אסור is because the far end of the chatzer is open directly across from the mavoy whereas Rav Yehuda allows such a mavoy which means he's **not** bothered by the opening at the far end. [24]
2. We assume that Rav was **not** concerned with the fact that the mavoy opened into an area that was אסור to it whereas Rav Yehuda seemingly **was** bothered by this, from the fact that he spoke about a case (rachba) which has no one living in it. In a case where someone **had** been living in it (for example, if it had it been a chatzer), then Rav Yehuda would presumably forbid such a case.[25]

The squiggly thick arrow labeled "2" indicates that the Gemara has a pshat in mind at this point but we will not see that until later.

Now, we follow the thick arrow labeled "3" down to the last row on the bottom. Rav Sheshes answers that our assumptions about Rav were wrong. He was **not** concerned with the fact that the far side of the chatzer was open directly across from the mavoy (just like Rav Yehuda was not concerned), rather he was bothered by the fact that the mavoy opened into the chatzer, which had people living in it who did not do an eruv with the mavoy. This too seemed to be the implication of Rav Yehuda's din.

(Thick arrow "4") And אין הכא נמי, if the people in the chatzer **did** do an eruv then the mavoy would have been permitted despite the fact that the chatzer opened directly across from it. (Thick arrow "5") And so too, if there was no one living there, for example in a rachba, then that would have been permitted.

---

[23] Picture in the top row of the chart, the left picture

[24] To demonstrate this in the chart, there's an "X" pointing to the opening in Rav's case while in Rav Yehuda's case there is an arrow in parentheses to show that it is irrelevant.

[25] To demonstrate this in the chart, the word "חצר" is in parentheses in Rav's case (to show it's irrelevant) while the word "רחבה" is underlined by Rav Yehuda (to show its importance). אין דיורין means no one is living there.

(Thick arrow "6") The Gemara then goes back and explains what it initially thought the machlokes between Rav and Shmuel[26] was.

There were actually two assumed machlokeses[27].

1. (In the chart: middle row, second picture from the left) נראה מבחוץ ושוה מבפנים, "[Walls] seen from the outside and flush (unseen) from the inside." The people in the chatzer can "see" the opening as it's framed on both sides by walls whereas the people in the mavoy cannot see those walls at all. To them, the mavoy appears to run directly into the chatzer with no mechitza. Rav holds that it has the din of a mechitza even for the people in the mavoy and therefore the chatzer people are cut off from the mavoy and do not assur it, despite the fact that they did not do an eruv with the mavoy. Shmuel holds that this is not a good mechitza and the chatzer assurs the mavoy.
2. (In the chart: middle row, picture all the way on the left) In a case where the mavoy opens into the chatzer with one wall of the mavoy being flush with the side of the chatzer, so that it appears to be one straight wall, is that considered to be a mavoy that opens directly into a reshus harabim or not? Rav says it is and therefore is assur whereas Shmuel says it's not and therefore it's muttar. (Here, we're talking about where they <u>did</u> do an eruv.)

The Gemara makes further adjustments to the dinnim but you'll have to work it out on your own as I haven't yet figured out how to neatly put that into the chart.

[26] When the Gemara initially said that the statement of Rav Yosef in the name of Rav Yehuda couldn't be that of Rav, it was assumed automatically that it must have been the opinion of Shmuel, since he was the other Rebbe that Rav Yehuda studied extensively under. See Rashi on ז'. ד"ה אי דר"י

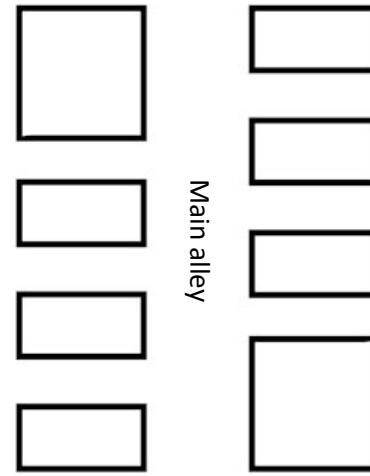
[27] I think in Yeshivish it's מחלוקתים and in Hebrew it's מחלוקות.

*

:ח - <u>Asymmetrical Centipedes</u>

The Gemara refers to a type of alley that has many small alleys coming off of the main one. The Gemara calls this type of alley a "נדל", a "centipede"[28] (which comes from the Latin word meaning "A hundred legs"[29]).

Tosfos understands that the alley's offshoot-alleys were on both sides of the main alley, and they were not aligned one across from the other but rather they were staggered.



Tosfos says, "And this is how they are like a centipede that there are two rows of legs, one on the right and one on the left, and they are not opposite each other."

What does Tosfos mean that they are not opposite each other? As far as I know, there is no known centipede species that has legs that are not aligned one side with the other.

I thought to explain that Tosfos means the way the centipedes walk; that some legs move forward while others move backwards.

---

[28] רש"י ויקרא (י"א:מ"ב) ד"ה **מרבה רגלים** זה נדל שרץ שיש לו רגלים מראשו ועד זנבו לכאן ולכאן וקורין צינטפיד"ש

[29] Although, it is interesting that there are no known species of centipedes that have exactly 100 legs. All known species have an odd number of pairs of legs, so a hundred legs would be 50 pairs, and 50 is not an odd number. Perhaps in the olden days they were counting the antennae? Or there were other species?

The passuk refers to them as "הולך", which might be an indication that they are seen essentially as "movers", a possible support for my teretz.

Additionally, the word נדל is related to the word נזל, which means a flow (-Aaron Kronenfeld)

Jastrow says the word שרץ means a creature that "moves" or "creeps" (-Yishai Rasowsky)

*

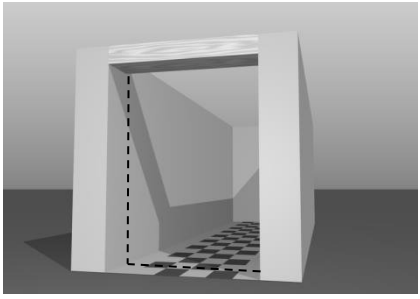
ח': - **Muttar Assur, Assur Muttar - Alley Beam**

There's a tefilla before learning that says that one should not (accidently) say about that which is מותר that it is אסור, or about that which is אסור that it is מותר.

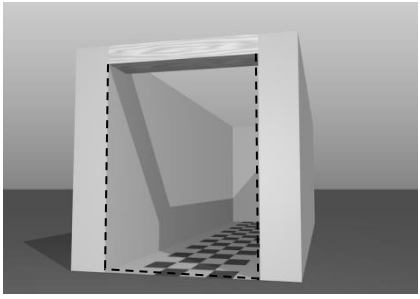
There is an exception here on daf ח'.

The Gemara asks what is the דין about whether one may carry in the area directly under the korah beam (and from there to the mavoy or back from the mavoy). One opinion says it's permitted while one says it is forbidden.

The Gemara's last understanding is that the argument is about whether it's the inside edge of the beam that makes a halachic mechitza under it or whether it's the outside edge of the beam that makes the mechitza.



Inside edge

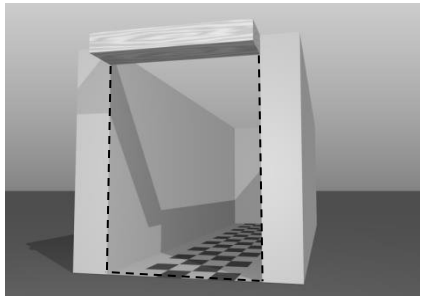


Outside edge

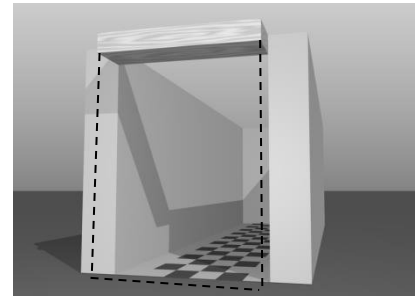
According to the one who says the inner edge creates the mechitza, it turns out that the area under the whole beam is outside that mechitza, and hence, outside the mavoy and is considered part of the reshus harabim.

The one who holds the outside edge of the beam creates the mechitza, the area under the beam is *inside* the mechitza and is within and part of the mavoy.

Rav Chisda then says that in a case where the beam was just outside the mavoy's airspace the halacha will be the opposite of the previous case[30]. The one who said the previous case is muttar will say that this case is assur, and vice versa, the one who said the previous case was assur will say this one is muttar.



Inside edge



Outside edge

The one who says the inner edge creates the mechitza views this mavoy as having a valid mechitza at its opening whereas the other opinion views the mechitza a bit of a distance away from the mavoy's opening which means that this mavoy effectively has no mechitza.

[30] Not exactly opposite. The first case was discussing the area under the beam whereas the second case is discussing the halachic status of the mavoy. We mean "opposite" in the way the dinnim come out: The one who allows the first case disallows the second, and the one who allows the second disallows the first.

*

:י - Leather Seat - Different Size Etzbaos, Solid vs. Space

The Gemara is describing a square piece of leather 1 tefach by 1 tefach with a hole in the middle of it. The leather, being a tefach square, should be able to act as an ohel to transmit tumah unless if the hole in the middle is too big. The Gemara has a discussion about how big the hole can be and still remain negligible.

Rav Dimi says the hole can be 2 etzbaos wide with a 2 etzbah-border of leather (See picture on next page, Picture #1).

Ravin says 1 etzbah for the hole with a 1.5 border of leather (Picture 2A). And Ravin clarifies that the total size of the leather is not different than Rav Dimi's since Ravin is using a bigger kind of etzbah, one that is 1.5 bigger than the usual small one (4 large etzbaos = 6 small etzbaos; see מנחות כ"א:). Therefore, Ravin's leather measures (1.5 + 1 + 1.5) * 1.5 = 6 smaller etzbaos (Picture 2B).

It comes out that the leather is the same according to both amorayim. The difference is only in how thick the leather border is in comparison with the hole in the middle. According to Rav Dimi they are the same size (2 etzbaos) while according to Ravin the leather is a bit wider than the hole.

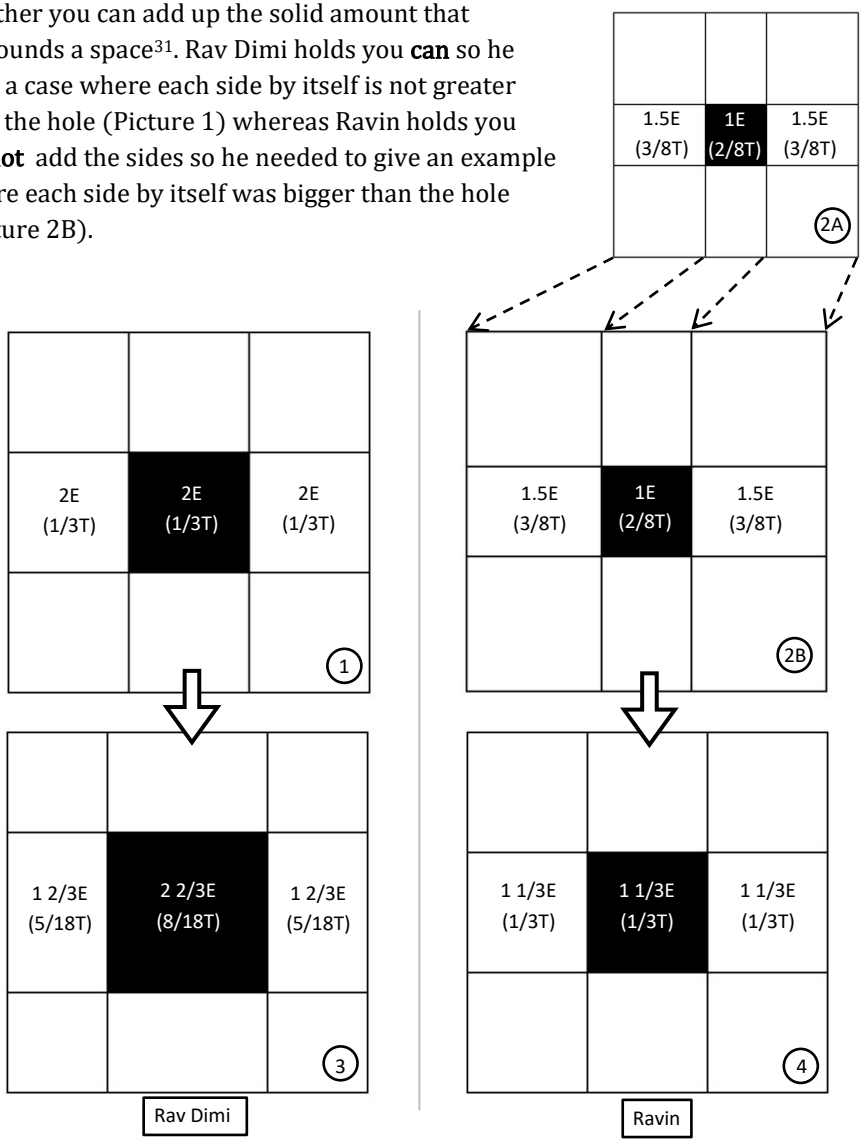
There is a din by mechitzos in general when you have gaps in it that there needs to be a sufficient solid:space ratio. That is, you cannot have too much "gap" compared to the amount of physical mechitza.

There are two aspects to this that we will discuss.

1. Whether you can add up the solid part from both sides of the space in determining the solid:space ratio, or perhaps each solid section by itself must be the required amount.
2. Whether the solid part needs to be <u>at least the same</u> as the space or whether it needs to be <u>more than</u> the space (regardless of whether you add from both sides or not).

The Gemara tries to understand our machlokes between Rav Dimi and Ravin in these two manners.

First, Abaya suggests that the machlokes is about whether you can add up the solid amount that surrounds a space[31]. Rav Dimi holds you **can** so he gave a case where each side by itself is not greater than the hole (Picture 1) whereas Ravin holds you **cannot** add the sides so he needed to give an example where each side by itself was bigger than the hole (Picture 2B).



[31] Abaya is assuming that everyone agrees that the solid amount must be **more** than the space. The only question is whether you can achieve that amount by adding two solid sides together or whether you need to have that with a single solid piece.

Rav Dimi responds that if that indeed was the machlokes then why would we give such small shiurim for the hole?! If I (Rav Dimi) wanted to show that two sides **can** add up together then I should have given an example where each side by itself is smaller than the hole! (Picture 3 on previous page)

And if Ravin wanted to show that the two sides **cannot** add up then he should have given a case where each side by themselves are the same as the hole's width[32]. (Picture 4)

Rather, says Rav Dimi, the machlokes is about whether you need the solid to be greater than the space (Ravin - Picture 2B) or whether it just has to be at least the same (Rav Dimi - Picture 1).[33]

*

ג:י"ג - Seeing Rebbe Meir

Rebbe said, "The reason I am sharper than my friends is because I saw Rebbe Meir from behind [as opposed to them who didn't see him at all]. Had I seen him from the front I would've been even sharper!"

Whenever I heard this Gemara said over, it sounded like Rebbe Meir was walking one time and Rebbe chanced to glimpse his back, once.[34]

However, Rashi[35] here says that it means Rebbe sat in the row behind Rebbe Meir when they were learning (when Rebbe Meir was giving shiur).

The implication is that Rebbe probably saw Rebbe Meir all the time, even his face, it's just that during the actual learning sessions he sat behind him.

[32] In Rav Dimi's response, he understands that it is only necessary for the solid to be at least the same as the space.
[33] But everyone agrees that the two sides don't add up
[34] וכן משמע מירושלמי ביצה כ:
[35] ד"ה דחזיתיה

Probably, the practical point is that by not seeing his facial expressions he couldn't fully grasp what Rebbe Meir meant.

The Maharsha puts it this way, "there is to understand ...[from] the winking of his eyes and with the hinting of his lips..."

Perhaps what Rebbe meant was that he sat right behind Rebbe Meir whereas his friends sat further back and couldn't quite see Rebbe Meir as well.

Possibly, there is a misugal point here, that during the time of learning there is a spiritual radiance that shines from a person's face and one who looks at it will gain spiritual energy, similar to idea of looking at a tzaddik's face or not looking at a rasha's face.[36]

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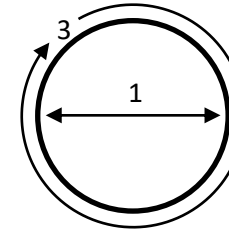
Truer Value of Pi - י"ג:- י"ד.



The Mishna says, "Whatever [round] object is 3 around is 1 across"

The Gemara brings a source for this ratio from the passuk in Melachim[37] where it says that Shlomo's pool was 10 amos across and 30 amos around.

If you measure a circle that is 10 across you will find that the circumference (measured around the circle) will be 10 * π, which comes out to 31.4159265358979...

The numbers in the passuk come out 1.4159... amos less than what we would expect from observation. This difference would be quite noticeable. According to the smaller shiur of Avrohom Chaim Na'ah[38], this would come out to a difference of about 68 cm (~2' $2\frac{3}{4}$").

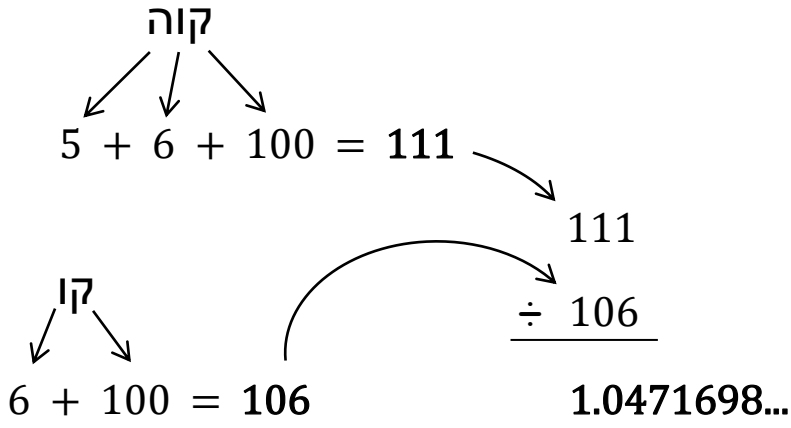
[36] וכן כתוב הבן יהוידע הכא וע' במגילה כ"ח:

[37] מלכים א' ז:כ"ג

[38] Who measures an amah to be 48 cm (~18.9 in.), see Appendix A.

There is an answer brought[39] that the passuk actually hints at a more precise number. The word in the passuk for "line" in the phrase "line around the pool" is written וקוה but is pronounced וְקָו.

Without the prefix "ו", the ratio between the gematrias of these two words is **1.0471698...**



The passuk is hinting that the circumference of the pool was **1.0471698...** times bigger than what is written (30) for a total of **31.41509...**[40]

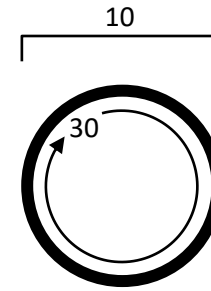
Now, the difference between what the passuk is saying and the measurement is

measurement:	31.415926...
Passuk:	−31.415094...
	.000832... amos
	x 24
	~.02 etzbah

[39] In the name of the Vilna Gaon, although the source for this seems elusive (dafyomi.co.il attributes Rabbi Max Munk or Rabbi Yaakov Levi).

[40] In other words, קוה is **1.04...** times bigger than קו. So too, the actual circumference of Shlomo's pool is **1.04...** times bigger than what's written in the Passuk.

According to the bigger shiur of the Chazon Ish[41], the difference comes out to a drop less than half a millimeter (about $\frac{1}{53}$ of an inch).



We could answer the discrepancy in a different manner. We could say that when the passuk says 10 amos across it is measuring from the outside of the pool's lip, and when it says 30 amos around it is measuring the inside of the pool's lip.

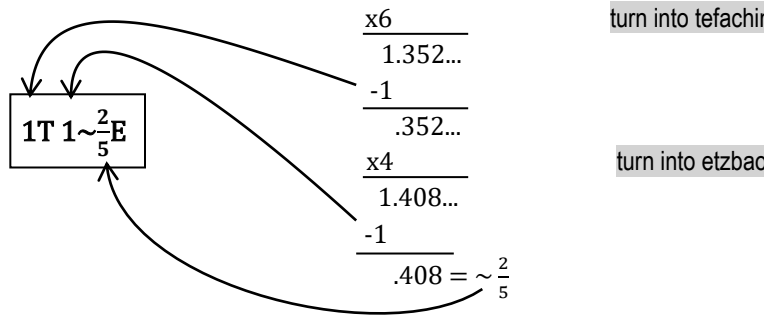
If so, we can calculate the inner diameter and the thickness of the lip as follows:

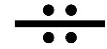
Inner Circumference (IC) $= 30$

Outer Diameter (OD) $= 10$

Inner Diameter (ID) $= \frac{\text{IC}}{\pi} = \frac{30}{3.14159\ldots} =$ **9.549... amos**

Lip Width[42] (LW) $= \frac{\text{OD}-\text{ID}}{2} = \frac{10-9.549}{2} =$ **.22535... amos**





[41] Who measures an etzbah to be $2\frac{5}{12}$ cm (~.95 in.), see Appendix A

[42] The lip width is the outer diameter minus the inner diameter divided by two (for each side of the pool).

The Ralbag gives this same answer but he point out that the passuk refers to the thickness of the pool's lip as 1 tefach, which would make the inner circumference "30 and a bit more than a third". (Inner Circumference = (10A - 2T) * π = 30.3687... amos). In other words, this teretz doesn't shtim with the passuk's description of the lip, unless if we say that either the tefach of the passuk or the 30 amos of the passuk is rounded off to the nearest tefach or amah, respectively.

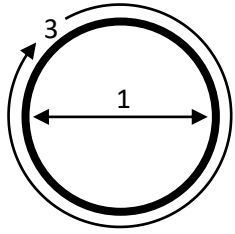
Either way, we'll be rounding off something.

Our Gemara says that the lip was a tefach near the bottom and that it tapered off to a sharp edge at the top. This would seem to rule out our suggested teretz.

*

.י"ד - <u>Tosfos Vihu'ikka - Torah Scroll In Aron</u>

The Mishna on :י"ג says that for every tefach across something round it will have a circumference of 3 tefachim (or amos or whatever the case may be).



The Gemara brings as a source for this דין the passuk that describes Shlomo's pool as 10 amos across and 30 amos around.

$$30 \div 10 = 3$$

The Gemara assumes that the circumference is measured on the outside edge of the pool's lip and the diameter is measured on the inside, in the hollow part of the pool[43]. If that's so then the thickness of the lip makes the total outside diameter more than 10.

If so, then dividing the 30 amos around by more than 10 will be *less* than 3!

[43] see Rashi והאיכא עובי שפתו

So the Gemara answers that the lip was very thin, like a sharp edge.

But still, asks the Gemara, that little bit means that the total diameter is at least a tiny bit larger than 10!

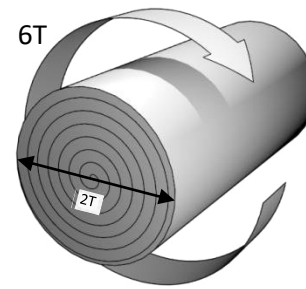
30 divided by slightly-larger-than-10 is a drop under 3!

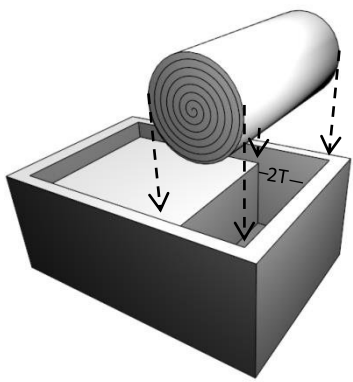
Until now, Tosfos assumed that the Gemara's value of 3 for pi was an approximation. However, once the Gemara asks that the lip must have had some thickness it sounds as though the Gemara's is taking the value of 3 for pi as exact but when we measure it we will find that it's actually a bit more[44].

Tosfos mentions a Gemara in Bava Basra :ד"ו where the Gemara there too implies that the ratio of "3" is exact.

There are 2 parts of the Gemara there that Tosfos quotes. I will start with the second part because that is easier to explain what's bothering Tosfos.

The Gemara there says that there was a Torah scroll that was rolled up and it had a circumference around its rolled up part that was 6 tefachim (see picture). The Gemara there says that since the circumference was 6 tefachim therefore the diameter was 2 tefachim ($6 \div 3 = 2$).





The Torah scroll was supposed to fit into the Aron (according to one opinion). The space in the Aron that was left for the Torah was exactly 2 tefachim wide (again, only according to one opinion). The Gemara at this stage assumes that an object cannot fit into a space the exact same width as the object. In this case, the space and the object are both 2T.

[44] See Appendix G about the Gemara's value of pi

Asks the Gemara, "How can an object of 2T fit into a space of 2T?"

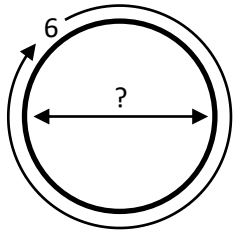
Obviously, this question is only valid if the object is indeed 2T. If it were less than 2 then this question would have no basis.

This is what is bothering Tosfos. The scroll really is less than 2! The true value of pi is about 3.14159... If we divide the circumference (6) by pi (3.14...) we will get **1.909858...**, which is plenty less than 2.

```
   2
-1.909858...
 .09014...  tefachim
      x4
 .36...     etzbaos
```

$= \sim\frac{1}{3}$ etzba

Diameter length determined by value of pi:

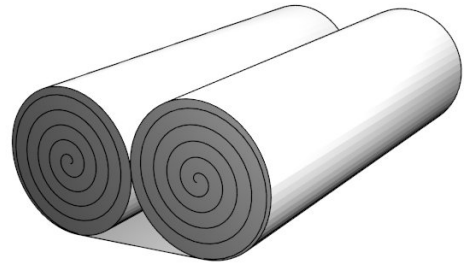


6 divided by ____ = diameter

3	π
6 ÷3 2	6 ÷3.14159... 1.9098...
Dividing 6 by 3 gives us exactly 2, and hence the Gemara's question: "How can an object of exactly 2 fit into a space of exactly 2?"	Dividing 6 by π gives us 1.9098..., which is plenty less than 2. So how are we to understand the Gemara's question? The Gemara asks, "How can an object of 2 fit into a space of 2?" It's *not* an object of 2! It's less than 2! It will fit *no problem!*

That question of Tosfos is clear.

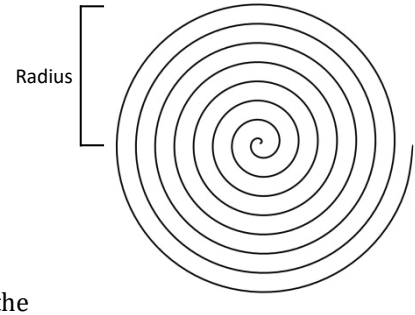
However, the first question that Tosfos has on that Gemara in Bava Basra is not so clear.
The Gemara at that stage understands that the Torah scroll was rolled to the middle, i.e., halfway.



If so, then the total width across both scrolls will be greater than 2.

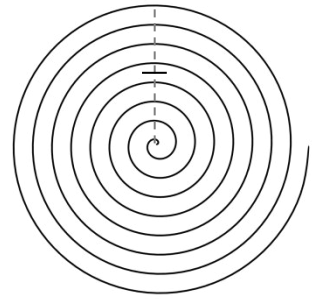
Let's see how this works.

Let's say that the big scroll is rolled up a certain number of times.

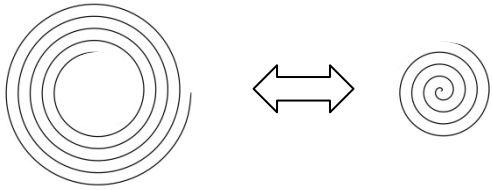


If we were to count from the outer edge until the center of the roll we will have a certain number of "layers" or rings. This is the radius of the scroll.

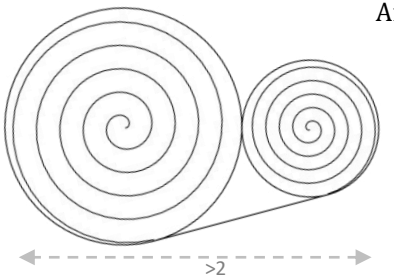
Let's make a mark halfway down the radius: →



We have the same number of rings above the mark as below the mark (until the center). However, the rings above the mark are much larger than the ones below.

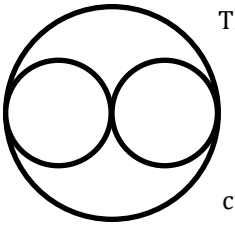


That means, if we were to take off all that outer parchment and roll it up into its own roll it will obviously be bigger than the little roll that was left behind.



And since the little roll has a diameter of 1 across (since it's half the diameter of the original roll) that means the bigger roll made up of the outer parchment is certainly bigger than 1 across, making the whole thing greater than 2!

The Pnei Shlomo[45] (who we will see again later) gives a simpler proof that the scroll rolled halfway (with the two little rolls side by side) is wider than the original scroll's diameter. He does so with the following illustration:



The outer circle is the original big roll (2 tefachim wide). The little circles are the little rolls if they were to truly be 1 tefach wide each. All of the area outside of the little circles within the big circle is all the parchment that *still* needs to be added to the little rolls! Which will make them wider than they currently are, and so they will be wider than 2.

A much neater proof.

Tosfos says that it makes a difference whether we use 3 for pi or the more accurate 3.14...

Tosfos says that the Gemara's question only makes sense if we use 3 but if we were to use 3.14... then the Gemara wouldn't have a question, meaning, the two little rolled-up scrolls would have a diameter less than 2.

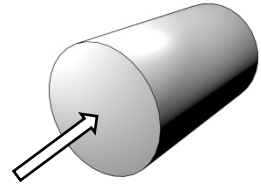
However, if we make the cheshbonos, we will see that it doesn't actually make a difference. This question of Tosfos is therefore difficult to understand.

Let us make the cheshbonos using both 3 and 3.14... and see what we get.

[45] בבא בתרא י"ד

Attempt 1: pi = 3

The area of the top of the scroll (the big circle) can be calculated as follows:

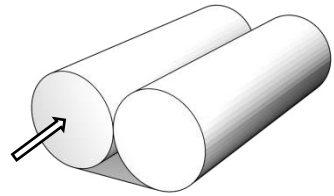


$A = \pi r^2$

The radius is 1 (Remember, we're using 3 as the value of pi so take the circumference 6 and divide it by 3 to get a diameter of 2. Divide that in half to get the radius of 1).

Therefore the area = $3*1^2 = 3*1 = \mathbf{3}$.

The areas of the little rolls will each be half of that, so little roll area = **1.5**



$A = \pi r^2$

$1.5 = 3r^2$

divide both sides by 3

$.5 = r^2$

square root both sides

.7071... = r

*2 to get diameter = 1.4142... $(\sqrt{2})$.

*2 for each roll = **2.828...** $(\sqrt{8})$, which is greater than 2 and hence the Gemara's question, "How can something greater than 2 fit into a space of 2?"

Attempt 2: pi = 3.14159265358979...

Big roll diameter: 6 ÷ 3.14159... = 1.909859...
Divide in half to get radius = .9549...

$A = \pi r^2$

$A = 3.14159... * .9549...^2$

$A = \mathbf{2.8647...}$

The areas of the little rolls will each be half[46] of that, so little roll area = **1.4323...**

▶ $A = \pi r^2$

⬇ $1.4323... = 3.14159...r^2$ divide both sides by 3.14159...

⬇ $.4559... = r^2$ square root both sides

■ **.6752...** = r

*2 to get diameter = 1.35...

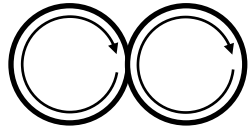
*2 for each roll = **2.7...** , which is much greater than 2 and so the Gemara can very well ask the question, "How can something greater than 2 fit into a space of 2?"

We see is that it makes no difference whether the value of pi is 3 or 3.14159...

Why then does Tosfos say that it makes a difference?

I was very happy to see that I was מכוון to the question of the Pnei Shlomo[47], who answers it in a מחודש way.

He says that when the Gemara there gives the circumference of 6, it's talking about *after* it was rolled up to the middle[48] and measuring the total circumferences of *both* rolls.

 = 6

[46] Since we're splitting up the big roll into two little rolls
[47] Thank you Rabbi Dovid Bloom for showing this to me!
[48] This approach is discussed in the Ramban

If together both rolls have a combined circumference of 6 then each roll has a circumference of 3.

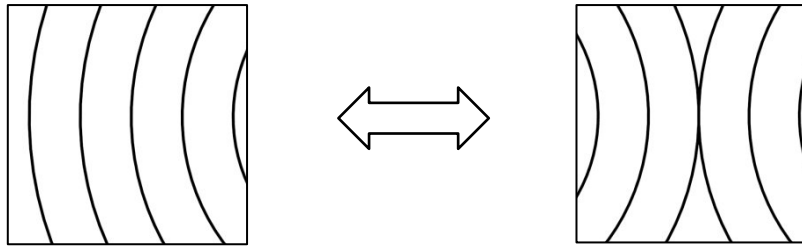
So now, if pi = 3 then the diameter of each roll will be 1, and together they will be 2 across. Plus, each roll needs a tiny bit more than its width to fit in a space the same as its width.

This is similar to the Gemara's next question except at this stage, the Gemara is trying to cram two 1-tefach-wide-objects into the space whereas later the Gemara is trying to cram a single 2-tefach-wide-object into that space. The Pnei Shlomo writes that 2 objects need more space than a single object, even if their total width is the same. A single scroll 2 tefachim wide can be crammed in "בדוחק" whereas 2 scrolls, each one a tefach wide cannot be crammed in, even בדוחק.

If however, pi = 3.14159... then these rolls are indeed smaller than 2 across and then we don't know what the Gemara is asking בכלל! This is Tosfos's question according to the Pnei Shlomo.

Let's cheshbon out how much it is. If each roll's circumference is 3 then each one's diameter is 3 ÷ 3.14159... = .9549... Double that for both rolls = 1.909859... which is less than 2, so there should be no problem to fit that into the Aron's space of 2. Even if we needed a bit of space for both rolls, there's plenty of space for each of them. This is why Tosfos is bothered by the Gemara's question of, "How can an object of greater than 2 fit into a space of 2?"

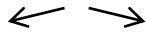
סתם, I don't really understand the difference between one 2-T-wide-roll and two 1-T-wide-rolls because at the end of the day, it's the same amount of parchment pressing against each other.



But now we have a new problem. If this indeed is the Gemara's first question, then what's the Gemara's next question? By the second question the Gemara says that the scroll was rolled to the end, meaning that it was a big roll. If so, then it must have been *smaller* than 2T across! Why? Because we just made an אוקימתא that the scroll was 2T wide when it was rolled into 2 small rolls. If so, then when it was rolled into one big one it will be bigger than 2T! And it won't matter if we use 3 for pi or 3.14159, which means now we don't know what Tosfos's second question is either!

Let's do the cheshbon.

Each little roll is 3T around.



$d = 3T \div 3 = 1T$ (diameter)	$d = 3T \div \pi = .95492...T$ (diameter)
$\div 2 = .5 =$ radius	$\div 2 = .47746... =$ radius
$A = 3r^2$	$A = \pi r^2$
$A = 3 * .5^2$	$A = \pi * .47746...^2$
$A = 3 * .25$	$A = 3.14159 * .2279...$
$A = .75$	$A = .716197...$
times 2 little rolls for the big roll's area	times 2 little rolls for the big roll's area
$= 1.5$	$= 1.43239...$
$A = 3r^2$	$A = \pi r^2$
now for the big roll's area	now for the big roll's area
$1.5 = 3r^2$	$1.43239... = 3.14159... * r^2$
divide both sides by 3	divided both sides by π
$.5 = r^2$	$.4559... = r^2$
square root both sides	square root both sides
$.707... = r$	$.6752... = r$
*2 for diameter = **1.4142...**	*2 for diameter = **1.35...**
Which fits into a space of 2, no problem.	Which *also* fits into 2, no problem.

This is a bigger problem than before. Before we made this אוקימתא (that the little rolls' combined circumference is 6) Tosfos had a problem with the Gemara's second question because it implied that the exact value of pi was 3 when it's really 3.14159...

But now, we can't understand the Gemara's question no matter how we slice it. The Gemara asks, "How can an object greater than 2 fit into a space of 2?" It's *not* greater than 2, whichever way you cheshbon it!

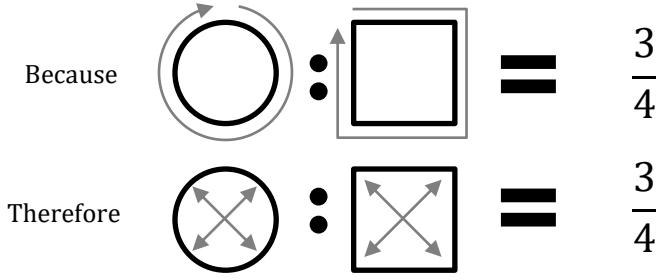
I guess what we would have to say is that אין הכי נמי, at this stage of the Gemara, the Gemara holds that the big roll was 6 around.[49]

*

:י"ד - Rashi's Proof of Circle's Area

The Gemara says (in a bigger context), "כמה מרובע יתר על העיגול רביע" "How much is a square['s area] bigger than a circle['s area of the same width]? One quarter [of the square's area]."

Rashi proves it like this: "Since every circle that is 1 amah wide has a circumference of 3 amos[50] and if [this shape] were a square of 1 by 1 amah it would have a circumference of 4, it turns out that a square's area is greater than a circle's by $\frac{1}{4}$."



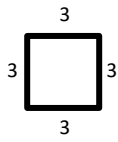
[49] In other words, the Gemara knows the circumference is 6. It's just not sure if that's one roll or two rolls.

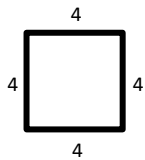
[50] משנה י"ג.

It sounds like Rashi is saying that if you have any two shapes, the ratio of their circumferences (i.e., one divided by the other) will be the same as the ratio of their areas.

I was baffled by this Rashi. I couldn't see what one has to do with the other.

And in fact, Yishai Rasowsky showed me a counter-proof. Let's say you had two squares, one of them was 3x3 and the other one was 4x4.





The circumference of the little square is **12** $(3 + 3 + 3 + 3)$

The circumference of the big square is **16** $(4 + 4 + 4 + 4)$

That means the ratio of the little square's circumference to the big one's is $\frac{\mathbf{12}}{\mathbf{16}}$

The area of the little square is **9** $(3 * 3)$

The area of the big square is **16** $(4 * 4)$

That means the ratio of the little square's area to the big one's is $\frac{\mathbf{9}}{\mathbf{16}}$

Here's an example where we have two shapes whose ratios of their circumferences and the ratios of their areas are not the same.

How can Rashi say that we know the area of the circle from the area of the square based on the differences between their circumferences when we see that it's not always true?

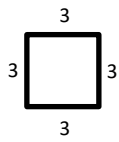
It *happens* to be true in the Gemara's particular case (a circle and a square with the same width) but we're trying to find a *proof* that it's true.

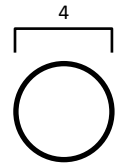
Tosfos in Sukkah[51] brings more counter-proofs to Rashi. Let's say you have three shapes.

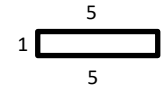
One is a square 3x3.

One is a circle with a width of 4.

And one is a rectangle 1x5.







The circumference of all these shapes is **12**

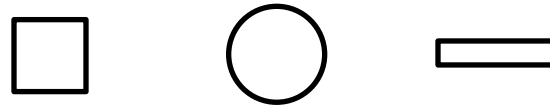
(The square is $3 + 3 + 3 + 3$, the circle's width is 4 so its circumference is 3 times that, and the rectangle is $1 + 5 + 1 + 5$)

And yet they all have different areas.

The square is **9** (3*3)

The circle is **12** $(4*4*\frac{3}{4})$

The rectangle is **5** (1*5)



Circumference: $\frac{12}{\quad} \neq \frac{12}{\quad} \neq \frac{12}{\quad}$

Area: $\frac{}{9} \quad \frac{}{12} \quad \frac{}{5}$

[51] ח'. ד"ה כמה

Nosson Kallenbach and Yishai Rasowsky suggested the following תרץ for Rashi.

(All terms will be defined more specifically afterwards.)

It can be proven that for any given radius, regardless of the number of sides of the shape, that the ratio of the circumference to the area will always be the same $\left(\frac{2}{r}\right)$.

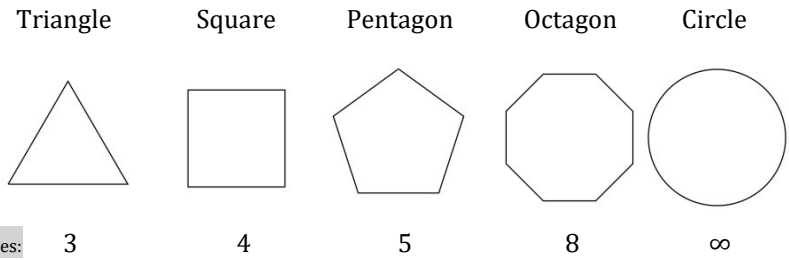
Therefore, it follows that if the ratio of the circumferences of a circle to a square (the same width) is $\frac{3}{4}$ then the ratio of their areas will also be $\frac{3}{4}$.

Whoa, that was a mouthful. Let's break that down.

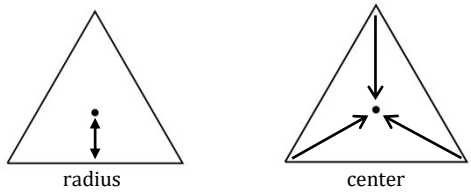
First off, let's define what we mean when we say "regardless of the number of sides of the shape".

What we are talking about are **Convex Regular Polygons**, that is, shapes that have all their sides the same size, and all the angles are the same, and it "bulges" out (as opposed to a "star" shape).

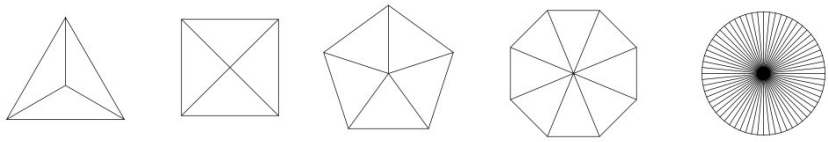
For examples:



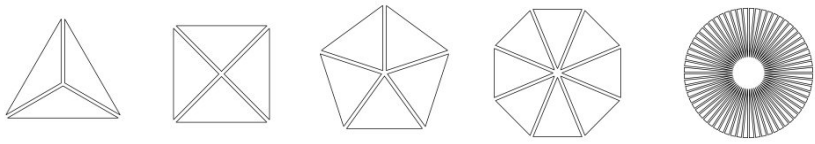
We're going to define the **radius** as the distance between the center of a side until the **center** point of the shape that is equidistant from every corner.



Let us relist our shapes again, each with the same radius. Let's cut each shape into a "ring" of triangles, with the peaks of the triangles meeting in the middle, and the bases of the triangles forming the edges of the shapes, like so:

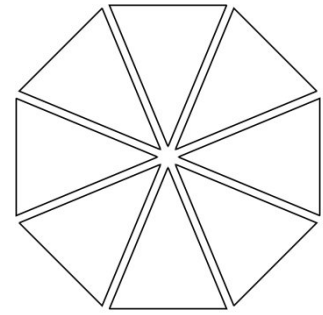


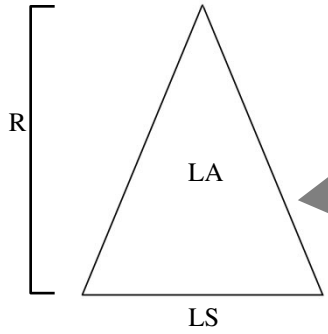
Do you see the little triangles inside each shape?



OK, great.

Now, let's look at each individual "triangle slice" and label and measure its parts.



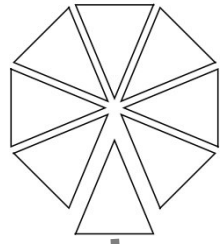


R = radius of the original "big" shape, the height of the little triangle slice.

LA = Little Area, the area of the little triangle slice, which is the big shape's area divided by the number of sides of the shape.

LS = side of little triangle slice, which is the big shape's circumference divided by the number of sides of the shape.

Now let's put everything into algebraic terms:

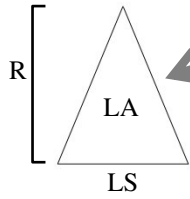


NOS = Number Of Sides of big shape

TA = Total Area, the area of the big shape = LA * NOS

TC = Total Circumference of the big shape = LS * NOS

The Little Area can be calculated:



LA = .5 * LS * R area of a triangle is .5 * W * H

Therefore, TA = .5 * LS * R * NOS

Therefore, the big shape's circumference to area ratio can be expressed as:

$$\frac{TC}{TA} \quad \text{...which expands into...} \quad \frac{LS*NOS}{.5*LS*R*NOS}$$

Now, we can cancel out the "LS" and the "NOS" both above and below the dividing line:

$$\frac{\cancel{LS} * \cancel{NOS}}{.5 * \cancel{LS} * R * \cancel{NOS}}$$

And we're left with:

$$\frac{1}{.5 * R}$$

Make it neater by multiplying above and below the line by 2:

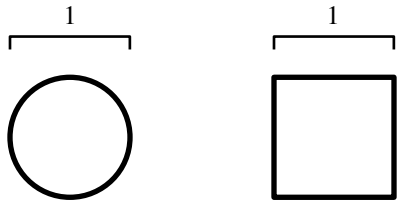
$$\frac{\mathbf{2}}{\mathbf{R}} = \frac{\textbf{Shape's Circumference}}{\textbf{Shape's Area}}$$

You see that we were able to factor out the number of sides. This proves that for a given radius, any particular shape will have the same ratio between its circumference and its area.

So now, back to Rashi.

If I have a circle and a square **the same radius**, the ratios of their circumferences to their areas will be the same.

So, let's say we have these shapes with a width of 1 amah (a radius of .5 amah)



Circumference:	3	=	4
Area:	?		1

Since we know that these ratios are the same we can figure out what should be below the 3 in the above diagram.

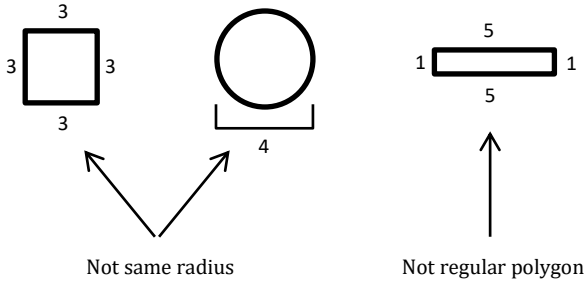
Since the circumference of the circle to that of the square's is **3 to 4** (3/4) therefore the area of the circle to the square's area must also be **3 to 4** (3/4).

Therefore, since the square's area is 1 the area of the circle must be 3/4 of that (a ratio of 3 to 4).

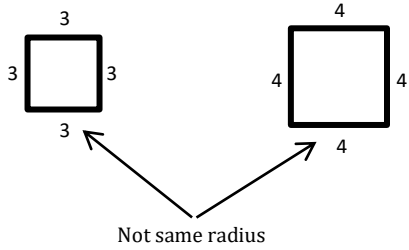
The counter-proofs that Tosfos and Yishai brought are irrelevant since we only proved this ratio with same-radii-convex-regular-polygons.

In the examples brought by Tosfos and Yishai, they were either not the same radius or were not regular polygons, or both.

Tosfos:



Yishai:



So, Rashi's words are right. Since the circumference of a circle to the circumference of a square [the same width] is 3/4, **therefore** the area of a circle to the area of a square **will be 3/4.**

This is the source of the Gemara's statement, "every square is 1/4 bigger than a circle."

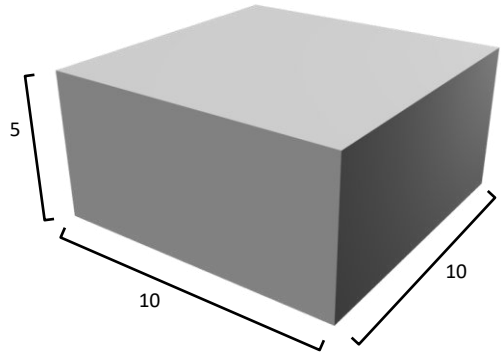
*

:"ד' - Shlomo's Pool - Round and Square

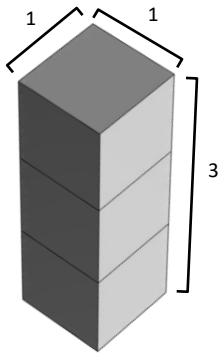
The Gemara's initial assumption is that the ים של שלמה, "Shlomo's Pool[52]" is a cube[53] 10 amos long, 10 amos wide, and 5 amos high.

The total volume is thus

$10 * 10 * 5 = \mathbf{500\ A^3}$



We have a Braisa from Rav Chisda that the pool could hold **150 mikvaos** (150 M).



A mikveh is 1 x 1 x 3 amos, for a total of **3 A^3**.

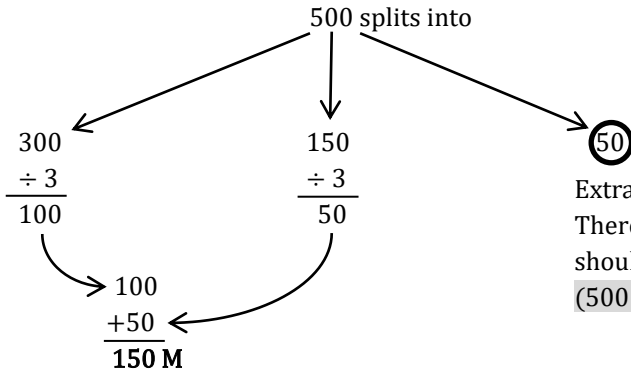
If the pool is 500 A^3 and a mikveh is 3 A^3 then how many mikvaos are there in the pool?

$$\begin{array}{r} 500 \\ \div\ 3 \\ \hline \mathbf{166\frac{2}{3}} \end{array}$$

[52] Literally: "Sea of Shlomo"

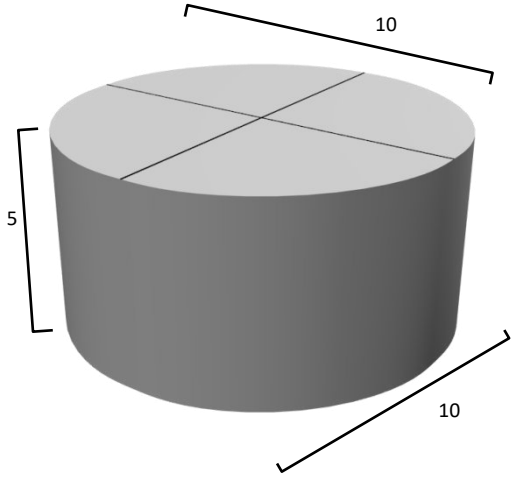
[53] I'm using the term "cube" in its loose sense, like a box. The more accurate term would be "cuboid" or "square cuboid".

Or the Gemara's method of doing it:



Extra!
Therefore, **450 A³** should be enough (500 - 50)

So, the Gemara assumes a new possibility, that the pool was round like a cylinder:



So now the volume can be calculated like this:

$V = \pi r^2 * height$

$V = 3*5^2*5$

375 A³ $= 3 * 25 * 5$

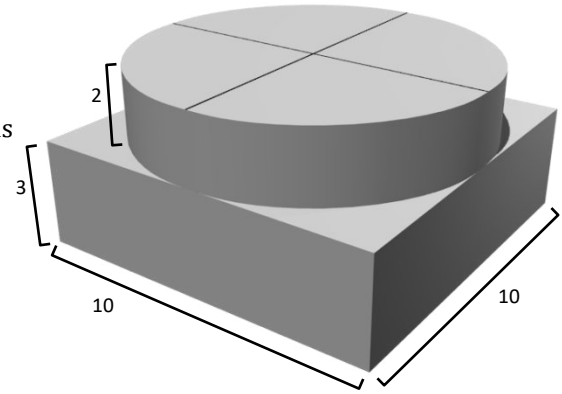
This doesn't seem to help. We need **450 A³** (150 M * 3 A³) but now all we have is **375**.

A cube shape is too much and a cylinder shape is not enough.

So now the Gemara gives us a combination answer. Part of the pool was square (like a box) and part of it was round (like a cylinder).

The Gemara says the lower part was square and was 3 high. The upper part was round and was 2 high.

Let's calculate the volume of this shape (which looks a bit like a container for cream).



$$\underbrace{\text{box}}_{l * w * h_1} + \underbrace{\text{cylinder}}_{\pi * r^2 * h_2}$$

$$10 * 10 * 3 + 3 * 25 * 2 = \mathbf{450\ A^3}$$

$$\div\ 3$$

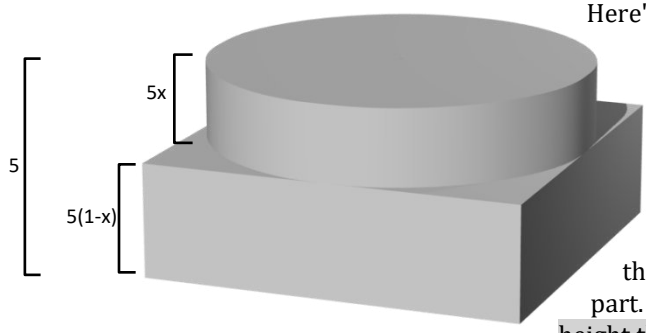
$$\mathbf{150\ M}$$

Excellent.

Now, here's the question. How did the Gemara know to give these numbers of 3 for the bottom and 2 for the top (assuming it didn't just guess and check)?

I'm going to demonstrate three methods I can think of. They're all basically the same thing just with slightly different approaches. Some people might relate better to one method over another. Besides, I like to work out everything from every direction. Sometimes I find interesting things that way!

Method 1 - Percentages



Here's what we know.

We know that the total height is 5.

We know that a certain portion of that height, let's call it "x" amount of the height, is for the round part. (If it was the entire height then x will be 1, so that 5 * 1 will equal the entire 5. If it was half the height then x will be .5 so that 5*.5 = 2.5. And so on. That's the idea.) What we're looking for is that "x" should be an amount between 0 and 1.

The remainder of the height will be "1 - x" times the entire height of 5, and that will be for the box part. That could be written out "5(1-x)", like in the picture.

Clear?

So now, we could write out the volume of the cylinder as: $v = \pi r^2 * 5x$

And the box part's volume can be written as: $v = l * w * 5(1-x)$

OK?

Now we know that the entire volume is supposed to be 450 A^3 (150 M).

So now we could add the volumes of the cylinder and the box and it should equal 450. Let's write that all out in algebraic form and then solve for x.

cylinder + box = 450

write out the formulas

$\pi * r^2 * h_1 + l * w * h_2 = 450$

fill in the numbers

$3 * 25 * 5x + 10 * 10 * 5(1 - x) = 450$

$75 \qquad 100$

$375x + 500 * (1 - x) = 450$

subtract 375x from both sides

$500(1 - x) = 450 - 375x$

divide both sides by 500

$1 - x = \frac{450 - 375x}{500}$

split the right side

$1 - x = \frac{450}{500} - \frac{375x}{500}$

simplify the right side

$1 - x = .9 - .75x$

add 1x to each side of the equation

$1 = .9 + .25x$

subtract .9 from both sides

$1 - .9 = .25x$

simplify the left side

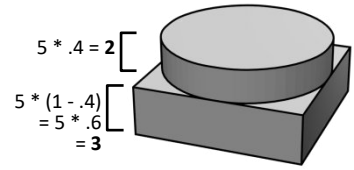
$.1 = .25x$

multiply both sides by 4 (to bring .25x up to 1x)

$.4 = x$

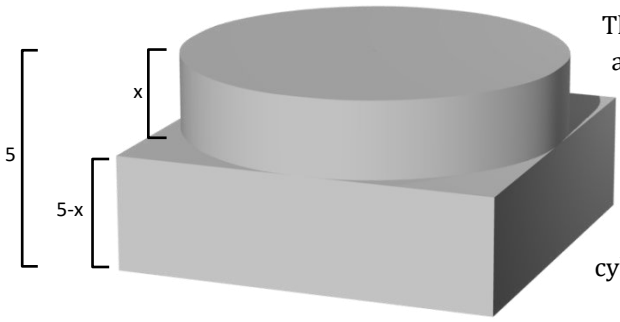
Now let's plug that into our original formula:

And we get cylinder height = 2
box height = 3



Amazing.

Method 2 - Integers



This time we're going to do it a little different.

This time, we'll take one of the measurements we're looking for, let's say the height of the cylinder, and just call it "x".

The box will then be whatever much of 5 that isn't x. Put another way, the box's height will be 5 - x.

So now, we could write out the volume of the cylinder as: $\mathbf{v = \pi r^2 * x}$

And the box part's volume can be written as: $\mathbf{v = l * w * (5 - x)}$

Now in algebraic form:

$\text{cylinder} + \text{box} = 450$	write out the formulas
$\pi * r^2 * h_1 + l * w * h_2 = 450$	fill in the numbers that we know
$3 * 25 * x + 10 * 10 * (5 - x) = 450$	multiply what we can
$75x + 100(5 - x) = 450$	subtract 75x from both sides
$100(5 - x) = 450 - 75x$	divide both sides by 100
$5 - x = \frac{450-75x}{100}$	split the right side
$5 - x = \frac{450}{100} - \frac{75x}{100}$	simplify the ride side
$5 - x = 4.5 - .75x$	

$5 - x = 4.5 - .75x$

add 1 x to both sides

$5 = 4.5 + .25x$

subtract 4.5 from both sides

$.5 = .25x$

multiply both sides by 4 (to bring .25x up to 1x)

$\mathbf{2 = x}$

cylinder height = **2**

box height = 5 - x = 5 - 2 = **3**

And there you have it.

Method 3 - Integers - Gemara's method

We're going to do the same thing as before (in Method 2) using "x" as the height of the cylinder and "5 - x" as the height of the box but we're going to write out the formulas a little different to match the way the Gemara presents it.

To calculate how many mikvaos are in a cylinder, multiply the length by the width and by the height and divide by 4.

The reason that works is because l * w * h is the volume of a cube. $\frac{3}{4}$ of that is a cylinder. $\frac{1}{3}$ of *that* is the number of mikvaos (since a mikveh is 3 A^3). $\frac{1}{3}$ of $\frac{3}{4}$ is $\frac{1}{4}$

To calculate the box just do l * w * h ÷ 3 to get the number of mikvaos.

We said before that Shlomo's pool held 150 mikvaos. So the formulas would be:

write out the formulas

$$\overbrace{l * w * h_1 \div 4}^{\text{cylinder}} + \overbrace{l * w * h_2 \div 3}^{\text{box}} = 150 \text{ M}$$

fill out what we know

$$10 * 10 * x \div 4 + 10 * 10 * (5 - x) \div 3 = 150$$

Now solve for "x".

$$10 * 10 * x \div 4 + 10 * 10 * (5 - x) \div 3 = 150$$

multiply what we can

$$100x \div 4 + 100(5 - x) \div 3 = 150$$

divide what we can

$$25x + 33\tfrac{1}{3}(5 - x) = 150$$

subtract 25x from each side

$$33\tfrac{1}{3}(5 - x) = 150 - 25x$$

divide both sides by $33\frac{1}{3}$

$$5 - x = \frac{150 - 25x}{33\frac{1}{3}}$$

split the right side in half

$$5 - x = \frac{150}{33\frac{1}{3}} - \frac{25x}{33\frac{1}{3}}$$

simplify the right side

$$5 - x = 4.5 - .75x$$

and now just like last time, add 1x to each side

$$5 = 4.5 + .25x$$

subtract 4.5 from each side

$$.5 = .25x$$

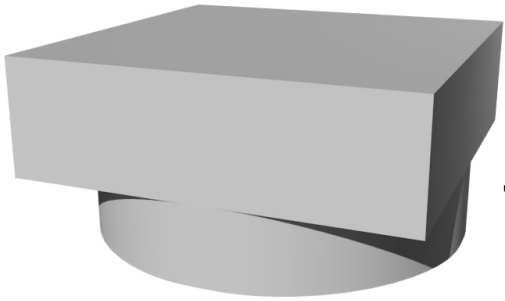
multiply both sides by 4 (to bring .25x up to 1x)

$$\mathbf{2} = x$$

Great!

By the way, if we did these methods using the true value of pi instead of 3 then the cylinder height would be about 2.33 amos and the box height would be about 2.67 amos.

Work it out.

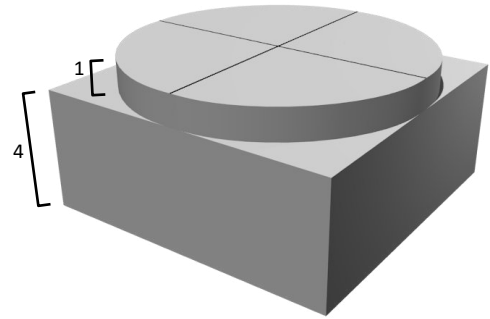


The Gemara points out that we can't reverse which is on top and which is on bottom (that is, to make the bottom part the cylinder and make the top part the box) because the Passuk calls the "lip" (the uppermost edge) a circle.

The Gemara then asks אימא חדא, maybe the cylinder part was only one amah tall (and the box 4)?

This question seems out of place. The Gemara already determined the heights of the box and the cylinder.

If indeed the box was 4 tall and the cylinder 1, then the total volume will be



$$v = \frac{3}{4} 10 * 10 * 1 + 10 * 10 * 4 = \mathbf{475\ A^3}$$

Divide by 3 to get $\mathbf{158\frac{1}{3}}$ **mikvaos**, which is too much.

The Gemara's response to this suggestion will inform us what exactly the Gemara was asking.

The Gemara responds by <u>proving</u> that the total volume of the pool was indeed 150 mikvaos. That means that the original question really was "**What is the source for Rav Chisda's 150 mikvaos?**"

The Gemara proves the volume of the pool from the Passuk.

The Passuk says that the pool held 2000 "baths"[54].

Each bath is $\frac{1}{10}$ of a kor[55].

Each kor is 30 sa'ah[56].

Which means that each bath is $\frac{1}{10}$ of 30 sa'ah, which comes out to **3 sa'ah per bath**.
(30 ÷ 10)

Which means that the pool held **6000 sa'ah.**
(2000 * 3)

Since each mikveh is 40 sa'ah, it comes out that there are **150 mikvaos** in the pool.
(6000 ÷ 40)

Shown another way:

1 Mikveh = 40 sa'ah
÷30 turn into kors
$1\frac{1}{3}$ kor
x10 turn into baths
$13\frac{1}{3}$ baths
x150
2000 baths

And another way:

Shlomo's pool	2000 baths
1 bath	.1 kor
1 kor	30 sa'ah
1 mikveh	40 sa'ah

The Gemara then brings a contradictory passuk[57] which says that the pool held 3000 baths.

3000 x 3 = 9000 sa'ah
÷ 40
225 mikvaos

[54] מלכים א' ז':כ"ו אלפים בת יכיל
[55] יחזקאל מ"ה:י"ד מעשר הבת מן הכור
[56] רש"י ד"ה מעשר הבת מן הכור
[57] דברי הימים ב' ד':ה'

This question is not even relating to the cylinder:box ratio, since even if the entire pool was a cube (10 x 10 x 5) it still wouldn't be 3000 baths (225 mikvaos).

$10 \times 10 \times 5 = 500\ A^3$

$\div 3$

$166\frac{2}{3}$ M

x 40

$6{,}666\frac{2}{3}$ sa'ah

$\div \quad 3$

$2{,}222\frac{2}{9}$ baths

The pessukim are contradicting each other.

The Gemara answers that the 2000 baths is referring to the "wet measure" while the 3000 baths is referring to the "dry measure".

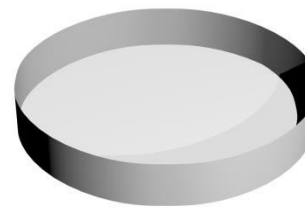
1 wet bath = 1.5 dry baths.

Rashi[58] says that a person can heap up [dry material above the rim of the container].

Rashi continues, "...therefore, the heap is one third, so a container whose height is about half its width [like Shlomo's pool]..."

The Maharsha explains Rashi (I'm paraphrasing, sort of, and adding in comments of the Ritva).

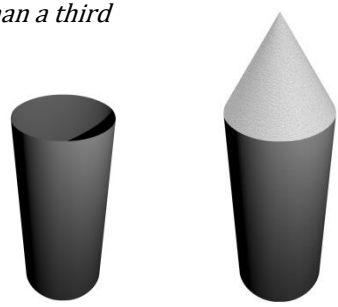
Let's say you had a container that held 1 sa'ah [of flour or some material] but it was very wide [and shallow].



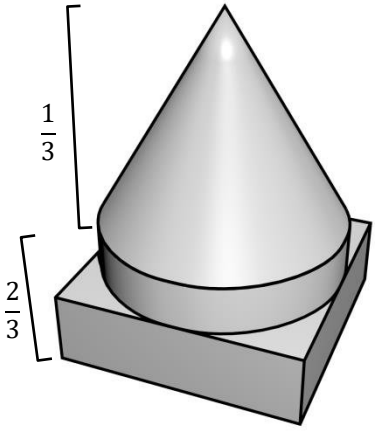
[58] ד"ה שהן כוריים ביבש

Obviously, the amount that can be heaped in such a keili will be far more than a third of the total volume that this kli can hold (within it plus the heap).

And if we had a kli that was very deep and narrow then the heap will be much less than a third of the total volume.



Rather, what we mean is that if the kli was made like Shlomo's pool, to scale, then the heap will be 1/3 the total volume.

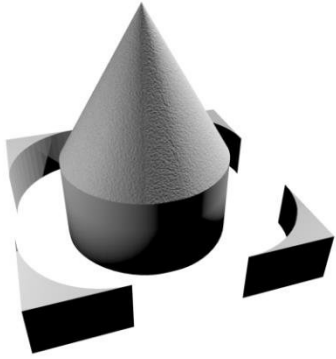


Rashi does not mean that any kli that is half as tall as it is wide will automatically have a heap that is 1/3 the total volume.

This can be proven very simply.

If a kli was round like a cylinder and its height was half its width, the heap will be the same as Shlomo's pool (since the circular top is where the heap forms) but the volume inside the kli will be less, since it is like Shlomo's pool except the corners have been shorn off the square part!

Rather, what we mean is that just like Shlomo's pool was about half as tall as it was wide, and some of it was round and some of it was square, so that its heap was 1/3 the total volume, so too any kli you make should have that same quality (that the heap is 1/3) whether it's all round or all square, even if the height:width ratio be different.



And this is what Tosfos meant "for buying and selling". It means that if someone buys a certain amount [of grain or flour[59]] or is hired to make a container, it should be done in such a way that the heap will be 1/3 the total volume. [That is, when buying or selling grain they should use containers that heap up 1/3, and when one is contracted to build a keili he should make it so that the heap will be a third[60].]

Until now, we have been saying that the dry measure with the heap is 3/2 the wet measure. (That is, the heap is 1/3 the total volume. Put another way: the heap is 1/2 the volume of what's inside the container itself.)

The way it seems the מפרשים are learning this is that Shlomo's pool itself, if it were filled up with dry material, would heap up 1/3 its total volume.

Let's calculate the dimensions of this heap.

We said the volume inside Shlomo's pool was 450 A^3. The heap should therefore be half of that, **225 A^3**, in order to be 1/3 the total volume (450 + 225 = 675, and 225 is 1/3 675). Let's solve for the height and the angle that the heap heaps up.

The formula for a cone is $v = \frac{1}{3}h * \pi * r^2$

Therefore, $225 = \frac{1}{3}h * \pi * 25$

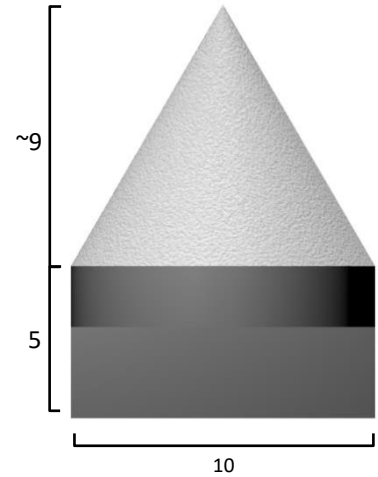
$\frac{1}{3}$ of 25 is $8\frac{1}{3}$

$225 = h * \pi * 8\frac{1}{3}$

divide both sides by $8\frac{1}{3}$

$27 = h * \pi$

If π is 3 then $h = 9$
If π is 3.14... then $h = 8.59...$



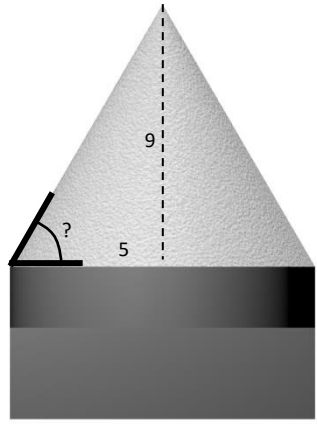
[59] I have not found a discussion concerning the type of material we're dealing with. This will be addressed somewhat in the next piece.
[60] Assumedly, this will change on the type of material being sold. See next piece.

The angle of the cone of heaped up dry material is called the **angle of repose**, that is, at what angle does dry material heap up.

Different materials have different angles of repose.

Let's calculate this cone's angle of repose.

Let's view the cone from the side and look at the left half the cone as a right triangle.



To calculate the angle, we use the following formula[61]:

$\text{angle} = \tan^{-1}\left(\frac{o}{a}\right)$

$\text{angle} = \tan^{-1}\left(\frac{9}{5}\right)$ using 3 as pi

$\text{angle} = \mathbf{60.945...}°$

If we use the true value of π to find the height (as we did on the previous page) then the formula would be:

$\text{angle} = \tan^{-1}\left(\frac{8.59...}{5}\right)$

$\text{angle} = \mathbf{59.81...}°$

Either way, it's pretty steep.

The Wikipedia article for "Angle of Repose" has a list of materials and their angles of repose.

Of particular interest to us are two entries:

Wheat flour - AOR = **45**° (max.)

Wheat granules - AOR = **27**°

Assuming the Gemara was talking about one of these, how do we reconcile these values with the calculations we just made about the heap pile?

[61] See Appendix J about trigonometry

Shall we suggest that the wheat in those days was of a different consistency, resulting in a significantly higher angle of repose? Or that they measured the heaps differently? Perhaps in more perfect conditions? Or maybe it wasn't wheat; maybe they were measuring Turkish coffee or cocoa powder?!

Had it not been for the Rishonim, I would've explained the Gemara like this: Not that Shlomo's pool filled up with dry material would produce a heap of 1/3 the total volume, but rather that the wet measure of 2000 bath in Shlomo's pool is equal to 3000 bath of dry measure.

In other words, there's a certain type of kli that when heaped up with dry material the heap will be 1/3 the total volume. Whatever total volume of dry material that keili can hold is the keili's "dry measure". Whatever amount of liquid that this same keili can hold is the keili's "wet measure".

We learn this ratio from the pessukim about Shlomo's pool. The Passuk tells us that the amount that Shlomo's pool holds in wet measure is 2000 baths, *which equals 3000 baths using the dry measure*.

And when you build a measuring kli for buying and selling dry material you need to do it in such a way that the heap will be 1/3 the total volume.

Assuming that the angle of the heap will be 45° (the angle of repose for wheat flour), let's calculate what ratio of width-to-height will be needed in a kli.

(I'm going to do the calculations assuming the kli is a cylinder. The calculations can actually be done with either a cylinder/cone or a box/pyramid and it will come out the same. Try it.)

Let's lay down some terms here:

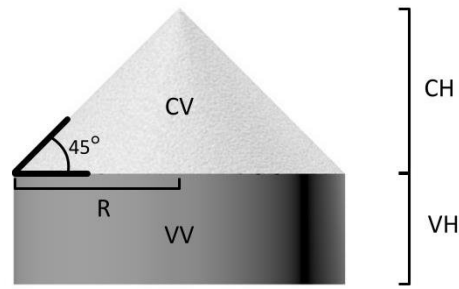
VV = Vessel Volume

VH = Vessel Height

CV = Cone Volume

CH = Cone Height

R = Radius (of cone base and of vessel width)

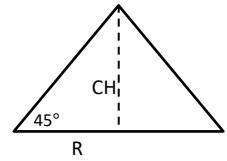


Now, we know that CV = .5VV (so that it is a third of CV + VV)

CH could be calculated as CH = tan(45) * R

$$CH = 1 * R$$

$$\mathbf{CH = R}$$



The Cone Volume can be calculated with the cone formula:

$$CV = \frac{1}{3}\pi * R^2 * CH$$

replace CH with R, as we proved above

$$CV = \frac{1}{3}\pi * R^2 * R$$

multiply the "R"s together

$$\mathbf{CV = \frac{1}{3}\pi * R^3}$$

The Vessel Volume can be calculated:

$$VV = \pi * R^2 * VH$$

Since we said CV = .5VV so we can turn what we got for VV into $.5\ \pi * R^2 * VH$

So now we can put the 2 formulas for CV together:

$$\frac{1}{3}\pi * R^3 = .5\ \pi * R^2 * VH$$

divide both sides by π and R^2

$$\frac{1}{3}R = .5\ VH$$

double each side (Radius becomes Diameter)

$$\mathbf{\frac{1}{3}D = VH}$$

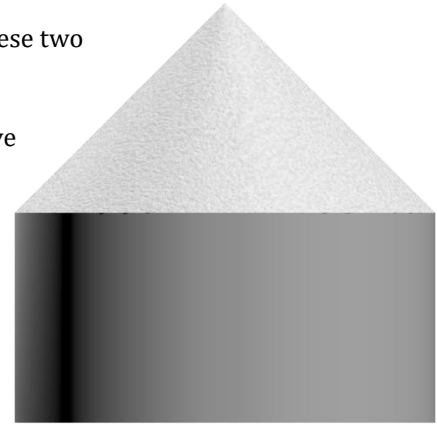
1/3 Diameter = Vessel Height

So, if you make a vessel with a diameter 1/3 the vessel's height (for example, 1 across and 3 high) then the heap of flour that heaps up will be 1/3 the total volume of all the flour that this vessel can hold (including the heap).

Aaron Kronenfeld suggested that perhaps the Torah is telling us that you can use one of two methods when buying or selling something.

For example, when a buyer asks for a dry measure of a sa'ah, the seller has the option of selling him 1.5 sa'ah in wet measure or he can use a kli that is half as tall as it is wide (like Shlomo's pool) with an inner volume of 1 sa'ah, and fill it up all the way with the heap. Now, even though this second method will give a little bit less than 1.5 sa'ah, that is the right of the seller (if you don't like it go buy from someone who sells the other way). The point is that the Torah is validating this method of sale and it's not considered overcharge.

What is the volume difference between these two methods?



A kli that is half as tall as its width will have an inner volume of

$$\frac{\pi}{4} * d^2 * .5d$$

The heap will be 1/3 of that for:

$$\frac{1}{3} * \frac{\pi}{4} * d^2 * .5d$$

Comparing the two formulas, we can see that the inner volume (of 1 sa'ah in our example) will be 3 times the volume of what's in the cone. So the cone is 1/3 of a sa'ah.

So the difference between selling the first method (1.5 sa'ah) and the second method $\left(1\frac{1}{3}\text{ sa'ah}\right)$ is 1/6 of a sa'ah $\left(1.5\ -\ 1\frac{1}{3}\right)$.

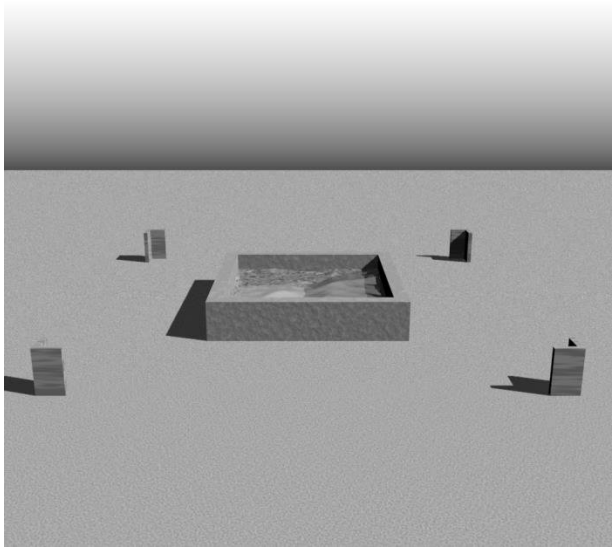
Put another way, buying with the first method will yield 9 portions of flour while buying with the second method will yield 8 portions of flour.

*

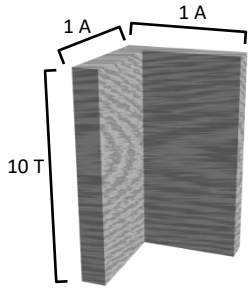
:"ט - Dyumad - Slope of 10 Tefachim Within 4 Amos

The Gemara is discussing a דיומד, "dyumad", which mean a "double-board". These are L-shaped boards that form part of a mechitza for the area surrounding a public water well, out in the open.

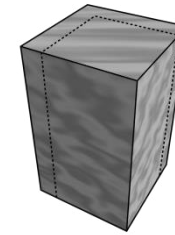
The dyumads are arranged in four corners delineating an area around the well.



Each dyumad needs to be at least 10 tefachim high and the individual sides of it need to be at least one amah wide.



Although the requirement is to have these boards in an "L" shape, there are ways they can be not an "L" shape, as long as theoretically an "L" shape could be cut out from it. For example, if you have a block of wood that is 1 amah wide by 1 amah thick by 10 tefachim high, that would be a kosher dyumad since a kosher "L" shape could theoretically be cut from it[62].



Taking this idea further, the Gemara wonders about a mound that is much bigger than a single dyumad, if we could "cut away" whatever isn't dyumad and make one.

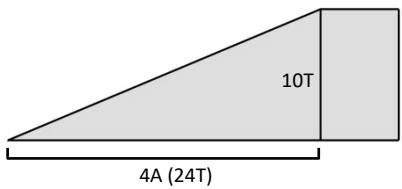
The Gemara's term for the dimensions of this mound is תל המתלקט עשרה מתוך ארבע, a mound that is **"gathered up" 10** [tefachim] **within four** [amos].

10 tefachim is referring to the height.

What does "4 amos" mean? I can think of two פשטים:

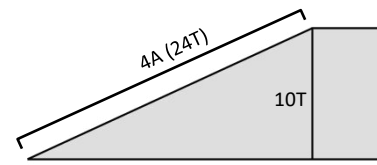
Either it means a horizontal distance of 4 amos (rise over run[63]) or it means the slope measures 4 amos (rise over slope), in which case the horizontal measurement will be less than 4 amos.

Rise Over Run



Slope angle $= \tan^{-1}\left(\frac{10}{24}\right) =$ **~22.6°**

Rise Over Slope



Slope angle $= \sin^{-1}\left(\frac{10}{24}\right) =$ **~24.6°**

I've replaced the "4A" with "24T" (since there's 6 tefachim per amah)

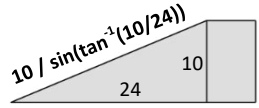
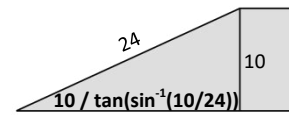
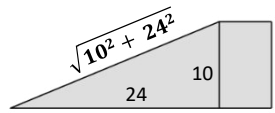
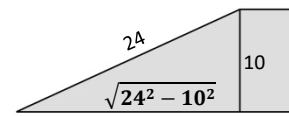
[62] When I say "cut from" I don't mean it has to physically be able to be cut. It means that this shape "contains" another shape within it, implicitly. I mean, "cut away" in your imagination.

[63] This is a math term that is used to define a slope. In this case it's 10T over 4A.

Let's calculate the remaining sides.

I'm going to present two ways of doing it, with trigonometric functions and with Pythagorean theorem. Don't worry, they take longer to pronounce than to perform. ;-)

Some people might prefer one method over another.

	Rise Over Run Solve for slope	Rise Over Slope Solve for run
Trigonometric Function		
Pythagoras' Theorem		
Answer:	**26**	**21.817...**

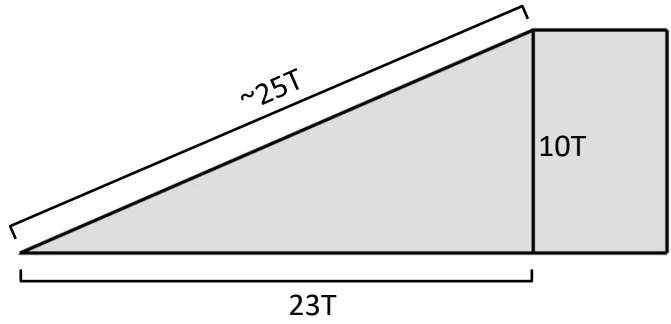
The trigonometric functions are simpler than they appear. For example, the one on the left is really just 10 ÷ the sine of the angle of the slope. We calculated that angle on the previous page as the arctangent of 10 ÷ 24. Embedded together it looks a little messy.

The main point of the Gemara is that if the mound is steep enough then we could view the area on top as a separate area from the ground at the bottom. If it's not steep enough then it is not considered a separate area and is just viewed as a "bump" of the surrounding area.

Now, what's the נ"מ between these two פשטים?

Give me a case where it will make a difference whether we say that "gathered up 10 within 4" means rise over run or rise over slope.

Let's say you had a slope that measured 23 tefachim horizontally, with a rise of 10.



The "run" is just under 4 amos (24T) while the "slope" is just over 4 amos.

According to the "Rise Over Run" פשט, this ramp is steep since it reached 10 amos within a horizontal run of 4 amos. Therefore, the flat area on top is viewed as a separate area.

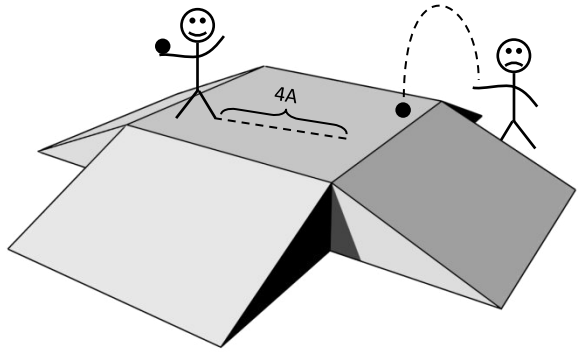
As far as the דין of dyumad is concerned, this is tall enough and steep enough to be a good dyumad[64].

The top of this mound is also considered a רשות היחיד and would have all the חומרות and קולות as such.

It would be מותר to carry on it even a distance of 4 amos (קולא). Conversely, one who throws an object onto it from a רשות הרבים would be חייב, even if he threw it less than 4 amos (חומרא). (Although, my son Yehuda wondered why the thrower would be חייב דאורייתא since the object went over 10T from the רשות הרבים, which means it passed through a מקום פטור. But from the Rambam we'll see on page 89, the guy would be חייב in such a case.)

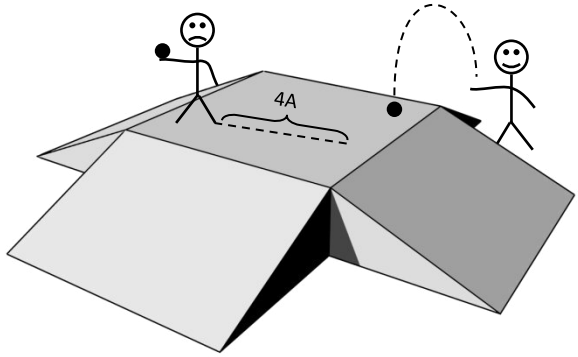
[64] Subject to another מחלוקת in the Gemara about how many times we can "cut" away surplus material.

Run of 23T - Rise Over Run Pshat:



On the other hand, the "Rise Over Slope" Pshat would say the very opposite. This is not a dyumad, and this is not a רשות היחיד, but rather is בטול to the area around it. For example, if the area around it is a רשות הרבים then the top of this mound is also a רשות הרבים. One who carries 4 amos on it is חייב (חומרא) and it is מותר to throw onto it from the רשות הרבים from a distance of less than 4 amos.

Run of 23T - Rise Over Slope Pshat:



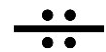
The Tur[65] sounds very much like the second pshat, rise over slope, when he writes:

ומתלקט עשרה מתוך ה' פירוש שבהילוך ה' אמות של שיפוע עולה עשרה טפחים אז... ואם הוא משופע יותר שבהילוך ד"א עולה עשרה טפחים אז...

"And [a slope] that gathers up 10 in 5, that is, by walking [or: going] 5 amos of the slope he [or:it] goes up 10 tefachim then (the din is such and such)... and if it is steeper, that is, with walking [or:going] (merely) 4 amos he [or: it] goes up 10 tefachim (then the din is such and such)..."

The Shuchan Aruch in the same siman says basically word for word like the Tur.

It sounds like the 4 amos is measured by the way the person walks, that is, up the slope. Or perhaps it merely means as it (i.e., the slope) goes up...



The Rambam[66] writes:

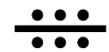
תל המתלקת גובה עשרה טפחים מתוך אורך ארבע אמות הרי הוא רשה"י ואם זרק מהרשה"ר ונח על גביו חייב

"A mound that is gathered up a height of 10 tefachim within a length of 4 amos behold it is a reshus hayachid and if he threw from a reshus harabim and it landed on it [he is] chayiv."

The Rambam's לשון is a little ambiguous. There's room to say either way. To me it sounds like rise over run ("height..." "length..."), although "length" could mean of the slope.

Markeves Hamishneh[67] writes that the Rambam means rise over run, though he proves it from elsewhere.

The שפת אמת[68] leaves the פשט in the Gemara as an open question.



[65] א"ח שצ"ט:ד'
[66] הילכות שבת י"ד:ט"ז
[67] הל' שבת י"ד:א' חלק ב' דף ל"ח Hebrewbooks.org book number 14647, page 76
[68] גמ' שבת ק'.

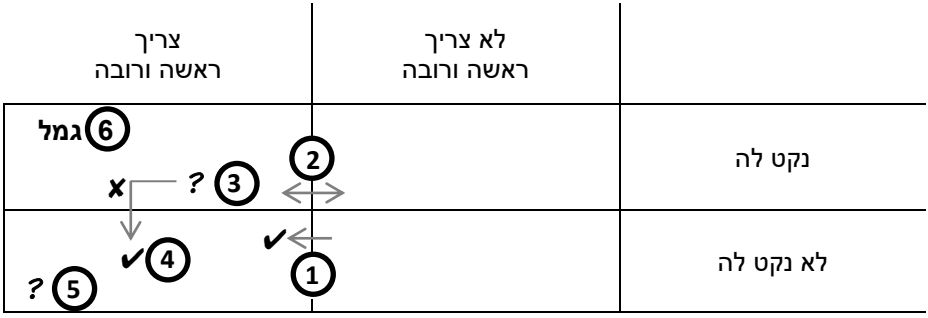
כ: - **Watering a Cow**

Here is another confusing שקלא וטריא.

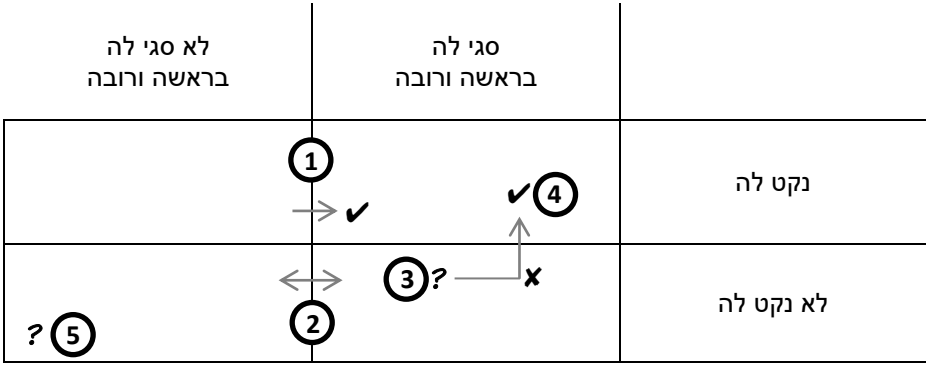
I wrote up a chart to organize the flow of the Gemara in a visually simple format.

First I will show the chart/s and then we will go through the Gemara step by step, keeping our eye on the circled numbers.

לשון ראשון:



איכא דאמרי:



Let me explain how each chart is broken up.

The two rows (labeled on the right side) are two separate cases, where he's holding the cow's head and where he isn't.

The two columns are the possible דינים.

A check mark (✔) in a particular square of the chart means that this case (row) is this דין (column). An X mark means this isn't the דין (or that it cannot be proven). A question mark means this דין for this case is being doubted.

The numbers show the flow of the Gemara.

The arrows show conclusions of the Gemara, or in the case of double arrows that the דין is questionable.

Lashon Rishon

The Gemara takes it for granted that in the case where he's not holding the cow that most of the cow needs to be in the same area as the water from which it is drinking (1).

The Gemara wonders about where he *is* holding the cow if most of the cow needs to be inside the area (2).

The Gemara brings a Mishnah which implies that the cow does need to be inside (3).

The Gemara rejects that proof by offering an alternative interpretation to the Mishna, that it's talking about where he's not holding the cow (4), like we said by (1).

The Gemara then brings a Braisa against this דין (that the cow needs to be inside in a case where he's not holding the cow) (5). And the Gemara answers that the Braisa is talking about a different case.

At this stage, the Gemara remains undecided what the din is where he's holding the cow.

Later, the Gemara will return to this stage and try to prove from another Braisa that where he's holding the cow the דין should be that most of the cow needs to be inside but the Gemara will reject that proof by saying that that Braisa is referring specifically to a camel (6) and has no bearing on the cow case.

Ikka Di'amri

The Gemara takes it for granted that in the case where he is holding the cow that it helps to have most of the cow in the same area as the water from which it is drinking (1).

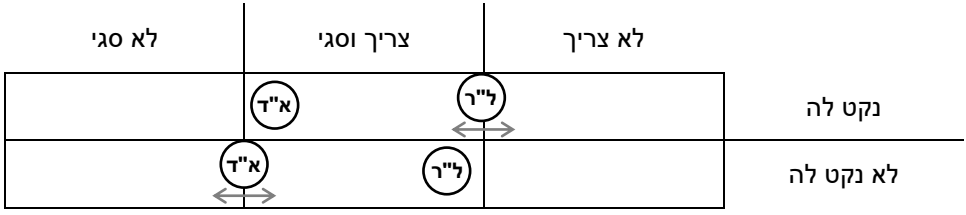
The Gemara wonders about where he is *not* holding the cow if it helps to have most of the cow inside the area (2).

The Gemara brings a Mishnah which implies that it does help to have the cow inside (3).

The Gemara rejects that proof by offering an alternative interpretation to the Mishna, that it's talking about where he is holding the cow (4), like we said by (1).

The Gemara then brings a Braisa to try to prove the דין that it doesn't help when the cow is inside in a case where he's not holding the cow (5). And the Gemara rejects this proof since this Braisa could be talking about a different case.

It would seem that we could combine these charts into one and show which לשון holds like what. The double arrows indicate inconclusiveness.



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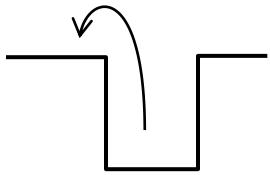
כ"א: - <u>Up and Down a Well</u>

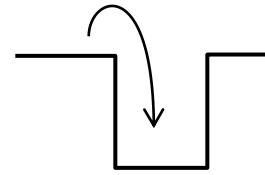
The Gemara brings a Braisa that says that a person should not drink from a well that has only the bare minimal mechitzos[69] by drawing the water out but rather he should מטפס ועולה מטפס ויורד "climb and go up, climb and go down".

When I first read these words I thought that the Braisa means he should come up out of the well and go down into the well.

Up (out):



Down (in):



This seems to be in the wrong order. First, one goes into the well and then comes out, unless if he lives in the well.

Rashi[70] says, "He goes in and goes out", which is the order that I would've expected.

Did Rashi change the order of the Gemara?

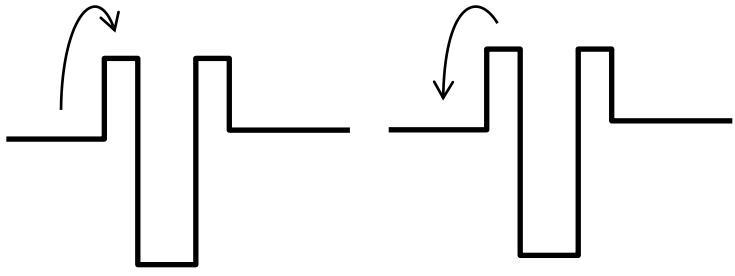
Or perhaps Rashi is defining for us what the Gemara's words mean. Maybe "up" means "in" and "down" means "out". Maybe we mean up the walls that surround the outside of the well[71].

[69] See picture on page 84

[70] ד"ה מטפס

[71] Yishai Rasowsky suggested a slightly alternative pshat, that it means up the well wall and down into the well. Artscroll also seems to learn this way.

Up (in)[72]: Down (out):



Mesoras Hashas on these words of the Gemara sends us to a Tosfos in Pesachim[73]. The Gemara there brings a statement by Rav Yehuda in the name of Rav that a person should always "go in when it's good; go out when it's good." Tosfos there wonders why the Gemara precedes going in to going out. Seemingly, Tosfos is bothered by the fact that one needs to go out of his home before coming in. Tosfos answers that "it is the style of the Gemara to precede going in to going out... like in Eruvin where it says he climbs and goes up; climbs and goes down."

When I first saw this Tosfos I thought that Tosfos is interpreting our Gemara in the second way we explained it, that "up" means "in" and "down" means "out". The way Tosfos would read would be: "It is the style of the Gemara to precede in to out like the Gemara does in Eruvin where it preceded in to out."

However, if that was the intent of Tosfos, then what sort of proof is Tosfos bringing from Eruvin? Tosfos is trying to prove that the Gemara precedes in to out even in cases where the opposite order would be more logical. If the Gemara in Eruvin *means* in and then out then Tosfos hasn't proven anything! If the order that appears in Eruvin is the logical order then we learn nothing about the style of the Gemara.

[72] A possible support for this פשט is the Gemara in שבת צ"ט: that says that the area on top of a wall enclosing a רשה"י has the דין of a רשה"י.
[73] ב'. ד"ה יכנס

In other words, you can't prove it's the style of the Gemara to precede in to out by bringing an example where logical, rather than stylistic, considerations are operative.

So, what then does Tosfos mean?

It could be that Tosfos means like the first way we learnt, that the Gemara says first go up (and out) and then it says go down (and in), which we suggested sounds like the wrong order. And it is exactly that point that Tosfos is making, that is, that *even though it is the wrong order logically* but that is the style of the Gemara, to put up before down, and to put in before out.

And that is what Tosfos means. Just like the Gemara in Eruvin lists up and down in the wrong order in order to satisfy the stylism, so too the Gemara in Pesachim puts in before out, even though that's not the "right" order in terms of time but rather because that is the style of the Gemara, to put in before out.

Perhaps.

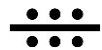
Eliezer Klopstien suggested a reason why the Gemara would want to precede up to down. Since the word "מטפס" intrinsically means "up", so once we start with that word we continue with the case that fits in best, linguistically.

The Yeivitz learns the first way that we did, out and then in, and he's bothered by the same question, that the order seems reversed.

He answers that the reason the Braisa taught it in this fashion is to teach us another דין, that it is permitted to go in and out many times.

So the Braisa should be read like this:

"He can go [down and] up and down [and up]."



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כ"א: - Half a Water Bucket

There's a Braisa that brings a story about Rebbe Akiva in jail. Every day, Rebbe Yehoshua Hagarsi would bring him water. One day, the guard saw Rebbe Yehoshua was bringing a lot of water and he suspected that they were trying to dig out an escape tunnel so the guard spilled out half and let him take half into the jail.

After Rebbe Yehoshua came inside and told Rebbe Akiva what happened, Rebbe Akiva asked to wash his hands. Rebbe Yehoshua responded, "There's not [even] enough in here to drink, [could it be that] there's enough to wash?!"[74]

I have a few questions on this story.

My main question is how much water did he bring in the first place? Right now he has half. This amount is not enough to drink nor to wash. When he first came in, he had double the amount. But that still couldn't have been enough to drink *and* wash! What did he expect Rebbe Akiva to do?

F = full bucket
D = amount needed to drink
W = amount needed to wash

$.5F < D$ "half the bucket" is less than the amount needed to drink

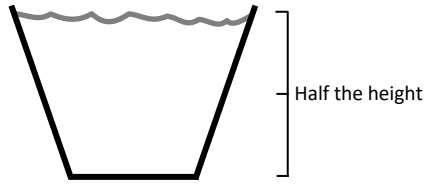
$.5F < W$ "half the bucket" is less than the amount needed to wash

therefore

$F < D + W$ "a full bucket" is less than drinking and washing together

[74] At first, I thought he said, "There's not enough to drink [but] there is enough to wash." Either way, we will have the ensuing questions. The only difference is .5F = W, but .5F is still < D and therefore it's still D + W < F.

One answer I thought of is that the bucket was narrow at the bottom and wide at the top.



So, when it says that the guard poured out "half" it means half the height, not half the volume. Half the height left over is less than half the volume.

So now, let's reassign our variables:

F = full bucket
D = amount needed to drink
W = amount needed to wash

$(<.5F) < D$ "less than half the bucket" is less than the amount needed to drink

$(<.5F) < W$ "less than half the bucket" is less than the amount needed to wash

therefore

$(<F) < D + W$ "less than a full bucket" is less than drinking and washing together

but it's still possible that

$F = D + W$ "a full bucket" is enough for drinking and washing together

Another question I had is what did Rebbe Yehoshua bring every day? Let's assume he was supposed to bring in half, but that wasn't even enough to drink!

We could say that he usually was allowed to bring in more than half and that was enough for drinking.

If so, then how did the guard expect that half would be enough to drink?

It could be he didn't but he was punishing them for bringing in a lot more today.

But then we should ask, what did Rebbe Akiva wash with every day?

We could answer that Rebbe Yehoshua usually brought in a full bucket and he wasn't noticed. On this particular day, he was noticed and the guard poured out half.

Yishai Rasowsky suggested a slightly different way of reading Rebbe Yehoshua's response: "There's not [even] enough to [just] drink, [could it be that] there's enough for [both drinking and] washing?!"

This answers the first question that I had in the beginning.

F = full bucket
D = amount needed to drink
W = amount needed to wash

$.5F < D$ "half the bucket" is less than the amount needed to drink

$? = W$ we don't know how much was needed to wash

therefore it could be that

$F = D + W$ "a full bucket" is enough for drinking and washing together

*

כ"ג: - Wrapping 50 Around 50

The Mishna on עמוד א' discusses the shape and size of a קרפף. If the קרפף exceeds certain dimensions then it takes on certain חומרות. The tannayim have different opinions as to what exactly constitutes a "small" קרפף that remains with certain קולות.

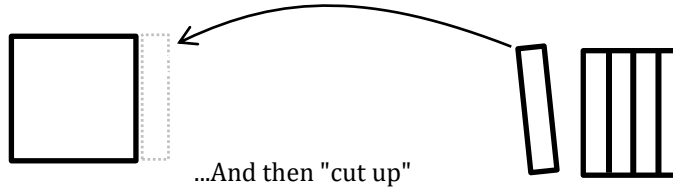
Our Gemara differentiates between two of the tannayim. Both of them seemingly require the קרפף to be no bigger than 5000 A^2, the same area as the Mishkan's chatzer which was 50 by 100 amos.

The difference between them is that according to the Tanna Kama, these 5000 A^2 are calculated exactly. So, if the קרפף was square then the sides must be no bigger than $\sqrt{5000}$, which is about 70.71 amos.

According to Rebbe Akiva there's a certain procedure we need to perform to know the amount of area that the קרפף can be. The Gemara brings פסוקים about the dimensions of the Mishkan's chatzer, which was 50 by 100 amos. Based on the פסוקים, we infer a measurement procedure where we measure an area that is 50 by 50 square, and we take another area of 50 by 50 and "slice it up" into thin strips of land and then "wrap" those around the initial 50 by 50, as many times as we can.

Like so:

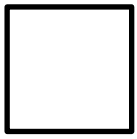
Put one square of 50x50 in the middle...

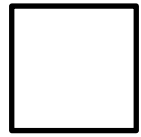


...And then "cut up" another 50x50 square into thin strips and "wrap" it around the first square.

Let's do this step by step while keeping cheshbon of how much we're starting with and how much we have left. I'm going to keep on the left side of the page the square that we're building by adding to it from the square on the right side of the page. What follows is Rashi's explanation[75].

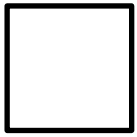
Stage 1

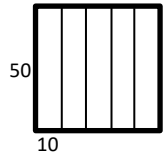
50x50 

50x50 

Stage 2

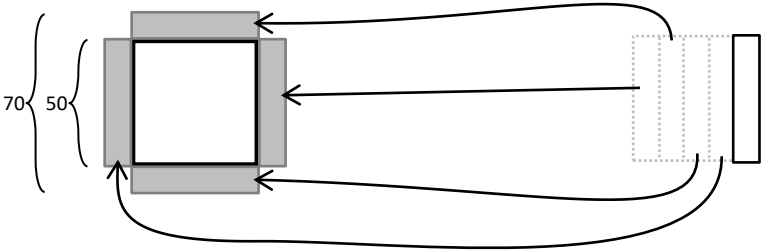
Slice up the box on the right into 5 strips that are each 50 x 10.





Stage 3

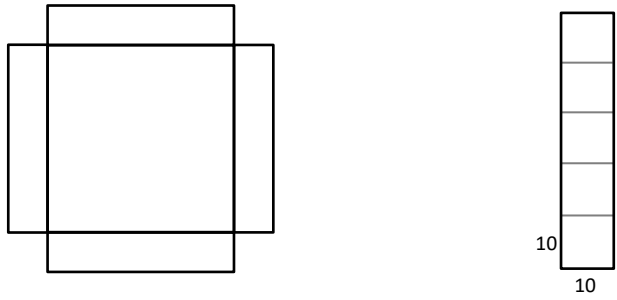
Take 4 of the 50 x 10 strips from the right side and lay them along the edges of the square on the left, leaving 1 strip of 50 x 10 on the right.



[75] ד"ה אלא אמרה תורה

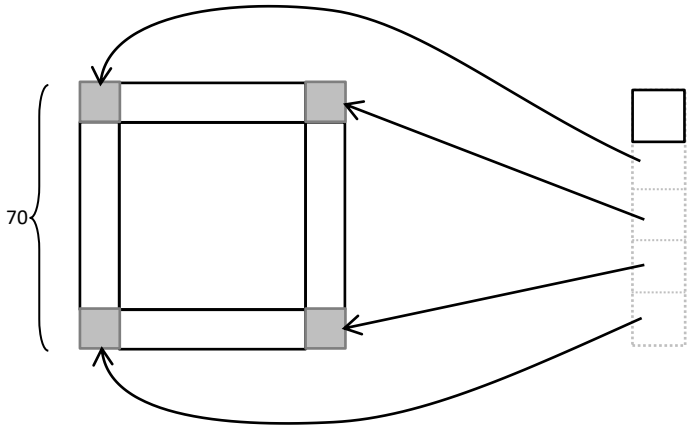
Stage 4

Cut up the 50 x 10 strip on the right into 5 squares of 10 x 10 each.



Stage 5

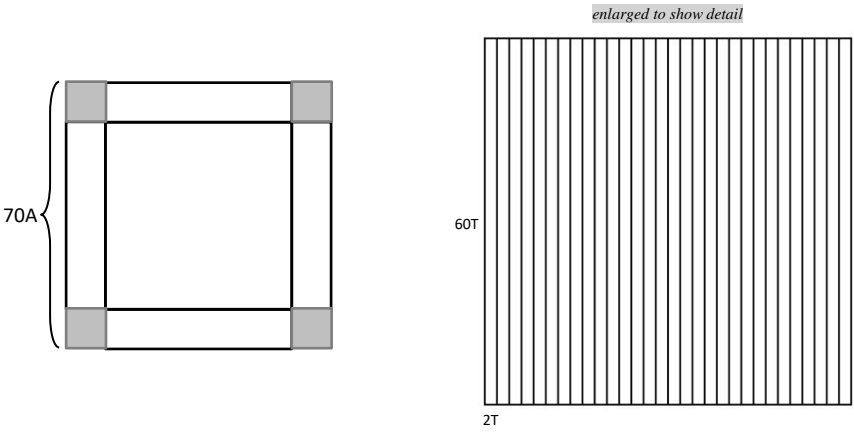
Put four of the 10x10 squares from the right into the corner gaps on the left.



What we're left with so far is a 70 by 70 amos square on the left, with a remaining 10 by 10 amos square on the right. Let's see how much more we can cut up that little 10x10 square into very thin strips to continue wrapping around the big square on the left.

Stage 6

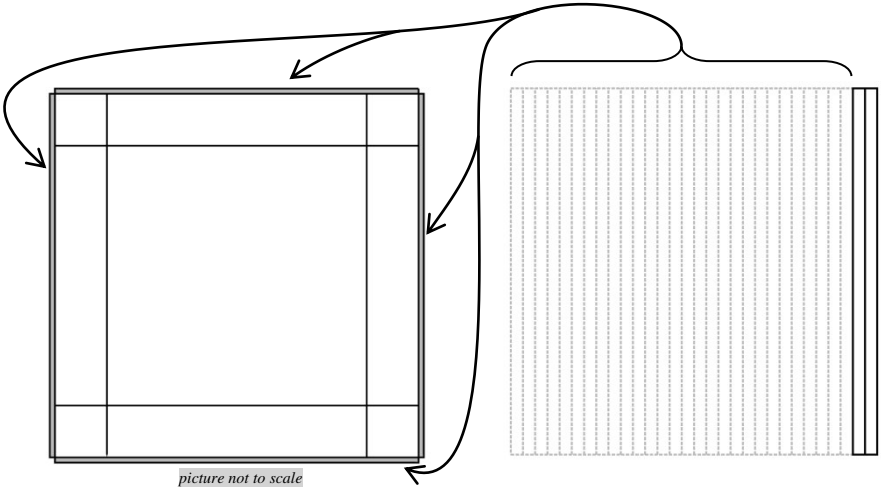
The 10 x 10 square on the right is 60 by 60 tefachim (6 tefachim per amah). If we slice it up into 30 strips, each strip will be 60T long by 2T wide.



Stage 7

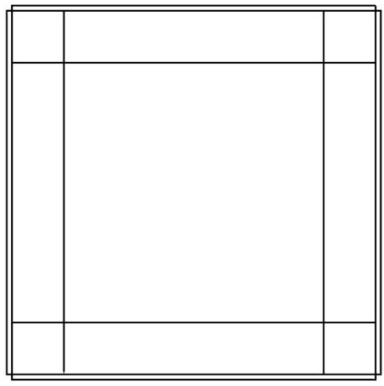
The 70 amos square on the left is 280 amos along the perimeter (70 * 4).

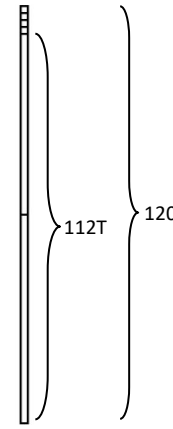
Each 60T x 2T strip on the right is 10 amos long (60 ÷ 6). 28 of them end to end would make a strip 280 amos long by 2 tefachim wide. Let's lay those along the edges of our 70A x 70A square on the left, leaving 2 strips on the right.



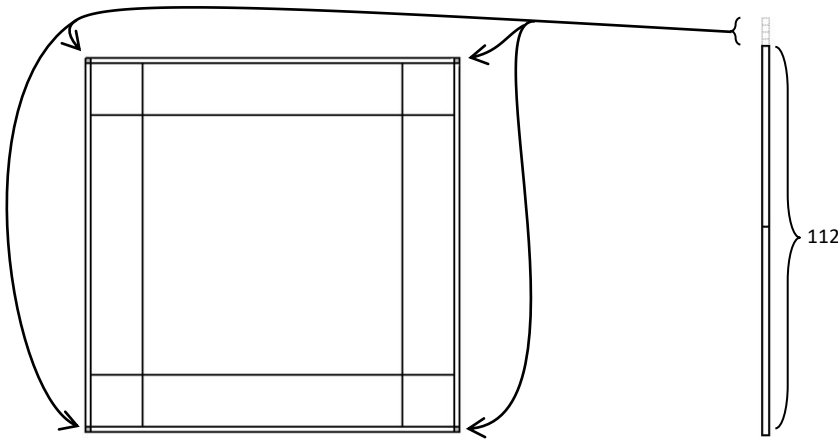
Stage 8

We need to fill in the corners. Each corner is a little square of 2T by 2T, for a total of **8T by 2T**. The remaining strips on the right were 60T by 2T each. Placed end to end they would be a total of **120T by 2T**. Taking 8T for the corners for the square on the left leaves us with 112T by 2T on the right.





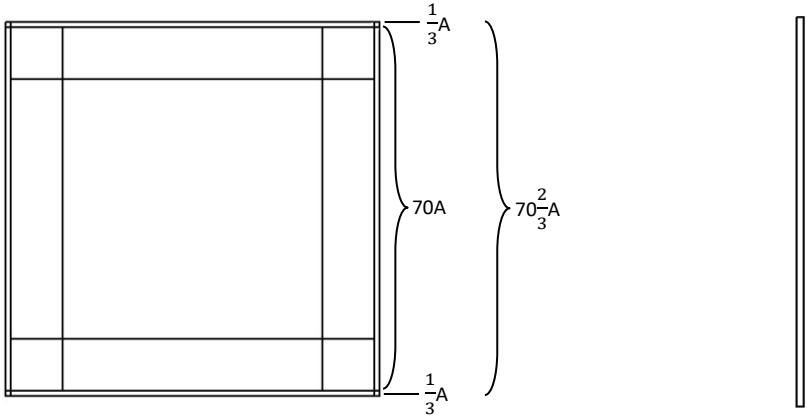
Move them over:



What remains is a strip that's 112T by 2T,
which is 224 T^2 $(112 * 2)$,

which is $\mathbf{6\frac{2}{9}\ A^2}$ $(224 \div 6^2)$ (6^2 is the amount of square tefachim in a square amah.)

The big square on the left is now $\mathbf{70\frac{2}{3}}$ **by** $\mathbf{70\frac{2}{3}}$ (since we added 2T on each side, for a total of 4T, which is $\frac{2}{3}$ of an amah (amah = 6T).



Rashi says that we cannot add anymore from the strip on the right since if we were to turn what was left $(6\frac{2}{9}\ A^2)$ into a strip thin enough to go all the way around the perimeter of the square on the left $\left(\text{the } 70\frac{2}{3} \text{ by } 70\frac{2}{3}\right)$ we would have to make it so thin that it "wouldn't even be 2/3 of an etzbah".

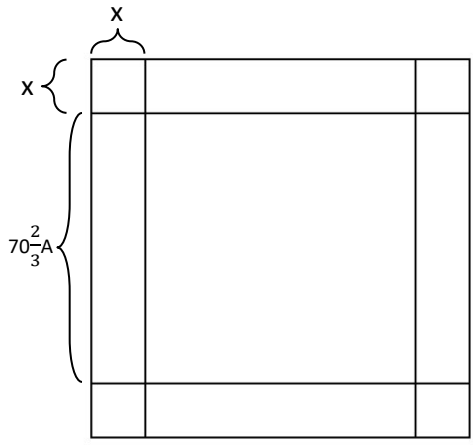
How much would it be exactly?

Let's write it out algebraically and then solve for x.

The picture that follows has in the middle, in the inside, the square we were making. It is $70\frac{2}{3}$ amos by $70\frac{2}{3}$ amos. We need to lay another 4 strips along the outside edges and then fill in the corners. Let's call the width of this new strip "x".

So, the long strips are $70\frac{2}{3}$ by x. And there's 4 of them.

The corners are x long by x wide, so they're x^2, and there's 4 of them.



not to scale

So, all together the whole formula would be:

(edge strips * 4) + (4 corners) = the area we have left to work with

length of edge strip * strip width * 4 + 4 * strip width * strip width = $6\frac{2}{9}$

$$\left(70\frac{2}{3}x\right) * 4 + 4x^2 = 6\frac{2}{9}$$

And now we solve for x.

▶ $\left(70\frac{2}{3}x\right) * 4 + 4x^2 = 6\frac{2}{9}$

when working with quadratic equations, it's good to make anything that is x^2 into $1x^2$, so in this case since it's $4x^2$, we'll divide everything by 4 (see page 378)

⬇ $70\frac{2}{3}x + x^2 = 1\frac{5}{9}$

add to both side the square of half of $70\frac{2}{3}$ (see Appendix F)

⬇ $70\frac{2}{3}x + x^2 + 1248\frac{4}{9} = 1250$

simplify the left side (factorization)

⬇ $\left(x + 35\frac{1}{3}\right)^2 = 1250$

square root both sides

⬇ $x + 35\frac{1}{3} = \sqrt{1250}$

subtract $35\frac{1}{3}$ from both sides

■ $\mathbf{x = \sqrt{1250} - 35\frac{1}{3}}$

Which is .0220057... amos, which (multiplied by 24) is .528... etzbaos.

And that's what Rashi means when he says that it won't even reach 2/3 of an etzbah.

÷

I'm assuming that for this purpose we cannot use an amount less than an etzbah.

÷

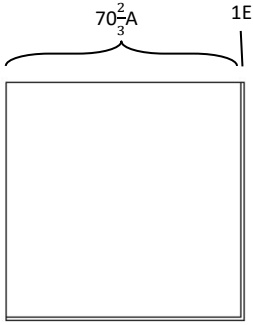
According to this method, the difference between the two opinions is $6\frac{2}{9}$ A^2.

Linearly, the difference is that Rebbe Akiva says the קרפף must be no longer than 70 amos and 4 tefachim, whereas the Tanna Kama says it needs to be no longer than $\sqrt{5000}$, which is 70.710678... amos, or **70A 4T 1$\frac{1}{\sim 18}$E**.

Start with 70.710678, take off **70** for the amos, and you're left with .710678
.710678 * 6 gives you 4.2640687..., take off the **4** for the tefachim, left with .2640687
.2640687 * 4 gives you 1.056..., take off the **1** for etzbaos, you're left with .056
1 ÷ .056 gives you 17.7699 which round off to about **1/18th**

÷

But I was wondering, why don't we lay 1-etzbah-wide strips along two adjacent sides?

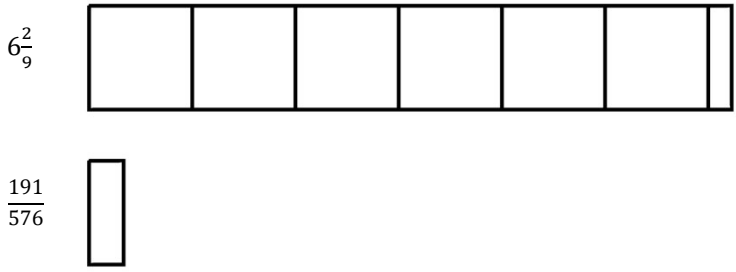


The amount of area we would be adding would be

long strip * etzbah * 2 + (1 x 1 etzbah corner) =

$$70\frac{2}{3} \quad * \quad \frac{1}{24} \quad * 2 + \quad \left(\frac{1}{24}\right)^2 = 5\frac{57}{64}$$

Taking that from what we had left before of $6\frac{2}{9}$ leaves us with $\frac{191}{576}$ which is about 1/3 of an amah squared.



Why then don't we do this? Why don't we add one etzbah to the width and length? That would make the מחלוקת between the tannayim much smaller!

I guess the answer is that the passuk is giving us a procedure to follow to make this cheshbon, "Take 50 and wrap it *around* 50!" You have to be able to make strips (with a minimal width of 1 etzbah) that could go <u>all the way around</u> the middle square, which we can't do with the amount we have left over.

The Hagaos HaGra writes:

> רש"י ד"ה אלא אמרה תורה טול כו' אין מגיע התוספת לרוחב וכו' נ"ב והוא אצבע וא' מי"ח באצבע בקירוב דע' על ע' וארבעה טפחי' ואצבע משתייר שליש אמה

"And that is close to $1\frac{1}{18}$ etzba since 70 by 70 and 4 tefachim and an etzba leaves over a third of an amah"

I don't know what exactly is the Gra's point but after the word "since" it sounds like he's describing exactly the cheshbon we made on the previous page, where we too came to a third of a square amah, about.

The beginning of the Gra's comment is also the same as what we cheshboned on page 106, where we too came up with 70A 4T and about 1 and 1/18 of an etzbah.

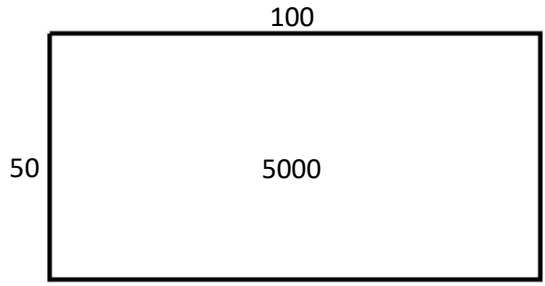
Another thing to point out is that this 1 and 1/18 is about double Rashi's "not even 2/3", since Rashi was describing the width of the strip we would need to wrap around to arrive at $\sqrt{5000}$, whereas this 1 1/18 is the total amount to be added to the square from both sides.

*

כ"ג: - <u>Tosfos Ikka - Rounded Rectangle</u>

The Mishna on עמוד א' has, among others, two opinions about the shape and size of an "oversized קרפף", i.e., a קרפף that takes on certain restrictions once it reaches certain dimensions.

As explained by the Gemara on עמוד ב', Rebbe Yosi and Rebbe Eliezer both say that the קרפף can be no longer than twice its width, and no bigger than $5000A^2$, like the chatzer of the Mishkan that was 100 amos long by 50 amos wide.



Both opinions seem to be describing the same size and shape, 50 amos by 100 amos.

The Gemara is מחלק between the two opinions by saying:

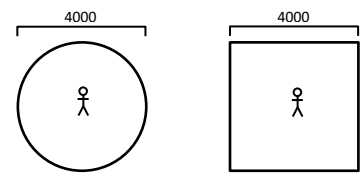
איכא בינייהו ריבועה דריבועה רבנן

"The difference between them is the square that the Rabbis squared."

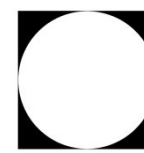
Tosfos[76] explains that this is based on the מחלוקת on :מ"ט between Rebbe Chanina ben Antigonus and the Chachomim regarding one who makes the center of his Techum outside the city (by standing there erev Shabbos until Shabbos comes in, thereby declaring that his "place" is where he is standing).

According to Rebbe Chanina, you get a circle with a diameter of 4000 amos.

According to the Chachomim, you get a 4000 amah wide square.

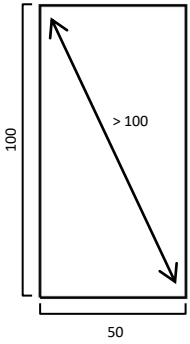


The difference between them is "that which the Rabbis squared".



[76] ד"ה איכא

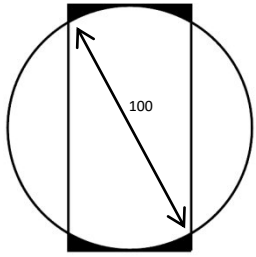
So too, says Tosfos, over here.



According to Rebbe Eliezer, you can't have a length of more than twice the width *anywhere* in the קרפף. Since the width is 50 you cannot have more than 100 amos in the קרפף, regardless of that length's position or orientation.

So, what we need to do is to limit this rectangle of 100 x 50 by cutting off from it any length longer than 100.

A way to do it would be to draw a circle with a diameter of 100 amos and superimpose it over the rectangle.



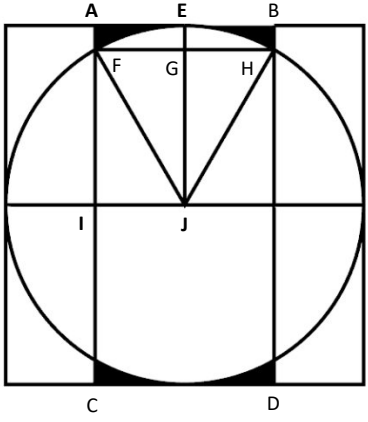
Anything outside of the circle needs to be cut off from the קרפף's shape (shown in black shading in the picture on the right).

According to Rebbe Eliezer, anything bigger than this shape is considered an "oversized קרפף".

Rebbe Yosi gives you the corners as well, like the chatzer of the Mishkan.

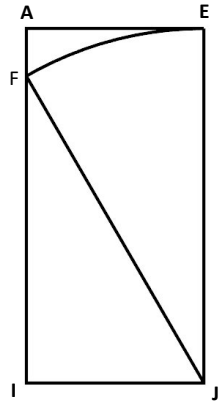
What is the total area difference between these two opinions?

Yishai Rasowsky showed me a relatively quick way to figure it out.



(We're going to calculate one quadrant of this shape, and then multiply by 4 for all 4 quadrants.)

Let's look at the small rectangle made up of **AEIJ**.

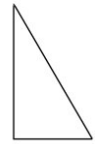


Let's break this quadrant down into 3 smaller shapes:

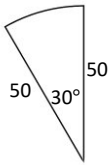
1) Curved Triangle AEF

2) Pizza Slice FEJ

3) Triangle FJI



Pizza Slice FEJ



JF and JE are each 50 — the radius of the circle

FH is 50 — the width of the rectangle (point "H" from previous page)

Therefore, angle FJH is 60° — since all sides of triangle FJH are the same

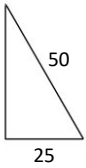
Angle FJE is half of that, and is therefore 30°.

So that means that the pizza slice is 1/12 of the whole circle. (360 ÷ 30 = 12)

Since the whole circle has a radius of 50, the whole area of the circle is 7853.98...

...divided by 12 is $\boxed{654.498\ldots}$

Triangle FJI



FJ = 50 — the radius of the circle

IJ = 25 — half the rectangle's width

Therefore, FI = **43.3...** — $\left(\sqrt{50^2 - 25^2}\right)$

Therefore the area of triangle FJI is $\boxed{541.26\ldots}$ (.5 * 43.3 * 25)

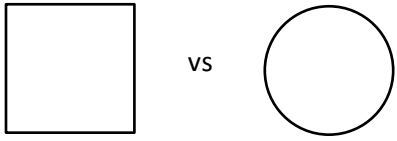
Curved Triangle AEF

Add the pizza slice and the other triangle together for a total of **1195.76...**

The whole quadrant is 1250 (50 x 25). Subtract 1195.76... and you get $\boxed{\mathbf{54.23...}}$

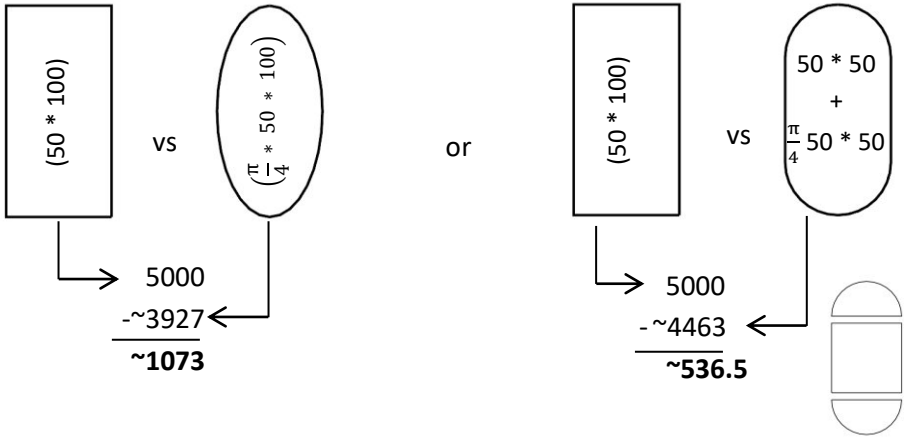
Now multiply by 4 for all 4 quadrants = $\boxed{\mathbf{216.94...A^2}}$

The first time I read this Tosfos, I read it superficially and thought what we're saying is that "just like by Techum there's a מחלוקת about whether the shape is a square or rounded off...



...so, too by קרפף, the מחלוקת is about whether it's square or rounded off."

I had two possible meanings for this:



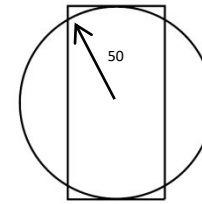
However, after reading Tosfos again carefully I realized these פשטים are wrong. Rather it's like the picture we showed above with the circle superimposed over the rectangle, as it is depicted in the sefer Pirush Chai.

The לשון of Tosfos is:

לר"א אי ארכה יתר ע"פ שנים ברכבה אפילו באלכסון אין מטלטלין...

"According to Rebbe Eliezer, if the length is more than twice the width even on the diagonal, one cannot carry..."

It is only the Pirush Chai's pshat that expresses this criterion.



What you have to do is take a rectangle 50x100 and then "shave off" whatever is farther than 50 from the middle, so that we don't have any total length longer than 100 amos in any direction.

The other פשטים I suggested (the oval and the rounded rectangle) are arbitrarily smaller than they need to be, since we can go into the corner areas up to 50 amos, like in the picture above-right.

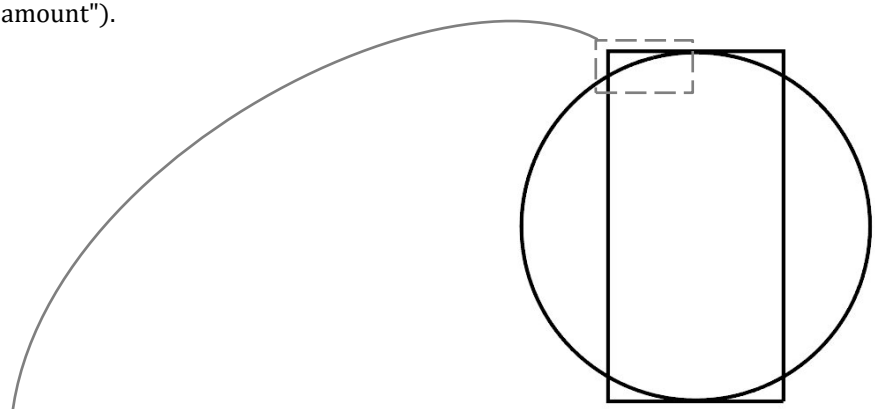
Tosfos[77] on דף כ"ה suggests that the rule that a קרפף cannot be more than twice its width is only true when the קרפף is exactly 5000 A^2 (בית סאתיים).

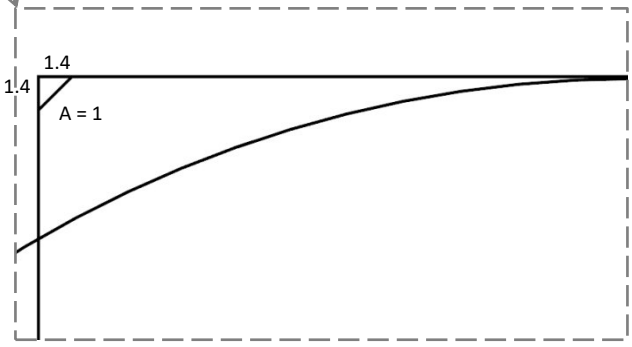
In our case, the קרפף described by Rebbe Eliezer is actually smaller than 5000 A^2, by about 217 A^2.

Accordingly, we should be able to make the קרפף bigger until it is just under 5000 A^2!

[77] ד"ה קרפף

Let Rebbe Eliezer make his קרפף 100 x 50, leaving off a tiny amount from the corner (in the example in the picture below there is 1 amah^2 left off by the corner). And now the מחלוקת between the two opinions is tiny (1 amah in this example but perhaps more or less depending on how you interpret Tosfos's word משהו, "any amount").



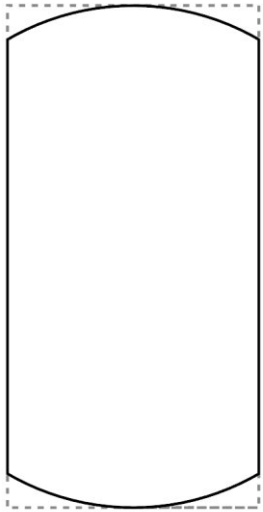


So, now the problem is when would Rebbe Eliezer ever say רבועה דרבועה דרבנן?

I guess we could answer אה"נ, he won't say that the rectangle with the shorn off corners is the maximum size for a קרפף. But if you ask him *why* a rectangle 100 x 50 is no good he'll answer you by saying that there are lengths in this קרפף that are more than twice the width.

In other words, the פסול of this 100 x 50 rectangular קרפף is the problem of having a length of more than twice the width (interpreted as he does, i.e., even when measured along the diagonal).

Aaron Kronenfeld suggested a different תרץ. When Tosfos on כ"ה says the קרפף can be more than twice the width when it's less than 5000 A^2, we still measure the area as though the corners were filled in, as far as this דין is concerned.

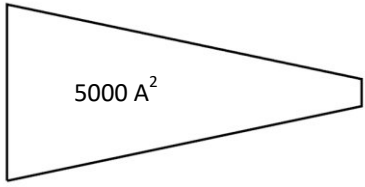


In other words, this קרפף (the one pictured on the left with the shaved off corners) is 100 by 50 amos.

If this had been a rectangle, then it would've been 5000 A^2. *Therefore*, we look at it as though it actually is 5000 A^2, inasmuch as it now cannot have a length of more than twice its width.

Anything bigger than this shape on the left will therefore be considered an "oversized קרפף".

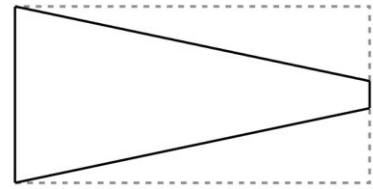
This idea that a קרפף might be viewed as large as it would be had it been a rectangle is from one of the תרוצים of Tosfos on כ"ה to a certain question (see there). The other תרץ describes a קרפף that is a trapezoid, with one end being much wider than the other.



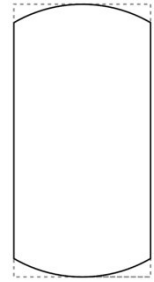
The entire area in the trapezoid is no more than 5000 A^2.

If we assume that each תרץ has its own approach then we could say that this very point is what the תרוצים are arguing about.

The first תרץ we discussed would not agree to this תרץ's case since we would measure the קרפף as though it were a filled-out rectangle, which would make it much more than 5000 A^2.



What we are suggesting is that in our case the קרפף is looked at as though it is 5000 A^2 as far as activating the דין that it cannot be longer than twice its width, and yet we do not include the imaginary corners (that we invoked to activate our דין) when calculating if indeed any part of this קרפף has a length longer than twice its width.



Perhaps, one could find a precedence for this in the Tosfos on דף נ"ו: ד"ה אביי where Tosfos suggests that the city's area is viewed as a circle as far as measuring the Migrash is concerned, and yet it is viewed as a filled in square as far as the Techum is concerned, even though both components are part of the same equation. ע"ש.

*

כ"ה: - Pi Tikra Yored Visoseim

As we go through the מסכתא, we find this term "Pi Tikra Yored Visoseim" over and over.

פי תקרא יורד וסותם "Pi tikra yored visoseim", literally means "The mouth of the ceiling goes down and seals."

What is the "mouth" of the ceiling? Is it the edge of the roof? The top surface of the roof? The bottom surface (i.e., the ceiling)?

I'm not going to provide any kind of סיכום here; I just want to gather some ideas and מארי מקומות. I'm going to present some possible meanings and/or criteria about when we do or don't say pi tikra yored visoseim.

Then I'm going to go through the מארי מקומות and point out which ones sound to me like which פשט.

All the מארי מקומות are originally in Hebrew, so I'll provide those as well so you can argue on my interpretations. The translations are loose paraphrasings.

כ"ה: רש"י ד"ה הכא כי אורזילא יציע גג משופע דליכא פה

ארטסקרול כי תקרה משופעת אינו נחשב כאילו הוא נמתח וירד עד הארץ שכן שפת תקרה כזו אינו מאונכת ואינו ניצבת כלפי הארץ כתחילתה של מחיצה.

חברותא ואין לו פה [עובי בגובה הקירוי]

מתיבתא שכיון שהוא משופע נמצא שקצה הקירוי שאותו אנו רוצים לראות כאילו הוא יורד וסותם אין בו עובי כלל. תוס' פ"ו: ד"ה גזוזטרא כתבו בשם רש"י שבכדי לומר פי תקרה יורד וסותם צריך שעובי התקרה לא יהיה פחות מטפח.

פ"ו: תוס' ד"ה גזוזטראו א"ת כי אין בה ד' אמות על ד' אמות לישתרי למלאות לפ"ה שפירש בסוף כל גגות (לקמן דף צד:) דאמר פי תקרה בכל ארבע הרוחות **(1)** וי"ל דהכא אייירי דעבידא כי ארזילא באלכסון כעין גגין שלנו דלא אמרינן פי תקרה יורד כדאמרינן בעושין פסין (לעיל דף כה:) **(2)** אי נמי כגון שהגזוזטרא אינה שוה ומליאה פגימות כל סביבותיה דאין פי תקרה (3) אי נמי כשאין טפח בעובי הגזוזטרא דמשמע מפירוש הקונטרס דסוף פרק קמא דסוכה (דף יח:) בשמעתא דאכסדרה שיש לה פצימין דלא אמר פי תקרה יורד וסותם אלא אם כן יש בעובי התקרה טפח

צ"ג. רש"י ד"ה לדידי והנך קורות יש להם פה דהיינו חודה שאינה עגולה אלא מרובעת

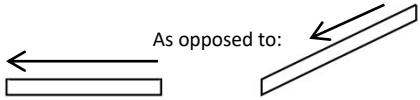
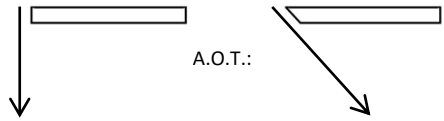
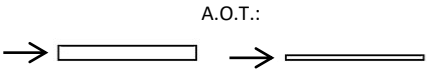
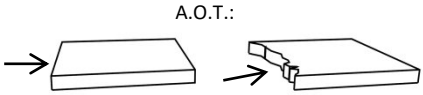
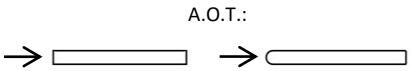
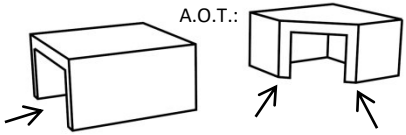
צ"ד: תוס' ד"ה וקירויו פי' בקונטרס בשיפוע כעין גגין שלנו דליכא פה היינו כי ארזילא כדאמרי' בעושין פסין ותימה מה דהוה ליה למינקט כי ארזילא כדנקט התם ואור"י דמיירי שנפרץ הקירוי באלכסון עם המחיצות בקרן זוית וכי האי גוונא לא אמרינן פי תקרה

שבת ק"א. מאירי ...במי שנעץ קנה ובראשו טרסקל וזרק ונח על גביו שפטור ואפי' הד' מחיצתא ר"ל שעומד הפוך על הקמה ראשו למטה ודפנותיו נוטות כלפי קרקע...

(קרפף) מ"ב שנ"ח:צ"ב ואפילו אם הקירוי משופע - כעין גגין שלנו שפיה באלכסון וקמ"ל שגם בפי תקרה כזו יורד וסותם ואף דלקמן בסימן שס"א ס"ב סתם המחבר דהיכא דפי התקרה בשיפוע לא אמרינן פי התקרה יורד וסותם חילקו האחרונים דשאני הכא דמן התורה רה"י גמור הוא לכו"ע שהרי מוקף ד' מחיצות אלא שחכמים גזרו בקרפף יותר מבית סאתים ולהכי הקילו בפי תקרה אף שפיה בשיפוע משא"כ התם דנפרץ מחיצה לא אמרינן יורד וסותם אלא א"כ פיה שוה

(גג סמוך לרה"ר) שו"ע א"ח שס"א:ב אבל פירצה שאינה בקרן זוית אמרינן בה פי תקרה יורד וסותם אפילו ביותר מי' והוא שלא יהא פי תקרה משופע דליכא פה

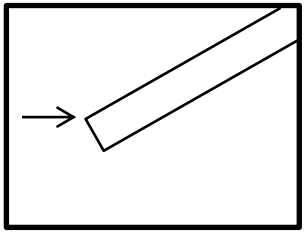
מ"ב שם ס"ק י"ד שלא וכו' משופע - ר"ל דלא אמרינן דיורד וסותם רק כשהתקרה חלק אבל לא כשהוא משופע כעין גגין שלנו שעומדים בשיפוע אם לא היה תקרה תחתיה

1) The top surface is horizontal 	2) The end surface is vertical 
3) The roof is thick 	4) The edge is unblemished 
5) The edge is square 	6) The edge is along a single wall 

Rashi 25b "Here" ...like an orzilla, [that is, a house] extension [with a] sloped roof, so it doesn't have a "mouth". Sounds to me like pshat #1

Artscroll ...because a sloped ceiling is not considered as if it stretches and goes down to the ground because the edge of such a ceiling is not perpendicular and doesn't head in the direction of the ground, like the beginning of a partition [would]. Sounds to me like pshat #2

Chavrusah "And it doesn't have a mouth" [i.e.,] thickness, in the height of the roofing. Pshat #3



Mesivta ...since it is sloped, the edge of the roofed area that we want to view as though it goes down and closes [what's underneath the roof] has no thickness at all. (Perhaps because the edge where the roof and the end of the roof meet is just a sharp point, see picture) Tosfos 86b "Gezustra" write in the name of Rashi that in order to say pi tikra yored visoseim [it] needs that the ceiling be no less than a tefach [thick]. Pshat #3. However, I don't think Tosfos is a good source for this pshat, as we will see.

Tosfos 86b "Gezustra" ...and if you will ask [even] when it [the "gezustra" platform] is less than 4 by 4 amos let it be permitted to draw water up through it according to Rashi who says at the end of the chapter Kol Gagos (later on daf 94b) that pi tikra works [even] on all four sides [of the platform]. -?-

One can answer

1) Here [by the gezustra] we're dealing with where it's like an orzilla, at an angle, like our roofs [in medieval France] about which we don't say pi tikra yored as we say in the chapter of Osin Passin (above on daf 25b).
Tosfos seems to be understanding that the pshat in orzilla is that the roof is sloped

2) Alternatively, for example, the [edge of the] gezustra is not even (i.e., straight) and is full of imperfections all around [the edge], therefore it doesn't have a pi tikra. Pshat #4

3) Alternatively, there is not a tefach in the thickness of the gezustra, as it is implied from Rashi at the end of the first perek of sukkah (daf 18b) in the discussion about a pavilion that has posts, that we don't say pi tikra yored visoseim unless if there is a thickness of a tefach in the ceiling.
This third teretz of Tosfos is the one that says that the roof needs to be a tefach thick in order to say pi tikra. This is *not* the pshat of orzilla, which was the first teretz of Tosfos. So, while it is true that the roof needs to be a tefach thick, one cannot say that this is Tosfos's pshat in orzilla

Rashi 93a "Lideedee" ...and these beams have a "mouth", that is, an edge that is not round but square.

Here, we are being told of another criterion for pi tikra, that is that the roof edge needs to be square. It is possible that this is only the opinion of Abaya who made this statement of "lideedee". It is also possible that this pshat is like pshat #1 or #2.

#1 said that the roof's top surface needs to be horizontal and not angled. It could be that since by the end of the roof it is round, the upper section of the roundness is like a sloped roof.

#2 said that the edge face needs to be perpendicular to the ground. Here, we are saying that a round edge is no good. Perhaps the reason is because the edge, being round, has no clearly defined face that can be considered perpendicular to the ground.

Tosfos 94b "Vikeerooyuv" Rashi explains, "with a slope like our roofs that don't have a mouth like an orzilla, as we say in the chapter of "Make Boards".

This is an astonishment! If it is as Rashi says then the Gemara here should've said "like an orzilla" as it did there!

The Ri explains differently, that we're dealing with where the roofing is opened at an angle with the walls, across a corner, and in such a case we don't say pi tikra yored visoseim.

This is Pshat #6 from the chart on page 118. Tosfos is saying that this is one of the criteria for pi tikra. At the same time, Tosfos is saying that this point is *not* the pshat in orzilla.

Meiri Shabbos 101a ...with someone who stuck a stick into the ground and put a basket on top of it and threw something that landed on top of the basket, he is exempt, even if it has four mechitzos, for example when the basket is upside down on top of the stick, with the basket's opening facing downward, and its sides are inclined towards the ground...

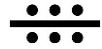
The sugya that the Meiri is discussing is not pi tikra but rather gud achis. However, I brought it as a possible support for pshat #2, to show that sometimes a mechitza is formed when the surface above it is perpendicular to the ground (as it sounds to me from the Meiri's words).

Mishna Berurah 358:92 (Laws of קרפף) "And even if the roof is sloped" Like our roofs (in 19th century Poland) whose mouths are sloped, and we're being told that even a "ceiling mouth" like this is yored visoseim... in a case of dirabbanan. However, by di'oraisah, we only say pi tikra when its mouth is even (horizontally straight).

To me, this sounds like pshat 1, or perhaps pshat 2.

Shulchan Aruch - Orach Chaim 361:2 (Laws of roof alongside reshus harabim) ...but a breach that's not in the corner, we say in that case pi tikra yored visoseim even more than 10 [amos wide], as long as the mouth of the ceiling is not sloped, since that doesn't have a mouth.
"...breach...not in the corner" is the criterion #6, like Tosfos said.
"as long as... not sloped" sound like #1 or #2.

Mishna Berurah (ibid.) - Orach Chaim 361:14 "...is not sloped" The Shulchan Aruch means to say that we only say "yored visoseim" when the ceiling is smooth (straight) but not when it's sloped like our roofs that are standing on an angle, if there is no ceiling beneath it (a flat roof under the sloped roof, like in an attic). Sound to me like #1.

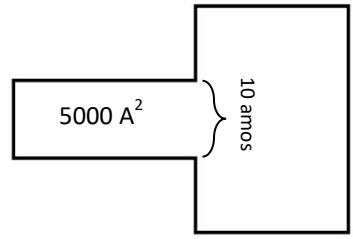


*

כ"ה: - Tosfos Karpef - Trapezoid

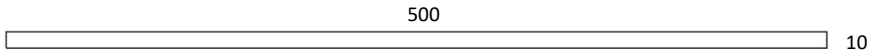
The Gemara discusses a קרפף (lot) that is 5000 A^2 (= בית סאתיים, beis sosayim) and it opens on one side into a courtyard. The opening is a "full" opening, i.e., there are no walls on that side of the קרפף.

Tosfos points out that the opening between the קרפף and the chatzer cannot be more than 10 amos wide[78]

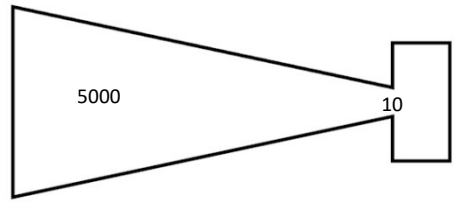


If so, says Tosfos, then the קרפף would be 500 amos long! (5000 ÷ 10)

That would make the קרפף 50 times longer than its width (500 ÷ 10), when it is only allowed to be twice its width[79]



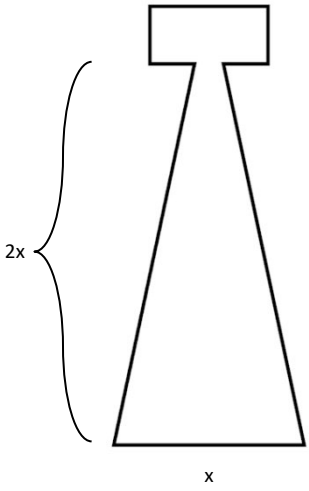
One of Tosfos's answers is that the קרפף is 10 amos wide on one side and it widens much more on the other side, so it ends up being a trapezoid. And now the length need not be more than twice its width, even if the total area is 5000 A^2.



[78] Since that would passul the chatzer, ע' במשנה ט"ו:
[79] ע' במשנה כ"ג.

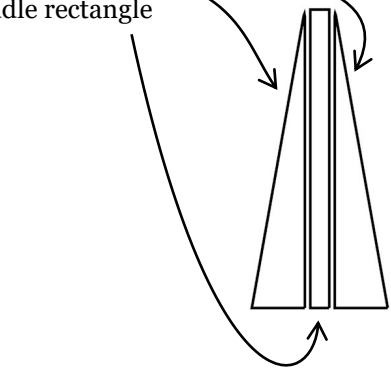
How wide would the other side be? (And by extension, how long would the קרפף be?)

We know that the קרפף is twice as long as it is wide, so we could say the width is x and the length is 2x.

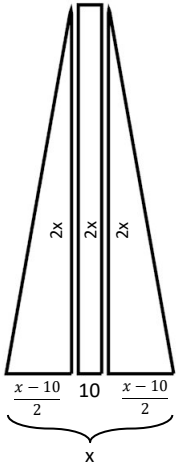


Let's break up the קרפף into 3 areas:

2 triangles
and a middle rectangle



Let's label what we know:



The whole bottom of the trapezoid is x, therefore the two triangle bases together are x minus the rectangle's width (10), so together they'd be x - 10. There's two bases, so each base is half of that, which would be: $\frac{x-10}{2}$

So now let's write it all up algebraically:

$$\frac{1}{2} * w * h * 2 + w * h = 5000$$

$$.5 * \frac{x-10}{2} * 2x * 2 + 10 * 2x = 5000$$

Now we solve for x:

$.5 * \frac{x-10}{2} * 2x * 2 + 10 * 2x = 5000$

.5 and *2 cancel each other out

$\frac{x-10}{2} * 2x + 10 * 2x = 5000$

the $\frac{}{2}$ and the *2 cancel each other out

$(x - 10) * x + 10 * 2x = 5000$

$(x - 10) * x = x^2 - 10x$

$x^2 - 10x + 10 * 2x = 5000$

$10 * 2x = 20x$

$x^2 - 10x + 20x = 5000$

-10x + 20x becomes +10x

$x^2 + 10x = 5000$

add 25 to each side[80]

$x^2 + 10x + 25 = 5025$

simplify the left side

$(x + 5)^2 = 5025$

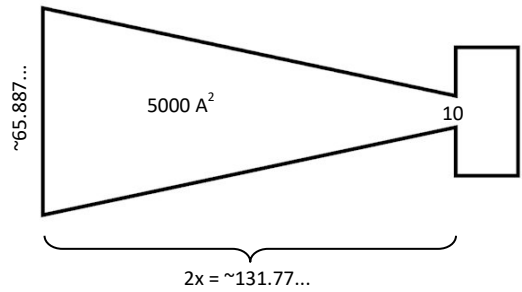
square root both sides

$x + 5 = \sqrt{5025}$

subtract 5 from both sides

$x = \sqrt{5025} - 5$

$x = 65.887...$

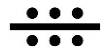


[80] since this is the square of half of 10 from the 10x; see Appendix F on quadratic equations

כ"ח. - **Bug Chart**

This is a simple chart to help you see the breakup of the number of איסורים that are assigned to each type of bug.

	כלל 2	מים 2	ארץ 3	עוף 1	Totals
פוטיתא	✔	✔	(?)		4 (7?)
נמלה	✔		✔		5
צירעה	✔		✔	✔	6



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ל"ז. - **Tosfos Asarah - Trumah, Maiser Rishon, and Maiser Sheni**

The Mishna[81] in Trumah says

שעור תרומה עין יפה אחת מארבעים בית שמאי אומרים משלשים והבינונית מחמישים והרעה מששים...

"The amount of Trumah: a generous person [gives] 1/40 (Beis Shamai says 1/30), an average person [gives] 1/50, and a stingy person [gives] 1/60..."

Trumah really has no shiur; any amount works. The Mishna lists the different amounts that are considered normal, high, or low.

After Trumah is given, 1/10 (10%) of what's left over is given as Maiser Rishon.

[81] תרומה ד':ג'

After Maiser Rishon has been given, 1/10 of what's left is given as Maiser Sheni.

Our Gemara brings a Braisa that describes someone giving 2 lugin[82] for Trumah, and then 10 lugin for Maiser Rishon, and then 9 lugin for Maiser Sheni.

Rashi says that the person started off with 100 lugin.

Tosfos points out that the Braisa's numbers of "10" for Maiser Rishon and "9" for Maiser Sheni are inexact since we started off with 100 (like Rashi said), so after giving 2 lugin for Trumah, we were left with 98.

Let's go through this step by step.

The person started with 100 lugin.

The average Trumah giver gives 1/50, which in this case is 2, so he's left with 98.

Maiser Rishon is 1/10 of that, which would be 9.8, which leaves him with 88.2

Maiser Sheni is 1/10 of that, which would be 8.82, which leaves him with **79.38** (which is a tiny bit more than the 79 that the Mishna implied).

How did Rashi know that we started off with 100 lugin?

Assuming that we're dealing with the average giver of Trumah (1/50)

x is the amount that we started with:

▶ $x * \frac{1}{50} = 2$

divide both sides by 1/50

⬇ $x = 2 \div \frac{1}{50}$

■ $x = 100$

[82] a certain measure, which is about half a liter; See Appendix A on שיעורים.

Reading the Braisa on my own, I would've thought of two other פשטים:

The Braisa's numbers are exact

Instead of starting with the assumption that the *average* giver gave 2 lugin for Trumah, let's assume that *someone* gave 2 and ended up with an amount whose Maiser Rishon will be exactly 10 lugin.

10 is 10% of what number? This should be easy. 100.

That means that *after* giving 2 lugin, the person ended up with 100 lugin. That means that he started off with **102 lugin.**

After Trumah, he then gives 10% of what's left. So that would be 10% of 100, which is 10, and then he's left with 90.

Maiser Sheni is 10% of that, which is 9, leaving him with **81**.

According to this approach, what fraction did the person give in the beginning for Trumah?

Well, we know he started with 102 and gave 2. What fraction of 102 is 2?

$102 * x = 2$

divide both sides by 102

$x = \frac{2}{102}$

simplify the fraction on the right by dividing each number by 2

$\mathbf{x = \frac{1}{51}}$

The person gave $\frac{1}{51}$ for Trumah, which is slightly less than the average.

The benefit of this pshat is that the Braisa's numbers are exact.

The problems with this pshat is that

1. we're not dealing with the average giver of 1/50 but only something close to that, whereas the normal assumption is that we are dealing with average people.

2. we're starting the person off with an almost random number of 102, instead of a nice round 100.

The Braisa's last 2 numbers are exact

In this alternative פשט, we will assume that the last two numbers of the Braisa (10 and 9) are exact and we are dealing with the average giver of 1/50.

We need to figure out how much the person started with and we will need to say that the Braisa's "2 lugin" for Trumah is inexact.

To put it into word form: The person started off with x, and then he gave 1/50 of that and was left with an amount of which 10% is 10 (i.e., he was left with 100, like we showed on the previous page).

$x - \frac{1}{50}x = 100$

x is the same as 1x which is the same as $\frac{50}{50}x$

$\frac{50}{50}x - \frac{1}{50}x = 100$

take off 1/50 x from 50/50 x

$\frac{49}{50}x = 100$

divide both sides by 49/50

$x = 100 \div \frac{49}{50}$

apply the rule that dividing by a fraction is the same as multiplying by that fraction upside down

$x = 100 * \frac{50}{49}$

turn the 100 into a fraction

$x = \frac{100}{1} * \frac{50}{49}$

multiply the tops and the bottoms of the fractions

$x = \frac{5000}{49}$

divide 5000 by 49 with a remainder

$\mathbf{x = 102\frac{2}{49}}$

So, the person started off with $102\frac{2}{49}$ and gave exactly 1/50 of that $\left(2\frac{2}{49}\right)$ for Trumah, which leaves him with 100. 10% of that is 10 for Maiser Rishon, which leaves him with 90. 10% of that is 9 for Maiser Sheni, which leaves him with **81**.

The benefits of this פשט are that we're dealing with the average giver of 1/50, and two of the Braisa's numbers are exact.

The problems with this pshat is that

1) one of the Braisa's numbers are inexact
2) we started off with a very random number of $102\frac{2}{49}$ instead of a nice round number like 100.

Recap of the three pshatim

	Starting amount	Fraction for Trumah	Trumah amount	What's left	Maiser Rishon amount	What's left	Maiser Sheni amount	What's left
Rashi	100	$\frac{1}{50}$	2	98	9.8	88.2	8.82	79.38
My 1st pshat	102	$\frac{1}{51}$	2	100	10	90	9	81
My 2nd pshat	$102\frac{2}{49}$	$\frac{1}{50}$	$2\frac{2}{49}$	100	10	90	9	81

Pros and cons of each pshat

	Nice, round starting amount	average giver	Braisa's numbers are exact: 1st (2)	2nd (10)	3rd (9)
Rashi	✔	✔	✔	✘	✘
My 1st pshat	✘	✘	✔	✔	✔
My 2nd pshat	✘	✔	✘	✔	✔

Assuming that Rashi considered all three options, why did he choose the 1st one?

I could think of two main tracks:

1) Having a nice, round starting number (like 100) is the most מסתבר, and there is only one pshat that has that.

2) It is most מסתבר to say we are dealing with the average giver (1/50)

So now we need to choose between the first and the third pshat.

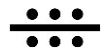
I could think of four sub-tracks:

a) Having a nice round number overrides having more numbers of the Braisa being exact[83].

b) Nice round number plus one exact number is better than two exact numbers.

c) Once some of the numbers of the Braisa are inexact then it doesn't matter how many, so having a nice round number like 100 (the first pshat) "wins".

d) Going in the order of the Braisa, the first number we encounter is "2" for Trumah. Assuming that's exact leads us to conclude that the following numbers are inexact.



[83] The third pshat has two numbers exact whereas the first pshat has only one.

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מ"א: - Accepting Difficulties Lovingly

The Gemara brings a Braisa that lists several difficulties that a person might experience in his life and the Braisa says about such people that they will not see the "face of Gihenom (purgatory)".

The Gemara asks, למאי נפקא מינה, for what practical purpose [do we need to know this]?

The Gemara answers, לקבולי מאהבה, to accept them lovingly.

Rashi explains, so that he will get a כפרה, a rectification. Seemingly, if a person suffers but doesn't accept his suffering with love then it doesn't effect the same kind of rectification.

I was wondering though, how is this a נ"מ for just these things? Shouldn't any suffering that a person experiences be accepted with love, if he wants the full purification?

I thought perhaps to answer that what we mean by נ"מ is that a person who is aware that his suffering can wipe out his flaws entirely, so much so that he won't even get a glimpse at Gihenom, will have a much easier time accepting the suffering, knowing that it will bring him a complete כפרה.

Anytime a person goes through any kind of difficulty, if he can be aware of the good that will come out of it then it will make it that much easier to bear the difficulty.

For example, if a person is running in a race and he is winning, so even though his muscles ache and he's out of breath he can still bear it, knowing that he will win an award at the end.

So perhaps it is just these three things which effect that full rectification that will make it so much easier to accept them with love.

*

מ"א: - Leaving the Techum Willingly

The שקלא וטריא of this Gemara is a little confusing so I made this chart to help follow the flow of the Gemara.

The Mishna says that one who was forced out of the Techum can only walk around within 4 amos.

If he was forcibly returned into the Techum then he gets to walk the entire Techum.

Rav Nachman said in the name of Shmuel that "One who left willingly can only walk around within 4 amos".

I'm going to present the chart and afterwards go through and explain the flow of the Gemara.

Taken out forcibly		Left willingly	
4 amos →		4 amos (1)	
Returned forcibly	Returned willingly	Returned forcibly	Returned willingly
Entire Techum →	4 amos (2)	4 amos (3)	

(Underlined in the chart is explicate in the Mishna; arrows indicate inferences)

1) At first, the Gemara takes Rav Nachman's words at face value, that one who leaves willingly only has 4 amos (#1 in the chart).

   To this the Gemara responds that we already know this from inference[84], since one who was forced out only gets 4 amos, certainly one who left on his

[84] and therefore, Rav Nachman's statement is unnecessary.

own will only get 4 amos! (Interestingly, the Gemara never entertains the possibility that such a person would not be allowed to walk around at all, even less than 4 amos.)

2) Then the Gemara suggests that what Rav Nachman meant is that one who was forced out and then went back in willingly can only walk 4 amos (#2 in the chart).

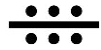
   This too, the Gemara says we know from inference. The Mishna says that one who was forced out and then forced back in gets the entire Techum, implying that only if he was forced back, but if he came back on his own then he only gets 4 amos.

3) The Gemara therefore suggests another interpretation of Rav Nachman's words: One who left willingly and was then forced back only gets 4 amos (#3 in the chart).

   This too, the Gemara contends can be known from inference. Only one who was **forced out** and then forced back gets the entire Techum, implying that one who **left on his own** and was then forced back only gets 4 amos!

   The Gemara rejects this inference (thus reestablishing the need for Rav Nachman's statement) by saying that one might have interpreted the Mishna's case of "Returned forcibly" as referring to where he left willingly. So one might mistakenly learn from the Mishna that such a case gets the entire Techum, comes along Rav Nachman and dispels such a notion.

Tosfos wonders why the Gemara didn't answer this way in the 2nd stage and answers[85] that this would have made Rav Nachman's statement and meaning quite apart from each other.



[85] perhaps one of two answers, depending on the גירסא.

מ"א: - **Fruit Left the Techum**

Here is another confusing שקלא וטריא that I thought I'd map out with some charts.

Let's start with the charts and then explain as we go through them.

To reduce clutter, I'm going to label the sections of the chart now and then you have to remember this format as you go through the rest of them.

במקומם In their place	לא במקומם Not in their place	
		מזיד On purpose
		שוגג By accident

All the charts are going to look like this:

and you have to remember which cell is what, without the labels.

Actually, there's going to be two charts side by side, the left side is going to be Rav Pappa/Tanna Kama and the right side will be Rebbe Nechemia & REB"Y.

Rav Pappa/Tanna Kama

Rebbe Nechemia & REB"Y

Sooooo, commit it to memory and let's go!

1) Rav Pappa's statement vs. Rebbe Nechemia

Rav Pappa			Rebbe Nechemia & REB"Y	
In their place	Not in their place		In their place	Not in their place
✔		on purpose	✘	✘
✔		by accident	✔	✘

2) אילימא If you will say...

Tanna Kama		Rebbe Nechemia & REB"Y	
✘	✘	✔	✘
✔	✔	✔	✘

3) אלא לאו Rather, is it not...

Tanna Kama		Rebbe Nechemia & REB"Y	
✔	✘	✘	✘
✔	✔	✔	✘

4) לא No!

Tanna Kama		Rebbe Nechemia & REB"Y	
✘	✘	✘	✘
✔	✔	✔	✘

5) והא But what about...

Tanna Kama		Rebbe Nechemia & REB"Y	
✔		✘	✘
✔		✔	✘

1) Rav Pappa starts off with a statement that fruit that left the Techum and were returned, even on purpose (and certainly by accident) have not lost their place, i.e., their original Techum allowance.

   Rav Yosef bar Shimaya asks on Rav Pappa from a statement by Rebbe Nechemia and Rebbe Eliezer ben Yaakov who say that **the fruit are always forbidden unless they are returned to their place accidentally**.

2) Rav Pappa responds that that quote is actually a machlokes tannayim in that Braisa and he was going like the Tanna Kama.

   The Braisa says **"Fruit that left the Techum: if by accident then they can be eaten; If on purpose then they cannot. Rebbe Nechemia says: In their place they can be eaten; not in their place they cannot"**.

   If you will say that "in their place" [that Rebbe Nechemia said is OK] means on purpose, that cannot be because we already said in Rebbe Nechemia [and Rebbe Eliezer ben Yaakov]'s name that such a case is forbidden!

3) Rather, is it not that when he said "in their place" he was referring to by accident?

   And then the Braisa must be read like this: **"Fruit that left the Techum: if by accident then they can be eaten; If on purpose then they cannot. This is when they are not in their place. When they're in their place, then even on purpose they can be eaten. Rebbe Nechemia says: Even though they're in their place, by accident they can be eaten; on purpose they cannot"**.

4) Rav Yosef responds, no! On purpose, in their place, everyone agrees that it is forbidden.

   The only debate is about by accident, not in their place. The Tanna Kama holds it is permitted, whereas Rebbe Nechemia holds that even though it is by accident, in their place it's permitted; not in their place it is forbidden.

5) Rav Pappa responds back, <u>but what about</u> the fact that Rebbe Nechemia and Rebbe Eliezer ben Yaakov have said that **the fruit are always forbidden unless they are returned to their place accidentally**? That implies that on purpose it would be forbidden, even in their place.

   That, in turn, would imply that the Tanna Kama *debates* that point, and holds that it is <u>permitted</u> on purpose in their place!

*

מ"ו. - <u>Tosfos Kah Mashma Lan - Sleeping Traveller</u>

There's a machlokes in the Mishna[86] regarding a travelling person who falls asleep before Shabbos comes in. Rebbe Yochanan ben Nuri says he gets 2000 amos in each direction from where he is whereas the Chachamim say he only gets 4 amos.

There is another machlokes[87] regarding ownerless keilim. Rebbe Yochanan ben Nuri holds they get their own shvisa, that is, they can only be brought within 2000 amos from their starting place[88] whereas the Chachamim hold they do not have a shvisa and can be brought as far as whoever picks them up can go.

This is seemingly the same machlokes, about whether someone or something can get a shvisa (Shabbos location) without daas (awareness).

Rebbe Yochanan ben Nuri holds yes whereas the Chachamim hold no.

By a sleeping person, Rebbe Yochanan ben Nuri is the lenient opinion, since he allows the person to walk 2000 amos.

[86] Second Mishna on מ"ה.
[87] Brought on מ"ה: and elsewhere
[88] Assuming, of course, that carrying is not an issue, for example in a רשות היחיד outside the תחום, or on Yom Tov.

By ownerless keilim, the Chachamim are the lenient opinion, since they allow the object to be carried further than 2000 amos (For example, when the keilim are situated at the end of the 2000 amos limit of the person who picked them up. He can take them back to his starting point and then for another 2000 amos for a total of 4000 amos).

Rebbe Yehoshua ben Levi made two statements that had seemed at first glance to be redundant (since we were only focusing on the case of a sleeping person).

1) The Halacha is like Rebbe Yochanan ben Nuri

2) The Halacha is like the lenient opinion.

Rebbe Zeira answers for each one:

> If all we heard was that the Halacha is like Rebbe Yochanan, I would have thought that's true both to be lenient (by a sleeping person) and to be strict (by ownerless keilim). Comes along the second statement that the Halacha is like the lenient opinion in either case.
>
> And if all we heard was that the Halacha is like the lenient opinion, I would have thought that's only true when an individual is arguing with an individual, or when a group is arguing with a group, but here (by a sleeping person) where it's an individual (Rebbe Yochanan ben Nuri) arguing with a group (the Chachamim) I would think to be strict like the Chachamim. Comes along the statement that the Halacha is like Rebbe Yochanan ben Nuri by a sleeping person, even though Rebbe Yochanan ben Nuri is an individual arguing with a group.

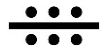
Tosfos points out that it is not a contradiction to be lenient by both a sleeping person and by an ownerless keili. Although they seem to be based on the same principle of whether someone or something needs daas to get a shvisa, in actuality, the sleeping person is better (more likely to get shvisa even without daas) since he has the capacity to have awareness when he is awake.

Therefore, says Tosfos, the statement that the Halacha is like the lenient opinion is enough to inform me that it is like the lenient opinion in both cases and I don't need to be told that the Halacha is like Rebbe Yochanan ben Nuri.

Now, perhaps this is obvious to everyone else, but when I saw this Tosfos I was confused, because if this statement about leniency is enough to tell me that the Halacha is like the lenient opinion, even by a sleeping person, then what's the Gemara's answer when it explains why we have to be told that the Halacha is like Rebbe Yochanan ben Nuri?

I think the pshat is that Tosfos is setting us up for the next stage of the Gemara. That is, without the chilluk that the Gemara provides that we would think to be michaleik when a yachid (individual) is arguing with a rabbim (group), we would indeed have enough information to know that the Halacha is *always* like the lenient opinion.

In other words, what Tosfos is telling us is that we shouldn't reject the idea that the Halacha is always like the lenient opinion based on what seems to us to be a contradiction. Rather, Tosfos tells us, it is logically possible that the Halacha could be that a sleeping person gets a shvisa and an ownerless object does not, and these laws are not mutually dependent.



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מ"ט: - <u>Size and Orientation of 4 Amos and Techum</u>

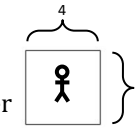
This piece is nothing but a collection of ideas that I've thought of or have come across. It is neither exhaustive nor conclusive at all. This should be viewed as a springboard for delving into the relevant subject/s according to your own time and abilities.

There are many ways that one could measure 4 amos, both for carrying and as far as Techum is concerned (no pun intended). There are also many ways of measuring the

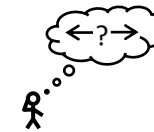
2000 amos of Techum. The size and shape of the Techum can also be affected by how we measure the 4 amos within it. Between all these factors, there are many permutations that are possible. I'd like to go through them all one day, but for today we will have to satisfy ourselves with this pockmarked and conjecturous presentation.

Are the 4 amos measured as

1) A square of 4 by 4 measured



מ"ה. רבי אליעזר a. with him in the center

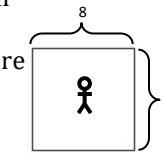


מ"ה. רבי יהודה b. how he chooses around him by

רש"י i. walking where he wants them to be

רבינו יהונתן ii. starting to walk in one direction

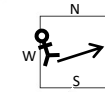
2) A square of 8 by 8 with the same options as before



מ"ה. חכמים

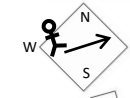
And whichever way it is, does he get

1) A square that is aligned



רבינו שמואל בתוס' נ"א. ד"ה כזה a. parallel

מ"ב שצ"ו ס"ק ו' b. diagonally

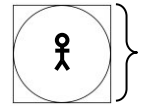


to the

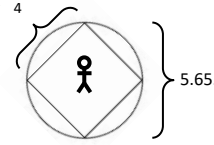
רבינו שמואל בתוס' נ"א. ד"ה כזה a. cardinal directions[89]

מ"ב שצ"ו ס"ק ו' b. direction he walks

2) A circle that is



a. 4 (or 8)

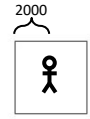


רבינו תם בתוס' נ"א. ד"ה כזה b. 5.65... (or 11.31...)[90]

[89] North, South, East, West

[90] Using $\sqrt{2}$ for the diagonal. Perhaps in some cases we will use 1.4 for the diagonal.

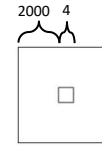
And when we measure the Techum, do we



1) start from the very center

2) start from the edge of his 4 amos (whichever way they are measured)

שו"ע שצ"ז:א'



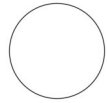
And whichever way it is, is the Techum

1) square

מ"ט: חכמים

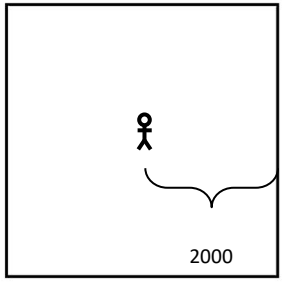
2) round

מ"ט: חנינה בנ אנטיגונוס

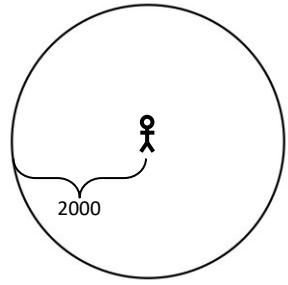


Here on מ"ט, we have a machlokes about whether the Techum is a square or a circle.

If we measure the 2000 amah radius from the very center then



Chachamim



Rebbe Chanina

the Chachamim's square Techum will be **16,000,000 A²**
$((2000 * 2)^2)$

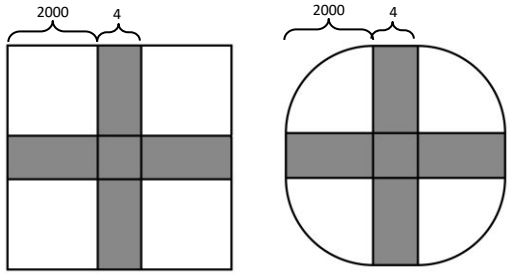
and Rebbe Chanina's round Techum will be **12,566,370.614... A²**
$\left((2000 * 2)^2 * \frac{\pi}{4}\right)$

The difference between them would be **3,433,629.385... A²**

Now, what if instead of measuring from the very center, we first put in 4 amos into the middle and then measured 2000 amos out from that?

If the middle 4 amos are square and its edges are oriented to the cardinal directions, the difference between the two opinions will be the same as before, regardless of how large these 4 amos are (whether they are 4x4 or 8x8).

The reason is because whatever is added due to the 4 amos is added the same way for both opinions.



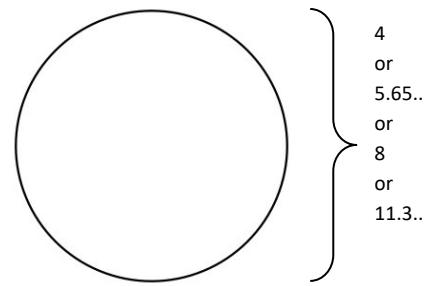
not to scale

The grey area in the picture is the amount that has been added due to the 4 amos, and it is the same amount of area whether the Techum is square or round.

The white areas in the corners are the same as they were before, they have just been shifted outwards without changing their sizes.

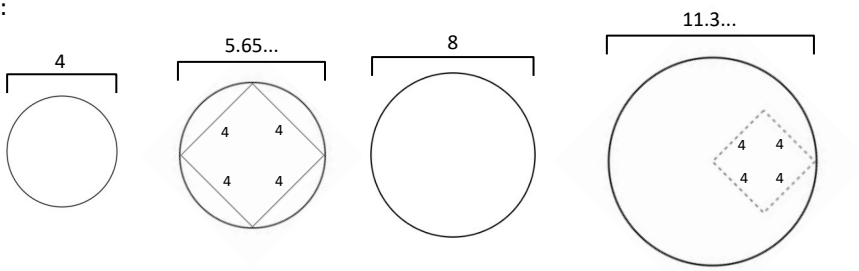
If, however, the 4 amos in the middle were a circle or a square rotated somewhat then that would change the difference between the Chachamim and Rebbe Chanina.

Let's first go through the possibilities of the 4 amos being a circle. We said before that there are at least four possible sizes (diameter) to the 4 amah circle:

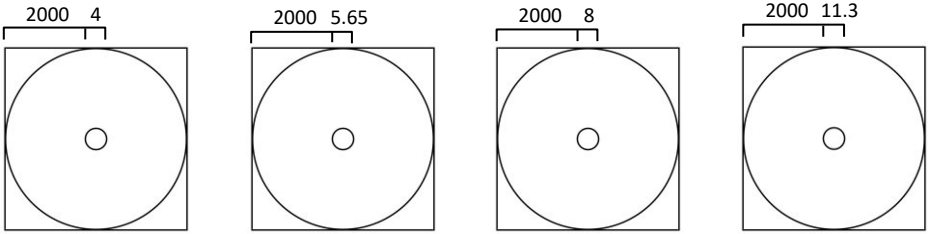


- 4 across
- 5.65... across (the diagonal of 4)
- 8 across (4 radius)
- 11.3... across (diagonal of 8)

4 amos:

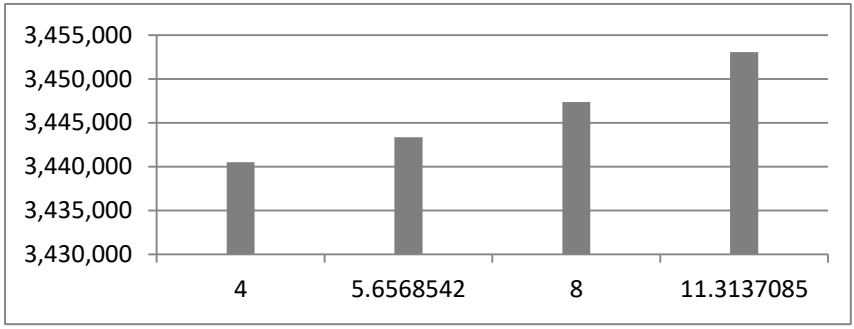


Techum:



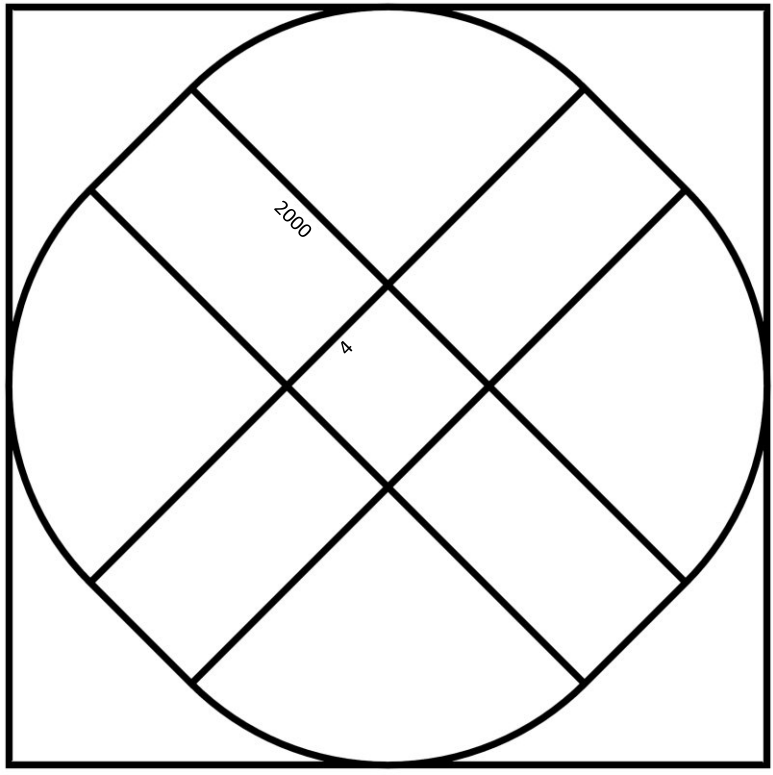
4 amos =	Square Techum	Round Techum	Difference
4	16,032,016	12,591,515.92...	3,440,500.078...
5.65...	16,045,286.83...	12,601,938.81...	3,443,348.023...
8	16,064,064	12,616,686.36...	3,447,377.637...
11.3...	16,090,637.66...	12,637,557.27...	3,453,080.395...

*Chart height does not begin at zero



מ"ט:

What if the 4 amos were a square rotated 45°, so that the corners were pointing towards the cardinal directions?



The square area is going to be the same as we calculated on the previous page, whether "4 amos" means the diagonal of 4 or of 8.

The formula would be $(4000 + 4 \text{ (or } 8) * \sqrt{2})^2$

- Diagonal of 4 is 5.65 so total Techum area = 16,045,286.83...
- Diagonal of 8 is 11.3 so total Techum area = 16,090,637.66...

The round Techum can be calculated like this:

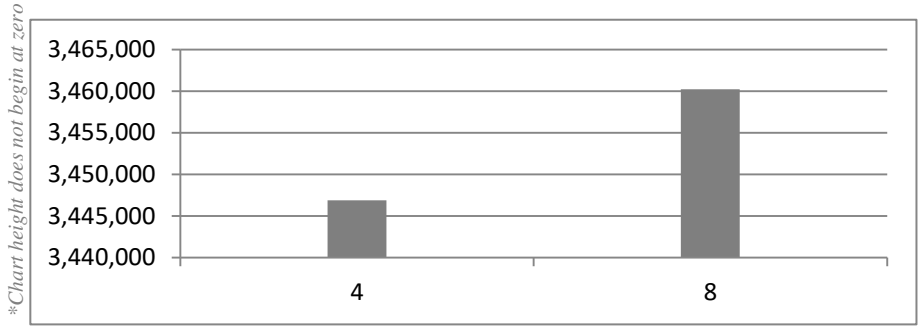
> The central diamond (from page 143) is 4^2 or 8^2 (depending on whether you count the 4 amos as a diameter or as a radius).
>
> The "arms" that stick out are 4 or 8 * 2000 (the extent of the Techum) * 4 (for the four "arms")
>
> The quarter circles between the "arms" can be viewed all together as a circle with a radius of 2000 amos, so they are $\pi 2000^2$.

All together the formulas would be

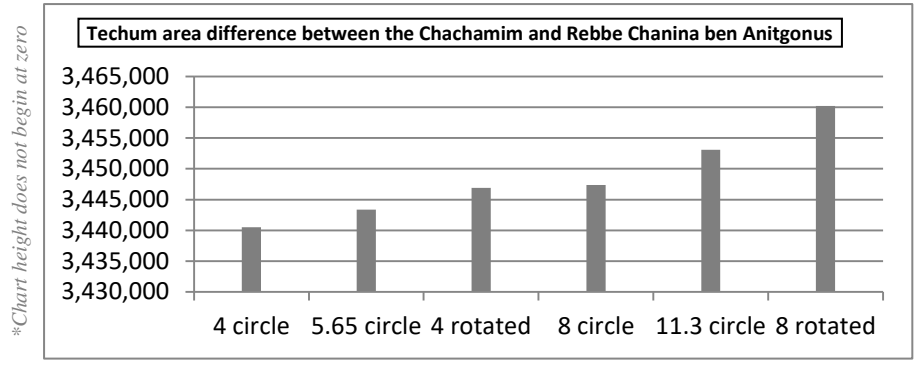
- 4 amah diameter: $4^2 + 4 * 2000 * 4 + \pi * 2000^2 = 12{,}598{,}386.61$
- 4 amah radius: $8^2 + 8 * 2000 * 4 + \pi * 2000^2 = 12{,}630{,}434.61$

And now the differences between the tannayim would be:

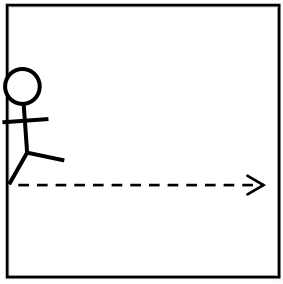
4 amos =	Square Techum	Round Techum	Difference
4	16,045,286.83...	12,598,386.61...	3,446,900.2196...
8	16,090,637.66...	12,630,434.61...	3,460,203.0536...



Comparison of all the pshatim we've done so far:



Shulchan Aruch 396:1 - The Rama brings two opinions regarding 4 amos whether we measure them

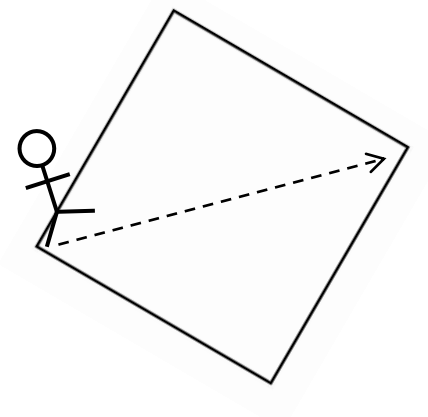


1) only on one side of the person (determined when he walks to that side[91])

The Mishna Berurah adds that we give him the diagonal, i.e., we orient the 4 amos to a 45° rotation in relation to the direction he is walking, so he can go a full 5.65... amos $(4 * \sqrt{2})$

Remember this point. We orient the 4 amos in whatever manner lets him walk the farthest.

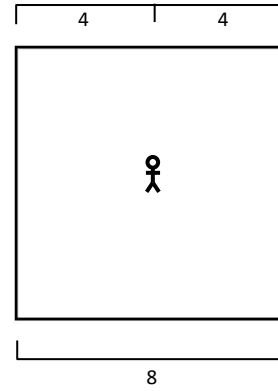
Once he's walked the full 5.65 amos then the orientation of his 4 amos has been set and he can no longer change it (for the rest of that Shabbos).



[91] מ"ב שם

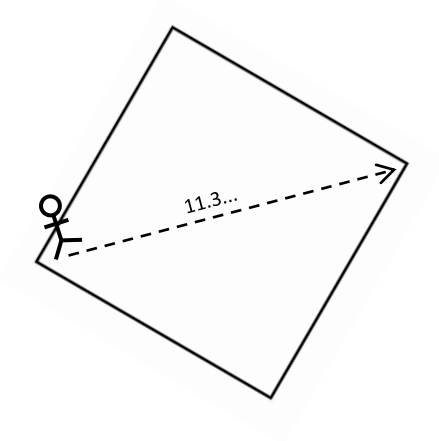
מ"ט:

2) he gets 4 amos on each side, for a total square of 8x8 amos.

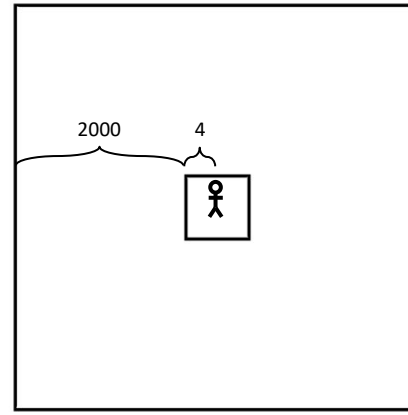


The Mishna Berurah[92] brings the Elya Rabbah who says that למעשה we can rely on this opinion.

I would assume that what we said above applies here as well, that we orient the square to his benefit.



In סימן שצ"ז:א', the Michaber writes that a person gets 2000 amos for his Techum "besides his 4 amos"[93].

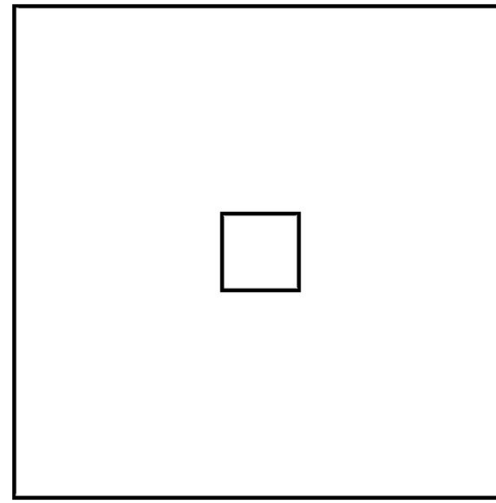


The Mishna Berurah explains that [according to the second opinion we brought above that you get 4 amos in each direction from the center] it comes out you get 2004 amos in each direction [for the entire Techum].

[92] ס"ק ט'

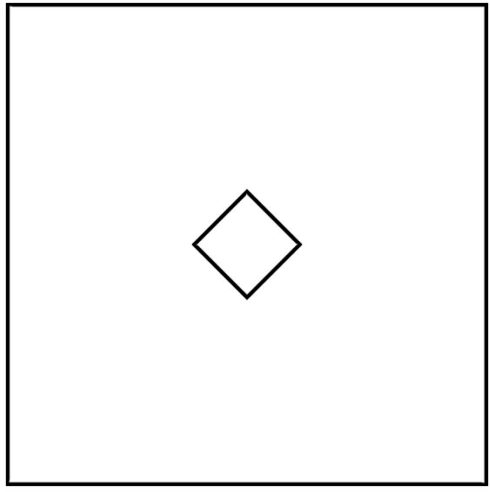
[93] חוץ מארבע אמותיו "Outside his 4 amos"

The Shaar Hatzion points out that according to the first opinion, once you've gone 2004 amos in one direction you can't go more than 2000 in the other direction.



According to the Shaar Hatzion, we see that we don't "fix" the person's 4 amos until he reaches the outermost place that he can go.

If that's so, then why don't we orient the 4 amos in the way that benefits him the most? Wouldn't it be better for him if we assume that the 4 amos are rotated 45° so that he gets an extra 1.65... amos in excess of the Techum?



This extra little bit would add up to a total of ~13,271 A^2 according to the first opinion[94], and a total of ~25,574 according to the second opinion[95].

?

[94] $(4000 + 4 * \sqrt{2})^2 - 4004^2$
[95] $(4000 + 8 * \sqrt{2})^2 - 4008^2$

Aaron Kronenfeld suggested that you can only rotate 4 amos when they're by themselves but when they're together within a Techum they need to be oriented the same way (since they're part of the same Techum "system").

I thought of a slightly different teretz. Since the determining factor that "sets" your 4 amos is the edge of the Techum so it automatically follows that Techum's orientation.

Rav Yehoshua Portowicz suggested that the question doesn't even begin since the term "4 amos" could be interpreted in many different ways (as we've have seen), so the Mishna Berurah only means that you get 2000 amos plus an additional "4 amos", which may in fact mean 5.65..., since that is the diagonal of 4 amos.

נ"א - Essential Eruv Method

Here is another confusing שקלא וטריא that I'm going to try to sort out.

We'll begin with the text of the Mishna, followed by the chart, followed by a step by step explanation.

The Mishna[96] says (among other things):

> **A person can make an eruv-center at a distance by declaring so.**
>
> **He may declare his eruv-center to be where he is standing.**

[96] מ"ט:

This is what they meant when they said a poor man can make an eruv-center where he is standing.

Rebbe Meir says this works only for a poor person.

Rebbe Yehuda says for both a poor and a rich person.

They only instituted making an eruv-center with bread to be lenient on the rich person so that he shouldn't have to go and make an eruv with his feet.

The Gemara goes through these statements of the Mishna and tries to figure out who says what.

Let's lay out the chart and then explain it.

Rav Nachman:

	By standing	With bread	"That place"
Rebbe Meir	only poor	essential eruv	only poor
Rebbe Yehuda	poor and rich essential eruv	for rich[97]	only poor

"This is what they meant... poor person... standing." -Rebbe Meir, referring to one who doesn't know how to make an eruv-center at a distance.

"They only instituted..." -Rebbe Yehuda

Rav Chisda:

	By standing	With bread	"That place"
Rebbe Meir	poor and rich essential eruv	for rich[97]	only poor
Rebbe Yehuda	poor and rich essential eruv	for rich[97]	poor and rich

"This is what they meant... poor person... standing." -Rebbe Meir, referring to one who makes eruv at a distance

"They only instituted..." -Everyone

[97] doesn't apply to poor person since, by definition, poor means without bread

Rav Nachman explains that the machlokes between Rebbe Meir and Rebbe Yehuda is by when a person makes an eruv where he is standing, "In my place", under his feet. Rebbe Meir holds that this only works for a poor person, that is, one who is trying to make an eruv at a distance (so he could get closer to the city) but doesn't know of a specific location within 2000 amos or he doesn't know the halacha (that he could do so).

I underlined the label of the column "By standing" to show that is where the machlokes is.

Rebbe Meir holds the essential eruv is with bread, thus doing it with one's feet is a leniency about which the Chachamim have said, "A poor person (in a tight situation) may make an eruv with his feet [but ideally it should be done with bread]."

Rebbe Meir could not have been the one who quoted "They only instituted... with bread to be lenient..." since he holds that bread is the essential and ideal method of eruv, and not a leniency.

According to Rav Chisda, the machlokes between Rebbe Meir and Rebbe Yehuda is about a person who makes an eruv at a distance.

They both agree that making an eruv in one's place (even without bread) is the essential type of eruv.

Thus, they both could have quoted the statement that to make an eruv with bread is a leniency.

The statement that "A poor person may make an eruv with his feet" cannot be referring to a person who makes an eruv where he is standing since that is the essential type of eruv, not a lenient one. Rather it is referring to one who makes an eruv at a distance.

This is how Rashi explains it. I have trouble understanding this since it comes out that when the person says "under my feet" it is specifically *not* referring to the case of one who makes it under his feet but rather one who makes it at a distance, at the base of a tree.

Perhaps, we are interpreting the words "under his feet" to be "under the tree's feet".

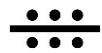
A little bit שווער, I admit.

The Mesivta had a better explanation, if I understood it correctly. There's basically two ways that a person can make an eruv: "With his feet" and "With bread".

"With his feet" has two methods: 1) "Literally with his feet, i.e., under his feet" and 2) "by declaring a distant location". The ideal method of eruv (according to this stage of the Gemara) is to do it under his feet. There is another way to do it that is less than ideal and that is to declare it to be at a distant location.

The way we are reading the Mishna then is "To declare an eruv at a distance is the kind of eruv-made-with-one's-feet that is a leniency for a poor person." As opposed to the ideal kind of making eruv-with-one's-feet, which would be literally under his feet, i.e., at his current location.

I think that fits better.



נ"א. - Tosfos Kizeh - Pillar's Orientation

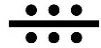
Tosfos brings a Rabbeinu Shmuel who says that we align a person's 4 amos to the cardinal directions (North, South, East, West), and that this applies to a square pillar as well, that is, if a pillar is 10 tefachim high and 4 by 4 tefachim wide, it only constitutes a reshus hayachid if the pillar is aligned along the cardinal directions.

On נ"ה:, we bring a Braisa that says that if the city is square but rotated then we don't "square it out". We see that a square area need not be oriented to the cardinal directions.

Why is it different by a pillar?

Perhaps, the difference is that the city is much larger or perhaps because the city is inhabited so it has more of a "set" area than the little, uninhabited pillar, which needs a minimal amount within the cardinal directions to make it substantial.

Or perhaps the difference is that by the pillar, we are trying to determine if it has the required dimension to <u>be</u> a certain kind of area, whereas by the city, we are just trying to determine an additional קולא <u>around</u> the city, so the city itself need not be oriented to the cardinal directions.



*

ב"נ: - <u>15 Amos Beyond the Techum</u>

In the Mishna, Rebbe Shimon says that you can come back into the Techum as long as you haven't gone farther than 15 amos out of it.

The second פרוש that Rashi brings is that when they measure the Techum they use 40 ropes (or one rope 40 times) that are each 50 amos long. The rope is held at both ends. Each person's hand that's holding the rope takes up 1 tefach and a half an etzbah of the rope (as he grasps the end of the rope his hand covers part of the rope). They only include the visible part of the rope in the total measurement.

So that means for each rope of 50 amos, they are actually not going the whole 50 amah distance but rather a little bit less, whatever amount that is not counted, since it was covered by the hands of the holders.

▶ Each hand is 1 tefach and a half an etzbah, which is $1\frac{1}{8}$ tefach...

⬇ ...Times 2 hands $= 2\frac{1}{4}$ tefachim per rope...

⬇ ...Times 40 ropes $= 90$ tefachim that are lost...

■ ...$90 \div 6$ (6 tefachim per amah) $=$ **15 amos**

*

נ"ג - Bnei Yehuda's Success In Learning

Rav Yehuda said in the name of Rav that the people of Yehuda (as opposed to the people of the Galil) were successful in their learning since they "spoke exactingly" (as the Gemara understands it at the end).

Rashi explains that it means that they said over what they heard from their רביים verbatim, and that they made סמנים, memory aids, for what they learnt, everything in its order.

The Gemara diverges from this point slightly and then returns to it on עמוד ב', asking, "Where do we see that the Bnei Yehuda were exacting?"[98]

The Gemara then proceeds to tell us about a certain Ben Yehuda who was selling his cloak in the market and the people asked him what color it was and he responded that it is "green like beet leaves".

As the Maharsha explains, the Ben Yehuda didn't leave anything ambiguous, since "ירוק" can mean green or yellow, so he said the specific shade of green.

When I saw this Gemara, I was a bit confused. According to Rashi, the meaning of "The Bnei Yehuda were exacting" is that they said over their Rebbes' teachings word-for-word. I would expect when the Gemara gives us an example that it would be an example of someone saying over their Rebbe's teaching word-for-word. What does that have to do with how they sold cloaks in the marketplace?

I think the answer is that you can't compartmentalize. It's like a מידה. The way you are in the Beis Midrash is the way you are in the שוק, and vice-versa. If you're careful to say over what you learned exactly then you will be careful to tell people the exact shade of color your coat is, if they ask.

This character trait of always speaking exactly will help you succeed in the Beis Midrash, and not speaking that way can interfere with your success, ח"ו.

We find a similar idea in Bava Basra[99]. The Gemara there describes the setup of a Talmid Chacham's table and bed, with each element thought out, in their proper

[98] Literally, "That the Bnei Yehuda were exacting, what is it?"

place, everything is done with a reason and a purpose. And the opposite with an Am Ha'aretz.

I believe we find this in all areas of Torah and middos. You need to always check that the way you are and act and think in any situation is congruent with a Torah attitude.[100]

My friend Yishai Rasowsky suggested another idea we see from this Gemara, and that is the power of the environment.

It would seem that the vast majority of the bnei Yehuda were like this, that they were exacting in their speech, and that the bnei Galil were the opposite, that they were sloppy.

It goes to show us what a vast influence the place we are in has on us.

---

[99] נ"ז:

[100] I once had a discussion with my Rosh Yeshiva, Rav Black from Zichron Yaakov, about bein hasedarim activities. One of the things that came up was the subject of chess. He said that chess is good game to play during bein hasedarim since you are relaxing and yet at the same time you are "using your mind. You have to hold kup and think ahead". The point is that even when a ben Torah relaxes, he should preferably do so in a way that furthers his life goal of being a ben Torah. In this example: to sharpen his mind, his analytical abilities.
Rav Dallah from Ohr Somayach would emphasize how important it is to use your "Gemara kup" in all situations. Whatever you are doing, wherever you are, you should be making your decisions based on the same methods that you use when trying to understand a Gemara or come up with a psak halacha.
This is called "thinking like a Jew".
I heard that the Brisker Rav would purposely set his clocks off by 5 minutes so that he would always have to think, not to do anything by rote, even checking the time.

*

נ"ו. - Determining Cardinal Directions

The Gemara says:

Rebbe Yosi says that to know the cardinal directions[101], see where the sun rises and sets on the longest day of the year, this is the northern direction. See where the sun rises and sets on the shortest day of the year, this is the southern direction. On the Vernal and Autumnal Equinoxes[102], the sun rises due east and sets due west.

Rav Misharshay says to disregard these rules because a Braisa taught that the sun never rises on the NE corner and sets on the NW corner. And the sun never rises on the SE corner and sets on the SW corner.

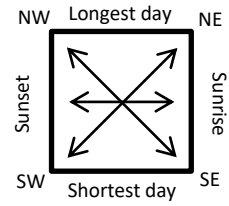
We had two ways of explaining this Gemara.

Aaron Kronenfeld explained it in the following manner (and perhaps the Meiri, as we shall see).

The Gemara starts off wanting to know the cardinal directions. Rebbe Yosi ostensibly provides us with a means of determining them.

By knowing any one of these six points on the compass, one could infer everything else:

- Northeast: Sunrise on the longest day
- Northwest: Sunset on the longest day
- Southeast: Sunrise on the shortest day
- Southwest: Sunset on the shortest day
- East: Sunrise on an equal day
- West: Sunset on an equal day



[101] The "square of the world", i.e., North South, East, and West

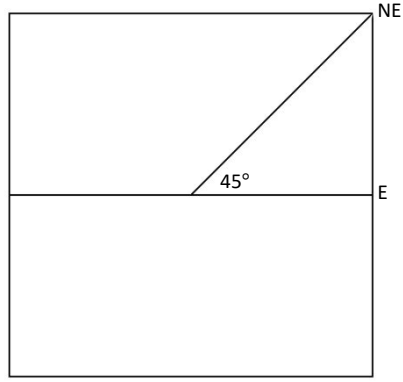
[102] תקופת ניסן ותקופת תשרי, the first day of Spring and Fall, that is, the days that have the same amount of day and night, about March 20th and September 23rd. See Appendix K for more information.

In other words, if you could know where the sun rises or sets on either the longest, shortest, or equal day then you should be able to determine all of the other directions.

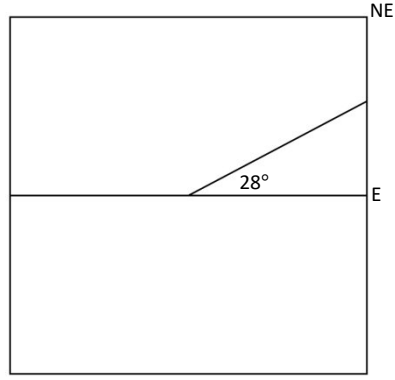
Rav Misharshay rejects four out of these six methods: NE, NW, SE, and SW (i.e., sunrise and sunset on the longest or shortest days).

The reason is because the Braisa quoted says that the sun will never rise exactly NE (or SE, or set NW or SW)

To rise NE, the sun would have to rise 45° north of due east:

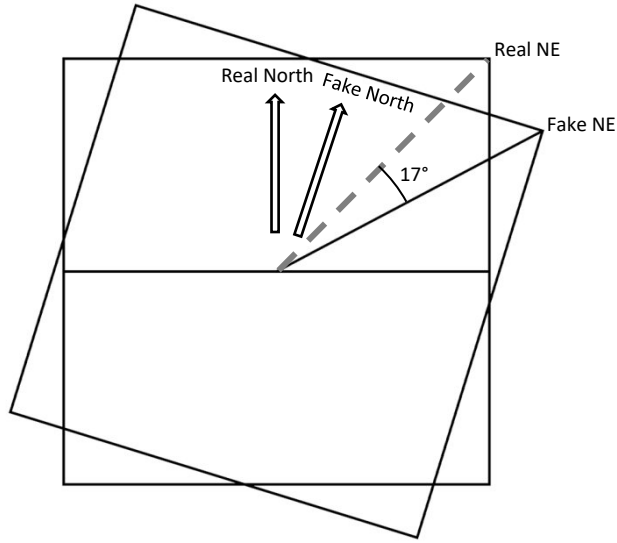


Due to a number of reasons, the most north that the sun will ever rise in Eretz Yisroel or Bavel is about 28° north of due east.[103]

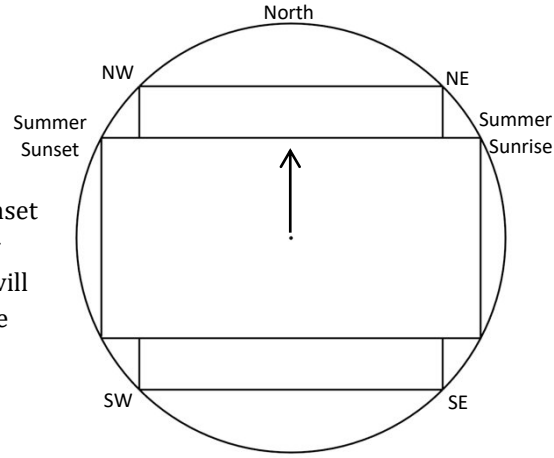


[103] See Appendix S.1 for calculations

If one were to extrapolate from this mistaken "NE" to determine all the other directions, the entire compass would be skewed by about 17° (45 - 28).



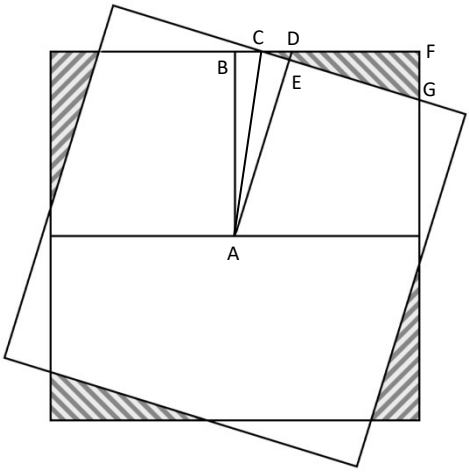
Therefore, the only valid methods for determining the directions is either sunrise or sunset on an equal day, or measuring *both* sunrise and sunset on the longest or shortest day[104] and finding the midpoint. That will show true north (or south, as the case may be).



[104] or any day, really

According to this, how much area would be mistakenly considered out of the Techum (and by extension, how much would be mistakenly considered in the Techum)?

Let's calculate how much area would be mistaken if one used the position of the longest sunrise as though that were truly northeast.



I'm going to use 17.14 degrees for the error.[105]

Lines $\overline{AB}$, $\overline{AE}$, and $\overline{BF}$ are half the Techum's width, so we'll call them .5

Angles $\angle BAE$ and $\angle FCG$ are 17.14°

$\angle BAC = .5 * \angle BAE$

Line $\overline{BC} = \overline{AB} * \tan(\angle BAC)$

Line $\overline{CF}$ is $\overline{BF}$ - $\overline{BC}$

Line $\overline{FG}$ is $\tan(\angle FCG) * \overline{CF}$

▶ Area of Triangle $\triangle FCG = .5 * \overline{CF} * \overline{FG}$

which in full would be

⬇ $= .5 * \left(\overline{BF} - \overline{BC}\right) * \tan(\angle FCG) * \left(\overline{BF} - \overline{BC}\right)$

which even fuller would be

⬇ $= .5 * (.5 - .5 * \tan(.5 * \angle BAE)) * \tan(\angle FCG) * (.5 - .5 * \tan(.5 * \angle BAE))$

which simplifies to

⬇ $= .5 * (.5 - .5 * \tan(.5 * 17.14°))^2 * \tan(17.14°)$

which equals

⬇ $= .027806912...$

x 4 times 4 for all 4 triangles at the Techum's corners

■ .1112..., or about **11%** of the total Techum area.

[105] For a more exact number, see Appendix S.1 on calculating Azimuth

A standard city of 2000 by 2000 amos would have a Techum of 6000^2, or 36,000,000 (Thirty six million) square amos. The error that we calculated above would result in **4,004,195.3...** square amos being misplaced (that much area in the Techum would be considered out, and that much area out of the Techum would be considered in).

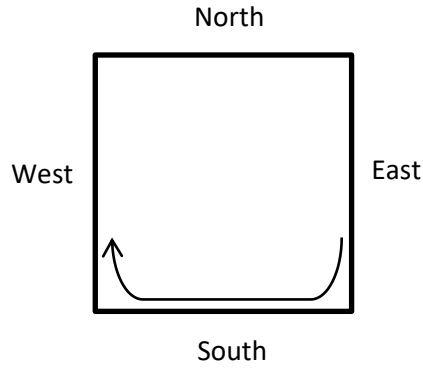
The actual distance one might walk in error is the same as how far it is from the line CG to the point F, which can be calculated as:

▶ $\sin(17.14) * \overline{CF}$ CF is being viewed as the hypotenuse of a right angle whose base is on the CG line

⬇ $\sin(17.14) * (.5 - .5 * \tan(.5 * 17.14°))$ which expands to

■ .1251... of the total Techum diameter, which for a city of 2000 amos comes out to **750.88...amos** which is beyond the Techum border!

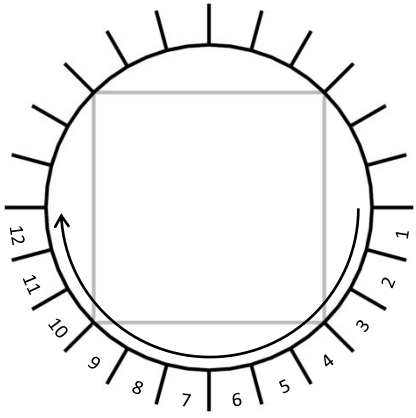
The Meiri describes the changes of where the sun rises and sets throughout the year, the way we described before like Rebbe Yosi, that the sun rises every day somewhere on the eastern horizon (more north in the summer, more south in the winter), and then traverses along the southern horizon, until it sets somewhere on the western horizon (again, more north in the summer, more south in the winter).



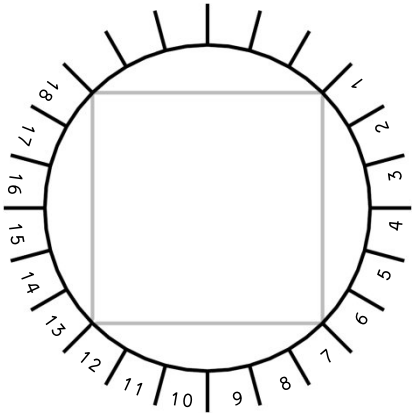
The Meiri views the horizon as cut up into 24 sections, one hour per section. Since we are also viewing the horizon as cut into quarters for the different directions, there are 6 hours for each direction $(24 \div 4 = 6)$.

He writes that by the equinoxes, since the sun rises and sets due east and due west therefore the day and night are of equal length.

In other words, since by the equinox the sun rises from the middle of the eastern horizon, so the sun travels half of the eastern horizon (3 hours), the entire southern horizon (6 hours), and then half of the western horizon before setting (3 hours) , for a total of 12 hours.



He writes that the Gemara explains how to find the northeast corner by seeing where the sun rises on the longest day (i.e., the summer solstice, the first day of summer).



The Meiri doesn't necessarily learn that Rav Misharshay rejects Rebbe Yosi's approach. He says that we can use this method "although it is not exact, as they said that the sun never rises on the northeast corner [exactly]..."

The Meiri proves another way that this is not exact, by saying that if it were true that the sun rose and set exactly on the corners then the longest day should be "18 hours of day and 6 hours of night... which it isn't."

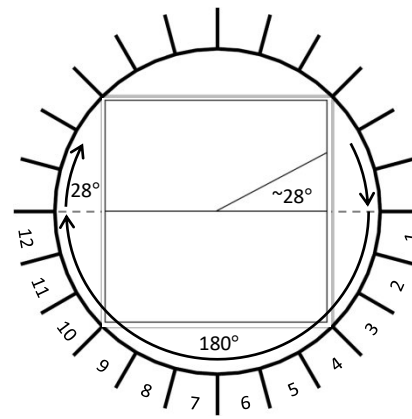
He says about that difference that it is OK because it is close, since the real number is about "15.5 hours of daytime and 8.5 hours of night".

I'm not sure what the Meiri means by 15.5 hours, because in Eretz Yisroel and in Bavel there are about 14 and a quarter hours of daylight on the longest day.[106]

I thought of two explanations for the Meiri.

1) One answer I thought of is that perhaps the Meiri is describing where he lived. He grew up in Perpignan, France, which has a longest day of about 15 hours and 20 minutes. Perhaps that is what the Meiri is referring when he writes "about 15 and a half hours".

2) Another answer might be as follows.



Consider that the sun rises from about 27.86° north of due east. Then, travelling south, the sun travels those 27.86 degrees until it reaches due east.

From there, the sun travels another 180 degrees until it arrives at due west.

And then it crosses another ~27.86 degrees until it sets.

Altogether, that's comes out to about 235.7 degrees.

Divide that number by 360 to get the percent of degrees of the entire horizon, and it comes out to ~.6547...

Multiply that by 24 hours and you get **~15.714...** hours, which might be what the Meiri means when he says about 15 and a half hours of daytime.

[106] See Appendix S.2, and for more exact calculations see Appendixes N, O, and P.

This whole approach is based on the sunrise and sunset positions in Eretz Yisroel or Bavel.

There are places in the world where the sun does rise and set at exactly 45° north of due east. As you go to places more north than Eretz Yirsoel, the positions of the rising and setting sun become more extreme throughout the seasons.

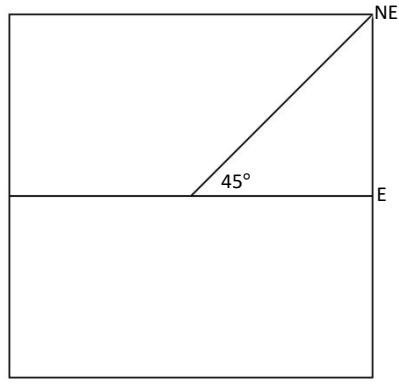
Yerushalayim is at about 31.8° north of the Equator (latitude). What latitude would have a summer solstice sunrise of 45° north of due east?

Here is a simplified formula[107]:

$$\text{latitude} = \cos^{-1}\left(\frac{\sin(\text{earth's tilt})}{\sin(45°)}\right)$$

which comes out to **55.829...**, which is about where Moscow is.

That means, if you were in Moscow, you would see the sun rising on the longest day at 45° north of due east. That direction is pointing to the "corner" of the "square of the world".

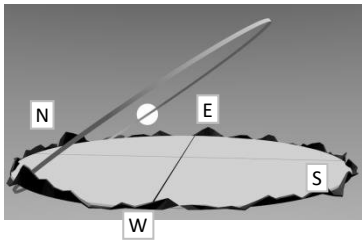


However, there is still a problem with this.

Even if you live in a place where the sun rises and sets at 45° north of due east, that doesn't mean that the day will be 18 hours long. Put differently, just because the sun travels through 3/4 of the horizon, doesn't necessarily mean that the sun is out for 3/4 of the day's entire 24 hour period.

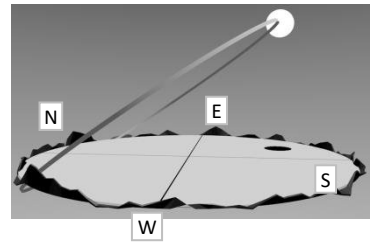
Let me explain.

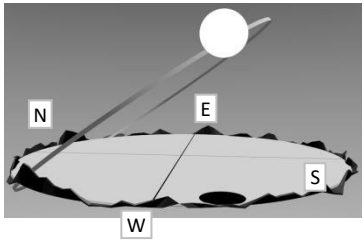
[107] This formula assumes the Earth to be a perfect sphere, and ignores altitude, atmospheric refraction, mountains on the horizon, the angular width of the sun, etc., although these factors are minor and will barely have any bearing on our discussion. See Appendix S.3 for an explanation of the formula shown here.



After the sun rises in the east, it arcs up into the sky as it goes towards the south[108],

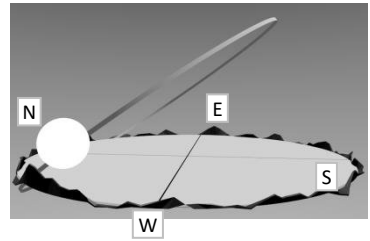
reaching its highest and southernmost point at midday (חצות היום, chatzos hayom)





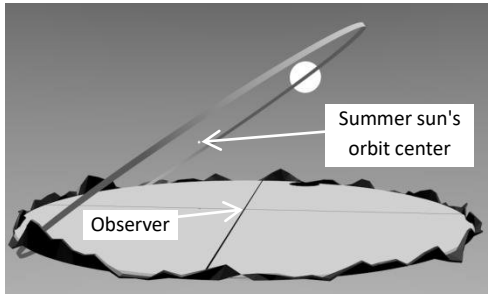
before heading towards the west, as it goes lower in the sky

until it sets somewhere along the western horizon.



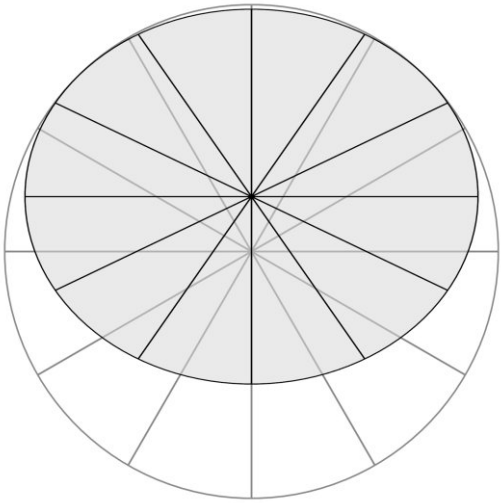
[108] in northern latitudes. This is all in the Topocentric view; see Appendix M.

Due to this tilt of the sun's orbit and to the fact that the center of this orbit (on the longest day) is not directly above the observer (see picture), the speed at which the sun's ground angle[109] changes will vary throughout the day. Either one of these factors alone will affect the consistency of the sun's speed along the ground. The two factors together compound the inconsistency.



This can be demonstrated simply.

Imagine we were viewing the ground from directly above the observer (bird's eye view, or better, spaceship eye view). The horizon can be drawn as a large circle around the observer. I have divided the circle into 12 segments, like a clock. (Although we are dealing with 24 hours, the principle is the same). The sun's orbit can be drawn as an oval[110] and off-centered[111]. I made it grey in the picture and have also divided it into 12 segments.



The exact shape and size of the sun's orbit relative to the horizon is

[109] The Azimuth. If you are standing on the ground watching the sun's movements in the sky, you can draw an imaginary line from the sun directly down until the horizon. You can then measure the number of degrees from that point to any other point on the horizon. For example, you can measure where above the horizon the sun is now and an hour ago. You can then measure the number of degrees between those two points and say "The sun's Azimuth has moved x number of degrees".

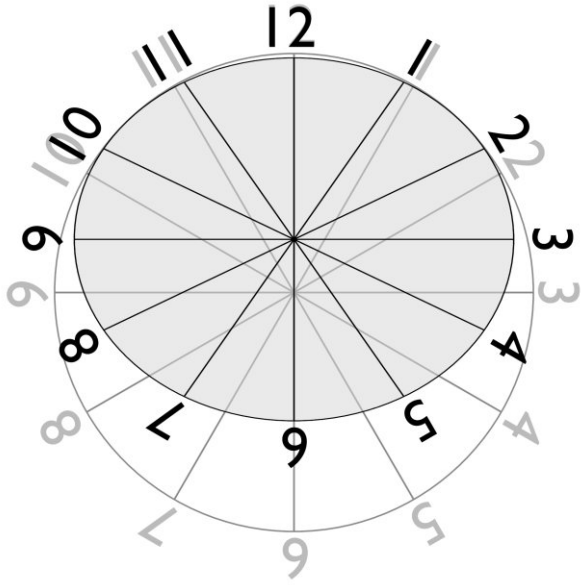
[110] since its orbit is tilted

[111] since the center of the sun's summer orbit is not directly above the observer

not relevant at this point because I'm just trying to demonstrate the principle.[112]

Look at the "clocks" now with the numbers in the picture below. Each "clock" will run according to its own internal speed. I'm going to call these the sun clock and the ground clock. The sun clock is like a clock tilted at an angle.

If you are standing in the middle (in the center of the larger circle) and you watch the sun going around you, its "hours" will not match up necessarily with the even divisions on the ground.

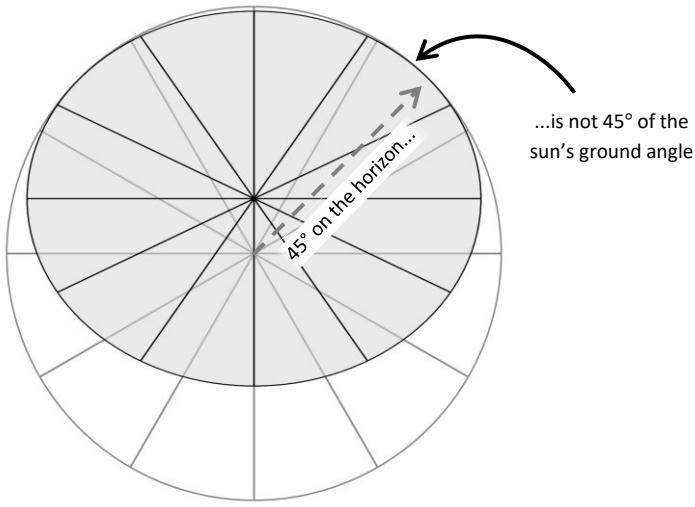


At 12 o'clock, both clocks show the same time. At 2 o'clock, the sun has dropped a bit behind the ground clock. At 6 o'clock, the sun caught up again. At 9 o'clock, the sun has moved ahead of the ground clock. And they are the same again at 12.

This principle is what happens in the sky with the sun relative to the horizon.

[112] If you are curious, the sun's orbit in the picture is the horizon scaled down by cosine(earth's tilt), and then along one axis by sine(latitude), and then off-centered along that same axis by sine(earth's tilt) * cosine(latitude). I'm using a latitude of 55.829 since that produces a 45° summer sunset, which we are assuming at this point will make an 18 hour day. See Appendix S.4 for full math.

So now, if we were at a latitude that produced a 45° summer sunset, although it is true that the sun will be above 3/4 of the horizon throughout the day (270° of the horizon) but it will not take exactly 18 hours but a little bit less.



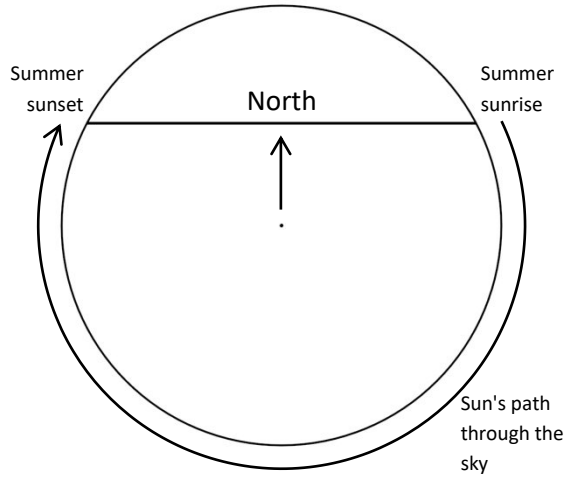
B'kitzur, it comes out to **17 hours 16 minutes and ~50 seconds**, for the longest day at latitudes with a 45° sunset.

To find what latitude would have an 18 hour day one can use the simplified formula[113]: Latitude $= \tan^{-1}\left(\frac{\sqrt{.5}}{\tan(\text{earth's tilt})}\right)$, which comes out to about **58.534°**, which is about as far north as Estonia.

[113] See Appendixes S.4 and S.5 for full mathematics of calculations on this page.

I initially had a different way of learning this Gemara.

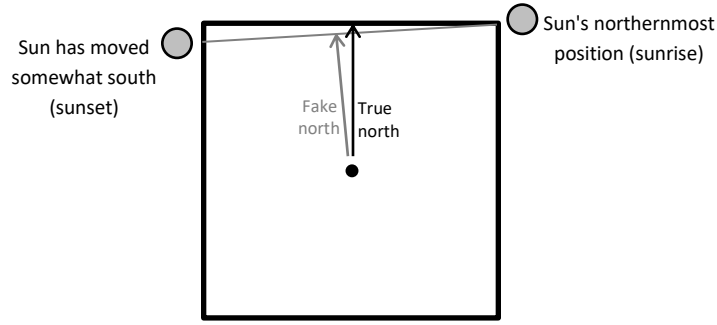
I thought that what Rebbe Yosi meant is that to determine the northern direction you draw an imaginary line between where the sun rises on the longest day and where it sets on that day and that line marks the northern face. That is, the midpoint of that line is due north.



The assumption of this pshat is that the point at which the sun rises and the point where the sun sets will be *exactly the same* amount of degrees north of due east and west, respectively.

Rav Misharshay responds that this method is inaccurate since the sun cannot both rise and set at the most northern position on the same day. The reason is because the exact moment that the sun reaches its most northern position it immediately begins to move south. If the sun reaches its most northern position at sunrise then it will have already moved somewhat south by the time of sunset.

Since sunrise and sunset are not at the same northern position, a line drawn between them will be lopsided and will not indicate true north.



This pshat is more borne out in the words of the Gemara than the first pshat we wrote, since the lashon of the Braisa that Rav Misharshay brought is:

לא יצאה חמה מעולם מקרן מזרחית צפונית ושקעה בקרן מערבית צפונית ולא יצאה חמה מקרן מזרחית דרומית ושקעה בקרן מערבית דרומית

"The sun never rose in the northeast corner and set in the northwest corner and the sun never rose in the southeast corner and set in the southwest corner."

It sounds like the first "rising" and "setting" are part of the same case, and the second "rising" and "setting" are part of another case.

According to the first pshat we brought, these are four separate cases.

How much of an error would this second pshat produce?

The formula (see Appendix S.6) to know the number of degrees north of due east or west a particular sunrise or sunset will be is as follows:

D = number of days after summer solstice

Y = number of days in a solar year (let's use the Gregorian 365.2425)

T = earth's tilt (23.4°)

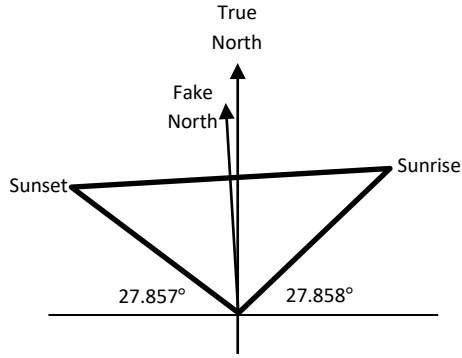
L = latitude

$$\text{angle} = \sin^{-1}\left(\frac{\cos\left(\frac{D}{Y} * 360\right)\sin(T)}{\cos(L)}\right)$$

Let's take Yerushalayim as an example. Yerushalayim's latitude is about 31.8° north. Since we're calculating sunrise as the moment of the solstice, we can put "0" as the number of days for "D" in the above formula, and so the sunrise angle north of due east will come out to **27.85864485...**°

The longest day in Yerushalayim is about 14 hours and 15 minutes, which comes out to .59375 days ((14+15/60)/24).

That's how long sunset is after the solstice. Using that value for "D" in the above formula will show that sunset will happen at about **27.85706515...**° north of due west.

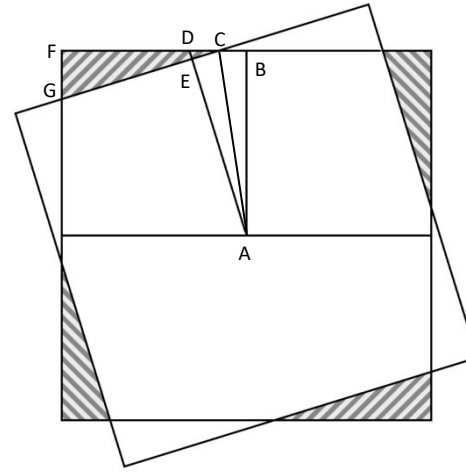


The number of degrees that fake north is from real north can be calculated[114] by measuring the degrees of the big upside-down triangle (in the picture) that's between sunrise and sunset. That triangle is 180° minus the sunrise and sunset angles combined. The triangle is therefore 124.28428998...°. Half of that (62.14214499) is the amount of degrees between the sunset and fake north. Add that to the sunset angle and you get 89.99921014.... Take that off of 90 and you get a whopping **0.00078985...**° that fake north will be off from real north.

The previous paragraph can be summed up in the following formula:

$$\text{Fake North Error Angle} = 90° - \left(\frac{180° - (\text{sunrise}° + \text{sunset}°)}{2} + \text{sunset}°\right)$$

So how bad is that?



Lines $\overline{AB}$, $\overline{AE}$, and $\overline{BF}$ are half the Techum's width, so we'll call them .5

Angles ∠BAE and ∠FCG are .00078985°

$\angle BAC = .5 * \angle BAE$

Line $\overline{BC} = \overline{AB} * \tan(\angle BAC)$

Line $\overline{CF}$ is $\overline{BF}$ - $\overline{BC}$

Line $\overline{FG}$ is $\overline{CF} * \tan(\angle FCG)$

Triangle $\triangle FCG = .5 * \overline{CF} * \overline{FG}$

[114] Or more simply: (sunrise° - sunset°) ÷ 2, but that is harder to visualize.

▶ Triangle $\Delta FCG = .5 * \overline{CF} * \overline{FG}$

which in full would be

⬇ $.5 * (\overline{BF} - \overline{BC}) * \overline{CF} * \tan(\angle FCG)$

which even fuller would be

⬇ $.5 * (.5 - .5 * \tan(.5 * .00078985°)) * (.5 - .5 * \tan(.5 * .00078985°)) * \tan(.00078985°)$

which simplifies to

⬇ $.5 * (.5 - .5 * \tan(.5 * .00078985°))^2 * \tan(.00078985)$

which equals

⬇ .000001723...

x 4 times 4 for all 4 triangles at the Techum's corners

■ .000006892..., or about **.00069%** of the total Techum area.

A standard city of 2000 by 2000 amos would have a Techum of 6000^2, or 36,000,000 (Thirty six million) square amos. The error that we calculated above would result in **248.13...** square amos being misplaced (that much area in the Techum would mistakenly be considered out, and that much area out of the Techum would mistakenly be considered in).

The actual distance one might walk in error is how far is the point F from the line $\overline{CG}$, which can be calculated as

CF is being viewed as the hypotenuse of a right angle whose base is on the CG line

▶ $\sin(.00078985) * \overline{CF}$

which comes out to

⬇ $\sin(.00078985) * (.5 - .5 * \tan(.5 * .00078985))$

which is

■ .000006892... of the total Techum diameter, which for a city of 2000 amos comes out to **.041356164...amos** beyond the Techum border, which comes out to .992547... of an etzbah, which comes out to about 2 centimeters or so.

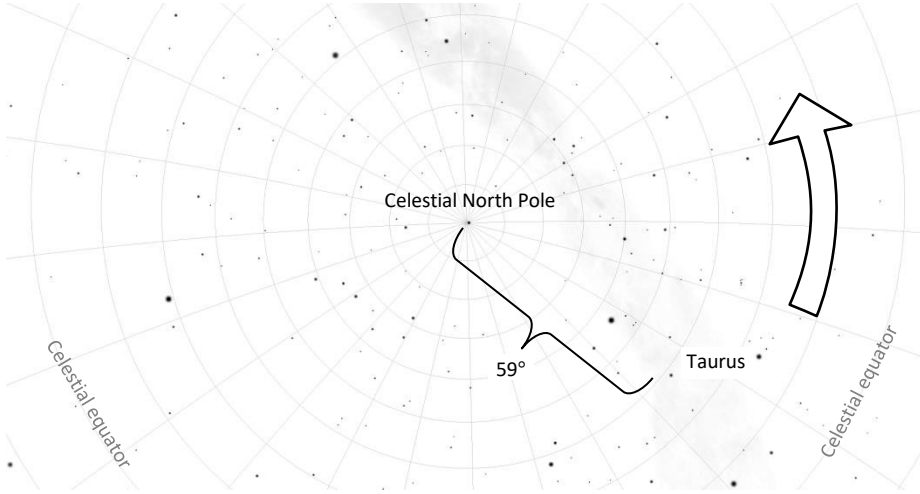
This margin of error is extremely small and it is therefore extremely doubtful that this is the pshat of the Gemara.

The truth is, I don't really know what margin of error is acceptable.

The Gemara initially told us that we could determine the directions by the locations of certain constellations whose positions vary far more than the degrees of error we are currently discussing.

For example, the Gemara gives Taurus as the direction for North. The most northern part of Taurus is (currently) about 31° above the celestial equator. As it rotates around the North Celestial Pole, it follows a circular path that is as wide as it is far from the center.

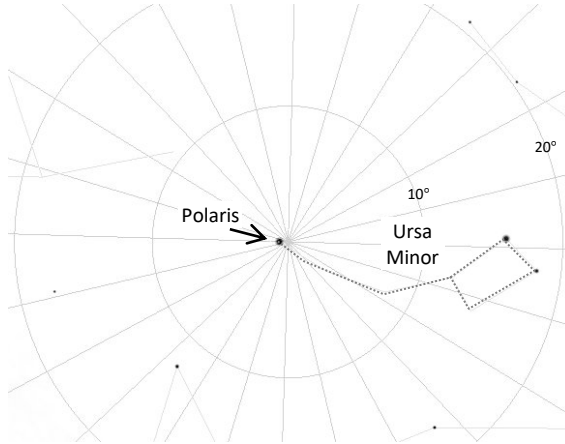
31° from the celestial equator is 59° (= 90° - 31°)from the pole. That means if one used Taurus as a sky-landmark to find north, he might be at least 59° off, which is way bigger than our errors.



The Meiri, in his first explanation, says that when the Gemara says עגלה, (which we have been reading as עֶגְלָה, "calf", presumably the Latin "Taurus"[115]) should be read as עַגָּלָה, "wagon" and he writes that it means the northern pole, i.e., the north celestial pole, i.e., the exact point in the sky around which all the stars revolve. That point is always due north.

[115] Similar to the Aramaic תור, from the Hebrew שור, both of which mean "bull".

The "wagon" is probably a reference to Ursa Minor (the Little Dipper), which was known as "the Wagon of Heaven" to the Babylonians[116], perhaps the same as Rabbeinu Chananel's עגלה בשמים.

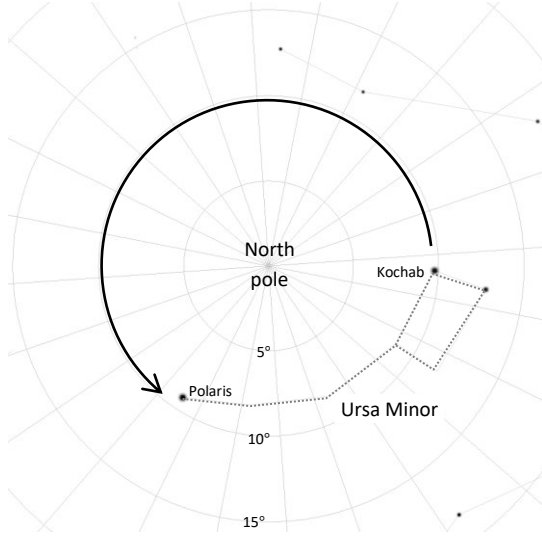


Currently (in 2020 C.E.) the center of the constellation is about 10° away from the pole.

The star Polaris (in Ursa Minor) is right next to the celestial north pole, about .7358...° away.

In the time of the Meiri[117], the center of the constellation was about 7.5° from the pole, with Polaris about 4.5° from the pole.

At the time of the writing down of the Gemara, the two main stars in Ursa Minor, Polaris and Kochab, were both about 9.5° from the pole, which would result[118] in a Techum area error of 7%, and distance error of about 454 amos beyond the Techum in some places.



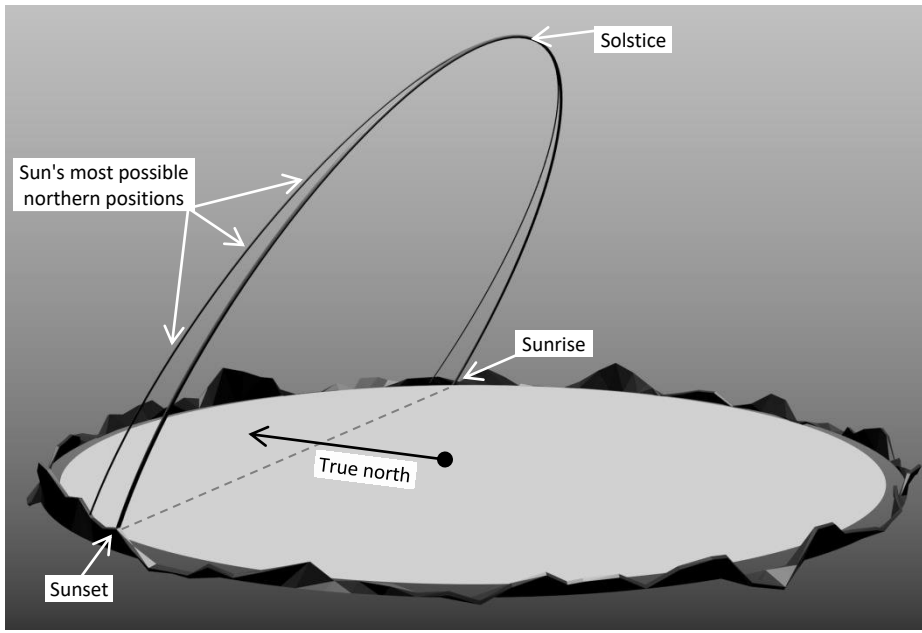
However, it could be that we look at the constellation as a circle around the pole and then it is much more accurate, at the time of the Gemara, see picture.

[116] Wikipedia "Ursa Minor"
[117] Circa 1300 C.E.
[118] Using the formulas from the previous pages

I can't tell you how accurate that would be because it is more of a sense-thing than an exact calculation.

Now, let's go back to the pshat we were discussing. We said that Rav Misharshay is saying that if the sun rises at the exact moment of the summer solstice then the sun won't set at its most northern position but just slightly south of that.

But what about if the summer solstice happens at midday? In such a case, the sun will rise and set at the exact same angle north relative to due east and west.



A line drawn between that sunrise and sunset will be directly aligned along the north direction.

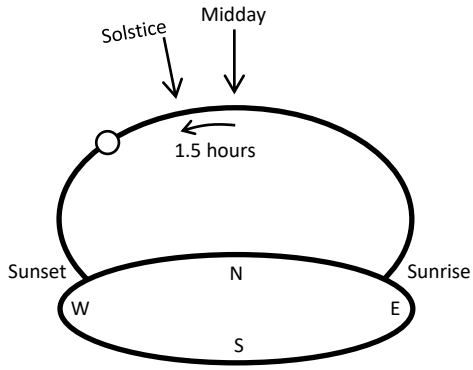
Couldn't such a method be used to determine the true cardinal directions?

The Gemara's follow up is a refutation to this suggestion.

The Gemara brings a Braisa that says that the summer solstice *never* happens at midday! The closest it gets there is an hour and a half after midday.

If one were to draw a northern border on such a day assuming that the sun rose and set the same number of degrees north of due east and west, his mistaken north will be off from real north by about .00016° (see Appendixes N to Q). This number is so exceedingly small and negligible that I have no doubt that this is not the pshat in the Gemara. But I wouldn't have known that had I not done the calculations.

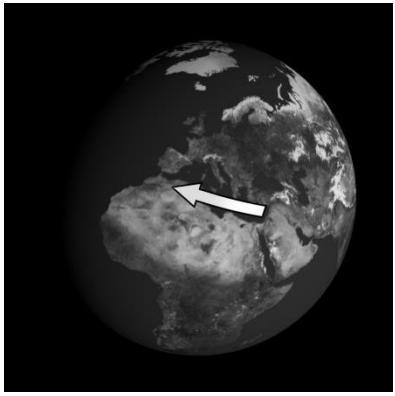
Anyway, assuming the Braisa's numbers are in reference to Eretz Yisroel, could we determine where on Earth the solstice could in fact happen at midday?



Since the solstice can happen at 1.5 hours after midday, we need to find a place on earth where it is midday while it is 1:30[119] in the afternoon in Eretz Yisroel.

Since in Eretz Yisroel, the sun has already passed overhead westward, that means that to get the sun over our head we would need to go west "1.5 hours out of 24", which would be 22.5 degrees $\left(\frac{1.5}{24} * 360°\right)$ west of Eretz Yisroel. Eretz Yisroel's longitudes

[119] These times are halachic times. They might refer to kavuos hours or zmaniyos hours (see Tosfos on this amud). For simplicity, I will assume we are talking about 60-minute hours (kavuos).



are between about 34.5E and 35.5E. Subtracting 22.5 degrees ends up in about Rome or Copenhagen.[120]

Now, since the Braisa said that the solstice can happen at intervals of 6 hours (1:30 or 7:30, AM or PM) that means that any place that has a longitude of 22.5° plus or minus a multiple of 90° $\left(\frac{6}{24} * 360°\right)$ can have a midday solstice.

This would include places like Washington D.C., the western coast of Alaska, and Malaysia.

Even if one were to assume such an approach, one would need to know which year of the four year cycle he is in (since the solstice can happen at 12 AM, 6 AM, 12 PM, or 6 PM, depending on the year) in order to know that the solstice happened at midday.

The same approach will work for determining the south direction at the winter solstice. The Gemara said that the winter solstice can happen at 4:30 or 10:30, AM or PM. That means that to find a place where it can happen at that place's midday we would need to travel 22.5° $(1.5 \div 24 * 360°)$ east, or that number plus or minus multiples of 90, which would end up in places like eastern Iran, eastern Japan or Australia, the western coast of the United States, and the eastern coast of Greenland or perhaps of Brazil.

[120] The latitude doesn't affect the time of midday.

*

נ"ו. - Tosfos Ein Bein - Shaos Kavuos and Zmaniyos

A few introductory remarks are in order.

The Seasons

Throughout the year, the days and nights get longer and shorter[121].

The longest day is called the **summer solstice**.[122] It happens on June 21st, give or take a day. The longest day also has the shortest night.

The shortest day of the year is called the **winter solstice**. It happens on December 21st, give or take a day or so. The shortest day has the longest night.

Right in between these two days are what's called the equinoxes. These are the "middle days", i.e., the two days in the year whose days and nights are of equal lengths.

The middle day as the days are getting longer is called the **vernal equinox**. It happens on March 20th or thereabouts.

The middle day as the days are getting shorter is called the **autumnal equinox**. It happens around September 22rd.

The periods between these points are the seasons.

Tekufas Nissan, spring[123], is the period beginning at the vernal equinox and ending at the summer solstice.

Tekufas Tammuz, summer, begins at the summer solstice[124] and ends at the autumnal equinox.

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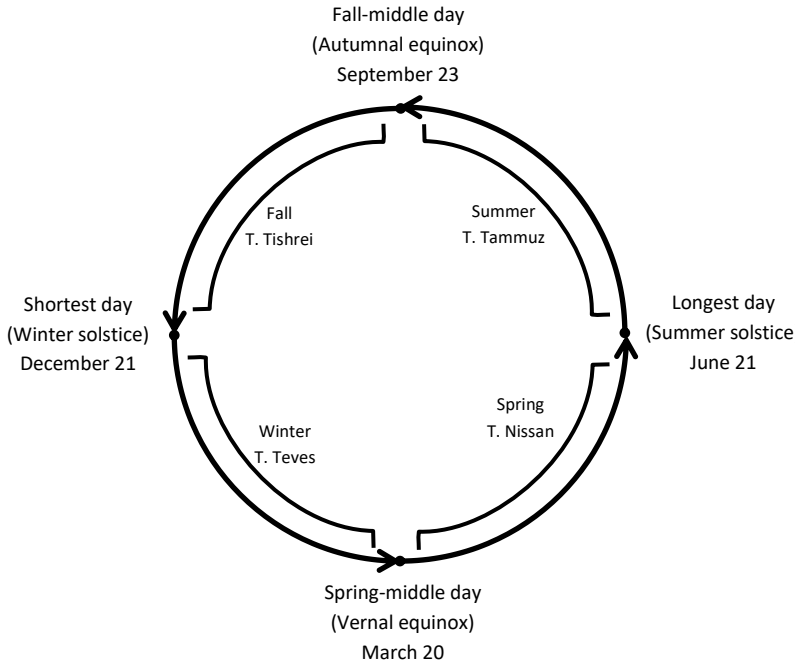
[121] See Appendixes K, L, and Q

[122] I'm generalizing here. Technically, the solstices and the equinoxes are single points in time. I'm equating the day on which they happen with the day itself.

[123] According to the astronomical definition in use in many countries.

[124] One would wonder why the summer only begins on the longest day. One would expect that day to be at the height of summer, when the sun has been and will continue to be shining for the most amount of time each day, making them the hottest days. The reason it is not so is because it takes a long time for the earth to heat up from the cold weather. After the sun has been out for long periods of time, the earth reaches its hottest temperatures after the longest days before the shortness of the days begin to cool the earth off. And vice versa for the winter.

Tekufas Tishrei, fall, begins at the autumnal equinox and ends at the winter solstice.
Tekufas Teves, winter, begins at the winter solstice and ends at the vernal equinox.



Tekufa	Season	Begins at...	...Usually on	Ends at...	...Usually on
Nissan	Spring	Vernal equinox	March 20	Summer solstice	June 21
Tammuz	Summer	Summer solstice	June 21	Autumnal equinox	September 22
Tishrei	Fall	Autumnal equinox	September 22	Winter solstice	December 21
Teves	Winter	Winter solstice	December 21	Vernal equinox	March 20

Each day has 24 hours. That includes the night and the day.

The hours that we use day to day are comprised of 60 minutes. This is what's called שעות קבועות, Shaos Kavuos, "Fixed hours". I will also sometimes refer to these hours as "Regular hours".

There are also hours called שעות זמניות, Shaos Zmaniyos, "Fluctuating hours". The length of these hours depends on the length of the day or night in which they appear.

The way shaos zmaniyos hours work is that each day or night is divided into 12 even divisions of time. If the day is long, then the hours are long. If the day is short then the hours are short. And the same with the night.

Let's say for example that the longest day is 18 hours long, from sunrise to sunset[125]. Altogether that's 1,080 minutes (18 * 60). Divide that by 12 gives us **90 minutes per hour**.

The longest day is also the shortest night. If the day was 18 hours long then the night is 6 hours long (24 - 18). 6 hours is 360 minutes (6 * 60). Divide that by 12 gives us **30 minutes per hour**.

And it's all the opposite for the shortest day. The shortest day's shaos zmaniyos will be 30 minutes per hour and the nights will have 90-minute hours.

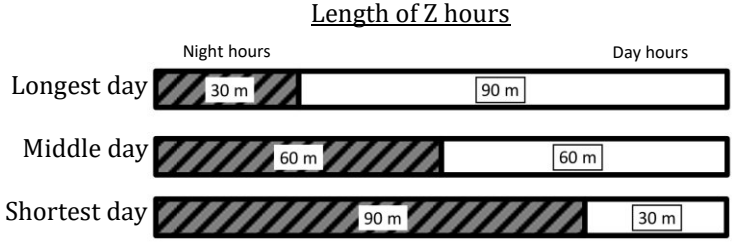
And the equal days will have zmaniyos hours that are 60 minutes long, just like the kavuos hours, since there's 12 regular hours for the day and for the night.

(I will here on in sometimes use the term KH for kavuos hours and ZH for zmaniyos hours.)

Tosfos uses this breakup of 18-6[126] for the longest/shortest day/night. We will see if that makes a difference to the points I want to bring out.

[125] There are different ways of determining the beginning and end of the day as far as shaos zmaniyos is concerned. I'm using a common opinion of sunrise to sunset.
[126] Like the Meiri. See page 159 and onward.

	Day			Night		
	Regular hours	* 60 = ...total minutes	÷ 12 = ...minutes per Z hours	Regular hours	* 60 = ...total minutes	÷ 12 = ...minutes per Z hours
Longest day	18	1080	90	6	360	30
Middle day	12	720	60	12	720	60
Shortest day	6	360	30	18	1080	90



There's one more idea I'd like to point out before starting with Tosfos, and that is - how does one define "a day"?

What I mean is that when we hear the term "In x number of days...", what time exactly will it be then relative to what time it is now?

Is "a day" 24 hours or the same part of the day as it is now?

Let's take as an example, it is now sunset at 6PM. Sunset in 30 days will be at 7PM. If I say (now at 6PM sunset) "in 30 days from now", am I referring to 6PM or 7PM in 30 days?

If "a day" means one 24 hour period then 30 days means thirty 24 hour periods and I am referring to 6PM, in 30 days from now.

If "a day" means from one part of the day to that same part the next day then 30 days means 30 "sunsets", which will be 7PM in 30 days.

In other words, we could differentiate between "Kavuos Days" (or 24 hours each) and "Zmaniyos Days" (from one day's events to the next day's).

Basically, a "Zmaniyah Day" goes from one Zmaniyah Hour to that same Zmaniyah Hour, the next day. In the winter and spring, as each day gets longer than the previous, zmaniyos days will be longer than 24 hours, since each sunset is later than the previous one. In the summer and fall, when each day is shorter than the previous, zmaniyos days will be shorter than 24 hours.

To avoid clutter, I will sometimes refer to these different kinds of days as KD and ZD.

One more point I should clarify is that I'll be using civil time in this piece, unless otherwise noted. That is, Monday begins at 12 AM and goes until 11:59 PM.

The Gemara uses a different system where midnight is "6 hours of the night", and midday is "6 hours of the day". On an equal day, sunrise happens at "zero hours of the day" or what might be called the "first hour of the day".

The hours of a non-equal day will be the subject of this piece.

The Gemara says that the length of each tekufa is 91 days and 7.5 hours long.

That number is derived by taking the year to be 365.25 days and dividing by 4, for the 4 seasons.

$$\begin{array}{r} 365.25 \\ \div \quad 4 \\ \hline 91\frac{5}{16} \end{array} \qquad \frac{5}{16} = 5 \div 16 = .3125$$

$$\begin{array}{r} \text{x} \quad 24 \text{ hours} \\ \hline 7.5 \text{ hours} \end{array}$$

In terms of days of the week, this number can be moduloed[127] down to **7.5 hours**.

That means that if one season begins on a specific day of the week, at a certain hour, the next season will begin that same day (or the next day if we cross midnight) of the week, 7.5 hours later.

For example, if tekufas Nissan begins on Monday at 6 AM then tekufas Tammuz will begin on Monday at 1:30 PM.

If tekufas Nissan begins at 6 PM on Monday then tekufas Tammuz will begin early Tuesday morning at 1:30 AM.

And so on.

This is point number one. We will return to this soon.

Point number 2:

The Gemara gives the time of day that each tekufa can fall out on.

Tekufas Nissan can fall out at 6 AM, 6 PM, 12 PM, or 12 AM[128].

Tekufas Tammuz comes out 7.5 hours later than tekufas Nissan. So, following the order we listed for Nissan, that would be 1:30 PM, 1:30 AM, 7:30 PM, and 7:30 AM.

Tekufas Tishrei: 9PM, 9AM, 3AM, and 3PM.

Tekufas Teves: 4:30AM, 4:30PM, 10:30AM, and 10:30PM.

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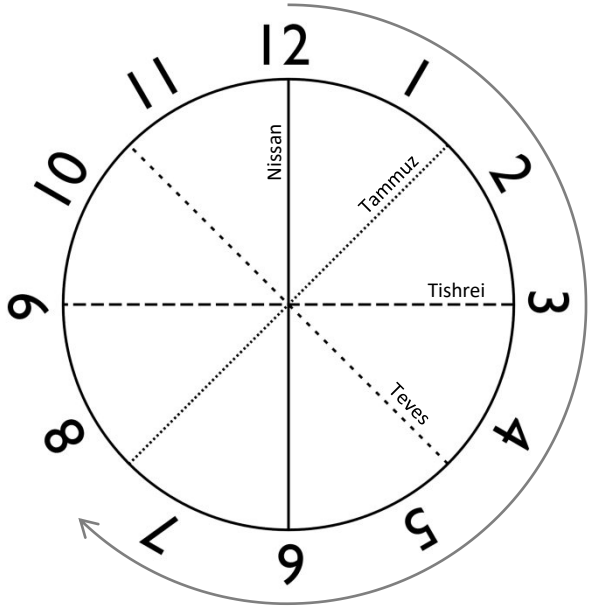
[127] Modulo isn't really a verb but I'm using it as such.
To modulo means to keep subtracting (or adding) a certain number until you reach a number between zero and that number.
In our example, we took 91 days and 7.5 hours and moduloed 7 days (for the 7 days in the week) and we were left with 0 days and 7.5 hours. Modulo is what Chazal and the Rishonim call "Throwing off a number", like "Throwing off sevens".

[128] Note that at this point it makes no difference whether we're using civil time or "6 and 12 hours of the day and night". Either way it will come out the same. The Gemara's list is "The beginning of the day, the end of the day, midday, and midnight".



Tekufa start:

Year	Nissan	Tammuz	Tishrei	Teves
1	6 PM	1:30 AM	9 AM	4:30 PM
2	12 AM	7:30 AM	3 PM	10:30 PM
3	6 AM	1:30 PM	9 PM	4:30 AM
4	12 PM	7:30 PM	3 AM	10:30 AM

Keep in mind that the Gemara doesn't use the term "4:30 PM" but rather "10.5 hours of the day" (which is 4:30 PM if the day starts at 6 AM and the hours are regular 60-minute hours). Use this chart more as a starting guide.

Tosfos wants to know whether the Gemara is using shaos kavuos (KH) or shaos zmaniyos (ZH), when giving us the start times of the tekufos and when giving us the tekufa lengths.

I want to know whether we're using kavuos days (KD) or zmaniyos days (ZD).

Combining all these factors gives us 4 possible permutations:

	KH	ZH
KD		
ZD		

We shall try all possibilities and see what comes out.

Kavuos Days - Kavuos Hours

I think this is what Tosfos means when he tries shaos kavuos, that it also means that the days are kavuos.

In this permutation, "one day" means the next day at the same time.

So, if it's 6 PM today at sunset, "one day from now" is tomorrow at 6 PM, even if sunset is later.

The problem that Tosfos has with this approach is that if tekufas Nissan falls out at 6 PM, then tekufas Tammuz falls out at **1:30 AM**, civil time. The Gemara says it will fall out "1.5 hours or 7.5 hours of the night". The night by tekufas Tammuz begins at 9 PM[129]. 1.5 hours after sunset is 10:30 PM and 7.5 hours after sunset is 4:30 AM. This is not the time that we expected, which was 1:30 AM (4.5 hours of the night)

[129] According to Tosfos's assumption that there are 18 hours in the longest day (See Appendix N). The question of Tosfos is valid no matter how many hours are in the longest day, as long as it is longer than the equal day and shorter than 24 hours.

I don't know why we can't just say the Gemara is using a form of "civil time", that is that "a day" begins at the average time of sunset, what we've been calling 6 PM, and then you count the hours of that day from then. So, for example, even though in the summer sunset is much later than 6 PM, maybe we're calling "1.5 hours of the night" or "7.5 hours of the night" from sunset, which would be 7:30 PM and 1:30 AM, respectively.

Zmaniyos Days - Kavuos Hours

I don't think this is what Tosfos means when he discusses kavuos hours, and I'll explain why as we go.

A zmaniyah day means you go from one day's zmaniyah hour to the same zmaniyah hour the next day (or number of days, as the case may be).

So "91 days and 7.5 hours" would mean you first calculate when are the 91 zmaniyos days up and then you add another 7.5 kavuos hours.

So, let's say tekufas Nissan begins at 6 PM, which is sunset. 91 zmaniyos days later means at sunset, which by tekufas Tammuz is at 9 PM. So just taking the days alone gets us to 9 PM. To which we need to add 7.5 kavuos hours, which ends up at **4:30 AM**.

4:30 AM is 7.5 hours of the night in tekufas Tammuz (which began at 9 PM).

Which is fine.

Except it isn't, because Tosfos said that it won't land on the right times. But we see it is landing on the right time. That means that Tosfos is understanding it differently than the way we are.

It seems to me that Tosfos understands that kavuos hours means you're also using kavuos days. You can't mix and match.

And it makes sense that you can't. Let me give you one reason. If I ask you at 6 PM tekufas Nissan to tell me what time it will be in 91 days and 7.5 hours, and I mean 91 ZD and 7.5 KH, you'll ask me which do I want to calculate first.

If I do the days first and then the hours, I'll end up at 4:30 AM. If I do the hours first and then the days I'll end up at 12:45 AM!

And what if I calculate some of the hours now and some of them after the 91 days?

The whole system breaks down.

OK, let's back up a little.

We said that if we're using ZD and KH, we'll get to tekufas Tammuz at 4:30 in the morning.

If we keep on going, we'll find that the times actually do get all misaligned with the times given to us in the Gemara.
(This chart is in 24-hour format)[130]

Year	Nissan	Tammuz	Tishrei	Teves
1	18	4.5	14.5	20.75
2	5.3333	10.1667	18.2778	22.9167
3	6.778	11.6667	19.2778	0.4167
4	7.7778	13.1667	20.2778	1.9167
5	8.7778	14.6667	21.2778	3.4167
6	9.7778	16.1667	22.2778	4.9167
7	10.7778	17.6667	23.2778	6.4167
8	11.7778	19.1667	0.2778	7.9167
9	12.7778	20.6667	1.2778	9.4167
10	14.3333	23	5.5	15.75

So, you might say that's good since that's what Tosfos said is going to happen!

No, it's bad, because there's another problem happening here that Tosfos *didn't* mention, even though he <u>did</u> mention it when discussing zmaniyos hours. The problem is that the tekufos are now not the same lengths as each other:

[130] Minutes are displayed as decimals of hours. For example, 4:30 is shown as 4.5. All numbers rounded to 4th decimal points. See Appendix U for Python code for charts.

Tekufa lengths:

91 kavuos days + ... kavuos hours

Year	Nissan	Tammuz	Tishrei	Teves
1	10.5	10	6.25	8.5833
2	4.8333	8.1111	4.6389	7.8611
3	4.8889	7.6111	5.1389	7.3611
4	5.3889	7.1111	5.6389	6.8611
5	5.8889	6.6111	6.1389	6.3611
6	6.3889	6.1111	6.6389	5.8611
7	6.8889	5.6111	7.1389	5.3611

Kavuos Days - Zmaniyos Hours

In this approach, we'll go from hour to same hour when calculating the 91 days and then add 7.5 zmaniyos hours, and let's see what happens.

Nissan, sunset, 6 PM.

91 KD later is Tammuz, afternoon, 6 PM.

Now let's add the zmaniyos hours. There's 3 hours left until sunset. That's 180 minutes (3 * 60). Since the day zmaniyos hours are 90 minutes each, that means that 2 ZH are left in the day (180 ÷ 90). We need another 5.5 ZH after sunset (7.5 - 2).

Sunset is at 9 PM. Night hours are 30 minutes each. 5.5 of those hours is 165 minutes (5.5 * 30), which is 2.75 regular hours (165 ÷ 60). 2.75 hours after sunset is **11:45 PM**.

Which is not one of the times listed in the Gemara as possible for tekufas Tammuz. Tosfos doesn't mention this problem which might indicate this is not what he meant by zmaniyos hours. However, we will see that even when we do ZD with ZH we'll still have this problem that the times are wrong. So either way, we have to wonder why Tosfos left out this point.

?

The issue that Tosfos does talk about by shaos zmaniyos is that the tekufos are of different lengths.

So, let's keep going. Tekufas Tammuz falls out at 11:45 PM. 91 KD later is Tishrei 11:45 PM. Sunrise is in 6 hours and 15 minutes (at 6 AM)[131], leaving us with another 1 hour and 15 minutes to fill up the 7.5 hours, gets us to **7:15 AM**.

First tekufa length is 91 KD and 5.75 KH.

Second tekufa length is 91 KD and 7.5 KH.

Not the same.

Tekufa times:

Year	Nissan	Tammuz	Tishrei	Teves
1	18	23.75	7.25	12.1667
2	19.6667	0.3056	7.8056	12.3519
3	19.8519	0.3673	7.8673	12.3724
4	19.8724	0.3741	7.8741	12.3747
5	19.8747	0.3749	7.8749	12.375
6	19.875	0.375	7.875	12.375
7	19.875	0.375	7.875	12.375

Tekufa lengths:

Year	Nissan	Tammuz	Tishrei	Teves
1	5.75	7.5	4.9167	7.5
2	4.6389	7.5	4.5463	7.5
3	4.5154	7.5	4.5051	7.5
4	4.5017	7.5	4.5006	7.5
5	4.5002	7.5	4.5001	7.5
6	4.5	7.5	4.5	7.5
7	4.5	7.5	4.5	7.5

[131] In Tishrei, the zmaniyos hours are like regular hours, 60 minutes each.

Interestingly, after a certain number of years the times and lengths of the tekufos converge to specific numbers.

Zmaniyos Days - Zmaniyos Hours

This means that we will go from one day to the next, to the same zmaniyah hour and then add another 7.5 zmaniyos hours.

So, let's start at tekufas Nissan, sunset at 6 PM.

91 days later is tekufas Tammuz, sunset 9 PM.

Since we're now in the night, 7.5 short hours is 225 minutes (7.5 * 30), which is 3.75 regular hours (225 ÷ 60). 3.75 regular hours after 9 PM is **12:45 AM**.

12:45 AM is not one of the times listed as a possible start for tekufas Tammuz. Tosfos doesn't mention this problem when he suggests that the Gemara is using shaos zmaniyos, whereas he does mention this problem when discussing shaos kavuos. I don't know why.

We said 12:45 AM at tekufas Tammuz is 7.5 shaos zmaniyos into the night. 91 days later is tekufas Tishrei. At that time of year, 7.5 hours into the night is 1:30 AM. Plus 7.5 hours (zmaniyos and kavuos are the same in Tishrei) is **9 AM**. At least that time is on the Gemara's list.

The tekufa lengths are:

Nissan: 6.75

Tammuz: 8.25

This is what's bothering Tosfos; that the tekufa lengths are different from each other.

Tekufa times:

Year	Nissan	Tammuz	Tishrei	Teves
1	18	0.75	9	14.25
2	0	5.25	15	21.75
3	6	14.25	21	6.75
4	12	21.75	3	11.25
5	18	0.75	9	14.25

Tekufa lengths:

Year	Nissan	Tammuz	Tishrei	Teves
1	6.75	8.25	5.25	9.75
2	5.25	9.75	6.75	8.25
3	8.25	6.75	9.75	5.25
4	9.75	5.25	8.25	6.75
5	6.75	8.25	5.25	9.75

Notice how the times and lengths repeat every 4 years, perhaps echoing our Gemara's 4 year pattern (although these charts are not exact as they are rounded to the nearest 4th decimal).

I don't really know what to make of Tosfos.

I understand that when he writes about kavuos hours, he understands that we mean kavuos days as well.

I don't know how to interpret zmaniyos hours, since whether it means in conjunction with zmaniyos days or kavuos days, either way the tekufa starting times will not be the same as the Gemara's list. Why then did Tosfos not mention this problem? We know such a problem bothers Tosfos since he mentioned it by shaos kavuos.

Maybe Tosfos holds that the different tekufa lengths is a bigger problem, and so he mentioned it by shaos zmaniyos, saving the starting time problem for shaos kavuos.

Perhaps.

למעשה, I don't know why the differing tekufa lengths bothers Tosfos. The מציות is like that.[132]

The seasons for 2020 C.E. vary from 88 days to 93 days. That's a lot more than the amount we're dealing with! The differences we cheshboned always come out to less than 10 hours.

Perhaps what's bothering Tosfos is that the Gemara is משמע that the tekufa times are the same as each other.

[132] due to the eccentricity of earth's orbit and other causes; see Appendix R.

While we're on this Tosfos, let's explain the last part. Tosfos writes that if we are using shaos kavuos then the Gemara in Shabbos[133] becomes difficult to understand.

The Gemara there gives a list of days on which one could or should not let blood.[134]

The days for letting blood are Sunday, Wednesday, and Friday.

The non-letting-blood days are Monday, Tuesday, and Thursday (and Shabbos).

The Gemara says blood should not be let on Monday or Thursday because those are the days that both the בית דין של מעלה[135] and the בית דין של מטה sit in judgment.

Not on Tuesday because the hour of Mars happens on an even-numbered hour of the day.[136]

The order[137] of the hours starting from Tuesday evening at sunset are:

---

[133] קכ"ט:

[134] Bloodletting itself is a dangerous activity, compounded by being in an even-numbered hour and in the hour of Mars. All these factors combine to make it dangerous enough that חז"ל felt it necessary to forbid it (or perhaps they sensed that it was dangerous enough to be forbidden ממילא).

[135] Rashi explains that the בית דין של מעלה judges every day. Since on Mondays and Thursdays both courts are ruling it is a day of judgment and so a person's sins are recounted.

[136] Both of which compound the danger.

[137] This order follows the longest to shortest sidereal periods of these bodies; that is, how long it takes them to come back to the same stars, as seen from earth.

1) Saturn 2) Jupiter 3) Mars 4) the Sun 5) Venus 6) Mercury 7) the Moon

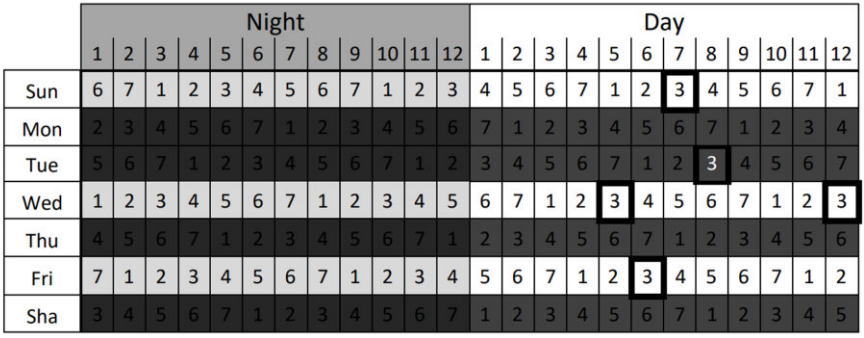
The first hour after sunset on Tuesday evening (the night going into Wednesday, when the planets were put into the sky) is the hour of Saturn.

The second hour is Jupiter.

And so on.[138]

	Night												Day											
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	4	5	6	7	8	9	10	11	12
Sun	6	7	1	2	3	4	5	6	7	1	2	3	4	5	6	7	1	2	3	4	5	6	7	1
Mon	2	3	4	5	6	7	1	2	3	4	5	6	7	1	2	3	4	5	6	7	1	2	3	4
Tue	5	6	7	1	2	3	4	5	6	7	1	2	3	4	5	6	7	1	2	3	4	5	6	7
Wed	1	2	3	4	5	6	7	1	2	3	4	5	6	7	1	2	3	4	5	6	7	1	2	3
Thu	4	5	6	7	1	2	3	4	5	6	7	1	2	3	4	5	6	7	1	2	3	4	5	6
Fri	7	1	2	3	4	5	6	7	1	2	3	4	5	6	7	1	2	3	4	5	6	7	1	2
Sha	3	4	5	6	7	1	2	3	4	5	6	7	1	2	3	4	5	6	7	1	2	3	4	5

For lack of space, I used numbers instead of planet names. I boxed in the first hour of the night of Wednesday since that is the "starting point". I boxed in all the "3"s in the daytime that represent "Mars".



[138] An interesting aside, the names of the week "Sunday", "Monday", etc. come from that day's first daytime hour. The first hour of Sunday day is 4, the Sun. The first hour of Monday is 7, Moon. And so on. The actual names of the week reflect various cultures' names of the planets.

Monday and Thursday are out because the Beis Din judges then, as we said.

Bloodletting is not allowed on Shabbos because it is a melacha.

Tuesday is out because Mars (3) appears in an even-numbered hour (8).

Friday is in (even though it has a "Mars" hour in an even hour (6)) because "people have trampled on this, and we say שומר פתאים ה', Hashem guards the foolish". In other words, even though it should technically be אסור, we have a principle that when something becomes common then it no longer has the same degree of danger (or perhaps, degree of איסור) as it would have otherwise.

Tosfos there[139] points out that Wednesday has a Mars hour in an even-numbered hour (12) and yet it's permitted to let blood on that day because it is at the very end of the day and Chazal were not worried since it is not normal for people to let blood so close to the night.

It sounds from Tosfos that the actual danger isn't in effect the whole day long but rather just in the hour that Mars is in an even number. If that's true then why did Chazal forbid bloodletting the entire Tuesday? Just forbid the even-hour of Mars.

It would seem to me that Chazal forbade the whole Tuesday because of the possibility that one might inadvertently let blood on the dangerous hour.

Alternatively, it could be that what Tosfos means is that a day is a dangerous day to let blood if there is within it a dangerous hour that is the normal time for people to let blood.

According to that, Tuesday is אסור because the whole day is dangerous.

Now, back to our Tosfos.

Tosfos says that if we are using shaos kavuos then the Gemara in Shabbos about the hours of bloodletting is difficult to understand because sometimes it will come out that the days that bloodletting is allowed will actually have a Mars hour in an even hour! And vice versa: Some of the days that we are <u>not</u> allowed to let blood will *not* have a Mars hour in an even hour!

[139] ד"ה דקאי מאדים בזווי

Let's see how that might work.

Let's take for example a day when there's 11 hours in the day, like a little bit before Nissan or after Tishrei:

	Night													Day									
	1	2	3	4	5	6	7	8	9	10	11	12	13	1	2	3	4	5	6	7	8	9	10	11
Sun														5	6	7	1	2	3	4	5	6	7	1
Mon														1	2	3	4	5	6	7	1	2	3	4
Tue	5	6	7	1	2	3	4	5	6	7	1	2	3	4	5	6	7	1	2	3	4	5	6	7
Wed														7	1	2	3	4	5	6	7	1	2	3
Thu														3	4	5	6	7	1	2	3	4	5	6
Fri	7	1	2	3	4	5	6	7	1	2	3	4	5	6	7	1	2	3	4	5	6	7	1	2
Sha														2	3	4	5	6	7	1	2	3	4	5

Here, Sunday and Wednesday are no good, but Tuesday is OK, and Friday should be OK even without the היתר of שומר פסאים ה', since Mars is not in an even hour.

I was thinking, we could answer Tosfos's question with one of the ways we explained the Tosfos in Shabbos about Wednesday, on the previous page.

We explained in our second explanation that the way it works is that whether a specific day is considered dangerous or not is based on if it is normal for people to let blood that day on a dangerous hour.

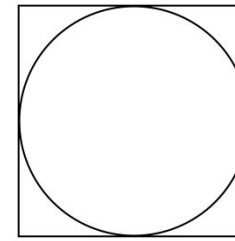
According to that explanation, Tuesday is a dangerous day because it *normally* has a dangerous hour in it, even if on this particular day it doesn't have that dangerous hour.

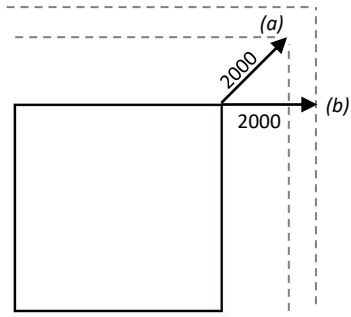
And the same with the other days. They are *not* dangerous because they are *normally* not dangerous.

*

נ"ו: - Abaya - Gaining on the Diagonal

The Gemara brings a Braisa that says that one who is "squaring a city" for the purposes of Techum, does so by first enclosing the city in a square (or rectangle, as the case may be), and then measuring out from there 2000 amos.





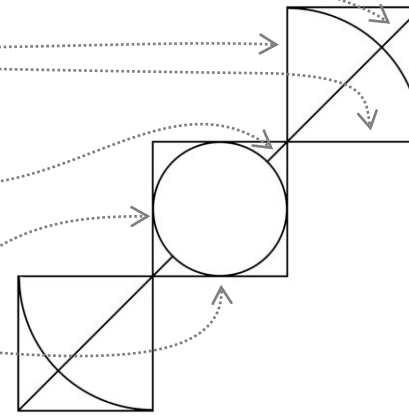
Then the Braisa qualifies that measurement by saying that one should not measure the Techum's corners by going out from the corners of the city's square (arrow "a") but rather they should be measured straight (parallel to the city, arrow "b") so that one will "gain on the diagonal".

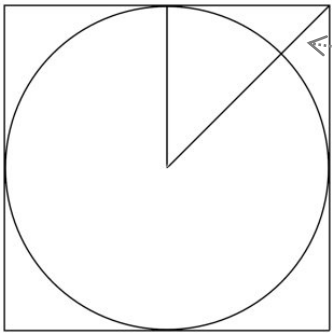
The Braisa says that with this method one will gain on the diagonal 400 amos in each corner of the city and he will gain 800 on the diagonal of the Techum for each corner.

The Techum gain of 800 is apparent from the fact that the corners are 2000 by 2000, and using the Gemara's ratio for diagonals as 1.4 of a square's side gives us 800 extra (2000 * 1.4 - 2000).

The city gain of 400 is not immediately apparent. Abaya says that we will find this in a situation where the city is 2000 by 2000.

How did Abaya know this?





The gain of the diagonal is 400 for each corner of the city.

That means that the total gain across the entire diagonal is 800 (400 *2).

The diagonal of the squared city, which is 1.4 times the city width, is 800 more than the city width.

If the city's width is "x" then the formula would be **1.4 * x = x + 800**.

▶ $1.4x = x + 800$

subtract 1x from both sides

⬇ $.4x = 800$

Divide both sides by .4

■ **$x = 2000$**

And there you have it.

נ"ו: - <u>Techum ÷ Migrash</u> = 4

The Gemara begins with a set of assumptions and reconfigures them as we go through the cheshbonos.

At certain points in the Gemara, the Gemara reverts to previous assumptions without informing us, thus making it confusing as to what type of cheshbonos the Gemara is making at any given point in time.

To simplify this, I will list the starting assumptions of the Gemara and then note when those assumptions change or revert back.

1. The city (and the Migrash) is square
2. The Migrash cheshbon <u>does not</u> include the city
3. The Techum cheshbon <u>does not</u> include the city

City:

Migrash:

Techum:

Migrash/Techum overlap:

The Braisa says that the Migrash is ¼ the Techum.

Whatever size you suggest for the city, the Migrash will be 2 (thousand) amos wider than that, since the Migrash is an additional 1 (thousand) in each direction.

Therefore, if "x" is the city's width then the Migrash is $x + 2$ wide. The area of the Migrash will be $(x + 2)^2$ minus the city's area (x^2).

The Techum is 4 wider than the city, since it is an additional 2 (thousand) amos in each direction from the city's edges.

Therefore, the Techum's width is $x + 4$, and the Techum's area is $(x + 4)^2$ minus the city's area (x^2).

Rava says that the city is 2 (thousand) amos wide.

The Gemara says that in that case the Migrash will be greater than 1/3 of the Techum.

City width = x (2000, which I am calling 2 for simplicity)

▶ Migrash ÷ Techum =

⬇ $((x + 2)^2 - x^2) \div ((x + 4)^2 - x^2) =$

⬇ $((2 + 2)^2 - 2^2) \div ((2 + 4)^2 - 2^2) =$

⬇ $(16 - 4) \div (36 - 4) =$

■ $12 \div 32 = \frac{3}{8}$

which is greater than $\frac{1}{3}$ $\left(\frac{3}{8} > \frac{1}{3} \text{ just like } \frac{9}{24} > \frac{8}{24}\right)$

So the Gemara adjusts assumption #3, to say that the Techum does include the city.

So now it's:

▶ $((x + 2)^2 - x^2) \div (x + 4)^2 =$

■ $12 \div 36 = \frac{1}{3}$

Which is no good still since we need it to be $\frac{1}{4}$

So the Gemara makes another adjustment. We change assumption #1 and make the city (and the Migrash[140]) into a circle.

[140] I don't know why the Migrash changes into a donut-shape (torus) when we change the city into a circle. Why isn't it always the same shape, square or circle?

The city, as a square, had an area of x^2. Now that it's a circle, we multiply that by $\frac{3}{4}$ (.75). That comes out to 3 "tiles", each 1000 by 1000 amah square (for a total of 3,000,000 square amos).

The Migrash, as a square, had an area of $(x + 2)^2 - x^2$. Now that it and the city within it are circles, its area is $.75(x + 2)^2 - .75x^2$, which comes out to 9.

So now, the Techum is 36 (as before, $(x + 4)^2$) and

$$9 \div 36 = \frac{1}{4}$$

Yay!

At this stage, when I cheshboned the Migrash, Yishai Rasowsky pointed out to me that I don't need to make it so complicated. He said that once we know how big the square Migrash is with the "hole" (the square city) in the middle, we can just multiply that amount by .75 to know how big it will be when it is round with the round hole (the city) in the middle.

He demonstrated this with the following formula:

If

Migrash = Total_Migrash_Area – City

Then it follows that

.75 * Migrash = .75 * Total_Migrash_Area - .75 * City

And one need not recalculate everything.

Assuming that Rava was working with the final adjustments that we made to our initial assumptions[141], let's work out how he may have arrived at his city width of 2 (thousand).

Let's write out algebraically what we know:

[141] That is, that the city (and Migrash) is round, the Migrash does not include the city, and the Techum **does** include the city.

▶ $T \div M = 4$

fill out the formula

⬇ $(x + 4)^2 \div (.75(x + 2)^2 - .75x^2) = 4$

expand the parentheses

⬇ $(x^2 + 8x + 16) \div (.75x^2 + 3x + 3 - .75x^2) = 4$

(+).75x^2 and -.75x^2 cancel each other out

⬇ $(x^2 + 8x + 16) \div (3x + 3) = 4$

move 3x + 3 to the other side and open parentheses

⬇ $x^2 + 8x + 16 = 12x + 12$

subtract 12x and 12 from each side

⬇ $x^2 - 4x + 4 = 0$

reverse FOIL

⬇ $(x - 2)^2 = 0$

square root both side (zero remains zero)

⬇ $x - 2 = 0$

add 2 to both sides

■ $\mathbf{x = 2}$

Great!

When the Gemara presents Abaya's suggested city size, it reverts to our original assumptions. That is, the Techum does <u>not</u> include the city, and the city (and the Migrash) is square.

(The Gemara, at the end, will make the city round again but it will keep the Techum definition as the initial assumption, that it does not include the city.

It could be that this is considered a benefit to this pshat, that it changes less of the Gemara's initial assumptions.)

Abaya says that the city is 1000 amos wide.

▶ Migrash ÷ Techum =

⬇ $((x + 2)^2 - x^2) \div ((x + 4)^2 - x^2) =$

⬇ $((1 + 2)^2 - 1^2) \div ((1 + 4)^2 - 1^2) =$

⬇ $(9 - 1) \div (25 - 1) =$

■ $8 \div 24 = \mathbf{\frac{1}{3}}$

To which the Gemara says is not the $\frac{1}{4}$ that we expected.

So the Gemara adjusts our assumption #1 and says that the city (and the Migrash) is round. So now it's:

▶ $.75((x + 2)^2 - x^2) \div ((x + 4)^2 - .75x^2) =$

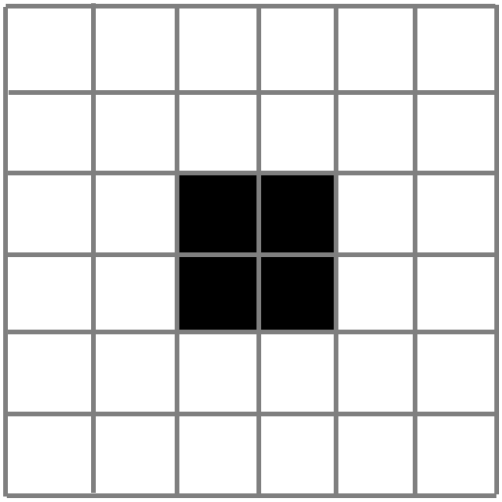
■ $6 \div (25 - .75) = \mathbf{\frac{1}{4.04\frac{1}{6}}}$

Which is very close to our target of one quarter[142]

When the Gemara moves on to Ravina's pshat, the Gemara seems to be reverting back to the point when Rava first gave his value of 2000 amos for the width of the city. At that stage, all the original assumptions of the Gemara are as initially stated, that the city is square, and that the Migrash and the Techum do not include the city in the size measurements.

Ravina says that the Migrash only includes the areas to the sides[143] of the city.

[142] See piece on Tosfos Abaya with the Gra, page 215.



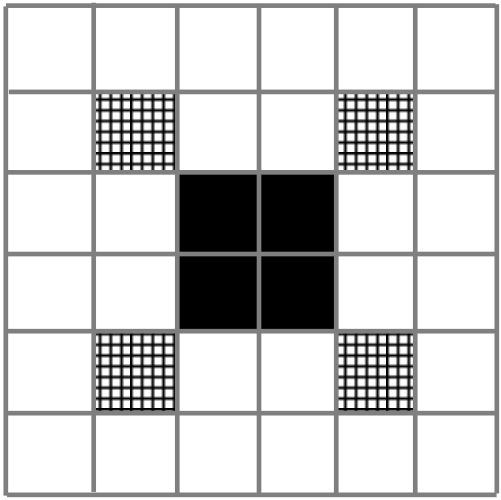
The city is 2 (2000) wide and there's four sides so there's 8 "tiles" for the Migrash (2 * 4).

The Techum is 32 ((city_width + 4)^2 – city_width^2), as before.

$$\text{Migrash} \div \text{Techum} = \mathbf{\frac{1}{4}}$$

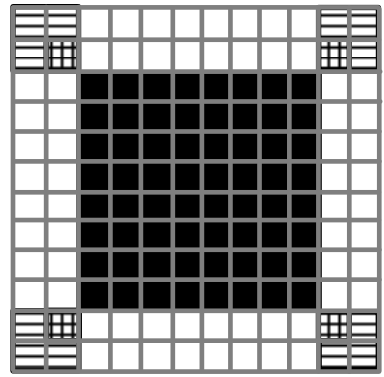
(Continuing with the same initial assumptions…) **Rav Ashi** says that we cheshbon only the corners for the Migrash and the Techum. That means that the corners of the Migrash (1 for each corner = 4 tiles) is $\frac{1}{4}$ of the corners of the Techum (4 for each corner = 16 tiles). And again:

$$\text{Migrash} \div \text{Techum} = \mathbf{\frac{1}{4}}$$



[143] What the Gemara calls the "Techum", not to be confused with the Techum we've been discussing until now which is the 2000 amah thick border that runs around the city.

Rashi at first says that Rav Ashi is using a city of 2 (thousand) by 2 (thousand), but in the next part he writes that whatever are the city's dimensions, the Migrash corners will always be $\frac{1}{4}$ of the Techum's corners.



At first glance, these explanations of Ravina and Rav Ashi seem strange. It would seem as though we are picking sections of the Migrash out to include in our calculations in order to make it fit. Perhaps that is what is bothering Tosfos when he explains that according to these amorayim, this <u>is</u> the shape of the Migrash.

According to Ravina, the Migrash only consisted of the side "flaps" of the city and there was no Migrash at the corners. (The Migrash was an area of uncultivated land. According to Ravina, the corner tiles were cultivated.)

And according to Rav Ashi, the Migrash consisted of only the corners, and the side "flaps" were cultivated.

One would wonder though, with what justification is Rav Ashi only including the corners of the Techum in the calculation? Even he agrees that the Techum included areas besides the corners!

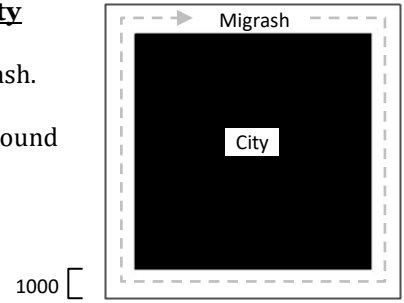
It could be that since the Migrash only consisted of the corners, it stands to reason that when we compare it to the Techum, we only compare the section of the Techum that is comparable to the Migrash, and that would be only the corner sections.

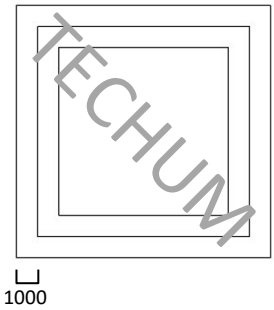
*

ב"ו: - Tosfos Mishkachas - 8000² City

A city has a band around it called the Migrash.

It is 1000 amos thick and it is a square around the city.





The Techum is a square whose border is an additional 1000 amos from the city's edges.

The Techum includes the Migrash and, at this stage of the Gemara, the city as well.

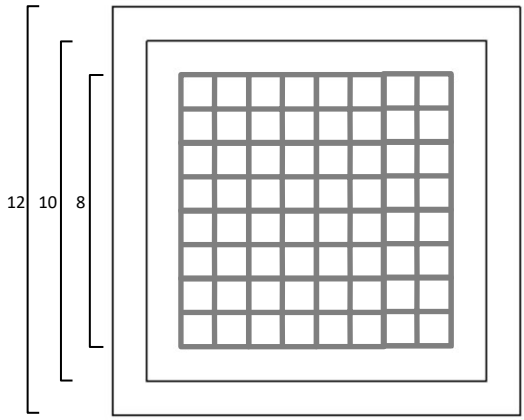
The Gemara is trying to find a city size that would make the area of the Migrash exactly $\frac{1}{4}$ of the area of the Techum. (Put another way, that the Migrash ÷ the Techum = ¼, or Techum ÷ Migrash = 4)

Tosfos suggests a city that is 8000 by 8000. The standard method used in the Gemara and Rishonim is to use single "tiles" that are 1000 by 1000. Therefore, this city is described as being "8 by 8".

Migrash = 36 $(10^2 - 8^2)$

Techum = 144 (12^2)

$\frac{\text{Techum}}{\text{Migrash}} = \mathbf{4}!$



Great.

Now, how did Tosfos know to use this number 8 (without having just guessed and checked)?

Let's write out algebraically what we know:

City Width = x

City Area = x^2

Migrash Width = City Width + 2

Migrash Area = Migrash Width^2 - City Area

Techum Width = City Width + 4

Techum Area = Techum Width^2

▶ Techum Area ÷ Migrash Area = 4

fill out the formula

⬇ $(x + 4)^2 \div ((x + 2)^2 - x^2) = 4$

expand parentheses using FOIL

⬇ $(x^2 + 8x + 16) \div ((x^2 + 4x + 4) - x^2) = 4$

on the right side, (+)x^2 and -x^2 cancel each other out

⬇ $(x^2 + 8x + 16) \div (4x + 4) = 4$

multiply both sides by (4x + 4)

⬇ $x^2 + 8x + 16 = 4(4x + 4)$

simplify parentheses

⬇ $x^2 + 8x + 16 = 16x + 16$

subtract 8x and 16 from both sides

⬇ $x^2 = 8x$

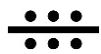
expand x^2

⬇ $xx = 8x$

one x on each side cancel each other out[144]

■ $\mathbf{x = 8}$

Amazing.



[144] Although, Yishai Rasowsky pointed out that x could also be "0" in this case.

*

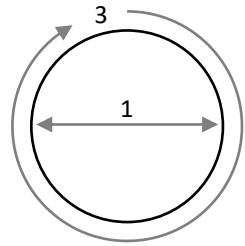
ב"ו: - <u>Tosfos Kamma - Wrapping Strings to Find Area</u>

The Gemara says that 1/4 of a square's area is how much larger a square is than a circle the same width. That means that the circle's area is 3/4 of the square's.

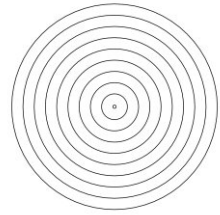
Tosfos demonstrates how that is derived assuming that the circle's circumference is 3 times its diameter.

Tosfos makes the following thought experiment:

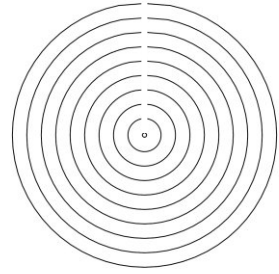
Take a circle that is 1 wide and 3 around:



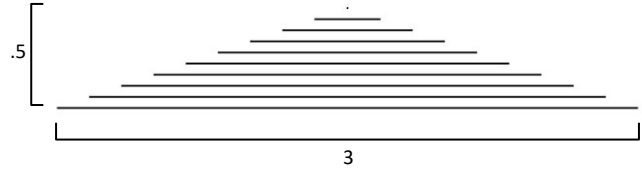
Fill it with concentric circles (what he calls "strings", or “ropes”):



Now cut all the “strings” from the top edge of the circle until the center of the circle.



Open the strings up until they form a wide triangle.



The base of the triangle is 3, since that was the circumference of the original circle, the largest string.

The height of the triangle is .5, since the original circle was 1 high and we cut it open from the top down to the center, leaving the bottom half uncut.

The triangle's area can now be calculated using the formula for triangles

▶ $a = .5 * w * h$

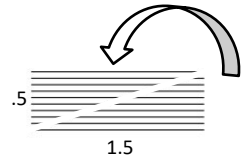
⬇ $a = .5 * 3 * .5$

■ **$a = .75$**

…which is thus the area of the original circle, ¾ of a 1 x 1 square.

Tosfos calculates the triangle using a more visual method where we cut the triangle down the middle…

…and then flip one side over and fit it into the empty space of the other triangle, and now we have a rectangle whose width is 1.5 (half the big triangle) and whose height is .5.



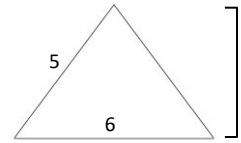
$1.5 * .5 = \mathbf{.75}$

(This is essentially the same thing as the formula method, just in a visual format.)

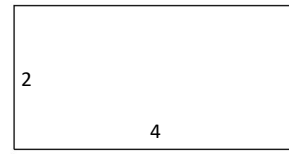
For the sake of clarity, I tried this entire "string theory" with other shapes to see if I could find their areas.

I started with two examples:

- A triangle with a base of 6, a height of 4, and legs (the slant) of 5



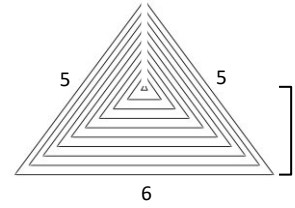
- and a rectangle that is 4 wide by 2 high.



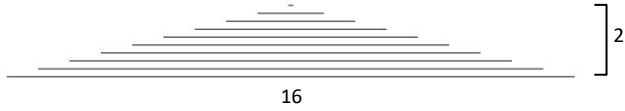
Let's fill these shapes with concentric rings, cut, open, flip, and calculate and see if we can get to the correct areas.

Let's start with the 4-5-6 triangle.

Cutting it down to the middle leaves a height of **2** for the final triangle.



Opening it up gives us a base width of **16** $(5 + 6 + 5)$ for the final triangle.

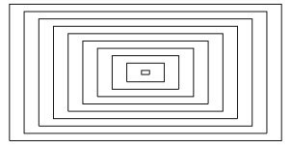


The area of this new triangle is **16** $(.5 * 16 * 2)$.

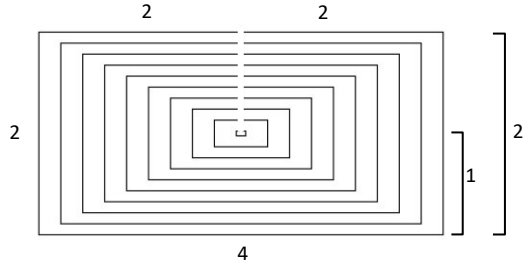
The original triangle was **12** $(.5 * 6 * 4)$.

Why aren't these numbers the same?

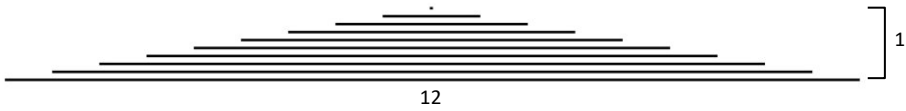
Let's try it with the rectangle.



Cutting it down the middle leaves a height of **1** for the final triangle.



Opening it up gives us a base width of **12** (2 + 2 + 4 + 2 + 2, the total length of the outermost string) for the final triangle.



The area of the final triangle is **6** (.5 * 12 * 1).

The original rectangle was **8** (4 * 2).

Again, why aren't these the numbers the same?

To paraphrase Tosfos[145] commenting on Rashi's explanation of this topic: "If it doesn't work in some cases then you can't use that method to prove other cases, even if it happens to work out sometimes."

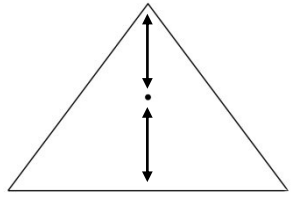
How then can Tosfos give us a method that doesn't always work?

[145] סוכה ח'.

Yishai Rasowsky provided the following answer.

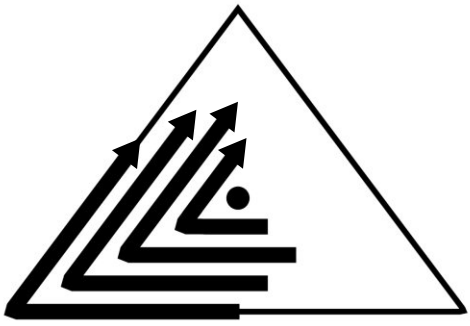
Let us look at our examples more closely.

We had assumed that we cut the shapes down the middle, halfway, like we did for the circle.



The problem with that is that leaves inconsistent area-widths for the "strings" to travel through.

Let's explain.

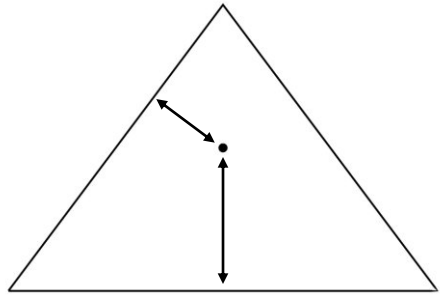


As the strings pass between each side and the center, there must be the same number of strings and they must be the same thickness.

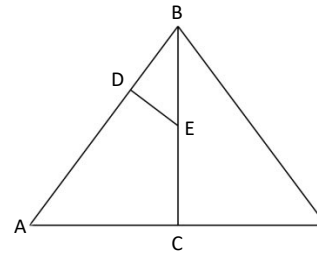
In other words, the "center" to which you will cut and around which the strings will wrap, must be positioned an equal distance from the sides of the whole shape.

In the example we tried with the triangle, the point which we called "the center" is not equidistant from all the sides.

You can see it in the picture but let's prove it by calculating the distances.



$\angle ABC$ is $\sin^{-1}\left(\frac{\overline{AC}}{\overline{AB}}\right)$ see Appendix J



$\overline{ED} = \sin(\angle DBE) * \overline{BE}$

And since $\angle ABC$ is the same as $\angle DBE$, we can put $\angle ABC$'s equation into $\overline{ED}$'s and we get:

▶ $\overline{ED} = \sin\left(\sin^{-1}\left(\frac{\overline{AC}}{\overline{AB}}\right)\right) * \overline{BE}$

sin(sin^{-1}… cancel each other out

⬇ $\overline{ED} = \left(\frac{\overline{AC}}{\overline{AB}}\right) * \overline{BE}$

fill in what we know

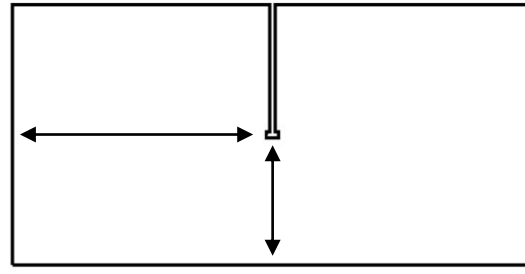
⬇ $\overline{ED} = \left(\frac{3}{5}\right) * 2$

■ $\overline{ED} = \mathbf{1\frac{1}{5}}$

…which is less than $\overline{EC}$, which is 2.

The same thing can be shown with the rectangle example.

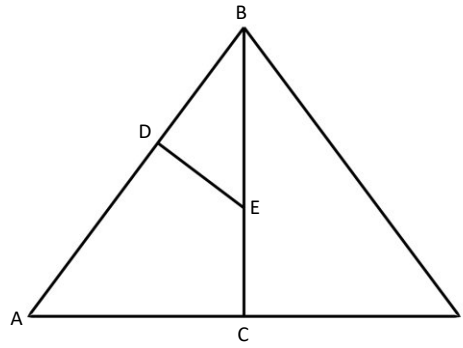
Besides being clear in the picture, we know the distances between the center and the bottom (half the rectangles height = 1) is not the same as the center to the side (half the rectangle's width = 2).



To find the center for our purposes, we need to find the point that is the same distance to each side.

Let's go back to our 4-5-6 triangle.

We want to make lines $\overline{EC}$ and $\overline{ED}$ to be the same.



Our givens are:

$\overline{ED} = \overline{EC}$

$\overline{AB} = 5$

$\overline{BC} = 4$

$\overline{AC} = 3$ (half the triangle's base)

Unlike by the last calculation where we knew the values of EB and EC, here we don't know them directly, but we do know that together they equal 4.

So we can go like this:

▶ $\overline{ED} = \sin(\angle DBE) * \overline{BE}$

replace BE with (4 - EC)

⬇ $\overline{ED} = \sin(\angle DBE) * (4 - \overline{EC})$

EC is the same as ED

⬇ $\overline{ED} = \sin(\angle DBE) * (4 - \overline{ED})$

sine of DBE = sine of ABC = AC ÷ AB

⬇ $\overline{ED} = .6 * (4 - \overline{ED})$

open the parentheses

⬇ $\overline{ED} = 2.4 - .6\overline{ED}$

add .6ED to each side

⬇ $1.6\overline{ED} = 2.4$

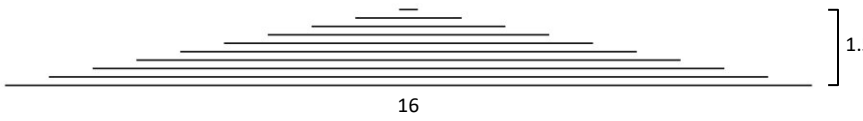
divide both sides by 1.6

■ $\overline{ED} = \mathbf{1.5 = \overline{EC}}$

Excellent!

נ"ו:

Now that we know how much to cut this triangle, let's open it up and see what happens:



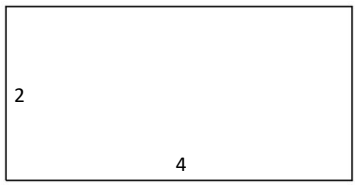
This new triangle is 16 wide and 1.5 high, which is

$$.5 * 16 * 1.5 = \mathbf{12}$$

…which is what we expected in the beginning (since our original triangle was 6 wide by 4 high, which is $.5 * 6 * 4 = \mathbf{12}$).

Great!

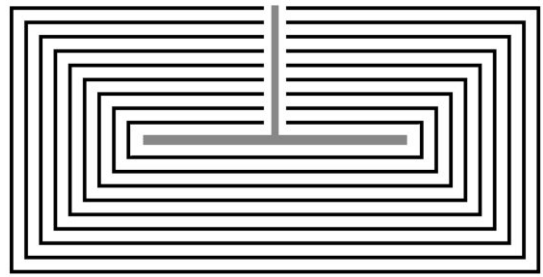
Now, let's look again at the 2 by 4 rectangle example.



Actually, if we look at this, we'll see there's no single straight cut we could make that would yield an evenly spaced path for the strings to travel through.[146]

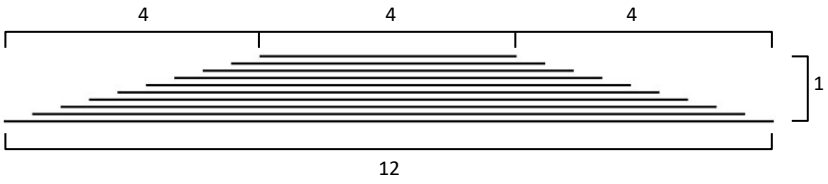
What we *could* do is to cut a "T" shape (upside down) into the rectangle, like in this picture on the right:

This way, all the strings will have the same thickness all the way around.

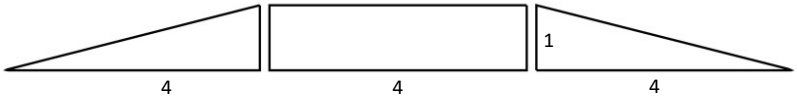


[146] For proof, see Appendix V

Now when we open that up, we got something else over here. This is a trapezoid.



If we break it into its discrete components, there's a triangle on the left, a central rectangle, and a triangle on the right.



They all have a width of 4 and a height of 1, so…

the triangles are each .5 * 4 * 1 = 2

…times two triangles = **4**

…plus

the rectangle of 4 x 1 = **4**

…equals **8**, which is what we expected in the beginning (4 x 2)!

Amazing!

Way to go!

*

נ"ו: - <u>Tosfos Abaya – 1000 Amah City</u>

At this stage of the Gemara, we are working with the following givens:

- City is round
- Migrash is 1000 amah thick border around the city, like a donut (does not include the city).
- Techum is a square that is 2000 amos larger than the city in each direction (east, west, north, and south). The Techum <u>does</u> include the Migrash but <u>does not</u> include the city.

(As is our style, we're using "1" for 1000, "2" for 2000, etc.)

- City_Width = x
- City_Area = $.75x^2$
- Migrash_Width = x + 2
- Migrash_Area = $.75(x + 2)^2$ - City_Area
- Techum_Width = x + 4
- Techum_Area = $(x + 4)^2$ - City_Area

- Techum area is 4 times bigger than the Migrash area $(T \div M = 4)$

Abaya says the city is 1000 amos wide.

Tosfos makes the cheshbon and finds that it is inexact.

Let's fill out the formula and cheshbon it out.

▶ $T \div M =$

fill out the formula

⬇ $((x + 4)^2 - .75x^2) \div (.75(x + 2)^2 - .75x^2) =$

⬇ $((1 + 4)^2 - .75*1^2) \div (.75(1 + 2)^2 - .75*1^2) =$

⬇ $(25 - .75) \div (.75 * 9 - .75) =$

⬇ $24.25 \div (6.75 - .75) =$

■ $24.25 \div 6 = \mathbf{4\frac{1}{24}}$ or $4.041\overline{6}$

Seeing how the cheshbon is inexact, Tosfos suggests that the Techum needs to be measured <u>as though</u> the city was a square, while keeping the city round as far as the Migrash is concerned.

Techum: Migrash:

So now the Techum $= (x + 4)^2 - \mathbf{x^2}$

And the Migrash $= .75(x + 2)^2 - \mathbf{.75x^2}$

And now $((x + 4)^2 - \mathbf{x^2}) \div (.75(x + 2)^2 - \mathbf{.75x^2}) = \mathbf{4}$ exactly.

▶ $((x + 4)^2 - x^2) \div (.75(x + 2)^2 - .75x^2) =$

⬇ $((1 + 4)^2 - 1^2) \div (.75(1 + 2)^2 - .75 * 1^2) =$

⬇ $(25 - 1) \div (.75 * 9 - .75 * 1) =$

⬇ $24 \div (6.75 - .75) =$

■ $24 \div 6 = \mathbf{4}$

The Gra[147] finds this explanation דוחק.

I assume the "דוחקness" is that we are measuring the same city in two different ways in the same cheshbon.

Or perhaps the דוחקness is that this explanation is not according to the simple meaning of the Gemara.

Whichever way it is, the Gra says that the Gemara meant to say that the city is **1 and $\frac{1}{4}$ wide**, and the Gemara was not concerned to give the exact number.

▶ $((x + 4)^2 - .75x^2) \div (.75(x + 2)^2 - .75x^2) =$

⬇ $((1.25 + 4)^2 - .75 * 1.25^2) \div (.75(1.25 + 2)^2 - .75 * 1.25^2) =$

⬇ $((5.25)^2 - .75 * 1.5625) \div (.75(3.25)^2 - .75 * 1.5625) =$

⬇ $(27.5625 - 1.171875) \div (.75 * 10.5625 - 1.171875) =$

⬇ $26.390625 \div (7.921875 - 1.171875) =$

■ $26.390625 \div 6.75 = \mathbf{3\frac{131}{144}}$ **or** $\mathbf{3.9097\overline{2}}$

[147] In the Hagaos HaGra on the side of the Gemara

Hmm. This number is also not exactly 4.

What city size would be exactly 4?

$((x + 4)^2 - .75x^2) \div (.75(x + 2)^2 - .75x^2) = 4$

expand the parentheses

$(x^2 + 8x + 16 - .75x^2) \div (.75(x^2 + 4x + 4) - .75x^2) = 4$

subtract $.75x^2$ from $(1)x^2$ on the left side, and expand the inner parentheses on the right

$(.25x^2 + 8x + 16) \div (.75x^2 + 3x + 3 - .75x^2) = 4$

(+)$.75x^2$ and -$.75x^2$ cancel each other out

$(.25x^2 + 8x + 16) \div (3x + 3) = 4$

move (3x + 3) to the right side of the equal sign

$.25x^2 + 8x + 16 = 4(3x + 3)$

expand the right side

$.25x^2 + 8x + 16 = 12x + 12$

subtract 12x and 12 from both sides

$.25x^2 - 4x + 4 = 0$

multiply everything by 4 (to bring $.25x^2$ up to $(1)x^2$)

$x^2 - 16x + 16 = 0$

subtract 16 from both sides

$x^2 - 16x = -16$

use the 16 from 16x, cut in half, square, then add to both sides[148]

$x^2 - 16x + 64 = 48$

simplify left side[149]

$(x - 8)^2 = 48$

square root both sides

$x - 8 = \sqrt{48}$

add 8 to each side

$x = \sqrt{48} + 8$

calculate square root of 48, which can be a positive or negative

$\mathbf{x = \pm 6.92820323... + 8}$

write out both possible values of x

$\mathbf{x = 1.07179677...}$ or $\mathbf{x = 14.92820323}$

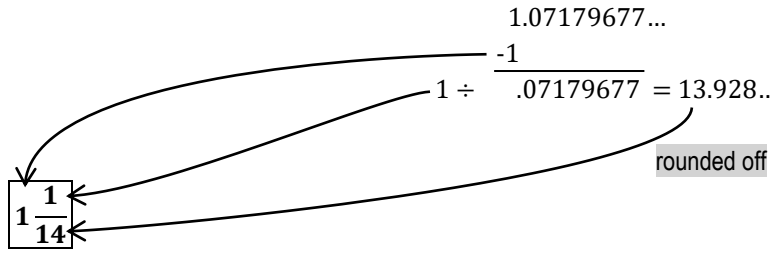
[148] See Quadratic Equations in Appendix F

[149] Reverse FOIL, see Appendix E

Tosfos[150] earlier on the דף already told us that a city that is 8 wide is too wide, so we can count out the 14.92… option.

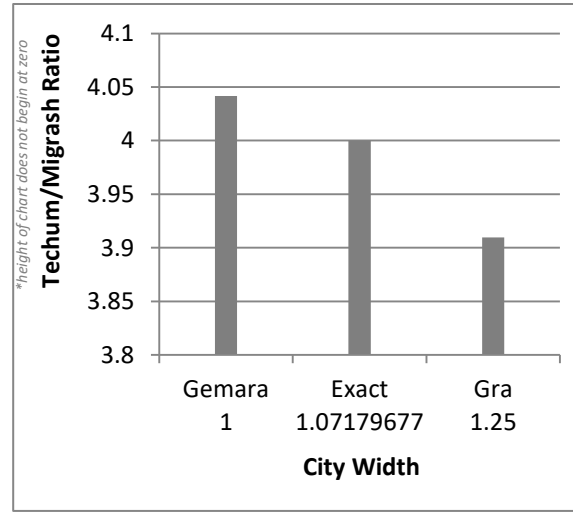
The exact city width would be 1.07179677…

To turn that into a fraction:



The fact that the Gra's number is inexact doesn't bother me so much. After all, he's explaining that the Gemara is being inexact.

What bothers me more is that if we compare the various values that we have for the city width, we will find that the Gra's number is *less* accurate than the Gemara's!



[150] ד"ה משכחת

As a fixed difference:

Gemara: .07179… (Exact size minus Gemara's size)

Gra: -.1782… (Exact size minus Gra's size)

As percentage:

Gemara: %107.17… (Exact ÷ Gemara) (about **%7** more)

Gra: %85.74… (Exact ÷ Gra) (about **%14** less)

Although it goes against the simple reading of the Gemara, I thought that perhaps the Gra was using the measured value of pi (3.14159…), but we shall see that that doesn't help, and if fact might be worse.

Let's do the Gemara's city width (x) of 1, the Gra's city width of 1.25, and then see if we could calculate the exact value, using pi (3.14...).

Gemara City Width of "1", using true value of pi for the circles

▶ $((x + 4)^2 - \frac{\pi}{4}x^2) \div (\frac{\pi}{4}(x + 2)^2 - \frac{\pi}{4}x^2) =$

⬇ $((1 + 4)^2 - \frac{\pi}{4}1^2) \div (\frac{\pi}{4}(1 + 2)^2 - \frac{\pi}{4}1^2) =$

⬇ $(25 - \frac{\pi}{4}) \div (\frac{\pi}{4}9 - \frac{\pi}{4}) =$

⬇ (25 – .785398163...) ÷ (.785398163... * 9 – .785398163...) =

⬇ 24.21460183... ÷ (7.06858347… – .785398163...) =

■ 24.21460183... ÷ 6.283185307… = **3.85387357…**

Gra's City Width of "1.25", using true value of pi for the circles

▶ $((x+4)^2 - \frac{\pi}{4}x^2) \div (\frac{\pi}{4}(x+2)^2 - \frac{\pi}{4}x^2) =$

⬇ $((1.25+4)^2 - \frac{\pi}{4}1.25^2) \div (\frac{\pi}{4}(1.25+2)^2 - \frac{\pi}{4}1.25^2) =$

⬇ $(27.5625 - \frac{\pi}{4}1.5625) \div (\frac{\pi}{4}10.5625 - \frac{\pi}{4}1.5625) =$

⬇ $(27.5625 - 1.22718463...) \div (8.2957681... - 1.22718463...) =$

■ $26.33531536... \div 7.06858347... =$ **3.72568499...**

...which is farther from 4 than the Gemara's value (which got to 3.85...).

Exact City Width , using true value of pi for the circles

▶ $((x+4)^2 - \frac{\pi}{4}x^2) \div (\frac{\pi}{4}(x+2)^2 - \frac{\pi}{4}x^2) = 4$

expand parentheses

⬇ $((x^2+8x+16 - \frac{\pi}{4}x^2) \div (\frac{\pi}{4}(x^2+4x+4) - \frac{\pi}{4}x^2) = 4$

expand inner parentheses on right side

⬇ $((x^2+8x+16 - \frac{\pi}{4}x^2) \div (\frac{\pi}{4}x^2 + \frac{\pi}{4}4x + \frac{\pi}{4}4 - \frac{\pi}{4}x^2) = 4$

(+)pi/4*x² and –pi/4*x² on right side cancel each other out

⬇ $((x^2+8x+16 - \frac{\pi}{4}x^2) \div (\frac{\pi}{4}4x + \frac{\pi}{4}4) = 4$

(4/4)x² – pi/4*x² becomes (4-pi)/4*x² on left, pi/4*4 becomes pi in middle parentheses

⬇ $\left(\frac{4-\pi}{4}x^2 + 8x + 16\right) \div (\pi x + \pi) = 4$

fill in pi

⬇ $(.2146...x^2 + 8x + 16) \div (3.14...x + 3.14...) = 4$

multiply both side by (3.14x+3.1)

⬇ $.2146...x^2 + 8x + 16 = 4 * (3.14...x + 3.14...)$

expand right parentheses

⬇ $.2146...x^2 + 8x + 16 = 12.566...x + 12.566...$

subtract 12.56...x and 12.56... from both sides

⬇ $.2146...x^2 - 4.566...x + 3.433... = 0$

multiply everything by 1/.2146 to bring .2146x² up to (1)x²

⬇ $x^2 - 21.278...x + 16 = 0$

add the square of half of -21.278 to both sides (see Quadratic Equations in Appendix F)

$x^2 - 21.278\ldots x + 113.19 + 16 = 113.19\ldots$

reverse FOIL on left side (see Appendix E)

$(x - 10.639\ldots)^2 + 16 = 113.19\ldots$

subtract 16 from both sides

$(x - 10.639\ldots)^2 = 97.19\ldots$

square root both sides

$x - 10.639\ldots = \sqrt{97.19\ldots}$

add 10.639… to both sides

$x = \sqrt{97.19\ldots} + 10.639\ldots$

calculate square root (could be positive or negative)

$\mathbf{x = \pm 9.858\ldots + 10.639\ldots}$

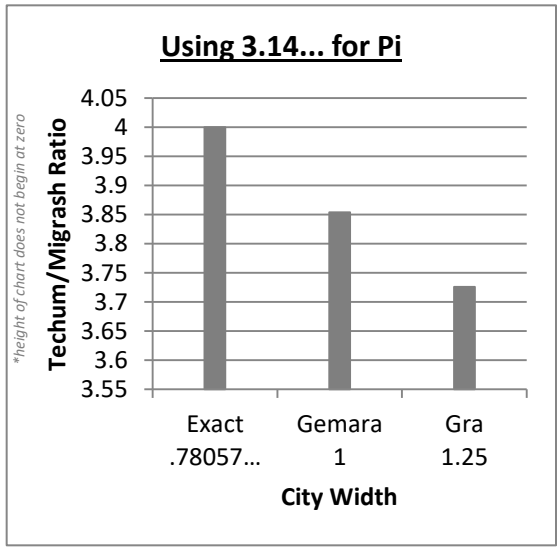
add together

$x = .780572865\ldots$ (or **$x = 20.49\ldots$** which we said[151] is too big)

Using pi not only doesn't help, but it makes things worse. Now the Gra's city width is not only less accurate than the Gemara's, but it's going in the wrong direction.

Besides, Tosfos's approach assumes 3 for pi. If Tosfos was using 3.14 then the Techum will be 3.819… times bigger than the Migrash.

Work it out.



$((x + 4)^2 - x^2) \div (pi/4(x + 2)^2 - pi/4x^2)$

[151] on page 219

Or, I thought, perhaps they're using a different value for pi.

Here again is the formula for Techum ÷ Migrash:

$$\left((x\ +\ 4)2 - \frac{\pi}{4}x2\right) \div \left(\frac{\pi}{4}\ (x\ +\ 2)2\ -\ \frac{\pi}{4}x2\right) = 4$$

Which, rearranged, becomes:

$$\pi = \frac{4(x+4)^2}{x^2 + 16x + 16}$$

Using the above formula and plugging in the various city widths we've been discussing will produce various values for Pi.

	City Width	Value of Pi
	.78057…	3.14159265…
Gemara	1	$\frac{1}{.33}$ or $3.\overline{03}$
	1.07179 …	3
Gra	1.25	$2\frac{562}{601}$ or 2.9351…

We can rearrange the above formula to work backwards; to show us how wide the city needs to be, based on whatever value we have for pi.

$$x = \frac{\mathbf{2(4\pi-8)-4*\sqrt{\pi(3\pi-8)}}}{\mathbf{4-\pi}}$$

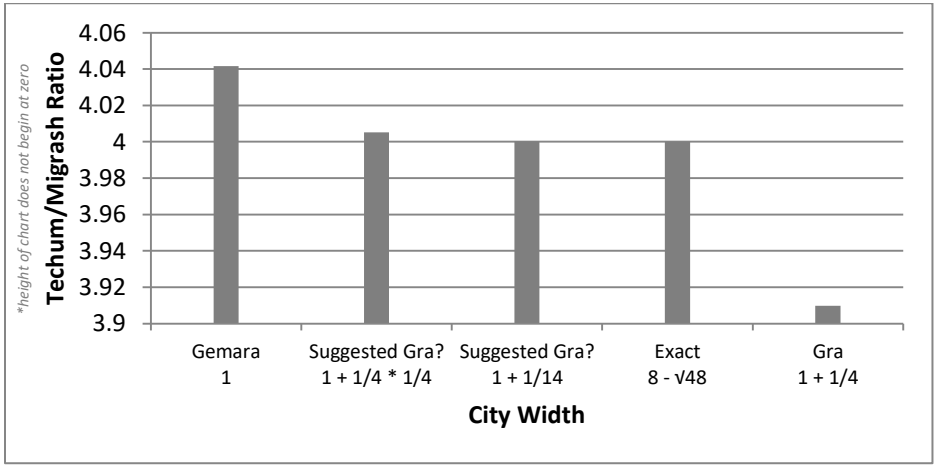
At the end of the day, I doubt very much that we are using different values for pi. The plain and simple reading of the Gemara is that we are using "3" for pi.

We will have to find some other explanation for the Gra.

Earlier[152], we showed that the exact city width should be 1.071…, which we rounded off and simplified into $1\frac{1}{14}$.

I would be really happy if someone told me that there was a טעות סופר in the Gra, and what he had really written was "One and a fourteenth" and it got changed into "One and a fourth", like this: אלף וא' מי"ד[153] and that became אלף וא' מד'.

Or even if he had written "One and a fourth of a fourth", which although it's a strange way to write it[154], and is not as exacting as we could be given that degree of exactness[155], at least it would be more exact than the Gemara, which is what the Gra seems to be doing.



[152] page 218
[153] In the same style he wrote on כ"ג:
[154] We do find in ש"ס (for example, סוטה ה'.) the term "an eighth of an eighth"
[155] In other words, if you're going to insist on a certain degree of exactness then you should at least be as correct as you can be to that degree.
For example, if I ask you where you live and you tell me the street but not the house number, that's OK. But if you do tell me the house number but you give me the wrong number, that's not OK.
I wish there was a term for this. Maybe "Overly exacting to the point of error"

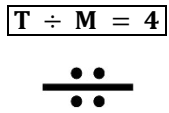
The נחל הערבים has a מחודש way of explaining the Gra[156].

The Gra wrote that the city is 1.25 wide.

That makes the Migrash 3.25 wide, and the Techum 5.25 wide.

The נחל הערבים calculates the areas as follows:

- City = 1
- Migrash = 6.3125
- Techum = 25.25



Let's go through his calculations one step at a time.

<u>City = 1</u>

The נחל הערבים writes that a round city that is 1.25 wide will have an area of 1.

I don't see how this comes out. If the city is 1.25 wide then a square around it would be 1.25^2, which is 1.5625.

To make it a circle in the Gemara's method, we would multiply that number by $\frac{3}{4}$, which would get us to 1.171875.

(Or, using $\frac{\pi}{4}$ it would come out to 1.22718…)

How did we get to 1?

Let's leave this for now and we'll come back to it later.

For now, let's just take for granted that the city area is 1.

[156] As understood by the יד בנימין, which I believe is the intent of the נחל הערבים.

Migrash = 6.3125

He writes "[The Migrash,] as a square of 3.25 * 3.25 comes out to 9.75".

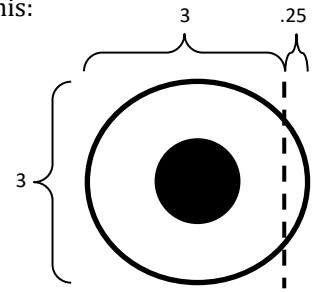
The יד בנימין explains the cheshbon like this:

As a square, the Migrash is:

3 by 3 = **9**

3 by .25 = **.75**

Together that's **9.75**



For some reason, we are only adding in the .25 along one direction to make that side 3.25. The other direction is being calculated as a plain 3.

Then, to circlify the Migrash, we multiply it by .75 and we get 7.3125.

Subtract the city of 1 and we arrive at the final **6.3125**.

Techum = 25.25

"5.25 by 5.25 comes out to 26.25"

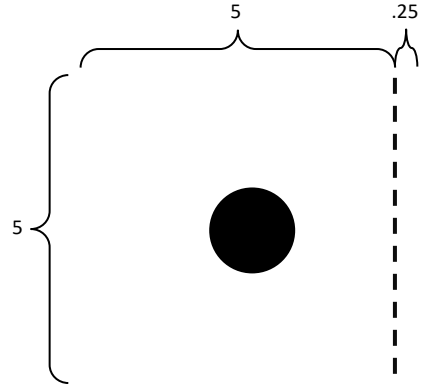
The יד בנימין explains this the same way as by the Migrash:

5 by 5 = **25**

5 by .25 = **1.25**

Together that's **26.25**

Subtract the city of 1 and we arrive at **25.25**.

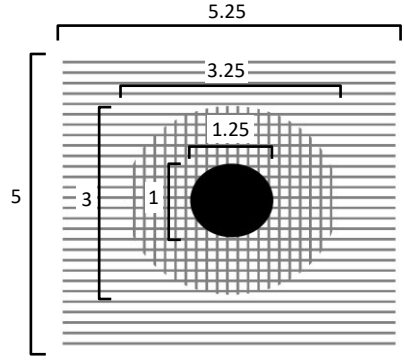


From the way the נחל הערבים makes the calculations, it sounds like the city is not the same width one way as it is the other.

It sounds almost as if the city was 1 by 1.25.

If so, that would explain why we calculate the overall sizes of the Migrash and the Techum the way we did.

If the city was indeed 1 by 1.25 then the Migrash would be 3 by 3.25, and the Techum would be 5 by 5.25.



However, it is difficult to explain that this was the intent of the נחל הערבים since he writes explicitly that the Migrash is 3.25 by 3.25, and that the Techum is 5.25 by 5.25.

Despite that, I would like to try to continue in this vein.

OK, so let's assume the city is 1 by 1.25. Does that help us with regards to the measurement of the city itself?

As a square, that would have an area of 1.25 (1 * 1.25).

.75 of that would be .9375. Hmm, now the city is too small.

What about with pi?

$\frac{\pi}{4} * 1.25 =$ **.981747704...**

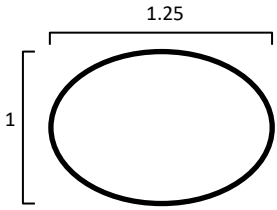
That's pretty close to 1.

(If pi were 3.2 then $\frac{\pi}{4} * 1.25$ would be exactly 1.)

But I think it's the wrong direction to go in, to say we're dealing with a value of pi other than 3, especially since the Migrash cheshbon is working with pi as 3.

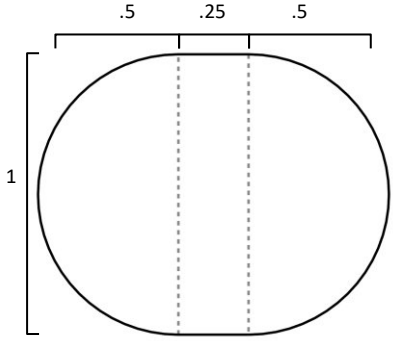
But perhaps we're assuming too much about the city's shape?

The way we've been cheshboning it was to make a square 1.25 by 1, and then to multiply it by .75, creating an ellipse (oval) shape:



*picture not to scale

Maybe the city's shape is a rounded rectangle, like a pill:



If so then the area of the city will be exactly 1!

The sides of the city are two halves of a 1 by 1 circle. 1 by 1, as a square, is 1. Multiplied by $\frac{3}{4}$ comes out to **.75**.

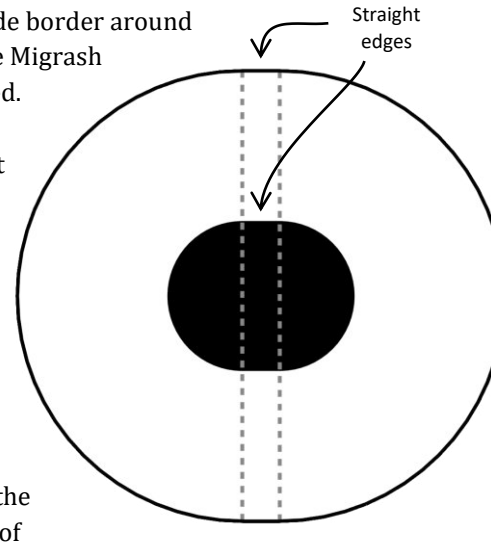
The rectangular middle strip is 1 by .25, which is **.25**.

Added together, that's **1**.

Great!

This would all be fine except now that we've changed the city shape it would seem that we would have to change the shape of the Migrash as well.

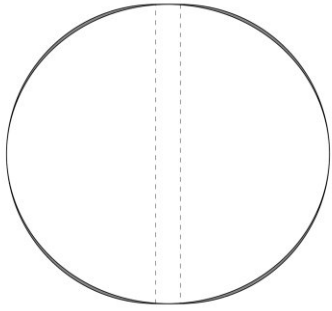
If the Migrash is a 1000-amah-wide border around the city then there are parts of the Migrash that will be straight and not curved. Since the city has some straight edges then when you measure out 1000 amos for the Migrash, the Migrash will have straight edges there as well.



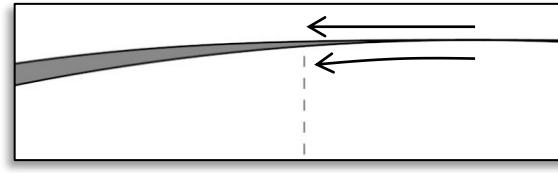
This change in shape will affect the total area of the Migrash.

In the picture below, you can see the difference between the two ways of measuring the Migrash.

The slight grey bulges by the corners show how much bigger the Migrash will be with a pill-shaped city within it, as opposed to an oval-shaped city within it.



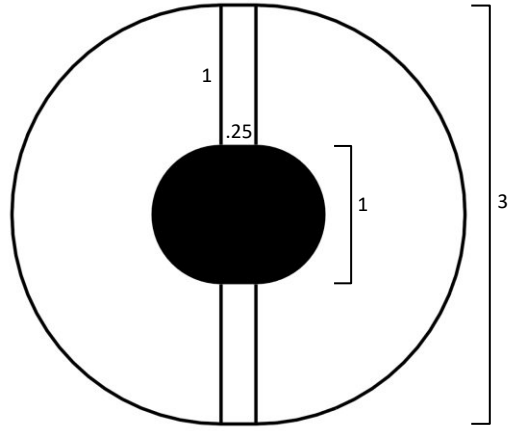
In this zoomed-in section, you might be able to see that the "pill-Migrash" is straight for a bit before it begins to curve. The "oval-Migrash" begins to curve immediately.



To calculate this new Migrash size, we can measure the two round extremities as two halves of a donut shape that is 3 (thousand) wide and across, with a 1x1 round hole in the middle.

That's:

$$.75(3^2) - .75(1^2)$$



The rectangular sections above and below the city are each .25 x 1, so for both of them that's:

$$.25 * 1 * 2$$

Altogether, this Migrash is

$$M = .75(3^2) - .75(1^2) + .25 * 1 * 2$$

...which comes out to **6.5**.

...which is a tad bit bigger than a quarter of the Techum.

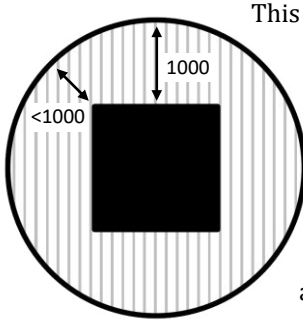
The Techum we calculated as being 25.25, and a quarter of that is 6.3125.

You might ask me, why do we need to reshape the Migrash? Let's just leave it as though it had an oval shaped city in the middle and זהו!

A proof that the Migrash should be measured in this way (if indeed the city is a "pill" shape) is the Gemara on דף נ"ז.

The Gemara there is asking on a certain stage of the שקלא וטריא where we are assuming that the city is a square inside a round Migrash.

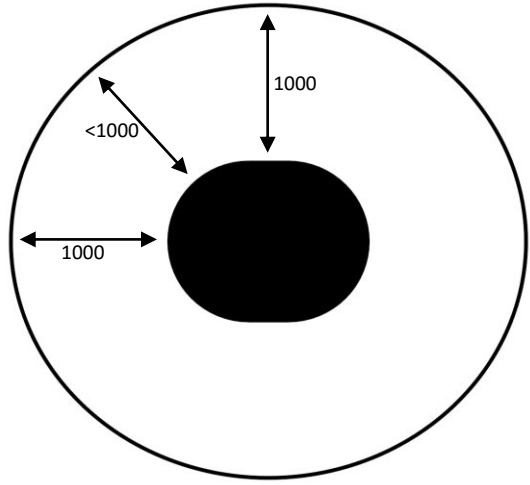
The Gemara asks that if that is so then it comes out that there are sections of the Migrash that are not 1000 amos wide!



This is what the Gemara calls "מורשא דקרנתא", the "bulge of the corner", meaning the corners of the square city bulge out into the Migrash, making those parts of the Migrash's thickness less than the parts that are by the flat edge of the city.

The Gemara's question takes for granted that the Migrash's outer border must be at least 1000 amos away from the city's border, all the way around.

If so, then by us it will be a problem that there are sections of the Migrash that are not a full 1000 amos from the city's border.

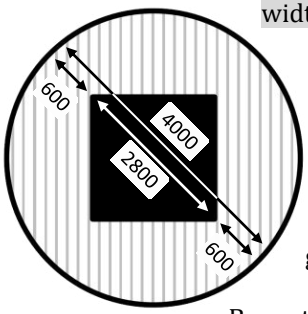


The narrowest point of the Migrash comes out to be about 970[157] amos.

[157] To see how I got that number, see Appendix T.

Now, we *could* differentiate between the Gemara on נ"ר and us by saying that over there, the narrowest part of the Migrash is much less than by us.

Over there, the city is 2000 wide, according to Rava. As a square, it would have a diagonal of 2800 (1.4 * 2000).[158] The Migrash is 4000 across (wider than the city's width of 2000 by 1000 in each direction). Taking out the city's diagonal from the Migrash's diameter is 1200 amos (4000 – 2800)[159]. Divide that number in half, one for each side of the city, leaves a Migrash width of **600 amos**[160]



Doing the same cheshbon with Abaya's city of 1000 gives us a Migrash width of **800 amos**[161]

By us, the narrowest part of the Migrash is about **970 amos**, lacking only about 30 amos.

Perhaps another way we can differentiate between our case and the case on נ"ו is that over there, the city cuts sharply into the Migrash whereas over here it is a more gradual encroachment.

In summary of our explanation of the Nachal Ha'aravim's explanation of the Gra's explanation of Abaya's explanation of the Braisa is as follows:

The city is pill-shaped 1 by 1.25 for a total city area of **1**.

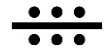
The Migrash is oval-shaped 3 by 3.25 for a total area of **6.3125**.

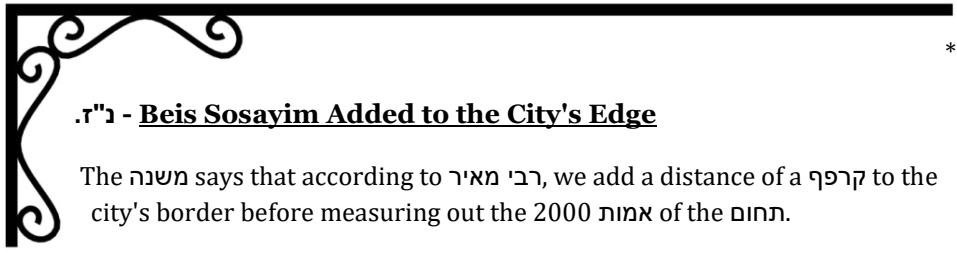
The Techum is a square 5 by 5.25 for a total area of **25.25**.

[158] Or 2828.427... using $\sqrt{2}$ for the diagonal
[159] Or 1171.57... using $\sqrt{2}$...
[160] Or 585.78... " "
[161] Or 792.89... " "

I think the simplest explanation of the Gra is to say that he meant to say 1 thousand and "a little bit more", and the way he wrote it was "1 thousand and a quarter".

At the end of the day, I don't feel like I have a satisfactory explanation for the Gra.





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נ"ז - <u>Beis Sosayim Added to the City's Edge</u>

The משנה says that according to רבי מאיר, we add a distance of a קרפף to the city's border before measuring out the 2000 אמות of the תחום.

We discussed earlier[162] that a קרפף can be as big as בית סאתיים and still retain the regular status of a רשות היחיד. Once it gets bigger than that then there are certain restrictions that might be applied.

A בית סאתיים is 5000 square אמות. As a square, each side of it would be either the square root of 5000 אמות $(70.710678...)$[163] or exactly $70\frac{2}{3}$ אמות $\left(70.\overline{6}\right)$[164].

Our משנה (on נ"ז) seems to be using the שיעור of $70\frac{2}{3}$, as it would seem from the end of the משנה, which we will see.

So, according to רבי מאיר, we measure out a קרפף that is $70\frac{2}{3}$ אמות from the city's border and then an additional 2000 אמות to find the edge of the תחום, for a total of $\mathbf{2070\frac{2}{3}}$ in each direction, for a total of $4141\frac{1}{3}$ for both directions.

The חכמים only measure 2000 אמות from the city's edge, in each direction, for a total of 4000 אמות in each direction.

[162] on page 99
[163] According to רבי יהודה בן בבא in the משנה there
[164] According to רבי עקיבא in the משנה there

Accordingly, the area of the תחום according to רבי מאיר is:

$$\left(\text{City_Width} + 4141\frac{1}{3}\right)^2 - \text{City_Width}^2$$

Whereas according to the חכמים the תחום area is:

$$(\text{City_Width} + 4000)^2 - \text{City_Width}^2$$

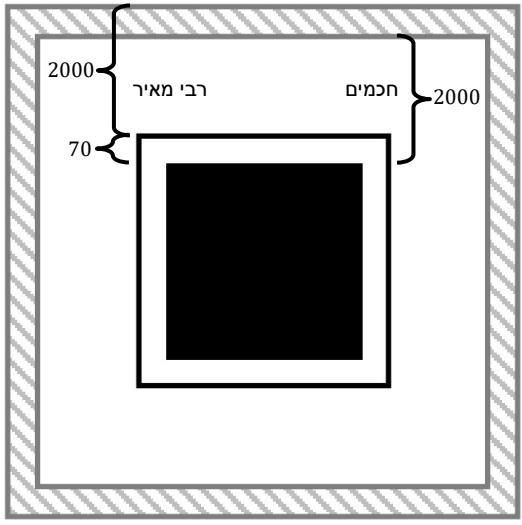
And the difference between them is therefore:

$$\left(\text{City_Width} + 4141\frac{1}{3}\right)^2 - \text{City_Width}^2 - ((\text{City_Width} + 4000)^2 - \text{City_Width}^2)$$

Which can be simplified to:

$$\mathbf{282\frac{2}{3}City_Width + 1{,}150{,}641\frac{7}{9}}$$

For a standard city width of 2000, that would be a difference of $1{,}715{,}975\frac{1}{9}$ A^2.



All of this is according to רבי עקיבא from the משנה on כ"ג:, who says that the side of a square קרפף is exactly $70\frac{2}{3}$ אמות.

According to the תנא קמא over there, רבי יהודה בן בבא, you calculate the sides of a 5000 sq. amah קרפף exactly, and that would be $\sqrt{5000}$, which comes out to 70.710678...

Accordingly, according to רבי מאיר here, one first measures out $\sqrt{5000}$ and then another 2000 אמות for the תחום, while the חכמים maintain that one simply measures out 2000 אמות starting from the city's edge.

The difference then between רבי מאיר and the חכמים can be expressed in the following formula:

$$\overbrace{((\text{City_Width} + 2\sqrt{5000} + 4000)^2 - \text{City_Width}^2)}^{\text{רבי מאיר}} - \overbrace{((\text{City_Width} + 4000)^2 - \text{City_Width}^2)}^{\text{חכמים}}$$

Which simplifies to:

$$4\sqrt{5,000} * \text{City_Width} + 20,000 + 16,000\sqrt{5,000}$$

Which is about:

$$282.84...\text{City_Width} + 1,151,370.849...$$

Which for a standard city width of 2000 comes out to about 1,717,056.2748... A^2.

Between רבי עקיבא and רבי יהודה בן בבא there's a difference between the מחלוקת between רבי מאיר and the חכמים (that is, the difference of the difference). This difference is the same as the difference in their מחלוקת about what רבי מאיר holds

(since the opinion of the חכמים is the same according to both תנאים). It can therefore be expressed in formula form as:

$$\overbrace{((\text{City_Width} + 2\sqrt{5000} + 4000)^2 - \text{City_Width}^2)}^{\text{רבי יהודה בן בבא}} - \overbrace{\left(\left(\text{City_Width} + 4141\frac{1}{3}\right)^2 - \text{City_Width}^2\right)}^{\text{רבי עקיבא}}$$

Which can be (somewhat) simplified to:

$$4\sqrt{5000} * \text{City_Width} + 16{,}000\sqrt{5000} - 282\frac{2}{3} * \text{City_Width} - 1{,}130{,}641\frac{7}{9}$$

Which, for a standard city width of 2000 אמות, comes out to about 1,081.16 A^2

	רבי יהודה בן בבא	רבי עקיבא
רבי מאיר	$(\text{City_Width} + \sqrt{20000} + 4000)^2 - \text{City_Width}^2$	$\left(\text{City_Width} + 4141\frac{1}{3}\right)^2 - \text{City_Width}^2$
חכמים	$(\text{City_Width} + 4000)^2 - \text{City_Width}^2$	$(\text{City_Width} + 4000)^2 - \text{City_Width}^2$

The חכמים do not measure a קרפף as part of the city as far as תחום is concerned.

They do, however, consider it part of the city in certain cases where there are two cities near each other. If the two cities are within 2 קרפפs of each other then we view both cities as one.

The משנה calls this distance "141 and $\frac{1}{3}$ אמות", reflecting the opinion of רבי עקיבא, who says that a square קרפף can be $70\frac{2}{3}$ $\left(\text{half of } 141\frac{1}{3}\right)$.

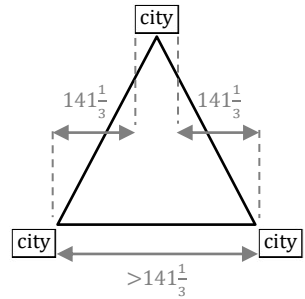
Or perhaps this could even be רבי יהודה בן בבא? He holds the קרפף to be $\sqrt{5000}$, so two of them would be $2\sqrt{5000}$ $\left(\sqrt{20000}\right)$, or about **141.421...**. Rounding to the closest third that comes out to $141\frac{1}{3}$, although it would seem more מסתבר to round to the closest half $\left(141\frac{1}{2}\right)$, since .421... is closer to a half than to a third.

	$\frac{1}{2}$	$\frac{1}{3}$
Subtract one from the other	.5 - .421... = **.0786...**	$.\overline{3}$ - .421... = -.088...
Divide one by the other	.5 ÷ .421... = **+%18.66...**	$.\overline{3}$ ÷ .421... = -%20.89...

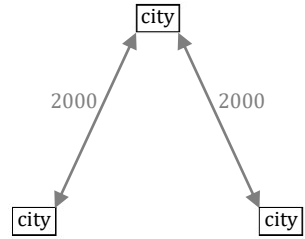
*

נ"ז: - <u>City Between Two Cities – Gaining On the Diagonal?</u>

The משנה on עמוד א' had said that although we need $141\frac{1}{3}$ אמות between two cities to connect them, it's possible to have them more apart somewhat if there is a third city in between them, even at the "peak of a triangle":



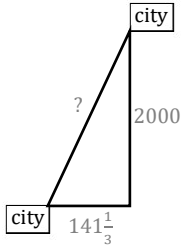
...as long as there is no more than $141\frac{1}{3}$ אמות horizontally between the middle city and the outer cities.



The גמרא makes another condition that there cannot be more than a 2000 amah distance between the outer cities and the middle one "so that one could walk there on שבת"[165].

[165] רש"י ד"ה אלפים וז"ל הואיל ויכול לבוא מזו לזו בלא עירוב

Which got me wondering. If we're talking about an angle, shouldn't one get more than 2000 אמות?[166] You can walk straight out of a city for 2000 אמות and the horizontal distance to the "middle" city is $141\frac{1}{3}$. What does that give you for the diagonal?



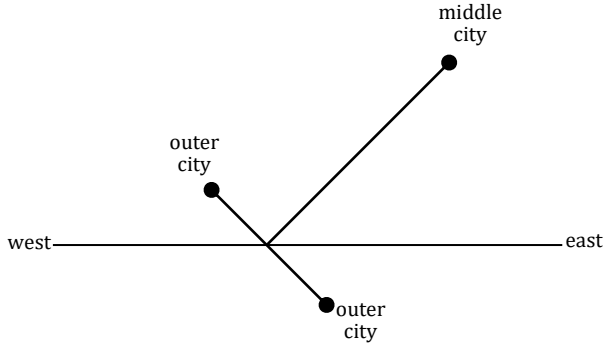
$$\text{hypotenuse} = \sqrt{a^2 + b^2}$$

$$\text{hypotenuse} = \sqrt{141\frac{1}{3}^2 + 2000^2}$$

$$\text{hypotenuse} = 2004.98...$$

Hmm, that doesn't seem to gain much. Perhaps the גמרא is rounding those 2004-and-something אמות to a nice round "2000".

But what if we were to rotate this whole triangular city-system 45° to give us the benefit of the diagonal[167]?



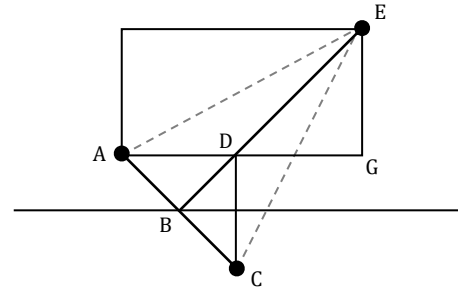
How far could you walk from the outer cities to the middle one? (While making sure the cities are within 2000 אמות of each other in the cardinal directions.)

[166] Like we said on נ"ו: (page 195).

[167] Since the diagonal of the city's תחום is 2800 amos (or more precisely $2000\sqrt{2}$).

$\overline{AG} = \mathbf{2000}$.

$\overline{AB} = 141$
$\left(\text{or more precisely } 2\sqrt{5000}\right)$ and $\overline{BD}$ must be the same (since it is the complementary leg of the 45° $\triangle ABD$)



$\overline{AD} = \sqrt{\overline{AB}^2 + \overline{BD}^2} = \sqrt{2\left(2\sqrt{5000}\right)^2} = \sqrt{40{,}000} = \mathbf{200}$.

$\overline{DG} = \overline{AG} - \overline{AD} = 2000 - 200 = \mathbf{1800}$, which must be the same values as $\overline{EG}$ (since $\angle EDG = 45°$, so $\overline{EG}$ is the same as $\overline{DG}$).

Now that we know $\overline{AG}$ and $\overline{EG}$, we can calculate $\overline{AE}$ quite simply with:

$$\overline{AE} = \sqrt{\overline{AG}^2 + \overline{EG}^2} = \sqrt{4000000 + 3240000} = \sqrt{7240000} = \boxed{\mathbf{2690.72480}\ldots}$$

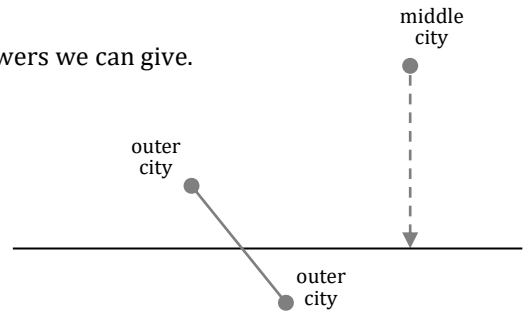
Which is way more than 2000 אמות.

So the question is, why does our גמרא say that there has to be no more than 2000 אמות between the outer cities and the middle one? We came up with a case where there can be several hundred more אמות!

?

So there would seem to be two answers we can give.

Either, אה"נ, we only "throw" a city along the cardinal directions, so in this case the 3rd city isn't between these other cities at all.



Or we can say that "2000 אמות" is a concise phrase that means the תחום, however you arrange it. If you can get more along the diagonal then so be it.

נ"ז:

The חזון איש[168] writes similarly:

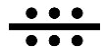
ג' עירות אחת לצפון ואחת לדרום ואחת כנגד אויר שביניהן משוכה למזרח רואין כל שאתה מטיל האמצעית ביניהן ואין אויר ביניהן רק קמ"א ושליש מכאן וקמ"א ושליש מכאן ואין בין האמצעית להחיצונות יותר מאלפים [נראה דרואין כאילו חוט מתוח מחיצונה לחיצונה ואין מן האמצעית עד קו הזה יותר מאלפים אף שיש באלכסון מן החיצונה לאמצעית יותר מאלפים] הרי כולן כעיר אחת ומרבעין אותן כעיר העשוי' כמשולש.

{My comments in curly brackets}

"*Three cities, one to the North and one to the South, and one opposite the space between them, to the East. We look at it as though the middle one is {actually} between them, and if there is only 141$\frac{1}{3}$ {אמות} on one side and 141$\frac{1}{3}$ {אמות} on the other side, and there isn't between the middle {city} and the outer {cites} more than 2000 {אמות} [it seems that we view it as though a string was stretched from one outer city to the other, and there isn't from the middle {city} until this line more than 2000 {אמות}, even if there is more than 2000 {אמות} on the diagonal from one of the outer {cities} until the middle {city}] then behold it is all considered one city and we make a square around them as though they were a {single} triangular city.*"

However, מגיד משנה[169] draws a picture with the 2000 אמות along the diagonal, so perhaps he doesn't learn like רש"י that it has anything to do with the ability to get there on שבת through the תחום.

פירוש חי brings רבינו ישמעאל בן חכמין and חות יאיר[170] who say like the מגיד משנה (2000 אמות even along the diagonal), and the מרכבת המשנה[169] who says like the חזון איש (2000 is vertical or horizontal, but אה"נ, you can get more along the diagonal).



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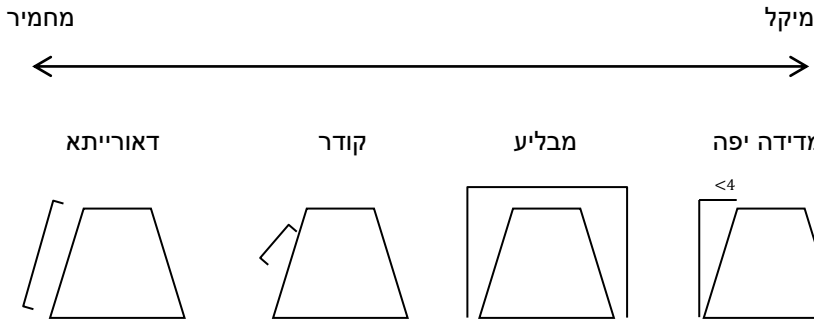
נ"ח: - <u>Tosfos Ein Modedin - Measuring Techum Over Mountains</u>

There are (at least) four ways you can measure the slope of a mountain, as far as תחום is concerned.

[168] הלכות עירובין סימן קי (ב) אות ט"ו ד"ה ג' עירות
[169] הילכות שבת כ"ח:ה'
[170] סי' קנ"ה

Let's list them with labels, picture them, and explain them:

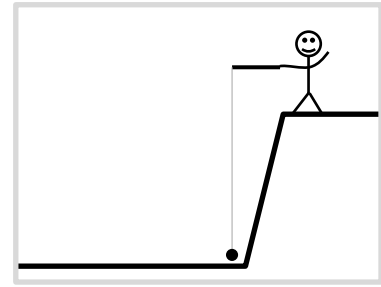


1) For a דאורייתא, you need to measure the slope directly, that is, the actual length of the slope.

2) קודר is when two people hold a 4-amah-long stick over the slope. The person on the lower part of the slope puts his end of the stick by his heart and the other person higher up puts his end of the stick by his feet. Usually, the stick will not be perfectly horizontal (like in the picture) and so that difference is lost when totaling the תחום length (since you're counting אמות until 2000, and you counted more אמות here than there really are).

3) מבליע is when a 50-amah-long measuring rope is held on top of tall poles, each held by a person on either end of the mountain/hill/mound. Effectively, this is no better or worse than had there been no mountain at all (we measure the mountain as though it were flat land)[171].

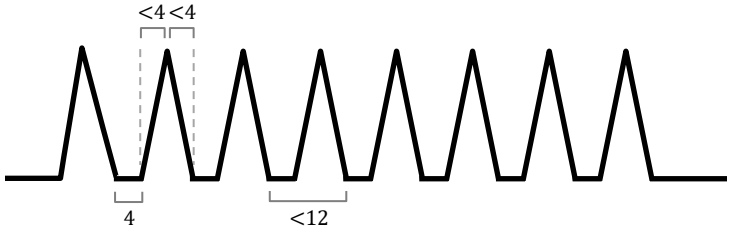
4) מדידה יפה is when the slope is so steep that the "run" of the slope is not even 4 אמות. This would be measured by standing at the top edge of the slope while holding out a 4-amah stick above the slope, with a string hanging down from the end of the stick, with a weight at the bottom of the string (to keep it straight). If the weight doesn't touch the slope at all, but rather the ground next to it then the horizontal length of the slope is less than 4 אמות, and then the entire slope is not counted at all as part of the תחום.



[171] Although, it would seem that a חומרא could come out of this since it would be very difficult to pull the rope taut by holding the bottom of the poles, thus counting less than 50 אמות as 50 אמות, making the overall תחום a little shorter.

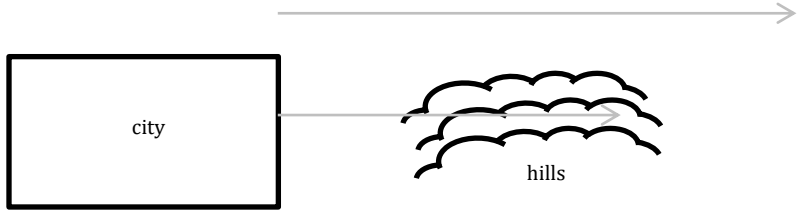
אגב, theoretically, you can have a series of steep mounds one after the other and stretch the תחום out indefinitely...

...unless if there is a minimal amount of flat ground necessary between mounds (let's say 4 אמות) then that would limit the amount you can stretch. But you could still get out to about 6,000 אמות, since each flat-rise-fall of almost 12 אמות only counts as 4, (see picture).



The משנה says that when you measure the תחום from the city, you measure "מן המומחה", which רש"י says means with an expert.

תוספות, quoting רבינו חננאל, says it means "Directly [outward] from the city, and not from the side, because sometimes there are mountains and valleys directly in line with the city that you should do מבליע or קודר".



קודר is understood, since one may end up having to be more מחמיר than if you had measured from the side. Measuring from the side would have given you 2000 אמות, as the crow flies, as the saying goes. But now that you're measuring from the city straight out, you'll be encountering all these mountains, and קודר ends up making the תחום somewhat closer than a direct 2000 אמות, as we explained in the beginning of this piece.

So, Tosfos's use of the word קודר is readily understood.

מבליע, however, is not so clear. As we explained in the beginning, מבליע measures the mountain as though the mountain were perfectly flat[172]. Thus, whether we measure the city directly out or from the side won't make any difference since both methods will produce the same length. So why does תוספות imply that it <u>will</u> make a difference?

Because of this problem, the מהרש"א prefers the גירסה of the ראש"ש that goes, "...in line with the city that you should do קודר[173] or **מדידה יפה**[174]".

As we explained earlier, מדידה יפה means you will sometimes not include the mountain's slope in the חשבון of the תחום at all, thus giving you <u>more</u> than 2000 אמות, as the crow flies.

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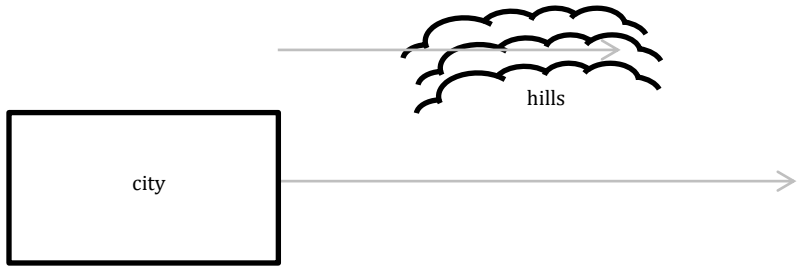
תוספות spoke about a case where the mountains were straight out from the city, whereas measuring from the side of the city would have been flat land.

What if the case was the other way around? What if the mountains were encountered when measuring from the side of the city, whereas measuring straight from the city yielded flat land?



[172] Unless you take into account what I wrote in the previous footnote #171, but the מהרש"א does not seem to be doing so.
[173] לחומרא
[174] לקולא

If you read תוספות the way I suggested[175], that מבליע leads to a חומרא as well, then it could be you always need to go לחומרא. In this case, that would mean that you <u>have</u> to measure from the side of the city, דווקא.

If תוספות is read like the מהרש"א then it would seem that you measure directly out from the city always, regardless of whether it results in a חומרא or קולא. In this case then, קודר leads to a קולא (since you're **not** measuring on the side where the mountains are, so you don't lose distance as קודר sometimes does), and מבליע leads to a חומרא (since, had you measured from the mountains on the side you would have been able to skip some steep cliffs but now you can't since you're measuring straight across the flat land).

Actually, the Maharsha's approach makes things much simpler. According to the "חומרא always" פשט, you'd have to measure from every possible place, in, near, and about the city, and find the most מחמיר measurement. That doesn't seem so מסתבר to me.

The שלחן ערוך[176] says to always measure directly out from the city regardless of whether you encounter mountains, valleys, or nothing, and then goes on to detail when to do מדידה יפה, when מבליע, when קודר, etc. It would seem that the ש"ע is learning תוספות like the מהרש"א.

[175] in footnote 171
[176] א"ח שצ"ט:ד

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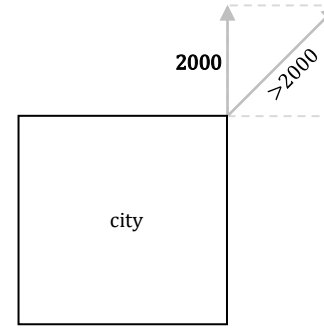
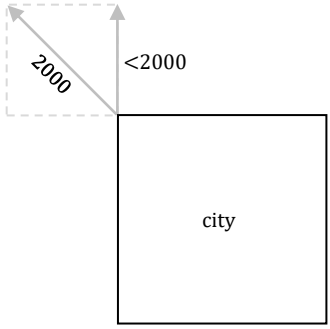
נ"ט. - Rashi Shelo Yarbeh - Measuring Techum Corner

The משנה on :נ"ח says that if, while measuring the תחום of the city, one of the surveyors measured out further than the other one then we go after the longer measurement.

אביי qualifies this, that as long as the difference between the measurements is no greater than "the measurement of the city with its diagonal".

רש"י explains that we assume that one of the surveyors measured from his corner of the city correctly, that is, straight out from the city for 2000 אמות, effectively making the תחום's diagonal measurement longer than 2000 אמות.





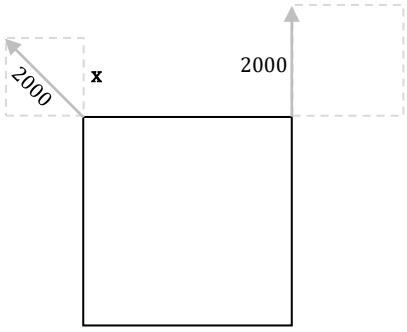
Meanwhile, the other fellow measured the other way around, accidently. That is, he measured the *diagonal* of the תחום's corner as 2000 אמות, thus making the תחום's straight border *less* than 2000 from the city's edge.

A difference in measurements within this range can be attributed to this difference of measurement methods.[177]

[177] I assume that if the difference is greater than this range then they either have to go after the shorter distance, or they have to measure again, perhaps with different surveyors, or perhaps they should inspect each other's work.

How much of a difference can these different methods yield?

Using 1.4 as the diagonal of a square:



x is the side of the little square at the top-left of the picture.

That square's diagonal is 2000.

x times 1.4 gives us the diagonal.

So...

$$1.4x = 2000$$

divide both sides by 1.4

$$x = \frac{2000}{1.4}$$

$$\mathbf{x = 1428\frac{4}{7}}$$

Now take that amount off of 2000 (the length of the bigger square on the right) to find the difference between the lengths of these two corner-squares.

$$2000 - 1428\frac{4}{7} = \mathbf{571\frac{3}{7}\ Amos}$$

Converting that into modern measurements[178]:

אח"נ		ח"א
$\sim 274\frac{2}{7}$ m	$< 571\frac{3}{7}$ Amos >	~331.42... m
~899.887... ft.		~1,087.364 ft

[178] 1 אמה = 48 cm -אברהם חיים נאה
= 58 cm -חזון איש See Appendix A

Let's do the חשבון again but using $\sqrt{2}$ for the diagonal of a square:

$$\sqrt{2}\,x = 2000$$

divide both sides by $\sqrt{2}$

$$x = \frac{2000}{\sqrt{2}}$$

x = 1,414.213...

2000 - 1,414.21 = **585.786... Amos**

אח"נ		ח"א
~281.177... m	<585.786... Amos >	~339.756... m
~922.498... ft		~1,114.685 ft

This measurement (somewhere within the range we just calculated) is the max amount of a difference we can have between the two surveyors and be allowed to use the larger one.[179]

That's the differences between measuring the 2000 straight out from the city versus measuring the 2000 אמות diagonally.

We showed that this difference itself depends on whether we use 1.4 or $\sqrt{2}$ for the diagonal of a square.

Thus, the differences of the difference is[180]:

אח"נ		ח"א
~6.891... m	< 585.786... - $571\frac{3}{7}$ Amos >	~8.32... m
~22.61... ft	< 14.357866... Amos >	~27.321... ft

[179] In the case where this max is exceeded, see footnote 177

[180] That is, subtract each cell in the second chart (at the top of this page) from each corresponding cell in the first chart (from the previous page).

*

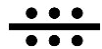
ס"ג: - Tosfos Miyad - Meaning of "Immediately"

The גמרא brings the story of יהושע בן נון who met the מלאך who told him that they weren't utilizing their time properly to learn Torah. The גמרא says that יהושע went to learn Torah, "immediately", "מיד".

תוספות points out that it wasn't literally immediately but rather when he had the opportunity, which was really much later.

I thought to use this תוספות to justify the way people speak when they say they'll do something in "one second" or "one minute" when in reality they know perfectly well that is will be much longer than that[181].

תוספות is telling us that the גמרא itself uses a similar type of גוזמא, exaggeration; it is in a manner of speaking.



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ס"ד. - Tosfos Shassah - Clarity In Prayer

The גמרא brings a מאמר from רבי יהודה אמר שמואל that says, "שתה רביעית אל יורה", "One who drank a quarter[182] of wine shouldn't teach [instruct]."

תוספות brings a גירסא that says "shouldn't daven (pray)," which תוספות finds difficult considering how רב נחמן rejected the above rule on the grounds that "my mind is not clear as long as I didn't drink a רביעית of wine".

תוספות concludes his question with, "What does clarity [of mind] have to do with prayer?"

[181] Or, how South Africans use the phrase "just now" when they mean at the next convenient time in the future.

[182] I.e., a quarter of a log. A log (pronounced "loog", like the short double o in "look", or more accurately like a long double o as in "food") is about 12 to 16 U.S. ounces (346 to 609.725 ml), so a quarter log is about 3 or 4 U.S. ounces (86.4 ml to 152.43125 ml).

This surprised me. Doesn't clarity of mind have everything to do with תפילה?

The יעב"ץ wonders the same thing and asks that the גמרא later[183] says, "whoever's mind is not settled, (שאין דעתו מיושבת עליו), shouldn't daven", and many other examples where the גמרא says one needs a settled mind (ישוב הדעת) and concentration (שיכוונו ליבם) for tefilla.

I thought maybe to explain תוספות like this. "צלותא", "clarity", is a function of analysis. When you're a פוסק (a יורה) or a judge, or a scholar plumbing the depths of a topic, you need to analyze everything with perfect "clarity", with the ability to consider everything, and combine different ideas together in every possible way until a conclusion or verdict crystalizes. That's צלותא.

ישוב הדעת is something else. ישוב הדעת is being calm, being focused, understanding things as they are, as they appear, without any intellectual acrobatics.

When someone is davening, they shouldn't be "analyzing" the words of the תפילה, like asking, "Why is this word here? Wouldn't that other word be more accurate? Why are these sections ordered this way? Etc." These types of questions should be thought about before or after davening, but not during.

During davening, the goal is to be "focused", concentrating on the meaning of the words (or the translation, if you're not familiar with לשון הקדוש), and that you are standing before Hashem, מלך מלכי המלכים, and that He is listening to your every word (and to your level of concentration!).

This might be what תוספות means.

רבינו פרץ writes, "ואי לתפילה מי בעי צילתא אדרבה מתוך אימה עדיף טפי".

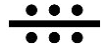
I don't know if this necessarily different than what I wrote. אימה could mean "awareness", like an electrician who's working with electricity has an "אימה" of electricity. It's not a "fear" like the way we usually understand it (as if it is something to run away from) but something to be aware of its power, to be focused on it. Not to be distracted by other thoughts and feelings.

I used to think of תפילה primarily as an active process; that is, a form of expressing yourself, and learning as more of a passive process; that is, taking in the information.

[183] 65a

But now I feel like it's the opposite. Now, I feel like תפילה is more of a passive process where you nullify the ego and allow Hashem to fill your whole existence. As Rav Shimshon Raphael Hirsch writes regarding the word "להתפלל", that it means to "Fill oneself with the unifying concept of G-d in all aspects of one's life (in health by רפאינו, in wealth by ברוך עלינו, etc.)".

And learning, the real aspect of learning, is more of an active process, to throw yourself into the סוגיא. To analyze, compare, contrast, dissect, to question, suggest answers, etc.[184]



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ס"ז: - Island Karpef

Here's another confusing שקלא וטריא that I found it helpful to make a chart. Let's start with the background גמרא.

עולא says in the name of רבי יוחנן: "A קרפף larger than בית סאתייס, not walled for domestic use (לא הוקף לדירה)[185]: one who throws from a רשות הרבים into it [or out of it] is חייב דאורייתא."

Meaning, it's a full-fledged רשות היחיד דאורייתא. And even though one is not allowed to carry within it ד' אמות, that is only a חומרא דרבנן, to treat it like a כרמלית.

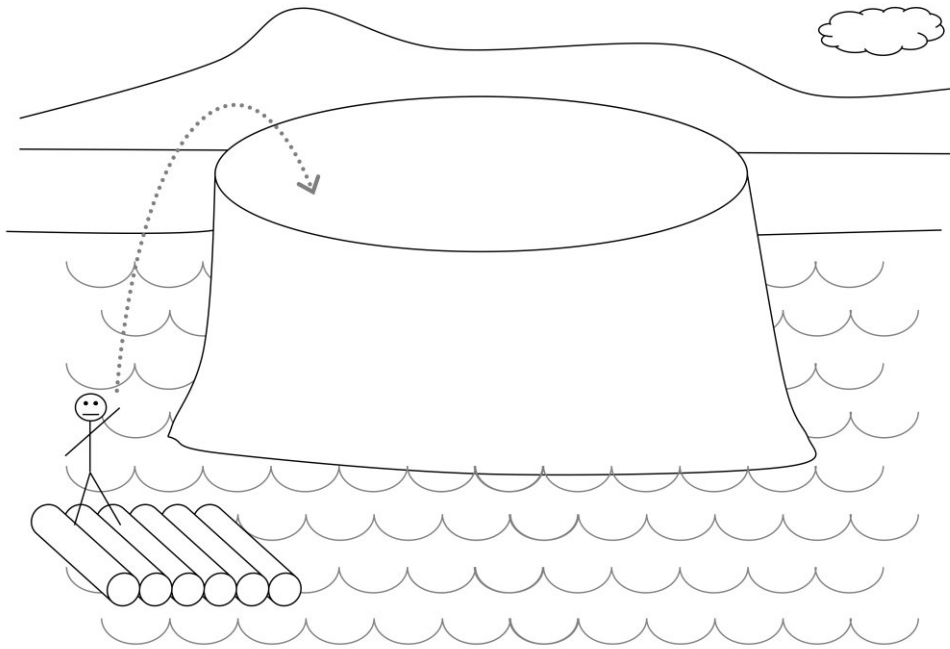
Now, the גמרא [רב הונא בר חיננה] is going to ask from a ברייתא on רבי יוחנן. Bear in mind that the ברייתא's case is a bit different than רבי יוחנן's since it is dealing with going from a כרמלית [the ocean] to a קרפף. So, the cases aren't exactly parallel but, for our purposes, it will suffice to demonstrate that the קרפף isn't a רשות היחיד דאורייתא. At least, we will attempt to demonstrate. Stayed tuned for exciting developments.

[184] See Rav S. R. Hirsch on Genesis 41:33 where he expands on these two types of thinking, though in a slightly different respect.

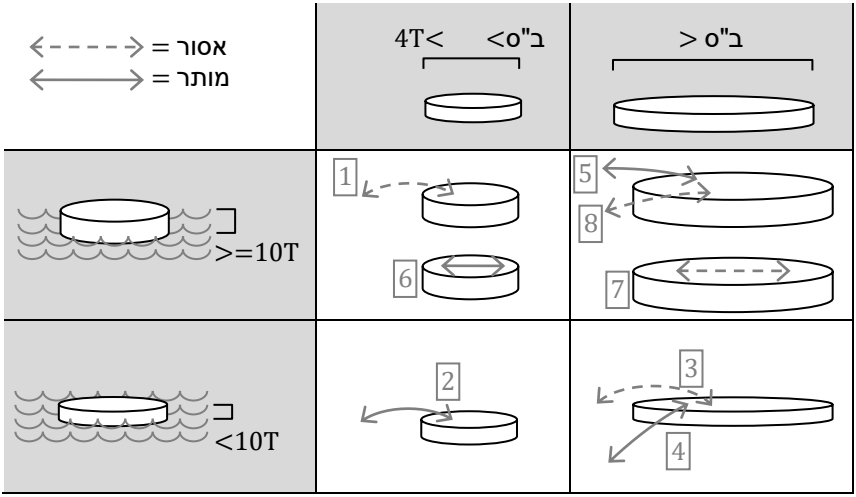
[185] I.e., It has walls, it's just that it wasn't constructed for the purpose of living within it.

(For our discussion, it makes no difference whether something is going into or out of the קרפף, so I will sometimes use only the one or the other case, but both are meant. Also, "throw" is לאו דווקא, since "move" works just as well. It just makes things more dramatic ;-) And anyway, that's how the גמרא talks.)

The ברייתא talks about a "טָלָע שֶׁבַּיָּם", that is, an island in the sea, with straight cliff-like walls (which could be a good מחיצה, if the walls are at least 10T tall).



Before continuing with the rest of the גמרא, let's lay out the chart, label it, then work through the שו"ט while referencing the chart, with explanatory notes.



בריית:א

<u>רישא</u>:

"If it [the island] is 10 טפחים high and at least 4 טפחים wide, one may not throw from it into the sea (or back)." [1]

<u>סיפא:</u>

"If it is less than 10 טפחים high, one may throw from it to the sea (or back)." [2]

<u>Outro:</u>

"This (?) is true until the island is a בית סאתיים."

The גמרא (and you, I'm sure) wonders what "This" in the end of the בריית:א is referring to.

It cannot be referring to the סיפא, since if it were so then the בריית:א would be saying that "It is only permitted to throw from the shorter-than-ten island if it is not as wide as a ב"ס." And what if it was as wide? Then it becomes אסור?! [3] Why should that be?! If it's less than 10 high then it is part of the surrounding area (the sea, a כרמלית), and should be permitted according to all opinions! [4]

Rather, the "outro" must be referring to the רישא, that one may not throw from a higher-than-10 island into the sea, until the island is a ב"ס wide.[1]

Implying that when it's greater than ב"ס it's permitted to throw back and forth! [5]

This is not like רבי יוחנן who said that such an area is a רשות היחיד גמור! Since, if that were so (that it is a רשות היחיד גמור) then it should be forbidden to throw from the ב"ס island into the sea (that is, from a רשה"י to a כרמלית)

[בשלמא, if you say not like רבי יוחנן, but rather that the ב"ס קרפף is nothing דאורייתא, and it was the רבנן who made it a regular כרמלית just like the sea, then it would be permitted to throw from one to the other, as the ברייתא implies[5]...]

?

The גמרא presents two answers, one that suggests an alternate דיוק of the ברייתא, and one that agrees with the initial דיוק but offers an explanation why this case is an exception to the general rule.

Answer #1 [alternate דיוק]:

רבא: The רישא said 10 high-4 wide, one may not throw from the island to the sea. [1]

An unspoken implication of that is that one may carry more than 4 אמות within the island [6] (since that's the cons and pros of being a regular רשה"י).

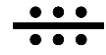
It is about this unspoken דיוק that the "outro" comes to qualify, that one may only carry 4 אמות within the island as long as the island is no larger than a ב"ס. Meaning, in a ב"ס or larger one may not carry within it (כרמלית דרבנן) [7], and yet one still cannot throw from there into the כרמלית sea[8], since this island is a רשות היחיד דאורייתא. And that shtims with רבי יוחנן.

Answer #2 [exception to the general rule]:

רב אשי: You're right, the ברייתא is saying that you may throw from a ב"ס, 10 high, into the sea.[5] And you're right that normally you cannot throw from a רשה"י into a כרמלית. But that דין itself is a דרבנן, and חז"ל made an exception in this case since they were afraid that if we didn't allow throwing into the sea in this case[8] people would conclude (correctly) that this island is a רשות היחיד, and they would come to carry within it 4 אמות, which is not allowed.[7]

So, you'll ask, what's the difference? Either way, it's a דין דרבנן. Why is one[7] worse than the other[8]? You're allowing one דרבנן so that people don't accidently transgress another דרבנן. What have you gained?

The גמרא answers that we are more concerned about the more שכיח case, which is to carry 4 אמות within the island.[7]



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ע. - Forgetful Inhabitants - Braisa Chart

Here's another confusing שקלא וטריא. Let's lay out a chart, go through the שו"ט while referencing the chart, and then suggest the logical order of the cases.

		בריתא					רבה				אביי				
A	אחד שלא עירב נותן רשותו לאחד שעירב		X	→	**O**	?	X	→	**O**	😵		X	→	**O**	😊
B	שנים שעירבו נותנים רשותן לאחד שלא עירב	**O**	**O**	→	X										
C	ושנים שלא עירבו נותנים רשותן לשנים שעירבו	X	X	→	**O**	**O**					X	X	→	**O**	**O**
C2		X	X	↛	**O**	**O**		↛					→		
D	...או לאחד שלא עירב	X	X	→	X										
E	אבל אין אחד שעירב נותן רשותו לאחד שלא עירב	?	**O**	↛	X		😊	**O**	↛	X					
F	ואין שנים שעירבו נותנים רשותן לשנים שלא עירו	**O**	**O**	↛	X	X									
G	ואין שנים שלא עירבו נותנים רשותן לשנים שלא עירבו	X	X	↛	X	X									

(Xs are those who didn't do an eruv; Os are those who did. Arrows means they can be mivatel; Arrows with an x on them means they cannot.)

A little background information.

Let's say you have a חצר with several houses in it. Usually, the inhabitants of the חצר need to do an עירוב together so that the houses (which are individually owned[186]) and the חצר (which is jointly owned) become a single רשות. If one of the inhabitants forgot to do the עירוב with the others then the חצר remains a separate רשות and no one can carry between the houses and the חצר. In that case, there is still the option of "ביטול", where either the forgetter or the other inhabitants give up their rights of the חצר to the other party thus making it a single רשות. Sometimes this can be done and sometimes not, which we will see as we go.

The גמרא starts off with a שאילה.

[186] מדאורייתא, the definitions of רשה"י and רשה"ר have nothing to do with ownership at all, but חז"ל recognized that people think in those terms, so when there are different ownerships, we need to differentiate so people don't get confused.

אביי asks רבה if the one who forgot to do the עירוב is מבטל his rights to everyone else, does he need to be מבטל to each one or not?[187]

רבה answers that it must be to each one (I.e., it cannot be given to one alone).

Throughout the גמרא, אביי defends the other position, that it <u>can</u> be given to only one.

$$\text{רבה} = X \twoheadrightarrow \mathbf{O} \mid \mathbf{O} \qquad \text{אביי} = X \longrightarrow \mathbf{O} \mid \mathbf{O}$$

אביי asks from the ברייתא (laid out in the big chart on the previous page). **איתיביה**:

(For the sake of brevity, I'm going to use an unconventional convention. Instead of referring to the characters as **One-who-didn't-do-עירוב** or **Two-who-did-עירוב** and the like, I'm going to refer to them as a **Did, Didn't, Dids,** and **Didn'ts**. Although initially this might seem funny, in the long run it'll save us precious computing brain power)

Row labeled A: The ברייתא says: **1 Didn't gives to 1 Did** $X \rightarrow \mathbf{O}$

> **אביי**: "With whom did the Did do עירוב? Obviously with someone else, and still the Didn't can give it to him alone."
>
> **רבה**: "The other guy died. Effectively, the Didn't is מבטל to "everyone" (i.e., whoever's left, that is, the one Did still alive)."
>
> **אביי**: "If so then you'd have to explain E like that as well, that the other guy died. If so, why can't the Did be מבטל?"
>
> **רבה**: No, that case is different. In A he died, in E he didn't.
>
> And I'll prove to you that you can't be מבטל to only one. In case C two Didn'ts give to two Dids, implying: but not to one Did (out of two) C2!
>
> **אביי**: No! That case C <u>is</u> where they gave to one. It means "...can give to [one of] two"!
>
> **רבה**: If so (that it means "gives to one of two" and that you can give to any 'one' in general) then C should've been taught together with D, like this: "Two Didn'ts can give to one Did C2 or to one Didn't D".
>
> **אביי:** Good question.

Then the גמרא goes through the whole ברייתא, considering what the חידוש is according to each אמורא:

[187] I understood the options are: to each of them individually or to one for all of them.

A: 1 Didn't can give to 1 Did X → **O**

אביי: Both Dids are alive (X → **O|** ☺) . חידוש is that you don't have to give to each individual one.

רבה: One Did died (X → **O|** ☹) . חידוש is we aren't גוזר that he might give to only one when both are alive.

B: 2 Dids can give to 1 Didn't **O O** → X

חידוש is we don't קנס the Didn't (לכאורה, by making the בִּיטוּל בָּטוּל, which will effectively make ineffective the entire עירוב, which would mean no one can carry between any of the houses and the חצר).[188]

C: 2 Didn'ts can give to 2 Dids X X → **O O**

רבה: חידוש is to let you know that the first case A must be where the other Did died. Since in this case C 2 Didn'ts can give to 2 Dids, but not to only 1 Did C2, unless if the other guy died A.

אביי: חידוש is that 2 Didn'ts can <u>give</u> even though it can't <u>be given to them</u> F & G, and we aren't גוזר C אטו F & G.

D: 2 Didn'ts can give to 1 Didn't X X → X

חידוש is that we don't קנס them, even though no one did an עירוב.[189] (ס"ד that we should קנס them so that תורת עירובין is not forgotten, קמ"ל).

E: 1 Did <u>cannot</u> give to 1 Didn't **O|O** ↛ X

אביי: The one Did was obviously one <u>of two</u> Dids (since it takes at least two to make an עירוב), to let you know that the first case A was also 1 <u>of two</u> Dids (and not that the other guy died, as רבה asserts).

רבה: There's no חידוש here. Since the ברייתא taught the case of A so now it teaches the inverse, which is the case of E.

In A, the ברייתא said one Didn't can give to one Did, when the other Did died.

But when both Dids are alive, the Didn't <u>can't</u> give to the one Did that's left.

And so too the reverse, E. Out of the two Dids, one alone cannot give to the Didn't.

---

[188] It would seem that the קנס would be on him for forgetting. Maybe the reason we don't קנס him is that it will end up affecting the other people in the חצר as well, since they will not be allowed to carry from their houses to the חצר.

[189] This is worse than case "B" since they all messed up (by not doing an עירוב), but even still we don't קנס them.

In other words, out of the two Dids, רשות cannot be given <u>to</u> [A] nor <u>from</u> [E] only one of them.

It would seem the reason ביטול wouldn't work here is simply because only one was מבטל but the other guy wasn't, so the חצר is jointly owned by the one who didn't מבטל and the one to whom it was מבוטל; which is again an area whose ownership is unique to the other houses, thus אסוריng carrying between the חצר and the houses.

F: 2 Dids <u>cannot</u> give to 2 Didn'ts **O O** ↛ X X

חידוש is that this never works, even if one of the Didn'ts goes afterwards and is מבטל to the other Didn't.

G: 2 Didn'ts <u>cannot</u> give to 2 Didn't s X X ↛ X X

חידוש is that the previous case [F] doesn't work even if the givers state that the receivers are getting the right to the חצר on condition that one of them will then give his right to the other receiver. (This is inferred from the fact that the ברייתא repeated this case [G] which we already knew from [F]. The repetition was obviously meant to be an emphasis of the previous case [F] that it never works, not even when there's a condition that seems to enforce it.

(It's like when the רבי says, "It'll never happen". And the student starts to ask, "What about..." and before he finishes, the רבי says "No, not even then". It is obvious that the רבי understood what sort of case the student may be thinking of and excluded it as well.)

These cases of [F] and [G] don't work as רש"י explains on the previous עמוד[190] that when there is a ביטול to 2 people, each one is still אסוריng on each other at the time of the ביטול (since this חצר is owned by these two guys jointly, which is distinctly different than their houses which are each owned by each person individually).

לכאורה, The reason why [C] and [D] <u>do</u> work (where 2 Didn't are מבטל) is because they could be מבטל at the same time, whereas by [F] and [G] there's another ביטול that needs to take place still.

I was wondering, according to אביי, what is the difference between [A] where we allow ביטול, and [E] where we don't? Either way, the Did not involved is only benefiting, no?

I had understood that [A] works because of the rule of זכין לאדם שלא בפניו[191], that before the ביטול, no one can carry into the חצר, and afterwards they all can carry. Since he's benefiting from the ביטול to him, we have it done to him on his behalf.

---

[190] ס"ט: ד"ה קמ"ל

[191] פ"א:

So let it work by **E** as well. Until now, no one can carry into the חצר, but after the ביטול, at least someone (the Didn't) can carry into the חצר.

The גאון יעקב writes:

לאביי תנא סיפא. פירש"י דלא איצטריך כו' ולא דמי למערב דא' מערב בשביל כולן התם זכות הוא להן ודמיא להא דתנן מערבין עירובי חצרות בין לדעתו בין שלא מדעתו הכא לא זכות היא לו דמבטל הא אסור:

"*...And this is incomparable to the case of making an* עירוב*, where one can make the* עירוב *for the sake of all of them, since there it is a* זכות *for them and is similar to the* משנה *that says*[191] *they can make an* עירוב חצרות *with or without his knowledge, but here it isn't a* זכות *for him*."

Is it that it's merely not a זכות and that's why it can't be done for him? Or does he mean that it is a חוב for him, and then we have the rule אין חבין לאדם שלא בפניו[191]?

Why would it be considered a חוב to the Did who's not here to have ביטול done on his behalf. What's he losing? Without the ביטול he also can't carry into the חצר!

Maybe, since before ביטול, it's possible for the Didn't to be מבטל to him, which would allow him to carry from his house into the חצר, even though right now before the ביטול he can't carry בפועל.

Also, he's losing his current right to carry to the other Did's house directly, not through the חצר (for example, through a window between their houses).

סתם, I was wondering, what is the logical order of the cases in the ברייתא?

I thought to say perhaps it goes like this:

The cases are ordered like a bullet list, with major and minor groupings.

First major grouping are cases in which you can be מבטל, and then all the cases where you cannot.

Within each of these groups, first deal with cases of 1 giver, then cases of 2 givers.

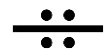
Within each of these groups, first where givers were מערב, and then where they weren't.

Within each of these groups, first where the receivers were מערב, and then where they weren't.

In bullet list/Tree form, it would look like the following:

(Although there aren't that many cases altogether since the tree form allows 16 cases ($= 2^4$, that is: 4 categories with 2 options each), but the ברייתא only lists 7 cases. Cases not represented in the ברייתא are faded out)

- Can
  - One Giver
    - Giver was מערב
      - Receiver/s was מערב
      - Receiver/s wasn't מערב
    - Giver wasn't מערב
      - Receiver/s was מערב **A**
      - Receiver/s wasn't מערב
  - Two Givers
    - Givers were מערב
      - Receiver/s was מערב
      - Receiver/s wasn't מערב **B**
    - Givers weren't מערב
      - Receiver/s was מערב **C**
      - Receiver/s wasn't מערב **D**
- Can't
  - One Giver
    - Giver was מערב
      - Receiver/s was מערב
      - Receiver/s wasn't מערב **E**
    - Giver wasn't מערב
      - Receiver/s was מערב
      - Receiver/s wasn't מערב
  - Two Givers
    - Givers were מערב
      - Receiver/s was מערב
      - Receiver/s wasn't מערב **F**
    - Givers weren't מערב
      - Receiver/s was מערב
      - Receiver/s wasn't מערב **G**



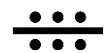
A friend of mine suggested an alternate organization method:

First major grouping: First those who can, and then those who can't be מבטל

2nd grouping: Ordered according to the number of forgetters, from 1 to 4

3rd grouping: First where the forgetter/s is מבטל, and then where the rememberer is (or tries to be) מבטל.[192] (Alternatively, this grouping could be according to the number of people being מבטל. -BG)

The benefit of this approach is that it assumes less categories.



[192] This is מסתבר, since מסתמא people are not forgetting.

ע'.

*

ע"א: - Rashi Mie Tiemuh - Mutually Dependent Opinions

The משנה on עמוד א' said that if a בעל הבית has two separate שותפיות (not for מבוי; just in general - רש"י) with two different neighbors, both with wine, then he doesn't need to make an additional עירוב with them. The גמרא brings the opinion of רבי אלעזר בן תדאי who says that it doesn't work. רב יוסף explains that the underlying מחלוקת is whether you can rely on שיתוף במקום עירוב, meaning, when you need to do an עירוב but you already have a שיתוף, whether you can use that existing שיתוף as the עירוב.

Now, here's the peculiar statement of רב יוסף (with Rashi's explanation). רב יוסף says, "How do I know that this is the underlying מחלוקת? Since רב held like ר"מ regarding שיתוף במקום עירוב, *and* רב also held like ר"א בן תדאי. Doesn't that tell you that it's the same מחלוקת?" In other words, since רב held like two different Rabbis in two different places it must be that those Rabbis were discussing the same topic.

?!

To quote the ריטב"א: What does one thing have to do with the other? Why can't ר"מ and the רבנן be debating one topic, while ר"א בן תדאי and the רבנן are debating a different topic, and רב held like one here and one there, and those topics have nothing to do with each other?! We find examples of this everywhere!

The ריטב"א answers that רב יוסף's proof comes from the fact that it is really strange that רב held like ר"מ when he is a דעת יחיד against a רבים, and it is also strange that רב held like ר"א בן תדאי, who is also a דעת יחיד against a רבים.

Now, consider the two alternatives: Either ר"מ and ראב"ת are discussing the same thing, which רב held like, or ר"מ and ראב"ת are discussing two completely different topics and רב held like each of them.

To say the first way is to say that רב held like a שיטה that isn't a דעת יחיד (since there are at least two Rabbis saying it).

But to say the second way is to say that רב held like a דעת יחיד (strange!) *twice!* (really strange!)

So, the first alternative is more logical.

Also, we have a כלל that we're usually מיקל in עירובין[193]. So, even though the first way goes against this כלל, the second way is worse because it goes against this כלל twice!

In the face of these issues, the first way is more reasonable, and that is רב יוסף's proof.

That's the question and answer of the ריטב"א.

But I want to focus on a specific point, since it made a big impact on me.

If you look back to where I finished quoting רב יוסף with רש"י, you'll see that I put a big fat (?!), meaning, a gong went off in my head. I was astonished. I didn't understand what רש"י was saying. But I didn't know *why* I didn't understand it! This question of the ריטב"א was in the back of my mind, but I hadn't <u>formulated</u> it; I didn't put what was bothering me into question form.

Only after seeing the ריטב"א, I was like, Oh right! *That's* what was bothering me!

I've experienced this many times in my learning. It's very important when something is bothering you to

1) Realize <u>that</u> something is bothering you and
2) Identify <u>what</u> is bothering you

Be specific. Formulate your question in an attack style. "I don't understand that because... because I see elsewhere that it's not true!" Or something like that.

This same thing happened to me with the רש"י on יבמות ד: where רש"י said "Since the ratio of the circumferences is 3/4, so the ratio of their areas is 3/4." There too, I merely asked, "How does that follow?" instead of the stronger forms of questions like תוס' and Yishai asked over there: "We see [specific examples] where that's <u>not</u> true!"

It will go a long way in answering your question if you know exactly what your question is. This is good advice in general. The more exactly you can identify a problem, the more likely a solution will appear.

This is a little אגב, but I think it's relevant.

I don't know if this is the פשט in the זהר but this is how I understood it. It says[194] that when the people of מגדל בבל spoke לשון הקודש, the language being exactly

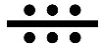
[193] מ"ו. וע' בדף פ'. בשם רבי ישמעאל

[194] זהר בראשית דף ע"ה:

accurate, whatever they wanted to do they expressed it, and since the language captured their intent exactly, it became obvious to them how to accomplish it[195].

As the English expression goes, "Knowing the problem is half the solution". In this case, it would be the *whole* solution!

This is similar to what I wrote above[196] about the בני יהודה being exacting in their speech.



*

"Transferring" Shituf to Eiruv, Eiruv to Shituf - ע"א: - ע"ב.

This was a bit of a complicated piece for me. I initially intended to just make a simple chart (the one on page 264) combining the מחלוקת of רבי מאיר and the חכמים with the מחלוקת אמוראים of פליגי ביין and פליגי בפת. But as I put the piece together, it got very complicated for me between Rashi and Tosfos, and then a misread comment in Artscroll which sent me into the Beis Yosef and the Bach. So now I'm just trying to sort it all out.

Let's try to present this as clearly as possible.

(For this piece I'm going to use more transliteration than usual to improve the flow of the text.

Also, the Gemara uses the word "לסמוך" "to rely on", as in: Can you "rely on" the Shituf in the place of Eiruv? I'm using the term "transfer", since that's one way to understand how this works. You're taking the חלות of the Shituf that exists in the Mavoy, and "transferring" that חלות to the Chatzer, giving you an effective or virtual Eiruv there.)

The גמרא is coming off of a sugya about if Shituf works in the place of Eiruv.

לכ"ע לכתחילה, an Eiruv between the houses of a Chatzer (or between two Chatzers) must be done with bread.

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[195] וזה לשונו של הזוהר מתורגם: וְאִם תֹּאמַר, לָמָּה הִתְבַּלְבְּלָה לְשׁוֹנָם? אֶלָּא מִשּׁוּם שֶׁכֻּלָּם מְדַבְּרִים בִּלְשׁוֹן הַקֹּדֶשׁ, אוֹתָהּ לָשׁוֹן עָשְׂתָה לָהֶם סִיּוּעַ. מִשּׁוּם שֶׁבְּמַעֲשֶׂה וּבְדִבּוּר הַפֶּה דְּבָרִים הַלָּלוּ תְּלוּיִים לְהַשִּׂיג כַּוָּנַת הַלֵּב, וּבָזֶה עוֹשִׂים סִיּוּעַ לְאוֹתוֹ מָקוֹם שֶׁרָצוּ לְהָקִים. וְעַל זֶה הִתְבַּלְבְּלָה לְשׁוֹנָם שֶׁלֹּא יָכְלוּ לְחַזֵּק רְצוֹנָם בִּלְשׁוֹן הַקֹּדֶשׁ. כְּיוָן שֶׁהִתְחַלְּפָה לְשׁוֹנָם, לֹא הִצְלִיחוּ בַּמַּעֲשֶׂה, מִשּׁוּם שֶׁהַצָּבָא שֶׁל מַעְלָה לֹא יוֹדְעִים וְלֹא מַכִּירִים - רַק אֶת לְשׁוֹן הַקֹּדֶשׁ. וּכְשֶׁהִתְבַּלְבְּלָה לְשׁוֹנָם, נֶחֱלַשׁ כֹּחָם וְנִשְׁבַּר תְּקָפָם.

[196] Page 153

לכ"ע, a Shituf of a Mavoy can be done with bread or wine[197].

What if the בני חצר forgot to do an Eiruv? Can they rely on the Shituf of the Mavoy? (At this point, I'm limiting the question to this; later we'll expand on it more.)

The ברייתא brings a מחלוקת between רבי מאיר and the חכמים (the רבנן). רבי מאיר says they can't whereas the חכמים say they can.

The question is, in which case are they arguing? What food item was used to make the Shituf? Bread or wine?

That is also a מחלוקת. An argument about an argument. Between ר' נחומי and רבה.

According to one of them, the argument between ר' מאיר and the רבנן is when the Shituf was done with wine (פליגי ביין).

According to ר' מאיר, it won't work because just like you can't use wine to make an Eiruv, so too you can't transfer the Shituf if it was made with wine.

According to the רבנן, you <u>can</u> transfer it. True, you couldn't have made the Eiruv לכתחילה with wine, but a transferred Shituf works, even if that Shituf was made with wine.

If it was done with bread then everyone agrees it can be transferred, since everyone agrees in principle that a Shituf can be transferred from the Mavoy to the Chatzer. The מחלוקת is only whether you can do that when the Shituf was made with a food item that wouldn't have worked לכתחילה for the Eiruv, i.e., with wine.

According to the other אמורא, the מחלוקת is when the Shituf was done with bread (פליגי בפת). The מחלוקת is whether transferring can be done at all, even if you had made the initial Shituf with bread.

ר' מאיר says no. Even though bread is good for an Eiruv, you can't transfer from one area to another.

The רבנן say you can. Transferring is allowed, as long as the food item used for the Shituf would've worked לכתחילה for the Eiruv, i.e., bread.

Let's make a concise chart showing the three possible opinions about this case, label where רבי מאיר and the רבנן are holding, according to both opinions of the אמוראים.

[197] Or any other food, although it seems that wine is the best. -תוס'

No, even if the Shituf was done with bread[198]	Only if the Shituf was done with bread[199]	Yes, even if the Shituf was done with wine[200]	
	ר"מ	רבנן	פליגי ביין
ר"מ	רבנן		פליגי בפת

פליגי ביין:

The רבנן say you can always rely on the שיתוף, even if the שיתוף was done with a food item that wouldn't have worked לכתחילה for עירוב, like wine.

ר"מ agrees in principle that you can use שיתוף for עירוב but it must have been done with a food item that would've worked for עירוב, i.e., bread.

פליגי בפת:

The רבנן hold that in principle you can use שיתוף for עירוב as long as the food item was one that could've been used for עירוב, i.e., bread.

ר"מ holds that you can never use שיתוף for עירוב, even if it was done with a food item that would've worked for עירוב (i.e., even bread), had you done so.

It seems that there are two separate concepts operating here.

A. סתם, the ability of a Shituf to transfer to the Eiruv.

B. (If you assume you can transfer) if you need a food item that would've worked לכתחילה for the other area.

[198] Perhaps since the intent was to use it for the Shituf so it cannot be used for the Eiruv. (Or perhaps since למעשה it wasn't used for the Eiruv, now it's too late.)

[199] It seems that according to this opinion, the Eiruv now is working upon the food item that was originally used for the Shituf. That being the case, wine is no good for an Eiruv.

[200] Vis-a-vis the Mavoy, the Shituf could've been with wine, which it was. Now that we have a good Shituf in place, we can use it for Eiruv and we don't care which food item it was made from. It could be that there's an underlying מחלוקת about which situation is likely to cause the kids to forget עירובין, as ר' יהודה said, "שלא לשכח תורת עירוב מן התינוקות"

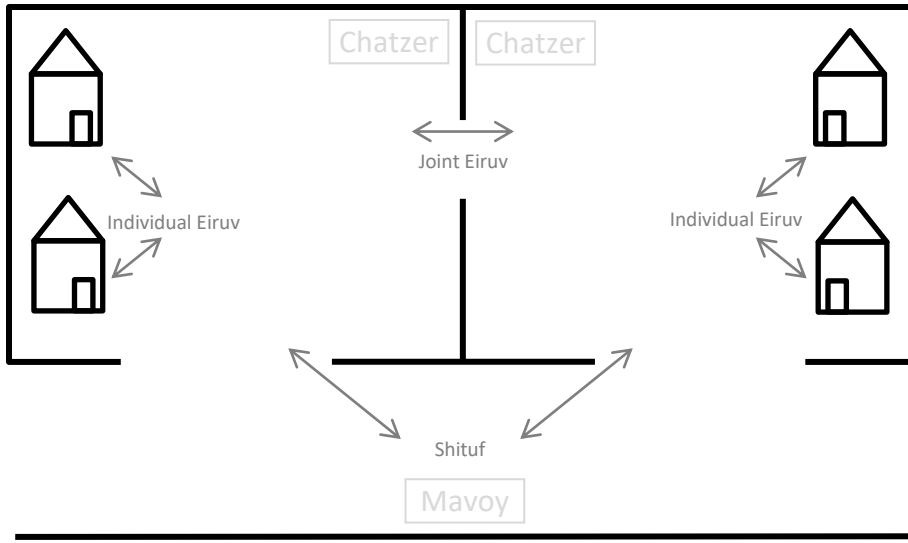
According to the first אמורה (פליגי ביין), "A" everyone is מיקל and agrees that it works, transferring is theoretically possible. The only מחלוקת is "B" whether you need the "good" food item or can you even use one that only worked here לכתחילה but not in the place to where you're transferring.

According to the second אמורה (פליגי בפת), "A" is the מחלוקת; "B" everyone is מחמיר and agrees it doesn't work (although ר' מאיר would never have a case where it actually matters, since there's no such thing as transferring).

(Theoretically, you can have a 3rd option, where they disagree on both points: ר' מאיר holds transferring is never possible and the רבנן hold not only can you transfer but even if the food item (wine of Shituf) wouldn't have worked by Eiruv. לכאורה, no one suggests that because we usually keep the מחלוקת as small as possible[201].)

Now, רש"י breaks up the cases into 3 parts:

1. Eiruv among the houses within each individual Chatzer. For short I'm going to call this **Individual Eiruv**, or internal Eiruv.
2. Eiruv between each Chatzer in the Mavoy to each other (through a direct opening between them, like a door or window, but not through the Mavoy itself). For short I'm going to call this **Joint Eiruv**.
3. Shituf of all the Chatzers within the Mavoy to each other, with and through the Mavoy. This will be called **Shituf**.



[201] ע' שיירי כנסת הגדולה בכללי התלמוד בסוף מסכת ברכות אות ע"ד

ע"א: - ע"ב.

First of all, רש"י clarifies that the whole discussion in our גמרא is about the **Shituf** and the **Joint Eiruv**. The **Individual Eiruv**, while relevant, isn't the Eiruv that the גמרא is discussing.

Secondly, רש"י expands the discussion to include not only transferring the Shituf to the Joint Eiruv, but also the other way, transferring the Joint Eiruv to the Shituf.

And, רש"י makes the מחלוקת dependent on whether or not there were Individual Eiruvs done within each individual Chatzer. How to understand it depends on how you read רש"י.

The first way to read it (which I think is the correct way) is that the whole מחלוקת is only when the individual Chatzers didn't do their own internal Eiruvs. If there were internal Eiruvs then everyone agrees that transference is possible either way, from Shituf to Eiruv and Eiruv to Shituf.

Another way to read רש"י is that according to רבי מאיר, you can never transfer Eiruv to Shituf, whether or not there were Individual Eiruvs. The only time it would make a difference is when trying to transfer from Shituf to Eiruv: If there were no Individual Eiruvs then רבי מאיר doesn't allow transference; if there were Individual Eiruvs then he does allow.

וזה לשון רש"י:

> **מערבין בחצירות.** הפתוחות מזו לזו אע"פ שעירבו דרך פתחיהן צריכין להשתתף במבוי להתיר המבוי ואם לאו חצירות מותרות מזו לזו ומבוי אסור דאין סומכין על עירוב במקום שיתוף ולא על שיתוף במקום עירוב היכא דלא עירבו בני חצר לעצמן שאם יסמכו על זה תורת עירוב ישתכח אבל עירבה כל חצר לעצמה ונשתתפו כולם במבוי אע"פ שלא עירבו חצירות זו עם זו שיתוף המבוי מחברן ומטלטלין מזו לזו בין דרך המבוי בין דרך פתחים דתנן שהמבוי לחצירות כחצירות לבתים מה עירוב החצר משוי לבתים חדא אף שיתוף המבוי משוי חצירות חדא:

> **(רבי מאיר:) We are ערב in Chatzers -** ...that are open to each other. [And] even though they were joined through their openings [to each other], still you need a Shituf in the Mavoy to מתיר the Mavoy. And if you don't [do a Shituf], so the Chatzers [remain] permitted from one to the other, but the Mavoy is אסור [i.e., you can't carry from one Chatzer into the Mavoy, or back], because [רבי מאיר holds] we can't use Eiruv in the place of Shituf, and you can't use Shituf in the place of Eiruv.
>
> [First way to read Rashi:] All this is talking about...
>
> [Second way to read Rashi:] This that you can't use Shituf in the place of Eiruv is talking about...

...where there weren't Individual Eiruvs done within each Chatzer. Since, if they would rely on this then תורת עירוב will be forgotten [since the kids only saw one Eiruv between one Chatzer and the next].

But, if each Chatzer *had* done their own internal Individual Eiruv and they did a Shituf for the Mavoy, even though they hadn't done Eiruv from one Chatzer to the next, still the Shituf of the Mavoy joins them all together, and you can carry from one [Chatzer] to the other, whether [i.e., even] through the Mavoy, whether [i.e., besides] through their [direct] openings [to each other], as we learned in the Mishna...

שיטת ר' מאיר:

- **No Individual Eiruv**:
  - And there was a **Joint Eiruv** but **no Shituf**:
    - Cannot rely on the Joint Eiruv in the place of Shituf. (That is, they cannot carry from one Chatzer to the next through the Mavoy. They can still carry from one Chatzer directly to the next through a door or window based on the Joint Eiruv.)
  - Or there was a **Shituf** but **no Joint Eiruv**:
    - Cannot rely on the Shituf in the place of the Eiruv. (That is, they cannot carry directly from one Chatzer to the next through a door or window. They can still carry through the Mavoy based on the Shituf.)
- There **were Individual Eiruvs**:
  - And there **was a Shituf** but **no Joint Eiruv**:
    - Can rely on the Shituf in the place of the Eiruv. (That is, they can carry directly from one Chatzer to the next through a door or window, besides being able to carry through the Mavoy.)
  - רש"י isn't explicate about what would be the דין if there **was a Joint Eiruv** but **no Shituf**. Would they be able to rely on the Eiruv in the place of Shituf? (That is, can they carry from one Chatzer to the next through the Mavoy, besides directly through a door or window?)

Whichever way you read רש"י, if they had done Individual Eiruvs within each Chatzer then even רבי מאיר allows transferring from Shituf to Eiruv (between the Chatzers).

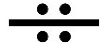
The only question is does רבי מאיר even allow transferring the other way, from Eiruv to Shituf. According to the first way we read רש"י, yes; according to the second way, no.

רש"י later[202] says that according to the שיטה of פליגי ביין, everyone (רבי מאיר and the רבנן) agrees that Eiruv can be transferred to Shituf, since everyone agrees that transference is possible; the only מחלוקת is if a "weak" food item (i.e., wine) used for a Shituf can be "upgraded" and work for the Eiruv as well. But Eiruv, which is always made initially with bread, of course can work for the Shituf, לכולי עלמא.

Now, the interesting thing is that רש"י here[203] is going on the line in the ברייתא about רבי מאיר before the גמרא brings the מחלוקת אמוראים. If so then we would expect Rashi's explanation of רבי מאיר to go like both אמוראים. But it seems as if רש"י is only going in פליגי בפת. For example, in the case where there were no Individual Eiruvs, and there was a Joint Eiruv but no Shituf, רש"י said that רבי מאיר won't allow transferring. But according to פליגי ביין everyone agrees that Eiruv can *always* transfer to Shituf!

So, what's going on?

Indeed, I found a footnote in the back of the מתיבתא that also points this out; that Rashi seems to be going in פליגי בפת.



We mentioned that רש"י writes that according to פליגי ביין, both רבי מאיר and the רבנן agree that transference is always possible from Eiruv to Shituf.

תוספות asks two questions on רש"י.

1) The לשון in רבי מאיר was "מערבין בחצירות ומשתתפין במבוי". According to רש"י, according to פליגי ביין, we would have to read Rebbe Meir's words as "Mi'arvin biChatzeros [even though you were already] Mishtatfin biMavoy". And that doesn't read so well in his words.
2) Later (73b), the גמרא brings a משנה about 5 Chatzers that were open to each other and to a Mavoy, and the inhabitants had done an Eiruv but not a

[202] ד"ה פליגי ביין
[203] ד"ה מערבין בחצירות

Shituf, which at this point רש"י in פליגי ביין would say is good according to everyone, even רבי מאיר, but the משנה says that only between the Chatzers is מותר, but the Mavoy is אסור, (which isn't like anyone here in רש"י in פליגי ביין) but the גמרא there says that that משנה is רבי מאיר. קשיא on רש"י. (תוס' suggests for רש"י that אה"נ, that גמרא is going in פליגי בפת)

תוס' brings the ר"י who disagrees with רש"י and says that even according to פליגי ביין, רבי מאיר doesn't allow transferring from Eiruv to Shituf. He says that the reason transference works from Shituf to Eiruv is because

3) the people who did the Shituf boosted up the power of their Shituf by doing it with bread. Shituf can normally be done with any food item; by doing it with bread they made it more like an Eiruv. So when it comes to transferring to the Eiruv, it's already on its way, so to speak. משא"כ when they set out to do an Eiruv: they did it with bread because they <u>had</u> to do it with bread; they didn't have a choice. So there's nothing in their action that shows that they intended on transferring this, so it doesn't work.
4) Another reason to differentiate is that Shituf is, by nature, more encompassing, since it usually joins Chatzers <u>and</u> a Mavoy, משא"כ an Eiruv, even a Joint Eiruv, is only usually joining houses or Chatzers together and not a Mavoy and so just because transference works from Shituf to Eiruv, it doesn't necessarily work the other way, from Eiruv to Shituf.

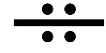
First of all, let's see how the 2 קושיות that תוס' asked on רש"י aren't relevant to the פשט of the ר"י.

The first question on רש"י was that the לשון of רבי מאיר doesn't flow so good. The way the ר"י is learning, רבי מאיר does indeed hold that "[even though one did] Eiruv for the Chatzers [one must still do] Shituf for the Mavoy", which is more the משמעות of the words.

My question on that is that פליגי ביין <u>means</u> they're arguing when wine was used, which is only possible by Mavoy, so however you explain פליגי ביין, whether like רש"י or the ר"י, you still need to read the words of רבי מאיר backwards, that is: "Make an Eiruv for the Chatzers [even though you had made] a Shituf on the Mavoy [with wine]".

?

The second question תוס' asked on רש"י, from the case of the 5 Chatzers, is good according to the ר"י since that's exactly what he holds, that both פליגי בפת <u>and</u> פליגי ביין hold that רבי מאיר doesn't allow Eiruv to work for Shituf in the Mavoy.



Now, according to the ר"י, what is שיטת רבנן regarding transferring from Eiruv to Shituf (in פליגי ביין)?

So תוס' writes that according to the ר"י, the רבנן do allow from Eiruv to Shituf, meaning, that they argue with רבי מאיר there as well. So, when פליגי ביין said they argue by wine and not by bread, he meant from Shituf to Eiruv. But from Eiruv to Shituf they argue by bread as well.

Let's see how each קשיא and ראיה that תוס' brought[204] has relevance to this last question (namely, does the ר"י hold that the רבנן allow Eiruv to Shituf in פליגי ביין?).

The first קשיא that תוס' asked on רש"י was that the לשון in רבי מאיר didn't read so well. לכאורה, according to the ר"י, the לשון doesn't bear on this question, since, like we said before, even the ר"י has to read these words of רב"י מאיר as going on Shituf to Eiruv, no? Unless if we want to say that what רבי מאיר meant was "[Even though you] had done Eiruv in the Chatzers [you still need to] do Shituf in the Mavoy [since the only time bread can transfer is when it was used in the Mavoy as a Shituf and you want to transfer it to the Chatzers]. But the רבנן disagree and say you can either [use bread and] do an Eiruv in the Chatzers [and transfer it to the Mavoy] or you can [even] do a Shituf [with wine and transfer it to the Chatzers]".

The second קשיא of תוס' on רש"י was from the case of the 5 Chatzers. The גמרא there said that only רבי מאיר would not allow transferring from the Eiruv to Mavoy, meaning that the רבנן would allow. And we brought ר"י שיטת to explain that גמרא according to both אמוראים, פליגי בפת and פליגי ביין, so לכאורה that גמרא is a ראיה that the ר"י holds the רבנן would allow transferring from Eiruv to Mavoy.

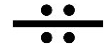
The third point we brought (labeled "3" above, which was the first ראיה of the ר"י) was a סברא to explain why transferring bread from Shituf to Eiruv was better than the other way, from Eiruv to Shituf, since the people "strengthened" their Shituf by using bread which was unnecessary. But that was רבי מאיר. The רבנן never required "boosting up" your Shituf; they allow it always. So what's the difference by Eiruv to Shituf? The reason the ר"י is saying that wouldn't work according to רבי מאיר is that it's lacking being boosted up; but according to the רבנן it's not necessary to boost it up, so of course it works.

The fourth point we brought (labeled "4", appropriately) was a סברא to differentiate in general between transferring from Shituf to Eiruv versus going the other way from Eiruv to Shituf. The סברא was that Shituf, by its nature, is a more encompassing mechanism. It joins together Mavoys and Chatzers. So, we understand why it should be transferrable to the Chatzers, areas that are usually included in a Shituf anyway.

[204] Which we labeled above as 1 to 4

But an Eiruv usually only joins Chatzers together (or houses), but not Mavoys. So where does this Eiruv get the power from to work in the Mavoy?

Now, לכאורה, this סברא is a good סברא whether you are רבי מאיר or the רבנן. So, I don't see how we could infer from this point alone what the ר"י holds regarding the רבנן. But I guess you could say that that גופא is the מחלוקת, whether we consider the Shituf as a more encompassing mechanism or not.



		פליגי ביין		פליגי בפת	
		Bread	Wine	Bread	Wine
No Individual Eiruv	Did Joint Eiruv, Can Transfer to Shituf?	✔ - רש"י ✘ – ר"מ - ר"י ✔ – רבנן	-	✘ - רבי מאיר ✔ - רבנן	-
	Did Shituf, Can Transfer to Joint Eiruv?	✔	✘ - רבי מאיר ✔ - רבנן	✘ - רבי מאיר ✔ - רבנן	✘
Did Individual Eiruv	Did Joint Eiruv, Can Transfer to Shituf?	✔ (or maybe ר"מ holds רש"י doesn't allow?)	-	✔	-
	Did Shituf, Can Transfer to Joint Eiruv?	✔	✔	✔	✘

Let's see the טור[205]:

> "We can rely on Eiruv for the Shituf and on the Shituf for the Eiruv. For example, if there were five Chatzers open to each other and to the Mavoy, and the Chatzers did an Eiruv through their openings between them they don't need a Shituf between them. And so too if they did a Shituf, even with wine (which isn't [usually] fitting to do an Eiruv within the Chatzers) they don't need to do an Eiruv between the Chatzers."

The טור for sure holds like פליגי ביין. He doesn't qualify whether there were Individual Eiruvs or not. If there weren't then he is going like the רבנן, and if there were then he could even be like ר"מ.

[205] א"ח שפ"ז

(The טור also brings the רמב"ם that requires bread to go from Shituf to Eiruv, which is either like the רבנן in פליגי בפת, or like ר"מ in פליגי ביין, assuming we're talking about where there were no Individual Eiruvs done. But if the רמב"ם was talking about where there were Individual Eiruvs done then he'd be going לכולי עלמא in פליגי בפת.)

בית יוסף:

> The ראש says that the ראב"ד holds like פליגי בפת, and the רז"ה (the בעל המאור) holds like פליגי ביין, and the ראש and the מרדכי go like the רז"ה,
>
> | פליגי ביין | פליגי בפת |
> |---|---|
> | בעל המאור
ראש
מרדכי | ראב"ד |
>
> and they both [i.e., the ראש and the מרדכי] continue to write that the הלכה is like רבי יוחנן, who says "נהגו" ("conduct") like רבי מאיר, and not "מורינן" ("teach") like רבי מאיר but rather like the רבנן, like הגהות אשיר"י..., and they continue to write: "הלכה like רבי מאיר". But in a situation of מצוה you don't need both (Eiruv and Shituf), like אביי said, "Ah, but they didn't do an Eiruv?" He said back, "We can rely on the Shituf." And also if they did Shituf with bread they can rely on that for the Eiruv.[206]"
>
> On this ברייתא (on the line of ר"מ): **מערבין בחצרות ומשתתפין במבוי,**
>
> רש"י explains ...[quotes רש"י] ... and these are the words of תוס' in פרק הדר (73b ד"ה כל שיתוף)...
>
> ...And according to this, according to everyone, every Chatzer needs to do an Individual Eiruv, and if they didn't then there isn't an opinion that we can rely on the Shituf..."

I'm not really sure how to understand this last line in the ב"י. It sounds like it's saying that every Chatzer needs to do an Individual Eiruv in order to transfer the Shituf to the Joint Eiruv. The way we were learning רש"י, the whole מחלוקת was גופה where there were no Individual Eiruvs done. If there <u>were</u> Individual Eiruvs then even <u>רבי מאיר</u> allows transferring the Shituf to the Joint Eiruv.

תאקע, I see that the הגהות והערות in the שירת דבורה Publication of the טור points out that it doesn't seem from רש"י like the ב"י, and that's not how the אחרונים are learning רש"י.

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[206] Is this a repeat of the previous sentence? Or does he mean in a place of non-מצוה you need bread to transfer the Shituf to the Eiruv?

> But רש"י and these Great Ones (רוובתא) that the ב"י brought seem to say that even ר"מ only argues where there weren't Individual Eiruvs nor Joint Eiruvs, because of the fear that the kids will forget תורת העירוב, but where there were Individual Eiruvs, just not a Joint Eiruv, where this problem of forgetting no longer exists, even ר"מ agrees that we can rely on Shituf in such a case. But the רבנן, who don't even start with this חשש, even <u>without</u> Individual Eiruvs, they can rely on Shituf for the Joint Eiruv...

Now, I'm not holding in all the שיטות, but סתם I'm wondering maybe the ב"י means that after taking into consideration all the פוסקים, it **comes out** that no one allows Shituf to Joint Eiruv without Individual Eiruvs, but אין הכי נמי, that's not necessarily how to learn the גמרא. Hmm?

I'm also wondering if the Artscroll understood the ב"י differently. They write (quoting the ב"י) that the Shituf can't ever help for the **Individual** Eiruvs. So then the ב"י meant "...there isn't an opinion that we can rely on the Shituf for the [Individual] Eiruvs".

Anyway, let's see the ב"ח:

> "This that the טור wrote means, if there was a **Joint Eiruv** then they can use that for the Mavoy. If there was no Joint Eiruv (either because they didn't join the Chatzers through their connecting doorways, or because there <u>were no</u> connecting doorways) then they have no Shituf, even if there were **Individual Eiruvs**."

If we assume that the last line ("even if there were Individual Eiruvs") means there <u>were</u> indeed Individual Eiruvs since the beginning of the paragraph then this could be רש"י לכולי עלמא (except for the second way we read רש"י above in which ר"מ in פליגי ביין won't allow transferring).

Or, it could be we were talking about where there weren't Individual Eiruvs; the ב"ח just means that when there is no Joint Eiruv and no Shituf, you have nothing, even if you change the case and make it that there were Individual Eiruvs, you still have no automatic Joint Eiruv nor Shituf.

If so then when the ב"ח says we can transfer the Joint Eiruv to the Shituf, and since the טור holds like פליגי ביין[207] then:

> If we hold like רש"י then this is like all the תנאים
>
> and if we hold like the ר"י then this is like the רבנן.

The ב"ח continues:

> "Also, where they had done Shituf [but no Joint Eiruv], also there the טור holds like the רבנן, that you don't need the Joint Eiruv, even if you had no Individual Eiruvs, (because if you had Individual Eiruvs, even רבי מאיר agrees you can rely on the Shituf for the Joint Eiruvs)".

So, now we see the ב"ח is learning that the טור was going like the רבנן the whole time.

> "Rather, only where there was no Eiruv, not Individual nor Joint, is there a מחלוקת, and we teach the הלכה like the רבנן, that it's enough for Shituf and we don't need Eiruv at all...
>
> and the בית יוסף wrote it inverted and it's a complete mistake that slipped under his pen, accidently."

In other words, we see the ב"ח clearly learning the first way we learned רש"י, that the whole מחלוקת is only when there was no Individual Eiruv, but if there were Individual Eiruvs then everyone (רבי מאיר and the רבנן) agrees that you can rely on Shituf for the Joint Eiruv, and even Joint Eiruv for the Shituf.

And it seems, לכאורה, that the ב"ח understood the ב"י the way we originally read it.

What's about the שלחן ערוך?

> At first, the שלחן ערוך writes not to rely on Shituf for the Eiruv, but then qualifies it that if they did the Shituf with bread then they don't need to do an Eiruv, since the kids recognize bread [as an Eiruv].

We'll see in a second that this is where there weren't Individual Eiruvs. So then according to our chart above, this is either רבנן in פליגי בפת or רבי מאיר in פליגי ביין.

> Then the שלחן ערוך brings a יש אומרים that you need Individual Eiruvs to rely on Shituf for the [Joint] Eiruv.

[207] As we wrote at the end of page 271

According to our chart, this would then seem to be רבי מאיר in פליגי בפת (since there is no other שיטה with bread, Shituf to Eiruv, that differentiates between No or Yes Individual Eiruvs). And indeed, the באר הגולה sources this in רש"י (the basis of our chart), but as we pointed out, that seems to be שיטת רבי מאיר.

But, the שלחן ערוך is לשיטתו, since he's learning that רש"י is talking even in the רבנן.

The ביאור הלכה brings ב"ח and "other אחרונים" who reject this יש אומרים, since "everyone allows transferring Shituf of bread even if there were no Individual Eiruvs."

לכאורה, that's in פליגי ביין.

The רמ"א brings another י"א that you can transfer a Shituf to the Chatzer even if the Shituf was done with wine.

This seems to be פליגי ביין. If no Individual Eiruv then this is the רבנן, and if this is with Individual Eiruvs then this would be like all the תנאים.

OK, I spent more than half a year on this piece on and off; I think it's time to move on. If someone can sit down with me and clarify whatever I've got wrong, you know, please, be my guest.

Speaking of which…

*

ע"ב - <u>Mechitza of Hotel Guests</u>

The משנה brings a מחלוקת between ב"ש and ב"ה regarding guests who stay over in an inn for שבת. The question is whether the various groups of guests in different rooms or sections of the hotel can be מערב together or whether each group needs its own individual עירוב.

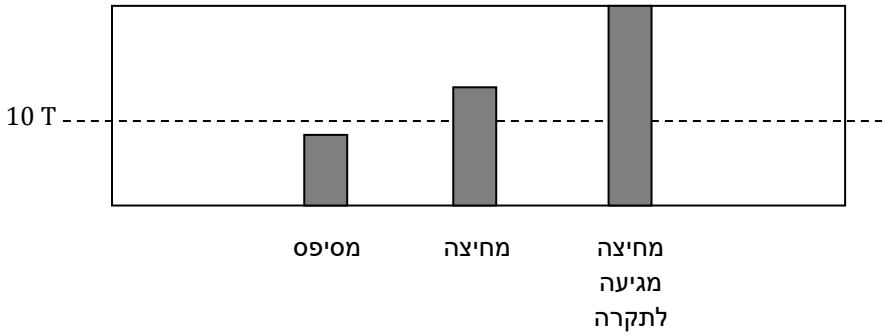
ב"ש says they each need to do their own עירוב whereas ב"ה says they can do a single עירוב for everyone.

The גמרא discusses three types of physical divisions between the groups:

1) מסיפס, a low partition, less than 10 טפחים high

2) מחיצה, a medium sized partition, at least 10 טפחים high but not reaching the ceiling

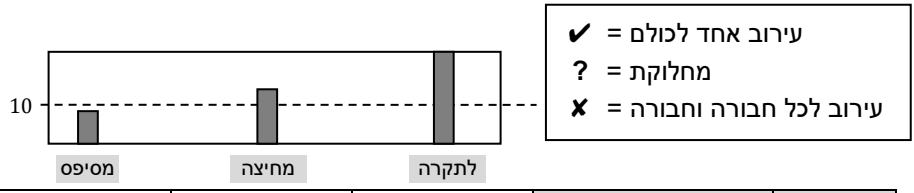
3) מחיצה מגיעה לתקרה, a high partition, reaching the ceiling



The גמרא then brings five interpretations of the מחלוקת between ב"ש and ב"ה vis-a-vis these different types of מחיצות.

The first 4 interpretations are of אמוראים, and the fifth one is that of a תנא. The גמרא compares each of the Amoraic interpretations to the Tanna's to see which ones are consistent with it or are contradicted by it.

Let's lay out the chart and then step by step through the גמרא while keeping an eye on the chart.



?	✘		רב נחמן #1	1
?	?		רב נחמן #2	2
	✔	?	חד אמר	3
	?	✘	וחד אמר	4
(✔) ← [?]	?	✘	רבי יהודה הסבר	5

1) First, the גמרא brings one version of רב נחמן who says that the מחלוקת of ש"ב and ה"ב is by מסיפס but by מחיצה של י' טפחים everyone agrees each חבורה needs their own עירוב[208].

2) Then, the גמרא brings an alternate version[209] of ר"נ who says that ש"ב and ה"ב argue in both cases, מסיפס and מחיצות of 10 tefachim.

Then, the גמרא brings two more opinions (#3 and #4), one of them being רבי חייא and the other one רבי שמעון ברבי.

3) One of them[210] says the מחלוקת of ש"ב and ה"ב is by מחיצות that go all the way up to the ceiling, but if they don't reach the ceiling (even if they are 10 T high[211]) then ש"ב and ה"ב agree that all the guests are considered one group and use a single עירוב.

4) The other one says the מחלוקת of ש"ב and ה"ב is when the מחיצות don't reach the ceiling, but when they do then ש"ב and ה"ב agree that these are separate groups and each group needs their own עירוב.

5) Then, the גמרא brings a ברייתא in the name of רבי יהודה הסבר who says just like #4 above, that the מחלוקת of ש"ב and ה"ב is by מחיצות י' טפחים, but by מגיעות לתקרה ד"ה עירוב לכל חבורה וחבורה.

Now, the גמרא compares and contrasts each אמורא to the statement of רבי יהודה.

#3 is out, since that interpretation disagrees with רבי יהודה on two accounts, both by מחיצות 10 T and מגיעות לתקרה.

#4 is good since it's the same as רבי יהודה.

#1, the first version of רב נחמן, clashes with רבי יהודה regarding מחיצות 10 T (ר"נ said כ"ע עירוב לכל חבורה וחבורה)

The גמרא suggests that maybe the second version of ר"נ (#2, the איכא דאמרי) clashes with רבי יהודה, since רבי יהודה said that ש"ב and ה"ב argue by מחיצות, which implies that by מסיפס they agree that the groups are all together and use one עירובת, whereas ר"נ said they argue there as well.

The גמרא answers that ר"נ would say that רבי יהודה never meant to make that implication but rather that ש"ב and ה"ב argue by מסיפס as well.

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[208] In other words, such a significant partition effectively separates the different חבורות into different groups

[209] איכא דאמרי

[210] The גמרא doesn't say who

[211] רש"י ד"ה אבל מחיצות

Ah, so why did רבי יהודה say they argue by מחיצות?[212]

To show you how far ב"ה is מקיל and is מתיר, that they even allow a single עירוב[213], even though there is somewhat of a מחיצה (of 10 T) between the rooms.

*

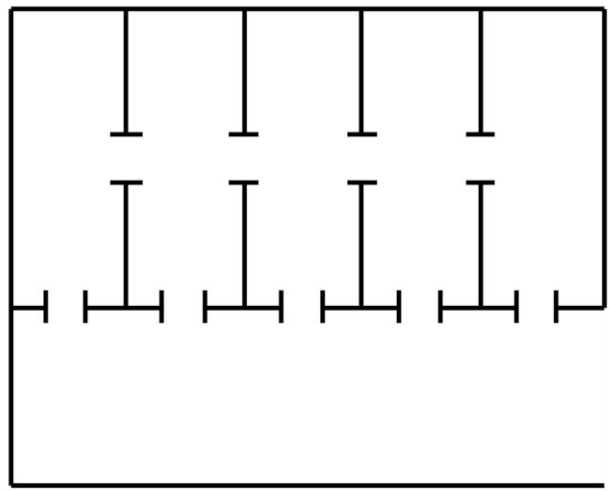
ע"ג. - Interconnected Courtyards

מהדורת נהרדעה has an introduction in which they trace the history of diagrams and pictures that appear in the text of the גמרא and מפרשים, and what might happen as you copy from one source to another. They point out that whereas by text it's a relatively straightforward process of identifying the letter and copying it, when it comes to pictures, you have to be very careful, since the meaning of the original picture can get garbled if salient aspects are not copied correctly.

Here, on this page, the point of the picture remains valid, but I just thought to point out one aspect of the picture that may cause someone to make a mistake or perhaps would change the דין.

The משנה is discussing a mavoy with five courtyards opening up into it, and these courtyards are also open directly to each other.

In my particular edition of עוז והדר (from 2006), they[214] have the following picture[215] in רש"י:



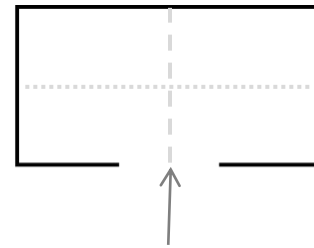
[212] If they argue by מסיפס as well then why didn't he tell us that חידוש?
[213] Which is less work to make, plus it allows the groups to carry from one room to the other
[214] Actually, I haven't seen a גמרא that shows it differently, except without the "serifs"
[215] I hope I'm not making the same error of copying it wrong!

I said, hey wait a second! Those aren't Chatzers; those are Mavoys! Chatzers need to be wider than their length, as it says on 12b:

אמר רב נחמן ... אי ארכה יתר על רחבה הוה ליה מבוי ומבוי בלחי וקורה סגיא ואי לא הוה לה חצר

> "Rav Nachman says... if its length is more than its width then it's a Mavoy, (and a Lechi or Korah is enough for a Mavoy), otherwise it's a Chatzer"

which I understood to mean that when you walk into them, their far wall needs to be closer to you than their right and left walls are from each other, in order to be a Chatzer.

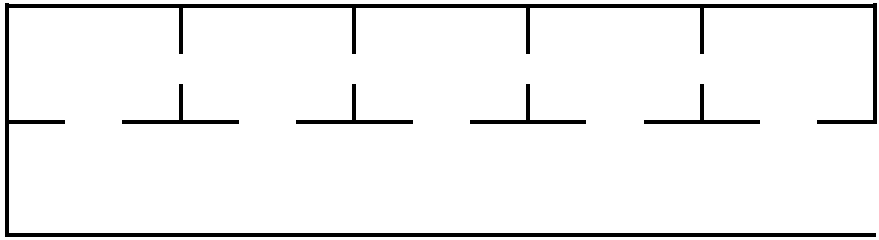


Indeed, the מגיד משנה on the רמב"ם הלכות שבת י"ז:ז' writes:

וכתב הרשב"א ז"ל זה הוא ארכו של מבוי זהו כניסתו ומה שאינו דרך כניסתו קרוי רחבו ואע"פ שרחבו יתר מארכו...

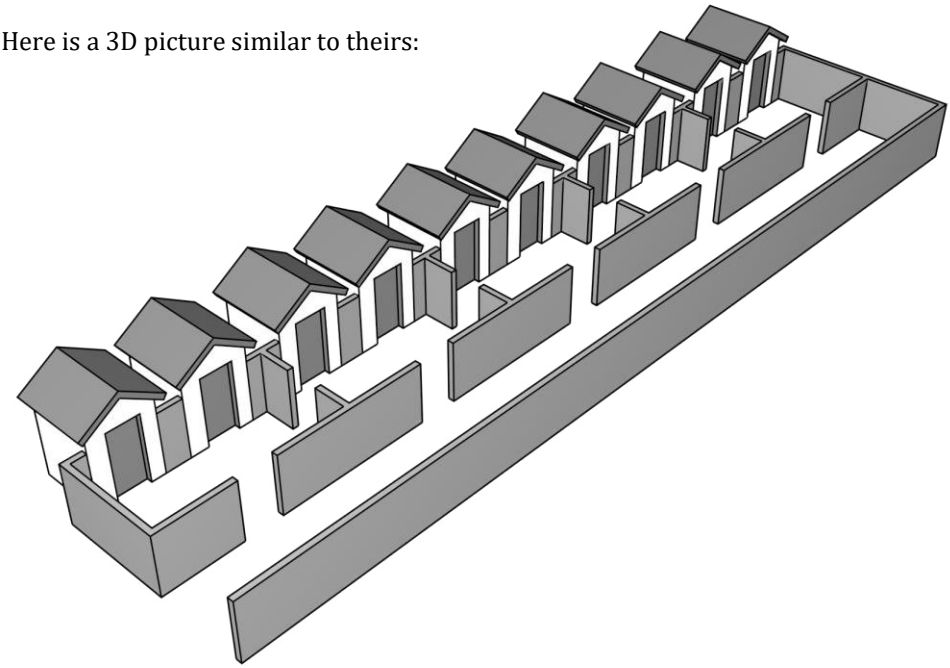
> "And the רשב"א writes that the length of a Mavoy is [determined by] its entrance, and that which isn't the way of its entrance is called its width, even if its width is more than its 'length'..."

Perhaps a more accurate picture would be:



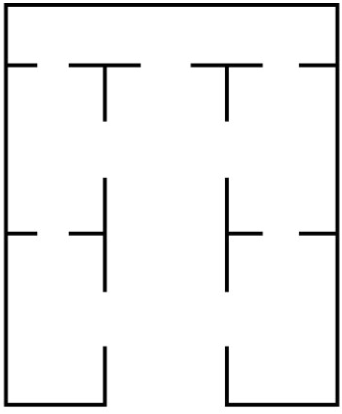
like עוז והדר has in the back in their 3-D pictures.

Here is a 3D picture similar to theirs:

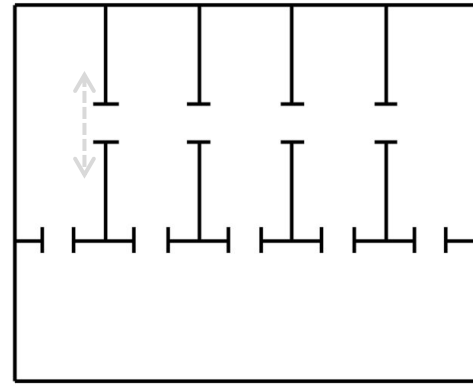


Or maybe we could draw it like this:

(I'm trying to keep the layout similar to like in תוס' 74b, as many of the picture books on Eruvin show Mavoys in general surrounded by Chatzers)



Or, perhaps in this case it's an exception since the courtyards have entrances in their sides. Perhaps we can look at the "width" of the courtyard as the right and left of those entrances.



Doesn't seem so מסתבר to me since it would seem that you'd determine the "entrance" by the way in from the מבוי, since it's the shortest route to the רשות הרבים in terms of how many areas you need to path through.

Would this affect the דין here? The משנה here is discussing if they forgot to do an עירוב or a שיטוף, if they can still carry through the various areas. The fact that a חצר is shaped like a מבוי, would that change how we can work with the various areas regarding עירוב and שיטוף?

For example, we said earlier[216] that a שיטוף of a מבוי can be done with wine. Would we be able to do a "שיטוף" with wine on a חצר that's shaped like a מבוי?

I haven't seen anybody discuss this.

Perhaps the whole difference between the shape of a מבוי or חצר is only relevant to the question of using a לחי or קורה.

[216] Above, on page 262, going on דף ע"א:

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ע"ה: - **Forgetful Inhabitants - Party Time-Machine Mashul**

The ברייתא says that if the outer חצר and the inner חצר did an עירוב together and placed the עירוב in the inner חצר, and one of the residents of the outer חצר forgot to join in, the דין is:

- רבי עקיבא says both courtyards are אסור,
- whereas the חכמים say the outer חצר is אסור but the inner one is מותר (I.e., the inner חצר is מעורבת and they can carry from their houses to each others' houses through the חצר).

The גמרא wants to know what's the difference between רבי עקיבא and the חכמים.

רבה בר חנן has a conversation with אביי that effectively goes like this:

- Why do the רבנן allow the inner residents?
  - Since they can close the door[217].
- According to רבי עקיבא, also close the door! (-?-)
  - The outer residents' עירוב keeps them in the picture[218]
- So the חכמים also! The outers' עירוב keeps them in the picture! (-?-)
  - The inners say to the outers, "We joined with you to help us; not to hurt us."[219]
- So let רבי עקיבא's inners say "We joined with you to help us; not to hurt us"! (-?)
  - No, since the outers could be מבטל now (and until they do, the inner courtyard remains אסור)
- So let the חכמים also require ביטול! (-?-)
  - No, since they hold you can't be מבטל from רשות לרשות. So we have no choice but to say מסתלק (that the outers are automatically removed), so already now the inners are מותר.

[217] That is, they remove the outer חצר from the picture (figuratively). - רש"י

[218] Literal translation: makes them prevalent

[219] The outers are אסור either way, so we remove them from the picture.

One way to explain this is that this is like a למפרע, that we never joined up with you in the first place.

It doesn't have to be למפרע ממש, it could be that the עירוב is made on condition that no one messes it up, and if someone does then that person's whole חצר is not included in the עירוב.

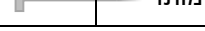
The reason that I prefer to explain that it's actually a למפרע ממש is because that's the whole point of the מחלוקת between רבי עקיבא and the חכמים, since activating a למפרע is a בדיעבד mechanism, we only do that if we can't do an "easier" method, like ביטול רשות. Since רבי עקיבא holds we <u>can</u> do a ביטול רשות so we cannot do a הסתקלות (למפרע), and the חצר remains אסור until we do a ביטול. The חכמים, on the other hand, hold we can't do a ביטול and so we need to come on to the הסתלקות למפרע, and therefore the inner חצר is מותר right away.

And now for the משל:

> Jonny (the חיצונים, the "outers") showed up to the party (was מערב with the inner חצר) and he spilled the punch (one of the outers forgot to join in the עירוב). So we say to him, "Hey, we didn't invite you to our party to mess things up so we're gonna flip this switch here on our time machine, which will go back in time and <u>not</u> invite you! (הסתלקות למפרע)
>
> One second guys. Before you start flipping switches and going back in time, do we have a mop? (Can he do ביטול?)
>
> If so, then we can't start messing with the fabric of time and uninvite him, rather, we have to clean up this punch (do ביטול) and until we do, this party is messed up (inner חצר is אסור).
>
> If we don't have a mop (ביטול מרשות לרשות no, שיטת חכמים) then we'll have to come on to the back-in-time-never-invited-you-to-mess-up-my-party switch[220] (הסתלקות למפרע).

ביטול is לכתחילה (if you hold it works at all), and as long as there is the possibility for ביטול, we don't say תסתלק, and both Chatzers will remain אסור until we do ביטול.

If you don't allow ביטול, then we say תסתלק, which works right away (at שקיעה, or whenever the עירוב is חל).

	עירוב made (one outer forgot)	Evening, חלות העירוב	Night	ביטול	
רבי עקיבא			אסור		מותר
חכמים			מותר		ביטול not possible

[220] Or B.I.T. N.I.Y.T. M.U.M.P.S. for short

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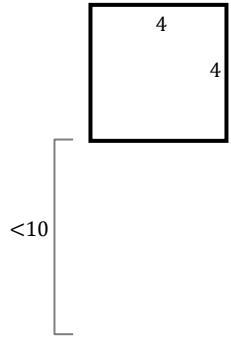
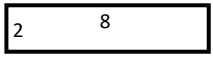
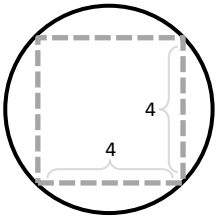
ע"ו. - Round Window



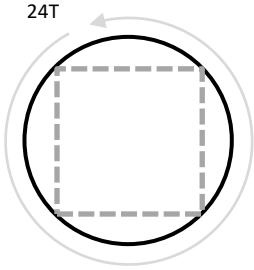
The משנה says you need to have a window[221] between two Chatzers if you want to be able to be מערב them. The window needs to be at least 4 x 4 טפחים, and that at least part of the window's opening needs to be within 10 טפחים from the ground.

It's not enough to have the same area as a 4x4 square (for example: 2x8); rather you need to have an actual 4x4 square (at least).





If, for example, the window is circular then it needs to big enough that it encompasses within it at least a 4x4 square.

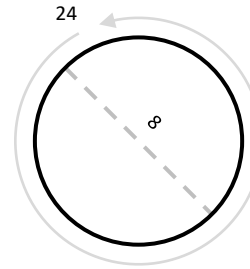


רבי יוחנן says that in order to achieve this, the round window needs to have a circumference of **24 טפחים.**

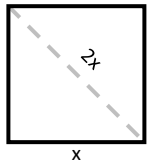
How so?

[221] At least, if you don't have any other opening like a proper doorway.

With תוספות, we get the following explanation:

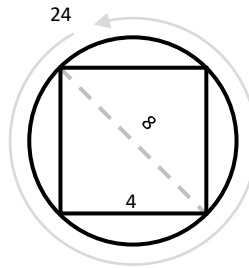


Using 3 for pi, a circle with a circumference of 24 will have a diameter of 8 (= 24 / 3).[222]



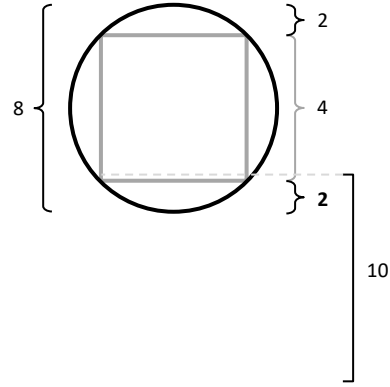
As the גמרא will conclude, רבי יוחנן is assuming that the diagonal of a square is always twice the length of any of its sides[223].

Therefore, the square inside a circle that's 24 around will have sides of 4.



רבי יוחנן continues that to ensure that the conceptual 4x4 square is within 10T from the ground we need at least **2T of the circle to be within 10T from the ground.**

תוספות explains that the 2 טפחים are the bottom 2T of the circle's height (which is the same as the diameter, which we said was 8T). If those 2T are within 10T from the ground then the bottom of the 4x4 square is also within 10T from the ground.



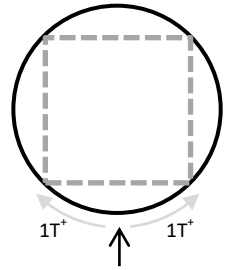
That's pretty straight forward.

[222] Using π for pi, we get a diameter of (24 / π =) 7.639...

[223] As opposed to the גמרא's usual value of 1.4; instead we are using a value of 2 here

Rashi's פשט confused me. From the beginning of רש"י we get the following description:

"When רבי יוחנן says we need 2T-and-a-bit (hereon referred to as $2T^+$) of the circle within 10T from the ground, it means $2T^+$ of the lowest part of the circumference of the circle. Measured from the bottom, in the middle, $1T^+$ to one side and $1T^+$ to the other."



At first, I thought רש"י means that the bottom quarter of the circle is only $2T^+$, which would mean the whole circle around is only $8T^+$. But it's supposed to be 24T around!

-?-

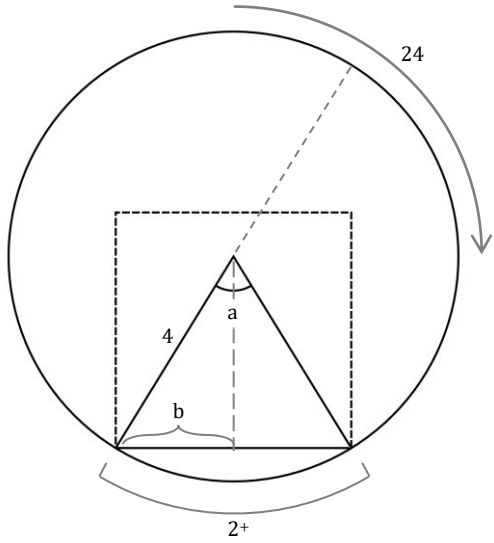
I thought perhaps רש"י means that since 24T is actually too big (considering the diagonal is really[224] less than 2) we need to know how much of that circle really needs to be below the 10T line in order for a 4x4 square inside it to also be below the 10T line.

So let's see if that works. Let's see how big of a box would sit within that curve at the bottom.

Let's draw a box whose sides come up from the ends of the bottom curve.

Let's put a triangle into the box as wide as the box and whose peak is the center of the circle.

Look at the diagram on the right.



[224] Like we'll see on page 293, from the גמרא in סוכה...

The left slant of the triangle is 4T long since it is half the diameter of the circle.

Angle a can be figured out by determining how much of the full circle are the 2^{+}T at the bottom. (From here on in I'll just be calling it 2 instead of 2^{+})

▶ $2 \div 24 = 1/12$ of the full circle.

⬇ Since the full circle is 360° then $1/12$ of that is 30° $\left(=\frac{360}{12}\right)$.

So $a = 30°$.

⬇ Divide that by 2 to get the left half of that angle (15°).

⬇ The value of line b is gotten from the formula[225] of $\sin(15°) \cdot 4$.

Which is 1.035...

⬇ Double that to get the whole length of the bottom edge of the box:

■ $$1.035... \cdot 2 = \mathbf{2.07055...}$$

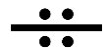
Which is way less than 4.

By the way, it comes out something funny. If we consider that if the bottom curve is only 2 then certainly the straight line inside is certainly less than 2. And yet we got a number greater than 2.

That's because we were using 3 as pi. If we use π as pi then the diameter is 7.639..., and the radius is 3.8197..., and following the חשבון we did above yields a box width of

1.977...

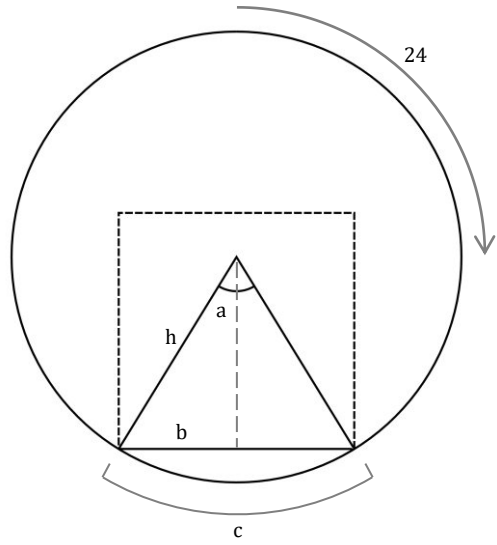
Which is indeed less than 2.



[225] See Appendix J on trigonometry

But anyway, this got some gears turning in my head.

If I had a circle 24 טפחים around, and I placed a 4x4 square into it, "resting" on the bottom curve of the circle, how long would that curve be?



Diameter = 24 / π

h = Diameter / 2

b = 2 (= half the 4T box width)

▶ $\angle a = \sin^{-1}\left(\frac{b}{h}\right)$

⬇ Double a to get the whole angle at the top of the triangle

⬇ Divide that by 360° to get the fraction of the whole circle

⬇ Multiply by 24 to get the length of the curve.

■ The whole formula together would be:

$$c = 2 \cdot \sin^{-1}(2 / (24 / \pi / 2)) / 360 * 24$$

Which simplifies to: $c = \frac{2}{15}\sin^{-1}\left(\frac{\pi}{6}\right)$

Using π as pi, this comes out to **4.20986...**

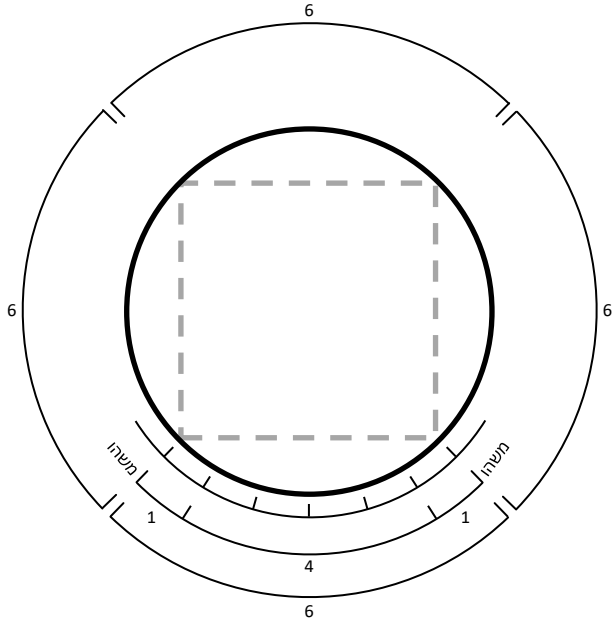
Interestingly, using 3 as pi, this comes out to exactly 4 (footnote[226]). Which is strange because it comes out that the straight line of the bottom edge of the box is the same length as the curve beneath it.

Let's get back to רש"י.

I was having trouble understanding what רש"י means. Maybe it's obvious to others but I didn't see it in רש"י until it was explained to me.

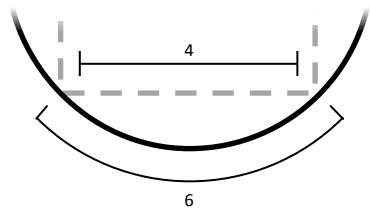
[226] See Appendix J for proof that $\sin^{-1}(.5) = 30°$

The ספר פירוש חי has the following picture and explanation (I still didn't get this until Rav Yehoshua Portowicz explained it to me):



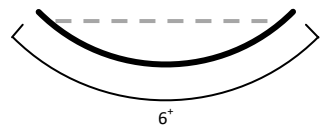
Imagine a square (4x4) bounded by a circle that is 24 around. If we were to divide the circle into four sections, each section between two corners of the square, each section would be a curve (arc) 6 long (= 24 / 4).

That means that the bottom curve, which is 6 long, is 2 longer than the bottom side of the square (= 6 - 4).



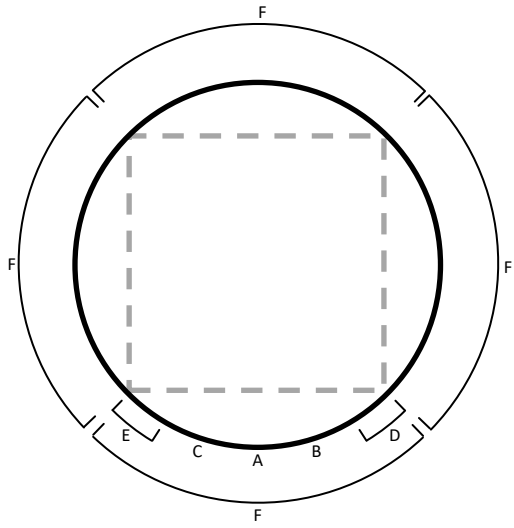
To make sure that the bottom of the square is *less* than 10T from the ground we need $6T^{+}$ of the circumference to be below 10T.

That's 2-and-a-bit T <u>more than</u> the 4T of the bottom of the square.



And that's what רבי יוחנן meant when he said that 2^{+} must be less than 10 from the ground.

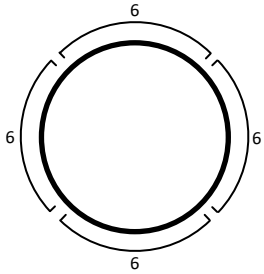
Now, let's paraphrase רש"י and see how he's saying this. (This point-by-point in רש"י is also done in the פירוש חי)



"The circle is low at the bottom [A], and goes up on each side [B, C]... You need 1T on this side [D] and 1T on that side [E] and a bit (in addition to the bottom 4T of [A, B, & C])...

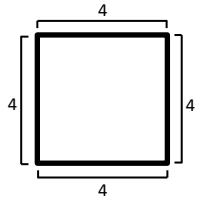
(Where do these 2T come from?) Since when you put a square inside a circle, the גמרא later says that the (circumference of) the circle is half (bigger than) the square (i.e., the circle's circumference is 1.5 times bigger than the square's circumference).

The square's circumference is 16 (4T · 4 sides) (so the circle is 24 (= 16 · 1.5).



Each quadrant of the circle is thus 6 (= 24 ÷ 4)...)

...which is 2 more than each quadrant of the square (= 6 - 4)."



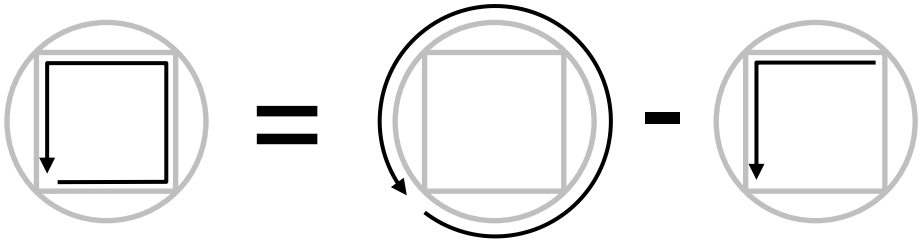
So we need 2-and-a-bit of the bottom part of the circle, besides the bottom 4, to be within 10T from the ground.

It's really not that complicated. I'm just trying to clarify every point in רש"י in רבי יוחנן.

ע"ו.

This whole שיטה of רבי יוחנן is based on the opinion of the "דייני דקיסרי", the "Judges of Caesarea" (or some say the "רבנן דקיסרי", the "Sages of Caesarea"), who say that the diagonal of a square is twice the length of one of the square's sides[227].

אגב, the way they say it is that ריבועא מגו עיגולא פלגא "a square inside a circle is half", which means that the circumference (perimeter) of a square inside a circle is equal to the circumference of the circle minus half the circumference of the square.



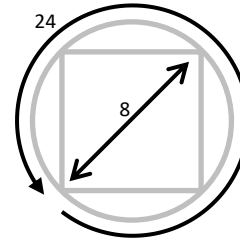
Rearranging the above formula yields:

$$\text{Circumference_of_Circle} = \text{Square's_Perimeter} + .5\ \text{Square's_Perimeter}$$

or simpler:

$$\text{Circumference_of_Circle} = 1.5\ \text{Square's_Perimeter}$$

For example, in our case, take a square 4x4. The perimeter is 16 (= 4 + 4 + 4 + 4), so according to the above formula the encompassing circle is 24.



Now, since the circle's diameter is taken to be 1/3, that comes out to be 8. That diameter is also the diagonal of the square.

And since the sides of the square are 4, it comes out that the diagonal is <u>twice</u> one of the sides.

And the Gemara leaves Rebbe Yochanan at that.

[227] And not 1.4 as we've doing in our גמרא, and not $\sqrt{2}$.

Now, when I learned this, I wondered where's this שיטה coming from? Where are they getting this number from, and why are they using a number that is so inaccurate?

So, I thought to invoke the classic "לא דק"[228], they weren't being exacting. They wanted to round the exact number (~1.41...) to an integer. They couldn't round down to 1 because the diagonal of a square is obviously longer than the side, so they rounded up.[229]

Or, they took the simple approach of adding the 2 sides of the square together[230].

Now, let's go to the גמרא in סוכה ז':-ח'. The גמרא there is almost parallel to our גמרא.

רבי יוחנן gives the circumference of a round סוכה as "24 people sitting around", which at some point in the גמרא the סוגיא takes to be 24 אמות around. The גמרא goes through the חשבון trying to get a square of 4x4 אמות inside the סוכה and finds that 24 אמות around is way too much. At some point, the גמרא says "All you need is 16.8 אמות around[231]" so why did רבי יוחנן say 24?

The גמרא suggests "לא דק", he wasn't being exacting.

The גמרא asks, would we say לא דק on such a large amount? (From 16.8 to 24)?

So the גמרא tries something else.

But I was wondering, what is that amount, that the גמרא didn't want to go with לא דק.

$$16.8 / 24 = \mathbf{.7}$$

That means, if the real number is 70% of the stated number, that's too much to say לא דק.

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[228] Which, incidentally, someone pointed out to me that the small גמטריא of that phrase are the same numbers as the beginning of π, 3.141... Cute.

[229] Although, you could counter the other way, that two adjacent sides of the square add up to 2, so obviously the straight path between them will be less than that, but perhaps that is not as obvious, so they took that as the better option.

[230] אגב, this may be a source for what I suggested (on page 36) that רש"י takes the diagonal of any rectangle that isn't square to be the sum of the length and width.

[231] If the square inside is 4x4 then its diagonal is 5.6 (= 4 * 1.4). The diagonal is also the diameter of the encompassing circle. The circumference is taken to be 3 times the diameter and that comes out to 16.8 (= 5.6 * 3). See first diagram on page 380.

So then I compared that to our גמרא. I was suggesting that Caesarea said 2 instead of 1.4 as a form of לא דק. Let's see, how much of a "לא דק" would that be? What's the ratio?

$$1.4 \ / \ 2 = \mathbf{.7}$$

Exactly the same! And the גמרא in סוכה questioned applying לא דק to such a ratio.[232]

So then we shouldn't be able to say לא דק here.

On the other hand, there is a difference between our גמרא in עירובין and the גמרא there in סוכה since the גמרא in סוכה also suggests what our גמרא suggests here, that רבי יוחנן is going like שיטת קיסרי, and that's how a 24-around-circle gets a square of exactly 4x4 inside it. But the גמרא in סוכה rejects this approach since "we see that it's not that much". Whereas the גמרא by us suggests שיטת קיסרי in רבי יוחנן and leaves it at that.

So maybe, there's a different degree of accuracy demanded by each סוגיא.

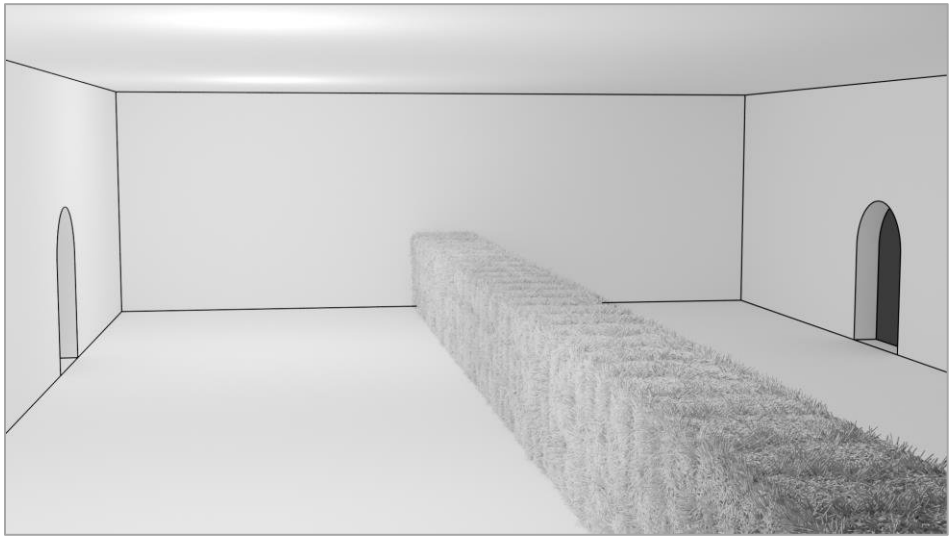
Perhaps because there it's discussing a דאורייתא, while here it's a דרבנן?

[232] Even if we take the diagonal of a square to be exactly $\sqrt{2}$ and the circumference of a circle to be exactly π, the final result of both Gemaras will still be exactly the same. $\left(4\sqrt{2}\,\pi \ / \ 24 \ = \ \frac{\pi}{6}\sqrt{2} \ = \ \mathbf{0.74048}\ldots\right)$

*

ע"ט: - **Tosfos Dichol - Straw Wall**

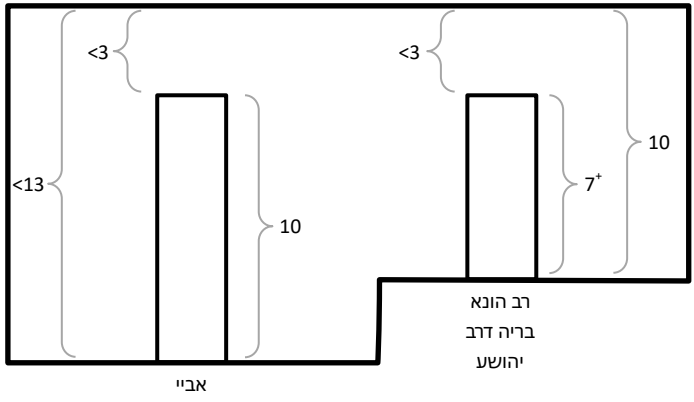
The בריית talks about a case of a wall made of straw running through the middle of a house, dividing the house in half. Each side of the house opens to separate courtyards.



The wall needs to be 10T tall to be considered a good מחיצה.

The גמרא has some discussion about it and at the conclusion says that the case is that the ceiling of the house needs to be within 3T from the top of the straw wall.

אביי understands the wall is physically 10T tall so that the ceiling is less than 13T from the ground.



Whereas רב הונא בריה דרב יהושע holds the case could even be where the wall is only 7T-and-a-bit tall and the ceiling is 10T off the ground.

Both אמוראים are

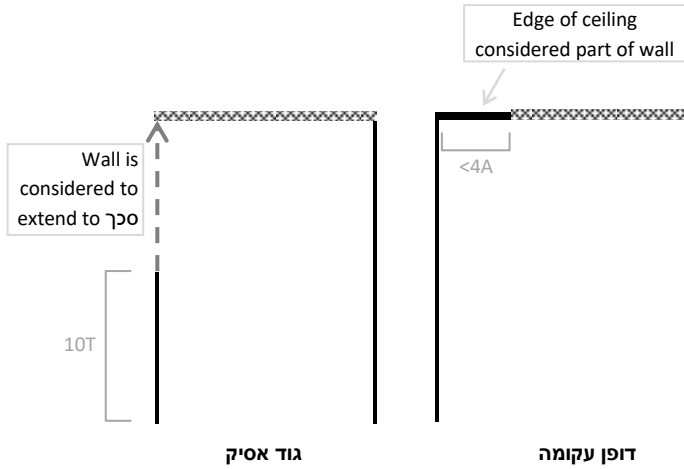
using לבוד to consider the wall to be touching the ceiling but רב הונא בר"י is also using לבוד to extend the wall up to the required 10T, whereas אביי does not use לבוד to extend the wall's height; rather he requires the wall to be physically 10T tall.

The question is why does אביי not use לבוד to extend the wall's height?

תוספות[233] asks that we indeed find that אביי elsewhere uses לבוד to make a wall the required height[234], so why is it different over here?

תוספות answers that the difference is that here the wall also needs to be touching the ceiling.

When I first read this, I thought that what תוספות means is that אביי holds that לבוד cannot accomplish two things at once, כעין like we say by סוכה that you cannot use גוד אסיק and דופן עקומה by the same wall, at the same time[235].

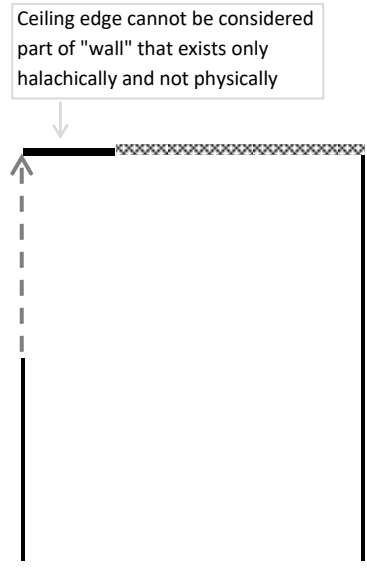


[233] ד"ה בכל פחות

[234] תוספות is pre-empting a possible answer that אביי simply doesn't hold that לבוד could extend something to a required height.

[235] משנה ברורה תרל"ב:ד', although he also brings another opinion that allows it

There's too many non-physical mechanisms working together at the same time.



גוד אסיק עם דופן עקומה

So too by our straw wall. We need לבוד to extend the wall height <u>and</u> we need לבוד to make the wall be considered touching the ceiling. It's too much work for one little לבוד.

The יד בנימין has a different פשט in תוספות. He writes first of all, there's a תוספות on the previous עמוד[236] that told us that it is only a temporary wall like this, of straw, that has the requirement that it reach the ceiling.

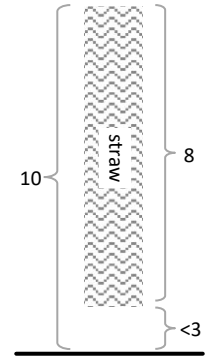
If so then אביי holds that such a "fake" wall cannot rely on לבוד to make the wall taller. It's like two weaknesses combining; A "fake" wall and לבוד.[237]

[236] ד"ה ש"מ

[237] However, לבוד <u>does</u> work here to consider the wall to be touching the ceiling.

This is similar to my פשט, but slightly different. I was saying that לבוד cannot perform two things at once, being a weak mechanism, whereas the יד בנימין is saying that לבוד cannot combine with a "weak" wall.

One נ"מ between the Yad Binyomin's pshat and mine would be concerning a gap of less-than-3T at the bottom of a straw wall.[238]



According to the יד בנימין, this is problematic since a "weak" wall cannot rely on לבוד to extend it.

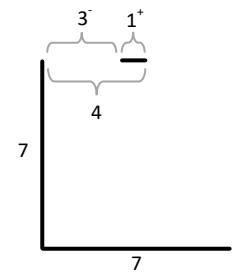
According to my פשט, this case is fine, since לבוד is only accomplishing one thing, namely, to close the gap.

Another נ"מ would be by a strong wall. According to the Yad Binyomin, לבוד would work here. According to my פשט, it would be no good.

Although, I may have found a proof against my פשט from סוכה.

(This is really from the גמרא but I'm bringing it from the ש"ע to show that it comes down להלכה).

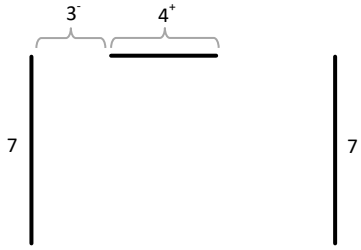
In the ש"ע (630:2), there's a case of a 7T x 7T סוכה with two full-sized (7T) walls at right angles to each other, in an "L" shape. A third wall is constructed by placing a tefach-plus-wide board within 3 tefachim from one of the full walls, for a total of 4T (adding the 1^+ and the 3^-), which is a majority of 7T. (And then a צורת הפתח is needed for the rest of that side but that is irrelevant right now.)



In this case, the added wall need only be 1^+ T wide.

[238] I'm assuming that the wall only need touch the ceiling and not the floor, although a סברה to differentiate would have to be found.

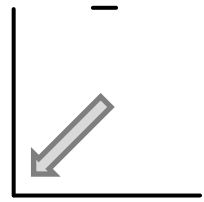
However, in a different case, where the two full-sized walls of the סוכה are opposite from each other, like this:

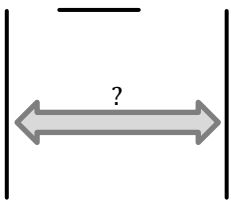


then the added third wall needs to be <u>at least 4T-wide-and-a-bit</u>, placed within 3 tefachim of one of the full-sized walls.

The משנה ברורה (14) wonders why in this case we need the third wall to be made with a board at least 4T wide whereas in the previous case, with the "L" shaped סוכה, it was enough that the added board be only 1T?

He answers that by the "L" shaped סוכה, we had two full walls next to each other...





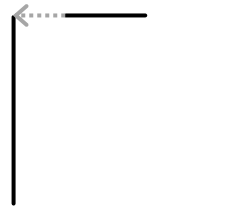
...whereas in the case with the full walls parallel to each other we didn't have two full walls next to each other.

Therefore, we need a more substantial wall here. Place a 4-tafechim-plus board within 3 tefachim of a full wall and now you have (effectively) a 7T wide wall near one of the other walls.

In this second case, we see לבוד accomplishing two things at once:

1) Making the wall larger and

2) Making the wall be considered connected (or near) another wall.



To paraphrase the משנה ברורה: "Now you have a 7T wide wall [=extension] near [=connected to] one of the full walls."

This would seem to rule out my pshat in תוספות, that לבוד cannot accomplish two things at once like this.

ע"ט:

Although, we could say that we don't paskin like אביי... but I think we do.

In the ש"ע (א"ח 372:19), the מחבר writes about this 10T high wall of straw, and the idea of having this ceiling be within 3T of the top of the straw wall. Nowhere in the discussion does anyone mention this היתר\קולה of רב יהודה בריה דרב יהושע that the straw wall can be 7T high. It sounds from there that the wall needs to be physically 10T tall, like אביי.

תיובתא לבומי תיובתא?

But maybe it's a combination of פשטים. Maybe, like the יד בנימין, only by a "weak" wall like one made of straw are we not saying לבוד to make the wall tall. But we <u>are</u> saying לבוד to make the wall considered connected to the ceiling! So maybe what we mean is that לבוד cannot accomplish two things דווקא by a weak wall. But אין הכא נמי, in other cases (like סוכה) it can yes multi-task.

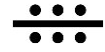
Now, we still have to be מחלק somehow in סוכה between the case of the two full parallel walls and the case of גוד אסיק and דופן עקומה. Why by the former do we allow לבוד to accomplish two things, whereas by the latter we don't allow these two mechanisms to work together?

We could say that the difference is that by אסיק\עקומה it's <u>two</u> הלכה למשה מסיני <u>mechanisms</u> working together, whereas by the parallel walls it's only <u>one</u>.

Or we could say that in general, דופן עקומה, גוד אסיק, and לבוד create "כאילו" realities, which work independently. מה שאין כן where the mechanisms are <u>extensions</u> of each other (NP"I) then they don't work.

In other words, דופן עקומה means we look at the ceiling as though it were a bent portion of the top of the wall. But in order to say that (or <u>see</u> that) there has to <u>be</u> an actual wall there! Otherwise, we're "bending" an imaginary wall created by גוד אסיק! The whole situation is too imaginary!

Whereas by the parallel walls case, the two requirements (or tasks) that לבוד is accomplishing have nothing to do with each other; they don't rely on each other. One is how tall is this wall, independent of whether it's connected to the ceiling or not. The other is that the wall be connected to the ceiling, independent of how tall the wall is.



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פ'. - Tosfos Ma'asseh - Being Mizakeh For Mavoys and Techums

The משנה on 79b says that when making a שיתוף מבואות how to be מזכה the other people in the alley (by having someone else make a קנין for them), which means that by מבואות, we need someone to be מזכה.

The משנה on 82a talks about making an ערוב תחומים for others and it doesn't say anything there about being מזכה the other people, implying that we don't need to be מזכה, by תחומים.

The גמרא on 80a brings a מחלוקת between רב and שמואל by both instances; by שיתופי מבואות and by ערוב תחומים.

(I'm going to present this out of order for simplicity.)

שמואל holds in both instances like the implications of the משניות we brought above, that by מבוי we need זיכוי, whereas by תחום we don't.

רב holds the exact opposite; that by מבוי you don't need זיכוי whereas by תחום you do.

	שתופי מבואות	ערובי תחומים
משנה ע"ט:	✔	
משמעות של משנה פ"ב.		✘
שמואל	✔	✘
רב	✘	✔

So, the גמרא says בשלמא שמואל, he's just like the משניות, but what about רב?[239]

So the גמרא answers that it's a מחלוקת תנאים, and brings the following story to prove it (this is the way רש"י learns it).

[239] Interestingly, the גמרא doesn't answer רב תנא ופליג. See ריטב"א ג'. ד"ה אמלתרא where he writes that the גמרא only gives this answer when any other answer is not possible, meaning this is a last resort. I wonder if this is what תוס' holds as well since on ב': ד"ה מידי he writes that the קשיא on רב from the משנה there cannot be answered with the ברייתא, since maybe the ברייתא argues on the משנה. Why doesn't תוס' just answer רב simply by saying תנא ופליג?
אלא מאי, Tosfos agrees with the ריטב"א that we only say רב תנא ופליג when we have no other answer. Seems to me.

Once upon a time, the daughter-in-law of רבי אושעיא went to the bathhouse (out of the תחום -רש"י) before שבת and forgot to partake of the ערוב תחומים, so her father-in-law tried to make an עירוב for her, but he wasn't מזכה with a קנין and רב חייא said it was אסור, while רבי ישמעאל said it was מותר.

So רבי חייא held by תחומים that you need to be מזכה like רב. That deals with the משנה on 82a of the משמעאת.

What about the משנה on 79b which is explicitly not like רב? רש"י says רב תנא ופליג.

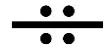
ערובי תחומים	שתופי מבואות	
	✔	משנה ע"ט:
✘		משמעות של משנה פ"ב.
✔		רבי חייא
✘		רבי ישמעאל
✘	✔	שמואל
✔	✘ – תנא ופליג	רב

תוספות asks on the way רש"י presents the גמרא (that the story was by תחומים and proved רב's position regarding תחומים) that the גמרא's main question was by the חצר (שיתופי מבואות), since that's where we have an explicate contradiction to רב's position.

Therefore, says תוספות, if this story about the daughter-in-law was by תחומים then we haven't answered the (original) question at all!

Based off this question, תוספות explains that this story was a question of עירובי חצירות (which, for this דין, is the same as שיתופי מבואות), and רב holds like the opposing opinion in that story, רבי ישמעאל.

ערובי תחומים	שתופי מבואות	
	✔	משנה ע"ט:
✘		משמעות של משנה פ"ב.
	✔	רבי חייא
	✘	רבי ישמעאל
✘	✔	שמואל
✔	✘	רב



At first, I didn't really understand what was bothering תוספות.

In the previous ד"ה (Rav Amar), תוספות said that רב's positions on each type of עירוב (or שיתוף) are based on one single principle, that is, whether the guy who's being included in the עירוב affects the status of the עירוב or not.

Since by שיתוף, I (the מערב) need you to be part of this עירוב (because otherwise your presence will make the עירוב not work and will אסיר the whole מבוי and I won't be able to carry from the חצר to the מבוי), so I'm "גמר ומקנה", that is, I have "complete" intention and therefore it works even without a מעשה קנין.[240]

By תחומים, on the other hand, my תחום status isn't affected whether you join in or not, so I'm not גמר בליבי with a "complete" intention to be מזכה you, so it doesn't work without being מזכה you with a מעשה קנין (either through you or through someone else).

So we see that according to רב these positions are תלי זה בזה! (Mutually dependent)

If so then by showing there's a תנא who holds like רב by תחום, we <u>automatically</u> have a support for רב by חצרות, since these positions are interdependent. And so that would answer תוספות's question on רש"י!

I think the answer to this question is that it is רב himself who holds the opinions to be mutually dependent. The תנא we're pinning him on, רבי חייא, can very well hold that you need to be מזכה both by תחומים <u>and</u> by חצירות.

And this is even in the לשון of רש"י himself[241] where he says that רבי חייא and רב are the same regarding תחומים but by חצירות he says that רב is a תנא in his own right, meaning, he has no other תנא to rely on, not even רבי חייא!

And all of this in רש"י is quoted by תוספות, so maybe that's exactly תוספות's point, that רש"י doesn't pin רב's opinion of חצירות on רבי חייא.

[240] See חזו"א ח"מ סימן כ"ב ד"ה כלל גדול where he writes that the קנין works with גמר בליבו and the (usually required) מעשה is just a catalyst for גמירת דעת, and sometimes all you need to get גומר בליבו is just אמירה בעלמא.
[241] ד"ה וזיל, after the first three words

*

פ"ג. - Vi'Odos Di'Rebbe - Getting to the Big Sa'ah

There's a lot on this עמוד, so I'm going to break this piece down into mini-articles. Let's start with a mini table of contents:

- שיעור חצי פרס
    - ועודות דרבי
    - Two גירסאות in רש"י
- Someone sent a סאה to רבי
    - Not big enough
- 3 sizes of סאה
- How to get $\frac{1}{5}$ quickly $\left(\cdot \frac{2}{10}\right)$
- Outside 6th vs. Inside 5th
    - Outside/Inside 24th
- From מדברית to ירושלמית
    - Rashi's Method
- גמרא is holding real number in mind ("173" = 172.8)
- From ירושלמית to ציפורית
    - Rashi's Method
- פשט in רש"י 207: "Not even .5 egg over"
- חלה
    - תוספות ?9 ?8
- פשט in "more than .5 egg"
- What is שיעור חלה, oh רש"י?
- ועודות דרבי
- גורס אלא?
- רש"י אי הכי טפי הוי להו
    - $\frac{1}{80}$ untenable, 3 pronged question.
    - How does 3rd point help?
    - All Possible Permutations (Awesome!)
- Rashi's alternate pshat $\frac{1}{20}$
- Why not $\frac{1}{120}$?

We're going to pick up in the middle of the גמרא, somewhat.

The גמרא is discussing the size of a חצי פרס.

1. One opinion says it is 2 eggs
2. One says it's 2 minus a bit
3. One says it's 2 plus a bit
4. רבי determined that it's 2 eggs "ועוד", "and a bit more" (perhaps synonymous with the 3rd opinion above who said it's 2 plus a bit).

The גמרא says that this "And a bit more" of רבי is $\frac{1}{20}$ of an egg.

Rashi brings two גירסאות of the גמרא, each with its own פירוש.

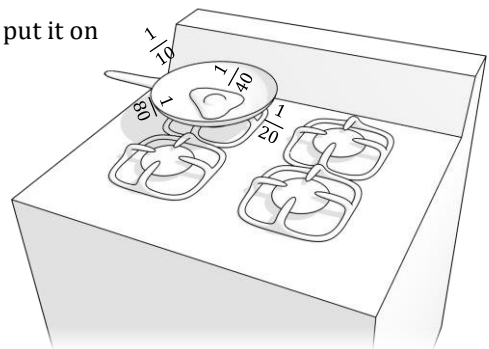
In the first גירסא, the "ועוד" that each egg is increased by is $\frac{1}{20}$ of an egg, so that each halachic egg is $1\frac{1}{20}$ of a physical egg.

Therefore, a חצי פרס (which is supposed to be 2 eggs) is 2 and $\frac{2}{20}$ physical eggs $\left(= 2 \cdot 1\frac{1}{20}\right)$, which is the same as $2\frac{1}{10}$.

In the second גירסא, the "ועוד" is $\frac{1}{40}$ of an egg. Here רש"י says that it doesn't mean of <u>each</u> egg but rather that the <u>total</u> increase of 2 eggs is $\frac{1}{40}$ of an egg. Which means that each egg is increased by $\frac{1}{80}$ to $1\frac{1}{80}$.

However, רש"י rejects this second גירסא, saying "One who is exacting in this לשון will find it incompatible with the גמרא later on, and I heard both versions".

We will come back to this later, For now, put it on the back burner.



The גמרא continues and says that a man named בוניוס sent to רבי[242] a container that presumably could hold a סאה. רבי measured it and found that it contained a volume equal to 217 eggs.

Remember that number well because that is going to be our end goal:

217 eggs

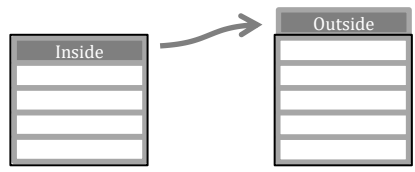
The גמרא tries to figure out which "סאה" this could be. Over time, a סאה increased in size. There were three different sizes in use at different times.

From smallest to biggest they are:

1. מדברית
2. ירושלמית
3. ציפורית

Each "סאה" type is $\frac{1}{5}$ bigger than the previous. That is, multiply a smaller סאה by $1\frac{1}{5}$ to get the next sized סאה.[243]

This is what the גמרא elsewhere[244] calls an "outside 6th", i.e., after a fifth of the original size has been added on, there are now 6 parts $\left(\frac{5}{5} + \frac{1}{5} = \frac{6}{5}\right)$, so after the fact, what we have added was 1/6 of the total $\left(\frac{1}{6} \text{ of } \frac{6}{5} \text{ is } \frac{1}{5} \text{of } \frac{5}{5}\right)$. (This is called "outside", meaning, only after it has been added on. As opposed to "inside" would mean before you add it on.)



[242] As a gift, I presume, or maybe he came across this container and sent it empty to Rebbe to determine if and what halachic significance it had.

[243] I.e., A bigger סאה = $\frac{6}{5}$ of a smaller סאה, or 1.2 * smaller סאה.
A quick and easy way to determine in your head what is $\frac{1}{5}$ of a number, take the starting number and double it and then bring the decimal point over to the left one digit. For example, let's take 65 as our starting number. Double it to get 130. Move the decimal over to the left to make it 13.0, which is 1/5 of 65. Then you can add that to the original number for a total of 78, which is 6/5 of 65.

[244] For example, ב"מ נ"ג:, it says איבעיא להו חומשא מלגיו או חומשא מלבר

Keep this concept in mind also for later when we talk about חלה, whose שיעור is $\frac{1}{24}$. There, as well, we will bring up this idea of "outside" and "inside" fractions.

In general, within each local "system" (מדברית, ירושלמית, and ציפורית) there are fixed ratios between different measurement sizes. For example, there's always 4 לוג in a קב regardless of how big these לוגים or קבים are. A קב מדברי has 4 לוגי מדבריים, just like a קב ירושלמי has 4 לוגי ירושלמיים, and a קב ציפורי has 4 לוגי ציפוריים.[245]

The exchange rates, or relative ratios between the different שיעורים are:

- 6 ביצים in a לוג
- 4 לוגים in a קב
- 6 קבים in a סאה

That's all we need to know for our current purposes.[246]

I'm going to use E for ביצים (eggs), L for לוג, K for קב, and S for סאה.

Now even though I said that each system retains the same internal ratios, the גמרא seems to be using מדבריות eggs throughout the discussion regardless of which system is being discussed, I guess in order to make a standard against which everything can be measured.

[245] By way of illustration, the United States of America and Canada both use the same ratios between their coins and bills, even though each system is a different size than the other. There are 4 quarters in a dollar, 5 nickels in a quarter, 5 pennies in a nickel, even though each country's value (buying power) of each individual coin differs from the other country. The current ratio is that a USA dollar is worth about $1.40 of Canadian money. That means it takes 1.4 Canadian money to buy the same thing with 1 of USA money. 1.4 Canadian quarters = 1 USA quarter; 1.4 Canadian dimes = 1 USA dime, etc.

[246] See Appendix A on Shiurim

So, a סאה מדברית then would have 144 eggs in it (= 6E · 4L · 6K). This number (**144 eggs**) is going to be our starting number and we're going to try, with various methods, to get from there to our target number of 217 (which we said before is the size of the סאה that רבי received).

$$144 \cdot \text{something} \cdot \ldots \cdot \text{something} = 217$$

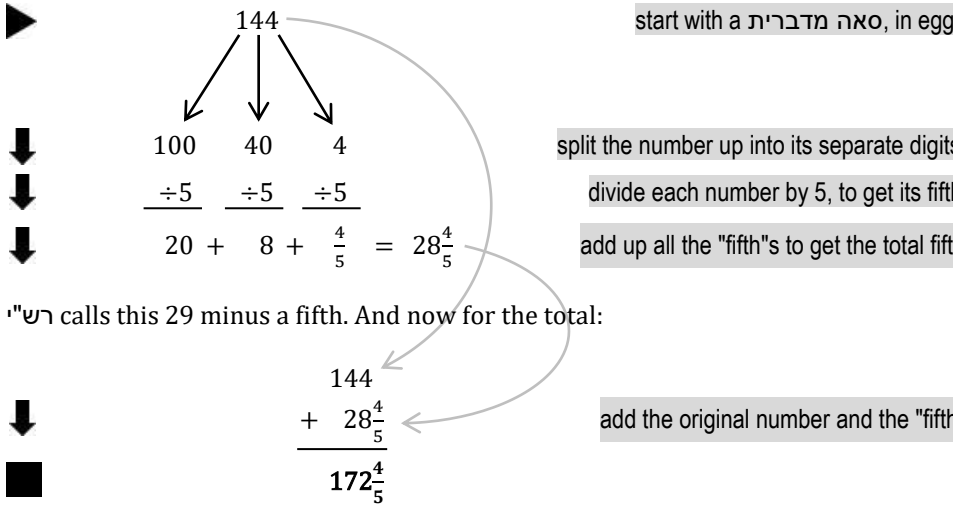
So, a סאה מדברית is not big enough.

So then the גמרא tries the next size up: ירושלמית

$$144\text{E} \cdot 1\tfrac{1}{5} = 172\tfrac{4}{5}\,\text{E}$$

Let's show how רש"י does the חשבון.

רש"י breaks up 144 into its separate digits and then divides each one by 5 (to get their "fifths") and then adds them all up (to get the total "fifth") and then adds that back to the original 144 (so that the entire number has increased by a fifth).



רש"י calls this 29 minus a fifth. And now for the total:

The גמרא calls this number 173. It will be obvious from the next step that the גמרא really has in mind the exact $172\frac{4}{5}$, as we shall see.[247]

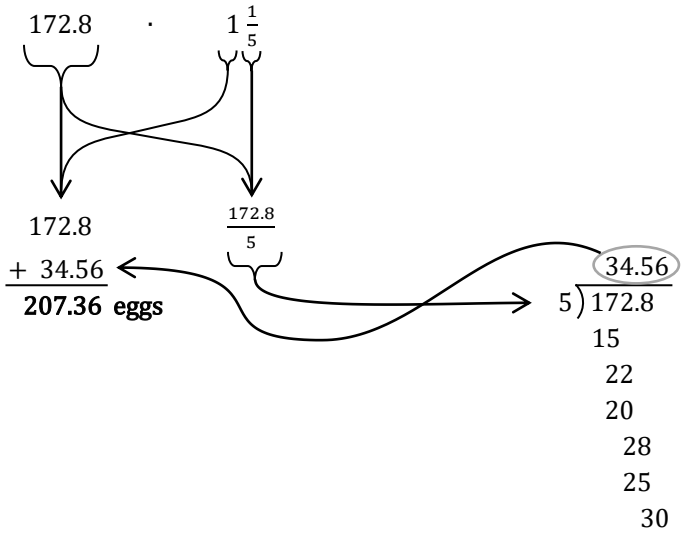
We still haven't gotten to 217.

So we go to the next size up: ציפורית.

ציפורית is $1\frac{1}{5}$ times bigger than ירושלמית. So that'd be:

$$172\frac{4}{5} \cdot 1\frac{1}{5} = \mathbf{207\frac{9}{25}}$$

Let's do it by hand (we're gonna use the decimal expression of 172.8):



Which the גמרא calls 207.

Now, I told you before that when the גמרא said 173 it actually meant $172\frac{4}{5}$ but was just rounding. So then it makes sense that when we get to this stage the גמרא call the ציפורית 207 since we could say that the גמרא was rounding off to the nearest whole number, again.

[247] At each stage of the גמרא and רש"י we'll be throwing around numbers, and sometimes there'll be a fraction of a number left over as a remainder. Sometimes we will discard the fraction in the discussion only to bring it back later for a more accurate result. It's a little tricky therefore getting through our גמרא since it's not always clear whether or not we're taking the fraction into consideration. I will try to make it clear as we go but you should expect it to be a little confusing (and so don't get overwhelmed...).

BUT, if the גמרא back then had really meant 173 exactly then the ציפורית increase would've been:

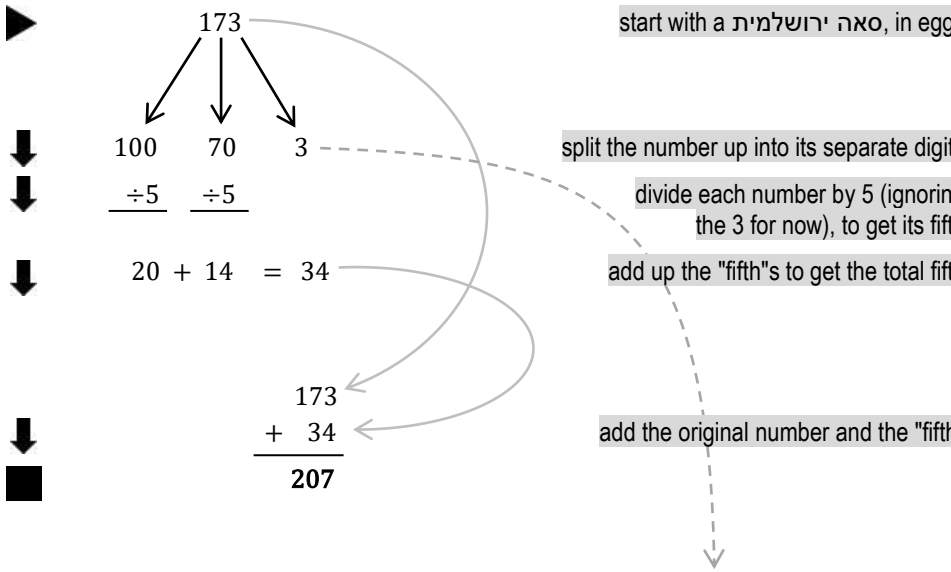
$173 \cdot 1\frac{1}{5} = \mathbf{207.6}$, which should've rounded off to 208, which the גמרא does not do. Ah?

Unless, if you'll suggest that the גמרא isn't rounding but is merely taking the whole number and ignoring the fraction.[248]

But anyway, רש"י and תוספות say the first way, that the גמרא is rounding when it says 173, but is keeping the exact number in mind when it goes on to the next חשבון.

Let's now do רש"י's method of getting to ציפורית:

He starts with the גמרא's previous rounded term of 173, and like before, breaks it into its separate digits, divides each digit by 5 to get their fifths, adds all the fifths together to get the total fifth, and then adds that to the original 173.



Here, רש"י asks why don't we take into account the fifth of the 3 eggs? Because it's really 3 - $\frac{1}{5}$, which is $2\frac{4}{5}$. Taking a fifth of $2\frac{4}{5}$ is "less than half an egg".

רש"י doesn't mean that a fifth of $2\frac{4}{5}$ is less than half an egg, because it isn't. It's .56 of an egg $\left(\frac{14}{25}\right)$.

[248] which the גמרא will actually do in the last stage

What he means is like what we said before, that when you add that fifth to the more accurate $172\frac{4}{5} + 34$, the total is less than half an egg above 207 (so we round down).

In formula form that'd be:

$$\frac{6}{5} \cdot \left(173 \ - \ \frac{1}{5}\right) < 207\frac{1}{2}$$

but

$$\frac{6}{5} \cdot 173 > 207\frac{1}{2}$$

Now, just to recap where we're holding.

The חשבון so far is:

6 eggs · 4 לוג · 6 קבים = 144 eggs = 1 סאה מדברית

$$144 \cdot \frac{6}{5} = 1 \text{ סאה ירושלמית}$$

$$144 \cdot \frac{6}{5} \cdot \frac{6}{5} = 1 \text{ סאה ציפורית}$$

In full: $6 \cdot 4 \cdot 6 \cdot \frac{6}{5} \cdot \frac{6}{5} = 1$ סאה ציפורית

Simplified: $144 \cdot \frac{36}{25} = \mathbf{207\frac{9}{25}}$

OK, so the גמרא tries to increase this סאה again (remember, our target is 217).

The גמרא says you need to חשבון in **חלה**. That means, that when בוניוס sent this container to רבי, he sent enough so that after חלה is taken off from it רבי would remain with a full סאה.

Effectively, what the גמרא is saying in formula form is:

$$217 - (\text{חלה} \cdot 217) = 207$$

Even though it's 207 and a bit, I'm calling it 207 for now just to get the basic idea; later we'll work with exact numbers.

Now, before seeing what the גמרא says, let's חשבון this on our own.

And let's not even assume that we know what the שיעור of חלה is. Let's calculate, based on the above formula, what the שיעור חלה should be in order to make the above formula work (regardless of what חלה actually is; we'll do that next).

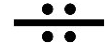
▶	$217 - (\text{חלה} \cdot 217) = 207$	write up the formula
⬇	$217 - 207 = \text{חלה} \cdot 217$	invert what is being subtracted from 217
⬇	$10 = \text{חלה} \cdot 217$	simplify left side
⬇	$\frac{10}{217} = \text{חלה}$	divide both sides by 217
■	$\mathbf{\frac{1}{21.7} = חלה}$	divide top and bottom by 10 (to get top to be 1)

Hmm. Is חלה 1 in 21.7 parts? No, חלה is 1 in 24 parts[249].

So let's do it the other way. Let's assume we know the rate of חלה $\left(\frac{1}{24}\right)$ and leave the total size as unknown, or x:

$x - (\text{חלה} \cdot x) = 207$	write out the formula
$1 \cdot x - \left(\frac{1}{24} \cdot x\right) = 207$	make x into 1x, change חלה with the שיעור חלה
$\frac{24}{24}x - \frac{1}{24}x = 207$	make 1x into 24/24x and remove parentheses
$\frac{23}{24}x = 207$	simplify left side
$x = 207 \div \left(\frac{23}{24}\right)$	divide both sides by 23/24
$x = \mathbf{216}$	(a trick to divide by a fraction is to multiply by that fraction upside down)[250]

which is one short of the elusive 217 that is so sought after. So we need to do something else.



[249] משניות חלה ב':ז'

[250] So instead of $\left(207 \div \frac{23}{24}\right)$ do $\left(207 \cdot \frac{24}{23}\right)$ which becomes $\left(\frac{4968}{23}\right)$ which becomes 216

By the way, let's do that again with the more exact $207\frac{9}{25}$ and see if that helps:

$$x - \frac{1}{24}x = 207\frac{9}{25}$$

$$\frac{23}{24}x = 207\frac{9}{25}$$ take off 1/24x from 24/24x

$$x = 207\frac{9}{25} \div \frac{23}{24}$$ divide both sides by 23/24

$$x = \frac{5184}{25} \div \frac{23}{24}$$ multiply 207 by the denominator and add to the numerator

$$x = \frac{124{,}416}{600} \div \frac{575}{600}$$ bring up all the fractions to their lowest common multiple

$$x = 124{,}416 \div 575$$ the common denominators cancel each other out[251]

$$\mathbf{x = 216\frac{216}{575}}$$ divide with a remainder (to keep as a fraction)

Still doesn't help. Still less than "half an egg" above 216 (so we can't even round up).

Now that we've done the חשבונות on our own, let's go back to the גמרא and see what it says.

We were holding so far that a סאה ציפורית is 207 eggs and the גמרא said that you can get to 217 if you add in the necessary חלה (an amount so that the total will be such a number that חלה taken from it will equal 207). The גמרא says that the amount of חלה taken would be **8**, which will give us a total of 215 (= 207 + 8).

In other words, if we start with 215 and take off חלה, that amount taken off would be 8.

Where does 8 come from?

רש"י says like this: Take 207 and divide by 24 and you get 8 with a remainder $\left(\frac{15}{24}\right)$.

$$\begin{array}{r} 8\frac{15}{24} \\ 24\overline{)207} \\ 192 \\ 15 \end{array}$$

[251] Since we just care what the numerators are relative to each other, it makes no difference how much you inflate or deflate them with a denominator

The remainder is more than a half, so the question then would be why don't we round it up to a whole egg (and call the חלה 9)?[252]

You could say we're just looking at the integers, ignoring the fractions or remainders.[248]

And that's what רש"י basically says, that we're calculating the חלה of 192, and ignoring the חלה of the 15 remaining.

For now.

Stay tuned.

But, there's a problem here. Did you catch it?

רש"י calculates the $\frac{1}{24}$ <u>of 207</u>. That's not how חלה is calculated!

You're supposed to <u>start</u> with a certain amount and <u>then</u> take off $\frac{1}{24}$ <u>of that amount</u>.[253] In our case, that would mean start with 217 (what בוניוס sent over) and take off $\frac{1}{24}$ <u>of 217</u> and then you're supposed to end up with 207.[254]

Which you don't, really. $\frac{1}{24}$ of 217 is $9\frac{1}{24}$. Taking that off 217 is $207\frac{23}{24}$.

Or even, if you want to end up with 215 as the גמרא does[255], then start with 215 and take off $\frac{1}{24}$ <u>of 215</u> . That would come out to $8\frac{23}{24}$.

What we expected רש"י to do was to calculate an "outside" 24th of 207 (since we need to start with that larger number, take off $\frac{1}{24}$ of it, and end up with 207).

That would be:

$$x - \frac{1}{24}x = 207$$

---

[252] And it certainly wouldn't help to use the more accurate $207\frac{9}{25}$ since that's a bigger number, so a 24th of that will be even bigger than the $8\frac{15}{24}$ we arrived at, and doesn't answer the question at all.

[253] ברוך שכיוונתי לדרך אמונה ה':ב' ס"ק ט' ציון הלכה מ' והוא השאיר את זה בצ"ע

[254] This is like the whole inside/outside fraction we spoke about on page 305.

[255] Actually, our גירסא doesn't give a number at all. רש"י is reading our גמרא as ending up with 215. תוס' apparently had the same גירסא but writes that the correct גירסא is like ר"ח, which has חלה as 9, and the total as 216, as we will write soon.

But we already did all this already[256]. We came up with a total of 216. So, now $\frac{1}{24}$ of that is exactly 9, taking us down to 207. This is actually the גירסא that תוס' goes with.

Maybe we can answer for רש"י, whether you calculate an inside or outside 24th it still comes out to less than 9 so it doesn't actually make a difference how you do it, so maybe רש"י just chose the simpler חשבון.

Also, it's simpler since doing it with an "inside" חשבון leads to a whole integer (216).

OK. So where we are holding now is that 1 סאה ציפורית times חלה is supposed to equal 217, and it doesn't.

The whole חשבון right now is:

$$6 \cdot 4 \cdot 6 \cdot \frac{6}{5} \cdot \frac{6}{5} \cdot \frac{24}{23} = 216\frac{216}{575}$$

Let's try to make this a little bigger.

So now the גמרא says we need to inflate the shiur by taking into account the "ועודות דרבי", the "And-a-little-bit-more"s that רבי mentioned.

If you remember at the beginning of this piece[257], the תנאים were trying to find a certain שיעור. The shiur was about 2 eggs. רבי said that it's "2 eggs and a bit". The גמרא called it $\frac{1}{20}$ of an egg, which רש"י explained means either each egg is increased by $\frac{1}{20}$ (which for 2 eggs that would be a total increase of $\frac{2}{20}$, or $\frac{1}{10}$ of an egg) or[258] by $\frac{1}{80}$ (which for 2 eggs would be a total increase of $\frac{2}{80}$, or $\frac{1}{40}$ of an egg, as we explained back there).

So too, over here. Let's take the amount of eggs we've arrived at so far and increase that amount by the "ועודות דרבי".

Now before we continue, it's important at this point to point out that there is another גירסא issue here, and that is whether you are גורס the word "אלא" or not.[259]

[256] back on page 311
[257] back on page 304
[258] in a different גירסא
[259] in the גמרא when it says "אלא אייתי ועודות דרבי"

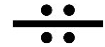
If you <u>are</u> גורס it, like our גירסא, then that means the גמרא is <u>rejecting</u> the previous stage of חלה. That would mean that the new חשבון is:

1 סאה ציפורית times ועודות דרבי

If you <u>aren't</u> גורס it then that means that the גמרא is <u>retaining</u> the חלה stage. If so then the חשבון is:

1 סאה ציפורית times חלה times ועודות דרבי

OK. Keeping that in mind, let's move on.



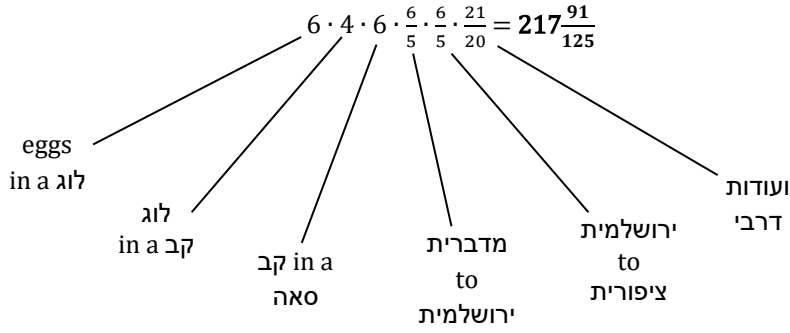
At the beginning, רש"י gave us two versions of how much were the "ועודות":

1. Each egg increased by $\frac{1}{20}$
2. Each egg increased by $\frac{1}{80}$

רש"י tries both פשטים and rejects the $\frac{1}{80}$ פשט. Let's see how that goes.

Now, pay attention to which גירסא in the גמרא is רש"י going with for each attempted פשט; with אלא or without.[260]

In the first פשט of ועודות, the שיעור is $\frac{1}{20}$ of an egg for each egg (so we multiply by $1\frac{1}{20}$, or $\frac{21}{20}$; same difference). And we <u>are</u> גורס the word אלא, which means we are <u>rejecting</u> the חלה stage, so the full חשבון would be:

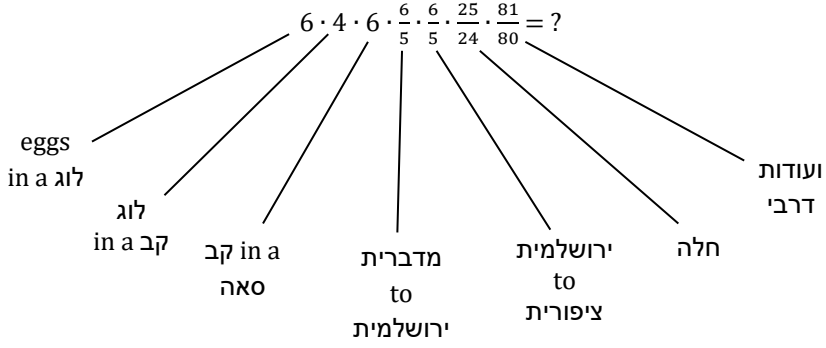


[260] The מלא הרועים points out this difference in רש"י

And then the גמרא asks, why are you calling it only 217, if it's a bit more? And on that the גמרא answers that we are not counting that leftover bit since it is less than a full egg.

Now let's do the other פשט: $\frac{1}{80}$

In this פשט, we <u>are not</u> גורס the word אלא, which means we are <u>retaining</u> the stage of חלה, so that the whole חשבון would be in one fell swoop:



(From this point on, we will be using רשי's way of חשבוning חלה as $\frac{25}{24}$ instead of the expected $\frac{24}{23}$.)

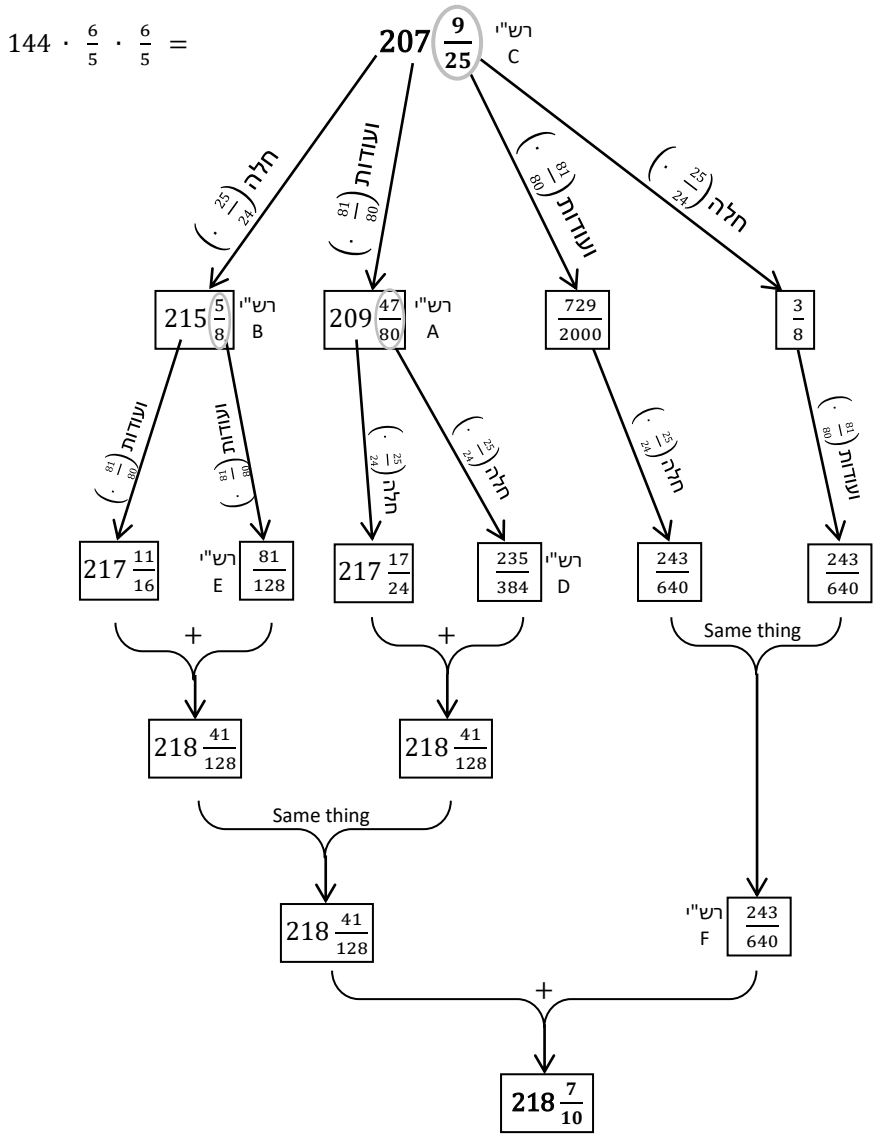
Now, if we calculate this right now we will get $218\frac{7}{10}$ which shows us right away that it's too much. Therefore, רש"י rejects this $\frac{1}{80}$ פשט.

But to follow through with רש"י is a little tricky since the calculation is not shown as I did it just now.

So, I'd like to break down the calculation into its itty bitty parts, and then show where the גמרא is holding and where רש"י has his problems (marked with letters A to F).

Let's start with a סאה ציפורית which is $144 \cdot \frac{6}{5} \cdot \frac{6}{5} = \mathbf{207\frac{9}{25}}$

Now we need to inflate this number by חלה and "ועודות". The way רש"י follows the גמרא, this all happens in separate steps, for each integer and each fraction separately. I'm going to fully map out what the steps look like, taking into account every possible permutation. You're ready? Take a deep breath...



רש"י says that if you're exacting you will find that the total amount is actually more than "an egg more than 217", i.e., at least 218.

רש"י gives us 3 points in the חשבון where on the way over to here we had rounded down when we had fractions left over but if you take these fractions together you get more than a whole egg.

A)

The first fraction רש"י mentions is when we took 207 and inflated it with ועודות, we got 2 more eggs ($207 \cdot$ ועודות $= 207 + \sim 2$ eggs).

However, that number "2" is a rounded down number. The real number is

$$207 \cdot \frac{81}{80} = 209\frac{47}{80}$$

which means that you need to take 207 and add $2\frac{47}{80}$

I'm going to write out the חשבון and then quote רש"י, and indicate by each word in רש"י where he is referring to in the חשבון (since it can be a little tricky following him).

$$\begin{array}{r} 2\frac{47}{80}\ (c) \\ 80\overline{)207}\ (a) \\ \underline{160} \\ 47\ (b) \end{array}$$

רש"י: בועודות[a] של מ"ז[b] ביצים שלא חשבת יש יותר מחצי[c] ביצה

Rashi: "The and-a-little-bits[a] of 47[b] eggs that you didn't take into account, comes out to more than half[c] an egg."

B)

When we calculated חלה, we took 207 and increased it by $\frac{1}{24}$ $\left(\text{by doing } 207 \cdot \frac{25}{24}\right)$ and we got 215, which is a rounded-down number because

$$\begin{array}{rl} 207 \cdot \frac{25}{24} = 215\frac{15}{24} & \\ \underline{\text{-}207} & \\ 8\frac{15}{24} & = \mathbf{8\frac{5}{8}} \text{ (as we wrote it in the chart)} \end{array}$$

Now with the חשבון written out and following the לשון of רש"י:

רש"י: ובחלתן[a] של ט"ו[b] ביצים שלא חשבת יש יותר מחצי[c] ביצה

$$24\overline{)207}\quad 8\tfrac{15}{24}$$
$$192$$
$$15$$

Rashi: "... and with the חלה[a] of 15[b] eggs that you didn't take into account there is an increase of more than half[c] an egg."

C)

And the number "207" itself is a rounded down number because:

$$144 \cdot \frac{6}{5} \cdot \frac{6}{5} = 207\frac{9}{25}$$

(See footnote[261])

לשון רש"י: וגם למעלה בתוספת ציפורית על ירושלמית נשארו ב' חומשין שלא חשבת שלא דקדקנו בהן

לשון רש"י:

> "And also there is above the increase of ציפורית upon the ירושלמית, there remains 2 fifths which you didn't take into account since we weren't being exacting in them."

I.e., when you increase from ירושלמית to ציפורית you get 207 and a fraction. That fraction "above" 207 is [about] 2 fifths, and we didn't take that into consideration.[262]

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[261] רש"י refers to this $\frac{9}{25}$ as "2 fifths", or $\frac{10}{25}$. We'll see later if that $\frac{1}{25}$ makes a difference.

[262] One might think that the 2 fifths Rashi is referring to is the exact $\frac{2}{5}$ of the סאה ירושלמית, i.e., $172\frac{2}{5}$ but that would be going in the opposite direction that רש"י is trying to say.
If רש"י meant that the 173 of ירושלמית was rounded up from $172\frac{2}{5}$ then the final count should be less than the total we got. Also, what would he mean by saying "the 2 fifths above the increase"? The $172\frac{2}{5}$ has nothing to do with the increase to ציפורית. And besides, it's clear from the earlier רש"י of מאתן ושבע הויין that we were exacting when we calculated this ירושלמית.

When I first saw this רש"י, I was a little confused since I did not see how these $\frac{2}{5}$ affect anything.

We have three factors that are supposed to add up to more than an egg:

A) $\frac{47}{80}$

B) $\frac{15}{24}$

C) $\frac{2}{5}$ (which is really $\frac{9}{25}$ rounded to nearest fifth)

Now, I see that $A + B > 1$ $= 1\frac{17}{80}$

But $A + C < 1$ $= \frac{79}{80}$

and $B + C < 1$ ...

So what's C doing here, בכלל? It doesn't help us at all!

Ah, you'll say $B + C$ is indeed greater than 1!? $\simeq 1\frac{3}{120}$

Yeah, but that's because you rounded $\frac{9}{25}$ to $\frac{10}{25}$! But carefully now: $\frac{15}{24} + \frac{9}{25} < 1$! $= \frac{197}{200}$

$A + B$:	$\frac{47}{80} + \frac{15}{24} = 1\frac{17}{80}$		
$A + C$: (rounded)	$\frac{47}{80} + \frac{2}{5} = \frac{79}{80}$		
$B + C$: (rounded)	$\frac{15}{24} + \frac{2}{5} = 1\frac{1}{40}$	$B + C$: (exact)	$\frac{15}{24} + \frac{9}{25} = \frac{197}{200}$

So, what's this $\frac{2}{5}$ doing here in רש"י? It's not relevant!

÷

The answer, I think, is that all of these numbers are themselves only steps in a larger equation.

D) The $\frac{47}{80}$ fraction that we got from ועודות דרבי needs to be inflated with חלה:

$$\frac{47}{80} \cdot \frac{25}{24} = \frac{235}{384}$$

E) The $\frac{15}{24}$ we got from חלה needs to be inflated with ועודות:

$$\frac{15}{24} \cdot \frac{81}{80} = \frac{81}{128}$$

F) The $\frac{9}{25}$ that we got by inflating ירושלמית to ציפורית needs to be inflated by <u>both</u> חלה and ועודות:

$$\frac{9}{25} \cdot \frac{25}{24} \cdot \frac{81}{80} = \frac{243}{640}$$

So now:

D + E = $\frac{235}{384} + \frac{81}{128} = \mathbf{1\frac{47}{192}}$

D + F = $\frac{235}{384} + \frac{243}{640} = \mathbf{\frac{119}{120}}$ ← Ooh, just under

E + F = $\frac{81}{128} + \frac{243}{640} = \mathbf{1\frac{1}{80}}$ ← Ah, just over

OK, so we could explain the relevance of C (fully inflated to F) is that when added to B (fully inflated to E) it gives us more than an egg.

And maybe that's what רש"י means when he says, "If you are exacting..." meaning, if you take all the inflations into account.

OK, so now let's return to the main subject. Because of these aforementioned problems, רש"י rejects this $\frac{1}{80}$ פשט for ועודות and goes on to explain the $\frac{1}{20}$ פשט.

This פשט reads the גמרא as <u>rejecting</u>[263] the חלה-step.

[263] With the גירסא of "אלא"

So, to repeat the full calculation:

$$144 \cdot \frac{6}{5} \cdot \frac{6}{5} \cdot \frac{21}{20} = \mathbf{217\frac{91}{125}}$$

So that's what the גמרא means when it said "217 and not another egg more".

So, to wrap up the story:

בוניוס send רבי a "סאה" with a volume of 217 $\left(\text{or}^{264} \text{ perhaps } 217\frac{91}{125}\right)$ eggs, which is an inflated שיעור based on "ועודות", whose initial שיעור would've been $207\frac{9}{25}$ $\left(= 217\frac{91}{125} \div \frac{21}{20}\right)$, $\left(\text{or perhaps } 206\frac{2}{3}\left(= 217 \div \frac{21}{20}\right)\right)$, which itself is the סאה of ציפורית, which is an inflated version of ירושלמית, which is an inflated version of מדברית.

Phew!

I have a question, please.

רש"י has 2 versions of how much "ועודות" are, either $\frac{1}{20}$ or $\frac{1}{80}$. And for each version, he reads the גמרא differently, one time as a follow-up of חלה, and one time reading that the גמרא fell off the חלה-פשט.

	מדבר'	ירושלמ'	ציפור'	חלה	ועוד'	
Version 1:	$144 \cdot \frac{6}{5}$	$\cdot \frac{6}{5}$		$\cdot \frac{25}{24} \cdot \frac{81}{80}$		רש"י, "Doesn't work."
Version 2:				$\cdot \frac{21}{20}$		רש"י, "Does work."

To me, it seems the whole question is whether we include חלה in the calculation or not. And apparently, both attempts are, at the outset, equally tenable.

So, the "with the חלה" פשט won't work at all with the $\frac{1}{20}$, since it's <u>way</u> off:

$$144 \cdot \frac{6}{5} \cdot \frac{6}{5} \cdot \frac{25}{24} \cdot \frac{21}{20} = \mathbf{226\frac{4}{5}}$$

[264] In other words, it's not clear which way to do the חשבון. You could do it from 144 up, which gets you to $217\frac{91}{125}$, or the other way, start with 217 and work down: $217 \div \frac{21}{20} \div \frac{6}{5} \div \frac{6}{5} = 144\frac{14}{27}$, which round off to 144 eggs, a קב.

So, fine. So we try the $\frac{1}{80}$ פשט and that came out a tad too big:

$$144 \cdot \frac{6}{5} \cdot \frac{6}{5} \cdot \frac{25}{24} \cdot \frac{81}{80} = 218\frac{7}{10}$$

So, רש"י rejects this whole "with חלה "פשט.[265]

But I wonder, who told us to use $\frac{1}{80}$ in the first place? Where did that number come from? Can't we find another number that would work?

Let's write out algebraically what we want. We want the final answer to be somewhere between 217 and 218 (since the גמרא said we end up with "217 and not another full egg".)

$$217 < 144 \cdot \frac{6}{5} \cdot \frac{6}{5} \cdot \frac{25}{24} \cdot \frac{x+1}{x} < 218$$

Simplify:

$$217 < 216 \cdot \frac{x+1}{x} < 218$$

If we divide all three parts of the equation[266] by 216 we will get:

$$\frac{217}{216} < \frac{x+1}{x} < \frac{218}{216}$$

Let's turn that into Unit-Fraction format:

$$\frac{217}{216} < \frac{x+1}{x} < \frac{218}{216}$$

$$\frac{216}{216} + \frac{1}{216} \qquad \frac{x}{x} + \frac{1}{x} \qquad \frac{216}{216} + \frac{2}{216}$$

$$1\frac{1}{216} < 1\frac{1}{x} < 1\frac{2}{216}$$

[265] And using the more accurate $\frac{24}{23}$ for חלה makes it worse:

$$144 \cdot \frac{6}{5} \cdot \frac{6}{5} \cdot \frac{24}{23} \cdot \frac{81}{80} > 219$$

[266] The rules for simplifying both sides of a ">" or "<" symbol is the same as for "=".

If we now simplify the right-side fraction (to make the numerator = 1) we will get:

$$1\frac{1}{216} < 1\frac{1}{x} < 1\frac{1}{108}$$

בקיצור, we see that any fraction in between[267] $\frac{1}{216}$ and $\frac{1}{108}$ would work.

Let's take a random fraction between these numbers, like $\frac{1}{120}$, and claim, by working backwards, that that is the "ועודות דרבי", and learn the פשט in the גמרא as a follow-up of the חלה פשט:

$$144 \cdot \frac{6}{5} \cdot \frac{6}{5} \cdot \frac{25}{24} \cdot \frac{121}{120} = 217\frac{4}{5}$$

and bam, we got it. 217 and not another egg.

So why doesn't רש"י give an amount of ועודות like $\frac{1}{120}$? Then he could explain the גירסא without the word "אלא" that learns חלה is part of the question! And where did רש"י get $\frac{1}{80}$ from in the first place?

Ah? Good question?

I'd like to suggest the following answer. The fact that רש"י rejects the $\frac{1}{80}$ פשט makes it sound like the whole פשט is a mistake, and I'd like suggest that it's <u>based</u> on a mistake.

What was the mistake?

When we first introduced this idea of ועודות[268], we showed that there's different ways to express the same idea. You could refer to the increase of <u>each</u> egg or the <u>total</u> increase.

We had a שיעור of 2 eggs and we wondered how much bigger would they be if we took into consideration ועודות.

One way we said it is that <u>each</u> egg gets bigger by $\frac{1}{20}$, for a <u>total</u> increase of $\frac{1}{10}$ of an egg $\left(2 \cdot \frac{1}{20} = \frac{1}{10}\right)$.

It's possible that one student heard $\frac{1}{20}$ of an egg and thought that refers to the <u>total</u> increase, which means each egg got bigger by $\frac{1}{40}$.

[267] but not including
[268] on page 304

And then when he repeated it over, he said $\frac{1}{40}$ as though that was the total, meaning that each egg increased by $\frac{1}{80}$.

So, there were two versions floating around the בית מדרש, one that worked with the גירסא of "אלא" that left חלה out of the equation, and one without the word "אלא" that only worked *with* חלה, since the ועודות only added 2. So, the חלה פשט needs another 8.

And so רש"י points out that this calculation doesn't work if you take into account all of the fractions.

I'm basing this answer on the fact that רש"י presents us with two versions, either $\frac{1}{20}$ or $\frac{1}{40}$, but the meaning of those versions differ from each other.

In the $\frac{1}{20}$ version, the fraction $\frac{1}{20}$ refers to the increase <u>of each egg</u>, whereas in the $\frac{1}{40}$ version, the fraction $\frac{1}{40}$ refers to the <u>total</u> increase of 2 eggs.

That got me wondering that perhaps this confusing aspect is what led to this untenable פשט.

And it could be that you don't even have to say that it was two separate mistakes.

> (Mistake #1: $\frac{1}{20}$ means total increase, when it actually means individual increase.
>
> Mistake #2: $\frac{1}{40}$ means individual increase, when it actually means total).

You can say that whoever confused it merely confused which is which; the individual with the total and the total with the individual (which is quite a reasonable mistake to make, as I can testify trying to get through this סוגיא!)

Like, he thought $\frac{1}{20}$ is smaller than $\frac{1}{40}$. So when he heard that $\frac{1}{20}$ is the smaller shiur (which it is, relative to $\frac{1}{10}$) he thought it was smaller relative to $\frac{1}{40}$. And when he was asked for the bigger shiur, he said $\frac{1}{40}$, but the listener (correctly) understood $\frac{1}{40}$ is bigger relative to $\frac{1}{80}$.

Perhaps.

(BTW, Tos' in Sukkah 8b D"H Ribuah uses a similar idea, about a lashon floating around the Beis Medrash which got reinterpreted over time.)

	Total	Each Egg
Version 1:	$\frac{1}{10}$	$\frac{1}{20}$
Version 2:	$\frac{1}{40}$	$\frac{1}{80}$

In other words, they knew that the difference between the total increase and each individual increase was a difference of a multiple of 2 (since there are 2 eggs in a חצי פרס).

And then the only question is where did חלה come from? Was the result of $\frac{1}{80}$ so small[269] that someone suggested חלה to make up the difference?

Just for fun, let's do that again with the more accurate value for חלה of $\frac{24}{23}$.

Let's write out the formula again, but with $\frac{24}{23}$ for חלה instead.

$$217 < 144 \cdot \frac{6}{5} \cdot \frac{6}{5} \cdot \frac{24}{23} \cdot \frac{x+1}{x} < 218$$

Simplify:

$$217 < 216\frac{216}{575} \cdot \frac{x+1}{x} < 218$$

If we divide all three parts of the equation by $216\frac{216}{575}$ we will get:

$$\frac{217}{216\frac{216}{575}} < \frac{x+1}{x} < \frac{218}{216\frac{216}{575}}$$

Let's turn that into Unit-Fraction format:

$$\frac{217}{216\frac{216}{575}} < \frac{x+1}{x} < \frac{218}{216\frac{216}{575}}$$

$$\frac{216\frac{216}{575} + \frac{359}{575}}{216\frac{216}{575}} \qquad \frac{x}{x} + \frac{1}{x} \qquad \frac{216\frac{216}{575} + 1\frac{359}{575}}{216\frac{216}{575}}$$

$$1\frac{\frac{359}{575}}{216\frac{216}{575}} < 1\frac{1}{x} < 1\frac{1\frac{359}{575}}{216\frac{216}{575}}$$

[269] $207.36 \cdot \frac{81}{80} = 209.952$

Turn left and right side into $1\frac{1}{\text{something}}$ by dividing the fractions by their numerators:

$$1\frac{1}{346\frac{202}{359}} < 1\frac{1}{x} < 1\frac{1}{133\frac{97}{467}}$$

We see that any fraction (after 1) in between $\frac{1}{346\frac{202}{359}}$ and $\frac{1}{133\frac{97}{467}}$ would work.

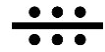
Let's take a random[270] fraction between these numbers, like $1\frac{1}{160}$, and plug it into[271] the formula:

$$144 \cdot \frac{6}{5} \cdot \frac{6}{5} \cdot \frac{24}{23} \cdot \frac{161}{160} = 217\frac{91}{125}$$

And there you have it.

אגב, this might strengthen the answer we gave earlier[272] about why did רש"י use an inside 24th when he was supposed to use an outside 24th. We suggested that since you could go through the גמרא with either method, רש"י chose what would be a simpler method. Based on the complications of these last two pages, we can add this complexity to the outside 24th method, which makes the inside 24th method relatively even easier.

Maybe.



[270] It's not totally random. I purposely chose a number that will yield a simple fraction result. But any random number would work.

[271] I've turned the Unit-Fraction $\left(1\frac{1}{160}\right)$ into a Fraction $\left(\frac{161}{160}\right)$, but it's all the same.

[272] On page 314

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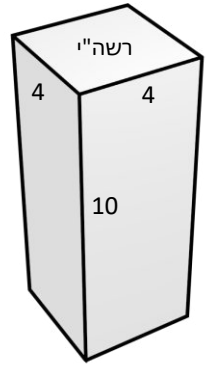
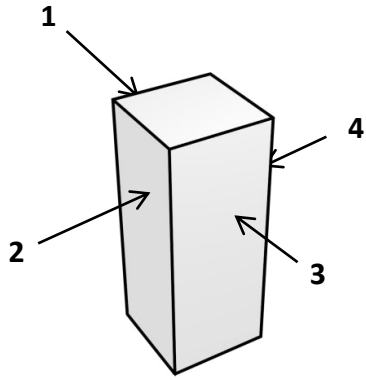
פ"ט. - **Rashi Kulei Alma - Amud Walls**



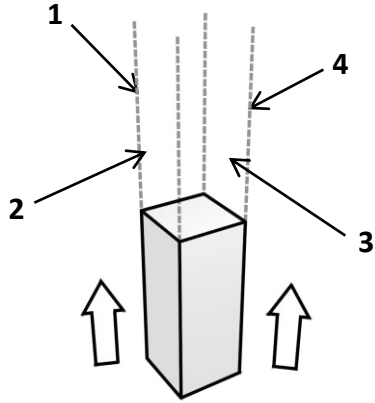
Let's say you have an עמוד in the middle of a רשות הרבים, and the עמוד is 10T tall and 4T wide, so the top of the עמוד is a רשות היחיד.



If you were to ask me where are the מחיצות for this רשות היחיד, I would've pointed to the sides of the pillar.

In other words, I would've said that the <u>difference</u> in height of 10T between the floor of the רשות הרבים and the top of the עמוד <u>is</u> the מחיצה.

Although, one could ask, if so then what is the purpose of the walls? Why can't the רשות היחיד just be a floating platform 10T high?[273]



So I would have to say something like the walls give a "קיום" to the distance, or something like that.

Anyway, this is not how רש"י is learning it. רש"י says over here[274] that the manner which down-below-מחיצות work is with גוד אסיק, that is, the 10T high walls down below create infinitely tall מחיצות above them, and those invisible side walls <u>above</u> the floor of the עמוד are the מחיצות of the רשות היחיד.

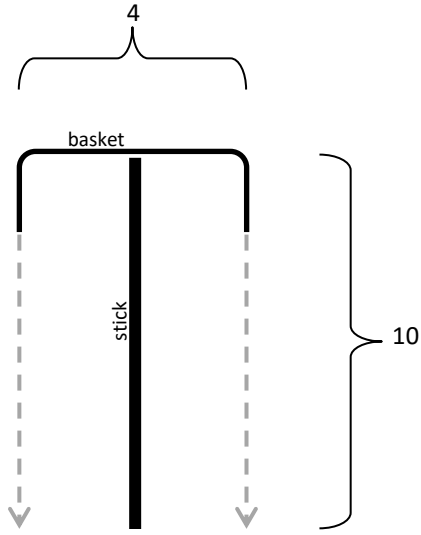
[273] Actually, that's a מחלוקת, but we hold that doesn't work. We'll discuss this in a minute.

[274] ד"ה כולי עלמא וזה לשונו: דדמו לעמוד ברשות הרבים דהוי רשות היחיד משום גוד אסיק משוך והעלה

רש"י also says this in מסכת שבת[275].

However, there's another גמרא in שבת[276] that implies the first way, that the מחיצות are the walls down below.

The גמרא brings a ברייתא that says that if there's a stick standing up in a רשות הרבים with a טרסקיל (a kind of large basket[277]) on top[278], there's a מחלוקת if we say גוד אחית (the walls of the basket extend downward).



According to רבי יהודה ברבי יוסי we do say the basket walls extend down and therefore the top of the basket is a רשות היחיד, whereas according to the חכמים we do not say that the walls extend down[279] and so the top of the basket is not a רשות היחיד.

Why talk about גוד אחית? Are we trying to make מחיצות down below? I thought we're trying to make מחיצות up above. So say גוד אסיק, going up!

Maybe, you need גוד אחית to create 10T high walls, and then those walls can do גוד אסיק upwards to surround the top of the basket with מחיצות?

Mmm, I'm not sure.

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[275] ו'. ד"ה קמ"ל גמורה וזה לשונו: ...וכן גדר דאמרינן מד' צדדין גוד אסיק פני המחיצה על ראשו ונמצא ראשו מוקף מד' צדדין וחללו ד'...

"like... a fence, we say from 4 sides גוד אסיק, the face of the מחיצה [goes] on its top. It turns out, its top is surrounded on 4 sides with a space between of 4 [טפחים]..."

[276] 101a

[277] The מאירי learns (שבת ק"א. ד"ה מחיצה) the basket is upside down, like in the diagram. The point of this is to give a "start" to the גוד אחית; the sides of the basket seem to "point downwards". See fuller quote on page 120.

[278] This is the "floating platform" we mentioned earlier

[279] Since "הגדיים בוקעים בה", "the little baby goats can walk under it", as the גמרא there explains (See picture on page 346). Meaning, there's a substantial space underneath which can be breached so it is difficult to view the walls as extending down.

Anyways, afterwards, I saw in Rabbi Ribiat's 39 Melachos[280] that he brings these two perspectives as a מחלוקת ראשונים. He brings this רש"י by us, and explains it the same way, that the walls of the רשות היחיד are the invisible walls <u>above</u> the עמוד that come up with גוד אסיק. And then he brings another שיטה, which is mainly the רמב"ן, that learns like we suggested in the beginning, that the מחיצות are down below, at the physical sides of the עמוד.

בש"כ וע"ש

This picture shows someone hanging up a picture frame on גוד אסיק walls in his roof-סוכה. The truth is, by סוכה the דין is more מחמיר. Although the רמב"ם holds גוד אסיק by סוכה, the שלחן ערוך brings two opinions, and the משנה ברורה doesn't allow it.

[280] In the section on הוצאה, footnote 23

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צ"ה: - צ"ו. Saving Tefillin - Complicated Chart

Here we have a מחלוקת between two תנאים, and the גמרא goes through a whole slew of possible understandings of what the מחלוקת is about, and on what points do the תנאים agree.

Once again, I have laid out a chart at the beginning of this piece that will serve as a map as we try to navigate through the גמרא, keeping track of all the different points.

(I'm only bringing in the elements of the גמרא that are relevant to the main מחלוקת.)

The rows are labeled on the left according to the stages of the גמרא.

The columns are labeled at the top by each דין, showing what the מאן דאמרים are holding for whatever stage of the גמרא we are up to.

To avoid clutter, I have labeled the right side of each row with a letter [A through O] and the bottom of each column with a number [1 through 7]. I will refer to specific squares in Battleship style, for those who know the game (or like a spreadsheet, I guess, but that's boring).

The slash in the box [✔/✘] means it's a מחלוקת; the top left part is the תנא קמא and the bottom right is רבן גמליאל.

✔ means he holds whatever that column is labeled and

✘ means he doesn't hold it.

Let's take for example G5. The "G" refers to the stage in the גמרא that says that כולי עלמא שבת זמן תפילין, and the מחלוקת is whether מצוות need כוונה, regarding whether there will be בל תוסיף or not. The "5" refers to the specific point of if מצוות need כוונה. In this particular row [G], the גמרא is understanding that whether מצוות need כוונה or not is the מחלוקת; the תנא קמא holds they do need כוונה whereas רבן גמליאל holds they do not.

As we go through it, I will try to point out what the גמרא says, what רש"י says, and my own comments.

	שבת זמן תפילין	תכשיט	רשבר"י מקום לב' תפילין	בל תוסיף בב'	מצות צריכות כוונה	בל תוסיף צריכה כוונה	לעבור שלא בזמנו צריך כוונה	
ת"ק	✔	✘						A
ר"ג אי שבת זמן תפילין	✔	✘						B
ר"ג אי לאו זמן תפילין	✘	✔						C
ר"ג לעולם לאו זמן	✘	✔	✔					D
לימא בדרב שמואל ב"ר יצחק קמפליגי	✘	✔	✘/✔					E
לא דב"ע איתא כרשבר"י ופלגינן אי שבת זמן תפילין	✔/✘	✘/✔	✔	✔/✘				F
אב"א כ"ע שבת זמן ופלגינן אי מצות צריכות כוונה	✔	✘	✔	✔/✘	✔/✘	/✔		G
כ"ע לאו תכשיט	✔	✘	✔	/✘	✔/✘			H
כ"ע תכשיט	✔	✔	✔	✘/	✔/✘	✔/		I
פלגינן אי תכשיט	✔	✘/✔	✔		✔/✘			J
מהרש"א פירוש של רש"י מחלוקת תכשיט	✔	✘/✔			✔			K
גיר' רש"י כ"ע שבת זמן ותכשיט ופלגינן צריכות כוונה	✔	✔	✔	✔/✘	✘/✔	✘/✔		L
גמ' איב"א כ"ע לצאת לא בעי כוונה ופלגינן אי בל תוסיף בכוונה	✔	✔	✔		✘	✘/✔		M
אי דסבירא לן שבת זמן דב"ע לא בעי כוונה לעבור ולא לצאת	✔				✘	✘		N
איב"א כ"ע לאו זמן ופלגינן אי בעי כוונה לעבור שלא בזמנו	✘	✔	✔	✔/✘	✘	✘	✘/✔	O
	1	2	3	4	5	6	7	

?רש"י

Let's start with the case and the מחלוקת itself.

משנה: A person finds תפילין out in a field (= כרמלית) where they might get lost/destroyed if he leaves them, so he wants to save them without being מחלל שבת (by carrying in a כרמלית = איסור דרבנן).

The תנא קמא says that the person should wear one pair of תפילין (i.e., one "תפילה" box on the head, and one on the arm).

רבן גמליאל says to wear 2 pairs (= 2 on the head and 2 on the arm).

And then, both תנאים say, the person should walk into the city, into a house and leave the תפילין there, and return for another pair (ת"ק) or two (ר"ג).

The גמרא wants to know what is the underlying מחלוקת between the תנאים. Why does the תנא קמא allow one pair and no more? Why does רבן גמליאל allow two and no more?

The possible factors that might be involved are:

1) Do they hold that שבת is a זמן for wearing תפילין?
2) Do they hold that תפילין can be considered a תכשיט (jewelry)?
3) Is there enough space on the head to halachically wear two boxes of head תפילין?
4) Is there a problem of בל תוסיף?
5) Do מצוות need כוונה to be fulfilled?
6) Does בל תוסיף need intent-to-be-מקיים-a-מצוה in order to be transgressed?
7) Can בל תוסיף be transgressed when it's not the time for the מצוה, without intent to do the מצוה?

These are the questions the גמרא will play with back and forth in trying to determine the rationale behind the opinions of the תנאים.

In the first stage [row "A"], the גמרא suggests that the reasoning of the תנא קמא is as follows: שבת is a זמן for תפילין. That means that מעיקר הדין, מן התורה, there's a חיוב to wear תפילין on שבת.[281]

Since there is a דין דאורייתא to wear תפילין, the תנא קמא says the person who found the תפילין should wear one pair and take it home (or to the city) and no carrying on שבת is being done, since he's wearing the תפילין.

The only possible היתר to wear more than one pair at a time would seem to be if תפילין can be considered a תכשיט, decoration, like jewelry, if not worn for the sake of a מצוה. The גמרא assumes, therefore, that the reason the ת"ק doesn't allow more than one pair must be because he holds that תפילין cannot be considered a תכשיט [A2], and therefore חז"ל had no way to allow more than one pair since דאורייתא it would be considered carrying.[282]

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[281] The reason we don't would then be a גזירה מדרבנן, a Rabbinical decree, similar to the דין that we don't blow שופר on ראש השנה if it falls out on שבת. See the beginning of the 4th פרק of מסכת ראש השנה.

[282] Even though we're talking about a כרמלית, which is only an איסור דרבנן, but חז"ל wanted to make the כרמלית as much like the דאורייתא as possible, and they felt that

The גמרא then tries to figure out the position of רבן גמליאל [rows B and C], specifically what he holds regarding if שבת is זמן תפילין and whether תפילין can be considered a תכשיט (which would allow more than the מצוה amount).

If the reason he allows is because שבת is זמן תפילין [B1] like the ת"ק, and not because of תכשיט [B2] then he should only allow one pair, just like the ת"ק!

And if he holds it's not זמן תפילין [C1] then the reason he allows any pairs at all must be because he holds it's a תכשיט [C2]. If so, he should allow <u>more</u> than two pairs (since they're all תכשיט, so allow as many pairs as he could fit on his head and arm)!

Either way, we're in a pickle.

So the גמרא explains that ר"ג holds שבת is <u>not</u> זמן תפילין [D1] and the reason he allows any at all is because he holds that תפילין can be a תכשיט [D2]. The reason he allows only two is because he holds like רב שמואל בר רב יצחק that there's space on the head for 2 תפילין (של ראש)[283] [D3].

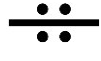
There are two points to this answer:

1) We limited תכשיט to only be applicable to an area of the head that is fitting for תפילין and

2) We expanded what we thought was the proper place for תפילין. Before, we thought there was only space for one תפילה של ראש whereas now we're saying there's space for two.

I was wondering why the גמרא has to say רבן גמליאל holds it's not זמן תפילין. Let it be זמן תפילין, one pair will be for the מצוה and the other will be for תכשיט (before the stage where we limit תכשיט to a specific area of the head).

רש"י[284] answers this by saying that if there was a מצוה on one pair of תפילין then there would be a problem of בל תוסיף on the other pair. This point will be discussed later in the גמרא.



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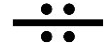
saving the תפילין two at a time wasn't urgent enough to warrant relaxing the restrictions of a כרמלית - so it seems to me.

[283] That could be considered תכשיט. More than 2 would bulge out too much. -רש"י

[284] ד"ה רבן גמליאל סבר

So now, the גמרא asks why don't we say that the תנא קמא holds שבת לאו זמן [E1] but the reason he allows only one pair is because he doesn't agree with this idea of רב שמואל בר רב יצחק that there's space on the head for 2 pairs (של ראש) but rather that there's only space for one [E3]? And the reason he allows any at all is because he holds that תפילין are a תכשיט in the place that's fitting for תפילין [E2]?



Now, I'm not sure what's the thrust of the גמרא's question.

1) It could be that the גמרא is asking on רב שמואל בר רב יצחק that there's a תנא (the תנא קמא) who disagrees with him. The force of this question would come from the idea that we normally assume the מחלוקת between מאן דאמרים to be as small as possible[285]. Since we *could* make the תנאים agree whether שבת is זמן תפילין, let's limit the מחלוקת to whether there is room on the head to properly wear 2 תפילין.

2) Or, the גמרא just סתם means how can you be sure that this is the way to learn and this is the דין, maybe you can learn it this other way and the דין is different?

   In other words, maybe both תנאים agree שבת is not זמן תפילין (and then that would for sure be the הלכה, since no one argues), AND תפילין is a תכשיט (and that too would for sure be the הלכה)...

Same idea, just a different thrust. Is the גמרא asking on רב שמואל or on what the דין might be.

3) Or, it could be, what's bothering the גמרא is that if indeed the תנא קמא holds there's enough space on the head for 2 תפילין then why doesn't he let wearing 2?

The follow-up of the גמרא seems to indicate this last way of learning by introducing בל תוסיף into the equation, as if until now we were unaware of (or did not take into consideration) בל תוסיף.

At any rate, the גמרא says no, the תנאים agree with רב שמואל בר רב יצחק that there's enough space on the head for 2 תפילין [F3][286]. They argue about whether שבת is זמן תפילין or not [F1] and whether תפילין can be considered a תכשיט [F2].[287]

[285] See footnote 201 on page 265

[286] The גמרא will stay with this אוקימתא for the rest of the סוגיא.

[287] This is identical with rows [A] and [D] except that this row [F] adds the point about בל תוסיף.

The תנא קמא holds it's not תכשיט [F2] and there would be בל תוסיף [F4] if the one who found the תפילין were to wear 2 (since there is a מצוה [F1] that activates the possibility of בל תוסיף), so he can only wear one.[288]

And רבן גמליאל holds it's not תפילין זמן [F1], so there's no בל תוסיף [F4] and there's space on the head for 2 תפילין, so the finder can wear that.

At this stage we're saying that לכ"ע it's not a תכשיט outside the normal area for wearing תפילין.

רש"י, in ד"ה ת"ק סבר שבת זמן, says "The Rabbis didn't allow more than the [required] מצוה because it's not תכשיט [F2] and... there will be בל תוסיף [F4] and that'll turn the תפילין into something carried [on שבת]."

I had understood רש"י means that if it <u>was</u> תכשיט then there <u>wouldn't</u> be בל תוסיף. Meaning, תכשיט precludes the problem of בל תוסיף (since it can be jewelry then it doesn't have to be that he's wearing it for another מצוה).

However, the next רש"י, in ד"ה ר"ג סבר זמן, says "The Rabbis allowed wearing the תפילין as תכשיט, and there won't be בל תוסיף [F4] <u>because there's no מצוה</u> [F1]." Why don't we say there's no בל תוסיף because it's a תכשיט [F2]?!

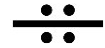
In other words, if תכשיט can make it not בל תוסיף then explain it like that and don't come on to an unnecessary argument of זמן תפילין!

So, I thought the פשט is that at this stage we're understanding that there's two possible options for allowing any תפילין at all, either because of מצוה or because of תכשיט. We start by trying each method on its own without the other and try to make sense of it [F1 & F2].

So we start with the תנא קמא. (What I mean is that we're building the argument step by step, and in this assumtion we get to the point that there's no בל תוסיף. We're not objectively determining what תכשיט would do; it's just that in this line of reasoning we got to a point where we don't need תכשיט to do anything since we're assuming there's no בל תוסיף on its own merit.) It's not תכשיט [F2] that's allowing the wearing of תפילין since if it was then he should allow two pairs. So it must be he's allowing because of the מצוה [F1]. And the reason he won't allow more than the מצוה must be because of בל תוסיף [F4]. It's not that תכשיט would stop בל תוסיף. Maybe it wouldn't. But we don't have it here for it to even be relevant.

[288] I'm not sure why we have to say the תנא קמא holds it's not a תכשיט; even if it was a תכשיט there'd be בל תוסיף, no? We'll discuss this point in a minute. Also, it could be this is what the גמרא will say later in row L.

And now for רבן גמליאל. It's not מצוה [F1] that's allowing the תפילין since that would only allow one. So it must be תכשיט [F2]. Ah, so why isn't there a problem of בל תוסיף [F4]? It must be because there's no מצוה. Even if תכשיט could preclude בל תוסיף, we don't have to come onto that, since there's no מצוה to begin with [F1].
And maybe this is what רש"י means when he says "And this is the reason the Rabbis allowed him..."



The מהרש"א says a little different.

He understands in רש"י ת"ק, that רש"י is giving two different ways to explain the תנא קמא.

In the first way, the reason the ת"ק doesn't allow more than one pair is because it's not a תכשיט.

In the second way, the reason he doesn't allow is because of בל תוסיף. And the מהרש"א writes, in this דרך it could even be that the ת"ק holds it is a תכשיט.

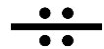
In other words, it could be across the board תכשיט cannot preclude בל תוסיף.

OK, back to the גמרא.

So now the גמרא suggests an alternate pshat [row G]: It could be that everyone holds שבת is time for תפילין [G1] and everyone agrees תפילין are not תכשיט[289] [G2], and the argument is whether מצוות need כוונה to be fulfilled [G5].

(According to our גירסא:) The תנא קמא holds מצוות need כוונה, so in order to allow wearing even 1 תפילה the finder will have to have in mind being yotzeh the mitzva (since it's not תכשיט, so he has no other recourse to mattir even 1 pair) and once he has in mind that he's doing the מצוה, (the assumption at this point is that) he'll automatically activate בל תוסיף on the second pair[290].

Whereas רבן גמליאל holds the finder doesn't need כוונה, so each pair is viewed as doing the מצוה, and בל תוסיף doesn't start since it requires כוונה-to-be-yotzeh to be a problem of בל תוסיף (or: he doesn't have in mind a specific pair so each pair could be the pair that's doing the מצוה! It's sort of like each pair "claims" to be the one doing the מצוה, and each one "claims" that it's the other one that's doing בל תוסיף, and since the finder isn't having כוונה on any specific one, so it can't be "proven" which one is the בל תוסיף, so it's like none of them are. Innocent until proven guilty.)



[289] רש"י will wonder about this point; stay tuned
[290] In other words, his כוונה to be יוצא the מצוה on the first pair automatically includes כוונה to be יוצא on the second pair as well, thus activating בל תוסיף.

רש"י isn't sure at this point what each תנא holds about תכשיט. רש"י tries 3 possibilities:

1) כ"ע לאו תכשיט [row H]
2) כ"ע תכשיט [row I]
3) מחלוקת [row J]

but no פשט works.

Let's start with כ"ע לאו תכשיט [row H]. רש"י's problem with this פשט is in the שיטה of רבן גמליאל. Even if you've solved the בל תוסיף problem, you haven't solved the שבת problem. In other words, you might be right that בל תוסיף needs to "identify" which תפילה is the extra one in order to be a problem, but שבת isn't as discriminatory. לגבי שבת, one pair is definitely not being worn for the מצוה (and it's not תכשיט [H2]) so there's no היתר to carry it! So how can רבן גמליאל allow 2 pairs?

Let's try the second פשט [row I], that כ"ע holds תכשיט [I2]. רש"י says, "If so then why doesn't the ת"ק allow two pairs? Don't have in mind the מצוה and you're not עובר on בל תוסיף [I4], and you can wear two for תכשיט!

(Here רש"י seems to contradict what he wrote above[291] that according to רבן גמליאל it's not בל תוסיף since there's no מצוה. Why make it a מחלוקת about whether it's a mitzvah? Let all agree it's a מצוה, and רבן גמליאל would say just don't be מכוון and there'll be no בל תוסיף!

I want to answer that at this stage the גמרא makes it a debate whether you need כוונה by מצוות [G5] and by בל תוסיף [G6], but at that stage the גמרא held that לכ"ע you don't need כוונה by מצוות [F5][292] nor by בל תוסיף [F6][292] so ממילא, the תפילין-finder is עובר whether he intends to do the מצוה or not [F4], if there is indeed a מצוה.)

So רש"י tries the 3rd option [row J], that תכשיט is a מחלוקת [J2]. This pshat is also untenable, says רש"י, because the גמרא should've made this the מחלוקת!

The מהרש"א explains that רש"י means let the גמרא make this the only מחלוקת [row K].

Rashi has another problem with this 3rd pshat, because it comes out that this line of the גמרא is synonymous with the earlier stage of לא דכולי עלמא אית להו דרב שמואל בר רב יצחק [row F].

The מהרש"א asks, how is this the same as [F]? There, it was a מחלוקת whether שבת was תפילין זמן or not [F1] while here it's כ"ע זמן [G1]! -?-

[291] ד"ה ר"ג סבר לאו זמן תפילין

[292] I didn't fill this in in the chart, to avoid clutter and confusion.

Additionally, asks the מהרש"א, before [row F] why doesn't the גמרא make the מחלוקת about תכשיט [F2], and not whether it's זמן or not [F1]?

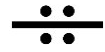
The מהרש"א leaves these questions בצריך עיון.

At any rate, due to these problems, רש"י suggests a different reading of the גמרא based on a different גירסא:

[row L] Everyone agrees שבת is זמן תפילין [L1], and everyone agrees תפילין can be a תכשיט [L2]. The argument is if מצוות need כוונה to be fulfilled [L5].

At this stage, the assumption is that whatever you hold regarding כוונה of מצוות [L5] is what you'll hold for כוונה of בל תוסיף [L6]:

> If you hold you don't need כוונה for מצוות then you also hold you don't need כוונה for בל תוסיף. That's the opinion of the ת"ק, so one pair you can wear but the second will be בל תוסיף (whether you intend to do a מצוה with it or not).
>
> Whereas if you hold that מצוות do need כוונה then you hold בל תוסיף also needs כוונה. And that's רבן גמליאל, so don't have in mind the מצוה and you won't be עובר בל תוסיף (but you can carry them because they are תכשיט).



The גמרא then suggests another פשט [row M] that you don't have to say that מצוות and בל תוסיף are necessarily connected regarding whether they need כוונה or not.

It could be that everyone holds מצוות don't need כוונה [M5] and they just argue about whether בל תוסיף needs כוונה [M6]:

- תנא קמא holds it doesn't (and therefore the תפילין-finder will be עובר if he puts on the second pair), whereas
- רבן גמליאל holds it does, so the finder just doesn't have כוונה to be מקיים the מצוה and so there's no בל תוסיף on two pairs.

Or, suggests the גמרא, a last possibility. (There seems to be גירסא issues here based on the ב"ח and the הגהות וציונים of the עוז והדר, so I'm going to present the גמרא as these גירסא-changers seem to be saying it.)

Maybe[293], everyone agrees that neither מצוות [N5] nor בל תוסיף [N6] need כוונה, and in fact if שבת was זמן תפילין [N1] then there'd be no מחלוקת. But in fact, שבת is not זמן תפילין [O1] and the only מחלוקת is whether to be עובר on בל תוסיף requires כוונה (to do the מצוה) when it's not even time to do the מצוה [O7].

- According to the תנא קמא, you don't need כוונה [O7] (so the finder is עובר on the second pair even though there's no מצוה now). The first pair is OK because... because...

Actually the גמרא will reject this explanation since there's no reason why the first pair should be OK.

whereas

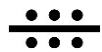
- רבן גמליאל holds to be עובר שלא בזמנו needs כוונה [O7] (so the finder simply doesn't have כוונה, no בל תוסיף [O4], and he wears two pairs as תכשיט [O2].

The גמרא rejects this explanation [row O] for two reasons:

1) because of the question I mentioned before that the ת"ק[294] shouldn't even allow a single pair (though what was the ה"א?!)

2) and because there cannot be a disagreement about whether בל תוסיף שלא בזמנו can be עובר without כוונה since we sit in the סוכה on the 8th day because of a doubt if it is still יום טוב, and our היתר is that we don't have כוונה (on the chance that it isn't יום טוב.) If it could be that to be עובר שלא בזמנו doesn't need כוונה then according to the תנא קמא there's no way to sit in the סוכה on the 8th day! Even if you don't have כוונה to be יוצא the מצוה (as we're saying in this stage would work to not be יוצא [O5]) but you might be עובר בל תוסיף if it actually turns out to not be סוכות today!

So, the גמרא concludes "מחוורתא כדשנינן מעיקרא", "More clear is like we answered initially", which רש"י says is the stage where we made the מחלוקת about whether שבת was זמן תפילין or not [row F].



[293] It could be that what's the driving the גמרא to say this is that it's מסתבר that כוונה is an all-or-nothing matter; If you don't need כוונה to be מקיים מצות תפילין then you shouldn't need it to be עובר on בל תוסיף either.

[294] Earlier, the גמרא says this could be רבי מאיר but continues to call him the תנא קמא. At this point, the גמרא actually calls him רבי מאיר.

*

ק"א: - <u>Lock and Key</u>

תנו רבנן "פתחי שערי גינה בזמן שיש להן בית שער מבפנים פותח ונועל מבפנים מבחוץ פותח ונועל מבחוץ מכאן ומכאן פותח ונועל כאן וכאן אין להן לא לכאן ולא לכאן אסורין כאן וכאן

וכן חנויות הפתוחות לרה"ר

- בזמן שהמנעול למטה מעשרה מביא מפתח מערב שבת ומניחו באיסקופה למחר פותח ונועל ומחזירו לאיסקופה
- ובזמן שהמנעול למעלה מעשר
  - מביא מפתח מערב שבת ומניחו במנעול למחר פותח ונועל ומחזירו למקומו דברי רבי מאיר
  - וחכמים אומרים אף בזמן שהמנעול למעלה מעשרה טפחים מביא מפתח מערב שבת ומניחו באיסקופה למחר פותח ונועל ומחזירו למקומו (רש"י: לאיסקופה) או בחלון שעל גבי הפתח אם יש בחלון ארבעה על ארבעה אסור מפני שהוא כמוציא מרשות לרשות"

מדקאמר וכן חנויות מכלל דבאיסקופת כרמלית עסקינן

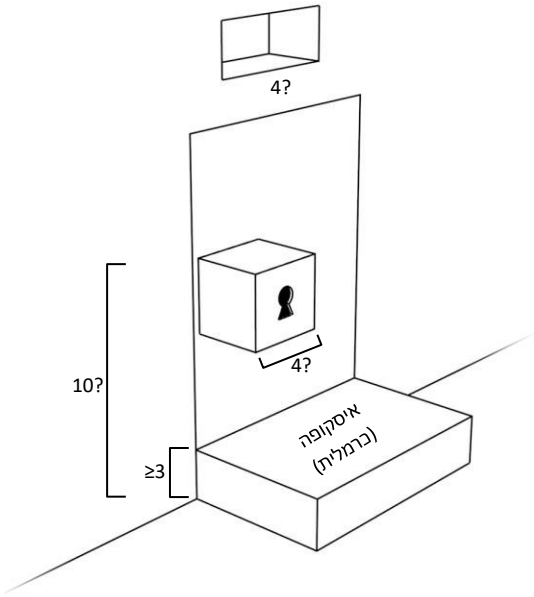
האי מנעול היכי דמי

- אי דלית ביה ארבעה מקום פטור הוא
- ואי אית ביה ארבעה בהא לימא רבנן "אף בזמן שהמנעול למעלה מי' מביא מפתח מע"ש ומניחו באיסקופה למחר פותח ונועל בו ומחזירו לאיסקופה או לחלון שעל גבי הפתח" והא קא מטלטל מכרמלית לרה"י

אמר אביי לעולם דאין בו ארבעה ויש בו לחוק ולהשלימו לד' ובהא פליגי דר"מ סבר חוקקין להשלים ורבנן סברי אין חוקקין להשלים

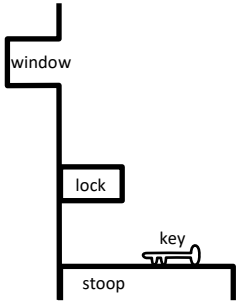
The ברייתא is discussing a door into a רשות היחיד (specifically, a door to a store; presumably the owner keeps food or other permitted items in there that he uses sometimes on שבת).

There is a stoop (an איסקופה) leading up to the door. The stoop itself has a din of a כרמלית (being at least 4T x 4T wide and at least 3T high but less than 10T high from the surrounding רשות הרבים).



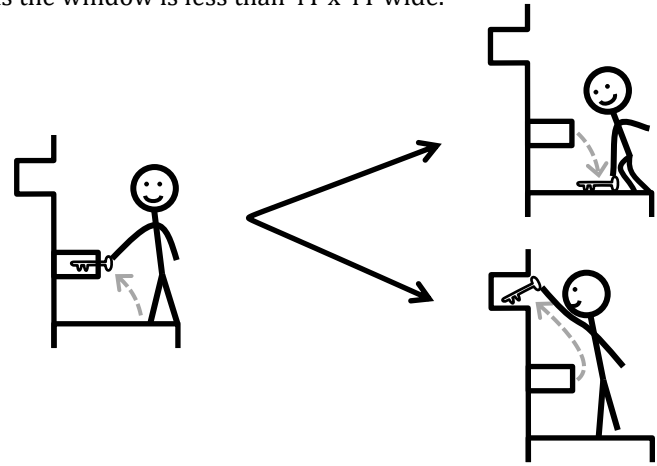
There is a window above the door. The ברייתא discusses both when the window is big enough to be a רשות היחיד and when it's too small and is therefore a מקום פטור.

And there is a lock for the door. The ברייתא and the גמרא will discuss various cases, including when the lock is below 10T and at least 10T in height (I'm not sure if that's measured from the ground or from the איסקופה), and whether the lock's width and length are at least or less than 4T.



The חכמים in the ברייתא say that even when the lock is above 10T, the owner can leave the key on the stoop before שבת.

On שבת itself, he can (while standing on the stoop) pick up the key from the stoop, unlock the door, and return the key to the stoop or place the key in the window above the door, as long as the window is less than 4T x 4T wide.



ק"א:

רבי מאיר does not allow moving the key from the stoop to the lock; rather he holds the key must be placed on top of the lock already before שבת. On שבת itself, you can take the key, unlock the door, and replace the key onto the top of the lock.

As we said before, the case is where the stoop is a כרמלית being at least 4T x 4T wide and at least 3T above the surrounding רשות הרבים.

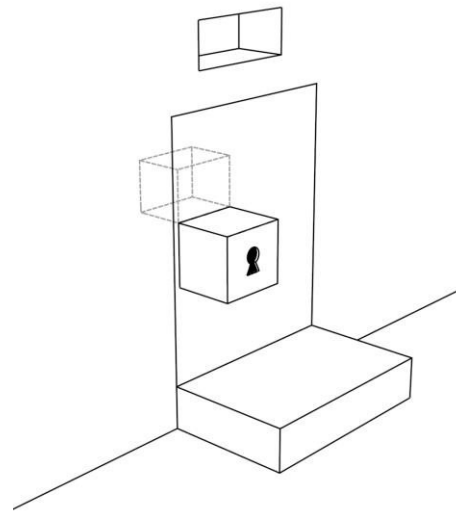
The lock itself might be either a מקום פטור or a רשות היחיד, depending on its dimensions.

If it's a מקום פטור then we could understand why according to the חכמים you can put the key in it from the כרמלית since it's going from a כרמלית to a מקום פטור. And we could understand why you can put the key afterwards above the door in the window since that too is a מקום פטור (at this stage of the ברייתא, which is dealing with a window that is less than 4T x 4T).

The גמרא's problem with this setup is that there should be no reason why רבי מאיר wouldn't allow moving the key from the stoop to the lock considering how the lock is a מקום פטור. So why doesn't he doesn't allow it?

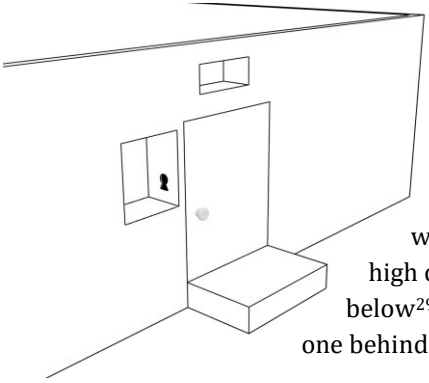
But, it also doesn't make sense to say the lock is a רשות ביחיד (by being at least 4T by 4T wide) since how could the חכמים allow placing the key there from the stoop? That would be going from a כרמלית to a רשות ביחיד, which is אסור!

So, the גמרא goes back and says we're talking about where the lock is less than 4T x 4T, a מקום פטור, and the reason רבי מאיר doesn't allow putting the key there from the stoop (=כרמלית) is that he holds "חוקקין להשלים"[295], that we "cut away" (conceptually) part of the door so that there is enough space to consider the lock to be 4T x 4T, to be a רשות היחיד.

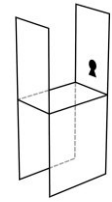


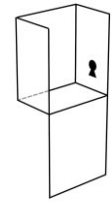
[295] like the גמרא say on 11b

Now, let's back up a step to when the גמרא suggested that this lock is a רשות היחיד, in that it is 4T wide by 4T long. How would that be enough to deem this lock a רשות היחיד? Where are the necessary 10T high walls surrounding it? (It's not enough to merely be 10T off the ground; it needs walls as well![296])



I thought to say that maybe the lock is on the side of the door, in a niche in the wall, with the keyhole going into the door from one of the side walls of the niche. The niche itself is 4T x 4T wide and has two walls 10T high on each side, with two walls below[297] the niche, one in front and one behind.

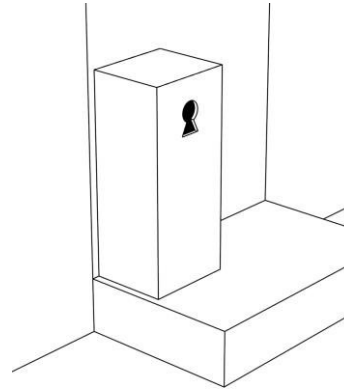




(Or 3 walls inside plus the one in front down below)

The keyhole would then be a רשות היחיד under the גדר of "חור דרשות היחיד", even if the keyhole itself lacks the dimensions of a proper רשות היחיד[298].

רבינו פרץ[299] suggests a different setup. He suggests that the lock protrudes forward from the door (like we first visualized it) but its sides extend all the way down to the ground. (I'm not sure if רבינו פרץ requires actual walls; just that it is 10T high is enough, as far as intrinsic dimensions are concerned. It is only לגבי גדיים בוקעין that he makes the walls actually extend down[300].)



[296] See page 328 and footnote 273 there.
[297] This works מתורת גוד אסיק; See piece on Rashi Kulei Alma which starts on page 328.
[298] According to some ראשונים. See משנה ברורה שמ"ה:ט'.
[299] And תוס' הרא"ש, and תוס' צ"ט:
[300] See footnote 279

However, רבינו פרץ[301] rejects this תרץ (and presumably mine as well) on the grounds that the ברייתא doesn't sound like you need more than what it says.

Although, I don't know why that's so bad. Often the משנה or ברייתא leaves out details that we need to fill in מסברה.

Be it as it may, רבינו פרץ gives another תרץ, that גדיים can't be בוקעין since there's an איסקופה.

I don't really understand why the איסקופה is relevant. Why can't גידיים be בוקעין? Because the איסקופה is too high for them? So pick them up and put them on the איסקופה! Why are we looking at the איסקופה as impeding the גידיים as though they were limited to the surrounding area? Why don't we just look at the איסקופה as the ground upon which they are standing?

I mean, if I had a סוכה whose walls were at least 3T off the ground but there was a fence around my סוכה so גידיים couldn't get there, would I say אין גידיים בוקעין?!

Let's suggest 2 answers:

1) Perhaps, what it means is that we look at this stoop as an <u>interruption</u> of the רשות הרבים. If you look at this <u>scene</u>, you see בעצם a רשות הרבים. The stoop is not part of this רשות הרבים since it is 3T high from it. Thus, when we judge the "בוקע-ness" of this lock, we view it as essentially in the רשות הרבים, and the goats of this רשות הרבים cannot be בוקעין through it. As opposed to a "standard" כרמלית like a desert, where it need not be 3T off the surrounding רשות הרבים (even if it <u>was</u> surrounded by a רשות הרבים, for example if it was bounded on four sides by highways) but rather it gets its identity of a כרמלית intrinsically due to its use and size but not its position.

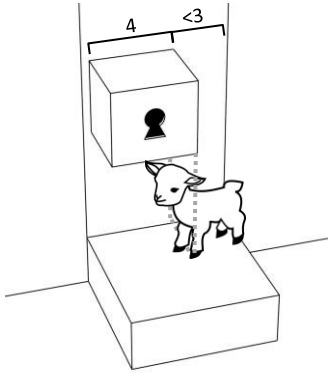
   The איסקופה, on the other hand, doesn't have enough components to be a רשות unto itself, so we look at it as something <u>in</u> the רשות הרבים.

   With this explanation, the case with the fence around the סוכה becomes clear. Since this רשות (i.e., the סוכה) is defined <u>intrinsically</u> (by its size and structure), the "goats" are viewed within it, and they are running under the hanging סוכה walls, thus invalidating them.

2) Or, perhaps there's a certain minimum area outside the hanging walls that the goats need to be considered "there" so much so that they could be considered "breaking through" from there as they run under the walls, and our ברייתא is dealing with a case where you don't have enough space for that.

[301] Only רבינו פרץ; תוס' and ש"הרא תוס' do not reject this תרץ.

So, for example, let's say the goats need an area at least 3T (not counting the area under the lock) upon which to stand. And let's say we had a stoop that was less than 7T wide.[302] There's not enough space for the גידיים to stand outside the גוד אחית of the lock, so we don't look at this כרמלית as one that could contain גידיים (unless if they were already under the lock, at least partly).



In other words, in order to be considered "breaking through" underneath the lock there needs to be an adequate space from which to break through.

It's similar to the idea that to be עובר הוצאה from רשות to רשות, each רשות needs to be an actual רשות unto itself and there needs to be an actual עקירה, הוצאה, and a הנחה.

Using those terms as a משל, we would say that the goats never had a proper "עקירה" (since they were never properly "out" of the hanging walls).

So, in my case with the fenced-in סוכה, if the fence was less than 3T away from the wall[303] then אין הכי נמי that wouldn't be considered גידיים בוקעין[304] if the fence was at least 3T high[305], whereas if the space between the wall and the fence was 3T apart from each other then we would say the גידיים are בוקעין, and that would be a non-kosher סוכה wall.

The ריטב"א[306] gives another reason why גידיים בוקעין is not a problem, since here the lock is essentially a roof that is 4T wide we can say פי תקרה יורד וסותם[307], and wherever we can say that then גידיים בוקעין will not be a problem, as we see from the סוגיא of אכסדרה[308].

[302] This explanation is still problematic since we've limited the ברייתא to a specific case. The whole דרך of רבינו פרץ is to explain the ברייתא without making אוקימתות.
[303] Again, I made up this distance. I'm just using it as a handle for the סברה.
[304] Besides the fact that לבוד will make the wall be considered attached to the fence, but that's beside the point.
[305] Otherwise, the goats would walk right over it, like רבינו פרץ implied.
[306] ד"ה וש"מ הדר
[307] Like תוס' says on פ"ו. ד"ה קורה
[308] On צ"ה., where we say פי תקרה יורד וסותם even though there is a large space underneath the roof.

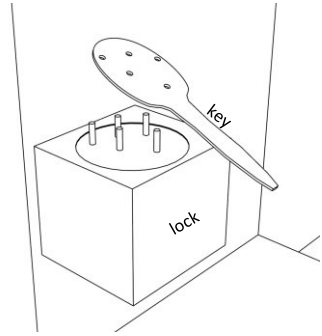
OK, so now we've managed to make the lock a רשות היחיד. But that's only the top of the lock! How does that affect the keyhole?

בשלימא, in my case with the side lock, the keyhole is a hole that opens directly into a רשות היחיד, thus giving it the same status as a חור ברשות היחיד. But in רבינו פרץ's case, the keyhole is in the airspace of the איסקופה! (Or perhaps of the רשות הרבים, if it sticks out enough!)

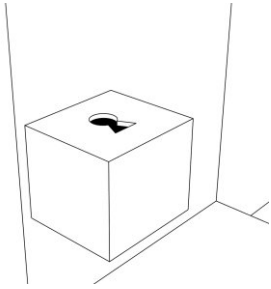
I was thinking of a few other ideas (although none would help רבינו פרץ since he doesn't want to make an אוקימתא in the משנה).

Maybe it's a huge lock with inner dimensions of a רשות היחיד.

Or maybe it's a kind of lock where the locking-mechanism itself was on top of the lock. →



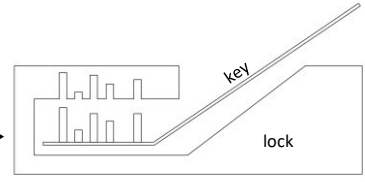
Or, it's a regular lock but the keyhole is facing up, so that the key goes down into it.

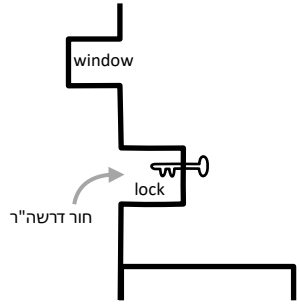


The keyhole would then have the status of חור דרשות היחיד.

In fact, I saw this picture somewhere in a פירוש on עירובין (but I cannot find where), that some types of ancient locks had a hole on top where you put the key in at a downward angle.

For example, in Britannica Encyclopedia, they have a type of ancient Egyptian lock that looks something like this: →





Or maybe we can suggest that the back of the lock opens up into the רשות היחיד.

ק"א:

Aaron Kronenfeld suggested to me a different type of answer, that perhaps the keyhole automatically gets a status of a חור דרשות היחיד since it <u>serves</u> the רשות היחיד (the store, by locking and unlocking it), as it sounds a little bit from רש"י in שבת ז: where he refers to the חורי רשות היחיד as "תשמישי רשות היחיד" והשלימו ד"ה.

(This answer avoids making an אוקימתא.)

If you explain the status of the keyhole as a חור דרשות היחיד (whichever way we suggested) then you would need to explain the גמרא according to the ראשונים[309] that hold that חור דרשות היחיד needs to be itself 4T x 4T, because otherwise how could the גמרא say that if it was less than 4T x 4T it would be a מקום פטור?

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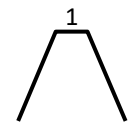
ק"ב. - **Slope of Bed Canopy**

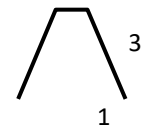
The Gemara brings a statement from רב in the name of רבי חייא that describes a sloped bed-canopy (shaped like a pup-tent), about whether it's מותר on שבת to spread it over the bed or unspread it, with the potential problem of בונה, building on שבת, or סותר, dismantling.

רב ששת בריה דרב אידי says it depends on three conditions or rules.

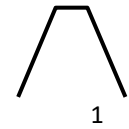
1) The "roof" of the tent (the top) cannot be a טפח wide



2) The slopes (the sides coming down) cannot be a טפח wide within 3 טפחים of the roof



3) The space under the slope cannot be a טפח wide



[309] Like תוס' brought in the ט:ה' משנה ברורה שמ"ה that we brought in footnote 298

The first rule is pretty straight forward, literally. (That's a pun)

The second rule isn't so clear cut, especially when taken together with the third rule.

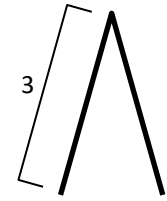
Let's discuss rule #2.

You've probably guessed by now that I'm wondering what "3 from the roof" means, and I wonder how you measure the one טפח.

I could think of two possibilities for each:

"3 from the roof" can mean:

measured diagonally along the slope



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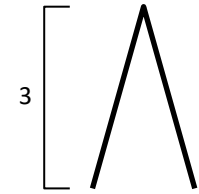
or vertically, directly up and down



And "one טפח" can be measured:

across the whole width

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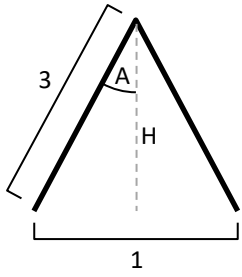
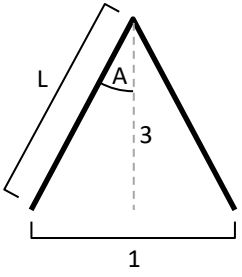
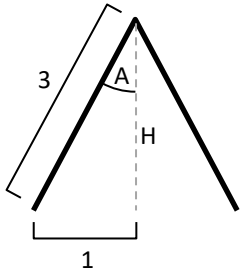
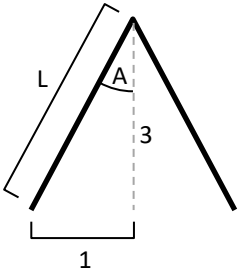
or under one slope only (one half of the total canopy width)[310]

[310] At the moment, I'll be ignoring the possibility that the "roof" of the tent might have some width to it

These two factors gives us 4 possible combinations (4 permutations):

(diagrams are not to scale)

	3 ⁄	3 |	difference
1 ⌐	 $A = s^{-1}\left(\frac{.5}{3}\right) = 9.59...°$ $H = \sqrt{3^2 - .5^2} = 2.958...$	 $A = t^{-1}\left(\frac{.5}{3}\right) = 9.46...°$ $L = \sqrt{.5^2 + 3^2} = 3.04...$	$A = .13...°$ $H = .04196...$ $L = .04138...$
1 ⌐	 $A = s^{-1}\left(\frac{1}{3}\right) = 19.47...°$ $H = \sqrt{3^2 - 1^2} = 2.828...$	 $A = t^{-1}\left(\frac{1}{3}\right) = 18.43...°$ $L = \sqrt{1^2 + 3^2} = 3.16...$	$A = 1.036...°$ $H = .17...$ $L = .162...$
difference	$A = 9.877...°$ $H = .1296...$	$A = 8.97...°$ $L = .12...$	

The chart shows that the difference is very little between whether 3 is vertical or on the slope.[311]

[311] See Appendix G about π and perceptible differences, and Appendix J on Trigonometry

Let's write out the גמרא's three rules again, expanding rule #2's possible meaning in bullet-list form, so we can easily refer to each possibility by name.

A) The roof cannot be a טפח wide

B) The slope cannot have a horizontal טפח within 3 טפחים from the roof

I. "Within 3T of the roof" can mean:

a. 3T along the slope

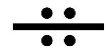
b. 3T measured vertically

II. and whichever way that is, the horizontal 1T is measured either:

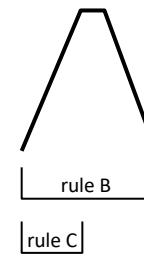
a. under both slopes

b. under one slope

C) There cannot be a טפח underneath one of the slopes.



בשלימא if you say B) II. a. then C) comes along and limits the total horizontal width to 2T.



In other words, B) II. a. means that under each slope is .5 טפח (since that's half of the total width of 1. I'm ignoring the "roof", for now). So then C) comes along and says that as the slope continues down and gets wider, it cannot get as wide as a טפח.

משא"כ if you say B) II. b. which says you cannot have a טפח under one slope so then היכי משכחת ליה where B) II. b. is considering C) as a possibility?

In other words, C) limits the horizontal width under each slope to 1T.
If so then B) II. b. is saying "And you can't have that within 3T from the roof". Um, I can't have that *anywhere* under the roof because of rule C)!

Maybe rule B) isn't a length requirement; it's an angle requirement. You can't have a slope where a טפח width is achieved within 3T. But אין הכי נמי, you can't *actually* have a טפח underneath this slope!

Meaning, this angle of slope is considered roof and not wall.[312]

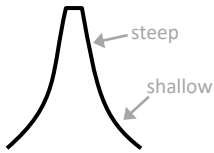
And what would be the נפקא מינה anyway, considering how neither roof (rule A) nor wall (rule B) can have a טפח width?!

Maybe, if it's considered roof then it's מצטרף with the top of the tent. So, let's say the top of the tent is 3/4 of a טפח, and the roofy slope is 1/4 of a טפח, that would add up to a טפח and be אסור.

But if it's considered wall then it's not מצטרף, and each (roof and wall) can be less than 1 טפח each.

This whole approach is a little שוער since the גמרא said גג לגג סמוך.

Maybe only there, slope angle needs to be steep, but below 3 can have wider angle.



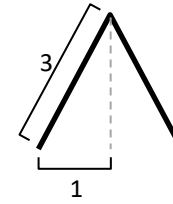
One may ask on this approach, if 3T/1T is an angle and not a length then why say טפח? Say אצבע or something else, like 3E/1E. Why טפח דווקא?

I'd answer that the rule before and the rule after are using טפח, so too here we use טפח, even though אין הכי נמי we could've used something else.

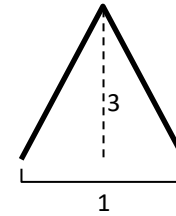
[312] This is very similar to the piece we wrote starting on page 84.

To me, it would seem that the two most likely permutations would be slope over half width B) I. a. & B) II. b. or vertical over whole width B) I. b. & B) II. a..

- **slope over half width** since the slope is being measured against the טפח. Meaning, the measurement of these 3 טפחים happens alongside the length of the bottom טפח.

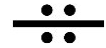


- **vertical over whole width** since the טפח is being measured as a certain distance from the roof (As opposed to slope over width where the slope 3 is the long way to measure the טפח's distance from the roof; Rather, it would make more sense to measure the shortest path.)



  Though, I'm still not sure about the טפח. Maybe, if it's an angle-requirement then it would make more sense to measure the טפח only under one slope.

  Not sure.



Let's see רש"י on the last two rules:

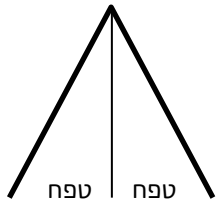
Rule 2) "It's a very narrow slope, so that when you measure the width below 3 to the roof, it's not a טפח wide. So it's all 'wall' and there's no 'אוהל' here at all. But if it's wide a טפח within 3 to the roof, it's like its roof is a טפח, and it's assur to set it up [this kind of cover]."

Rule 3) "Corresponding to the middle of its roof until where the edge of the sheet ends there, it doesn't widen a טפח, so it's all 'wall' and there's no 'אוהל' here at all. But if each wall's slope widens a טפח, that slope (even though it's at a slant) is considered 'אוהל'..."

Now, there are pictures embedded in the text of רש"י but it's not clear where these pictures are going on.

I don't think the first picture is relevant to us for this discussion.

Here's the second picture:



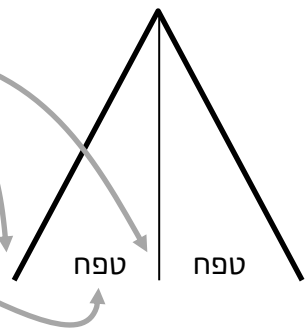
This picture appears on the side of the page where רש"י is explaining the first two rules, so it might be going on Rule B. If so then it determines B.I.b. and B.II.a. (i.e., vertical 3 over half-width טפח)

If it's going on C) (which I think it is) then the interpretation of B) remains undetermined, and the picture is on the wrong side of the page, since the פירוש of רש"י on C) appears on the next page, after you turn the page. If so then the vertical line illustrates the words of רש"י ד"ה בשיפועה:

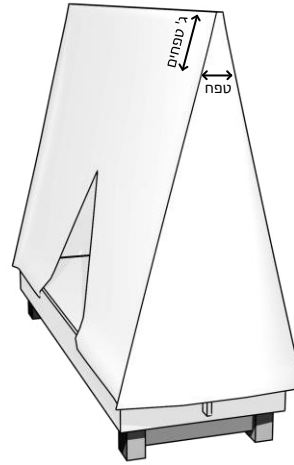
"Corresponding to the middle of its roof...

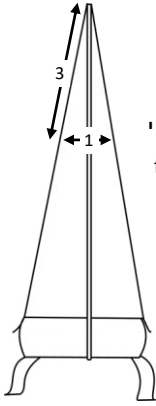
until where the edge of the sheet ends there...

it doesn't widen a טפח..."

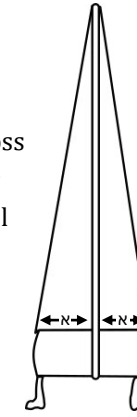


In the back of the עוז והדר גמרא they have a picture showing 3 along the slope and the טפח under the whole width, B) I. a. & B) II. a..

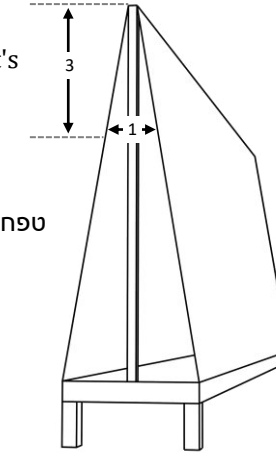


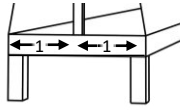


פירוש חי shows for rule B, 3 on the slope and the טפח across the whole width (like עוז והדר), and for rule C, they show like we explained רש"י that the טפח is under half the total width, meaning, under one slope.



Artscroll shows 3 measured vertically and 1 across the whole width B) I. b. & B) II. a.. (Various versions of Artscroll show slightly different pictures, 2D or 3D, but it's the same thing.)





And they show rule C like פירוש חי, a טפח on each side.

טאנצן

Appendixes

Shiurim, Sizes, and Conversions

Linear (Length):

Imperial:

12 inches (in. ") in a foot

5,280 feet (ft. ') in a mile (mi.)

Metric:

10 millimeters (mm.) in a centimeter

100 centimeters (cm.) in a meter

1000 meters (mtr.) in a kilometer (km.)

Halachic:

4 etzbaos (sing. etzbah) in a tefach

6 tefachim (sing. tefach) in an amah

2,000 amos (sing. amah) in a mil

4 milin (sing. mil) in a parsa'ah (plu. parsaos)

Conversion:

2.54 cm per inch

2 cm per etzbah (Rav Avrohom Chaim Na'ah) (= 48 cm per amah)

$2\frac{5}{12}$ centimeters per etzbah (The Chazon Ish) (= 58 cm per amah)

For example, to convert 1 mil to 1 mile, according to Rav Avrohom Chaim Na'ah:

▶	1 mil · 2000	= 2000 (amos)
⬇	· 6	= 12,000 (tefachim)
⬇	· 4	= 48,000 (etzbaos)
⬇	· 2	= 96,000 (cm)
⬇	÷ 2.54	= 37,795.27... (inches)
⬇	÷ 12	= 3149.60... (feet)
■	÷ 5,280	= 0.5965... (miles)

1 mil = .5965... of a mile, according to Rav Avrohom Chaim Na'ah

Area:

Beis Sa'ah = 50 amos by 50 amos (or some other shape that is 2500 sq amos)

Beis Sa'asayim = 100 amos by 50 amos (or some other shape = 5000 sq amos)

Volume:

Metric:

1 milliliter = 1 cm · 1 cm · 1 cm (= 1 cm^3 = 1 cc)

1 liter = 10 cm · 10 cm · 10 cm (= 1,000 cm^3)

Halachic:

6 beitzos in a loog

4 loog in a kav

6 kav in a sa'ah

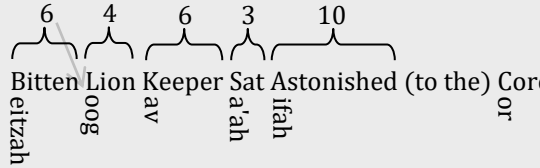
3 sa'ah in an aifah (pronounced AYE·fuh)

10 aifah in a cor (pronounced CORE)

Here's a mnemonic for remembering the exchange rates. Each word in the mnemonic represents one of the volumes. The first letter in the word is the same as the volume it represents (for example, the word "bitten" starts with a "b" like the word "Beitzah"). The number of letters in the word represents the exchange rate to the next volume. For example, the first word is "Bitten" which represent a "Beitzah". The word bitten has 6 letters in it, so there are 6 Beitzos in the next volume, a loog. The whole mnemonic is:

Bitten **L**ion **K**eeper **S**at **A**stonished (to the) **C**ore

That is:



Some other volumes:

4 rivi'is in a loog[313]

3 sa'ah in a bath (bas)

40 sa'ah in a mikveh

A British imperial fluid ounce is 28.4130625 mL. A US customary fluid ounce is 29.5735295625 mL. A US food labeling fluid ounce is 30 mL. I never use ounces except in my baby's bottle.

By the way if you've noticed, I prefer to remember a single string of conversions that are easy to remember but might take longer to perform than a giant chart that might be faster but much harder to memorize.

[313] Which is why it's called a rivi'is which means "a fourth", although sometimes the word rivi'is might refer to the loog itself being a fourth of a kav. See Tos' on 29b, paragraph headed ויעמוד על רביעית

Difference between 5000² Amos and 5000 Amos²

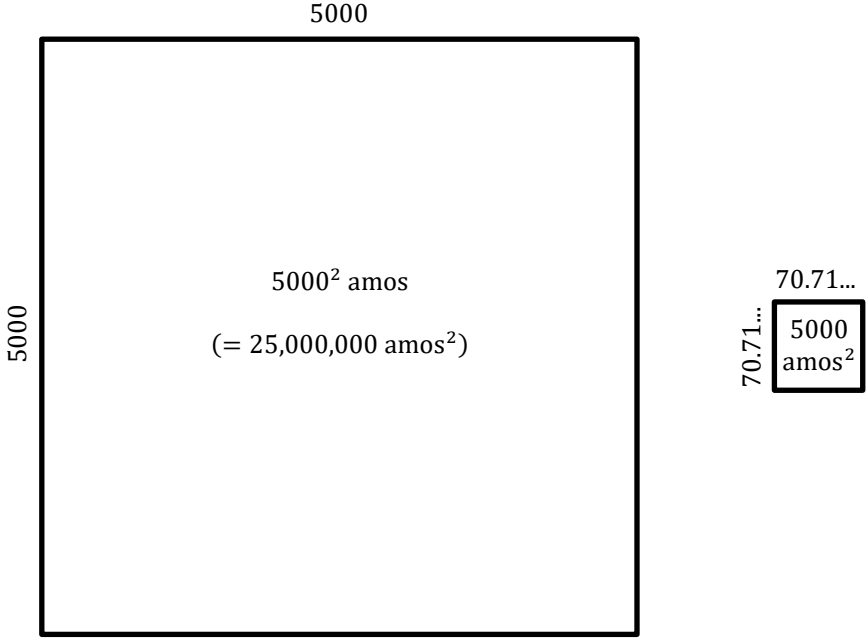
You gotta be careful when people use the word "square" since they might mean one thing but are saying something completely different.

"5000 square amos" is not the same as "5000 amos squared", or even more ambiguous will be "5000 squared amos". Is that 5000 squared which happen to be in amos or is that 5000 little amos-squares? Big difference.

5000 squared (5000^2) means $5000 \cdot 5000$, which is a big number. Something like 25,000,000.

Square amos are little squares with sides one amah long.

Usually in this sefer, we will use 5000 $amos^2$



Diagrams not to scale!

How to Calculate Square Roots

Write down the number you want to square root inside the square root symbol:

$$\sqrt{751.6983}$$

Now, starting from the decimal point and working your way outward, break up the number into 2-digit chunks:

$$\sqrt{7|51|.|69|83}$$

Each chunk of 2 digits will get a single digit above it, above the square root symbol, which we will soon show how to do (the decimal point goes straight above the line):

$$\begin{array}{c} 9\ \ 9\ .\ 9\ \ 9 \\ \sqrt{7|51|.|69|83} \end{array}$$

First, take the chunk all the way on the left (in this case the number "7"), and find what is the highest number that can be squared ($?^2$) and not go over this chunk.

3^2 is 9 so that's too much. 2^2 is 4 which isn't over 7, so write "2" above the line, above the "7".

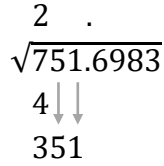
$$\begin{array}{c} 2\ \ . \\ \sqrt{751.6983} \end{array}$$

Now, take the 2 and square it (=4) and write the answer under the "7", and then subtract 4 from 7 (=3) and write the answer under the "4".

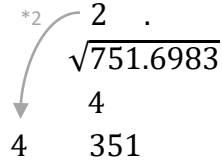
$$\begin{array}{c} 2\ \ . \\ \sqrt{751.6983} \\ 4 \\ 3 \end{array}$$

Until now is the starting method. From now on, we will do a slightly different stage, and we will repeat this new stage over and over until we've solved as many digits as we want.

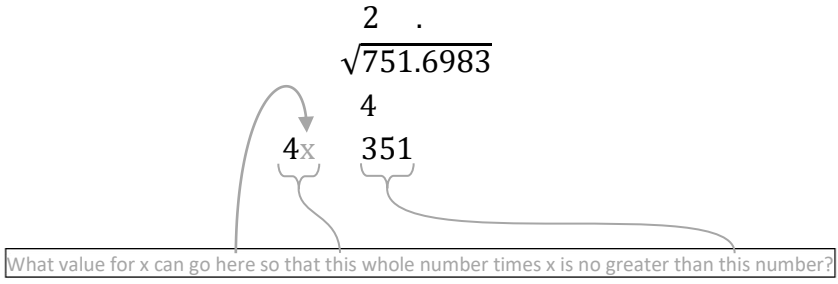
Bring down the next chunk of 2 digits next to the 3 (if there are no more digits, bring down "00")



Now take the number at the top (in this case "2"), ignoring the decimal, **double it** (=4), and write it down to the left of the bottom number ("351").



Now here's the shtikiest part. You need to find a single-digit number, let's call it "x", that you can place after the doubled number you just wrote down, so that when taken as a whole (the doubled number and x making up the digits of a new number) multiplied by x will be no greater than the number at the bottom of our calculations (351 in our case). Phew!



Usually, the way I do this is I divide 351 by 40 (in other words, I divide by the smallest number, which here would be 4 followed by a zero.) and see what that is (in this case it's 8.something) and I try the integer of that ($48 \cdot 8 \leq 351$?). If that doesn't work (which it doesn't in this case) then I use the integer below that. $47 \cdot 7 = 329$.

Now, write the integer you found in the next spot above the square-root symbol, and write the multiple you got underneath the last part of the calculation (in this case, under the 351), and subtract what you just wrote down from the line above it (=22).

```
      2 7.
    √751.6983
      4
47    351
      329
       22
```

Now, continue by following the instructions again starting from the beginning of the previous page, by bringing down the next chunk of 2 digits. Rinse and repeat as necessary.

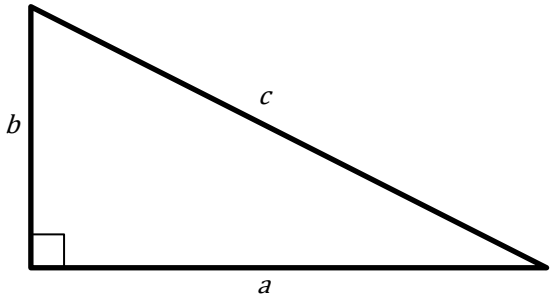
After solving several digits, your calculation should look like this:

```
      2 7.4 1 7 1
    √751.6983000000000
      4
47    351
      329
544    2269
       2176
5481     9383
         5481
54827    390200
         383789
548341     641100
           548341
            92759
```

At some point it starts becoming tedious, especially if you're doing it by hand.

Pythagorean Theorem

Taking any triangle where one (and only 1!) of the angles is 90° (a right angle, shown as a little box in that corner), there is a relationship between the sides of the triangle which enables us to determine any of the sides if two of the other sides are known.



The sides are labeled *a*, *b*, and *c*.

c is the hypotenuse, that is the side across from the right angle.

The other two sides are labeled *a* and *b*, and it doesn't matter for this formula which is which.

The formula is:

$$a^2 + b^2 = c^2$$

This can be rearranged in various ways, depending on what you're trying to solve.

For example, if you know *a* and *b* and you want to know *c*, you can rearrange the formula as follows:

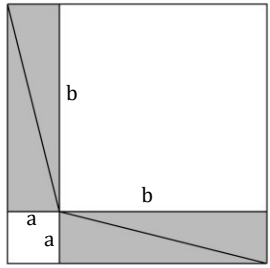
$$c = \sqrt{a^2 + b^2}$$

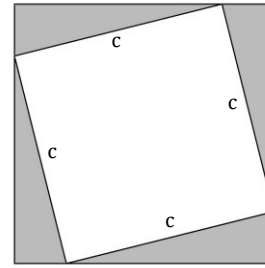
Or, let's say you know *a* and *c* and you want to know *b*, you can do this:

$$b = \sqrt{c^2 - a^2}$$

Pythagoras (the Greek guy who it's named after) offered the following logical proof:

Imagine you had four right triangles, all the same size and shape (shaded grey in the pictures). Now, arrange them in the following manners, like in the pictures below.





In the picture on the left, you can see the white space created by a^2 (the little white square) and b^2 (the bigger white square).

In the picture on the right, c^2 (the big slanty white square) is that same amount of area, just rearranged. It's obviously the same area since it's the space in the encompassing square (i.e., the large square made up of the "a" and "b" sides) minus the grey space taken up by the four triangles.

Hence,

$$a^2 + b^2 = c^2$$

FOIL and Reverse FOIL (or whatever it's called[314])

Here's an idea that pops up over and over again in this sefer. It's called FOIL which stands for **F**IRST **O**UTSIDE **I**NSIDE **L**AST, which as far as I can tell is a totally arbitrary way of doing it. Perhaps they made this up just to make sure that every permutation is accounted for.

Anyways, the idea is like this. When you're multiplying two (or more[315]) sets of parentheses, where inside the parentheses are plusses and minuses, you need to multiply each term in one parentheses with each term of the other parentheses.

For example, let's take $(3 + x) \cdot (5 - y)$.

In this case, the parentheses on the left has two terms: +3 and +x.

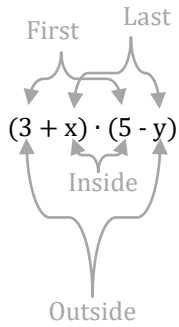
The parentheses on the right also has two terms: +5 and -y.

+3 has to be multiplied separately by +5 and by -y.

And then +x also has to be multiplied separately by +5 and by -y.

And then all the results need to be added up (or subtracted down) to (or from) each other.

Before we do it, let's mark off which terms are the "First"s, the "Outside"s, etc.



[314] I'm not looking it up because I don't care, and if I don't care why should I make you care? Ah, I see in my own book, page 106, it's called factorization. Oh, well.

[315] The specific acronym FOIL is specifically for 2 sets of parentheses with 2 items each, but the idea behind it applies to any number of sets with any number of items.

First we multiply the "First"s together, that'd be +3 times +5 = 15

$$(3 + x) \cdot (5 - y)$$
$$15$$

Then we multiply the "Outside"s together, and add it to the result:

$$(3 + x) \cdot (5 - y)$$
$$15 \quad -3y$$

Now the "Inside"s:

$$(3 + x) \cdot (5 - y)$$
$$15 -3y +5x$$

And the "Last"s:

$$(3 + x) \cdot (5 - y)$$
$$15 -3y +5x -xy$$

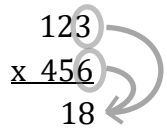
Technically speaking, when you do multiplication of multi-digit numbers, you're doing a type of FOIL operation (though not necessarily in that order).

Let's take for example $123 \cdot 456$:

$$\begin{array}{r} 123 \\ \underline{\times\ 456} \end{array}$$

What you have here is essentially the same as $(100 + 20 + 3) \cdot (400 + 50 + 6)$. And you need to multiply each number from the first group with each number from the second group.

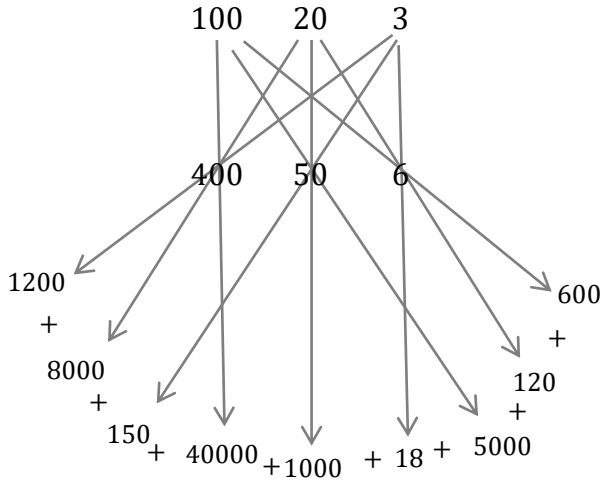
The way I was taught in school is you start with the last digit of the last number, in this case "6", and multiply it by 123, step by step which is by 3, and then by 20, and then by 100 so it looks like this:



The "1" of the 18 is going to be carried, which means it's going to be added (as a 10 really) to the next group of multiplication, which will be 6 · 20. And so on.

Eventually, each digit of each number to be multiplied (each "factor"), will be multiplied with each digit of the other number, and then they will all be added up to each other.

In the end, there will be nine multiplications happening with all the results added up.



Now, while we're at it, I want to teach you something I learned from a guy named Scott Flanders. He demonstrated that if you get this concept down pat, you can use it to multiply large digit numbers in your head, very quickly.

If we consider the above diagram, we will find that there are groups of multiplications that share the same digits-place.

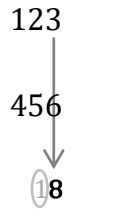
For example, 3 · 50 shares the same digit place in the final answer as 6 · 20, that is, the 10s place. Both multipliers have a singles digit and a tens digit, which means their answers will be in the ten digits.[316]

[316] Since 1 · 10 = 10, so replace the "1"s with other numbers and the zero will remain the same. Example: 5 · 70 = 350. The 5 and 7 become 35 and the zero remains the same.

So, instead of doing it the long way they taught us in school which is to multiply until you have 3 new numbers that you'll then add all up together, instead just group together similar digits and then add them up on the spot, in your head.

I'm going to show you the steps one by one. (This may seem a little complicated at first but once you get the hang of it its structure becomes very clear and the process is much faster. When you get really good at this you can do the whole thing in your head in seconds.)

First the "One"s digits:

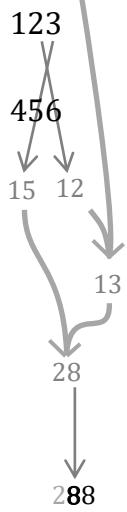


Don't write down the "1" of 18. Just hold it in your mind. Now add that "1" to:

$2 \cdot 6$ (= 13) that is: $2 \cdot 6$ + the "1" we were holding onto

and $3 \cdot 5$ (=28) that is: $3 \cdot 5 = 15$. Then add "13" from previous step

Write down the "8" of 28 in the 10s place in the answer.



Don't write down the "2" of the 28. Hold it in your mind for the next operation.

Now for the hundredth place. This is the funnest part.

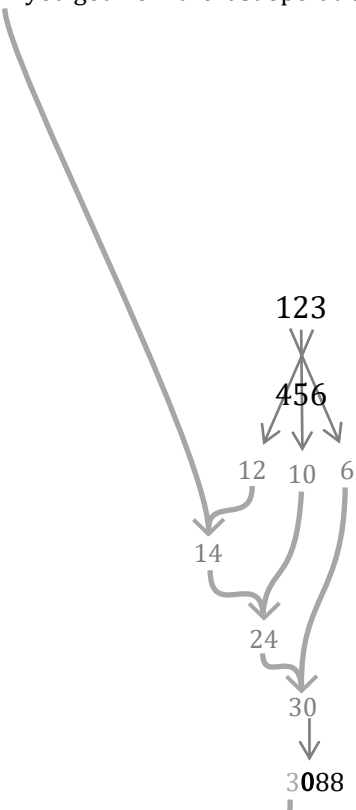
Take the "2" you got from the last operation and add it to:

1 · 6 and

2 · 5 and

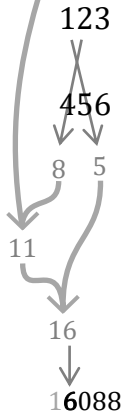
3 · 4

Which is:



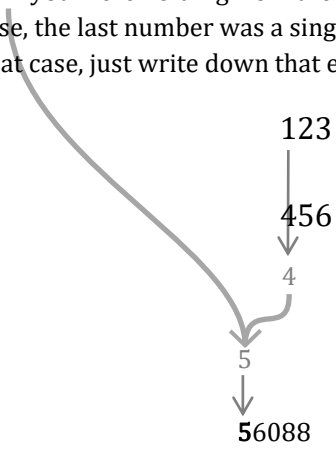
Write down the "0" from 30 and keep the "3" in your head for the next part.

Add the "3" to: 2· 4 and 1· 5 for a total of 16. Write down the "6" of 16 and hold the 1.



And the final step:

Take the "1" you were holding from the last step and add it to $1 \cdot 4$, which totals 5. (In this case, the last number was a single digit (5) but sometimes it'll be a double digit. In that case, just write down that entire double-digit number in the final place)



And there's your answer.

With practice, you should be able to do this very quickly, much faster than writing out 3 large multiplications and then adding them up.

Eventually, you should be able to do all this in your head, and from left to right (which is a little harder since you need to "update" your answer as you go), and then you can wow your friends with your ability to multiply multiple digit numbers in your head in seconds.

You can also do this with any number of digits, though more will be harder.

The reason why I'm going through all this is because I'm trying to show you that it doesn't matter what order you do FOIL in; the point is just that each possible way of multiplying is accounted for.

Now, a situation where we use FOIL a lot in this sefer is when we have two numbers added up (or subtracted down, whichever), within a parentheses, and the whole parentheses has a little ^{2} symbol at the end, which means we're gonna multiply this parentheses by itself.

For example: $(x + 4)^2 = (x + 4) \cdot (x + 4)$

Using FOIL we get $x^2 + 8x + 16$.

Now, REVERSE FOIL will be where we'll take that previous jumble: $x^2 + 8x + 16$ and revert it back to its simpler form $(x + 4)^2$.

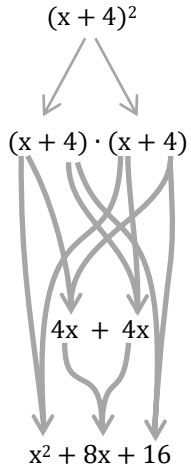
This will be useful when we get to solving quadratic equations.

The trick here is that if you take the number next to x, which in this case is 8, and cut it in half, and then square it, you get the number on the right.

$8 \div 2 = 4.$

$4^2 = \mathbf{16}$

If that happens then you can simplify the jumble the way we did. The reason it works is because when we go the other way, from $(x + 4)^2$ and expand it out, we'll end up with two 4x's (=8x) and one 4^2 (=16), besides the x^2.



It happens to work out in this case, but it won't always. But it will still be useful when we solve quadratic equations because we're going to *make* these numbers work out!

Stay tuned.

Quadratic Equations and Completing the Square

What on earth are quadratic equations?

It's a very fancy way of saying $x^2 + x + 1 = 0$.

Quadratic means it's made up of 4 parts. So, if I was speaking English instead of Latin I would've said something like "4 part thing". Doesn't have the same ring to it.

Quadratic Equations are just the 4th type of polynomial (whatever that is!)

Many numbers. Or, many *types* of numbers, where each type cannot be simplified with the other types.

So, let's say you have +5 and +7. Together you have +12, very nice.

But let's say you had +5 and +x? What would you do now? You have no way to simplify it, so you're just stuck with +5 +x, or $5 + x$.

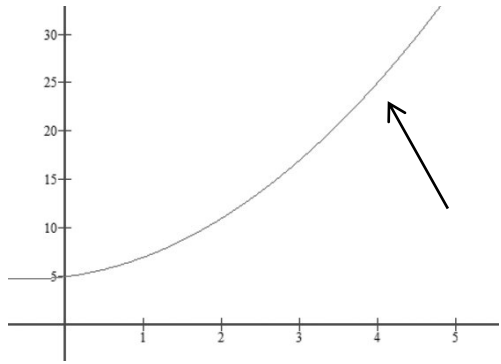
That's 2 types of unmixable numbers. The 5 and the x.

Let's do 3 types. We could do $5 + x + y$ but then we're going off on a tangent. Let's keep it neat.

$x^2 + x + 5$. No way to simplify that, huh? Ah, but let's say we happen to know what this added up to... for example:

$x^2 + x + 5 = 25$, now we have what to work with.

By the way, before we solve this, let's stick the left side of the equation into a graph:



We can see from the graph that the answer is about 4. Let's try that:

$4^2 + 4 + 5 = 25$. Ah, yes, that works out. But let's do it the long way now.

So, if you remember from Appendix E on FOIL, we took a case of addition within a parentheses which the whole thing was squared

$$(x + 4)^2$$

When we expanded the parentheses and multiplied everything out, we ended up with:

$$x^2 + 8x + 16$$

We showed that if we have such an expanded formula we can revert it back to its simpler form of $(x + 4)^2$.

So our job is to make sure that this formula is in the right state to revert to that simple parentheses.

For example, let's say we have the following:

$$x^2 + 8x + 15 = 99$$

Oh, it's so close! So let's make it right by adding 1 to each side of the equal sign:

$$x^2 + 8x + 16 = 100$$

Ha ha! We got it. That was sooooo cheating. OK, let's continue simplifying.

▶	$x^2 + 8x + 16 = 100$	
⬇	$(x + 4)^2 = 100$	reverse FOIL
⬇	$x + 4 = \sqrt{100}$	square root both sides
⬇	$x = \sqrt{100} - 4$	subtract 4 from each side
⬇	$x = 10 - 4$	square root 100
■	$x = 6$	subtract

OK, Bumy, so maybe you want to explain what we do in a different situation?

Certainly.

What we're doing is called "Completing the Square", which we'll explain in a minute.

We said back at the top of page 372, "The trick here is that if you take the number next to x... and cut it in half, and then square it..."

That's what we're trying to do. Because that will give us a quadratic equation that can be reduced easily.

Let's do a not-so-obvious example.

$$x^2 + 20x + 27 = 99$$

What do we do here?

Let's first blindly follow the formula. Take the number next to x, which is 20, and cut it in half. That gives us 10. Square that and you got 100.

So what we *want* is: $x^2 + 20x + 100$

These numbers by themselves will simplify into (reverse FOIL):

$$(x + 10)^2$$

Problem is we don't have + 100, we have + 27.

So we need to add 100 to each side of the equation:

$$x^2 + 20x + 100 + 27 = 99 + 100$$

Or simpler, add what is needed to each side of the equation so that we'll have 100 on the left side:

$$x^2 + 20x + 27 = 99$$

add whatever it takes to get this to 100 (+73)...

...add that same amount (+73) to the other side

$$x^2 + 20x + 100 = 172$$

And solve:

take off 100 from each side

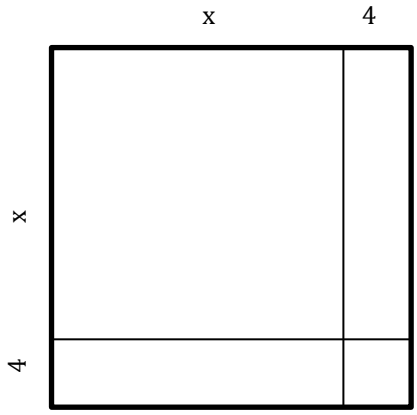
⬇ $x^2 + 20x = 72$

etc.

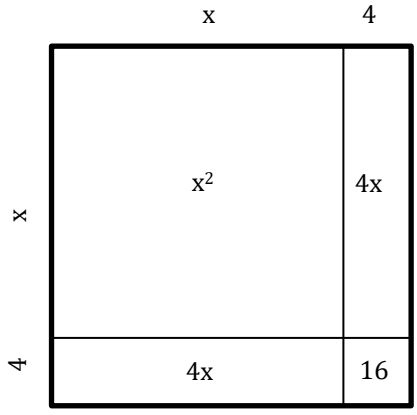
Now, I said I was going to explain it more, so let's go to some visuals!

(Turn the page)

Imagine (or look at the picture) the sides of a square whose sides are x + 4 long:



We can write out the values of each smaller area within the (large) square as follows:



Which is how we get $x^2 + 4x + 4x + 16$,

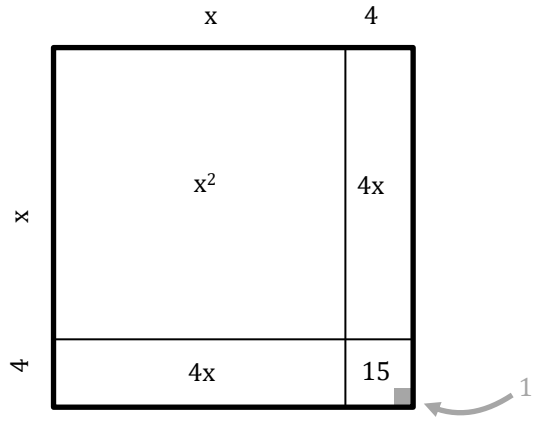
or $x^2 + 8x + 16$.

By the way, you see why we're cutting the 8 in half? Since there's always going to be 2 of those long rectangles with a number and x, so we're always looking for half the doubled number.

But let's say we don't have 16, or we have more than 16. If we can add or subtract until we get to 16 we will then have "completed the square", ready for simplifying.

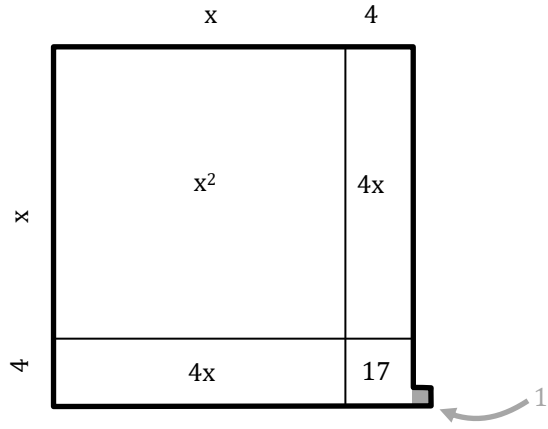
Let's visualize that.

Let's take $x^2 + 8x + 15$.



You can see in the diagram, the "1" that is missing (the little grey square) that would "make the square complete". So, if we just add 1 to the whole equation we can simplify easily with reverse FOIL.

Or let's say you had too much. $x^2 + 8x + 17$.



Now you can see the "1" that makes it too much.

If we just subtract 1 from the whole equation (from both sides of the equal sign), we will have a complete square that we can simplify with reverse FOIL.

All this is fine and dandy if you happen to be lucky enough that the x^2 in your formula is just that. But what if it was different?

For example, on page 106 we have the formula:

$$\left(70\frac{2}{3}x\right)\cdot 4 + 4x^2 = 6\frac{2}{9}$$

The comment on the side of the page says

> when working with quadratic equations, it's good to make anything that is x^2 into $1x^2$, so in this case since it's $4x^2$, we'll divide everything by 4

Which means, we'll divide everything in the entire equation by whatever number is modifying x^2.

In this case it's 4, so we'll divide everything by 4:

$$\underbrace{\left(70\frac{2}{3}x\right)\cdot 4}_{70\frac{2}{3}x} + \underset{\downarrow}{4x^2} = \underset{\downarrow}{6\frac{2}{9}}$$

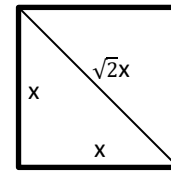
$$70\frac{2}{3}x + x^2 = 1\frac{5}{9}$$

And then we'll continue by completing the square.

1.4 or $\sqrt{2}$? 3 or π?

Perceptible Differences

The diagonal of a square will always be $\sqrt{2}$ times bigger than the square's width.

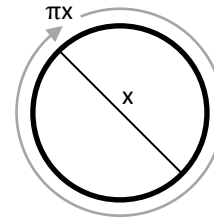


The value of $\sqrt{2}$ is **1.414213562...**

This number goes on and on without any discernible pattern.

The Gemara uses the value of **1.4** for the diagonal of a square.

Another discrepancy between what the Gemara says and what measurement bears out is the value of a circle's circumference relative to its diameter. This amount is always the same and is known as pi (π).



The value of π is **3.14159265358979...**

This number also goes on and on without any discernible pattern.

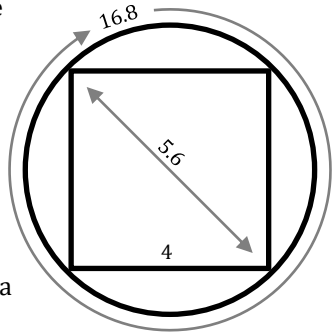
The Gemara uses the value of **3** for pi.

As an example where we find these two values together is the Gemara on ע"ו where the Gemara discusses a window between two courtyards. The window needs to be big enough to be considered an opening between the courtyards to allow them (or require them) to make an eruv together.

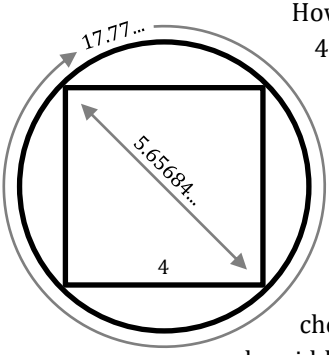
The window needs to be 4 tefachim wide by 4 tefachim high.

The Gemara then discusses a round window. The round window must be big enough that it contains within it a space of 4 by 4 tefachim. To achieve that, the Gemara says you need a round window with a circumference of **16.8** tefachim.

This number is derived as follows. Since the square within it is 4 by 4, that means that the square's diagonal is 5.6 $(4 \cdot 1.4)$. The square's diagonal is the same length as the round window's diameter. If the diameter is 5.6 then the circumference must be 16.8 $(5.6 \cdot 3)$. That is how the Gemara arrives at this figure.



All this was derived by using 1.4 as the diagonal of a square, and 3 as the circumference of a circle.



However, when we measure a circle that encompasses a 4 by 4 square we find that the circle's circumference is **17.77...** $(4 \cdot \sqrt{2} \cdot \pi)$, which is almost a whole tefach more!

That means that a circumference of 16.8 is not big enough to contain a 4 by 4 square.

Tosfos[317] regarding π leaves this as a kasha. "The cheshbon is not exact according to the chachmei hamiddos, the measurement scholars."

Regarding $\sqrt{2}$, Tosfos[318] proves that the real number is a bit more than 1.4.

There's different approaches to this, both in terms of pshat in the Gemara and halacha limeisah. I would like to just present a few opinions that probably encompass all the others between them.

Regarding the pshat in the Gemara, the Tosfos Harosh[319] starts by taking the Gemara's question of "From where do we know that a circle's circumference is 3 times its diameter?" and asking on that, why are you looking for a passuk? Just go out and measure!

He answers that since π is actually greater than 3, where did Chazal get the *right* to use a rounded number of 3?! It's too small! And on that the Gemara brings a proof from Shlomo's Pool. Just like the Torah used 3 for π, we see that we can too.

[317] 14a D"H Vihu'ikka
[318] 57a D"H Kol
[319] 14a D"H Kol

The Tashbeitz[320] brings this Tosfos Harosh[321] and has a whole discussion about it and sums up his view with two possibilities:

"אלא יש לנו לומר אחד משני דברים:

- או שקבלתן כך היתה ללכת ע"פ דרכים אלו ואע"פ שיש בהם קירוב דהא שיעורין הלכה למשה מסיני הם... והטעם בזה לפי שלא ניתנה התורה למלאכי השרת... ושמא כך נמסרה להם הלכה שיתנהגו על עיקרים אלו אע"פ שיש בהם קירוב כאלו הם מדוקדקים ויש סמך בזה מים שעשה שלמה שהלך בו הכתוב על דרך קירוב...
- או שנאמר שהם כשנשאו ונתנו בזה על עיקרים אלו עשו זה לקרב ההבנה אל התלמידים לפי שלעולם ישנה אדם לתלמידו בדרך קצרה... אבל לענין מעשה יש לנו לדקדק הענין ע"פ הדקדוק האמתי..."

(Loosely translated)

"Either we say:

- They had a tradition to calculate like this, even though they are rounded numbers, since measurements in general are a tradition given by Hashem to Moshe on Har Sinai[322]...since the Torah wasn't given to angels... so maybe here too it was a tradition... and we see that we can rely on this from the way the Torah describes Shlomo's Pool
- or, when they were teaching the students these concepts they did it using rounded numbers, since a person is always supposed to teach his students concisely... but when it comes to halacha limeisah we need to use the exact numbers..."

OK, so now halacha limeisah, what do we do?

The Rambam in Hilchos Eruvin[323] writes briefly about the round window, "It must be big enough to contain a square that is 4 by 4".

The Maggid Mishna explains that the Gemara's values are inexact and "that is why the Rambam just wrote simply that 'it needs to be big enough to contain a square of 4 by 4'".

[320] 1:165
[321] in the name of Tosfos, but I didn't see Tosfos say all this
[322] Eruvin 4a
[323] פרק ג' הלכה ב'

I understand the Maggid Mishna to mean that according to the Rambam, one must use the measured values of the diagonal of a square $(\sqrt{2})$ and of the circumference of a circle (π). (This seems to be how the Tashbeitz we just quoted also understands the Rambam in Pirush Hamishnayos; see Tashbeitz there in full).

And this is regarding a דין דרבנן. How much more so when we are dealing with דיני דאורייתא.

On the other end of the spectrum is the Shaar Hatzion[324], who writes (loosely translated) that

> "even though the measured values come out bigger than the Gemara's values, still we don't need to be so exacting since Chazal relied on these simplified values by דיני דאורייתא since it is difficult to be exacting with the differences. And perhaps they had a tradition from Sinai that we could rely on these values, even by דיני דאורייתא, so certainly by דיני דרבנן..."

This seems to be like the Tosfos Harosh, as explained by the Tashbeitz. There seems to be two reasons why we could use the Gemara's values.

1. The Torah was not given to מלאכים[325]. In other words, we are only required to be as exacting as is normal for regular people, a principle we find throughout ש"ס.

2. When Hashem gave the Torah to Moshe, He told him that we can use the rounded שיעורים of 1.4 for the diagonal and 3 for the circumference, for halachic purposes[326].

[324] שו"ע שע"ב אות י"ח

[325] קידושין נ"ד. ושאר מקומות

[326] One could find examples like this in contemporary society. For example, when calculating the price of an item, the exact value sometimes ends in a fraction of a penny. Instead of always demanding that difference be paid by the buyer or seller, the law allows a certain degree of rounding to the nearest penny. So too, Hashem doesn't demand, so to speak, an overly exact degree of accuracy in Jewish Law, in certain contexts.

Throughout this sefer, I will use both “systems”, depending on the context.

For פשט in the Gemara, I will almost always use 1.4 and 3, since these are the numbers we are given to work with and the פשט in the Gemara is almost always dependent on them. Sometimes, I will try to see if and how the Gemara can be understood using $\sqrt{2}$ and π. See for example page 50; the piece on Tosfos והאיכא.

When I am discussing distance, area, and volume as far as they come out as a נ"מ, either by themselves or by comparing different שיטות, I will almost always use $\sqrt{2}$ and π.

Now, I would like to point out a couple of things about pi.

First of all, when people discuss the value of pi vis-a-vis the Torah, I always like to ask them, when was the last time you actually used pi in your real life (not in academics)?

The only time I've used it in the last 20 years (outside of academia and programming) is when I've opened a container of plaster and had some left over, and needed to seal the opening tight until the next time I might use it, so I need to wrap some tape around it. How much tape? Well, I open the tape across the top of the round container, and then I open it again that distance, and then again, and then a bit more to wrap it over itself, for 3+ diameters worth. Then I wrap the tape around the lip of the container. Tada! I've used pi. 3 and a bit.

In other words, people make it like the lack of pi's accuracy clashes with their everyday life, and it doesn't. You don't need more accuracy than "3 and something".

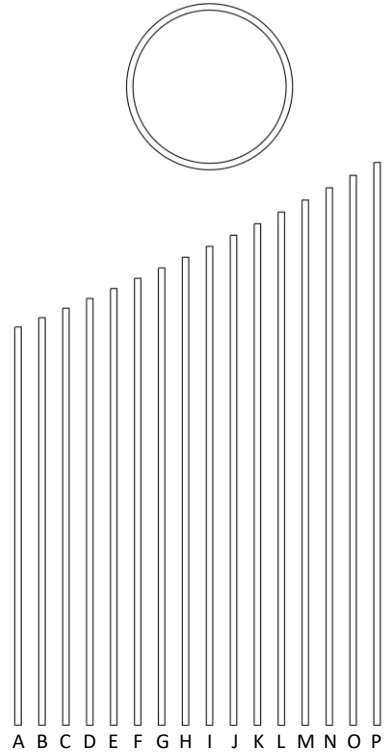
And then I ask people, well how much is pi, anyway? And they say 3.14. That's it. And I'm like, "and?" And they have to look it up.

So there's two things here. One, they're complaining about what the Gemara says when they don't even know it themselves.

And two, what do they want the Gemara to do? Give a trillion digits of pi? And then stop or keep going? In other words, when you ask the people themselves what is pi, they stop after 2 digits. So the Gemara stops at the decimal point. You're rounding, the Gemara is rounding.

Now, I'd like to do a little experiment, bear with me.

I'm going to show you a circle, and I want you to show me which straight line is the same length as the circle's circumference. You ready?



One of these lines is 3 times the circle's diameter, and one of them is π times.

I cannot tell from just looking at them. Maybe I could tell that the shortest one is too short, and the longest one is too long, but more accurate than that is hard.

And the point is that lo bashamayim hi. We tend to think of the world as existing and then man shows up and then the Torah comes along... but that's totally backwards. The Torah comes first, then the man who needs to fulfill it, and then the heicha timtza for man to fulfill it, that is the world. ɹǝʇɟ ǝɥʇ ɔıɥǝ ʇɐɯʇıɹɔ ǝɥʇ ʇɐɥ ɯɐɯ ɐ ʇɐ ʇɟıɟnɟ oʇ ʇ ɹǝ ɥɔ

That means, when Hashem is designing the man who will fulfill this Torah, He creates him with the perception bounds of what is necessary to keep the Torah.

What's beyond the perception of man is halachically negligible.

For example, some bugs are so tiny that they can't be seen with the un-aided eye. It's not that since we can't see them therefore they are not a problem, rather, it's because they're not a problem is why we can't see them. The fact that we can't see them is a סימן that they're not a problem, not the סיבה. They were never included in the plan to begin with! (Why they're there at all is another discussion.)

It might be the same idea when the Gemara uses the term "mashehu", "a tiny amount". Is there some shiur to a tiny amount?

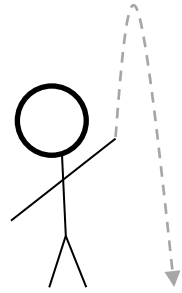
When checking an esrog for black dots, the halacha is[327] that you hold the esrog in your hand at a normal distance from your eye, as you would any other object, and

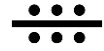
[327] Mishnah Brurah 648:46, as explained by Rabbi Black, see next footnote.

take a glance. If you can't see a black dot like that then it's not a black dot, even if you could see it when inspecting it closer[328].

I[329] apply this principle to many areas. We[330] were just learning recently in Sanhedrin about someone throwing a rock up in the air and it comes down either "in the same place" or "to the side", and we wondered how do you define "the same place"?

I wanted to say, lechora, this same principle. Whatever the distance is, but there's some distance that would be perceivable as being "a different place", and whatever isn't that is called "the same place".





[328] My Rosh Yeshiva from Zichron Yaakov, Rabbi Shabsai Black, wondered why people bring loupes (strong magnifying glasses) to the arba minim sale. If you need the loupe to see the dot then it's not a dot. He explained that what they're probably doing is if they do indeed see a dot at a normal glance then they need the loupe to check it more closely, to see if indeed it's a discoloration on the esrog skin or maybe it's something else, like dirt.

[329] I'm not a posek; I'm just a regular guy trying to understand the sugya.

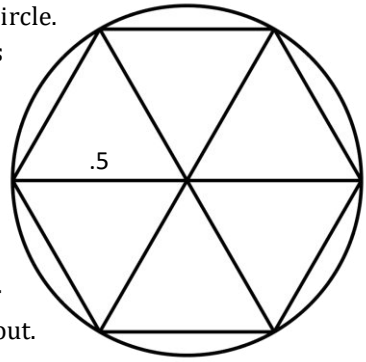
[330] In Rabbi Liff's Yeshiva in Beit Shemesh, Atteres Yaakov, Summer zman 5785.

Calculating π

First off, a simple proof that π is greater than 3. (I didn't make this up; it's all over the place; I think the Tashbeitz[320] brought this, if I understood him correctly.)

Consider the following diagram:

We have a hexagon (six sided) surrounded by a circle. The hexagon is made up of 6 equilateral triangles (each side the same length). Each triangle's sides are .5 long. The diameter of the circle is thus 1.



If we add up the outside edge of the hexagon we will see it is 3 long (.5 · 6). The circle that encompasses it must be longer than that in order to have that extra length to be able to go farther out.

Therefore, the circle's circumference must be greater than 3.

Tada!

÷

Next.

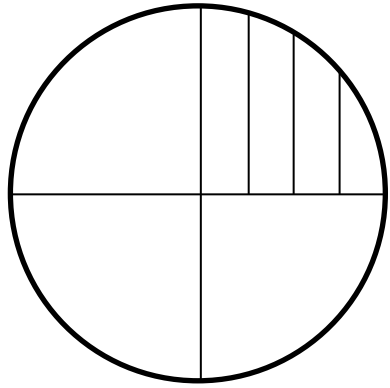
How can we calculate the value of π?

This is the way I would do it. Let's first start with a "lower bound", that is, an amount that pi is for sure not less than.

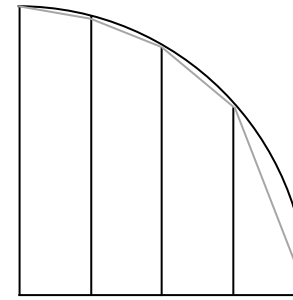
Let's start with a circle with a radius of 1.

Now let's take one quarter of the circle, let's say the top right quarter. Now divide it into a specific number of columns, the same width.

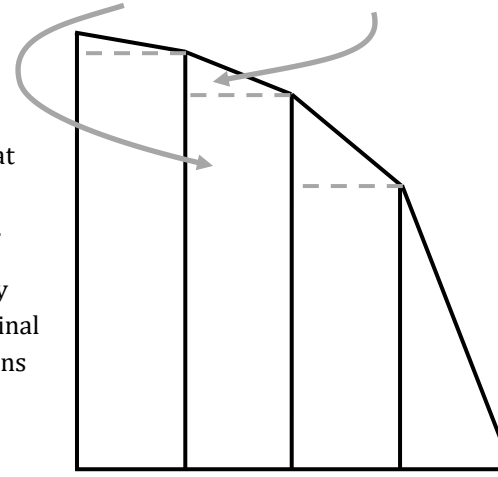
Let's start with 4 columns.



Now, if we were to connect straight lines from the tops of the sides of the columns to each other, we should then be able to measure the lengths of these lines using the Pythagorean Theorem ($a^2 + b^2 = c^2$).



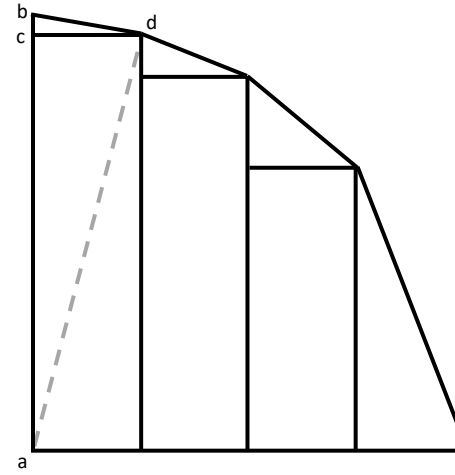
Let's divide each column into two pieces, a tallish rectangle, and a triangle at the top where the sides of the column are different heights (see diagram)



We need to determine the different heights of the rectangles, and from that information we should be able to determine the heights of the triangles.

The widths of the triangles are already known; they are the radius of the original circle divided by the number of columns we made ($1 \div 4 =$ **.25**)

(Please note that I'm referring to lines by their end point letters, like "ad" means line $\overline{ad}$)



ab = 1

ad = 1

cd = .25

$ac = \sqrt{ad^2 - cd^2}$

bc = ab - ac

$bd = \sqrt{bc^2 + cd^2}$ **= .252...**

Now for the next column

be = ac from previous calculation

ad = 1

cd = radius / #_of_total_columns

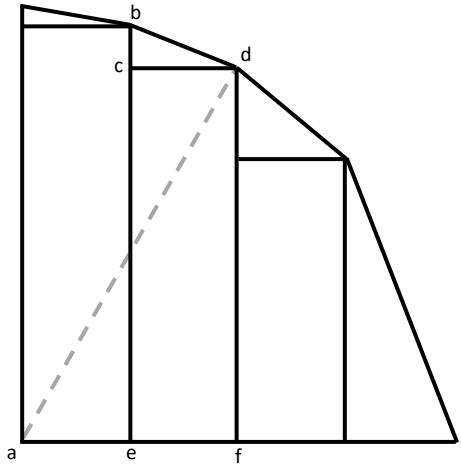
af = radius / #_of_total_columns * this_column_# (=1 / 4 * 2 = .5)

$df = \sqrt{ad^2 - af^2}$

ce = df

bc = be - ce

$bd = \sqrt{bc^2 + cd^2}$ **= .27...**



Now, we can continue in this fashion and get the lengths of the rest of the slanty lines.

Then, we can add them up to get the length of this whole quarter of the circle.

Multiply by 4 to get the whole "circle" length (it's a little less than the actual circle length since we measured lines beneath the actual curve of the circle, but that is the point of a "lower bound"; to get a number that pi is for sure no less than).

Divide by the diameter of the circle (which is 2, since the radius is 1).

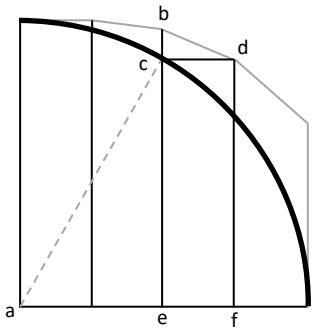
Then, if you wanted a more accurate number, do this whole process again with more columns, what would be a higher "resolution".

Since we are looking for a lower bound, when calculating the square roots, round down to whatever decimal point you are calculating.

upper bound

This whole method can be done in a similar way to find an "upper bound" as well.

We will again cut the quarter circle into vertical columns, but this time we'll make sure each column is taller than the curve of the circle, and we'll need to add the last vertical line all the way on the right side.



Lower Bound value of pi in python:

```
from math import sqrt

r = 400000
#resolution
#Each quarter circle is divided into this number of vertical strips

be = 1

total = 0

for x in range(r):

    ad = 1
    #radius of circle

    cd = 1 / r
    #radius divided by resolution, number of columns

    af = 1 / r * (x + 1)
    #column width times this column number
    #(adding 1 to x since python starts from zero)

    df = sqrt(ad**2 - af**2)
    # ** in python means power, or in this case, square

    ce = df

    bc = be - ce

    be = ce
    #need this value for the next column

    bd = sqrt(bc**2 + cd**2)

    total += bd
    #add this slanty line to the total

circle = total * 4

pi = circle / 2
#since radius is 1 therefore diameter is 2

print(pi)
```

The particular resolution I ran in this program is 400,000 (which is way too much for me to do by hand!). It produces a value of pi that is accurate to 8 decimal points.

3.1415926524276867...

Upper Bound value of pi in python:

```
from math import sqrt

r = 400000
#resolution
#Each quarter circle is divided into this number of vertical strips

be = 1

total = 0

for x in range(r):

    ac = 1
    #radius of circle

    cd = 1 / r
    #radius divided by resolution, number of columns

    ae = 1 / r * x
    #column width times this column number

    ce = sqrt(ac**2 - ae**2)

    bc = be - ce

    be = ce
    #need this value for the next column

    bd = sqrt(bc**2 + cd**2)

    total += bd
    #add this slanty line to the total


total += be
#adding last vertical line

circle = total * 4

pi = circle / 2
#since radius is 1 therefore diameter is 2

print(pi)
```

Upper Bound value result accurate to the fifth decimal:

3.1415976496326006...

If you are calculating by hand, round the square roots up to whatever decimal point you are calculating, since we are looking for an upper bound limit.

While researching this topic, I came across these other methods for solving for pi. I have no idea why or how they work.

This method is much simpler than mine, though it takes longer (at least at the resolution I was running it).

It's called the Gregory-Leibniz method.

$$\pi = \frac{4}{1} - \frac{4}{3} + \frac{4}{5} - \frac{4}{7} + \frac{4}{9} - \frac{4}{11} \ldots$$

In python:

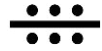
```
p = 4.0
f = -1

for x in range(3,100000000,2):
    p += f*(4.0/x)
    f *= -1

print(p)
```

This next one is called the Nilakantha method; also have no idea why it works, and it's much faster than my method:

$$\pi = 3 + \frac{4}{4 \cdot 3 \cdot 2} - \frac{4}{6 \cdot 5 \cdot 4} + \frac{4}{8 \cdot 7 \cdot 6} - \frac{4}{10 \cdot 9 \cdot 8} \ldots$$

Python:

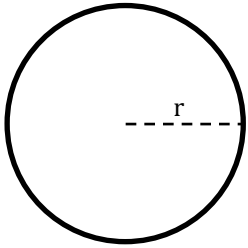
```
p = 3.0
f = 1

for x in range(4,50000,2):
    p += f*(4.0/(x*(x-1)*(x-2)))
    f *= -1

print(p)
```



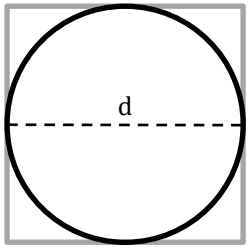
$\pi r^2 = \pi/4\ d^2$

(or: Contemporary Formula For Calculating a Circle's Area Is the Same As Chazal's, Just Reconfigured)

The contemporary formula for finding the area of a circle is πr^2, where "r" is the radius.



The Gemara and the miforshim use the formula of $\frac{\pi}{4}d^2$, where "d" is the diameter. (Although, the Gemara uses "3" for π but that is not relevant to the point right now.)



In reality, this is the exact same formula.

How so?

The diameter is twice the radius, so we could replace the Gemara's formula with:

$$\frac{\pi}{4}(2r)^2$$

which expands to:

$$\frac{\pi}{4}2r * 2r$$

Since we'll be diving π by 4 and then multiplying it by 2 and then 2 again, we can cancel out the "4" with the two "2"s, and we'll be left with:

$$\pi rr$$

which is the same as:

$$\pi r^2$$

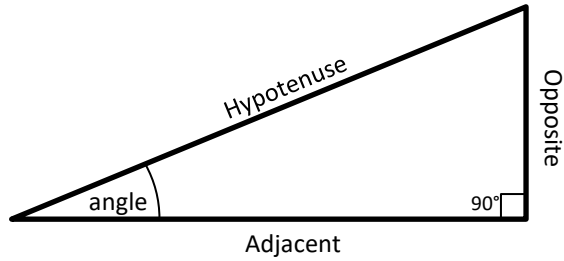
I

SOHCAHTOA - Sines and Cosines

The basic premise of this idea is that for any right triangle (= a triangle that has one corner that is 90°), there is a predictable relationship between the other angles and the lengths of its sides.

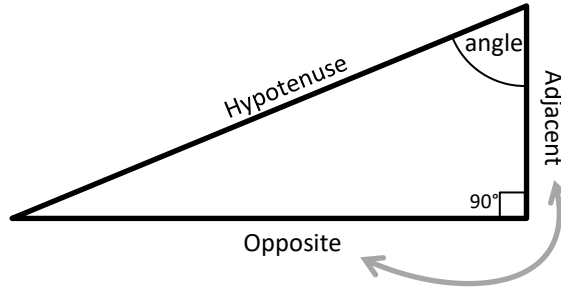
The way we're going to be doing it is we'll compare two sides and one angle. If two of these three factors are known, we'll be able to figure out the third factor as well.

Let's first get some labels down.



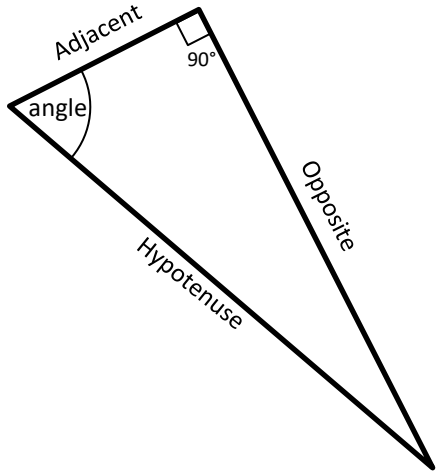
There's the "angle" we'll be working with. The slanty side of the triangle is called the "Hypotenuse"[331] The side across from the angle is called the "Opposite", and the side next to the angle (but not the hypotenuse) is called the "Adjacent". Remember these names because it will be important for when we commit the formulas to memory.

Now, if we'd be working with the other angle, in the other corner, then we'll need to rename our sides accordingly. It will still follow what we said above but notice how in the diagram the sides changed labels. The Opposite is always across from the angle, and the Adjacent is always next to it.



[331] pronounced "hie POT niss", depending where you're from. And not Hippopotamus.

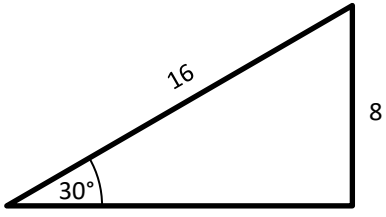
Also, bear in mind that it doesn't matter how the triangle is oriented. The Hypotenuse will always be across from the 90°, and the Opposite will always be across from the angle we're working with, and the Adjacent is always next to it.



Now, we will learn three terms: Sine, Cosine, and Tangent.

The Sine of an angle is a certain value (depending on the angle), and its value is the Opposite side length divided by the Hypotenuse length.

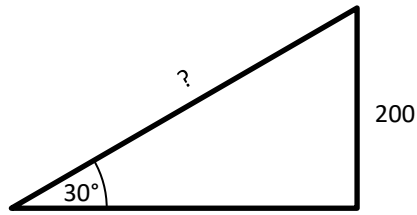
For example, let's take a right-triangle whose one known angle is 30°, and whose known sides are: the Opposite which is 8 long, and the Hypotenuse which is 16 long.



Since by definition the Sine of an angle is the Opposite divided by the Hypotenuse therefore, the Sine of 30° is $8 \div 16 = .5$

With this information, we can determine the angles or lengths of other similar triangles, if we know at least some other information about that triangle.

For example, let's take another right-triangle whose known angle is again 30°, and whose Opposite is 200 long, but the Hypotenuse is unknown.



Can we figure out, based on the previous triangle, how long the Hypotenuse is?

Well, since we determined with the previous triangle that the Sine of 30° is = .5, and we know that that value is the same as the Opposite divided by the Hypotenuse, we can write up the following formula:

$$\text{Sine}(30°) = .5 = \frac{200}{\text{Hypotenuse}}$$

We can rearrange this formula (leaving out the "Sine(30°)") as follows:

$$\text{Hypotenuse} = \frac{200}{.5}$$

Dividing 200 by .5 = **400**.

And that is the length of the Hypotenuse.

Now, let's get the other definitions.

The **Cosine**[332] of an angle is equal to the Adjacent Side divided by the Hypotenuse.

The **Tangent**[333] of an angle is equal to the Opposite divided by the Adjacent.

We will use these three formulas all the time when solving the angles and sides of various right-triangles. Which one we will use in any particular case will depend on what factors are known.

[332] pronounced "COE sine"
[333] pronounced "TAN jint"

All three formulas are written more concisely as:

$$\text{sine(angle)} = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

$$\text{cosine(angle)} = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$

$$\text{tangent(angle)} = \frac{\text{Opposite}}{\text{Adjacent}}$$

Or even more concisely as:

$$S = \frac{O}{H}$$

$$C = \frac{A}{H}$$

$$T = \frac{O}{A}$$

(Pronounced: Sine Equals Opposite over Hypotenuse, etc.)

The classic way people remember these formulas is by the acronym S.O.H.C.A.H.T.O.A., which is **S**ine (equals) **O**pposite (over) **H**ypotenuse. Cosine etc.

But that's never worked for me since I can't remember if it's SAHCAHTOA or SAHCOHTOA or SACK-A-TOE-UGH or some other combination.

So, my friend Asher Burrows shared with me an awesome mnemonic that he heard in Rochester from his 10th grade math teacher, Mr. Diskind, that goes:

"Some Old Hippie, Caught Another Hippie, Tripping On Acid"

And that I'll never forget.

Sometimes these terms are shortened as sin(x), cos(x), and tan(x), or sometimes we'll be doing it even simpler as s(x), c(x), and t(x)

Now, we'll sometimes be working with triangles that aren't right-triangles. In that case, we will sometimes cut the triangle into two separate right-triangles and solve each one independently, and then afterwards put them back together at the end.

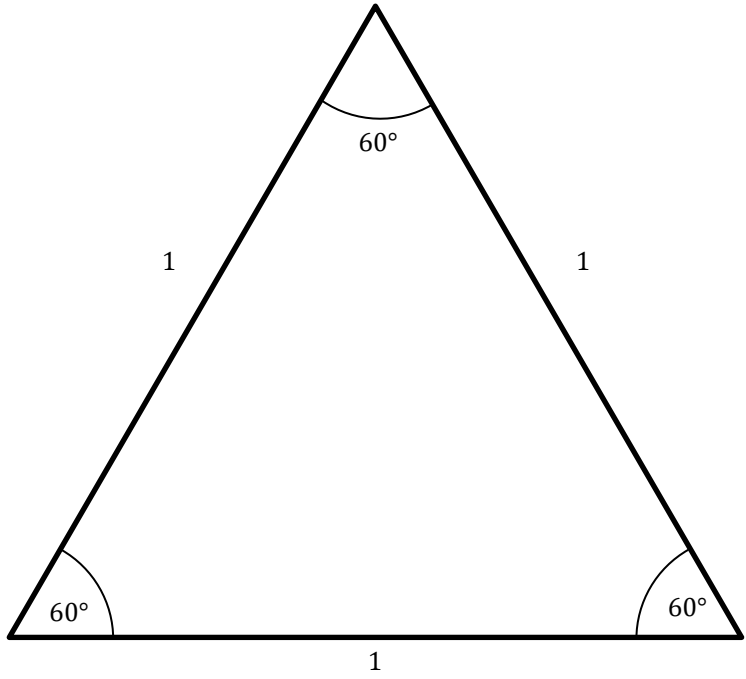
Now, how do we know[334] the values of Sines and Cosines and all that?

What I'm going to show you now is based on the Chazon Ish's method[335].

Let's start with an equilateral triangle (that is, a triangle whose sides are all the same length).

Let's assume the sides all have a length of 1 (this is what's called "Unity" in math terms).

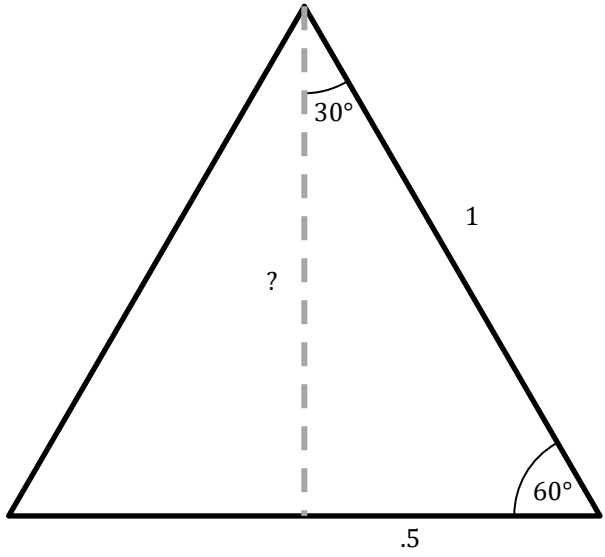
An equilateral triangle's angles are all 60° (=180° / 3).



[334] You don't need to know all this; this is just to prove the point. Once we (or some other reliable person) compiles a Sine table, we can refer to it. Many calculators have the Sine values built into them.

[335] Hilchos Kiddush Hachodesh, Siman 115:7

Let's cut this triangle vertically in half, and then let's focus on the triangle on the right and determine its angles and sides.



The upper angle is 30° since it's half of what it was before, and for the same reason the bottom side of this new triangle is .5.

The height of the original triangle can be determined by the Pythagorean Theorem as

$$? = \sqrt{1^2 - .5^2} = \sqrt{.75}$$

Now that we know all the sides and angles of this new triangle, we can begin to compile a Sine table.

Let's start with the 30° at the top of the triangle.

The Opposite side is .5 and the Hypotenuse is 1. Since the Sine of an angle is Opposite over Hypotenuse therefore we know that the

$$\sin(30°) = \frac{.5}{1} = \mathbf{.5}$$

And the cosine is the Adjacent over the Hypotenuse, so that comes out to:

$$\cos(30°) = \sqrt{.75}$$

Now, since Sin $= \frac{O}{H}$ and Cos $= \frac{A}{H}$, so if you divide the Sin by the Cos you get:

$$\frac{O}{H} \div \frac{A}{H}$$

Which is the same[336] thing as:

$$\frac{O}{H} \cdot \frac{H}{A}$$

Which is:

$$\frac{OH}{HA}$$

And if you cross out the H's you get:

$$\frac{O}{A}$$

Which is the formula for Tangent. So you can always get the Tangent of an angle by dividing the Sine by the Cosine.

So, dividing the Sine of 30° by the Cosine of 30° gives us the Tangent of 30° which is

$$\frac{.5}{\sqrt{.75}}$$

Which comes out to $\sqrt{\frac{1}{3}}$ somehow[337].

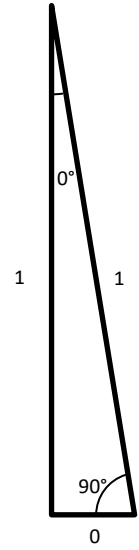
Now let's do the 60°. His Sine is $\sqrt{.75}$ (because that's the opposite side), and his cosine is .5 $\left(\text{or } \frac{1}{2}, \text{the adjacent side}\right)$, and his tangent must be $\frac{\sqrt{.75}}{.5}$, which comes out to $\sqrt{3}$, somehow[337].

Let's fill in the table as much as we know.

Angle	Sine	Cosine	Tangent
30°	$\frac{1}{2}$	$\sqrt{\frac{3}{4}}$	$\sqrt{\frac{1}{3}}$
60°	$\sqrt{\frac{3}{4}}$	$\frac{1}{2}$	$\sqrt{3}$

[336] Someone taught me this, that dividing by a fraction is the same as multiplying by the upside-down version of that fraction, something to do with reciprocals.
[337] I don't know how; I just checked it on a calculator.

What about the sines of 90° and 0°?



Now, I won't be able to actually draw such a triangle, due to its thinness, but we can make an exaggerated diagram and work with it:

If one angle is 0° then the Opposite side must be 0 long (since there's no width to the angle).

And since the Adjacent is starting and ending in the same places as the Hypotenuse, they must both be the same length, 1.

Therefore:

$\text{Sin}(0°) = 0$

$\text{Cos}(0°) = 1$

$\text{Tan}(0°) = 0$

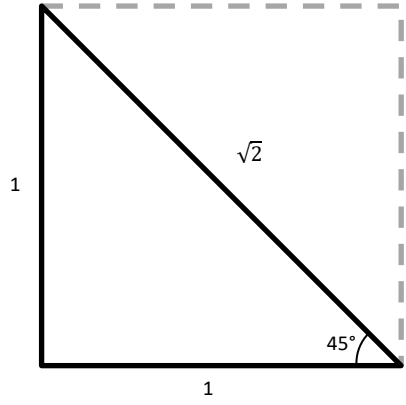
And

$\text{Sin}(90°) = 1$

$\text{Cos}(90°) = 0$

$\text{Tangent}(90°) = \text{infinity}$ (or however you calculate dividing by zero)

Now, what about 45°?



By 45°, the Opposite and the Adjacent are the same length, since they are the triangulated half of a square (Because a square's angles are all 90°, so when that is cut in half, each half-angle is 45°, and the sides remain the same).

$\text{Sin}(45°) = \frac{1}{\sqrt{2}} = \sqrt{.5}$

$\text{Cos}(45°) = \frac{1}{\sqrt{2}} = \sqrt{.5}$

$\text{Tan}(45°) = 1$

Let's fill in our table more with what we know so far:

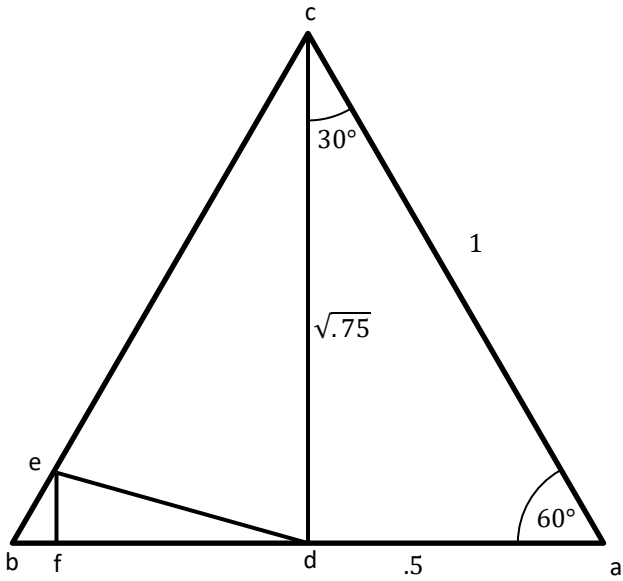
Angle	Sine	Cosine	Tangent
0°	0	1	0
30°	$\frac{1}{2}$	$\sqrt{\frac{3}{4}}$	$\sqrt{\frac{1}{3}}$
45°	$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{2}}$	1
60°	$\sqrt{\frac{3}{4}}$	$\frac{1}{2}$	$\sqrt{3}$
90°	1	0	∞

Now, I want to do one more step and then I think we've got the idea, and we could continue the process on our own, if we so chose to. Otherwise, we're going to have to rely on the Sine tables that are out there, or within our calculators. The Chazon Ish has a Sine table accurate down to the single degrees increments. Sometimes, our calculations will take us into fractions of degrees, so a general table might not help us.

Let's take our triangle from page 398 and continue cutting it up into pieces. I'm going to put letter-labels on it to make it easier.

We're adding new lines here. First, draw a line from d until point e on line bc, so that the length of ce is the same as cd $(\sqrt{.75})$.

Now draw a vertical line down from e to point f, which will create two right-triangles to the right and left of line ef.



So now:

bc = 1

ce = $\sqrt{\frac{3}{4}}$

be = bc - ce = $1 - \sqrt{\frac{3}{4}}$

ef = sin(ebf) · be $\left(\text{rearranged from } \sin(ebf) = \frac{ef}{be}\right) = \sqrt{\frac{3}{4}} \cdot \left(1 - \sqrt{\frac{3}{4}}\right) = \sqrt{\frac{3}{4}} - \frac{3}{4}$

bf = cos(ebf) · be = $\frac{1}{2} \cdot \left(1 - \sqrt{\frac{3}{4}}\right) = \frac{1}{2} - \sqrt{\frac{3}{16}}$

df = bd - bf = $\frac{1}{2} - \left(\frac{1}{2} - \sqrt{\frac{3}{16}}\right) = \sqrt{\frac{3}{16}}$

dfe = 90°

ced = cde

And since the three inner angles of triangle cde add up to 180°, and we know angle dce = 30°, so the remaining angles are each half of (180° - 30°), which comes out to:

ced = 75°

And since angle bef = 30°, so def must be 180° - ced - bef...

def = 75°

Therefore, angle edf must be:

edf = 15° (what remains in that triangle out of the total 180° that every triangle has.)

$$de = \sqrt{ef^2 + df^2} = \sqrt{\frac{3}{2} - \frac{3}{2}\sqrt{\frac{3}{4}}}$$

Once all the sides and angles of triangle def is known, we can extract the Sines from them.

Angle	Sine	Cosine	Tangent
0°	0	1	0
15°	$\frac{\sqrt{\frac{3}{4} - \frac{3}{4}}}{\sqrt{\frac{3}{2} - \frac{3}{2}\sqrt{\frac{3}{4}}}}$	$\frac{\sqrt{\frac{3}{16}}}{\sqrt{\frac{3}{2} - \frac{3}{2}\sqrt{\frac{3}{4}}}}$	$\frac{\sqrt{\frac{3}{4} - \frac{3}{4}}}{\sqrt{\frac{3}{16}}}$
30°	$\frac{1}{2}$	$\sqrt{\frac{3}{4}}$	$\sqrt{\frac{1}{3}}$
45°	$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{2}}$	1
60°	$\sqrt{\frac{3}{4}}$	$\frac{1}{2}$	$\sqrt{3}$
75°	$\frac{\sqrt{\frac{3}{16}}}{\sqrt{\frac{3}{2} - \frac{3}{2}\sqrt{\frac{3}{4}}}}$	$\frac{\sqrt{\frac{3}{4} - \frac{3}{4}}}{\sqrt{\frac{3}{2} - \frac{3}{2}\sqrt{\frac{3}{4}}}}$	$\frac{\sqrt{\frac{3}{16}}}{\sqrt{\frac{3}{4} - \frac{3}{4}}}$
90°	1	0	∞

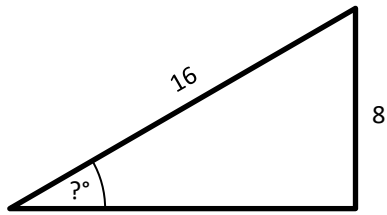
Now, you can keep going in this fashion and find finer and finer divisions of angles and their respective sines and cosines. I'm not going to continue with it but I just wanted you to understand the principle.

Now there's one more thing we need to speak out and that is the reverse of these formulas.

Some calculators call this the "inverse".

And that is when you don't know the angle but you know the value of its sine (or cosine or tangent), then you refer to the **arcsine** to extract the angle, which is sometimes written like this: $\sin^{-1}$(some value).

For example, let's go back to the triangle from page 394:



Let's say you knew the Opposite side was 8 long and the Hypotenuse was 16.

If $\frac{O}{H}$ is .5, and we know from our mnemonic[338] of formulas that $\frac{O}{H}$ is the Sine of the angle, then we can say the angle is the arcsine of .5.

$$\text{angle} = \sin^{-1}(.5)$$

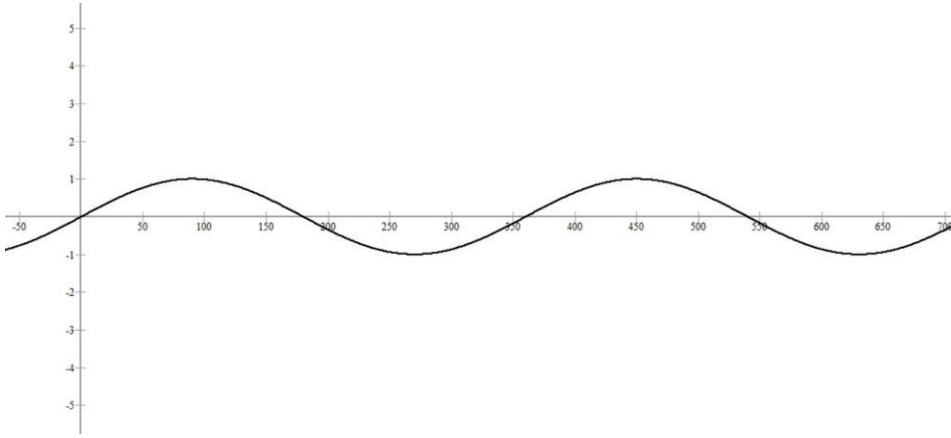
Checking our tables (or plugging that into a calculator) tells us that the unknown angle is 30°

The same can be done with arccosines $\left(\cos^{-1}\left(\frac{A}{H}\right)\right)$ and arctangents $\left(\tan^{-1}\left(\frac{O}{A}\right)\right)$.

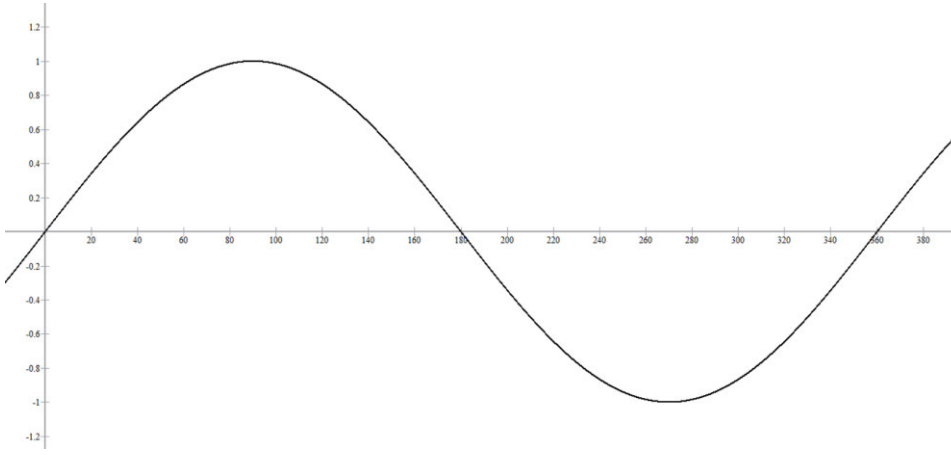
Now, let's see how this stuff looks on a graph

[338] From page 396

If we fill in all the sine values between 0° and 360° and beyond, we will get the following curve (what is referred to as a "sine wave"):



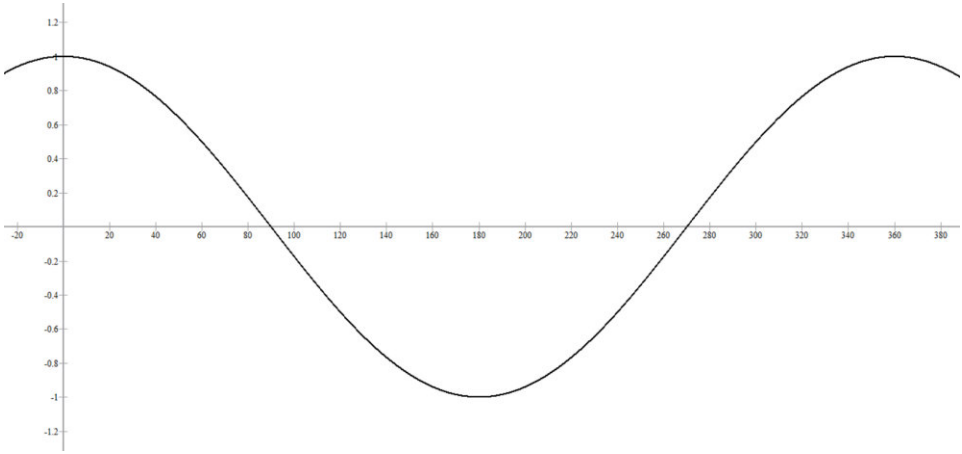
What will most concern us is the area of the x axis between 0° and 360°. Zoomed in it looks like this:



Starting from the left side, the wave begins at $x = 0$, $y = 0$. As we move to the right (towards 90°) the wave goes up at a sharp angle, and then begins to taper off, until at a height of 1 it flattens out (at $x = 90°$), and then begins the plummet down, rocketing through the x-axis at 180°, and then slowing down and flattening out at $x = 270°$ ($=90° \cdot 3$) at a low point of $y = -1$, and then driving up again at full speed through the x-axis at 360°, starting the cycle over again and again, repeating every 360°.

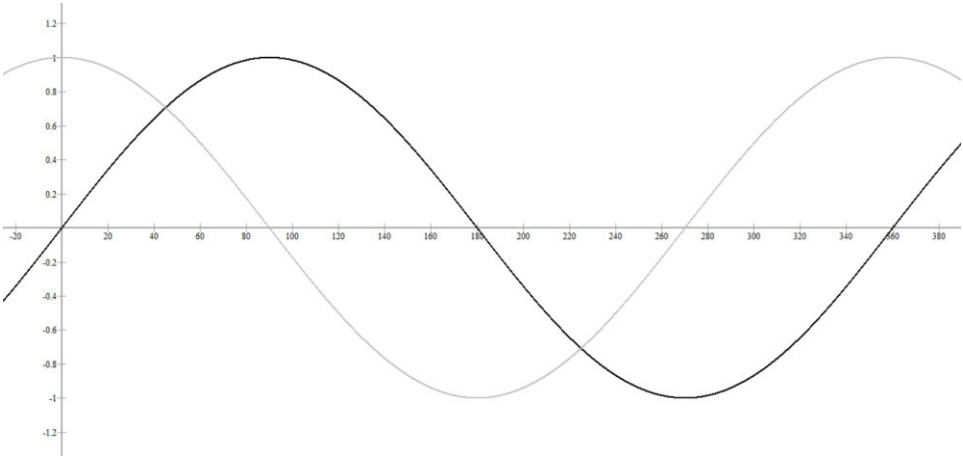
Remember this shape very well because it will come up over and over again in trigonometry and in the next few appendixes.

Now, a cosine wave looks very similar to the sine wave, except that it starts at a different position:



The cosine begins at 1 and slowly begins to fall through a very similar type of cycle, coming through the x-axis at 90°, and then tapering off at the bottom at 180°, rocketing through back up at 270°, and meeting the first height point again at 360°.

Together, the two waves look like this:



I find this image very useful, as I'm always trying to remember if I should be using sines or cosines, when it comes to spinning angles and circles and all that fun stuff we're going to do in the next few appendixes with spherical trigonometry.

Now let's manipulate these waves a little bit.

J

These last waves were put into a graphing calculator[339] as

$y = \sin(x)$

and

$y = \cos(x)$

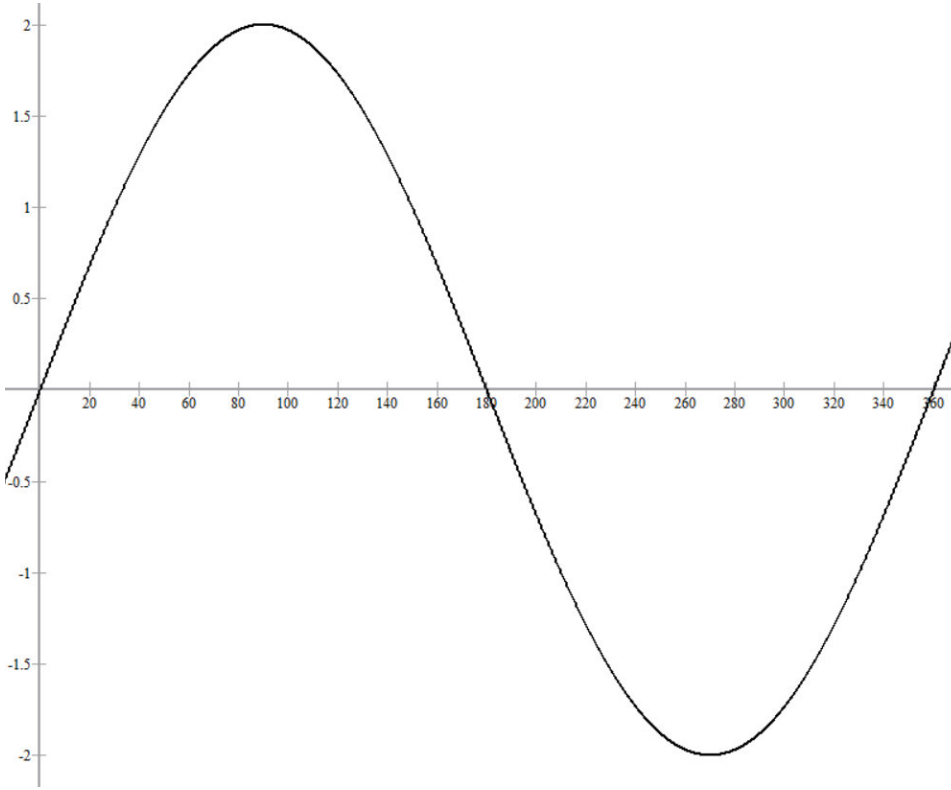
But let's say for whatever reason you want to make these waves taller, or shorter, or narrower, or wider. How can we do that?

Let's first make a sine wave twice as tall. We want that when 90° are (is?) plugged into the algorithm, we want it to return 2 (instead of the usual 1) according to the left hand vertical axis.

Very simple.

$y = \sin(x) \cdot \mathbf{2}$

And now y is twice as tall (and as low underneath the x-line).

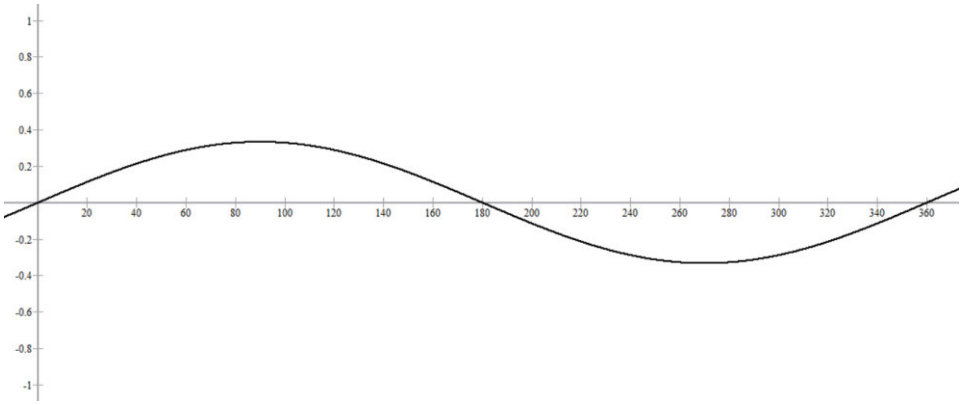


Notice the y-values on the left hand side.

[339] I'm using the program Graph, by Ivan Johansen

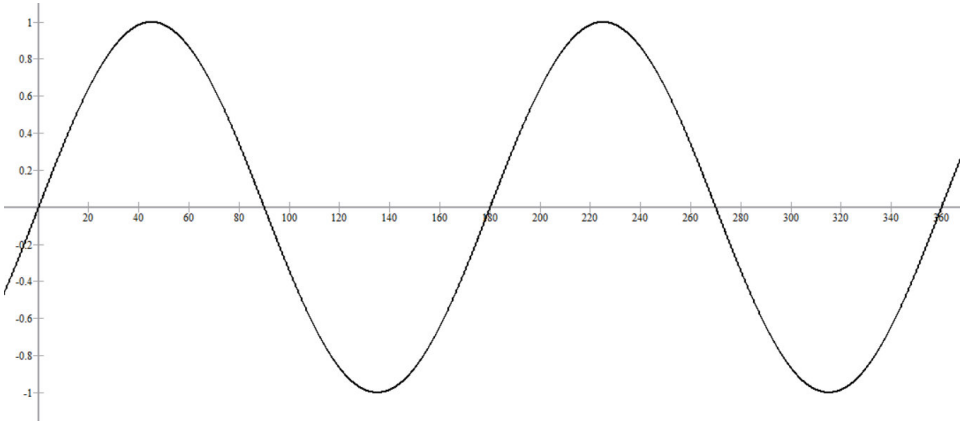
Making the wave 1/3 its height? You got it.

$y = \sin(x) / 3$



What if we wanted to make it skinnier (narrower)?

$y = \sin(x \cdot 2)$



This one is counter-intuitive, since we would expect that if we made x twice as big then it should stretch out along the x-axis, making the wave *wider*, not narrower.

The reason why it does this is because for the *present value of x*, it treats x as though he were from further down the line.

For example, when x is 30, "sin(x)" (i.e., sine of 30) would return a y-height of .5

When x is 60, it'll return a y-height of sine of 60, which is $\sqrt{\frac{3}{4}} (\simeq 0.866)$

But when the input is $\sin(x \cdot 2)$, when the program gets to $x = 30$, it doesn't give the sine of 30 (=.5) but the sine of 60 (.886...) since the formula says to consider double the value of x.

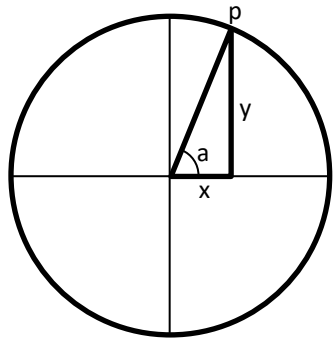
So that pretty high value of .866 shows up already where, with the simpler formula of sin(x), it would've been only .5.

You can play around with a graphing calculator to get the hang of it.

Now, let's show how this relates to circles. This section will be very relevant when we get to measuring astronomy related stuff.

Consider the following circle.

If we want to know the "x" and "y" of point "p" for a given angle "a", we take the radius of the circle to be 1 (unity). The hypotenuse of the triangle is the same as the radius, 1. This makes the trigonometric formulas simpler, since we will be doing Opposite over Hypotenuse, and Adjacent over Hypotenuse, and since the Hypotenuse is 1, we can just scratch it out of the formula.



The side of the triangle labeled "y" is sin(a) since:

$$\text{sine of angle} = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

and since Hypotenuse is 1, so anything divided by 1 is that thing, so we can scratch "over Hypotenuse" from the formula and we're left with:

$$\sin(\text{angle}) = \text{Opposite}$$

And for the same price: side "x" = cos(angle).

Now, let's apply this to measuring angles on the earth (and the sun in the sky).

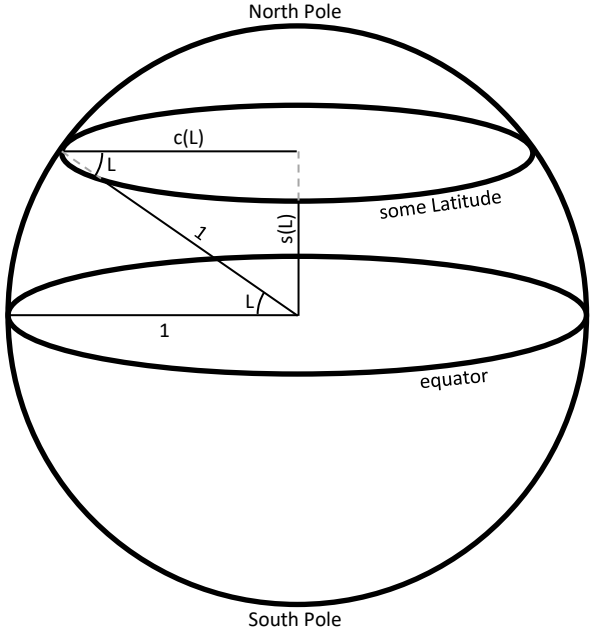
In the next diagram, we show a see-through earth.

The large outer circle is the earth's surface.

The central oval (labeled "equator") is a (3D view of a) flat disc cutting the earth into top and bottom halves, split at the equator.

The upper oval (labeled "some Latitude") is a smaller disc parallel to the equator at some latitude, L.

We draw a triangle starting from the center of the central disc up to the center of the upper disc (the side labeled "s(L)"). Then we draw (at a 90° angle) another line from the center of the upper disc to that disc's edge (the side of the triangle labeled "c(L)"). And then a third line from the disc's edge back to the starting point, the center of the central disc.



Taking the radius of the earth to be "1", we can measure the sides of this triangle.

First, we know the angle between the latitude disc and the diagonal line is the same as the angle between the diagonal line and the central disc, both angles being L. (We know this due to one of those funny laws of parallel lines and angles; I'm not explaining that here now).

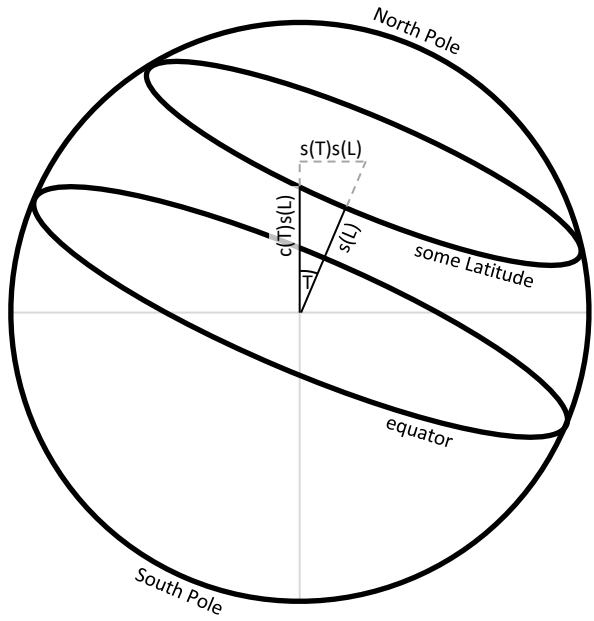
Since the Hypotenuse of this triangle is 1, we know the other sides of this triangle, as we explained earlier in this appendix.

The vertical line, the "height" of the upper oval, is sine(latitude), or simply s(L)

And the radius of the upper circle is c(L).

Get used to these types of calculations since we will be doing this quite often when we get to astronomy measuring angles on the earth and also when measuring the sun's paths in the sky; it will have the same tzurah.

Now, let's take this diagram one step further, and rotate the whole thing by T, tilt:



So now let's make and measure a new right triangle whose Hypotenuse is the central line between the "equator disc" and the "latitude disc", which in the previous calculation we showed to be s(L).

The other two sides are:

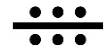
2) a vertical line from the equator's disc center (the center of the earth) straight up until the height of the Hypotenuse

3) and a horizontal line drawn connecting the previous two lines.

The angle between the Hypotenuse and the vertical line is T, earth's tilt (23.4°).

Following the same methods as before, we can determine that the vertical line is = c(T) · the Hypotenuse, and that the horizontal line is = s(T) · the Hypotenuse.

Using these methods, we will make many triangular calculations within the sphere of the earth and of the sky (the "celestial sphere").



General Astronomy

We'll be using 3 viewpoints when making the calculations for astronomy:

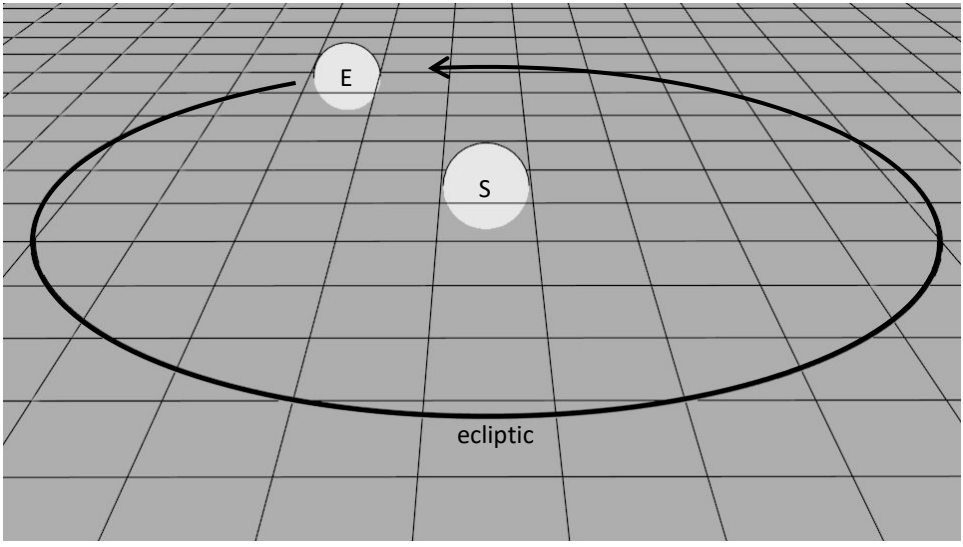
- Heliocentric (Sun-centered)
- Geocentric (Earth-centered)
- Topocentric (Ground/Observer-centered)

The calculations for each of them are exactly the same as any other one. Which one we will use at any given time will depend on which one makes the calculations more intuitive.

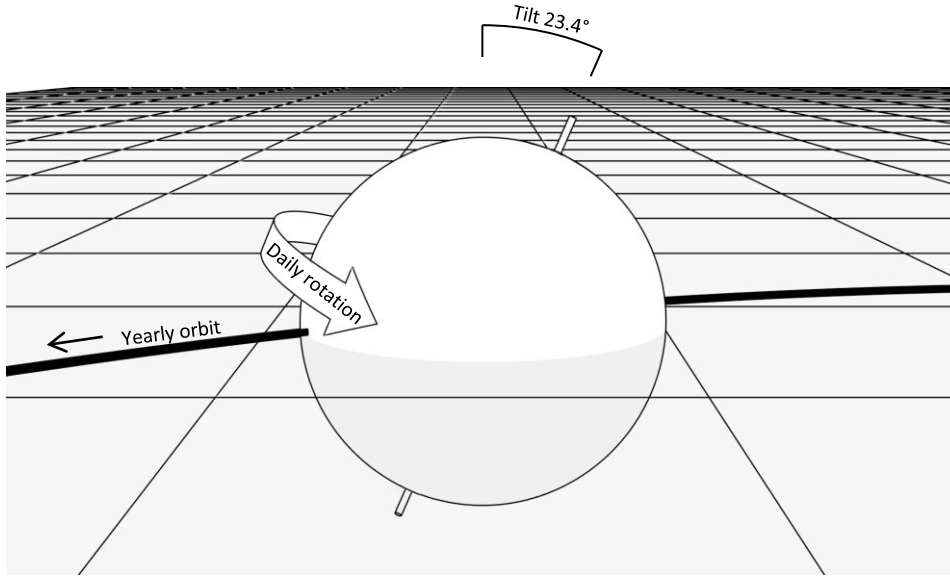
We will start with Heliocentric since that is what most people are used to. We will move to Geocentric since the factors needed for the calculations are easier to extract. And then we will move to Topocentric since it's the most relevant perspective and it will make the more complex calculations simpler, once you get used to it.

In the Heliocentric view, the earth revolves around the sun once a year, along a path called the ecliptic.

The ecliptic can also be viewed as a plane, with the center of the sun in the plane, and the center of the earth revolving around the sun along the plane.



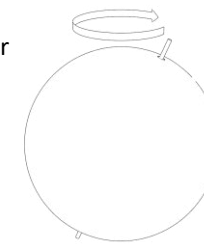
The earth's daily rotation is tilted[340] relative to this plane of its orbit by about 23.4°.



The poles of the earth (which aren't actually sticking out of the earth, fashteit zich) stay pointing to the same[341] distant stars, even as the earth revolves around the sun.

[340] The Rambam in Hilchos Kiddush Hachodesh 19:4 writes that the tilt of the earth is "about $23\frac{1}{2}$ degrees". Throughout this sefer, we will be using 23.4° for the tilt, abbreviated to T.

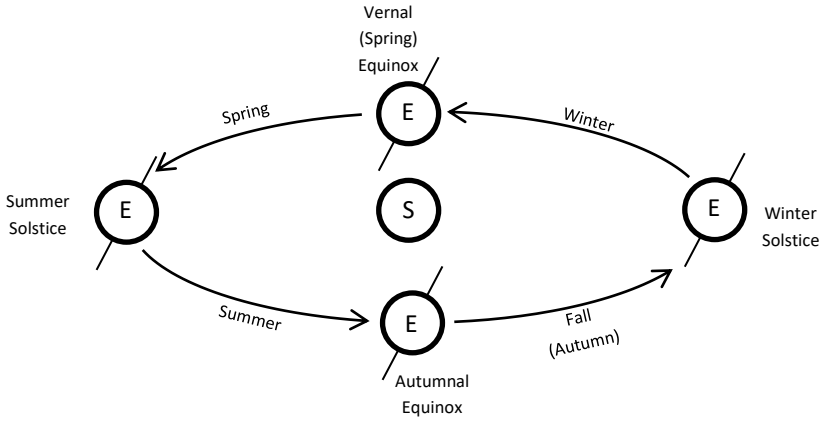
[341] Not exactly. There is a cycle called axial precession in which the tilt of the earth itself rotates slowly over time, so that the particular group of stars that the earth is tilted towards at any given time changes.



The Rambam in Hilchos Yesodei HaTorah 3:7 gives the speed of axial precession as "The amount the Sun moves in one day is what the stars move in about 70 years". In Hilchos Kiddush Hachodesh 12:1 he writes the speed of the sun in 100 days as 98° 33' 53'', so one day is $0.985647\overline{2}$°. The full axial precession cycle is 360° divided by that amount times ~70 years ≃ 25,567 years. Current scientific estimate is about 25,765 years (-Irv Bromberg). But axial precession isn't relevant to our discussion since we will be defining a year by the relative positions of earth's tilt to earth's position in its orbit, irrespective of the positions of the distant stars. This definition of a year is called the Tropical Year, i.e., the year of the seasons. The reason I'm calling the tilt relative to the stars is because it gives people a context within which to understand the concept.

This yearly cycle of going around the sun while tilting in the same direction is what gives rise to the seasons.

When the earth is pointing[342] towards the sun, that is the start of summer. When it is pointing away from the sun, that is the start of winter. In between summer and winter, when the tilt of the earth is sideways relative to the sun, that is the start of fall, or autumn. And in between winter and summer, when the tilt is the other way sideways (see diagram), that is the beginning of spring.



The beginning of summer is called the Summer Solstice, and is usually around[343] June 21. The exact moment of the summer solstice is when the earth is pointing directly at the sun, and is when the sun is at its most northern position, which we will explain more soon. The day that the solstice happens is the longest day of the year (and the shortest night). We will get to the exact calculations soon.

The beginning of fall (autumn) is called the Autumnal Equinox, and is usually around September 23. The length of the day and night are 12 hours (about, which we'll explain more soon). The word equinox means "equal night", meaning when the day is the same as the night.

The beginning of winter is called the winter solstice, and is usually around December 21. That day is the shortest day, and the longest night.

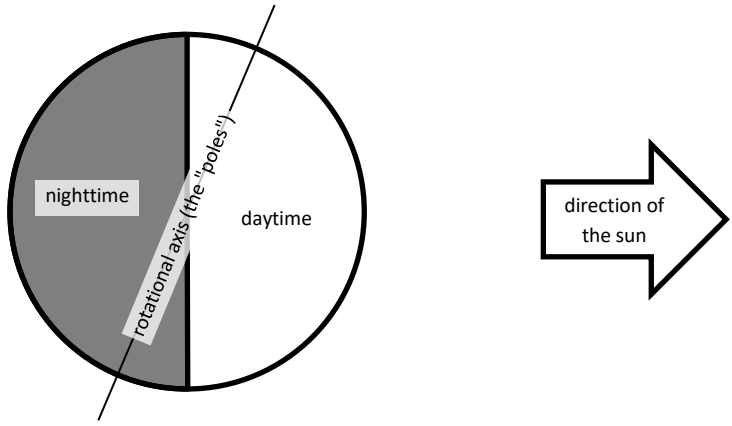
The beginning of spring is called the Vernal (or spring) equinox. It happens around March 20. At that time, the days and nights are the same length.

[342] When I say "pointing" or "top", I'm referring to the pointing of the north pole or viewed from on top of the north pole since I come from a north-oriented culture.
[343] The Gregorian calendar (the "goyish" or "English" calendar) was designed to keep the days of the year aligned with the seasons. It is accurate to about 1 day within 3,000 years. But even within that accuracy, the exact day that a specific solstice or equinox happens fluctuates a day this way or that, due to various reasons, among them the arbitrary year that gets the leap day of February 29.

Length of Day - Geocentric

Let's show how the day length changes based on the direction and angle of earth's tilt.

The first thing to consider is that the side of earth facing the sun is lit up, the "sunny side of earth" where it is daytime, and the other side is dark and is nighttime.



The way we will be doing these calculations is to take a particular latitude which is a circle a certain number of degrees above the equator (measured from the center of the earth), and measure how much of that circle is within the light side of earth (the side facing the sun), and we'll then convert that amount into time. (For clarification, see diagram on page 420. Latitude is angle DAF. Amount of that latitude in the sunny side is arc EDGH)

The calculation will look something like this:

> Latitude x has 240° (for example) on the sunny side of earth.
>
> That is $\frac{2}{3}$ of the whole circle ($= 240° \div 360°$)
>
> It takes 24 hours to go around the whole circle.
>
> It will take $\frac{2}{3}$ of 24 hours to go around the part of the circle that is in the sunny side.
>
> Therefore, the length of the day at that latitude is 16 hours $\left(= \frac{2}{3} \cdot 24 \text{ hours} \right)$

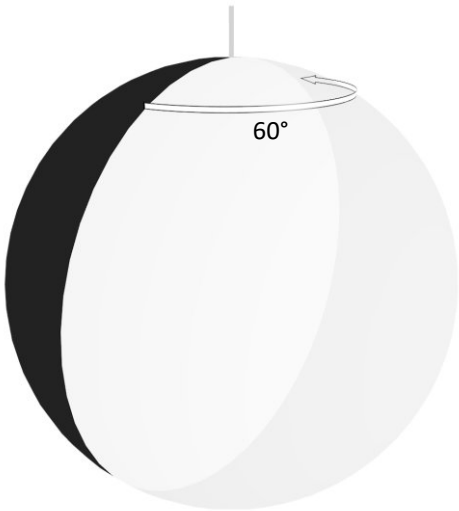
For the equator, it won't make a difference what season it is; the daytime and nighttime will each always be 12 hours[344].

Whether the earth is tilted towards, away from, or sideways to the sun, exactly half of the equator's circle will always be in the sunny side of earth.

towards *away* *sideways*

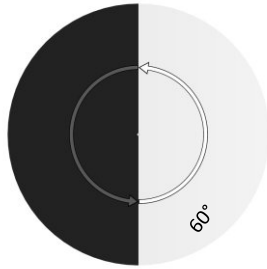
As you move north or south of the equator, the time of daytime begins to change throughout the year. The further you go, the more drastic the change.

Let's take 60° latitude as an example.



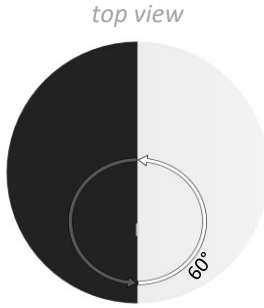
If the earth wasn't tilted, 60° latitude would always have a 12 hour day. Although it is a smaller circle, half of it would be in the sunny side and half in the dark side.

top[342] *view*



[344] At least according to the method we present in this particular appendix. In the next few appendixes, we will fine-tune our calculations and there will be some difference between the seasons, even at the equator, although it will be slight.

And this is essentially what happens when the earth is at the equinox; every latitude spends half the day in the sun and half in the dark.



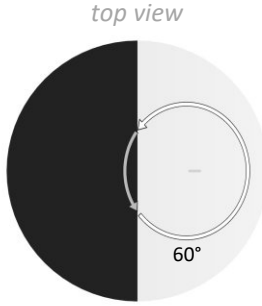
It is the other days of the year where non-equator latitudes have different amounts of day and night.

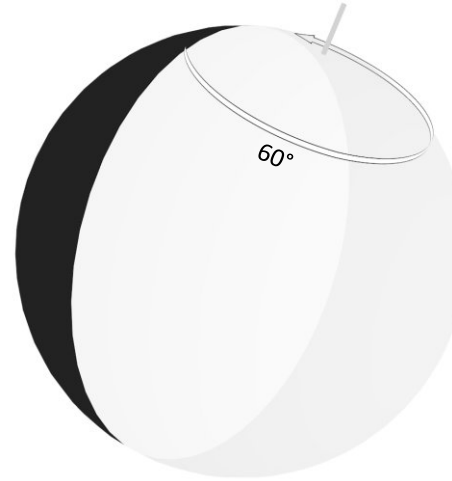
Let's continue with our example of 60° latitude and see how it looks at the summer solstice, when the earth is pointing towards the sun.

Here, we see that more than half of the circle is within the sunny side. That means that it takes longer than half a day to get through the sunny side.

Which means, the day is longer than 12 hours.

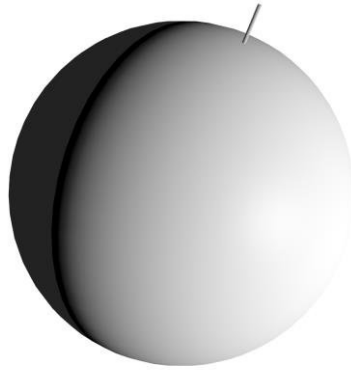
How much longer?





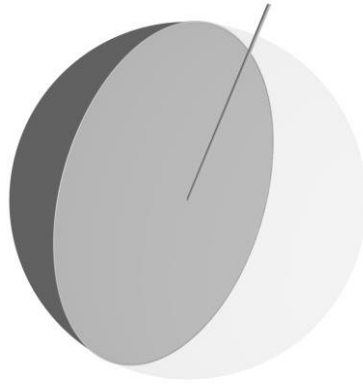
In order not to lose the audience, we're going to build up a 3D diagram one step at a time, so you can follow where we're going.

Let's start with a blank earth, half light - half dark, tilted towards the sun.

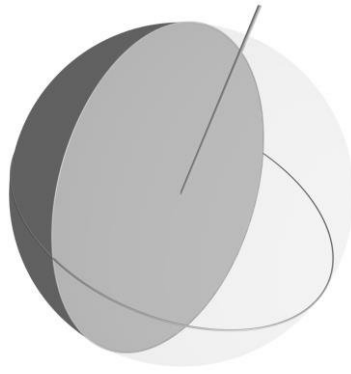


Now, let's make the entire light half of the earth hollow, so we can get to the kishkes inside ("Geocentric", or "Earth Centered"), to make the calculations easier.

In the image, you can see the pole going through the earth to the center of the earth.



Now, let's add the equator; the line halfway between the poles.

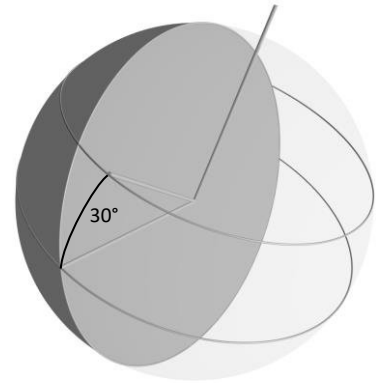


L

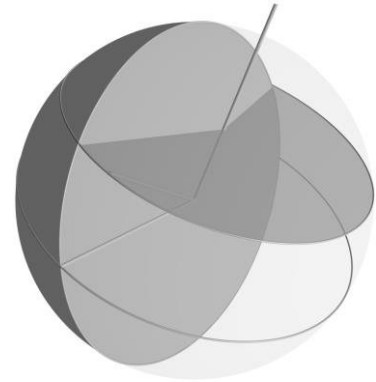
Now, let's add a latitude.

At this point, it doesn't matter which one, so let's just take a random latitude of 30°. That's the circle 30° north (towards the north pole; the "upper" pole) measured from the center of the earth, relative to the equator.

The picture shows the latitude and the angular lines measuring the 30°.

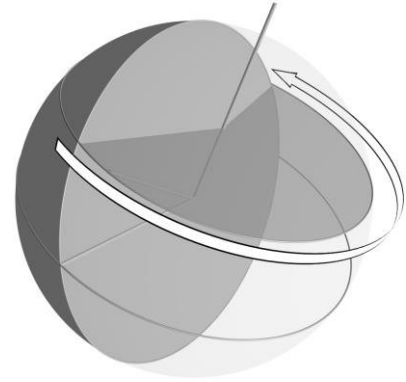


Now, we want to make a "pie" (not pi) that is along the latitude, and goes from the point (on the left) where the light and dark meet (sunrise) through the daytime, until the other point where the light and dark meet (sunset).

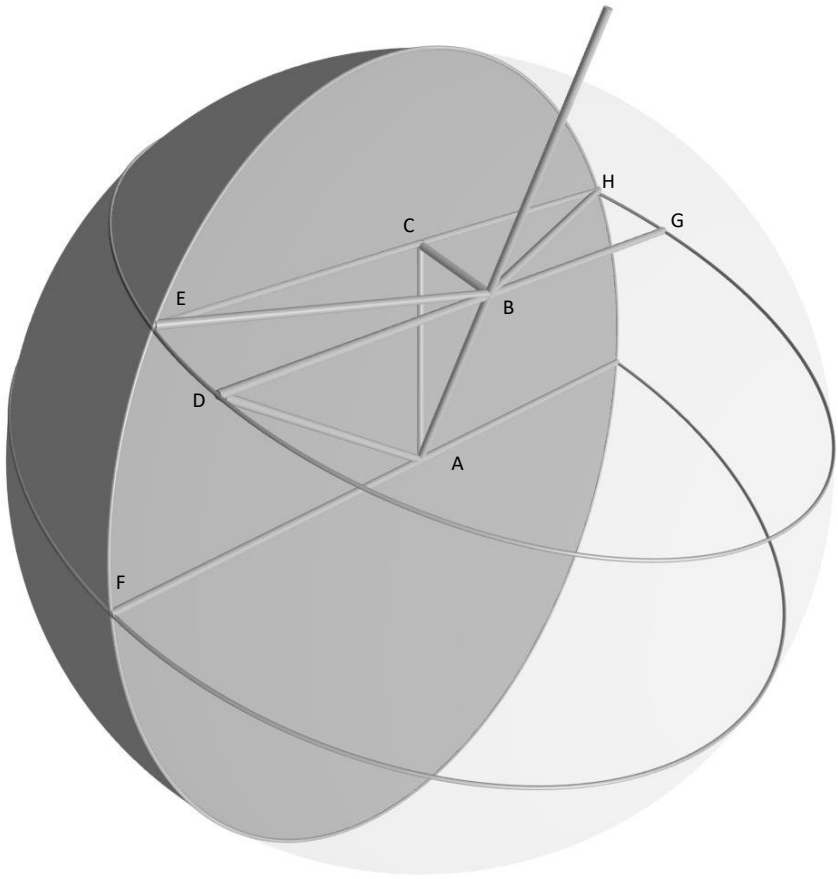


This represents the angle a person on this latitude would travel through the day, as he just sits there and the earth rotates.

We're going to call this the "day angle". Throughout these calculations, we will take the day angle, divide it by 360, then multiply by 24, to get the day length in hours.



$$\text{Day length} = \frac{\text{day angle}}{360} \cdot 24$$



$AF = 1$	the radius of the earth, taken as "Unity"
$AD = AF$	1, the radius of earth
$BAC = T$	the tilt of the earth; we'll be using 23.4°
$DAF = L$	latitude of the observer; we'll be using 30 at the moment
$BDA = DAF$	since mirror angles between parallel lines
$AB = \sin(L)$	the "height" of the latitude, as we showed on page 410
$BD = \cos(L)$	the radius of the latitude, also on page 410
$BE = BD$	the radius of the latitude

$$BC = \tan(BAC) \cdot AB$$ since tan(BAC) = BC / AB

$$BEC = \sin^{-1}\left(\frac{BC}{BE}\right)$$ since sin(BEC) = BC / BE

$$DBE = BEC$$ since mirror angles between parallel lines

which expands to:

$$DBE = \sin^{-1}\left(\frac{\tan(T) \cdot \sin(L)}{\cos(L)}\right)$$

And since[345] sine divided by cosine = tangent, we can simplify this formula to:

$$DBE = \sin^{-1}(\tan(T) \cdot \tan(L))$$

Now, since we want to know the entire angle of the "pie" (that is, from point "E" all around the latitude on the "light side" until point "H"), we need to double DBE (in order to include the angle of GBH which is the same angle as DBE) and then to add 180° (in order to include the half of the circle from "D" around the light side until "G")

Altogether, our formula would be:

$$\text{Day angle} = \sin^{-1}(\tan(T) \cdot \tan(L)) \cdot 2 + 180$$

And to turn that into time we will follow the formula we brought on page 419:

$$\text{Day length} = \frac{\sin^{-1}(\tan(T) \cdot \tan(L)) \cdot 2 + 180}{360} \cdot 24$$

And since dividing by 360 multiplied by 24 is the same as dividing by 15, we can simplify the above formula to:

$$\text{Day length} = \frac{\sin^{-1}(\tan(T) \cdot \tan(L)) \cdot 2 + 180}{15}$$

Using this formula with a latitude of 30° yields a day length of

13.929... hours

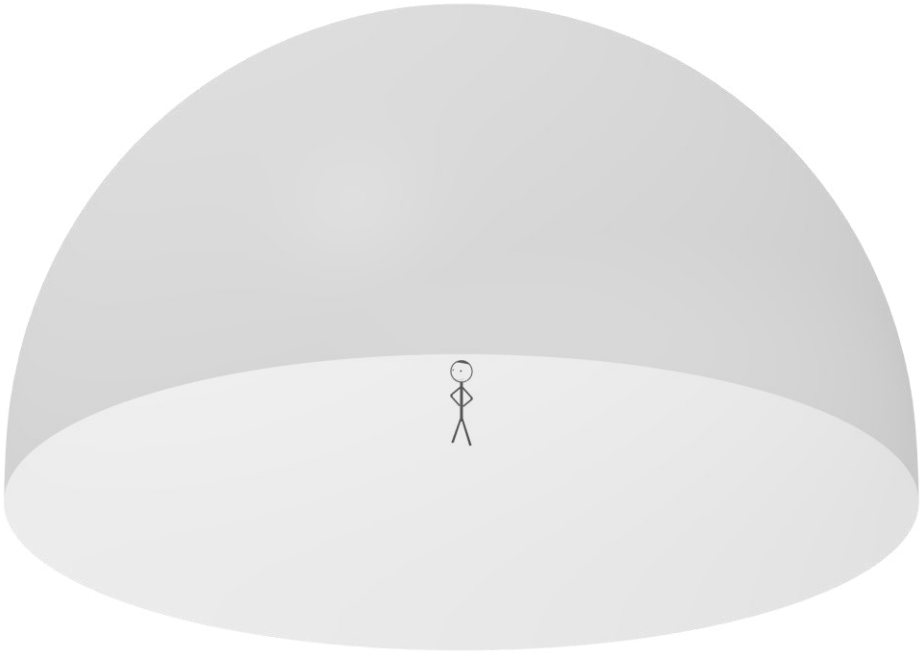
which is

13h 55m ~45s

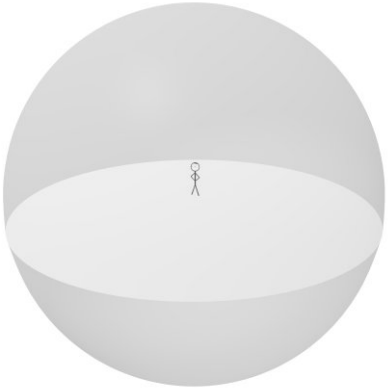
[345] As we showed on page 399

Topocentricity - The Viewer's Perspective

In the Topocentric view, we look at the observer as though he were standing on a large circle covered by a huge dome[346].



Sometimes, we will view the sky as a complete sphere that entirely surrounds the observer and his "ground circle".

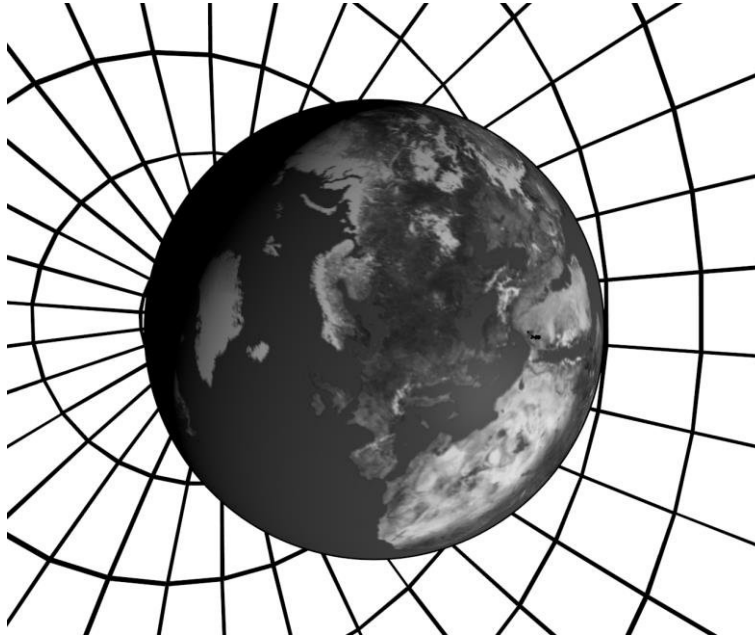


This sphere is called the Celestial Sphere.

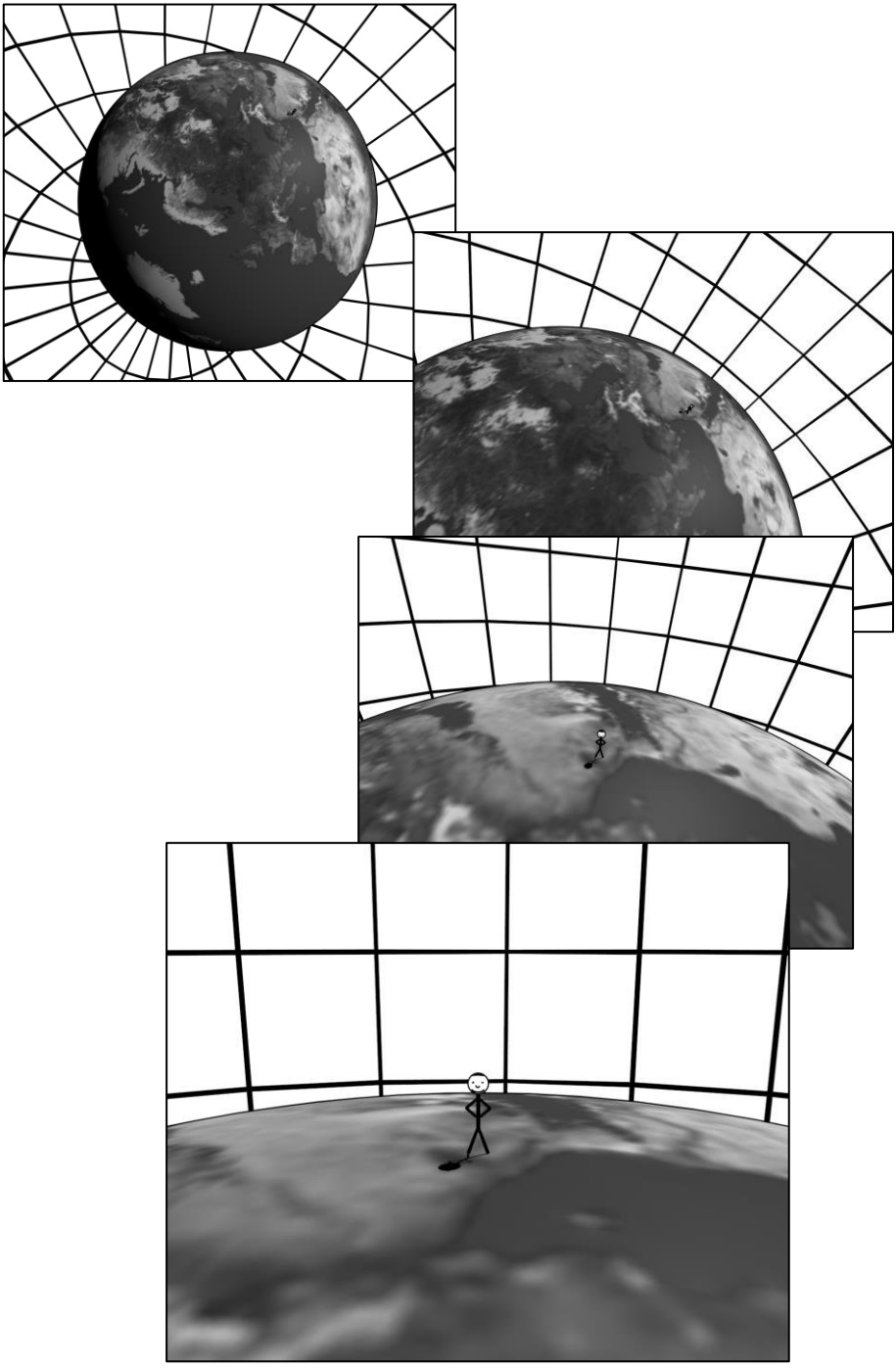
[346] Topocentric just means measured from the surface of Earth (or any other sphere), but we will be specifically focusing on the observer stationed there, since he will help us orient ourselves for the calculations needed.

To show how this perspective comes about, let's start with a camera shot far away from earth

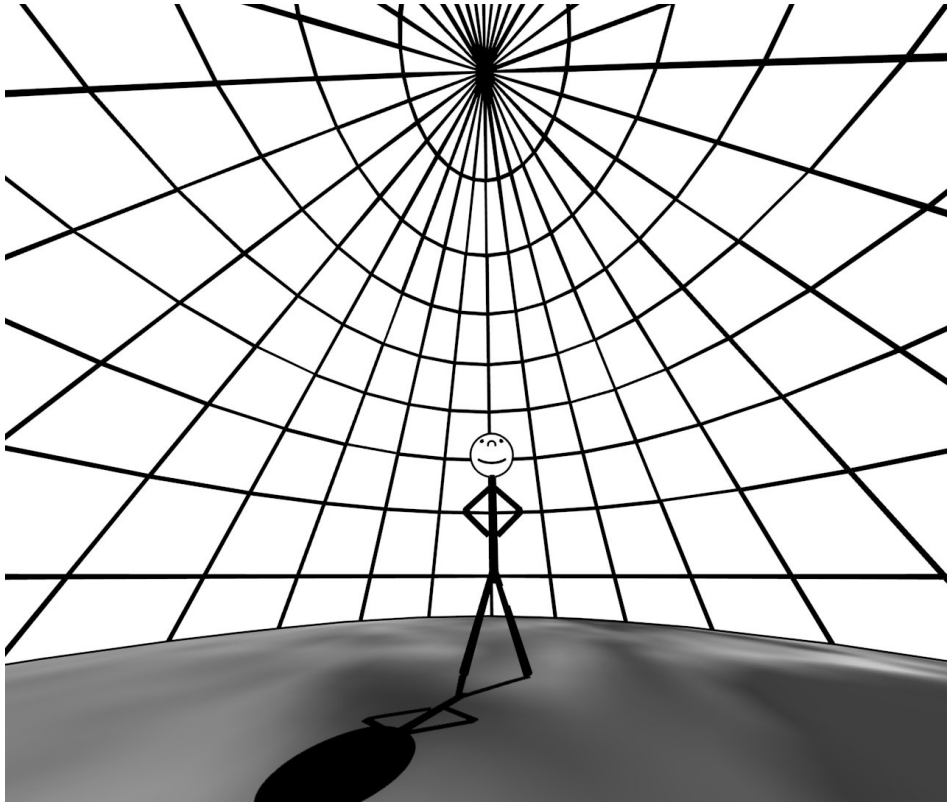
Let's fill the sky with a spherical grid that is oriented to the observer:



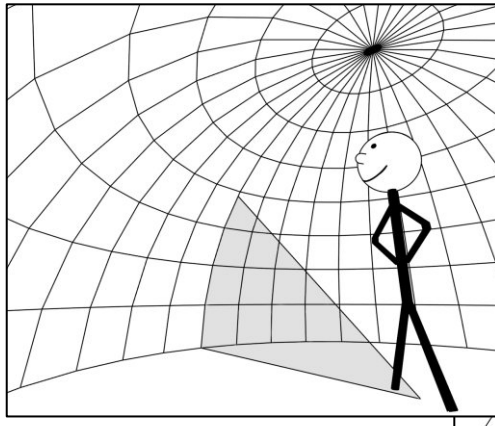
Let's zoom in to the man:



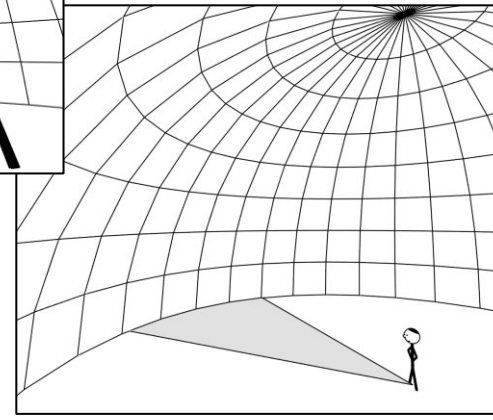
M



The purpose of this perspective is to create an environment that most resembles the experience of an observer on earth.



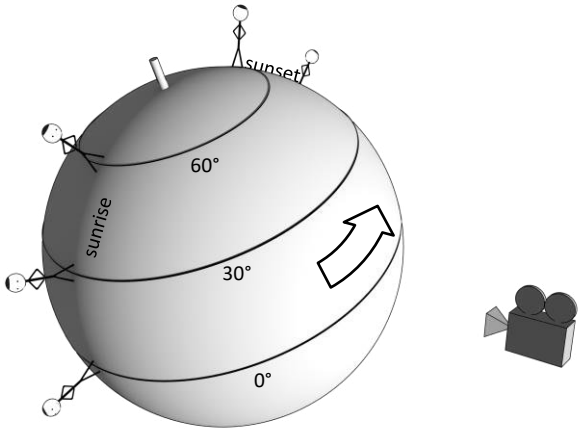
Some of the other useful reasons to use this view is to measure objects in the sky by their angular measurements relative to the observer



or to make angular measurements on the ground (the Azimuth[109]).

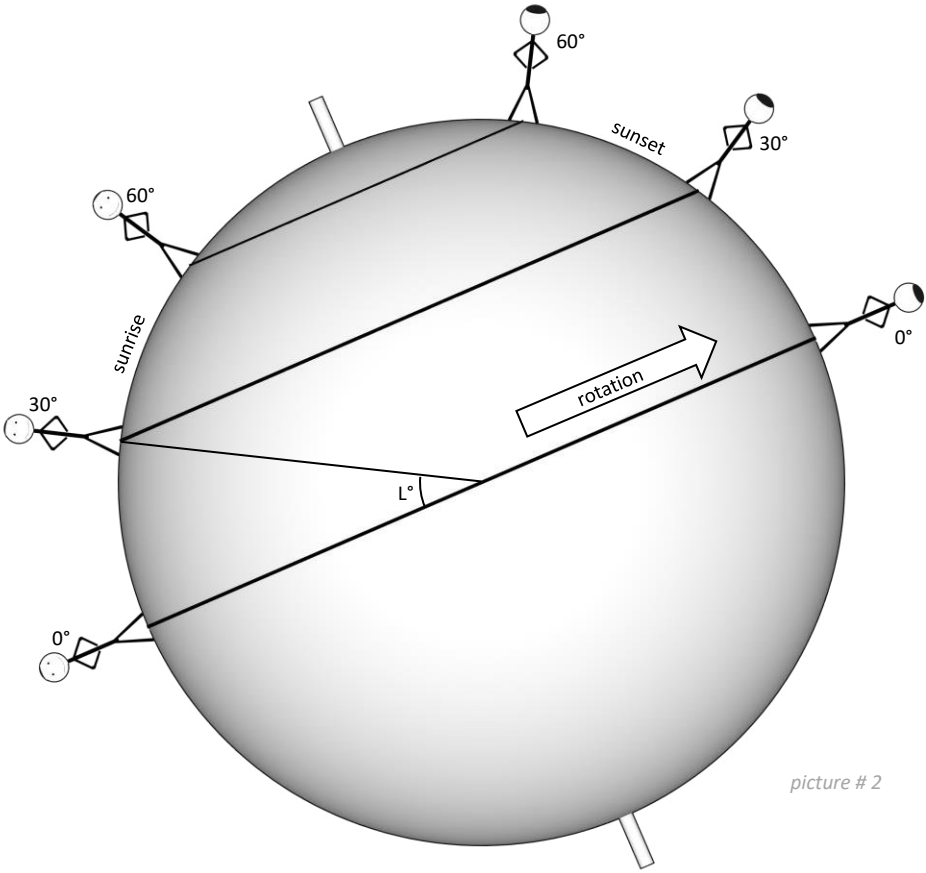
Also, I find this perspective makes the calculations more intuitive.

Let's begin with the time of year at the equinox. That is, the beginning of Spring or Fall, when the earth's tilt is sideways relative to the sun. We'll have everyone face East and we'll note where the sun is relative to them, based on their latitude and the time of day. We'll show the calculations using the fall equinox but it'll come out the same by the spring equinox.



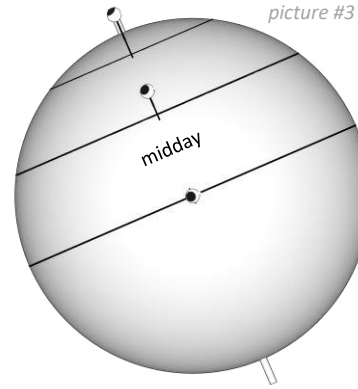
picture #1

If we position the camera looking from the direction of the sun, we'll see the people on earth at **sunrise** (the left side of earth) are looking directly at the sun (see picture #2), and at **sunset** (the right side of earth) the sun is directly behind them, regardless of their latitude.



picture # 2

Now let's do **midday**.

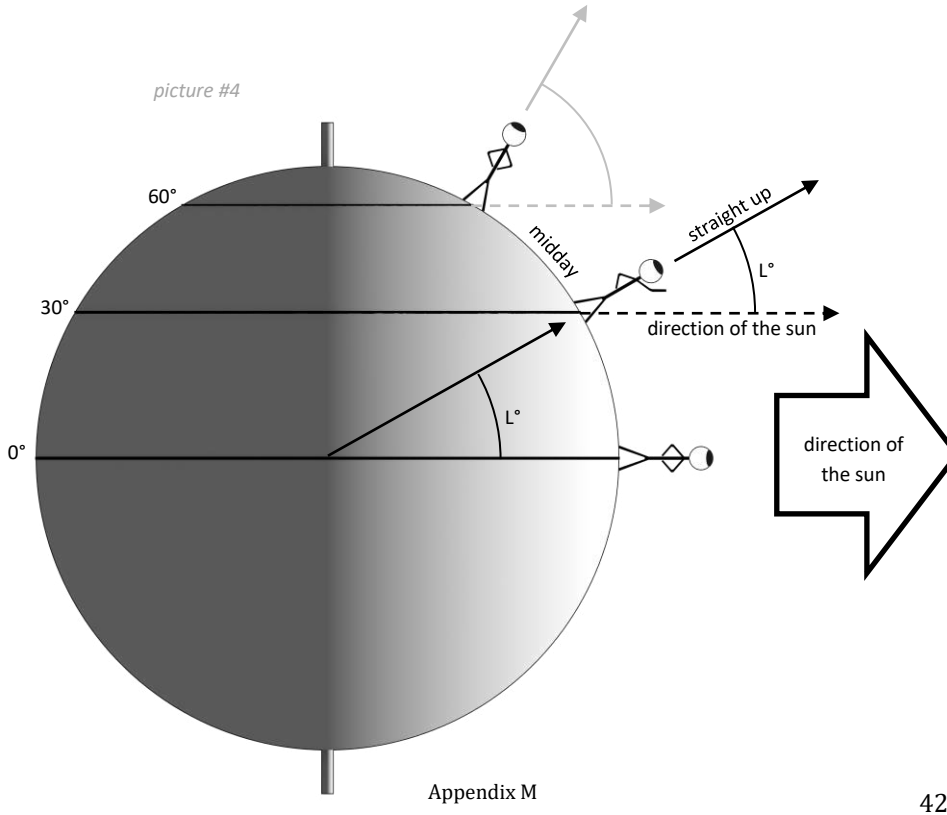


If we now position the camera on the side of the earth, looking at the half light - half dark, lowering the camera a bit so that it can look up a bit towards the equator:

We can now measure the angles in the sky that the sun would appear to each person based on his latitude.

We can see in picture #4 that the angle between straight up and the sun is the same as the observer's latitude.

We can also see that the angle is to the observer's right-hand side (for latitudes north of the equator).



Now, let's see how this plays out topocentrically.

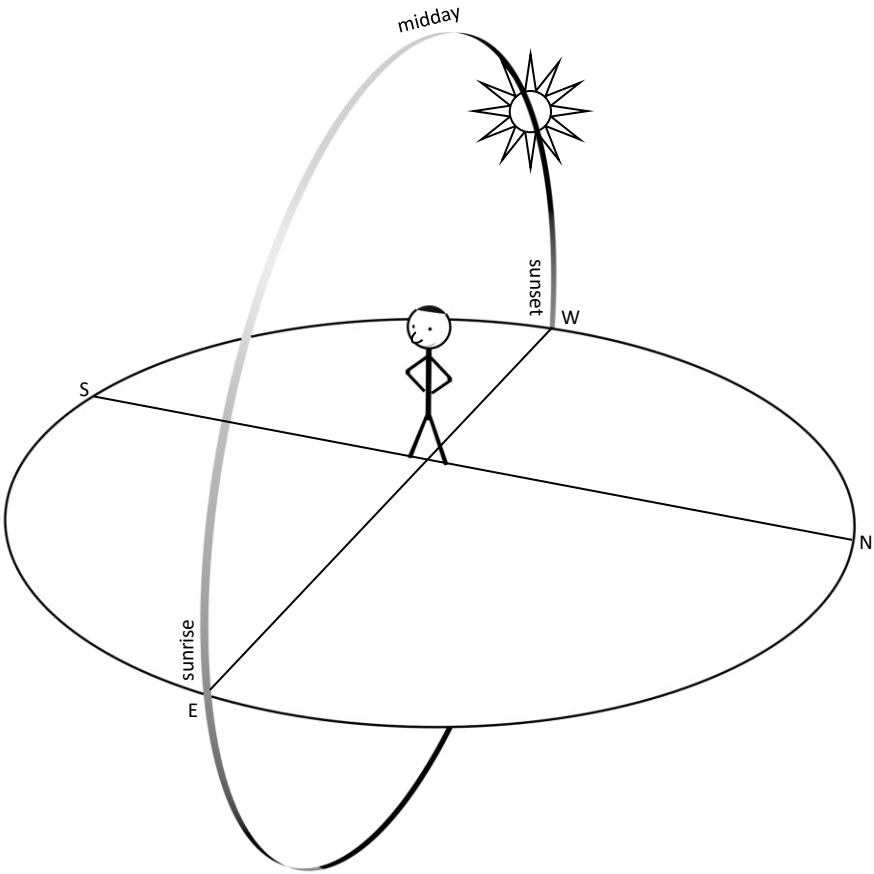
Here's a man standing at the equator, facing East.

At sunrise, the sun is directly in front of him, as we showed on page 426, picture #2 (at equator = 0° latitude, sunrise, on the left side of the earth)

We also showed there that at sunset, the sun is directly behind him.

And at midday, the sun is directly overhead (see page 427, picture #4, the guy at the bottom right).

In topocentric view, it would look like this:



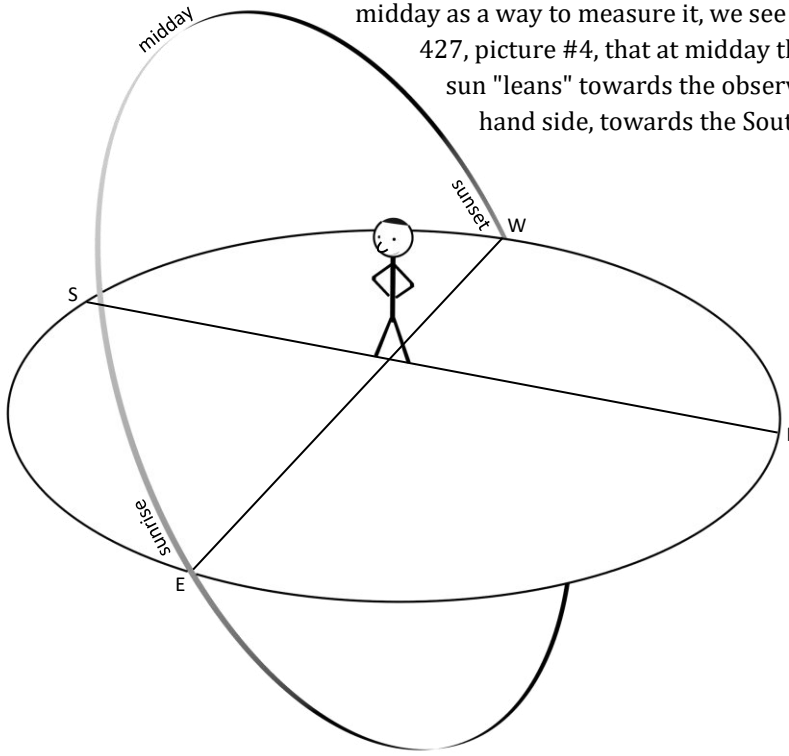
(In the picture, the sun is in the afternoon)

M

Now, let's go to latitude 30° (north), with the man facing due East.

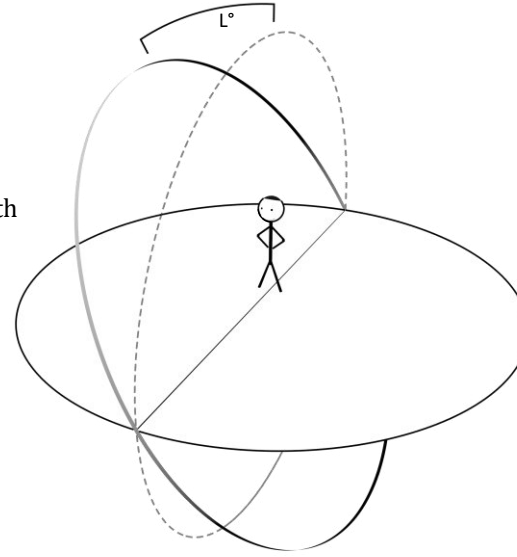
From page 426, picture #2, we see that at sunrise the sun is directly in front of him, and at sunset the sun is directly behind him, just like at the equator.

The rest of the day is where the sun will be different than at the equator. Using midday as a way to measure it, we see from page 427, picture #4, that at midday the path of the sun "leans" towards the observer's right hand side, towards the South.



The amount of degrees that the path tilts is equal to the degrees of latitude of the observer, as we showed there.

In this case, that would be 30°.

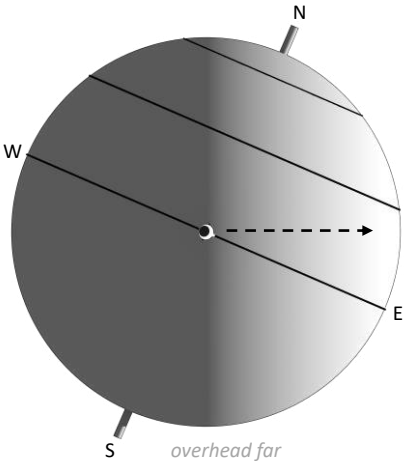


This line in the sky that the sun follows at the equinox can be called simply the equinox, or a term more commonly used is the Celestial Equator.

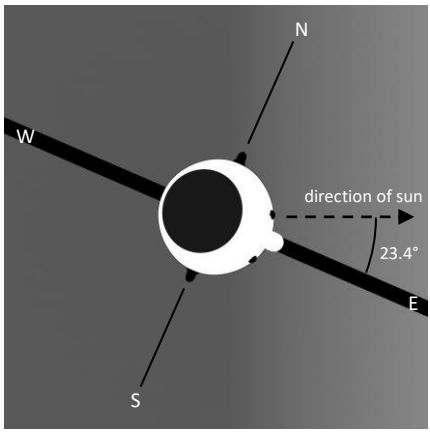
This is because if you were to take the equator of earth and project it outwards onto the celestial sphere, this line would be synonymous with the sun's path during the equinox.

Now, let's see what happens during the rest of the year. At this point, I just want to show what happens by the Solstices, so we can lay down the foundations for doing topocentric calculations, and later[347] we'll show how to calculate other days of the year.

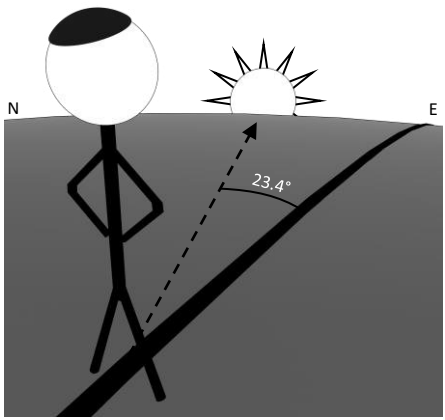
Let's go to the Summer Solstice[348], at the equator. Let's start with sunrise:



overhead far



overhead close

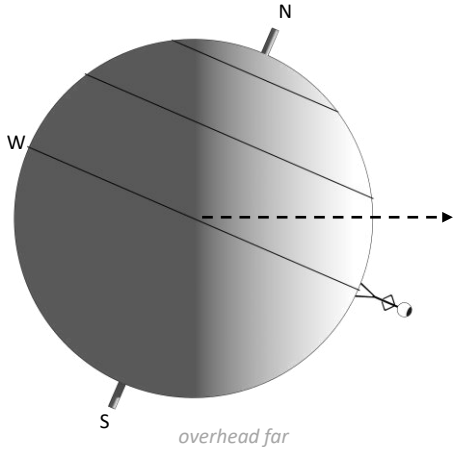


topocentric

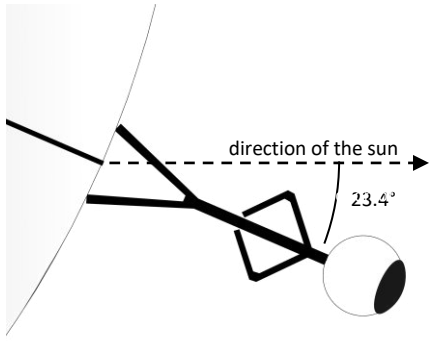
[347] by Appendix Q

[348] The Winter Solstice will be the same type of calculations, just in reverse directions. See later, page 434.

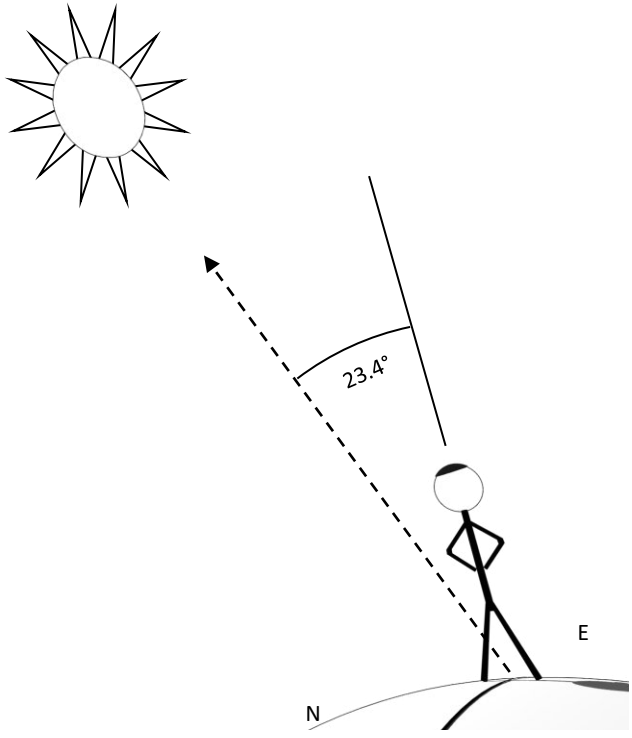
Now let's see what happens at midday (noon):



overhead far

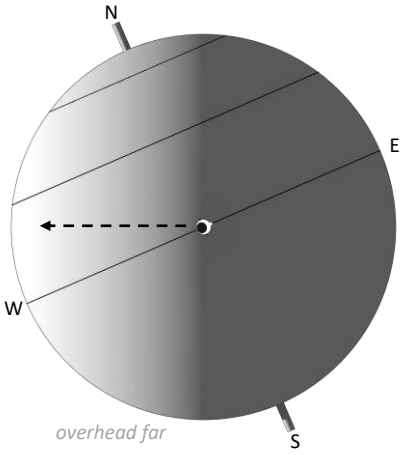


overhead close

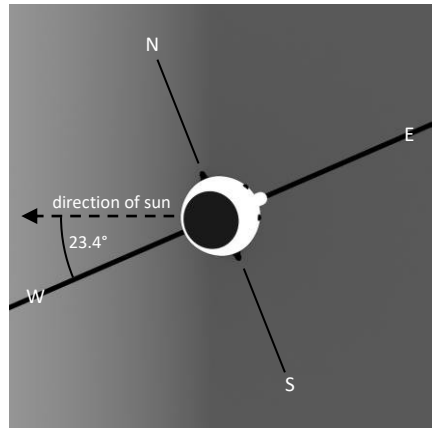


topocentric

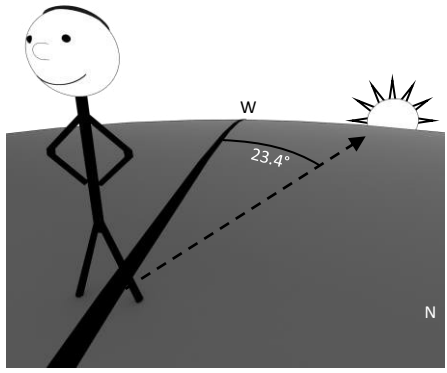
Now let's go to sunset. For this picture, we need to go to the other side of the earth, so we're looking at it from the other side now.



overhead far



overhead close

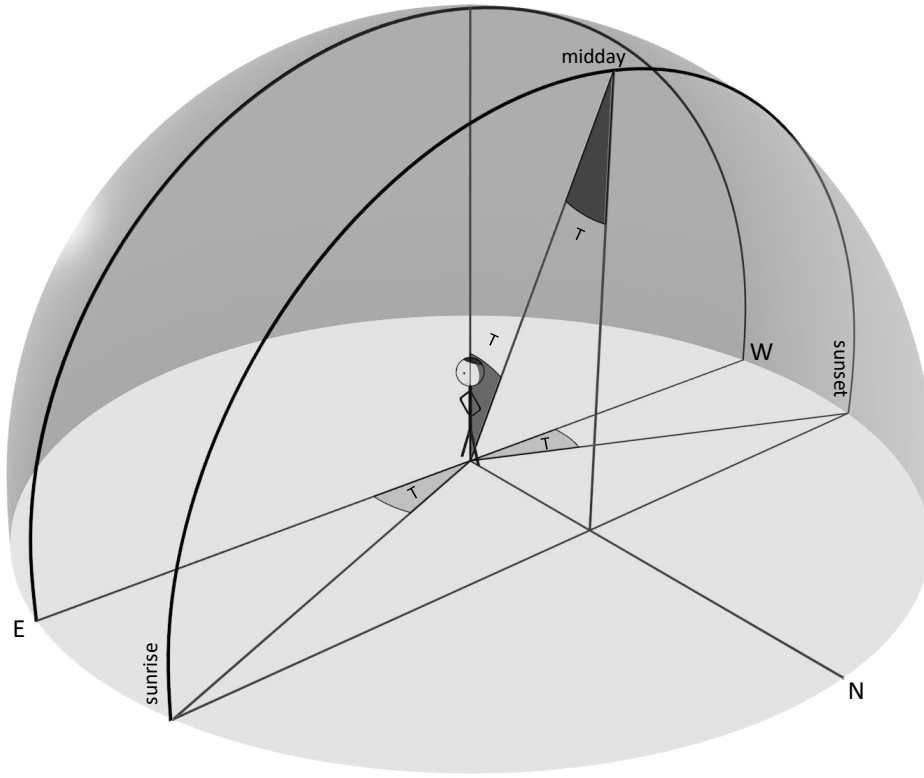


topocentric

From these pictures in the last three pages, we can extract from all of them the entire path of the sun during the summer solstice, topocentrically.

Let's put them all together on the next page.

I'm going to replace 23.4° with T.



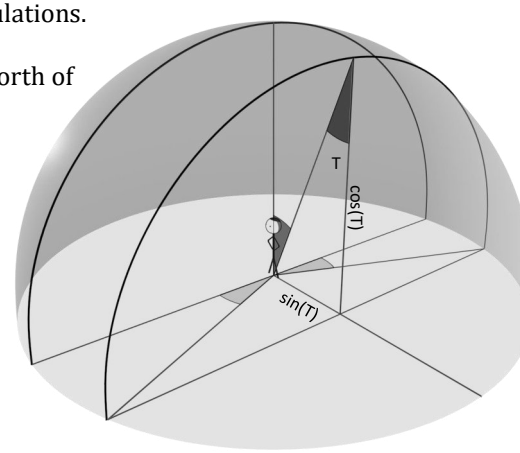
Starting from the sunrise, the sun rises out of the horizon at T° north of due east.

It then rises along the Celestial Sphere, towards the top, until at midday when it is north of the observer, T° lower than straight up. I've drawn a vertical line from the point of the sun at midday straight down, and the angle between that and a line drawn from the sun to the observer make the same angle of T. We will use that angle to measure the distances of the sides of the triangle that those lines create, in order to help us with some of the later calculations.

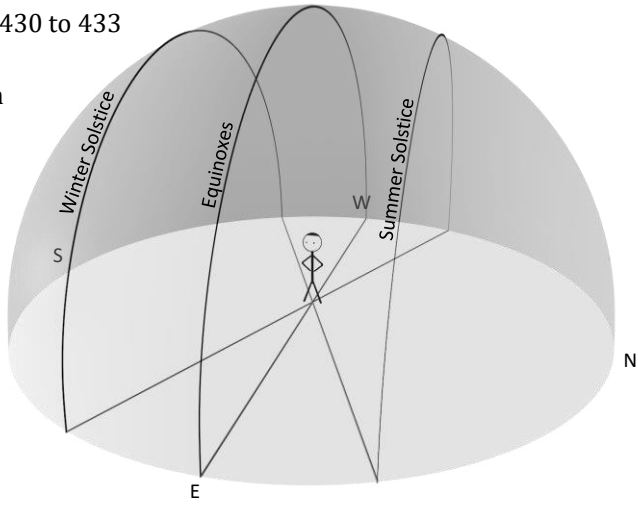
The sun then sets at the horizon, T° north of due West.

Notice that this circle that the sun follows during the Solstice is smaller than the circle of the Equinox.

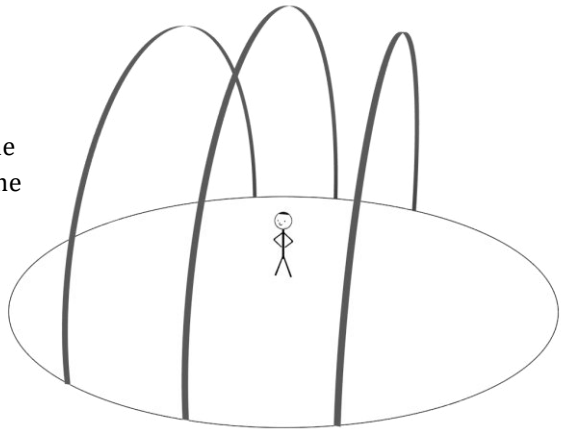
Taking the radius of the Celestial Sphere to be 1 we can determine that the radius of the Solstice circle is cos(T), and that its distance to the Equinox circle is sin(T).

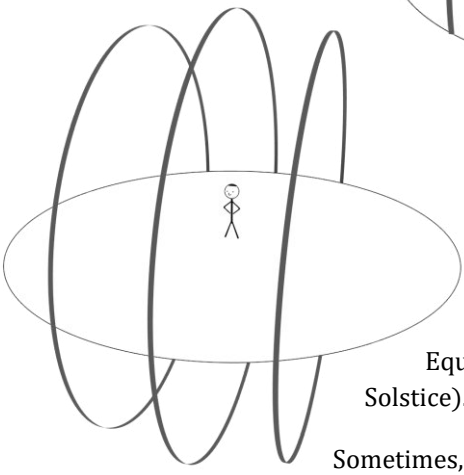


If you do all these diagrams and calculations again that we did from page 430 to 433 but for the Winter Solstice instead, everything will be in reverse and we will end up with a topocentric dome with the Winter Solstice circle on the South side of our observer.



Now, let's remove the dome and the floor angles in order to unclutter the image a little bit.



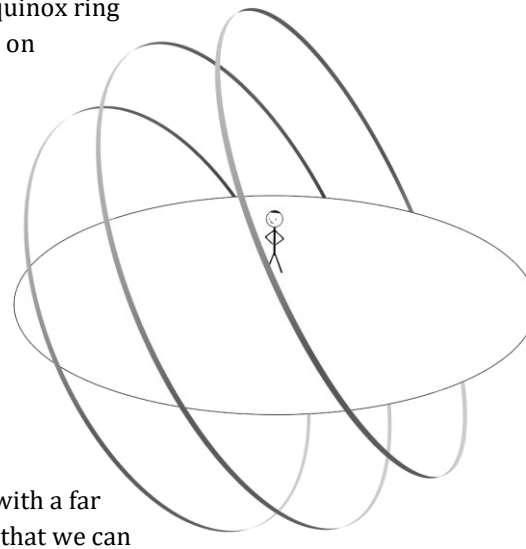


And, let's complete the circles under the earth to get the main topocentric method of calculations, which will involve these three circles or rings (The Equinoxes, Summer Solstice, and Winter Solstice).

Sometimes, this is called the Sun Path Diagram.

M

We showed on page 429 how the equinox ring rotates relative to straight up based on the observer's latitude.

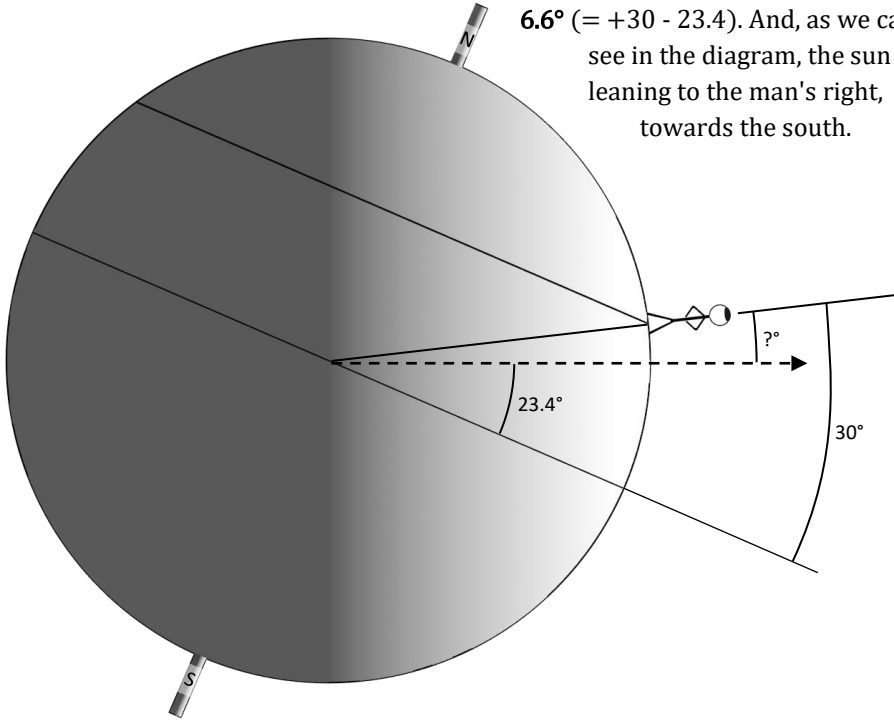


Now that we've introduced the Solstice rings, let's show how they rotate as one unit together with the Equinox ring, based on latitude.

Let's start with midday. We'll start with a far view looking at the side of earth, so that we can see half the earth lit up and half dark, with the sun on the right side.

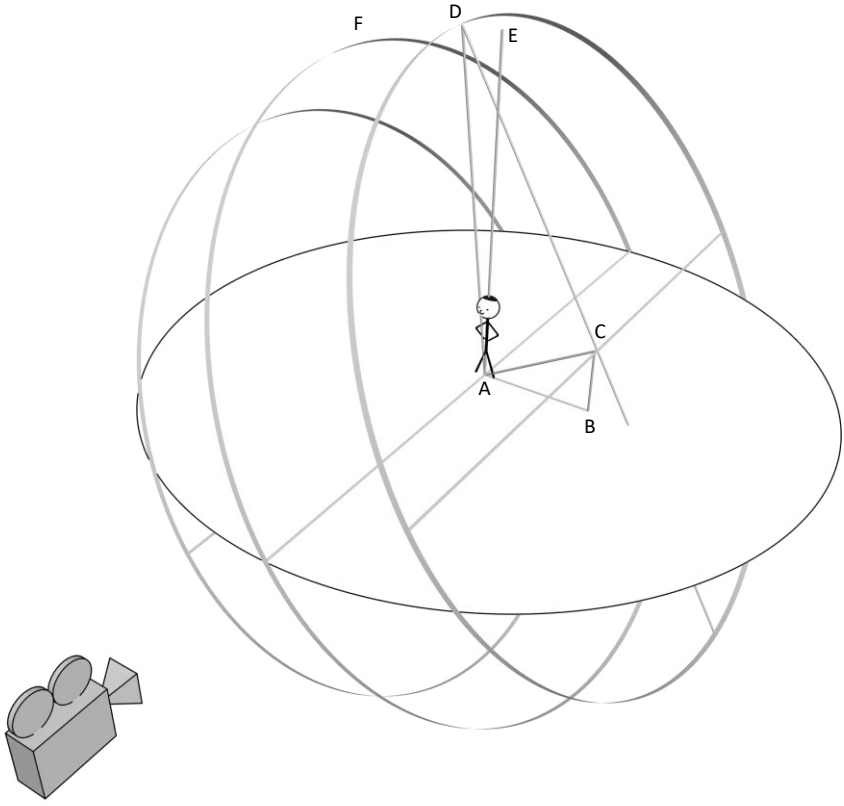
We'll lean the earth T° (23.4°) towards the sun, and we'll put our man at 30° latitude, at midday as far to the right as he can go.

The man is 30° above the equator but the equator has been lowered by 23.4 degrees, so the man's sum degrees is **6.6°** (= +30 - 23.4). And, as we can see in the diagram, the sun is leaning to the man's right, towards the south.

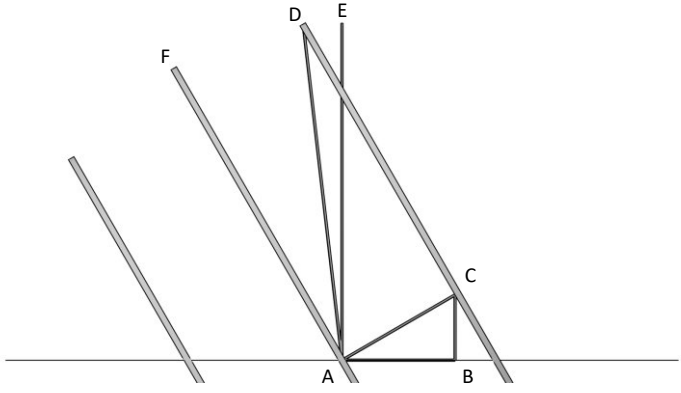


So, we can sum up the sun's position as being on the right side of the man, lowered down from straight up by **L° - T°**.

Now, let's go to the topocentric view.



Let's place the camera on the eastern side, looking towards the observer, but we'll take him out of the picture to avoid clutter.



A is the Observer

C is the center of the Summer Solstice ring.

B is a point directly below point C, on the ground.

D is the highest point of the Solstice ring, the point where the sun will be by midday.

E is a point directly above the observer.

F is the highest point of the Equinox ring (where the sun is at midday during the equinox).

BAC = L° the Solstice rings rotate south L° around a line drawn from due east to due west

EAF = L° same as BAC, just rotated 90° southward (counter-clockwise in picture)

DAF = T° as we showed on page 433

DAE = EAF - DAF

which is

$$DAE = \mathbf{L° - T°}$$

which is the same formula we came up with at the beginning of the previous page.

The truth is, we don't need the points B and C but we'll be using those points in later calculations, so we might as well get used to them now. (They might have different names then, don't worry :-))

That's midday.

Now let's show the sunrise and sunset points.

If you go to page 421, you will see we calculated the angle between sunrise and when the sun reaches the next quarter of the circle, angle DBE, as $\mathbf{\sin^{-1}(\tan(T) \cdot \tan(L))}$.

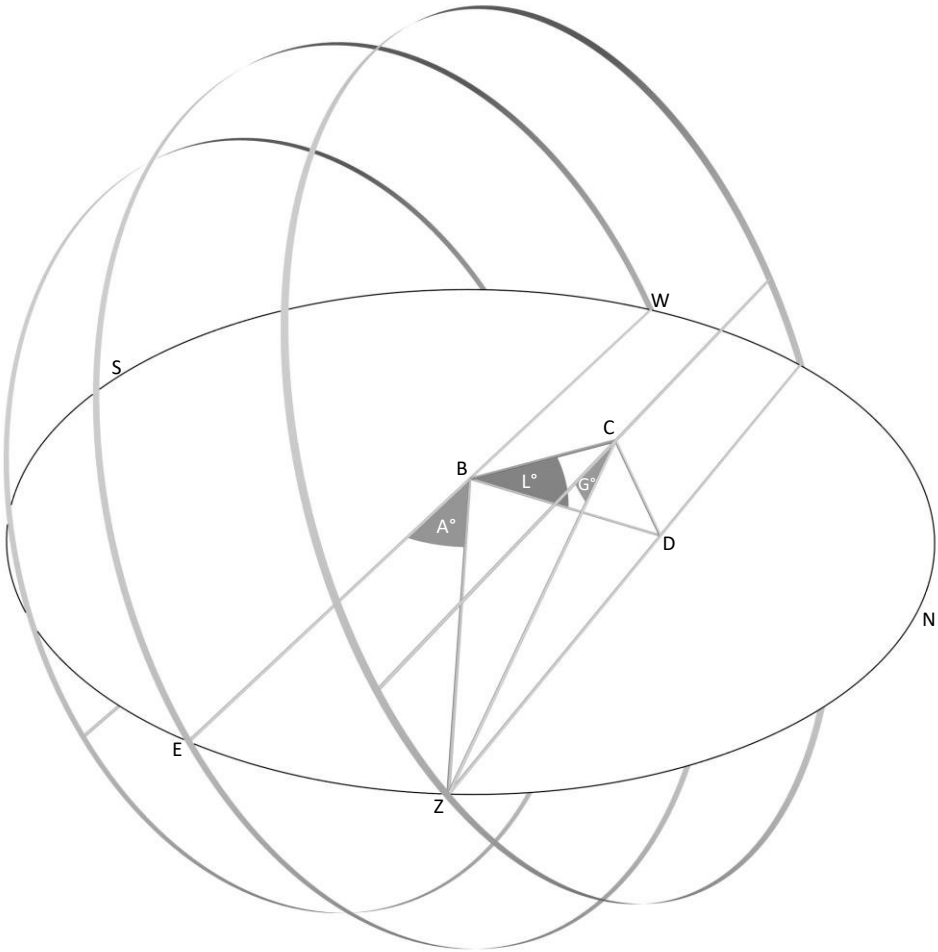
If you go ahead to page 441, you'll see we get the same formula from the topocentric view, showing the angle along the Solstice ring from sunrise until the upper half of that ring.

It's the same thing[349].

In the next appendix, we'll make all the calculations and then some, using Topocentricity.

[349] Although the manner in which we got to it was different. In Geocentric, the smaller ring size and distance is based on latitude, and its rotation is based on tilt. In Topocentric, the smaller ring size and distance is based on tilt, while its rotation is based on latitude.

Azimuth, Latitude, Day Angle, and Day Length Conversions



Some Constants:

$BZ = 1$	"Unity", the radius of the Celestial Sphere.
$T = 23.4°$	(For Summer Solstice. For other days in the year; See Appendix Q)
$BC = \sin(T)$	the distance of the Solstice ring, as we showed on page 433
$CZ = \cos(T)$	the radius of the Solstice ring, as we showed on page 433
$CBD = L$	same angle as BAC on page 436

Input **A**zimuth[109] (North of Due East), Return **L**atitude

$BD = \sin(A)$ since A = BZD, and sin(BZD) = BD / BZ, and BZ = 1 so sin(BZD) = BD

$L = \cos^{-1}\left(\frac{BC}{BD}\right)$ since cos(L) = BC / BD

$$\mathbf{L = \cos^{-1}\left(\frac{\sin(T)}{\sin(A)}\right)}$$

Input **L**atitude, return **A**zimuth (North of Due East)

$BD = \frac{BC}{\cos(L)}$ since cos(L) = BC / BD

$BZD = \sin^{-1}(BD)$ since sin(BZD) = BD / BZ, and BZ is 1 so sin(BZD) = BD

$A = BZD$ since opposite angles between parallel lines

$$\mathbf{A = \sin^{-1}\left(\frac{\sin(T)}{\cos(L)}\right)}$$

Input **L**atitude, Return **G** - Summer Solstice Day An**g**le (which gives Day Length)

On the equator, or during the Equinox, the sun rises (and sets) at the half-points of its daily path, where its path meets the Observer's ground or horizon. At other latitudes on other days of the year, the sun rises below or above these half-points, resulting in days that are longer or shorter than the average 12 hours, since the sun needs to travel through more or less angles than the average 180°. Once we know this angle, we'll turn that into day length, in hours.

We'll start by calculating the angle between sunrise and the half-point of the sun's path (this might result in a negative number). Then we'll double this angle to get both sunrise and sunset. Then we'll add 180° for the half-circle that is normally above the ground. Then we'll divide that total angle by 360° (to get the percentage of the whole day) and then multiply that by 24 hours, to see how much of those 24 hours are daytime.

$CD = BC \cdot \tan(L)$ since tan(L) = CD / BC

$CZD = \sin^{-1}\left(\frac{CD}{CZ}\right)$ since sin(CZD) = CD / CZ

$G = CZD$ since opposite angles between parallel lines

$G = \sin^{-1}\left(\frac{\sin(T) \cdot \tan(L)}{\cos(T)}\right)$, which can be simplified[350] to

$$\mathbf{G = \sin^{-1}(\tan(T) \cdot \tan(L))}$$

[350] Since sine ÷ cosine = tangent, as we showed on page 399

Then to turn G into day length, in hours:

Multiply G by 2	for sunrise and sunset
Add 180	for the half-circle usually above the ground/horizon
Divide result by 360	to get percentage of the whole circle
Multiply by 24	to get amount of hours out of a whole day

In short it looks like this:

$$\text{Day Length} = (G \cdot 2 + 180) \div 360 \cdot 24$$

Or simpler:

$$\textbf{Day Length} = \frac{2}{15}G + 12$$

Input Summer Solstice Day Angle - **G** (or day length), Return **L**

(Starting from length of day)

$$G = (\text{Length_of_Day_in_Hours} \div 24 \cdot 360 - 180) \div 2$$

this is the reverse of what we did above

or simpler

$$\mathbf{G = (Day\ Length - 12) \cdot 7.5}$$

Now, to turn that angle into latitude:

$CZD = G$	since opposite angles between parallel lines
$CD = \sin(CZD) \cdot CZ$	since sin(CZD) = CD / CZ
$CBD = \tan^{-1}\left(\frac{CD}{BC}\right)$	since tan(CBD) = CD / BC
$L = CBD$	as we showed on page 437 by BAC

$$L = \tan^{-1}\left(\frac{\sin(G) \cdot \cos(T)}{\sin(T)}\right)$$

Which is the same[336] as

$$L = \tan^{-1}\left(\frac{\sin(G) \div \sin(T)}{\cos(T)}\right)$$

Which simplifies to since sine / cosine = tangent, as we showed on page 399

$$\mathbf{L = \tan^{-1}\left(\frac{\sin(G)}{\tan(T)}\right)}$$

Input **A**zimuth, Return **G** - Summer Solstice Day An**g**le (which gives Day Length)

$BZD = A$ since opposite angles between parallel lines

$DZ = \cos(BZD)$ since cos(BZD) = DZ / BZ, and BZ = 1

$CZD = \cos^{-1}\left(\frac{DZ}{CZ}\right)$ since cos(CZD) = DZ / CZ

$G = CZD$ since opposite angles between parallel lines

$$\mathbf{G = \cos^{-1}\left(\frac{\cos(A)}{\cos(T)}\right)}$$

And again, to turn that into day length in hours, follow the formula from the top of the previous page: $\mathbf{Day\ Length = \frac{2}{15}G + 12}$

Input Summer Solstice Day An**g**le - **G** (or day length), Return **A**zimuth

(Turn Day Length into G like on previous page: $\mathbf{G = (Day_Length - 12) \cdot 7.5}$)

Now to turn G into azimuth:

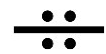
$CZD = G$ since opposite angles between parallel lines

$DZ = \cos(CZD) \cdot CZ$ since c(CZD) = DZ / CZ

$BZD = \cos^{-1}(DZ)$ since c(BZD) = DZ / BZ, and BZ = 1

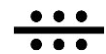
$A = BZD$ since opposite angles between parallel lines

$$\mathbf{A = \cos^{-1}(\cos(G) \cdot \cos(T))}$$



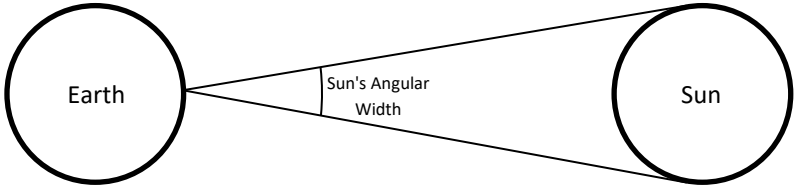
Turn / into

	Azimuth	Latitude	Day Angle
Azimuth	-	$A = \sin^{-1}\left(\frac{\sin(T)}{\cos(L)}\right)$	$A = \cos^{-1}(\cos(G) \cdot \cos(T))$
Latitude	$L = \cos^{-1}\left(\frac{\sin(T)}{\sin(A)}\right)$	-	$L = \tan^{-1}\left(\frac{\sin(G)}{\tan(T)}\right)$
Day Angle	$G = \cos^{-1}\left(\frac{\cos(A)}{\cos(T)}\right)$	$G = \sin^{-1}(\tan(T) \cdot \tan(L))$	-



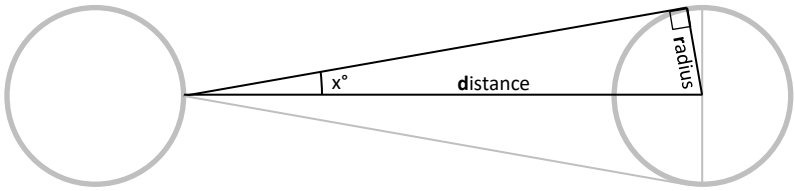
Sun Width and Refraction, Degrees Below Horizon

The sun takes up a certain amount of space when viewed from earth. This is what's called the sun's "angular size", i.e., how much space it takes up, in degrees, when measured from earth.



The Sun is (about, on average) 150,000,000 kilometers away. The Sun's diameter is (about) 1,400,000 kilometers across.[351]

How much angle space does the sun take up?



$r = .5 \cdot 1{,}400{,}000 = 700{,}000$

$d = 150{,}000{,}000$

$\sin(x) = \frac{r}{d}$

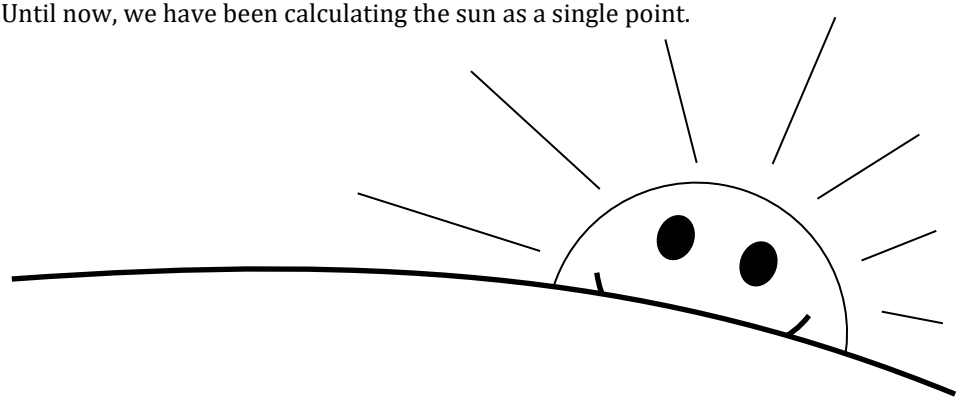
$x = \sin^{-1}\left(\frac{r}{d}\right)$

$x = \sim 0.267...$

Angular Width $= 2x =$ **~0.53°** doubling x, since x is only half the width

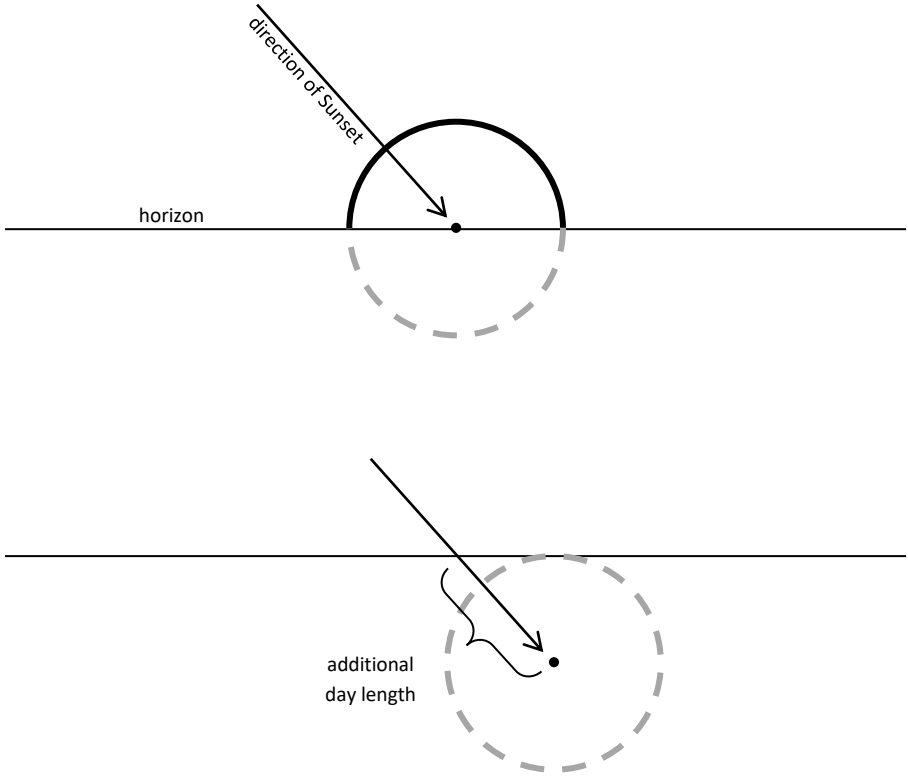
[351] Stellarium

Until now, we have been calculating the sun as a single point.



However, the definition[352] of Sunrise (and Sunset) is when the top of the sun is at the horizon.

That means, when calculating day length, you need to add in the additional time it takes for the top half of the sun to come up at sunrise, and go down at sunset.

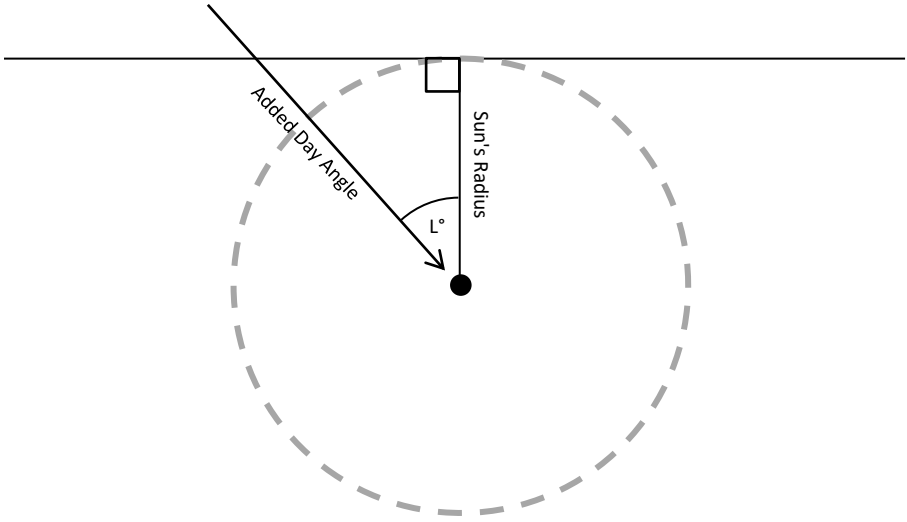


[352] Both halachically and scientifically - Dvar Yom page 185, 217, 218. Also see, https://en.wikipedia.org/wiki/Atmospheric_refraction

How long is this extra time?

We could do a quick approximation if we viewed the lines involved as straight edges of a triangle, and took the sunset angle to be directly proportional to the latitude (which it is, at the equinoxes).

The calculation would then be:



$$\text{Cos(L)} = \frac{\text{Sun's Radius}}{\text{Added Day Angle}}$$

so: $\text{Added Day Angle} = \frac{\text{Sun's Radius}}{\text{Cos(L)}}$

Rounding[353] the Sun's Radius to .25°, and using a Latitude of 32° would yield an added Sunset length of ~**0.29°**.

In time-terms that would come out to about **70 seconds**. = ~.29 / 360 * 24 * 60 * 60

Doubling that to include Sunrise as well comes out to about 140 seconds, or **2 minutes and 20 seconds** added daytime length.

[353] The exact size will change based on Earth's distance to the Sun, as well as refraction, and maybe other factors that I'm not aware of.

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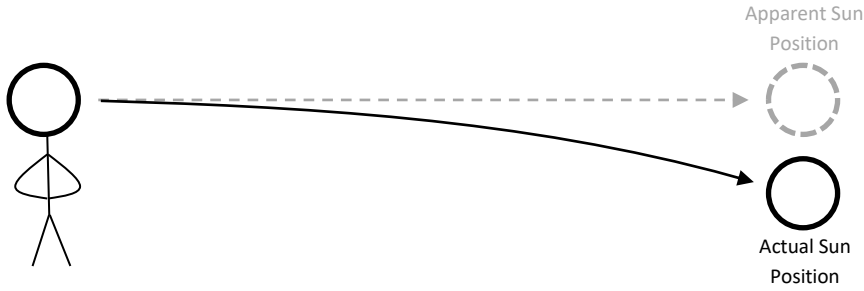
Another factor that affects the time of sunset is refraction.

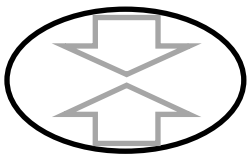
The air in the atmosphere bends the light coming from the sun, causing it to appear higher than it actually is. (Halacha is concerned only with where the sun appears, not where it actually is[354], ke'ein what we wrote on page 384 about the perception of man)

We need the sun to set far enough below the horizon so that it doesn't even *appear* over the horizon.



The exact amount of refraction varies by the temperature between the horizon and the observer, so it is hard to predict, but in general it causes the sun to appear about 1/2 of a degree higher than it actually is.

Refraction also causes the sun itself to appear squashed (so that its height is about 5/6 its width), so that makes the day length slightly shorter than had the sun been a perfect circle.



Combining the radius of the sun and the effects of refraction, we need to add to the sun's position a total of about **5/6 of a degree below the horizon** when calculating sunrise or sunset time.

So now, let's update the previous page's formula as:

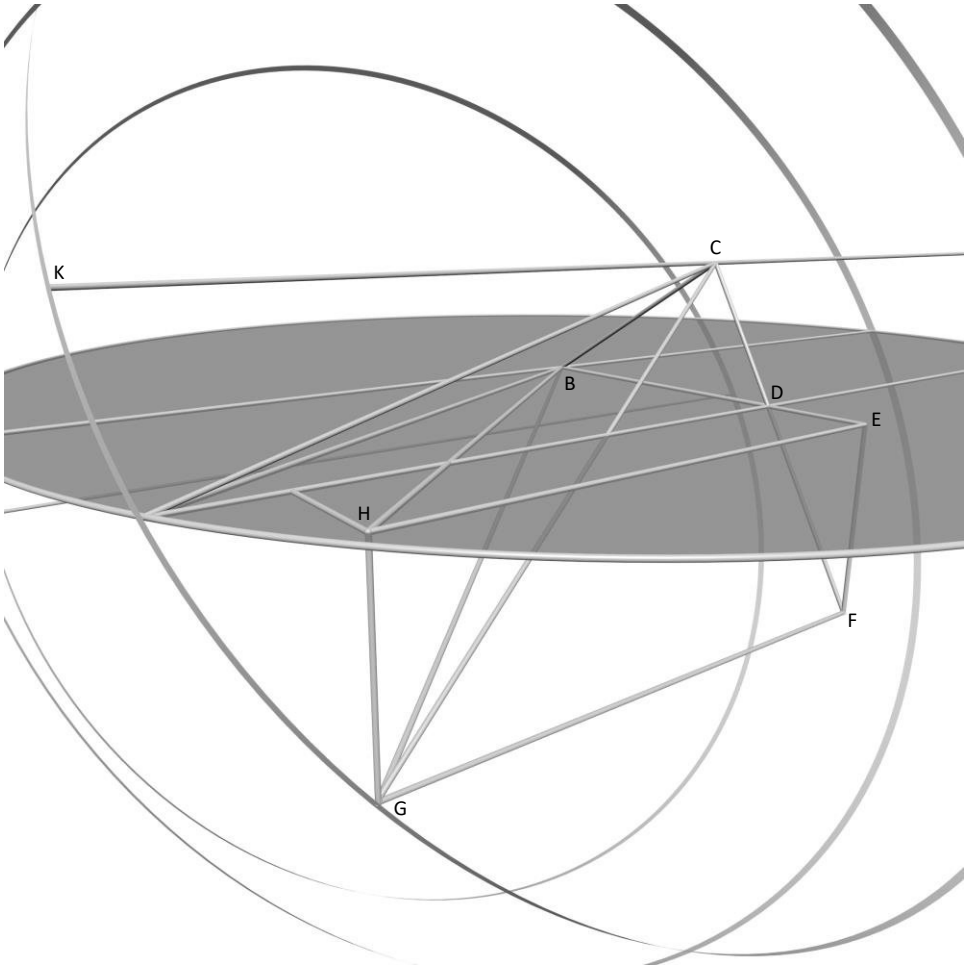
$$\text{Added Day Angle} = \frac{\frac{5}{6}}{\text{Cos(L)}}$$

$$=0.982...°$$

$$=3\text{m} \sim 56\text{s}$$ for each sunrise and sunset. Together that'd be **7m ~52s**

Let's now do a more exact calculation, since the triangle we calculated on the previous page was flat, but the actual triangle is bent on the surface of a sphere.

[354] Dvar Yom page 214



We'll be calculating the angle that is more than the half-way point of the whole day; that is angle GCK in the image. We'll use this angle to give us the total day length.

Angle GBH will be the degrees below the horizon relative to the observer.

B is the observer

C is the center of the Summer Solstice ring

K is the half-way point along the Summer Solstice ring

T is the earth's tilt, which we'll take as 23.4° For other days in the year, see Appendix Q

L is the observer's latitude

GH is a vertical line from the sun-below-the-horizon straight up to the plane of the observer

$BC = \sin(T)$ the distance of the solstice ring, as we showed on page 433

$CD = \tan(CBD) \cdot BC$ since tan(CBD) = CD / BC

$GH = \sin(GBH)$ since sin(GBH) = GH / BG, and BG = 1, the radius of the sphere

$EF = GH$ since opposite ends of a rectangle

$CBD = L$ same angle as BAC on page 436

$DFE = CBD$ these angles are reflections of each other, at point "d"

$DF = \frac{EF}{\cos(DFE)}$ since cos(DFE) = EF / DF

$CF = CD + DF$ the larger line made up of two smaller segments

$CG = \cos(T)$ the radius of the solstice ring, as we showed on page 433

$CGF = \sin^{-1}\left(\frac{CF}{CG}\right)$ since sin(CGF) = CF / CG

$GCK = CGF$ since opposite angles between parallel lines

$GCK = \sin^{-1}\left(\frac{CF}{CG}\right)$

$$\mathbf{GCK = \sin^{-1}\left(\frac{\tan(L) \cdot \sin(T) + \frac{\sin(GBH)}{\cos(L)}}{\cos(T)}\right)}$$

Taking:

$L = 31.778$ (the latitude of the makom hamikdash, according to Google Maps)

$GBH = \frac{5}{6}$

yields: $GCK = 16.661...$

Turn into time: $(GCK \cdot 2 + 180) \div 360 \cdot 24 = 14.221567...$ hours

= 14 hours, 13 minutes, ~18 seconds

Itim Libinah 5784 says (if Jerusalem was flat) June 21, 2024 sunrise = 5:34:15, sunset = 7:47:30, which is **14h 13m 15s** (they round off sunrise and sunset to nearest 15 seconds, so we'd expect an error of ±15 seconds (±7.5 each). Also, even

on the longest day, the sun begins to move back south (see page 167). According to our calculations here and on page 459, one day away from the equinox will make a difference of about 2 seconds.

Using the original simpler formula we did on page 439 (Input Latitude, Return Summer Solstice Day Angle) without sun's width and refraction would've yielded a longest day of 14h 4m ~24s, which is almost 9 minutes shorter than after taking into account the sun's width and refraction.

If you wanted to do this all in reverse, that is, for a given latitude, tilt (at the longest day of the year = 23.4, or for some other day of the year, see Appendix Q), and angle after the halfway point (point K, that is angle GCK), tell me how much the center of the sun is below (or above, if result is in the negative) the horizon relative to the observer (angle GBH), we could do:

$CGF = GCK$	since opposite angles between parallel lines
$CG = \cos(T)$	radius of Solstice, as we showed on page 433
$CF = \sin(CGF) \cdot CG$	since sin(CGF) = CF / CG
$CBD = L$	same angle as BAC on page 436
$BC = \sin(T)$	the distance of the Solstice ring, as we showed on page 433
$CD = \tan(CBD) \cdot BC$	since tan(CBD) = CD / BC
$DF = CF - CD$	the smaller segment of the larger line
$DFE = CBD$	these angles are reflections of each other, at point "d"
$EF = \cos(DFE) \cdot DF$	since cos(DFE) = EF / DF
$GH = EF$	two ends of a rectangle
$GBH = \sin^{-1}(GH)$	since sin(GBH) = GH / BG, and BG = 1

$$\mathbf{GBH = \sin^{-1}(\cos(L) \cdot (\sin(GCK) \cdot \cos(T) - \tan(L) \cdot \sin(T)))}$$

Amazing!

Once we have these methods, we can use it for other "degrees below horizon", like Alos.

0

Elevation

As was pointed out in Itim Libinah[355], elevation can only affect length of day if the elevation of your horizon is different than your elevation. The reason is because we are making all the calculations as though the earth has a radius of "1". 1 what? 1 radius. Which assumes the same distance from the center of the earth to every point on its surface.

If there's a difference between your elevation and your horizon's then we will see a difference in sunrise and sunset time.

In the Itim Libinah calendar, for Yerushalayim, there are two times listed for sunset:

1. In the main calendar page, it lists the time of sunset "as seen" in Yerushalayim, meaning, since Yerushalayim has an elevation of about 800 meters[356], and its western horizon is at sea level[357], you would see sunset a bit later (since you're higher and can see the sun even after it dips lower) than if Yerushalayim was lower.

2. On the other side of the page of the main weekly calendar listings, there is a more detailed chart, including sunset "as if Yerushalayim was flat", meaning, as if it had no elevation; was at sea-level.

For sunrise, there is no difference between these 2 types of calculating since the eastern horizon of Yerushalayim are the hills of Moav in Jordan, which are (about) the same elevation as Yerushalayim.

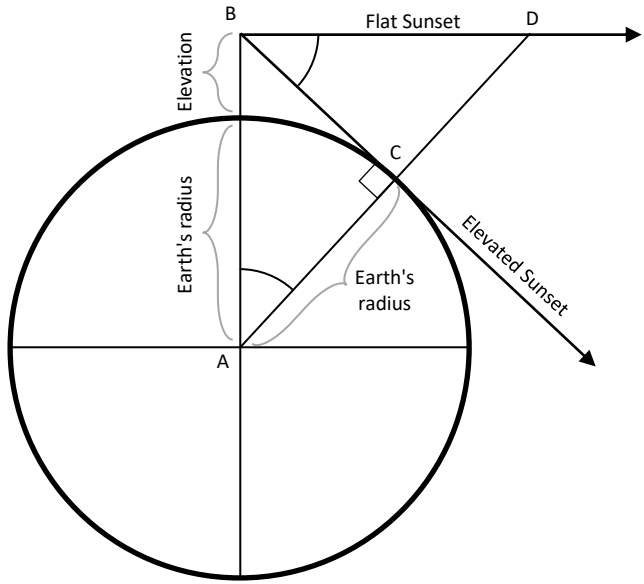
Let's now make the calculations for elevation.

---

[355] in the extended kuntros they print in the back of their non-leap year calendars, chapter 8, Calculating Hours of the Day From Sea-level Sunrise To Sunset, חשבון שעות היום מהנץ לשקיעה המישוריים

[356] As they write in the kuntros, that they use 800 m for Yerushalayim

[357] Itim Libinah calls the western horizon of Yerushalayim as "open". I guess that there must be some places in Yerushalayim where you can see the western lowlands. Dvar Yom says so on page 204 in footnote 68, at least for some days of the year.
But even if you can't see the sun setting, he brings a whole debate about whether you erase mountains that are close to you, like the Kastel Mountains that he brings on page 200 footnote 53.
I know when I was in Neveh Yeshiva in Telstone, a bit west of Yerushalayim (and west of Kastel), I used to watch from the window the airplanes taking off and landing in Tel Aviv, which is basically at sea-level.



Consider the above diagram.

Imagine someone standing at point "B". His distance to the center of the earth is earth's radius + the elevation (= line AB).

AB = Earth's Radius + Elevation

AC = Earth's Radius

The angle that the sun has to set extra below the "normal" horizon (the line labeled "Flat Sunset", line BD) is angle CBD.

CBD = BAC since ABC = 90 - BAC, and CBD = 90 - ABC, so CBD = 90 - (90 - BAC), which is the same as 90 - 90 + BAC, and the 90s cancel each other out

$BAC = \cos^{-1}\left(\frac{AC}{AB}\right)$

$$\mathbf{CBD = \cos^{-1}\left(\frac{Earth's\ Radius}{Earth's\ Radius + Elevation}\right)}$$

Let's take the elevation of Yerushalayim as 800 meters (.8 kilometers)[358].

[358] That's the number Itim Libinah uses[355].

Circumference of Earth from Equator over the North Pole = 40,008 kilometers[359]

Circumference of Earth Around Equator = 40,075 kilometers[360]

Radius of Earth = Earth's Circumference ÷ π ÷ 2

(Using[361] Equator Circumference):

Radius of Earth = 6378.134... Kilometers

Plugging those numbers into our formula from the previous page yields:

CBD = ~0.907...°

Adding that to GBH from our formula on page 447 (since we need this additional number of degrees below the "horizon") and crunching that formula anew yields longest day in Yerushalayim with elevation:

Extra Day Angle for Sunrise = 16.661...°	no difference between altitude and horizon
Extra Day Angle for Sunset = 17.879...°	including elevation
+ 180	for half of circle
÷ 360	to get the "part"[362] of the circle
· 24 =	to get hours

14h 18m ~10s

(Equator to North Pole radius: 14h 18m 10.14...s = same)

---

[359] The meter was originally meant to be 1 ten millionth the distance from the north pole to the equator, which would've made the entire earth's circumference 40,000 kilometers. I think they updated the definition of a meter at some point (shrinking it a bit), to match some more objective standard, so that now the circumference measured from the equator over the North Pole is 40,008.

[360] Since the Earth bulges in the middle, due to some dieting issue or something.

[361] Using equator-to-north-pole radius: CBD = 0.908...°. This will make a difference of a fraction of a second in the end of the whole calculation. And I think that measuring the circumference any other way will just lead to a result in between these two. I think. Anyway, the distance from the observer to the horizon is over a distance of the Earth's curve that can be measured with angle BAC ÷ 360 (to get the "part" of the circle) times Earth's Circumference which ≃ 100 km, so I think the exact circumference is negligible.

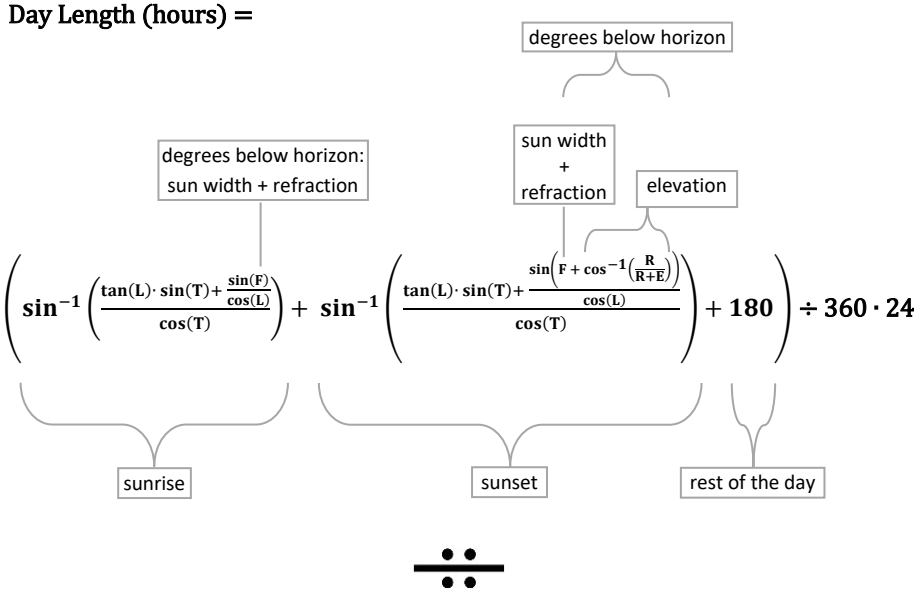
[362] The actual term for this is "turn", which is another system of measuring angles. We've been using degrees, which divides a circle into 360 divisions. Another popular system is Radians, which divides a circle into 2π divisions (So that 2π = a whole circle). The Turn system divides a circle into 1 division, meaning 1 = the whole circle, 1/2 = 180°, 1/4 = 90°, etc.

Just for fun, let's write out the entire formula in expanded form:

L = Latitude = 31.778°
T = Tilt = 23.4°
F = Combined Angle of Sun Radius + Refraction Effect = 5/6°
E = Elevation = .8 (km)
R = Radius of Earth = 40,075 ÷ π ÷ 2 (km)

Day Length = (Sunrise_Added_Angle + Sunset_Added_Angle + 180) ÷ 360 · 24

Day Length (hours) =



In Python:

```
from math import sin, cos, tan, asin, acos, atan, pi, radians, degrees

def day(latitude=31.778,tilt=23.4,radius=40075/pi/2,elevation=.8,refraction=5/6):
    L = radians(latitude)
    T = radians(tilt)
    R = radius
    E = elevation
    F = radians(refraction)

    sunrise = degrees(asin((tan(L)*sin(T) + sin(F) / cos(L)) / cos(T)))
    sunset = degrees(asin((tan(L)*sin(T) + sin(F + acos(R/(R+E)))/cos(L))/cos(T)))

    length = (sunrise + sunset + 180) / 360 * 24

    print(length)
    h = int(length)
    m = (length - h) * 60
    s = (m - int(m)) * 60
    print(h, int(m), s)
```


P

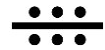
Let's compare our results to other calendars.

We can expect the numbers to be off if we take into consideration rounding to nearest (or more machmir) unit of time, different values of Earth's tilt, latitude of Jerusalem (Makom Hamikdash, Southern, or Northern tip of Jerusalem), whether or not we take into account nearby or distant obstructions, differing refraction rates (yearly average vs. day-specific average, global average vs. location specific), and probably other factors that I'm not aware of.

But really we'd be going beyond the scope of what we need to cover for the purposes of these appendixes.

Longest day of the Year, according to different calendars:

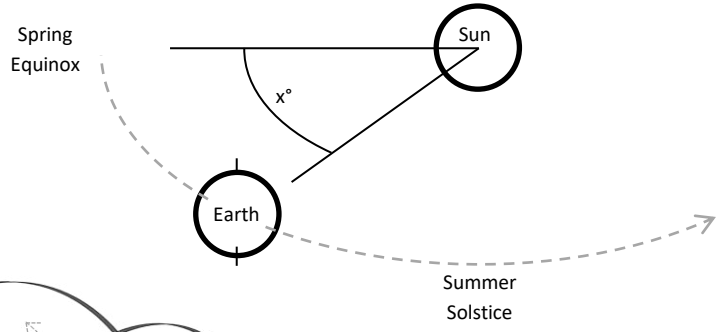
Calendar	Length of Longest Day (Yerushalayim, June ~21)	Notes
	no elevation	
My Cheshbon	14h 13m 18s	
Itim Libinah	14h 13m 15s	Within expected range of ±15s
MyZmanim	14h 12m 59s	Using Southernmost Coordinates for Jerusalem of 31.736 gets my cheshbon within 6 seconds of 14h 12m 59s
Kaluach	14h 13m 35s	Using Northernmost Coordinates for Jerusalem of 31.8449 in my cheshbon yields 14h 13m 38s
Stellarium	14h 13m 32s	Gives Elevation as 756m but shows no difference if Jerusalem was flat; See previous note
	with elevation	
My Cheshbon	14h 18m 10s (800 meters)	
Itim Libinah	14h 18m 30s (800 meters)	Expected Rounding Error ±15s; raising tilt to 23.42, or latitude to 31.8449 in my cheshbon yields within ±15s of 14h 18m 30s
MyZmanim	14h 17m 56s (650 meters)	At 650m my cheshbon yields 14h 17m 41s; Raising Tilt to 23.44 yields 14h 17m 56s; Raising Latitude to 31.8449 yields more than 14h 18m 2s

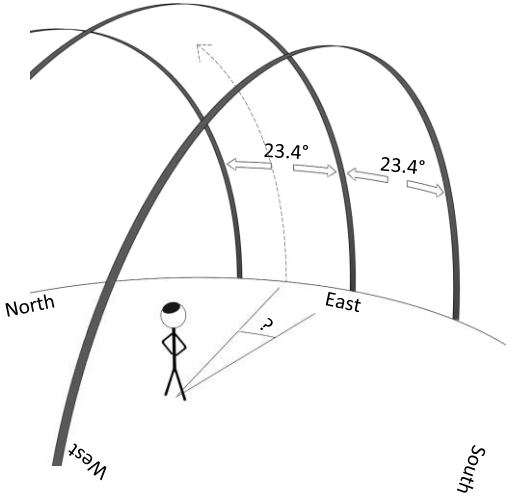


Other Days of the Year

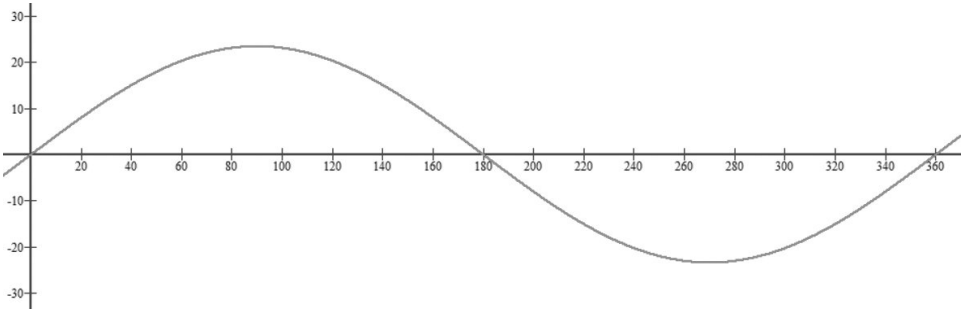
Initially, I thought that the Sun's path topocentrically, from Spring equinox to summer solstice, to winter solstice, and back to equinox, was sining its way through the year.

That is, I thought you take the number of degrees earth has gone past the Spring Equinox, take the sine of that and multiply that by T (tilt) and you get the angular distance between the topocentric equinox ring and the sun's path ring for that particular day.





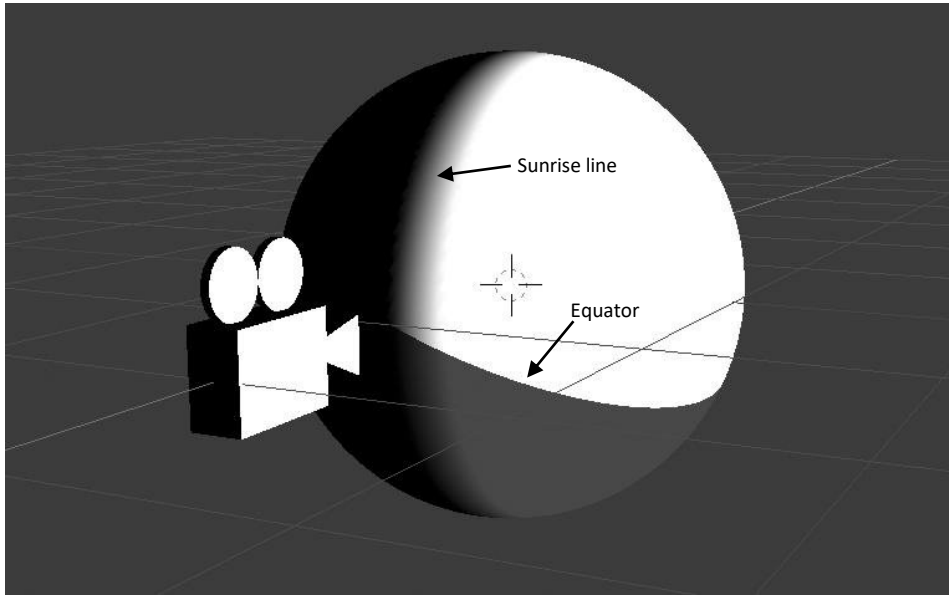
$$? = \sin(x) \cdot 23.4$$



But I discovered that it is incorrect; that this is just an approximation.

The first experiment I did was to take a camera in the 3D program Blender, and try to position it to be directly over the equator at sunrise throughout the year.

The way I did this was I made a sphere to represent the earth. I colored the bottom half of it to represent south of the equator. I rotated the earth 23.4° towards a spotlight that represented the sun. (This then would be the moment of the Solstice). Then I positioned a camera to look at the exact spot where the equator and the "sunrise" line meet.

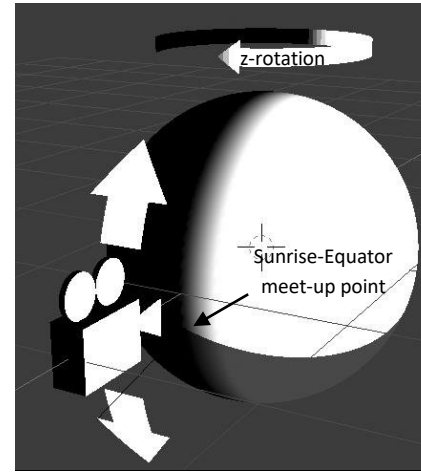


Next, to show the changes in earth's tilt relative to the sun, I rotated the earth along the z-axis (along a line running straight up and down) clockwise when looking down at the earth.

Notice how the point where sunrise and the equator meet has gone down somewhat.

What I want to do is to raise and lower the camera (pivoting around the center of the earth, where the plus-shaped cursor is) along the sunrise line, basing the rotation amount on some formula controlled by the earth's z-rotation.

We'll know if we put in the right formula if the camera stays directly over the sunrise-equator line.



So first of all, this rotation itself I also assumed follows the formula we suggested on page 454 (that is, sin(x) · T, just in the negative) but the results returned a rather wobbly inaccuracy. Raising the value of T made the inaccuracy more obvious.

So then I thought, let's break down this calculation to its bare bones.

(The picture is actually upside down but it's much easier to follow that way)

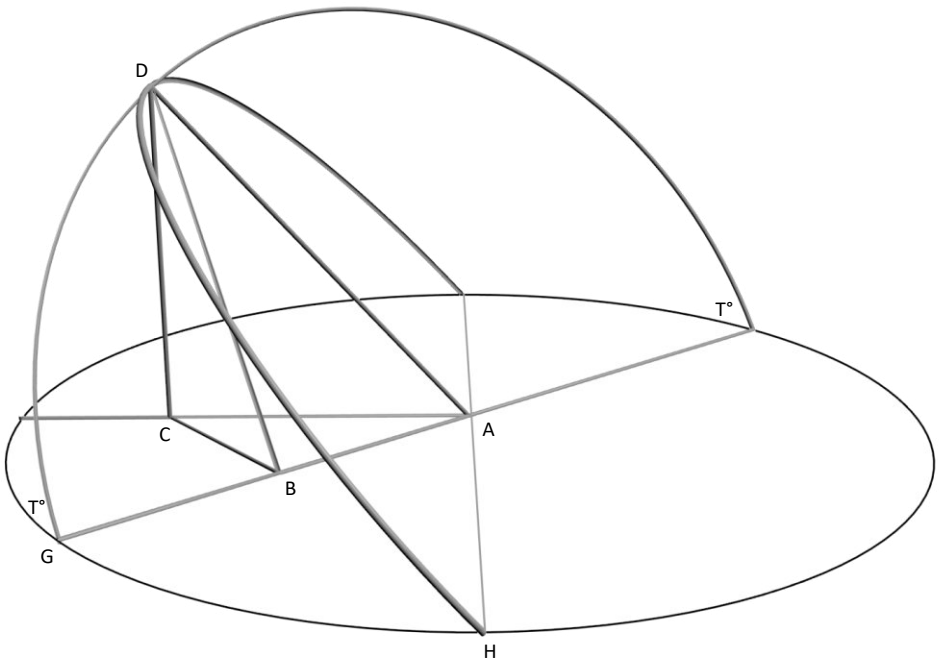
Let's make a flat circle representing a cross section of earth aligned with its orbit (the Ecliptic). We'll label the center of this circle "A".

Now let's copy that circle and rotate it T° along one of its diameters to represent the Equator. (The ring in the diagram along the letters "D" and "G")

We're going to make an angle sweeping around point "A", starting by pointing at "G", and moving clockwise until point "C" to represent earth's tilt "rotating" throughout the year, as the earth revolves around the sun. Angle CAG or BAC is synonymous with angle x° on page 454, except we're starting from the moment of Summer Solstice.

Point "D" is where we'd like our camera to be, directly over the point where the sunrise line and the equator meet.

We need to find angle CAD in order to know where to place the camera.



Input Degrees-After-Summer-Solstice, Return Angular-Distance-Between-Ecliptic-And-Sunrise/Equator-Meeting-Point

Or, using the letter labels in the diagram: Input **BAC**, Return **CAD**

$AD = 1$

$CBD = T\ (= \text{tilt})$ the same angle as the equator ring

$AB = c(BAD)$ since c(BAD) = AB / AD, and AD = 1

$BD = s(BAD)$ since s(BAD) = BD / AD, and AD = 1

$CD = s(CBD) \cdot BD$ since s(CBD) = CD / BD

$BC = c(CBD) \cdot BD$ since c(CBD) = BC / BD

$x = BAC = t^{-1}\left(\frac{BC}{AB}\right)$ since t(BAC) = BC / AB

which expands to:

$BAC = t^{-1}\left(\frac{c(T) \cdot s(BAD)}{c(BAD)}\right)$ which simplifies to:

$BAC = t^{-1}(c(T) \cdot t(BAD))$ since[345] dividing sin by cos = tan

Inverting that becomes:

$t(BAC) = c(T) \cdot t(BAD)$ which can be rearranged to:

$\frac{t(BAC)}{c(T)} = t(BAD)$ which can be inverted to:

$t^{-1}\left(\frac{t(BAC)}{c(T)}\right) = BAD$ or:

$BAD = t^{-1}\left(\frac{t(BAC)}{c(T)}\right)$

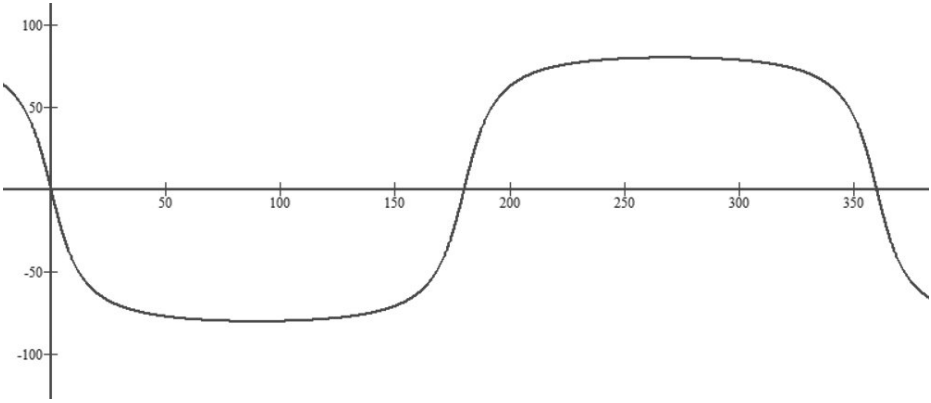
$CAD = s^{-1}(CD)$ since sin(CAD) = CD / AD, and AD = 1, which expands to:

$CAD = s^{-1}(s(T) \cdot s(BAD))$ which expands to:

$$\mathbf{CAD = s^{-1}\left(s(T) \cdot s\left(t^{-1}\left(\frac{t(BAC)}{c(T)}\right)\right)\right)}$$

Now using this formula in Blender, the camera stays exactly[363] over the sunrise/equator point.

When T = 23.4, it looks like an upside down sine wave but when you raise up the value of T, let's say up to 80°, it looks more like a square wave (castle ramparts):



The closer T is to 90°, the more square this wave will appear.

Next.

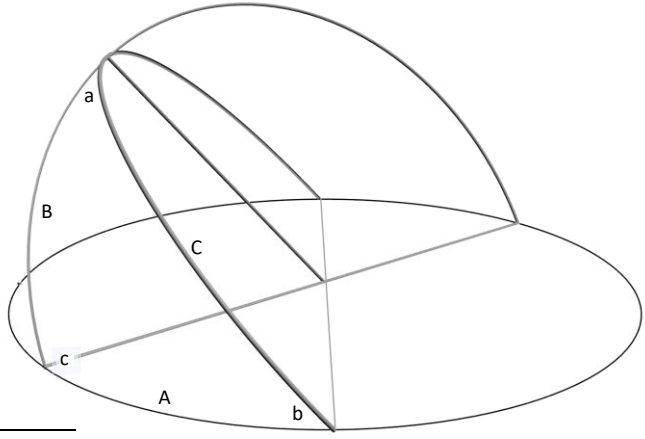
Now that we've got the camera over the right spot, we need to rotate it along its local z-axis (along a line coming straight forward out of the camera) so that the equator remains perfectly straight in the camera's sight. This rotation then would equal the sun's azimuth[109] as it rises for an observer at the equator, and this would give us the value of ? that we wanted to know from page 454.

For this, we have to use the "rule of sines" from spherical trigonometry which goes:

$$\frac{s(a)}{s(A)} = \frac{s(b)}{s(B)} = \frac{s(c)}{s(C)}$$

I don't know how it works so I can't explain it, but it will give us what we need.

This only works with "Great Circles", i.e., circles that have a radius of 1 along the edge of the "sphere".



[363] Although, when x is between 90° and 270°, the result is flipped in the negative, due to the fact that arcsines and arctangents only work within a certain range. To fix that you can multiply the result by abs(cos(x)) / -cos(x)

Q

Notice that we're labeling this diagram a little different than what we've been doing before.

Here, we're not labeling the points; we're labeling the angles.

The lower case letters are the inside angles of a spherical triangle "wrapped" along the outer edge of the sphere.

The upper case letters are the lengths of the sides of the triangle, measured in degrees viewed from the center of the sphere.

So now, filling in the rule of sines from the points of the diagram from page 456, we get:

$$\frac{s(GDH)}{s(GAH)} = \frac{s(DHG)}{s(DAG)}$$

which can be rearranged to:

$$GDH = s^{-1}\left(s(GAH)\frac{s(DHG)}{s(DAG)}\right)$$

and since GAH = 90 - x, you can change s(GAH) to c(x):

$$GDH = s^{-1}\left(c(x)\frac{s(DHG)}{s(DAG)}\right)$$

$$DHG = CAD$$

$$DAG = BAD$$

So, altogether that's:

$$GDH = s^{-1}\left(c(x)\frac{s\left(s^{-1}\left(s(T)\cdot s\left(t^{-1}\left(\frac{t(x)}{c(T)}\right)\right)\right)\right)}{s\left(t^{-1}\left(\frac{t(x)}{c(T)}\right)\right)}\right)$$

Which can be slightly simplified to: since sine and arcsine cancel each other out

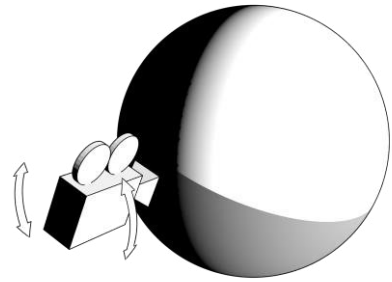
$$GDH = s^{-1}\left(c(x)\frac{s(T)\cdot s\left(t^{-1}\left(\frac{t(x)}{c(T)}\right)\right)}{s\left(t^{-1}\left(\frac{t(x)}{c(T)}\right)\right)}\right)$$

Which can be further simplified to:

$$\mathbf{GDH = s^{-1}\left(c(x)\cdot s(T)\right)}$$ since the denominator cancels out same in numerator

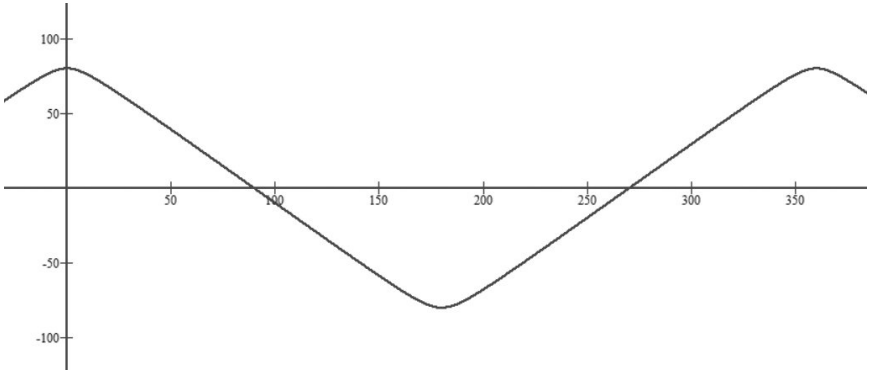
(The amount this simplifies makes me wonder if I perhaps overcomplicated the whole thing, again)

This is the rotation of the camera, which is identical to the observer's facing direction (east) relative to the sun at sunrise. See related diagrams on page 430.

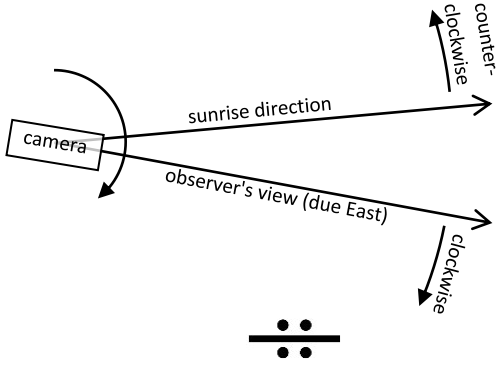


When tilt is 23.4, this last formula we derived on the previous page will look basically like a regular cosine wave on a graph.

When tilt is raised to a high number, like let's say 80°, then the wave begins to look like a triangle wave. The closer T is to 90°, the sharper the wave will be.



Now, keep in mind that when going from winter to summer we're rotating our camera clockwise (bird-eye view) to keep it aligned with the observer, but from his perspective the sun rising position is rotating counter-clockwise (along the ground), and from summer to winter it'll be in reverse.



So now, taking any of our original formulas for day length, azimuth, etc., replacing "T" with the above formula will give us other days in the year. This will replace the original Solstice Ring with the sun path for a given degree after the Summer Solstice. This path is the grey dashed arrow in the middle diagram on page 454 (with the topocentric view).

Let's take for example the last formula on page 439:

$$\text{(Day Angle) } G = \sin^{-1}(\tan(T) \cdot \tan(L))$$

And let's incorporate that into the first formula on page 440 (filling in "G"):

$$\text{Day Length} = \frac{2}{15} \cdot \sin^{-1}(\tan(T) \cdot \tan(L)) + 12$$

And now let's incorporate into this formula the last formula we made on page 459 (GDH) for degrees_after_equinox (x), using 40° as an example, replacing T in the above formula with $\mathbf{s^{-1}(c(x) \cdot s(T))}$: that formula filled in looks like: $s^{-1}(c(40) \cdot s(23.4))$

$$\mathbf{Day\ Length = \frac{2}{15} \cdot \sin^{-1}\left(\tan\left(\sin^{-1}(\cos(40) \cdot \sin(23.4))\right) \cdot \tan(31.778)\right) + 12}$$

$$\mathbf{Day\ Length = 13h\ 31m\ \sim 17s}$$

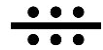
We've done this entire cheshbon on the equator but it won't change for other latitudes, just like the 3 topocentric rings don't change relative to each other regardless of latitude, as we showed starting from page 435.

I mentioned above that degrees can be roughly translated into days. The basic method would be:

$$\text{Days} = \frac{\text{degrees}}{360} \cdot \text{days_in_a_year}$$

but we'll see in the next appendix that it's not so simple.

Stay tuned.



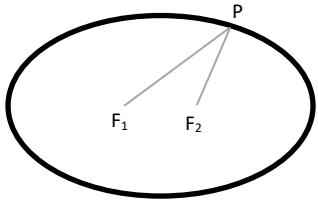
Varying Season Lengths

The shape of Earth's orbit around the sun is an ellipse (an oval). An ellipse is defined as a curve around two points called the foci (the focuses), such that the distance from one focus to any particular point plus the distance from the other focus to that same point, adds up to the same sum, for every point along the curve.

The way this is usually demonstrated is by tacking a loose string at both ends to a writing surface and then pulling the string taut to the side with a pencil point, and then dragging the pencil around the tacked ends, so that the pencil draws an ellipse.

P = any point along the ellipse

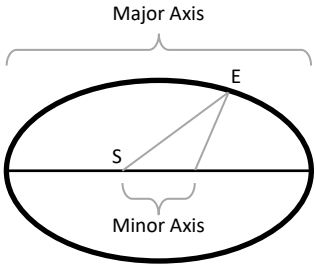
Some Value = F_1 to P + F_2 to P



Kepler's laws of Planetary Motion state that earth moves in an elliptical orbit, with the sun[364] at one of the foci, and nothing at the other. The speed of the earth as it moves along this orbit changes depending on its distance to the sun. The closer, the faster; the farther, the slower. The exact mathematical control is that a line from the sun to the earth sweeps an equal amount of area per a given amount of time[365].

The fact that the gravitational pull controlling earth (i.e., the sun) is not at the center of earth's orbit makes this orbit eccentric, which means off-centered.

The amount of eccentricity is measured by dividing the length of the minor axis[366] of the orbit by the length of its major axis[367].



[364] Not exactly. More correct would be the barycenter of the sun-earth system, that is, the point between the sun's center and earth's center that is gravitationally balanced between the two. This point is somewhere within the sun's body, but is not at its center. This level of exactness is not relevant to the purpose of this appendix.

[365] This is all based on the strength of gravity which is cut down by 1/4 for every double distance; termed "inverse squared", which we mentioned on page 9; see the graph there.

[366] The distance between the two foci

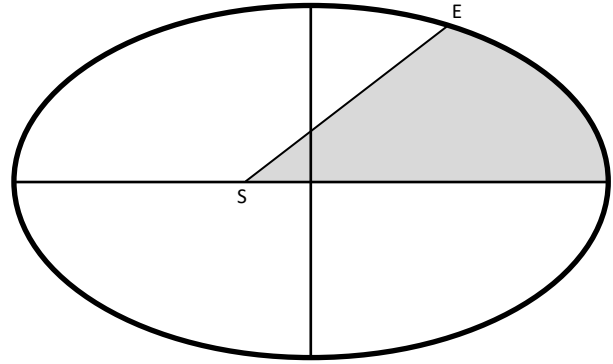
[367] The longest diameter of the ellipse

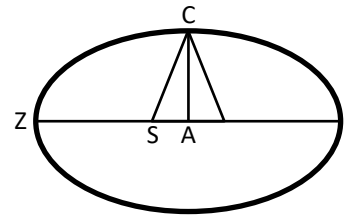
Wikipedia[368] gives earth's eccentricity as = .0167086

Let us calculate for a given amount of degrees the earth goes around the sun how long that would take.

First, let's turn degrees into area:

Input angle ESZ, Return area ESZ





Minor Axis = 2AS

Major Axis = 2AZ

Taking AZ as 1:

$$\text{Eccentricity} = .0167086 = \frac{\text{Minor Axis}}{\text{Major Axis}} = \frac{2AS}{2AZ} = \frac{AS}{AZ} = = \frac{.0167086}{1}$$

$AS = \text{Eccentricity} = .0167086$ since eccentricity = AS / 1

$AZ = 1$ unity

$2CS = 2AS + 2SZ$ in other words, point C is just as far from the two foci together, as is (either) point Z is from the two foci together

$CS = AS + SZ$ divide each side of the equal sign in half

$AZ = AS + SZ$ since that line is made up of those lines

$CS = AZ$ since they are both the sum of AS + SZ

$AC = \sqrt{CS^2 - AS^2}$ a rearrangement of the Pythagorean Theorem

[368] https://en.wikipedia.org/wiki/Earth%27s_orbit

Now, we need to extract an "Inflation rate", that is, the amount we can multiply the short side of this oval (line AC) so that it equals the long side of the oval (line AZ), to turn this oval into a perfect circle, which will help us continue with the calculations. After we calculate some vertical lines in the circle, we will revert them to the oval by shrinking them with the inflation rate.

But doing it this way will be unnecessarily clumsy since it will give us an inflation rate of 1 over something. Instead of doing it that way, let's do it in a way that we can do something over 1, and then we can just erase the over 1, so that we're just left with the something. So for that we need the reciprocal, that is, the amount we can divide the short side by so that it equals the long side. The way to find this is to first divide the short side by the long side.

We want an inflation rate so that:

$$\text{Short_side} \div \text{Inflation_rate} = \text{Long_side}$$

To find that, we rearrange above to:

$$\text{Inflation_rate} = \text{Short_side} \div \text{Long_side}$$

$$\text{Inflation rate} = \text{AC} / \text{AZ}$$

$$\text{I} = \text{Inflation rate}$$

$$\text{I} = \text{AC} \qquad \text{since AZ = 1}$$

(from here on, refer to the diagrams on the next page)

$$\text{AG} = \tan(\text{ESZ}) \cdot \text{AS} \qquad \text{since tan(ESZ) = AG / AS}$$

$$\text{AF} = 1 \qquad \text{the radius of the circle that was inflated from the oval}$$

$$\text{AH} = \text{AG} \div \text{I} \qquad \text{meaning, what would AG be if it was stretched like the circle?}$$

$$\text{ASH} = \tan^{-1}\left(\frac{\text{AH}}{\text{AS}}\right) \qquad \text{since tan(ASH) = AH / AS}$$

$$\text{JS} = \cos(\text{ASH}) \cdot \text{AS} \qquad \text{since ASH is same as ASJ, and cos(ASJ) = JS / AS}$$

$$\text{AJ} = \sin(\text{ASH}) \cdot \text{AS} \qquad \text{since ASH is same as ASJ, and sin(ASJ) = AJ / AS}$$

$$\text{FJ} = \sqrt{\text{AF}^2 - \text{AJ}^2} \qquad \text{a rearrangement of the Pythagorean Theorem}$$

$$\text{FS} = \text{JS} + \text{FJ} \qquad \text{since that line is made up of those two lines}$$

$$\text{BF} = \sin(\text{ASH}) \cdot \text{FS} \qquad \text{since ASH is same as BSF, and sin(BSF) = BF / FS}$$

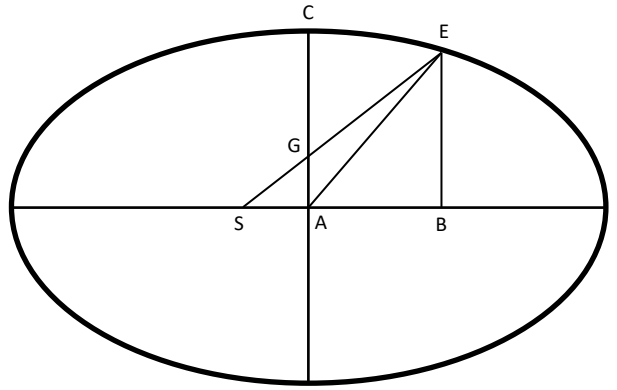
$\text{BAF} = \sin^{-1}\left(\frac{\text{BF}}{\text{AF}}\right)$ since sin(BAF) = BF / AF

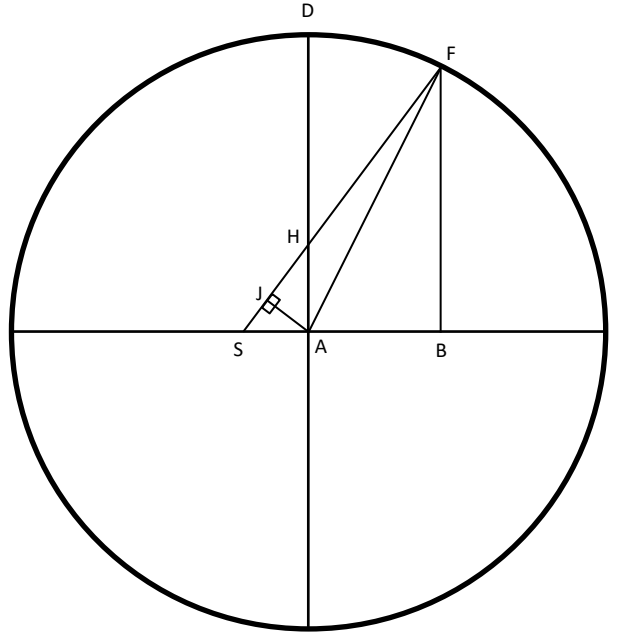
$\text{area FAZ} = \frac{\text{BAF}}{360} \cdot \pi r^2 = \frac{\text{BAF}}{360} \cdot \pi$ area FAZ = part of the circle's area, which is π * r^2

$\text{area AFS} = .5 \cdot \text{AS} \cdot \text{BF}$ area of triangle AFS = .5 * width * height

$\text{area FSZ} = \text{area FAZ} + \text{area AFS}$ area FSZ is made up of sector FAZ and triangle AFS

$\text{area ESZ} = \text{area FSZ} \cdot \text{I}$ the inflation rate reshrinks everything back to the oval





R

area ESZ fully expanded:

$$\left(\frac{s^{-1}\left(s\left(t^{-1}\left(\frac{t(ESZ)}{\sqrt{1-AS^2}}\right)\right)\cdot\left(\left(c\left(t^{-1}\left(\frac{t(ESZ)}{\sqrt{1-AS^2}}\right)\right)\cdot AS\right)+\sqrt{1-\left(s\left(t^{-1}\left(\frac{t(ESZ)}{\sqrt{1-AS^2}}\right)\right)\cdot AS\right)^2}\right)\right)}{360}\cdot\pi+.5\cdot AS\cdot s\left(t^{-1}\left(\frac{t(ESZ)}{\sqrt{1-AS^2}}\right)\right)\cdot\left(c\left(t^{-1}\left(\frac{t(ESZ)}{\sqrt{1-AS^2}}\right)\right)\cdot AS+\sqrt{1-\left(s\left(t^{-1}\left(\frac{t(ESZ)}{\sqrt{1-AS^2}}\right)\right)\cdot AS\right)^2}\right)\cdot\sqrt{1-AS^2}\right)$$

To keep this a bit tidier, let's define angle ASH once and keep it unexpanded in the final formula:

$$ASH = t^{-1}\left(\frac{t(ESZ)}{\sqrt{1-AS^2}}\right)$$

area ESZ =

$$\left(\frac{s^{-1}\left(s(ASH)\cdot\left(c(ASH)\cdot AS\right)+\sqrt{1-(s(ASH)\cdot AS)^2}\right)}{360}\cdot\pi+.5\cdot AS\cdot s(ASH)\cdot\left(c(ASH)\cdot AS+\sqrt{1-(s(ASH)\cdot AS)^2}\right)\cdot\sqrt{1-AS^2}\right)$$

That's a little more manageable.

But, there's two problems with this formula.

One problem is that once angle EAZ becomes greater than 90°, the size of FAZ begins to shrink instead of growing larger (due to the max range of arcsines), until FSZ becomes greater than 90°.

The other problem is that once ESZ becomes greater than 90°, the size of AFS flips into the negative, since line AH would now have a negative length (since it's based on a tangent of an angle that's greater than 90°, which is outside the range of tangents.)

R

I don't really want to fix those right now because I want to get to the point.

Let's compare the amount of time it takes earth to move through two separate areas of its orbit, that are the same amount of degrees relative to the sun.

ESZ is the earth's position relative to the farthest point of earth's orbit (when earth is farthest from the sun, what's called aphelion, which means "farthest from the sun", point Z on the right side of our diagrams).

We'll describe earth's position in degrees, and it refers to ESZ in our diagram/formula.

1st movement = 0° - 15°

2nd movement = 65° - 80°.

Both movements cover a range of 15°.

How long does each movement take?

How much area was covered in the first movement? Plugging into our formula 15° for the angle ESZ (within the formula) gives us the total amount of area earth "swept", the area of ESZ, which comes out to about **0.1353** (of the total oval).

For the second movement, we need to calculate the area of ESZ twice, once using the angle of ESZ as 65°, and the other using the angle of ESZ as 80°, and find the difference between them.

angle ESZ = 80°:	area ESZ =	0.7145
angle ESZ = 65°:	area ESZ =	0.5824
		−
Difference:		**0.1321**

Which is less area and is therefore slightly faster than the first movement.

Turning this into time. Let's take the Gregorian year as a working example:

Year Length = 365.2425 days

Oval area = $\pi \cdot I$ really $\pi r^2 \cdot I$, but r is 1 so that goes away

Movement_%_of_Oval = Movement / Oval area

Movement Days = Movement_%_of_Oval · Year_Length

First Movement: 15 days 17 hours ~34 minutes

Second Movement: 15 days 8 hours ~39 minutes

Difference: **~9 hours**

Now, if we tried to do all of this backwards, that is to input time and return degrees, we're going to run into a problem. Because now we're going to have to take area ESZ and figure out angle ESZ but how would we do that?

Turn time (number of days) into area by:

$$\text{area ESZ} = \frac{\text{days}}{\text{year}} \cdot \pi r^2 \cdot I$$

$$\text{area FSZ} = \frac{\text{days}}{\text{year}} \cdot \pi r^2 = \frac{\text{days}}{\text{year}} \cdot \pi$$

$$\text{area FSZ} = \text{triangle AFS} + \text{sector FAZ}$$

Now, if only BF were known, we could solve this equation.

$$\text{triangle AFS} = .5 \cdot AS \cdot BF$$

$$\text{sector FAZ} = \text{angle FAZ} \div \pi r^2 = \text{angle FAZ} \div \pi$$

$$\text{angle FAZ} = \sin^{-1}(BF)$$ since $\sin(\angle FAZ) = BF / AF$, and $AF = 1$

so, $$\text{area FSZ} = .5 \cdot AS \cdot BF + \sin^{-1}(BF) \div \pi$$

Which is basically unsolvable in a direct way, since it's like saying $x = \sin(x)$, now go and solve for x.

So, the way I would go about solving this is to write a computer program that runs what's called a binary search, that is, the computer basically guesses and checks, then readjusts its guess, getting closer and closer to the right answer.

But that's not really the point here now so we won't go there.

At the end of Appendix Q, we wrote that one might assume that the amount of degrees the earth moved around the sun was directly proportional to the amount of time that has passed, but we see here that it's not so.

Let's use the example from page 467.

We initially assumed that 15° of earth movement would take $\frac{15}{360} \cdot$ days_in_a_year.

Taking a year to be 365.2425 days yields:

$$15° = 15 \text{ days } 5 \text{ hours } \sim 15 \text{ minutes}$$

Which is pretty close but as we showed on the previous pages, not exact.

So, you need to take this into account if you wanted to be super accurate.

If you want to make the cheshbonos yourself, take into account that the Spring Equinox is when angle ESZ = 257.105°, in the year 2000 according to Wikipedia[369], but be aware that this angle is slowly getting smaller[370].

According to the Rambam[371], this angle gets smaller by 9''' per day[372].

On page 190, we wrote:

> "למעשה, I don't know why the differing tekufa lengths bothers Tosfos. The מציות is like that."

We have shown in the last few pages how to calculate the מציות.

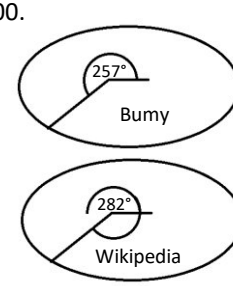
Ah, maybe you'll say Tosfos wasn't aware of the different lengths of the seasons?

Maybe. But the Rambam certainly was. (See Hilchos Kiddush Hachodesh chapters 11 and 12).

---

[369] https://en.wikipedia.org/wiki/Apsis#Earth_perihelion_and_aphelion. If I extracted this number correctly, from Perihelion = 282.895° in the year 2000.

[370] The way we're showing it is that earth's tilt is rotating clockwise relative to the orbit; the way the Rambam describes it the orbit is rotating counter-clockwise relative to earth's tilt; the way contemporary science describes it the orbit is rotating counter-clockwise relative to the stars while earth's tilt is rotating clockwise relative to the stars. Everyone is describing the same thing, just in different terms.



[371] Hilchos Kiddush Hachodesh 12:2

[372] that is, 9 / 60 / 60 / 60 degrees per day. Wikipedia says 62'' per year, which = 62 / 60 / 60 degrees per year. The Rambam's number would come out to ~55'' per year

Appendix **S**

Specific Cases

In this section, we're going to use the knowledge we gained from the previous appendixes and apply them to specific cases we came across in the main body of the sefer, or expand on a topic that we briefly touched on earlier.

Appendix **S.1**

Most Northern Sunrise in Eretz Yisroel or Bavel

On page 156, we wrote:

> "Due to a number of reasons, the most north that the sun will ever rise in Eretz Yisroel or Bavel is about 28° north of due east."

Let's see how that works.

On page 439, we demonstrated a formula for Input **L**atitude, Return **A**zimuth as:

$$A = \sin^{-1}\left(\frac{\sin(T)}{\cos(L)}\right)$$

Using:

$T = 23.4$

$L = 32$

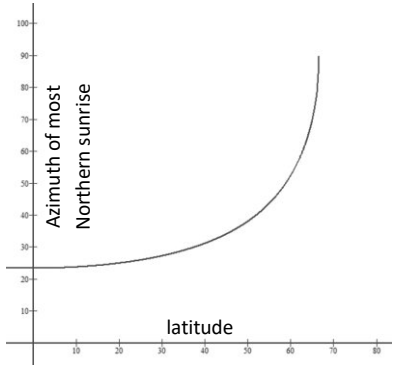
Returns:

27.92...°

Which rounds to 28°.

(The more north you go, the more northern sunrise you'll get.

Northern Eretz Yisroel and the northern Babylonian city of Pumbadisa (present day Fallujah) are about 33° north, which results in ~28.26° sunrise)



Then, on page 158, we wrote:

"I'm going to use 17.14 degrees for the error."

This number is derived by doing the above formula, using:

$T = 23.4$

$L = 31.778$ (Dome of the Rock)

Which results in:

$A = 27.851439...°$

If one mistakenly assumes this "A" to be 45° (exactly North-East), he will be off by:

$45 - A =$ **17.14856...**

Then I cut off after 2 decimal places to get 17.14.

Using the more accurate calculations that include sun width, refraction and elevation doesn't really change the results much, at least not enough to make a difference in the point we are trying to make in the sefer, which is to determine the pshat of the Gemara.

Adding in elevation:

Elevation = .8 km

CBD from page 451 = ~0.907...°

+ 5/6 (sun width + refraction)

Gives us a total of ~1.74...° the sun needs to be below the horizon

Let's use this angle for GBH in the diagram from page 446.

We need to know angle BHE (which is synonymous with the angle at point B, between point H and due east (off the left side of the page), the Azimuth of Sunrise).

GBH = ~1.74...

BG = 1 — the radius of the "sphere"

BH = c(GBH) — since c(GBH) = BH / BG, and BG = 1

$BE = BD + DE$ since that line is made up of those lines

$BC = s(T)$ as we demonstrated on page 433

$CBD = L$ same as angle BAC on page 436

$BD = \frac{BC}{c(CBD)}$ since c(CBD) = BC / BD

$DE = t(CBD) \cdot EF$ since CBD is same as DFE (see page 447), and t(DFE) = DE / EF

$GH = s(GBH)$ since s(GBH) = GH / BG, and BG = 1

$EF = GH$ two ends of a rectangle

$BHE = \sin^{-1}\left(\frac{BE}{BH}\right)$ since s(BHE) = BE / BH

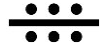
$$BHE = \sin^{-1}\left(\frac{\frac{s(T)}{c(L)} + t(L) \cdot s(GBH)}{c(GBH)}\right)$$

$BHE = \mathbf{29.09267...}$

So now, the amount of error produced by assuming this angle to be exactly north-east:

Amount of error = 45° - BHE = **15.9073**...°

Following the formulas on page 158 and 159 we will get **%10.546...** out of the techum, which can get as far as **707.36... amos** beyond the Techum edge.



Appendix **S.2**

Longest Day: Eretz Yisroel, Perpignan

On page 161 we wrote:

> "...in Eretz Yisroel and in Bavel there are about 14 and a quarter hours of daylight on the longest day."

In Appendixes O and P, we showed how to calculate the longest day. We showed on page 447 without elevation the longest day came out to about 14h 13m 18s, and we showed on page 451 with elevation came out to about 14h 18m 10s.

We continued on page 161:

> "...in Perpignan, France, which has a longest day of about 15 hours and 20 minutes."

Perpignan is located at about 42.7° north latitude.

Using the formula from page 447:

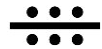
$T = 23.4$

$L = 42.7$

$GBH = 5/6$

Longest day on Perpignan, France ≃

15h 19m 7s



Appendix **S.3**

Sunrise 45° North of Due East

On page 162, we wrote:

> "What latitude would have a summer solstice sunrise of 45° north of due east?"

On page 439, we have a formula called: Input **A**zimuth (North of Due East), Return **L**atitude

Which goes:

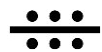
$$L = \cos^{-1}\left(\frac{\sin(T)}{\sin(A)}\right)$$

$T = 23.4$

$A = 45$

Returns:

L = 55.829...



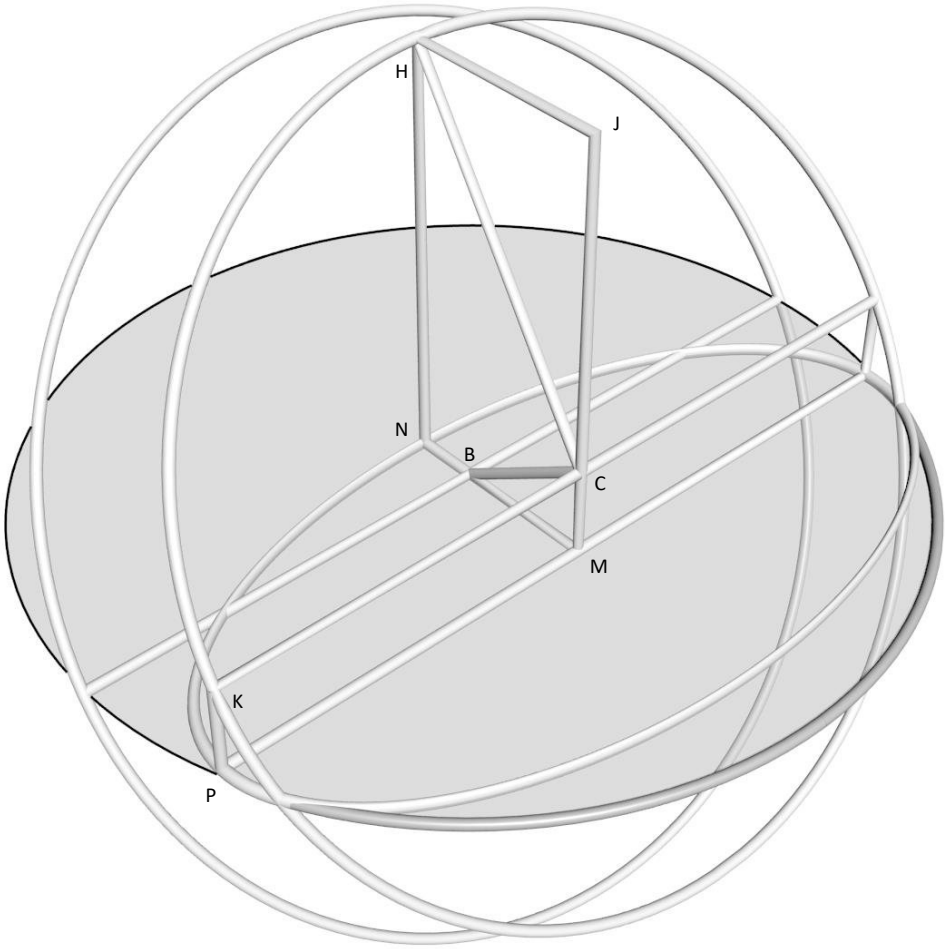
Projecting Sun Path Onto Ground

On page 165, we projected the Sun's Topocentric solstice path flat onto the ground.

We wrote in the footnote:

> "...the sun's orbit in the picture is the horizon scaled down by cosine(earth's tilt), and then along one axis by sine(latitude), and then off-centered along that same axis by sine(earth's tilt) * cosine(latitude)"

Let's see how we got that.



We're measuring the flat oval in the diagram (whose central long axis runs from P through M, and whose short axis runs from N to M)

> "...the sun's orbit in the picture is the horizon scaled down by cosine(earth's tilt)..."

In other words, the radius of the solstice ring (lines CK and CH) = c(T), as we showed on page 433, so

$CK = c(T)$

$CH = c(T)$

$MP = CK$ two sides of a rectangle

> "...and then [scaled down] along one axis [line MN] by sine(latitude)..."

That is, line MN is smaller than line MP by a factor of s(L)

Because:

$CBM = L$ same as angle BAC on page 436

$JCH = CBM$
since BCH = 90, and BCM = 90 - L, and since all the angles along CM, BC, CH, and CJ make up 180° so JCH = 180 - BCM - BCH, so JCH = 180 - (90 - L) - 90 = 90 - (90 - L) = 90 - 90 + L = L

$HJ = CH \cdot s(JCH)$ since s(JCH) = HJ / CH

$MN = HJ$ two sides of a rectangle

$MN = MP \cdot s(L)$ since MP is the same as (CK and) CH

> "...and then off-centered along that same axis [line BM] by sine(earth's tilt) * cosine(latitude)"

since:

$BC = s(T)$ as we showed on page 433

$BM = BC \cdot c(CBM)$ since c(CBM) = BM / BC

$BM = s(T) \cdot c(L)$

On page 166, we wrote:

> "B'kitzur, it comes out to 17 hours 16 minutes and ~50 seconds, for the longest day at latitudes with a 45° sunset."

How did we get that?

On page 441, we gave a formula to find how long the day is based on Sunset Ground angle (the Azimuth) as:

> Input **A**zimuth, Return **G**
>
> $G = \cos^{-1}\left(\frac{\cos(A)}{\cos(T)}\right)$

And as we wrote there to turn that into time do:

> $\text{Day Length} = \frac{2}{15}G + 12$

Which altogether would be:

> $\text{Day Length} = \frac{2}{15}\cos^{-1}\left(\frac{\cos(45)}{\cos(T)}\right) + 12$

Which comes out to:

> $\text{Day Length} = 17.280461112...$

Which is:

> **Day Length = 17h 16m 49.66...s**

Appendix **S.5**

Latitude with 18 Hour Longest Day

We wrote on page 166:

> "To find what latitude would have an 18 hour day one can use the simplified formula:
>
> $$\text{Latitude} = \tan^{-1}\left(\frac{\sqrt{.5}}{\tan(\text{earth's tilt})}\right)$$ "

How did we get that?

On page 440, we showed the formula for inputting Day Angle and returning Latitude.

With started with a formula to turn Length of Day into Day Angle:

$$G = (\text{Day Length} - 12) \cdot 7.5$$

$$G = (18 - 12) \cdot 7.5$$

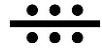
$$\mathbf{G = 45}$$

Then to turn that into latitude we did:

$$L = \tan^{-1}\left(\frac{\sin(G)}{\tan(T)}\right)$$

and we know that $\sin(45) = \sqrt{.5}$ (as we showed on page 400), so that's how we got this formula.

We wrote that this is a simplified formula, since we have shown in Appendix O and P that it can get more complicated, but that is going beyond the scope of the point that we were trying to make.



Appendix **S.6**

Solstice Offset of North Face

On page 168, we were discussing the idea that when the Solstice happens exactly at sunrise then by the time sunset comes around, the earth has already moved a bit past the point of the Solstice in its orbit, causing the Topocentric sun path ring to move a little closer (southward) towards the equinox ring, which will cause the sun to set at a slightly more southern position than that morning's sunrise. (See last few appendixes in case anything I just said didn't make sense.)

If you go to page 433, you'll see there diagrams that show the Solstice ring's angular distance from the Equinox ring as being equal to T (the tilt of the earth, 23.4°).

On page 459, we came up with a formula that you could use to replace the value of T as the angular distance to the Solstice ring to get the sun path for a different day of the year. The term we used there was GDH, which will replace T in the regular formulas of Appendix N and thereabouts. The formula for GDH was:

$$GDH = s^{-1}(c(x) \cdot s(T))$$

x being the number of degrees earth has moved past the Summer Solstice.

Since we are using the Summer Solstice as our starting point, at the Summer Solstice itself x would be = 0°. Plugging that into our formula returns exactly 23.4.

So, for sunrise, we will use the regular value of T to determine the sunrise ground angle (the Azimuth).

Using the formula from page 439 labeled Input **L**atitude, Return **A**zimuth which was:

$$A = \sin^{-1}\left(\frac{\sin(T)}{\cos(L)}\right)$$

Using:

$T = 23.4$ as we showed with GDH above

$L = 31.8$ approximation of Yerushalayim's latitude

Yields:

$A = 27.85864485...$

Now for sunset.

As we wrote on page 168, the longest day in Yerushalayim is about 14 hours and 15 minutes. That itself is a rounded number, as we have been showing all these last appendixes, and it must also be shortened by a tiny bit since the sun is already moving southward (as we are showing now), so the day will be a tiny bit shorter.

But these are tiny differences, and we were trying to show in the text that the entire result will end up being really tiny, so these tiny adjustments to the calculation are going to affect the result in an even tinier way, so it doesn't change the point that we were trying to make, which is that the results are so tiny that it is hard to accept that this was the problem the Gemara was talking about.

I wrote out this whole paragraph instead of just making the tiny adjustments because the whole approach is recursive, meaning, each tiny adjustment of the day

length will affect moment of sunset, which will affect position of sunset, which will affect the day length, and the cycle will continue forever into an infinitely tiny difference.

So now, taking the day length as 14h 15m, and ignoring everything we wrote in Appendix R (since we would be overcomplicating the point, if we haven't already), and rearranging the last formula on page 461, we could say the earth moved:

Degrees = 14.25h / 24h / days_in_year · 360

Degrees = 0.58522762... using the Gregorian Year length of 365.2425 days

Using the last formula from page 459 (to find GDH), we can turn those degrees into the sun path's angular distance from the Equinox ring

$$GDH = s^{-1}(c(Degrees) \cdot s(T)) = 23.39870664...$$

Using this value instead of "T" in the formula on page 439 labeled Input Latitude, Return Azimuth results in

$$A = \sin^{-1}\left(\frac{\sin(GDH)}{\cos(L)}\right) = 27.85706515...$$

The whole formula altogether would be:

$$A = \sin^{-1}\left(\frac{c\left(\frac{Days}{Year} \cdot 360\right) \cdot s(T)}{\cos(L)}\right)$$

So now viewing the sunrise and sunset azimuths side by side:

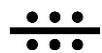
Sunrise = 27.85864485...

Sunset = 27.85706515...

Difference:

0.00157970...

Fake North Error Angle = 0.00078985...



Pill-Shaped City, Narrowest Migrash Point

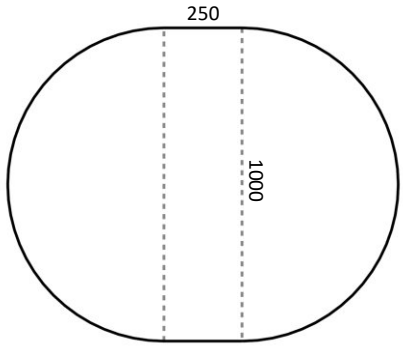
On page 231, we were discussing the possibility of having a pill-shaped city surrounded by an oval-shaped Migrash.

We wrote that the narrowest distance between the city's edge and the Migrash's edge is "about 970 amos".

Let's see how we got that number.

The city's central area is a rectangle that is 250 amos wide by 1,000 amos "tall".

The side half-circles of the city have a radius of 500 amos (i.e., each half-circle is 1,000 amos tall by 500 amos wide).



Let's mathematically describe the top-right curve of this circle, so we can view it on a graphing calculator.

If that quadrant (=quarter circle) was centered over the center of the graph, its radius could be viewed as the hypotenuse of a triangle whose width is x and whose height is y.

Using the Pythagorean Theorem, we would write out the triangle as:

$a^2 + b^2 = c^2$

Which is our case would be

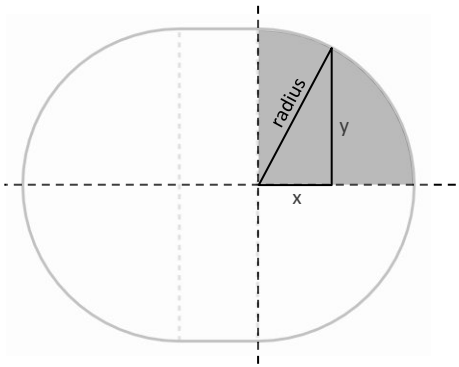
$x^2 + y^2 = \text{radius}^2$

So now, let's rearrange this so we can plot this circle on a graph, in the format of y = something

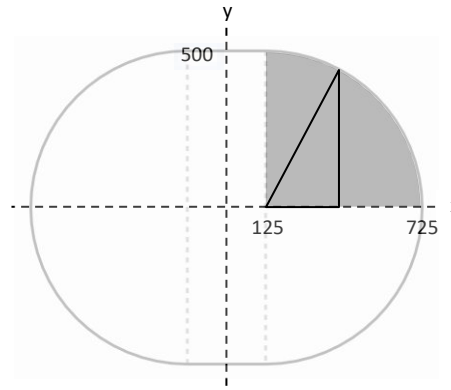
$y^2 = \text{radius}^2 - x^2$

$y = \sqrt{\text{radius}^2 - x^2}$

$\text{radius} = 500$



Now, because the quadrant is moved along the x axis by 125, we need to modify x in the formula to x - 125 (so that the x's that come later will act as though they were x's from earlier).



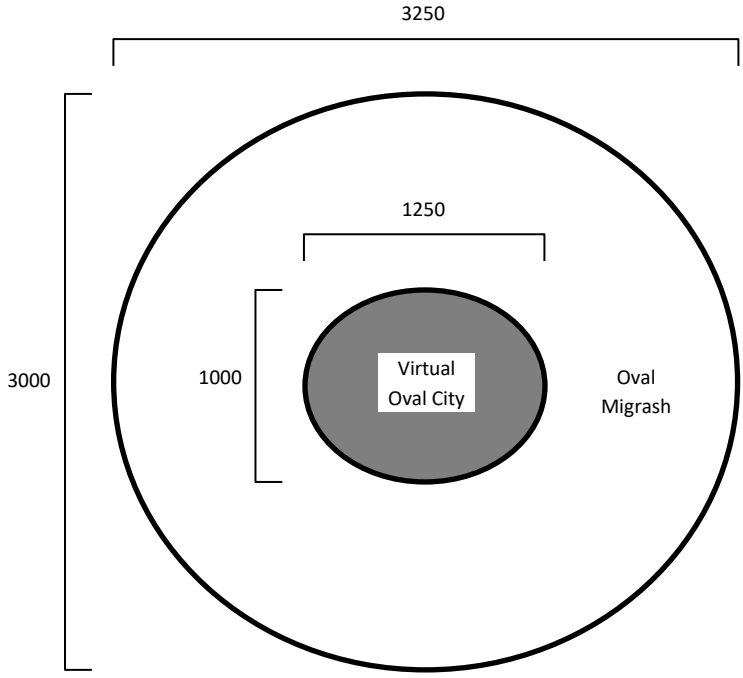
Pill-City Quadrant:

$$y = \sqrt{500^2 - (x - 125)^2}$$

Now let's graph out the Migrash.

The Migrash is based on a virtual city that is oval shaped with a width of 1250 and with a "height" of 1000.

The Migrash itself is 1000 amos more in each direction, so that's an oval with a width of 3250 and a "height" of 3000.



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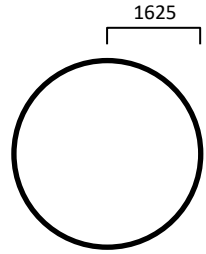
To graph this out we need to first make a circle that is 3250 wide. That would be a radius of 1625. Following the method that we did for the pill-city, we would get a formula of:



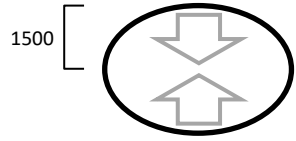
$$y = \sqrt{1625^2 - x^2}$$

Now, we need to squash this along the y-axis an amount that the resulting vertical radius will be half of 3000, i.e., 1500.

initial_vertical_radius · squash_rate = target_vertical_radius

1625 · squash_rate = 1500

squash_rate = 1500÷ 1625

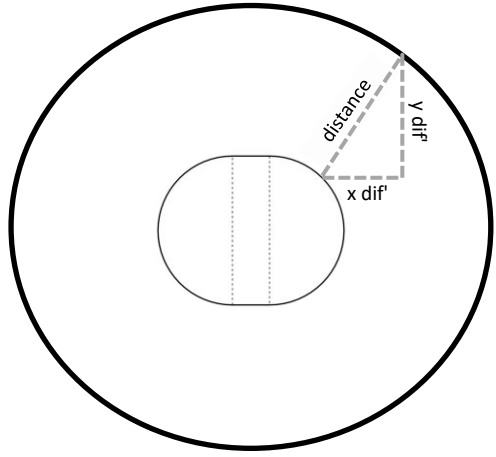


So the final formula will be:

Migrash: $y = \sqrt{1625^2 - x^2} \cdot 1500 \div 1625$

Now, I don't know how to directly solve for the narrowest point between these two curves so I'm going to do it the lazy way and write a computer program that will go through each point of the pill-city's curve, and for each point will then measure the distance to each point in the Migrash curve, keeping a record of the shortest distance.

For each point along the city's curve and for each point along the Migrash's curve, make a triangle whose sides are made up of a horizontal line that is the difference between the x positions of each point, and a vertical line that is the difference between the y positions of each point, and a diagonal line that is the hypotenuse of those first two lines, which is the distance between the city edge and the Migrash edge, measured from those specific points.



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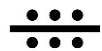
Python code:

```
from math import sqrt

min_distance = 1000 #starting record of minimum distance

for city_x in range(125,625):

    city_y = sqrt(500**2 - (city_x - 125)**2)

    for migrash_x in range(125,1625):

        migrash_y = sqrt(1625**2 - migrash_x**2) * 1500 / 1625

        distance = sqrt((migrash_x - city_x)**2 + (migrash_y - city_y)**2)

        if distance < min_distance:
            min_distance = distance

print(min_distance)
```

Result: **969.69...**



Appendix **U**

Python Code for Tosfos Ein Bein

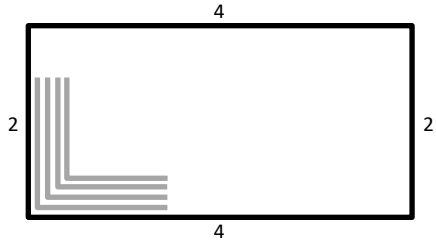
Here's the Python code used to generate the lists from pages 185 to 189.

The first two lines defines the day and hour type, which can be kavuos or zmaniyos. You'll need to change those to see the other types of charts. The charts display 40 seasons (you can change the number of seasons by changing the value of "n" on the 5th line). Each season will be displayed on a single row. The first item in the row is the starting time of day for that season. The second item is the length of that season. And the third item is which season (0 = Nissan, 1 = Tammuz, 2 = Tishrei, 3 = Teves).

(This code prints each season length on the next line with the next season, after it calculates the starting point of the next season.)

```
d = 'z' #day type, 'k' or 'z'
h = 'k' #hour type, 'k' or 'z'
t = 18 #starting time
s = 0 #starting season
n = 40 #number of seasons to add

L = 18 #length of longest day

season = [[6,18,12,12]] * 4 #Nissan and Tishrei [sunrise,sunset,daylength,nightlength]

season[1] = [(24-L)/2,24-(24-L)/2,L,24-L] #Tammuz

season[3] = [6+(24-L)/2,18-(24-L)/2,24-L,L] #Teves

def zhours(t,s): #convert from k hours to z hours
    if t % season[s][1] > season[s][0]:
        return (t - season[s][0]) / season[s][2] * 12 + 6
    else:
        return ((t - season[s][1]) % 24 / season[s][3] * 12 + 18) % 24

def khours(t,s): #convert from z hours to k hours
    if t % 18 > 6:
        return (t - 6) / 12 * season[s][2] + season[s][0]
    else:
        return ((t + 6) % 24 / 12 * season[s][3] + season[s][1]) % 24


def addseason(t,s): #add a single season of (91 days and) 7.5 hours

    #add days
    if d == 'z': t = zhours(t,s) #for zmaniyos days, convert time to z time
    s += 1 #add 1 season
    s %= 4 #keep seasons within 0 to 3
    if d == 'z': t = khours(t,s) #for zmaniyos days, convert time to k time

    #add hours
    if h == 'z': t = zhours(t,s) #for zmaniyos hours, convert time to z time
    t += 7.5
    t %= 24
    if h == 'z': t = khours(t,s)#for zmaniyos hours, convert time to k time

    return t, s

def hours(t,s): #add many seasons and print, including previous season length
    th = t #time hold, so we can subtract next season's time to find length
    print(str(t)+'\t-\t'+str(s))
    for x in range(n):
        t,s = addseason(t,s)
        length = str(round((t + 24 - th) % 24,4)) #calculate previous season length
        th = t #assign "new time" to "time hold" for next round
        print(str(round(t,4))+'\t'+length+'\t'+str(s))


hours(t,s) #run program
```

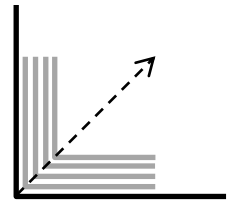
Proof For Tosfos Kamma, No Center Point For Rectangle



On page 213 we wrote: Actually, if we look at this [2x4 rectangle], we'll see there's no single straight cut we could make that would yield an evenly spaced path for the strings to travel through.

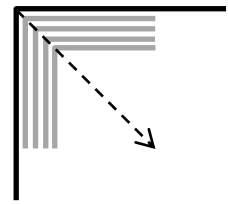
When strings are running along the edge, they will turn with the outer edge. The strings that are further in from the edge will turn by that corner as well, just further in.

We could draw a line between the two edges, a line that bisects the corner's angle, and that line represents all the possible points to which a straight line could be cut to from some other edge of the rectangle.

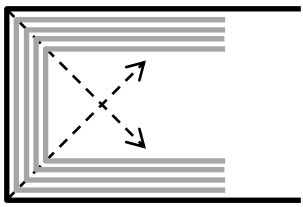


Let's show that for the bottom left corner.

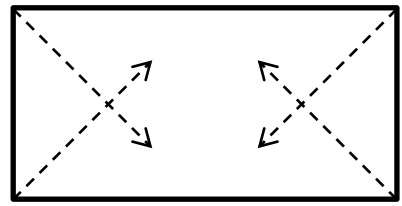
So now, if we could bisect another corner of this rectangle, say the top left, that new line would represent all the possible points to which a straight line could be cut to so that all the strings in the shape would wrap uniformly around it.



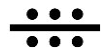
Taking these two corners together gives us two lines that represent all the possible end points <u>for the strings of that corner</u>. Since we're looking for a single point in this whole shape, the point where these two lines intersect would be the one point we could cut to that would satisfy the requirements of the strings in both corners.



However, because this is a rectangle (and not a square), we're going to have a problem with the other side, the right side. Since there too we would draw two corner-bisecting lines that meet on the right side, that means our <u>one single point</u> needs to be there.



So, because we can't have a single point on two separate sides of the rectangle, that proves that there's no single cut we could make on it that would allow all the strings to go around it in a uniform manner.



Rav S.R. Hirsch On Melacha of Carrying

The meaning of the law of carrying on Shabbos

Rav Shimshon Raphael Hirsch writes[373] about the differences between carrying and all the other melachos.

The other melachos make a significant change to some object. Even gathering and trapping, although there's no physical change to the object, it is an essential change, as you brought something from a state of being wild into a state of domestication.

Carrying, however, makes no such change. The item is brought back or forth between the individual's and public's domain, or at least 4 amos within the public domain. The item remains the same. It does not change ownership (necessarily), it does not become more or less free. It is merely the change of an item's position vis-a-vis the individual's and/or the public's domains.

Rav Hirsch writes that these two categories (all the other melachos and carrying) represent two aspects of the world.

There is **the material world**, represented by the other melachos which demonstrate man's mastery over the physical world; his intelligence re-forming the raw materials of the earth for his benefit.

And then there is **history**, or **the social world**, which deals with man's interpersonal relationships, his own personal life, his relationship to the public, and the relationships within the public. All of this is represented by the laws of carrying. Taking something from one's private domain to the public represents the idea of the individual contributing to the public. Taking something from the outer public domain to his own inner personal space represents the individual "absorbing" from the public, what the public provides for him. And then there is carrying at least 4 amos within the public domain which represents man, as an individual (4 amos) making social changes (within the public domain).

These two aspects of man's mastery over the world are the two aspects of what Shabbos represents. By abstaining from these types of "mastery" on Shabbos, the Jew demonstrates that it is not he who is control but rather it is Hashem.

We find these two aspects expressed in the Torah as the reasons provided as to why we are keeping Shabbos.

---

[373] Shemos, beginning of Chapter 35

"As a remembrance to the **6 days of creation**" and "As a remembrance to the **Exodus from Egypt**".

The 6 days of creation is Hashem's demonstration of his total control over the **material** world.

The Exodus demonstrates Hashem's total control over **nations and history**.

Rav Hirsch continues that this is why when Yirmiyahu was warning the people about the imminent fall of the Jewish State, he exhorted them specifically to keep the laws of **carrying**, since this law represents the State which - in its most essential form - is the organization of the relationship between individuals and the public.

"So has Hashem spoken to me: Go and stand in the gate of the children of the people, through which the kings of Yehuda will come in and through which they will leave, and in all the other gates of Yerushalayim. Say to them: Hear the word of Hashem, O kings of Yehuda, and all of Yehuda, and all the dwellers of Yerushalayim who are coming through these gates, so has Hashem said: Guard with your soul and don't carry a burden on the day of Shabbos, to bring it in through the gates of Yerushalayim, and don't bring a burden out from your houses on the day of Shabbos, and all work don't do, and you shall sanctify the day of Shabbos as I commanded your fathers. But they did not listen to Me, and they did not incline their ear, and they hardened their necks, to not hear and to not accept ethical teaching. And it will be if you will truly listen to Me, the word of Hashem, to not bring in a burden through the gates of this city on the day of Shabbos, and to sanctify the day of Shabbos, to not do in it any work, then kings and princes who sit on the throne of Dovid will come in through the gates of this city riding in chariots and on horses, they and their princes, each man of Yehudah and the dwellers of Yerushalayim, and this city will remain settled forever. And they will come from the cities of Yehuda, and from the suburbs of Yerushalayim, and from the land of Binyamin, from the lowland, and from the mountain and from the desert, bringing offerings of Olah, Shelamim, Mincha, incense and Todah to the House of Hashem. But if you will not listen to Me, to sanctify the day of Shabbos, and to not carry a burden and come through the gates of Yerushalayim on the day of Shabbos, then I will light a fire in its gates that will destroy the palaces of Yerushalayim and it will not be extinguished."[374]

May we see Yerushalayim built soon, in our days, Amen.

[374] ירמיהו י"ז:י"ט - כ"ז